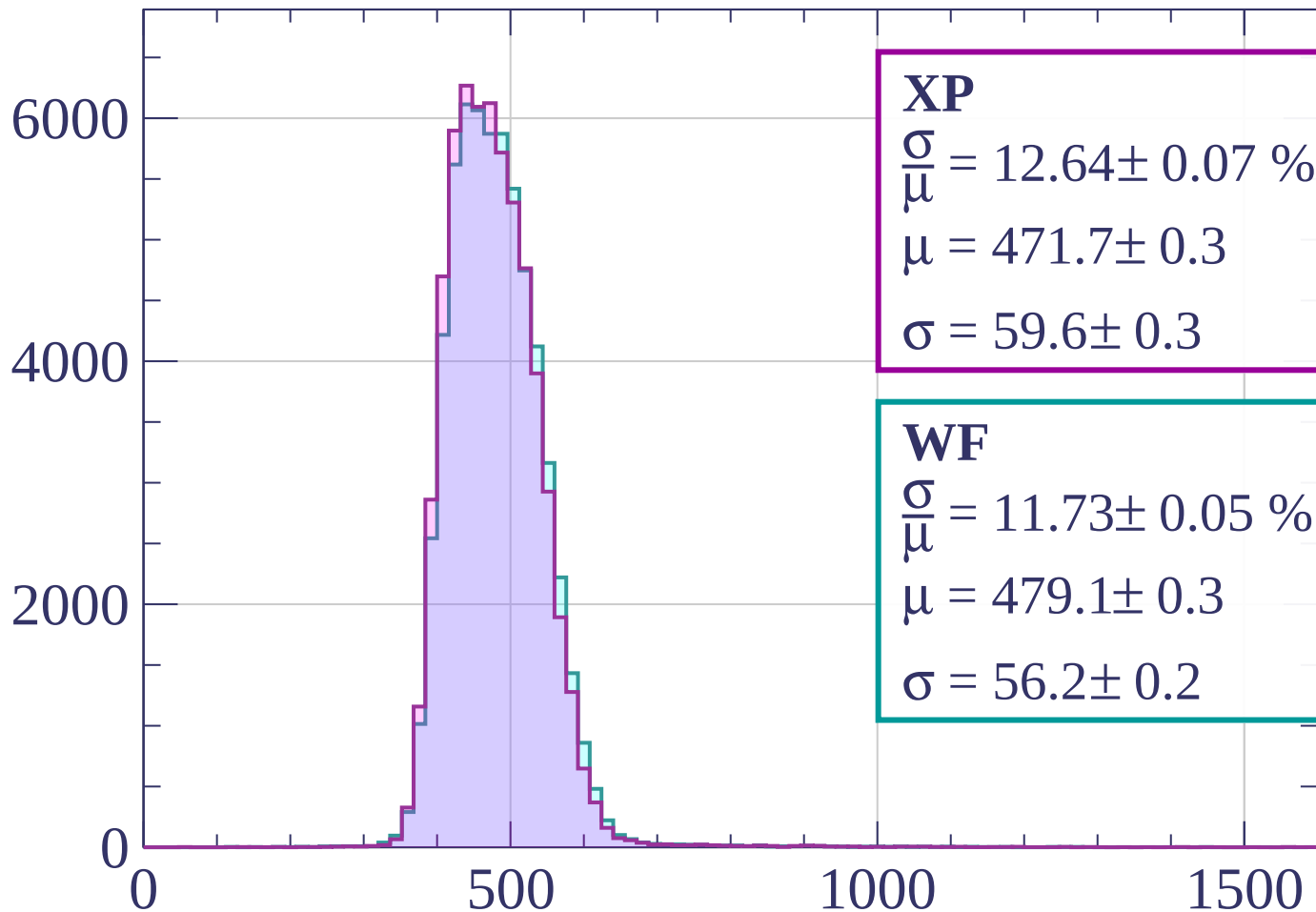


Count



XP

$$\frac{\sigma}{\mu} = 12.64 \pm 0.07 \%$$

$$\mu = 471.7 \pm 0.3$$

$$\sigma = 59.6 \pm 0.3$$

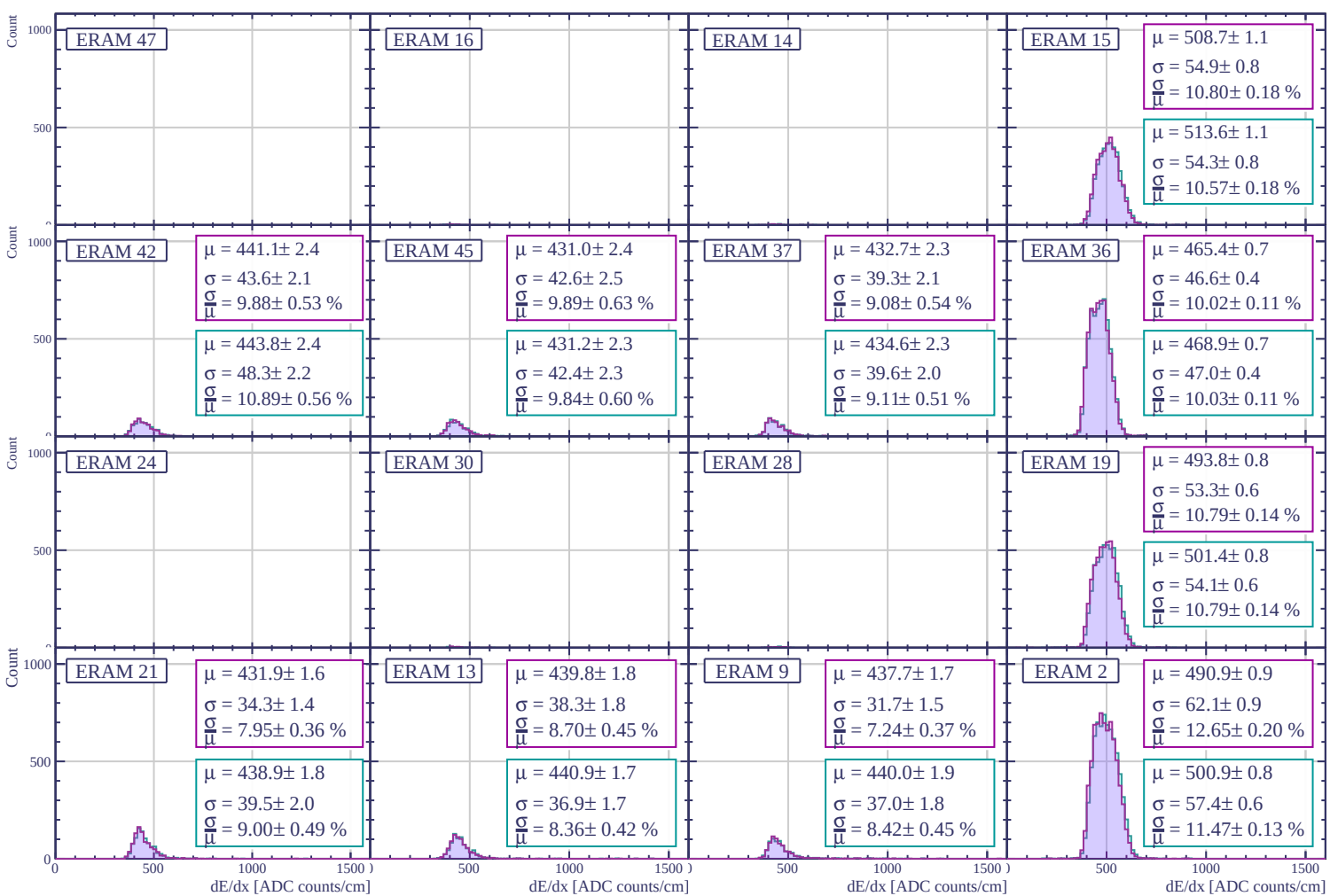
WF

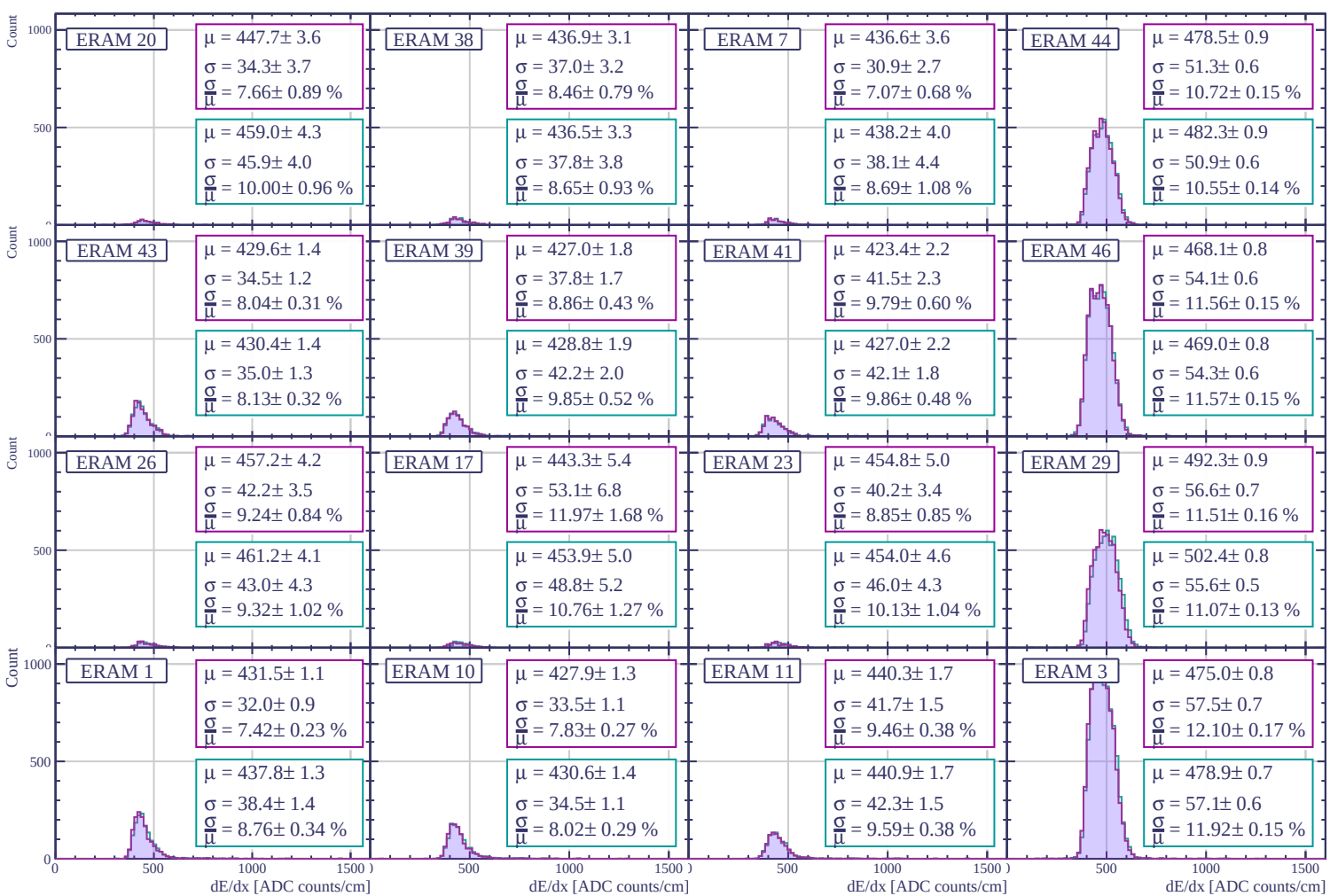
$$\frac{\sigma}{\mu} = 11.73 \pm 0.05 \%$$

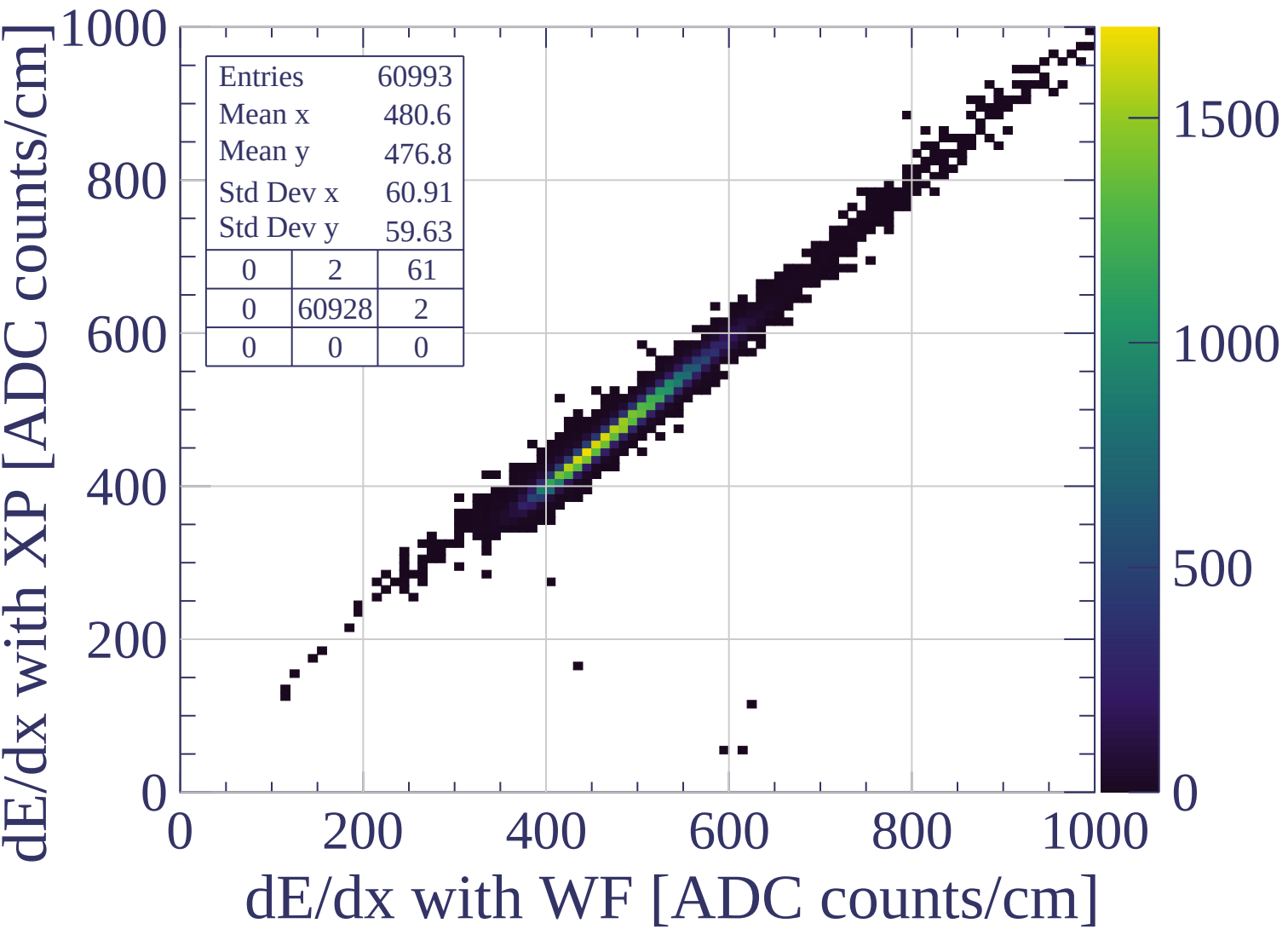
$$\mu = 479.1 \pm 0.3$$

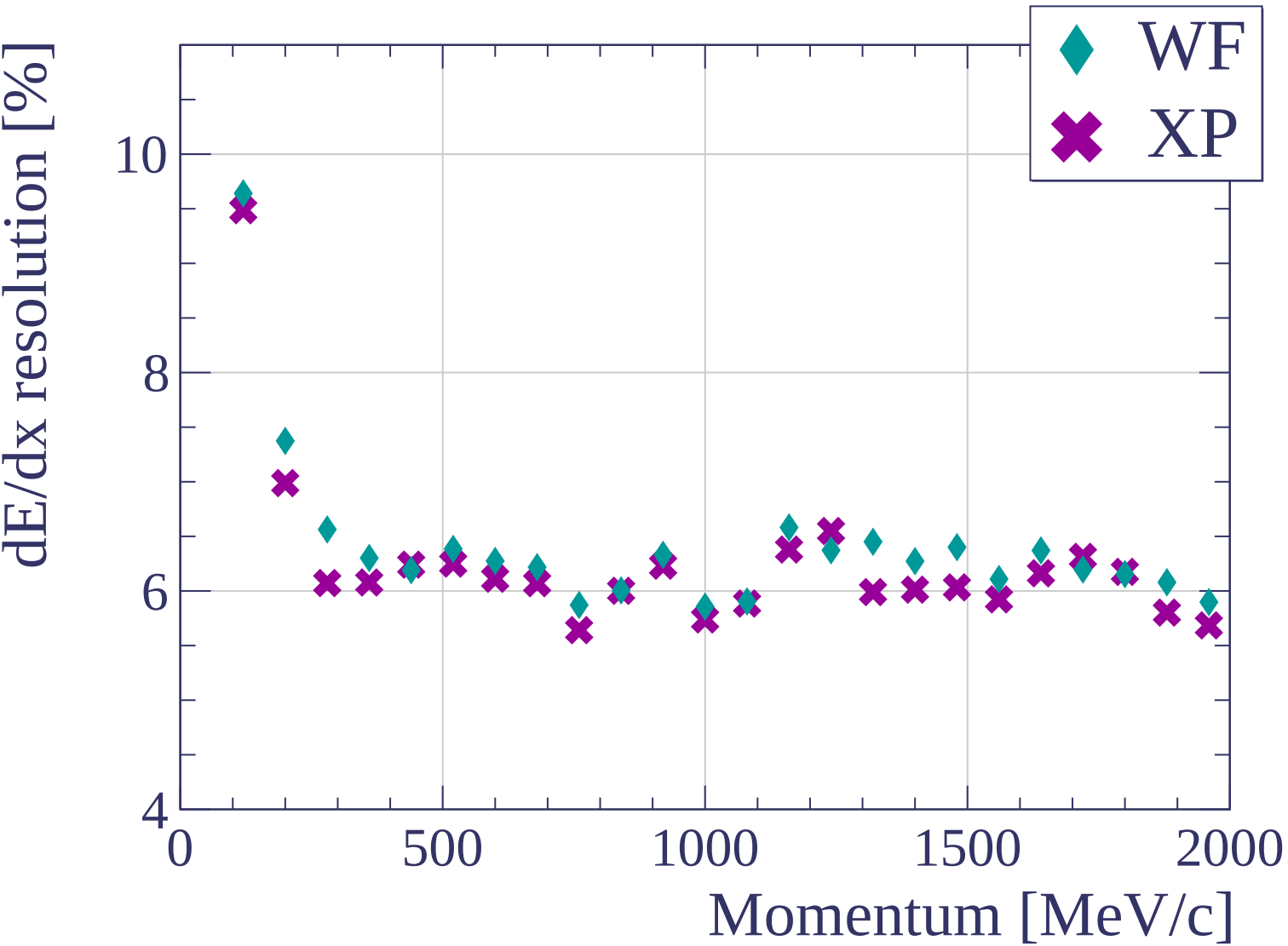
$$\sigma = 56.2 \pm 0.2$$

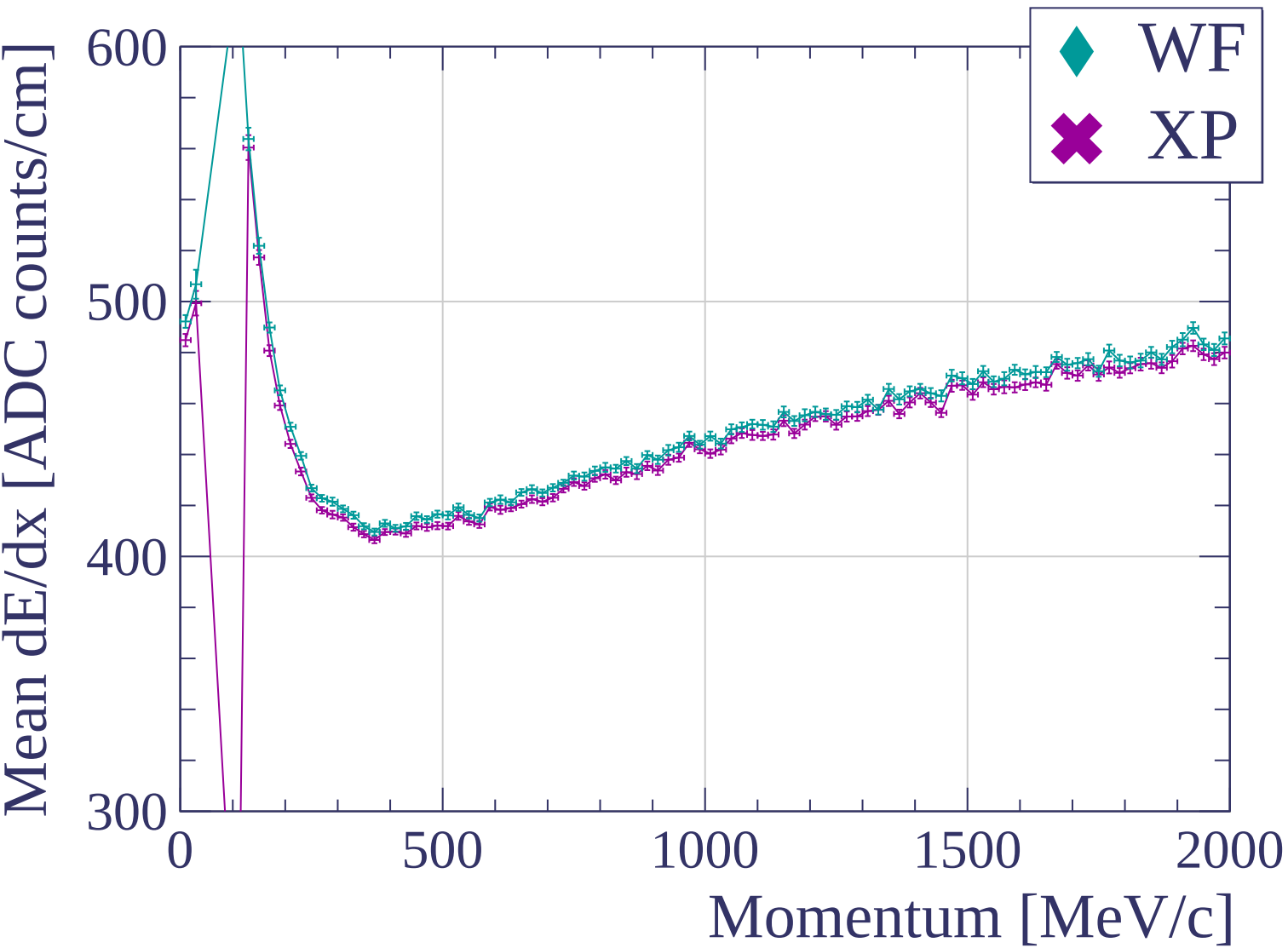
dE/dx [ADC counts/cm]

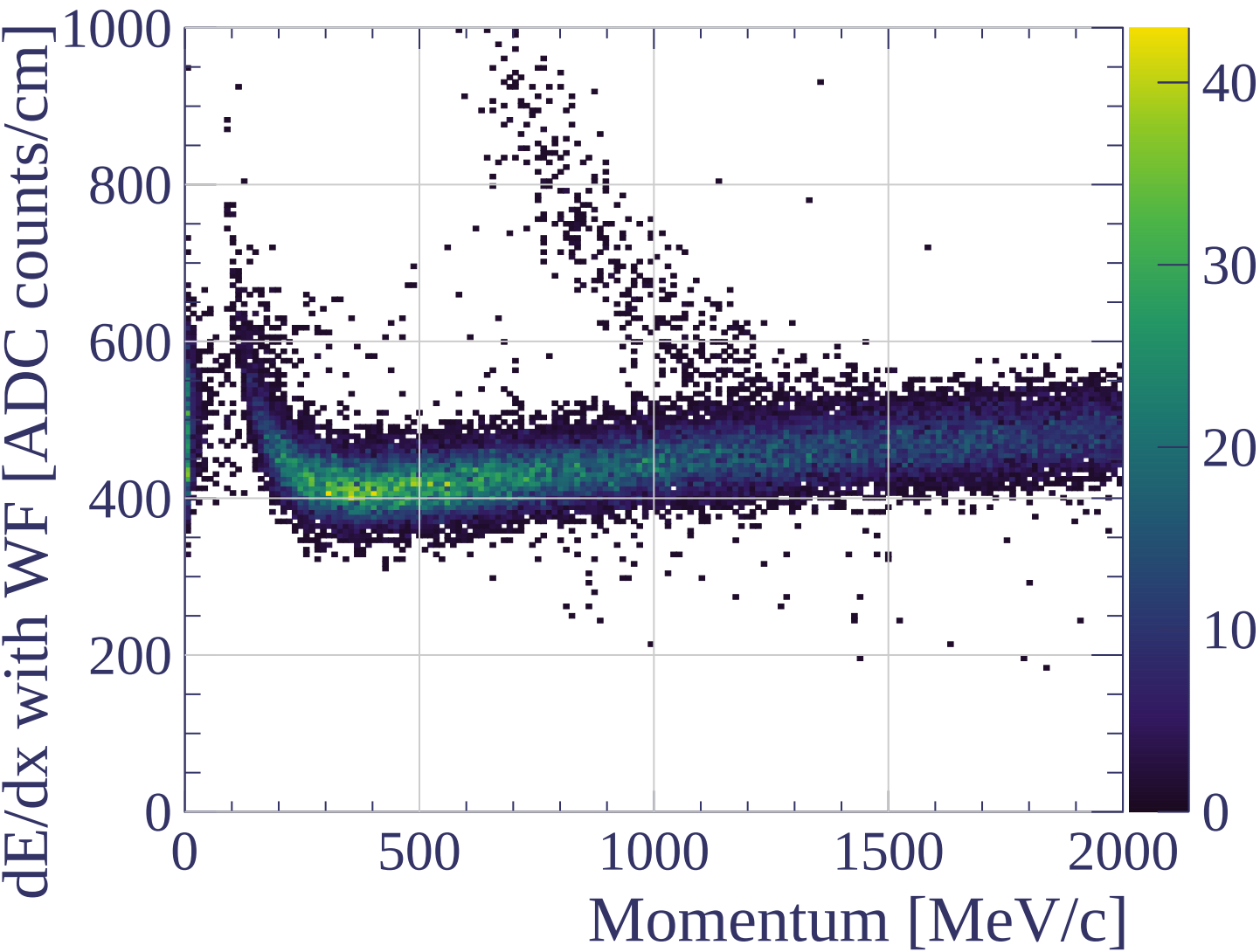


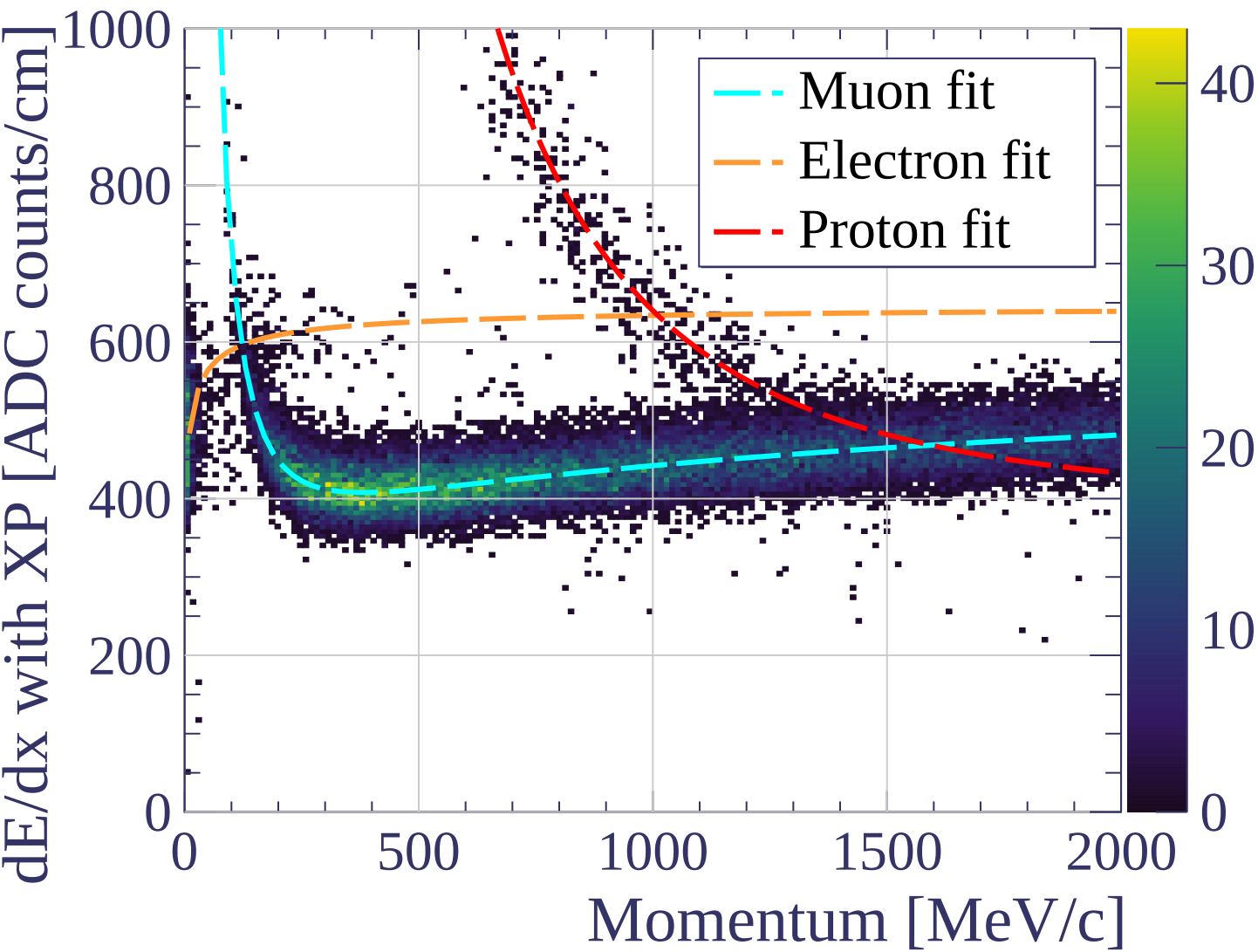




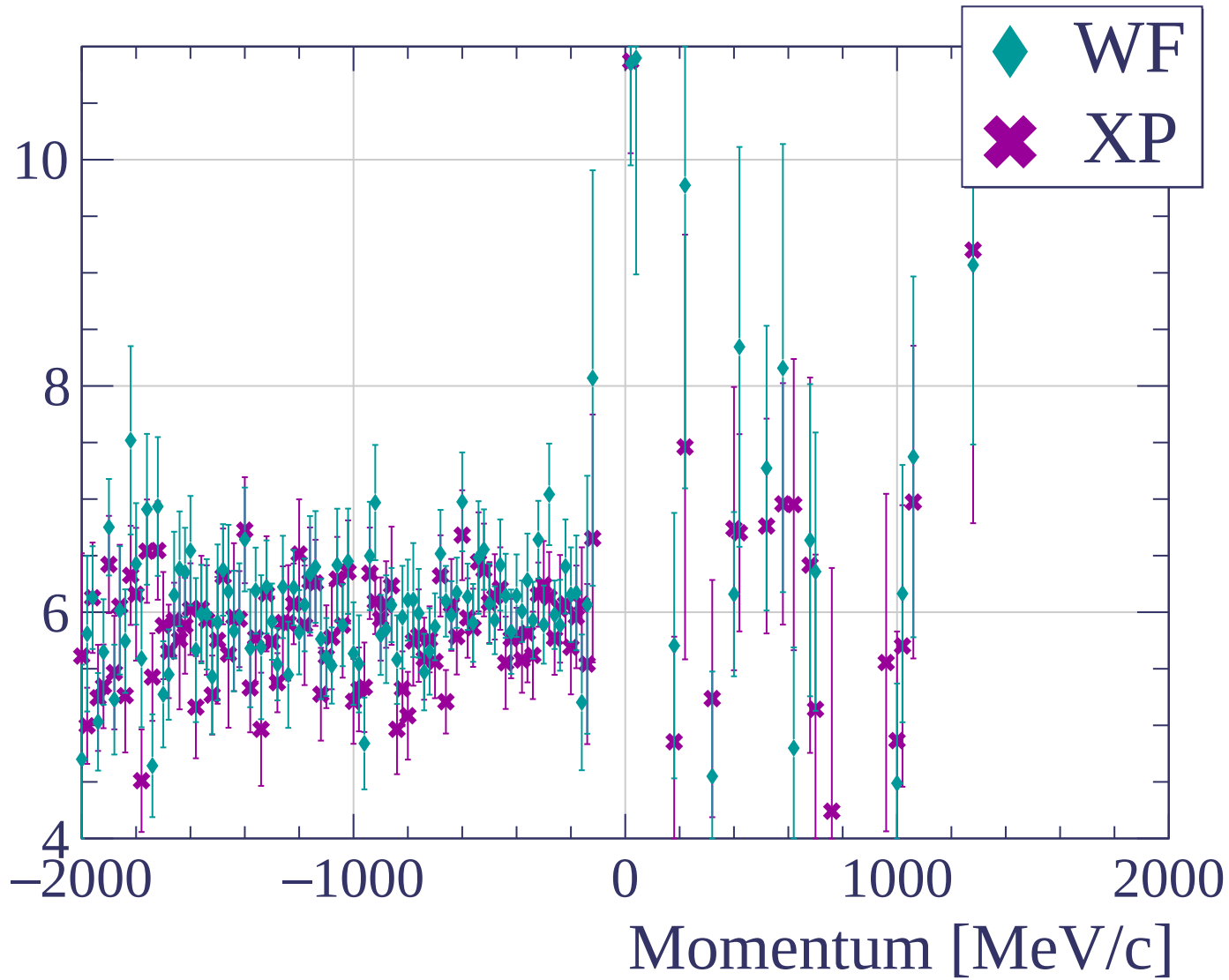


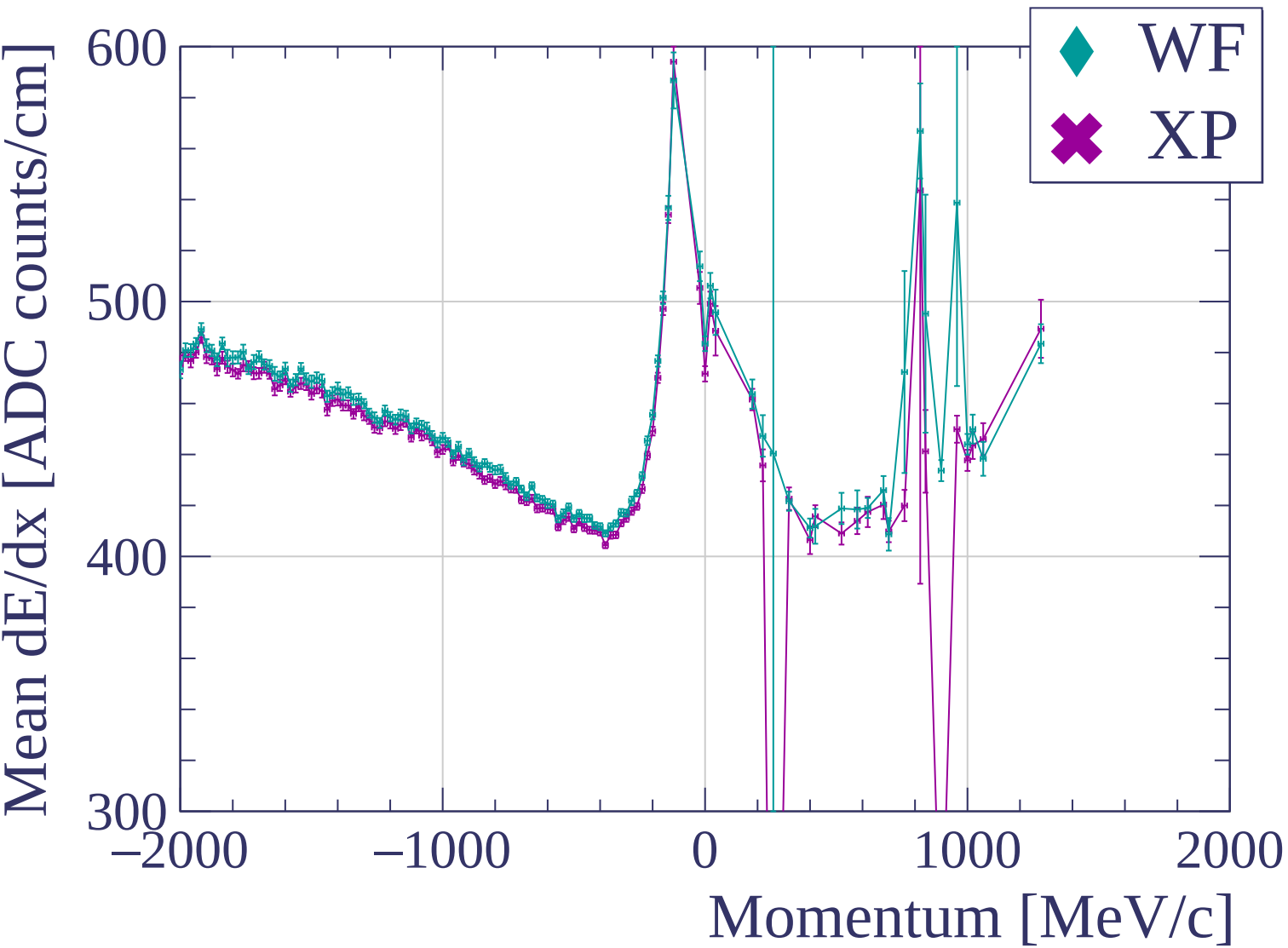


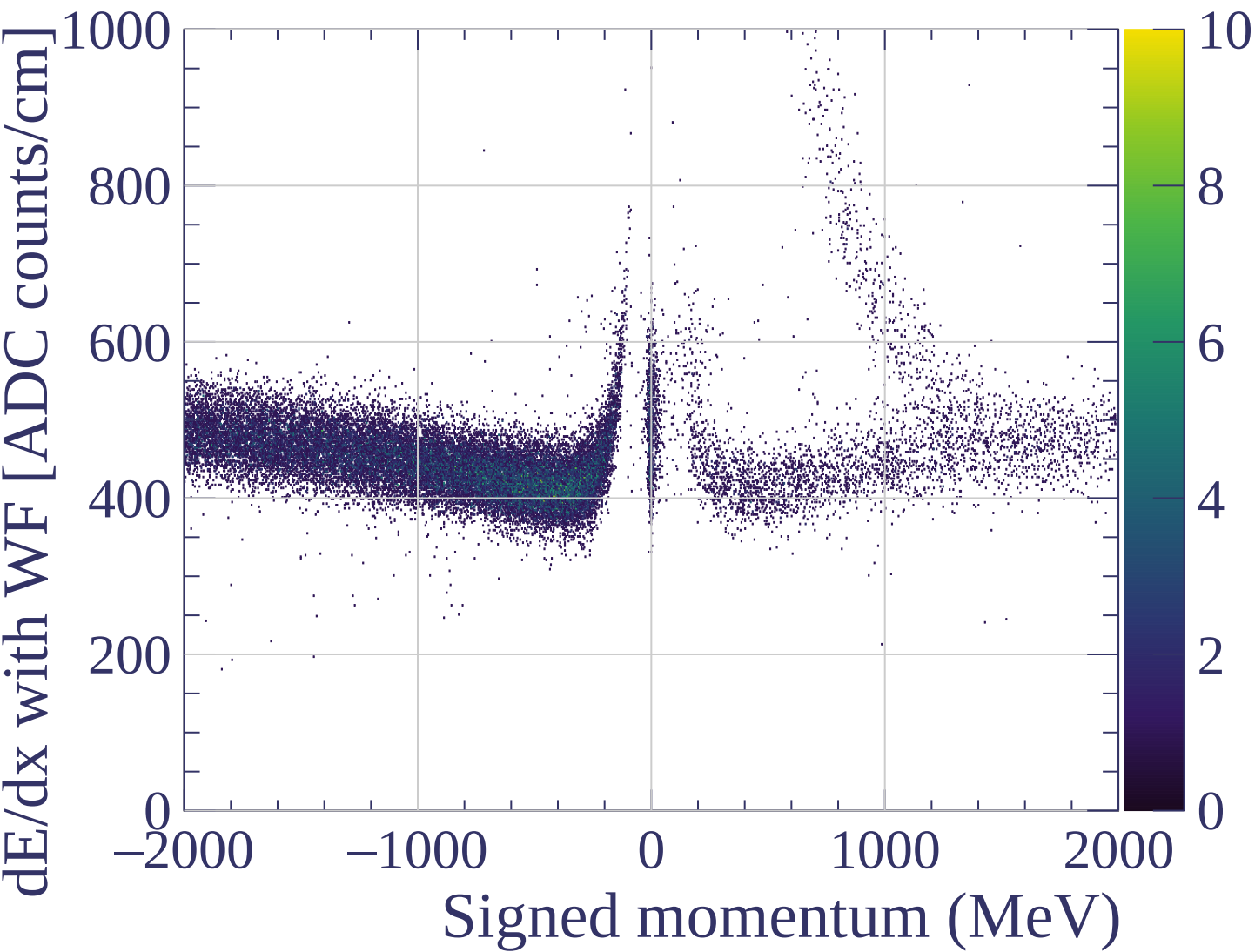


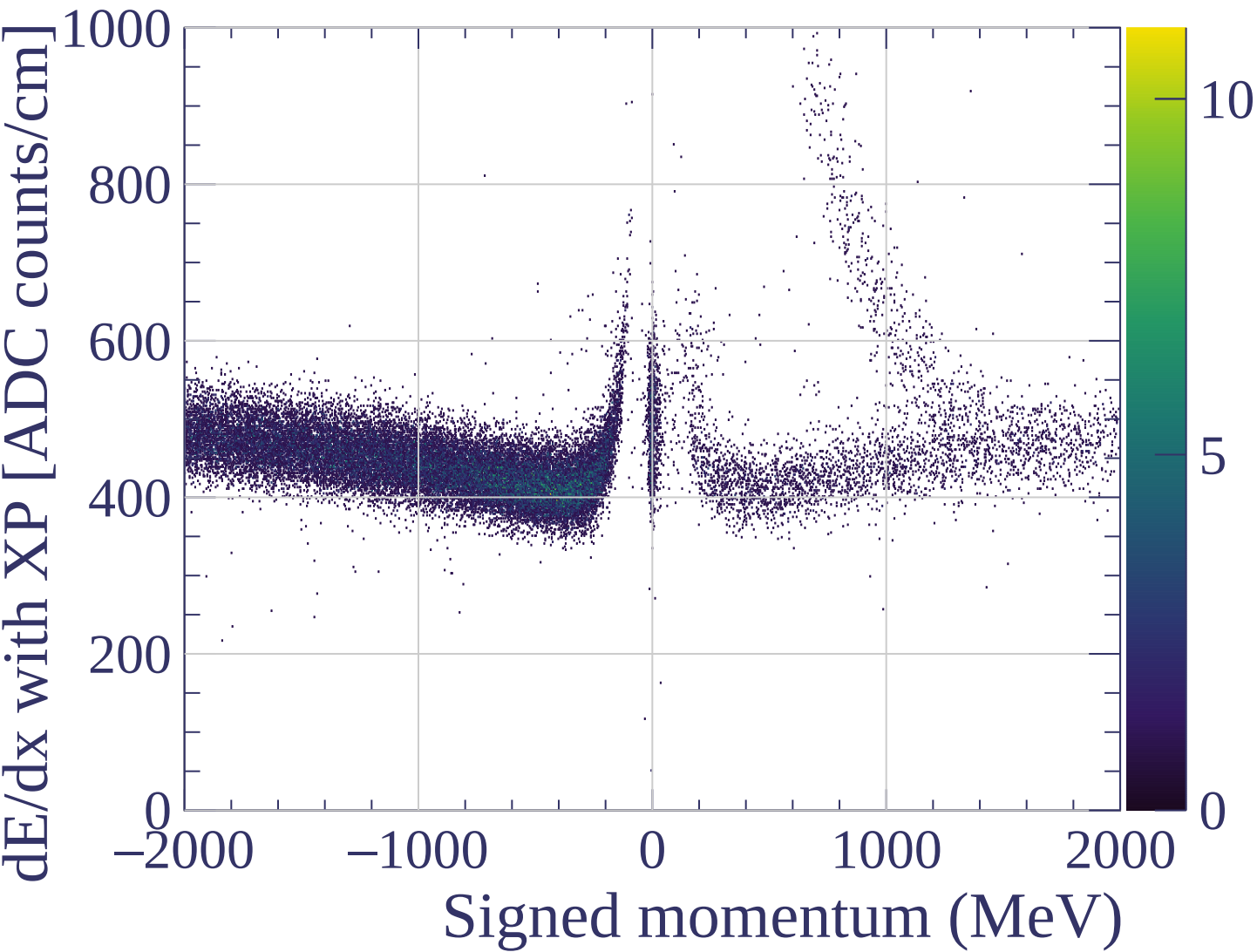


dE/dx resolution [%]

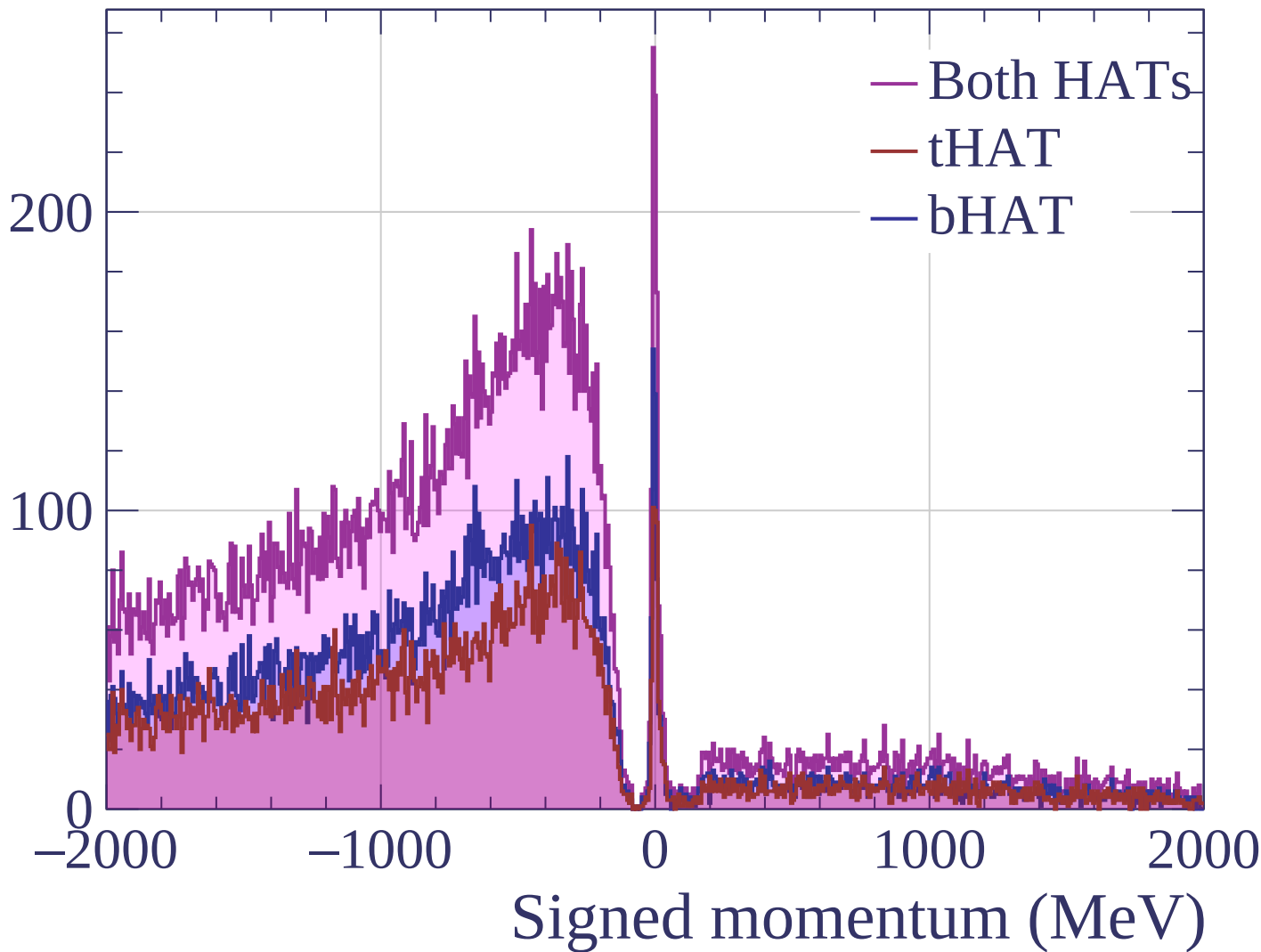


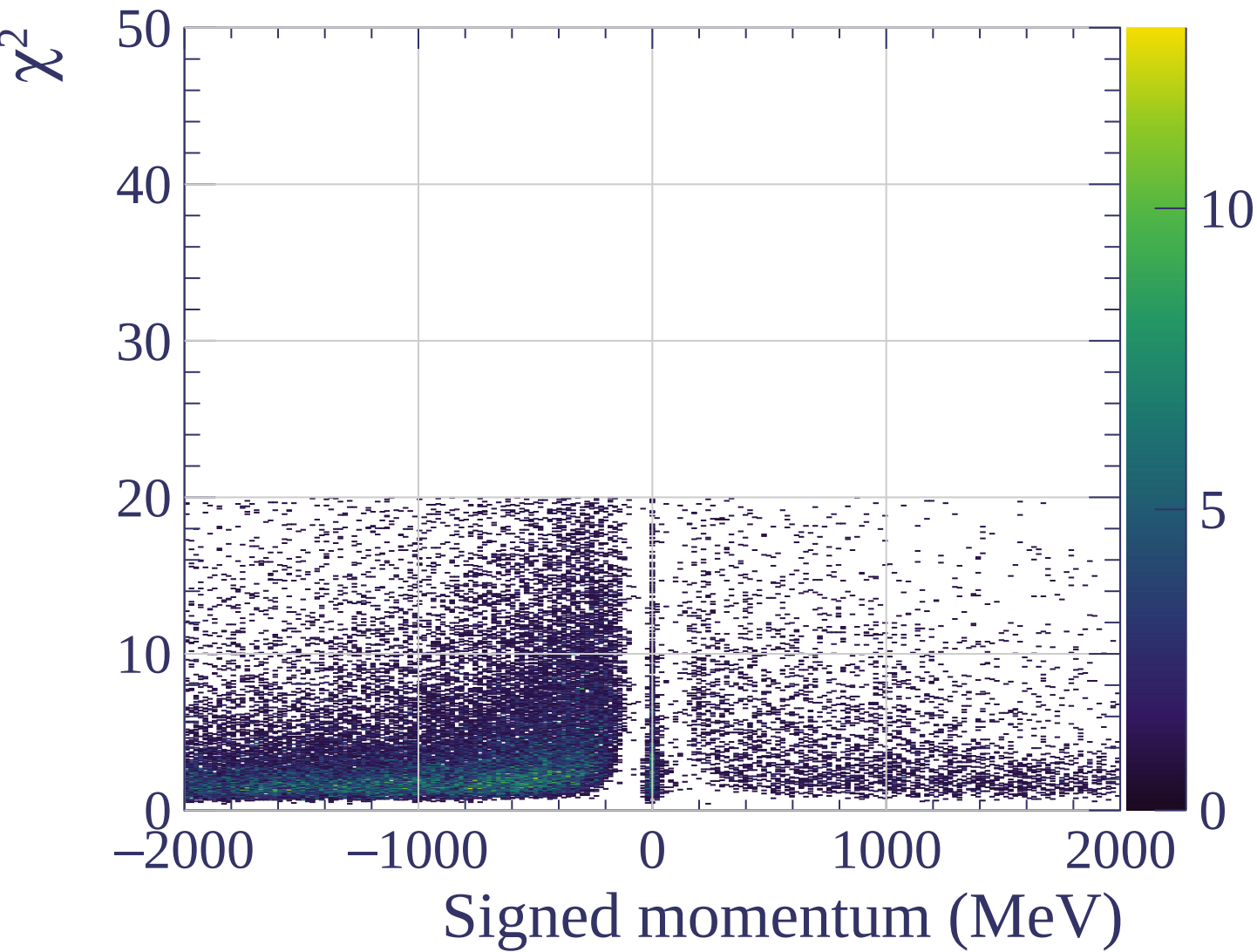


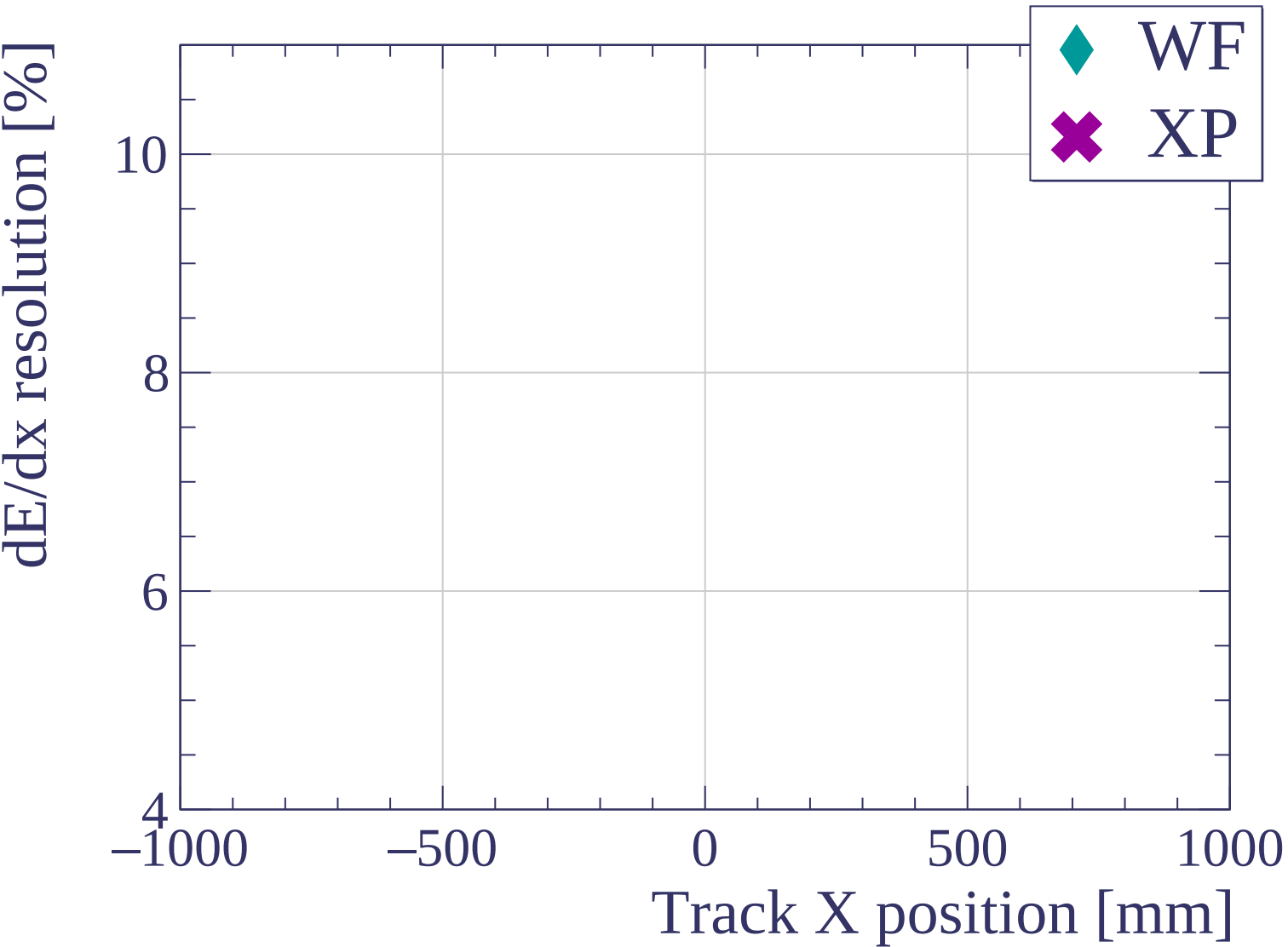


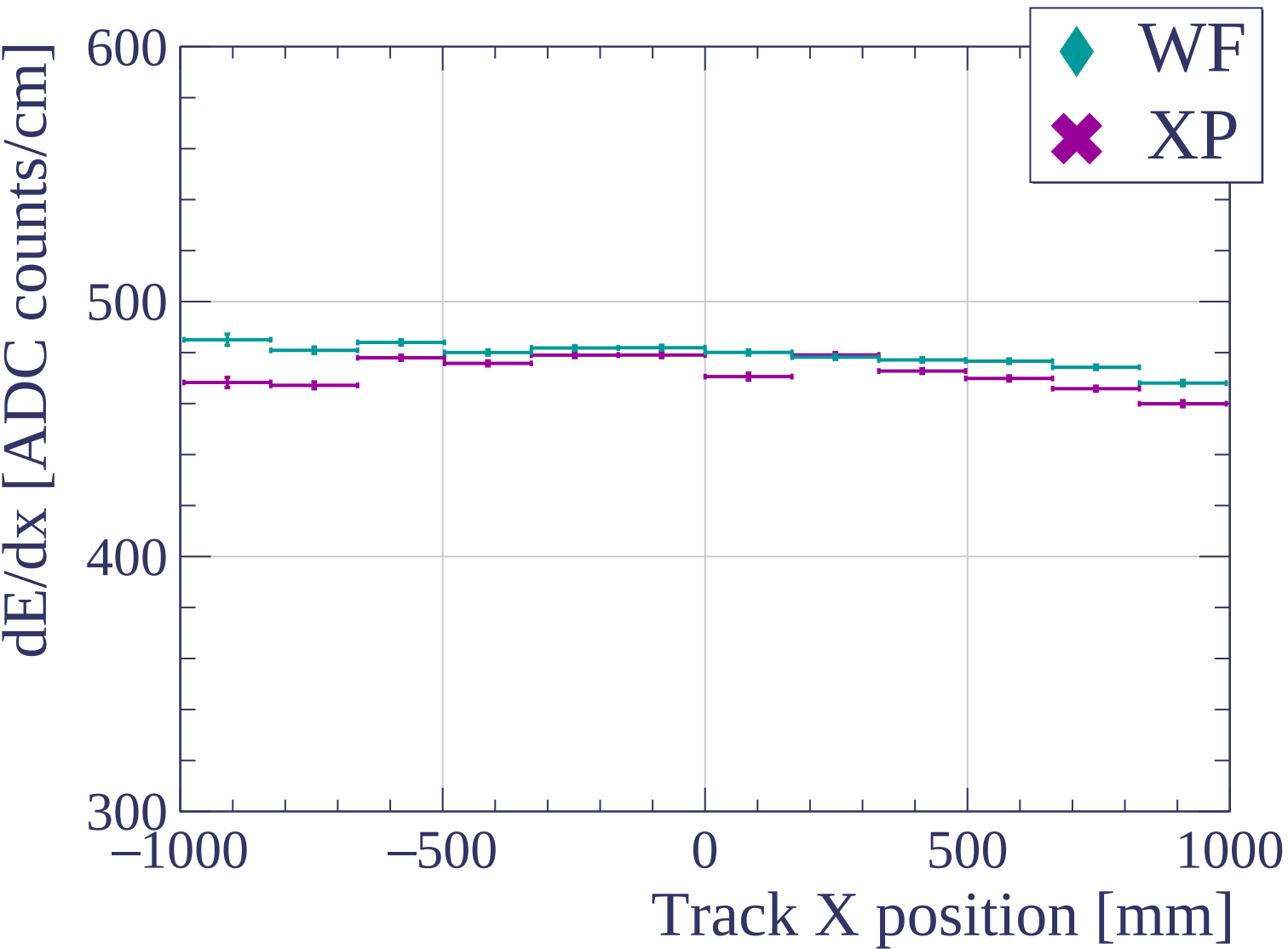


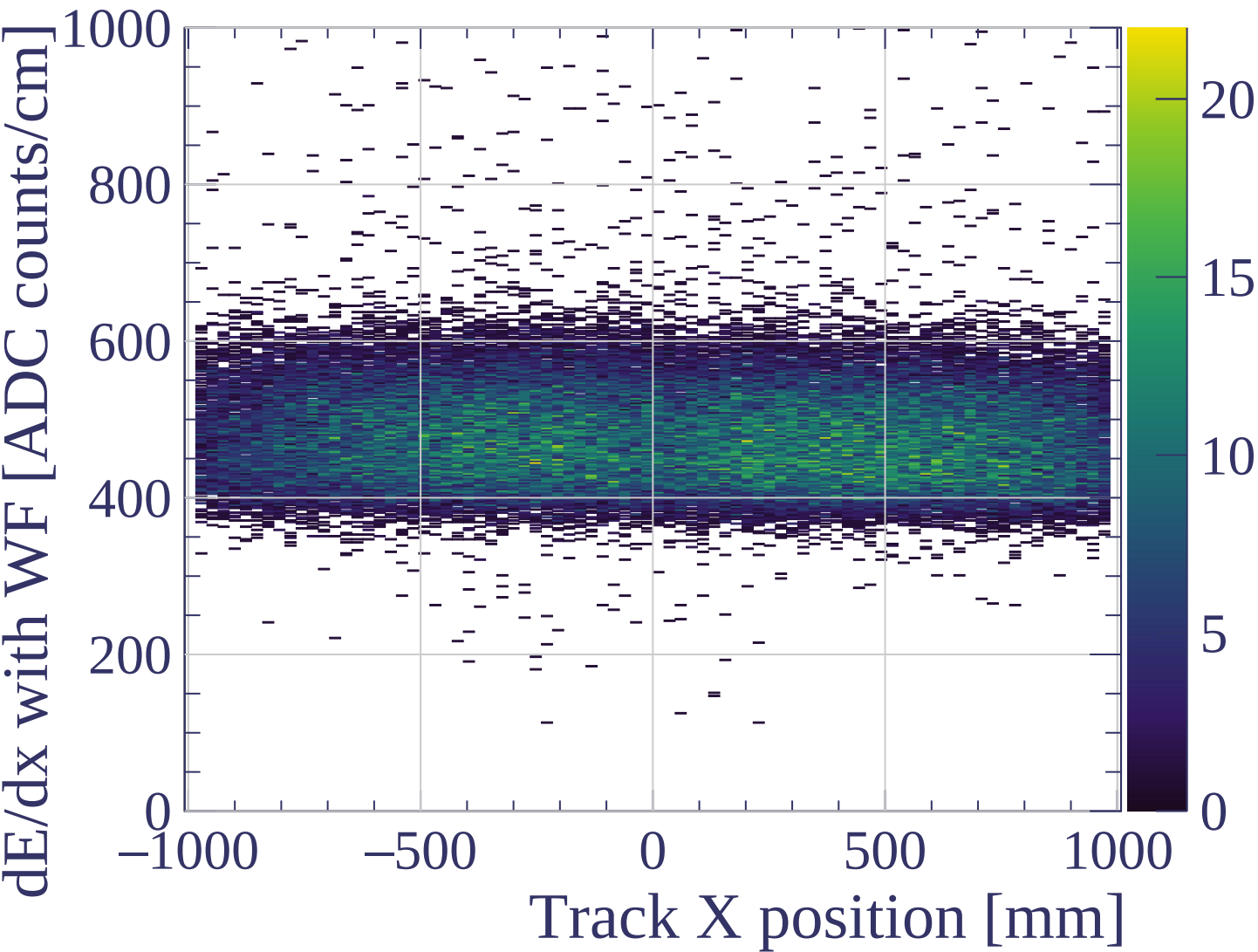
Count

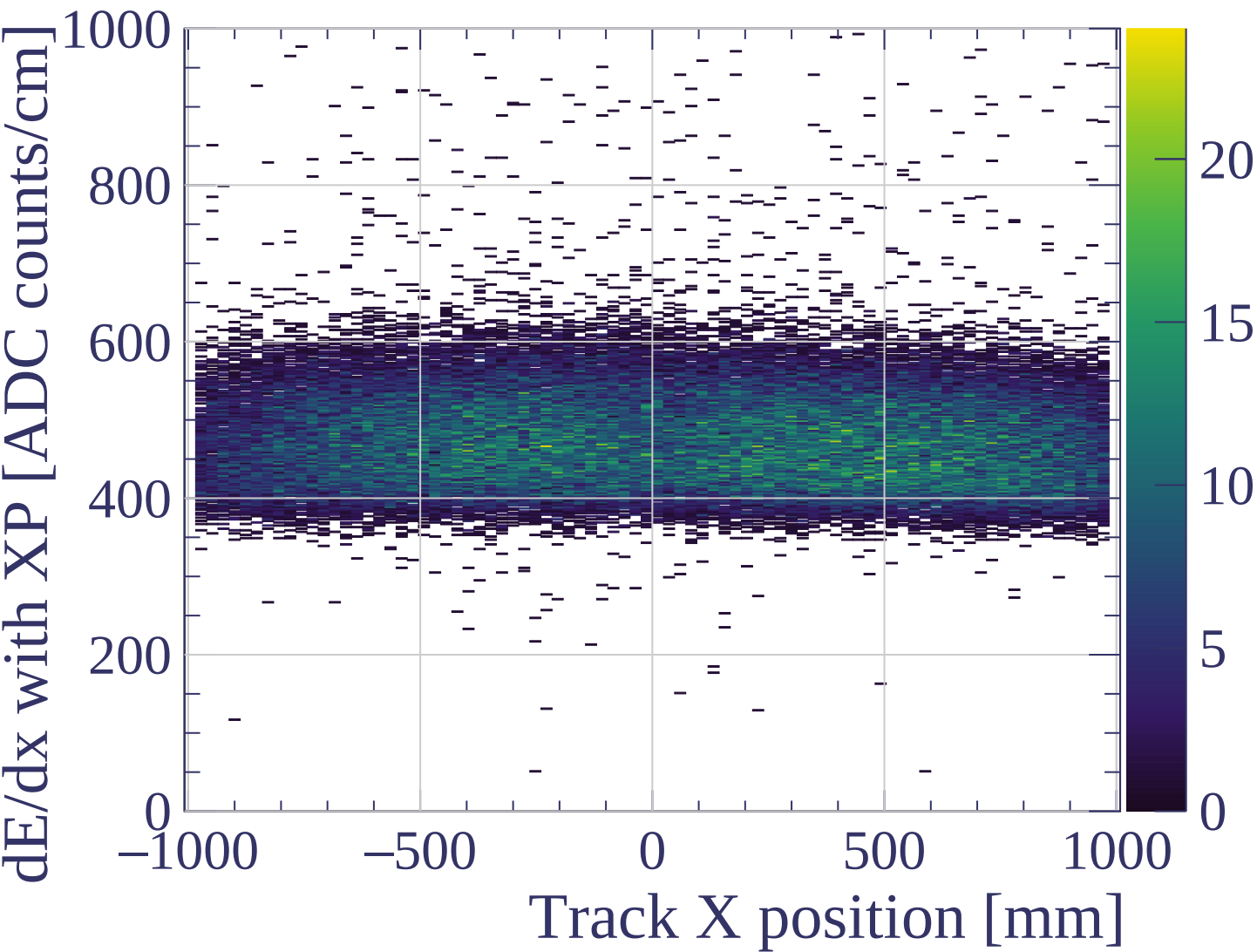




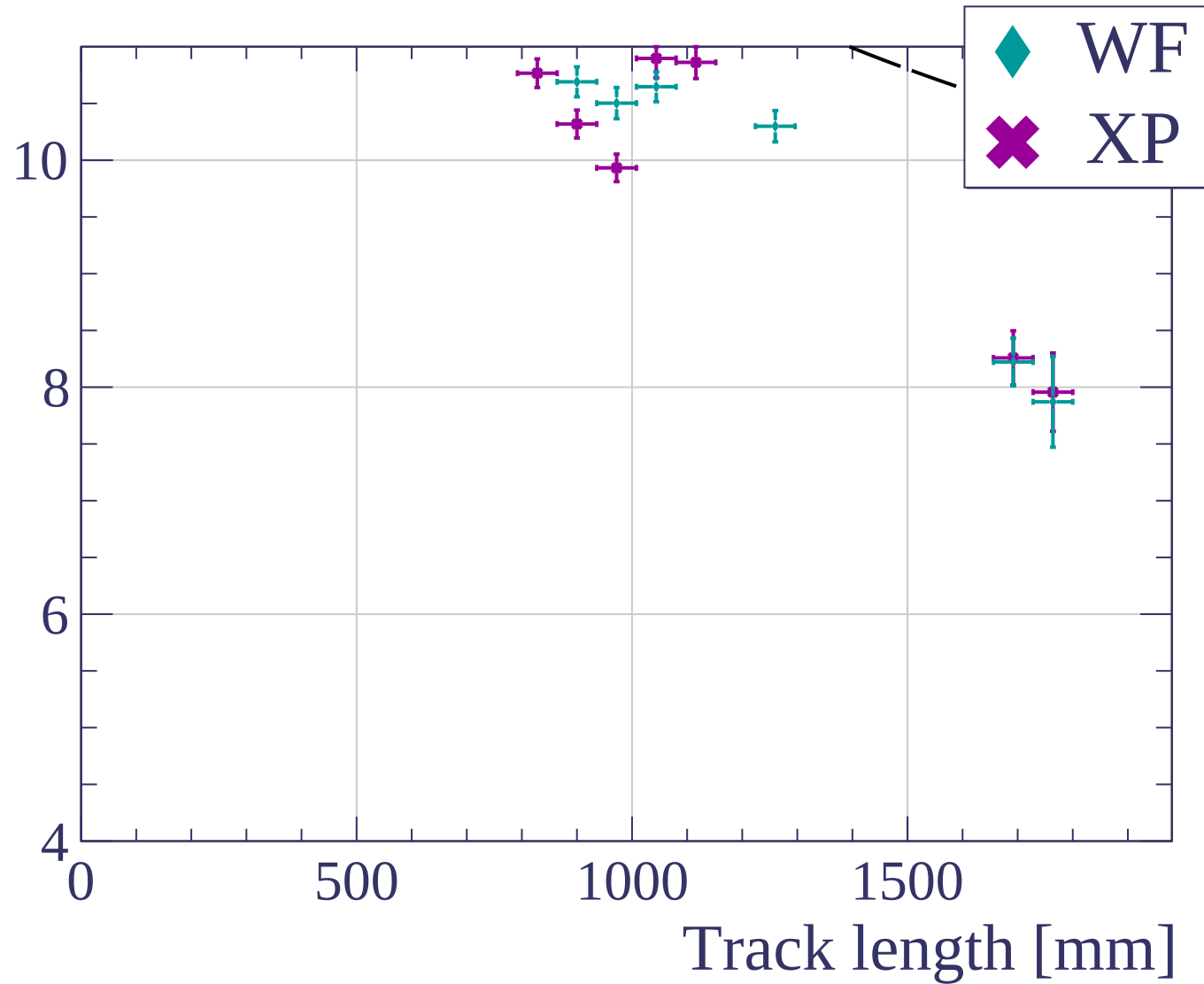


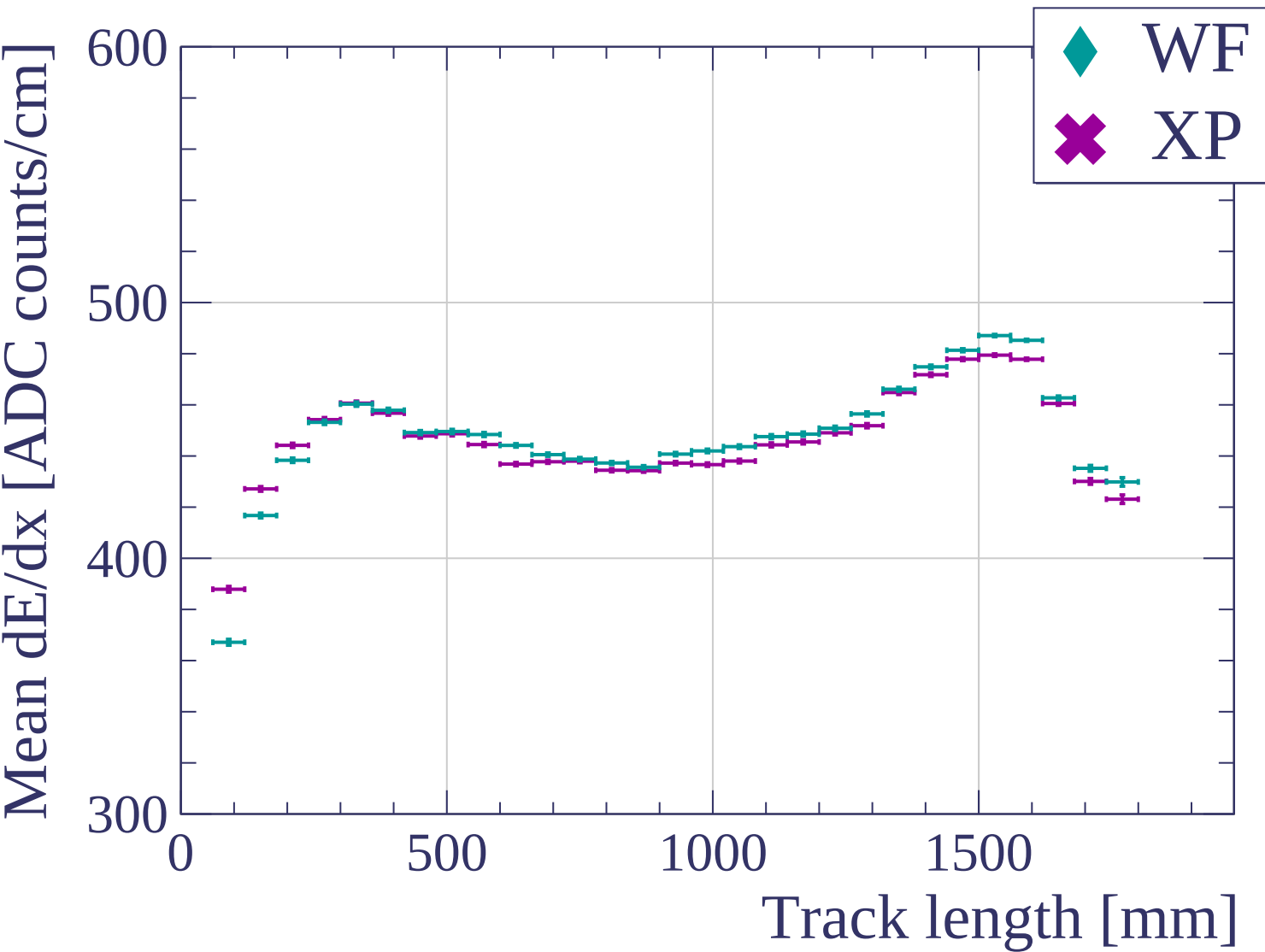


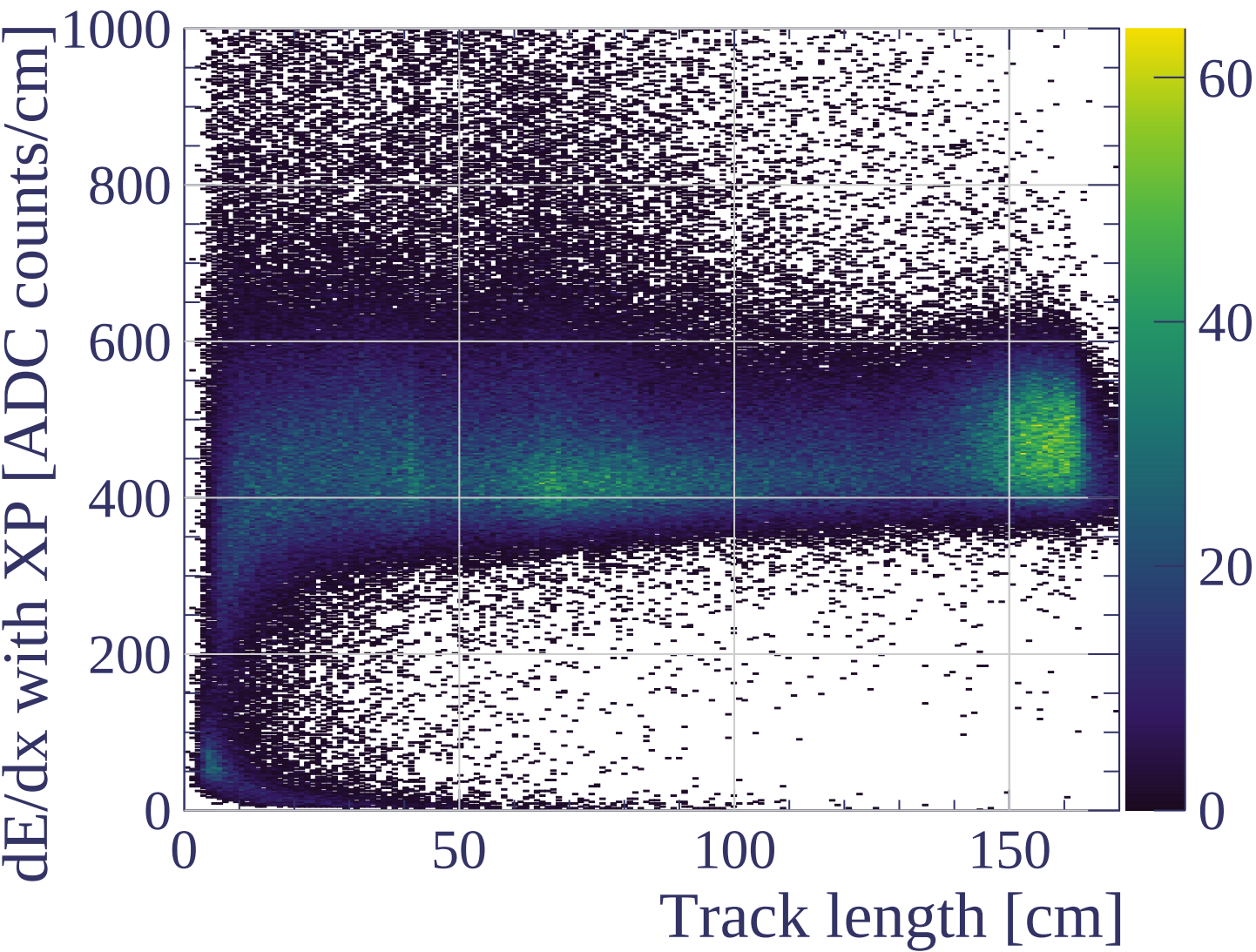


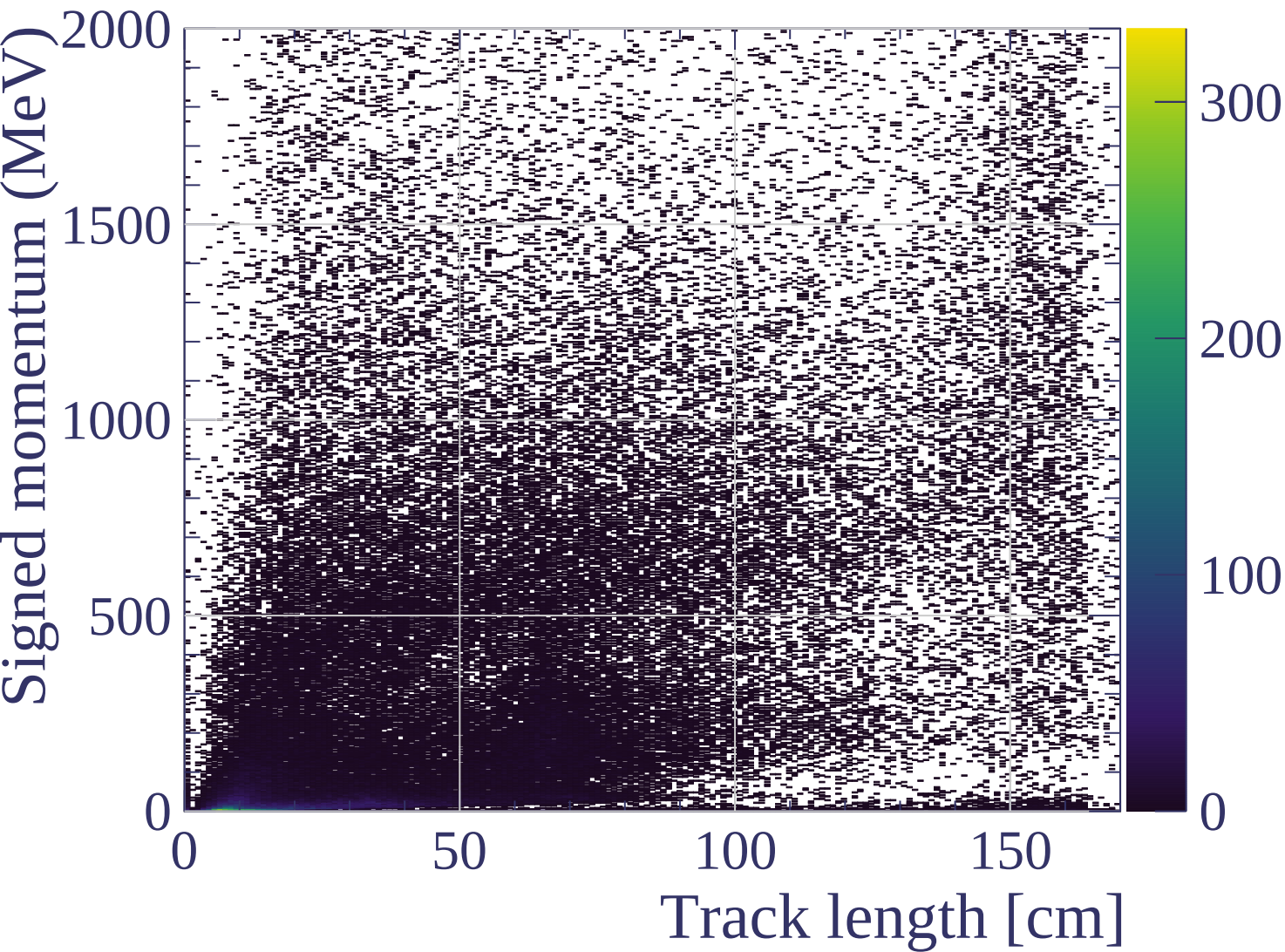


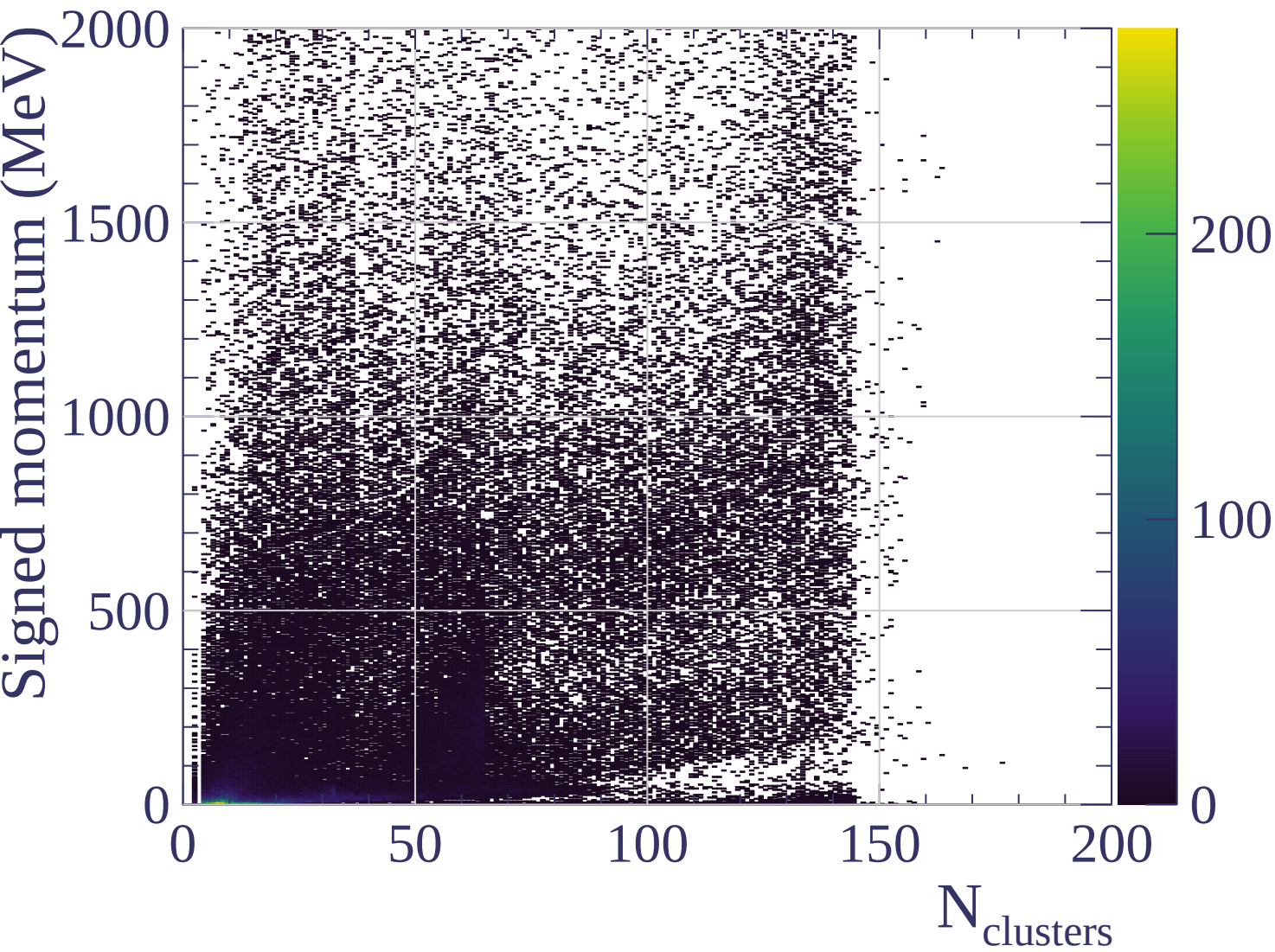
dE/dx resolution [%]



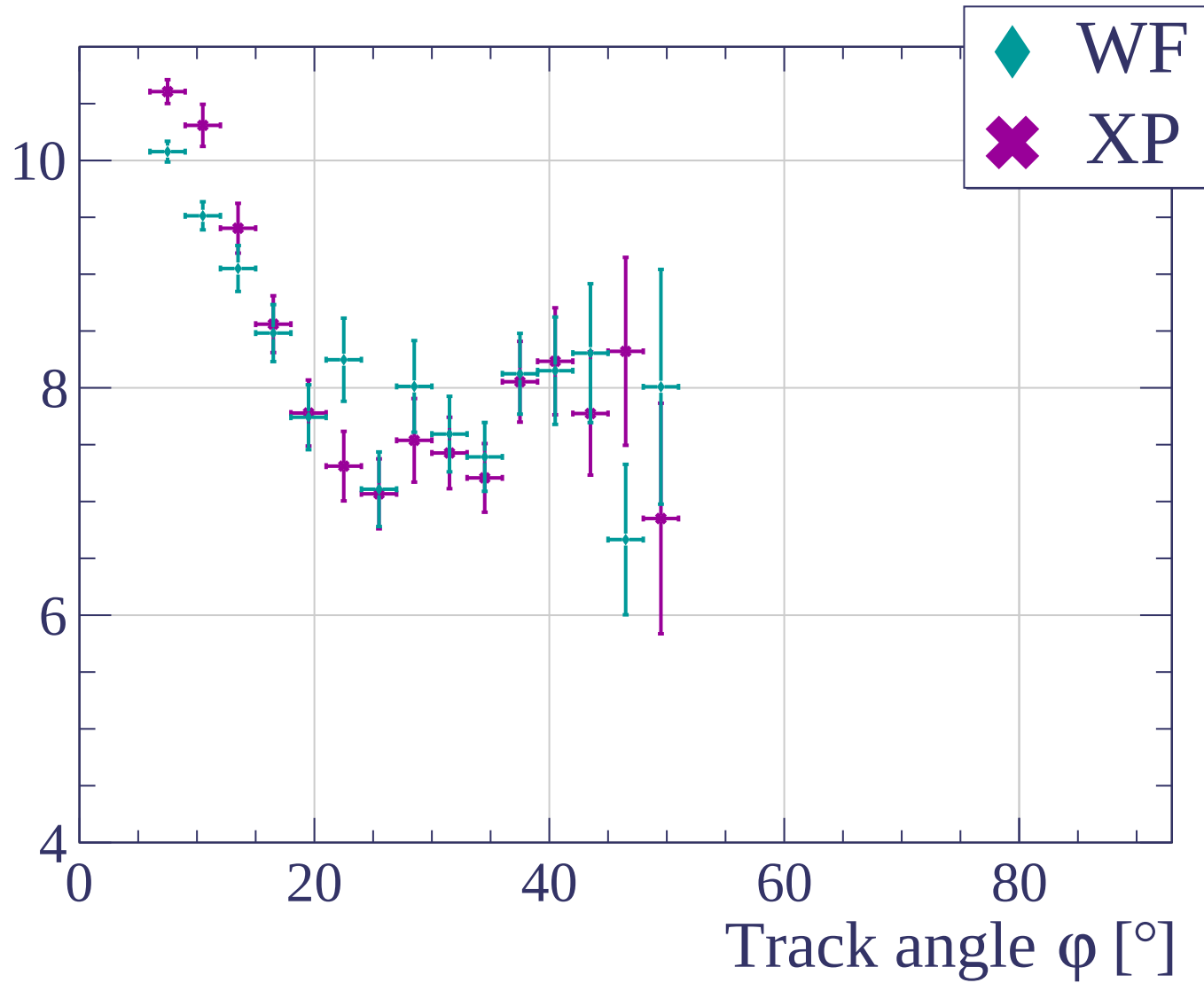


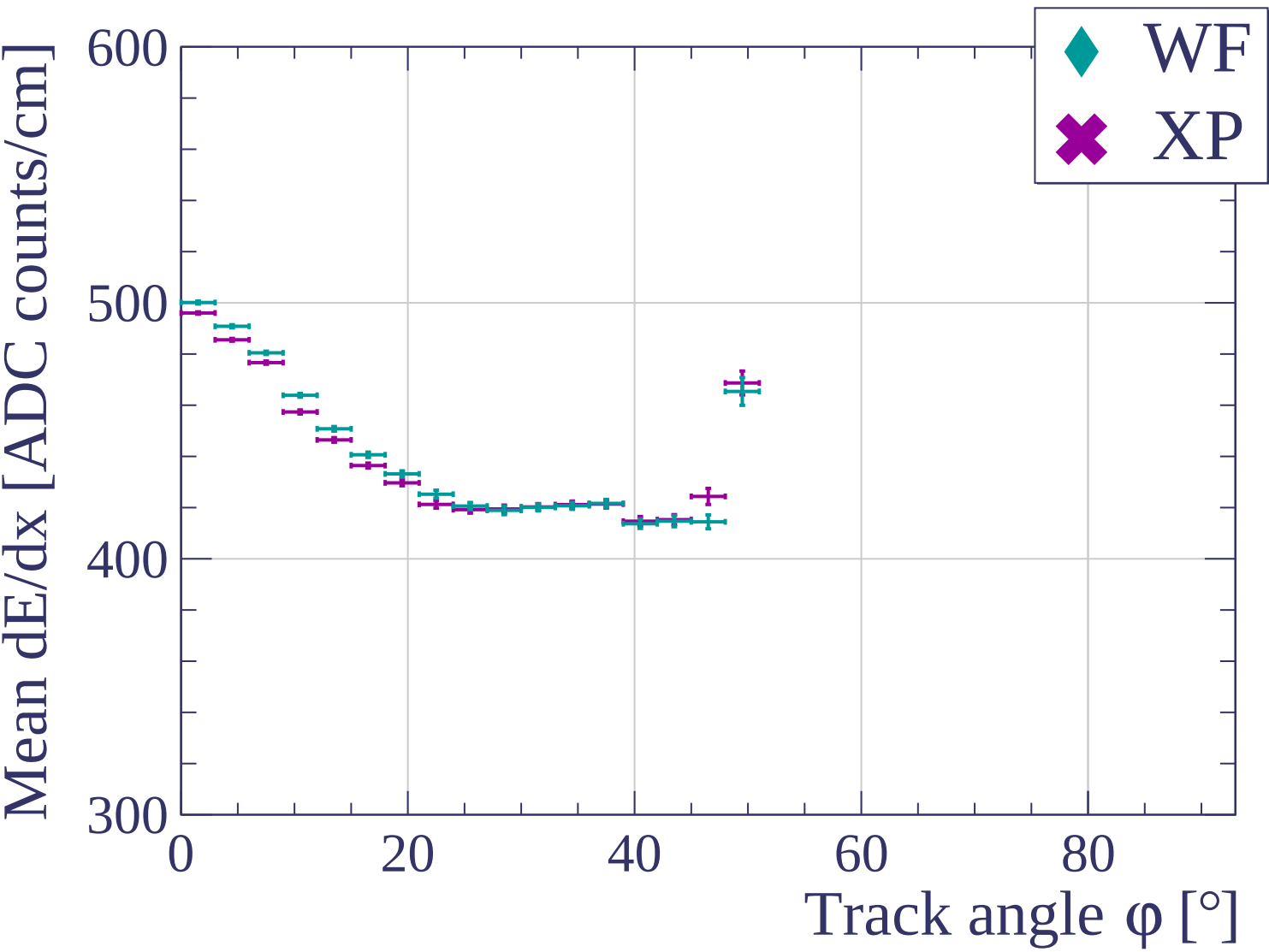




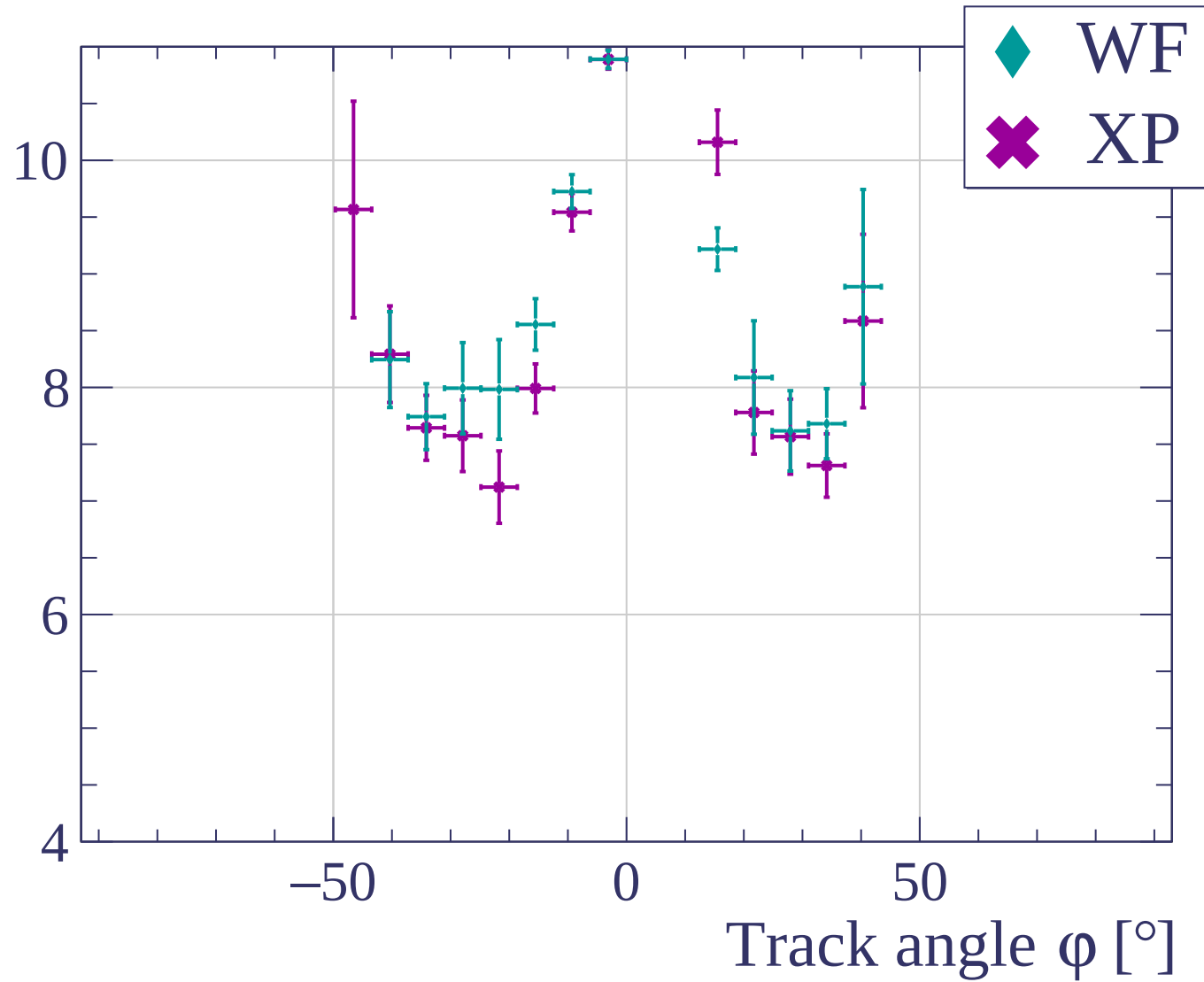


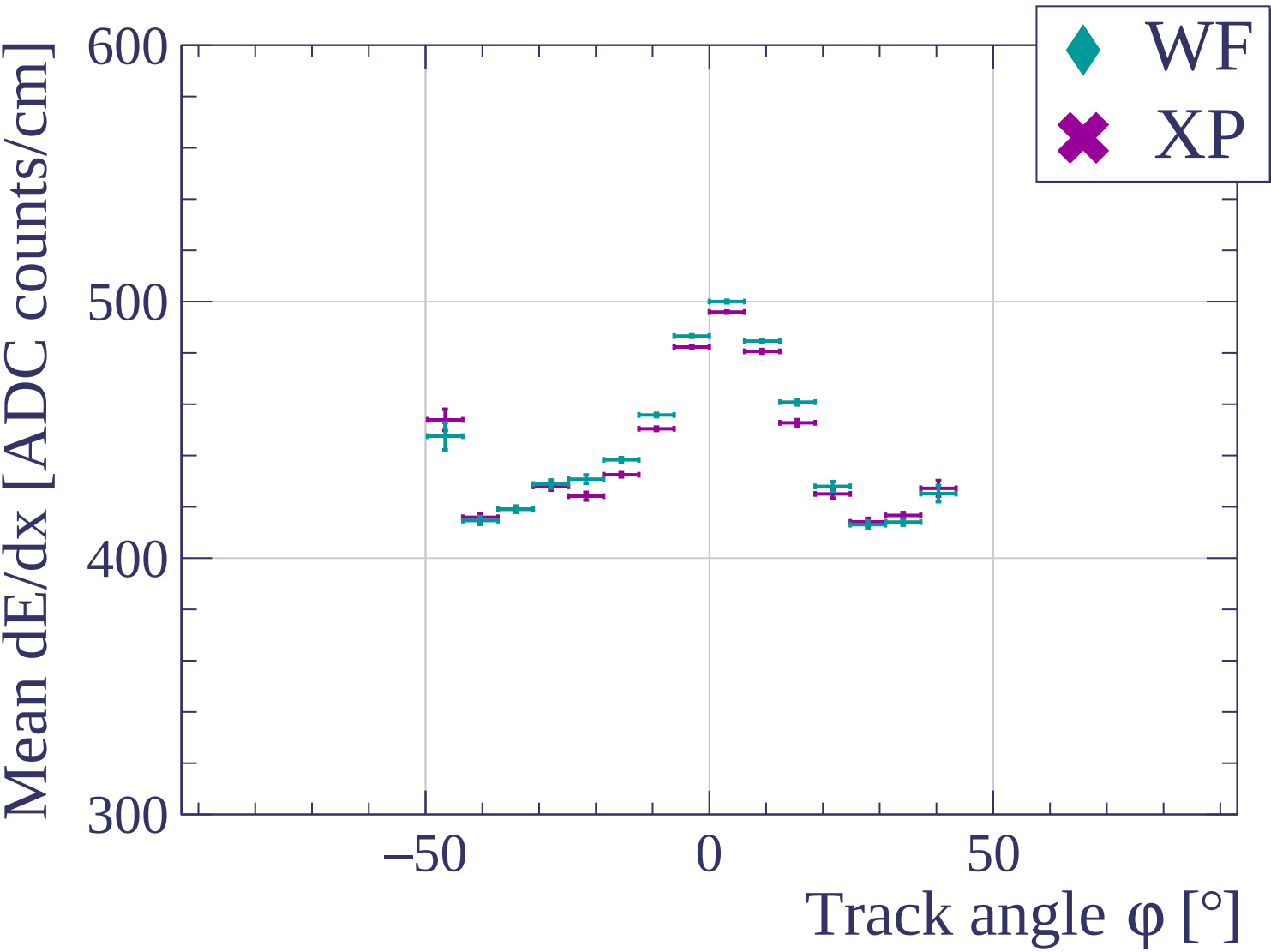
dE/dx resolution [%]

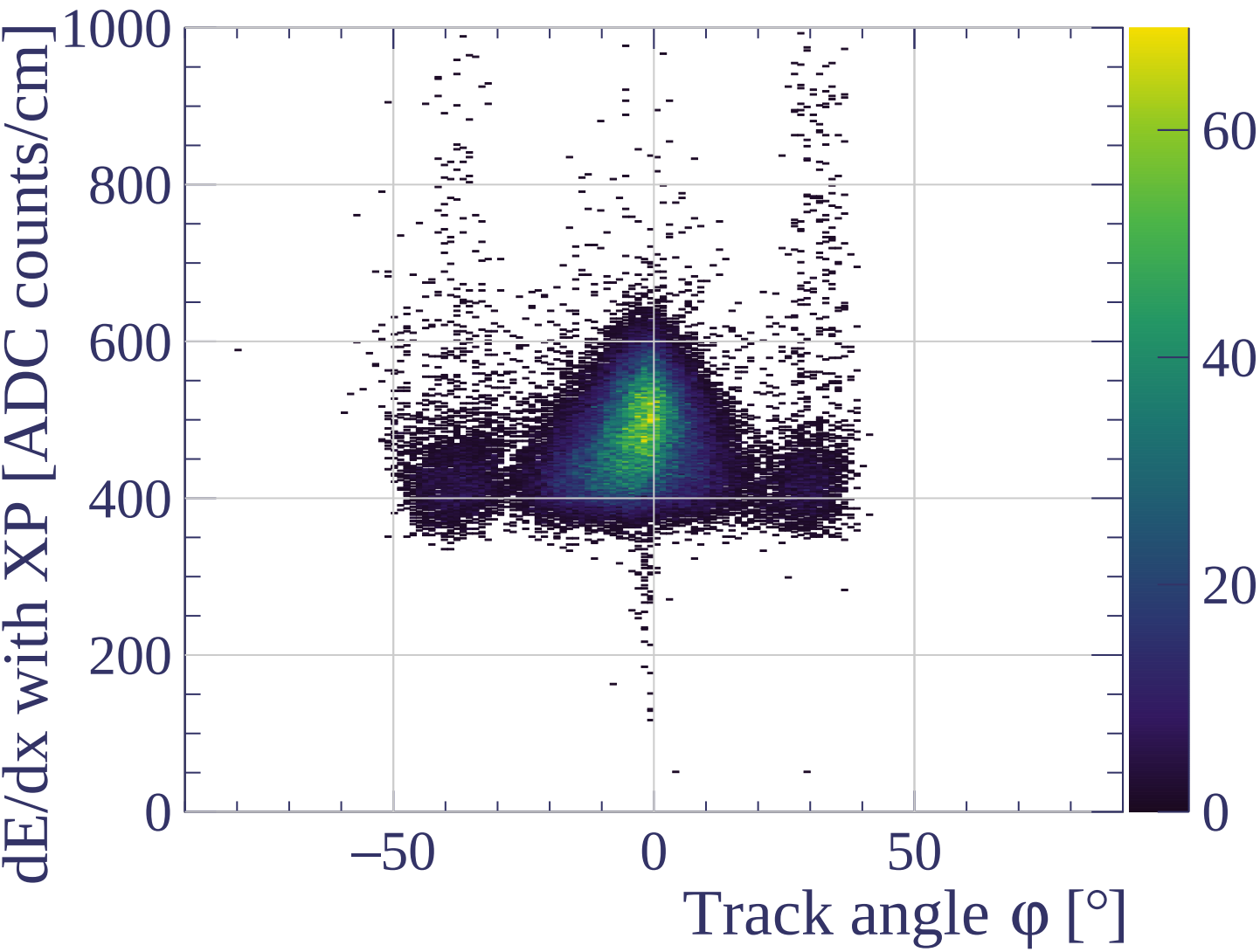




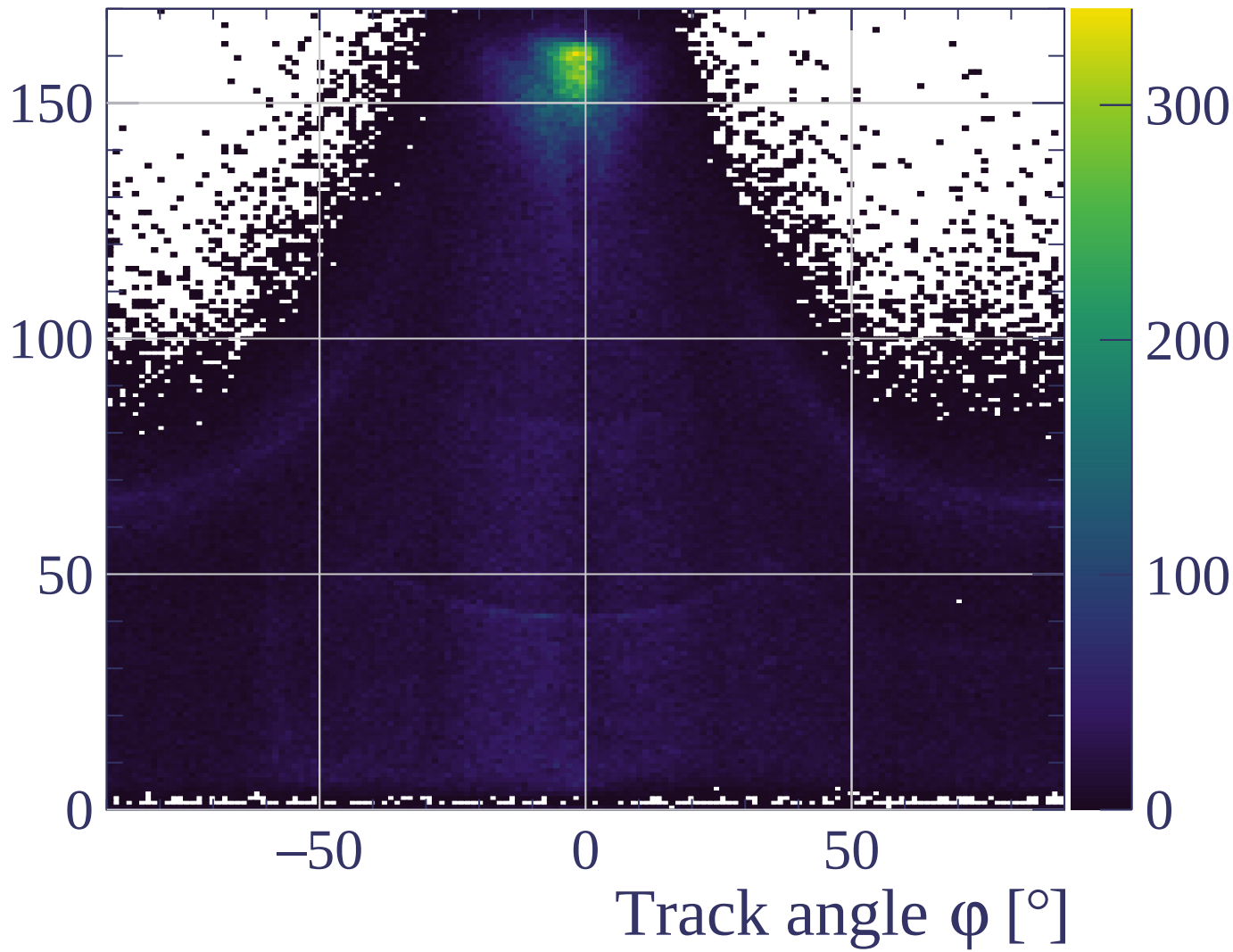
dE/dx resolution [%]

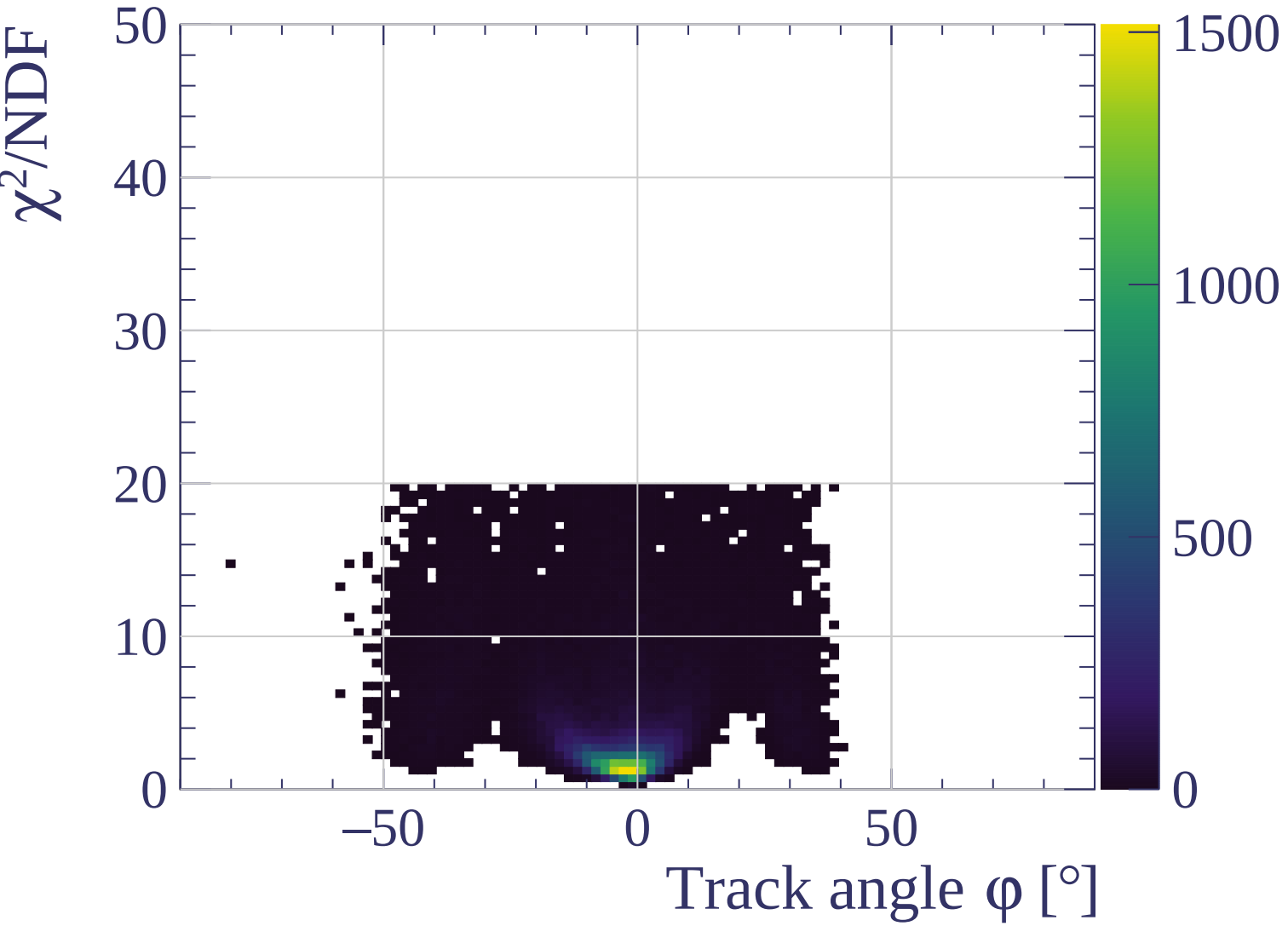


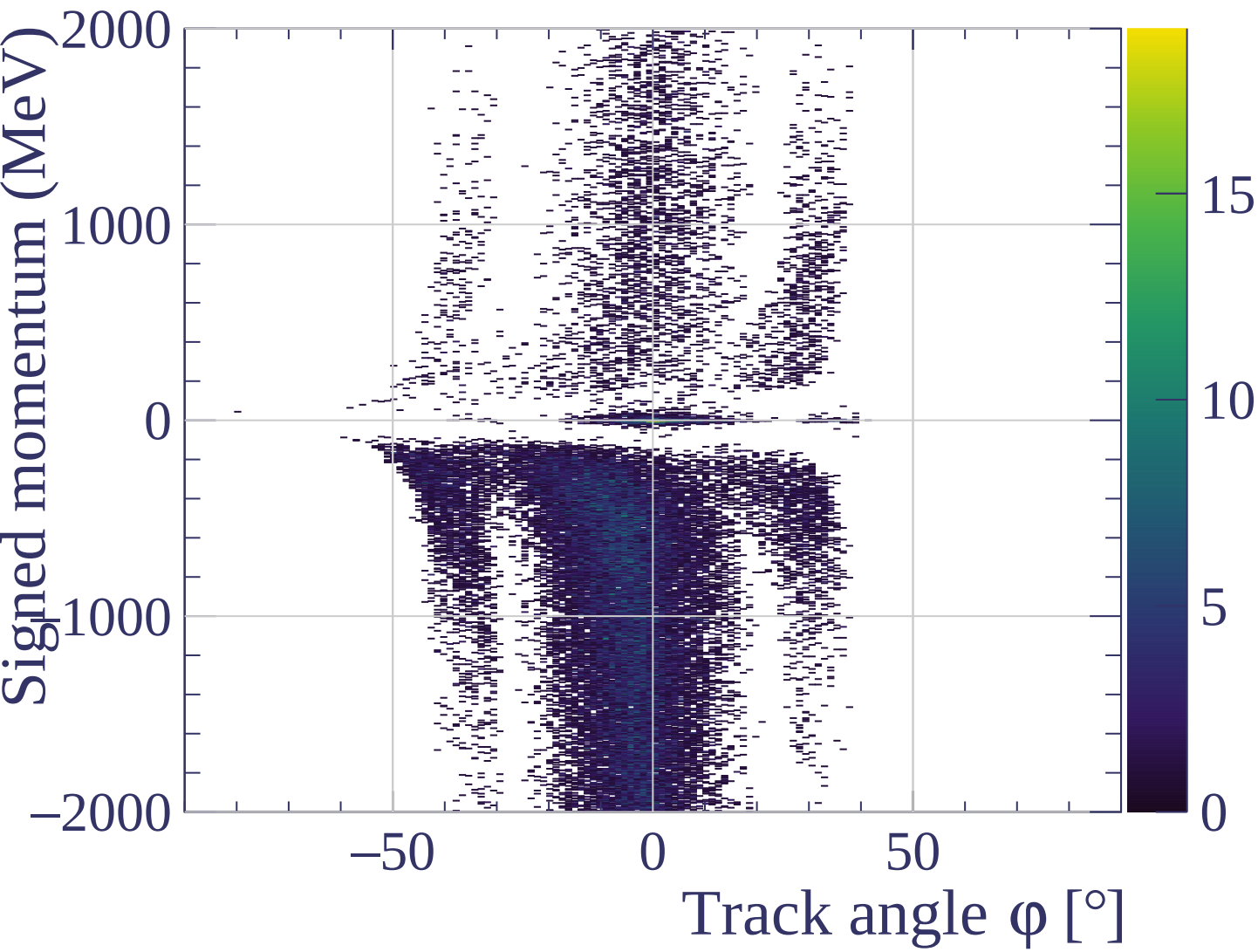


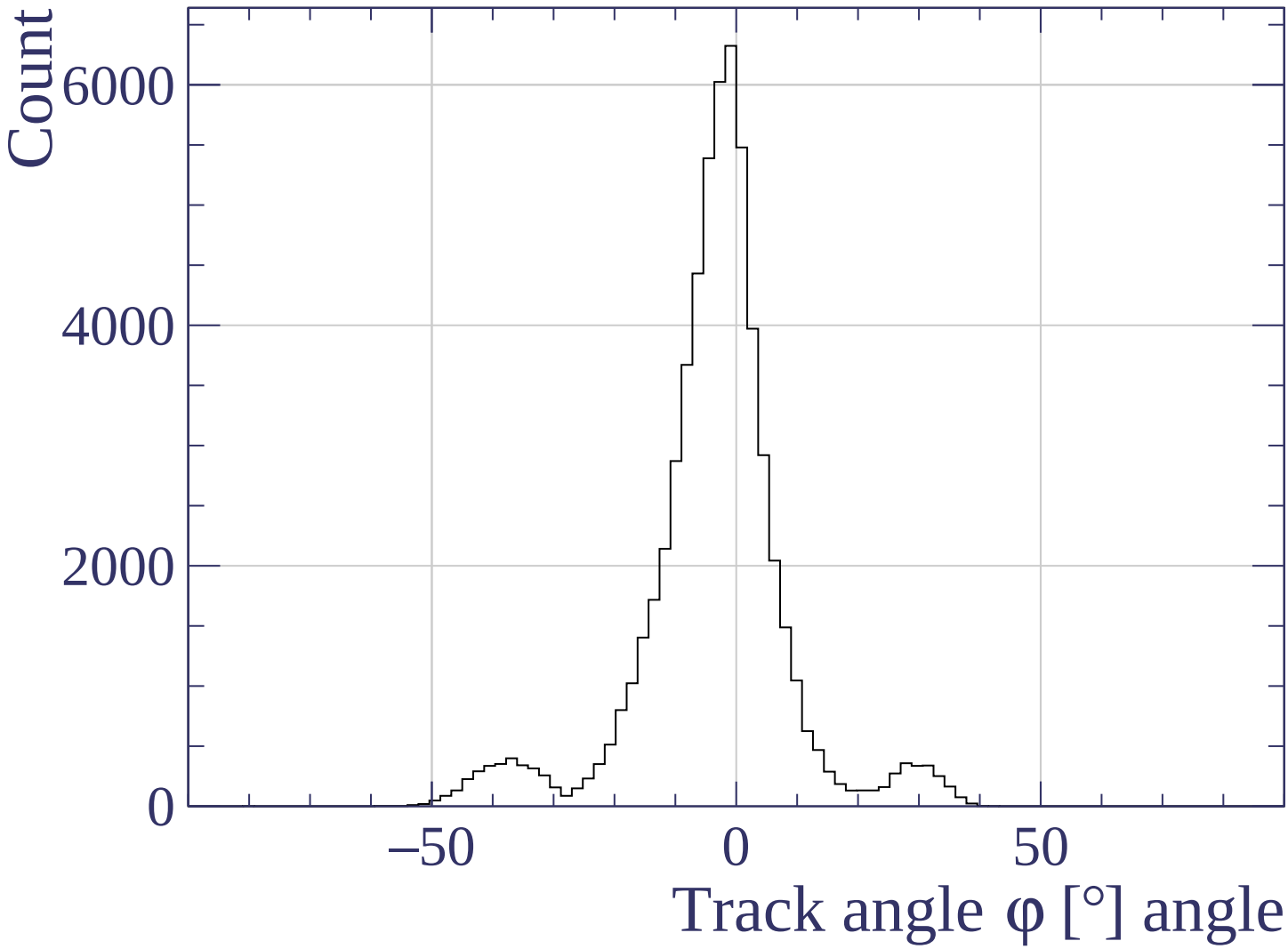


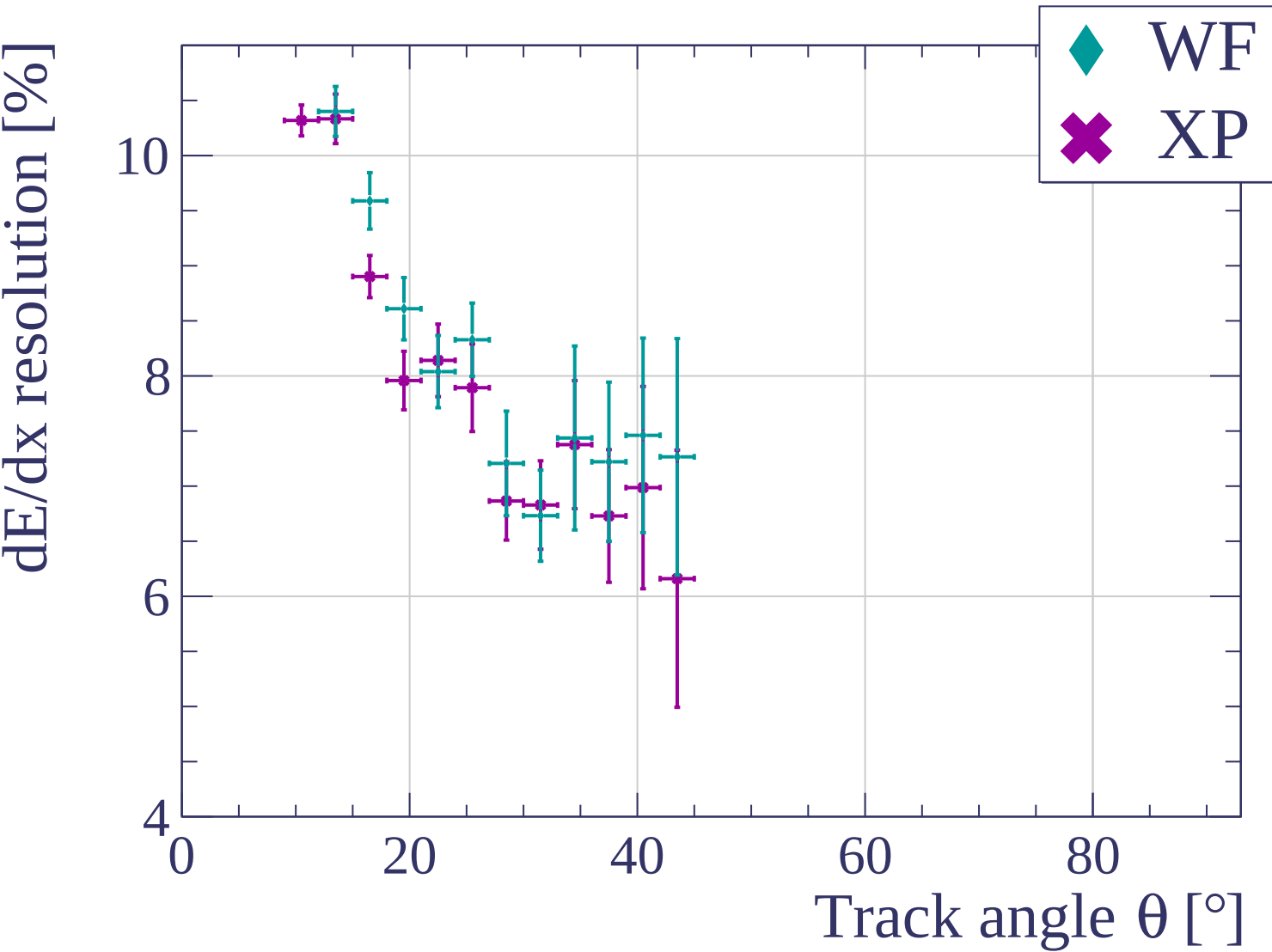
Track length [cm]

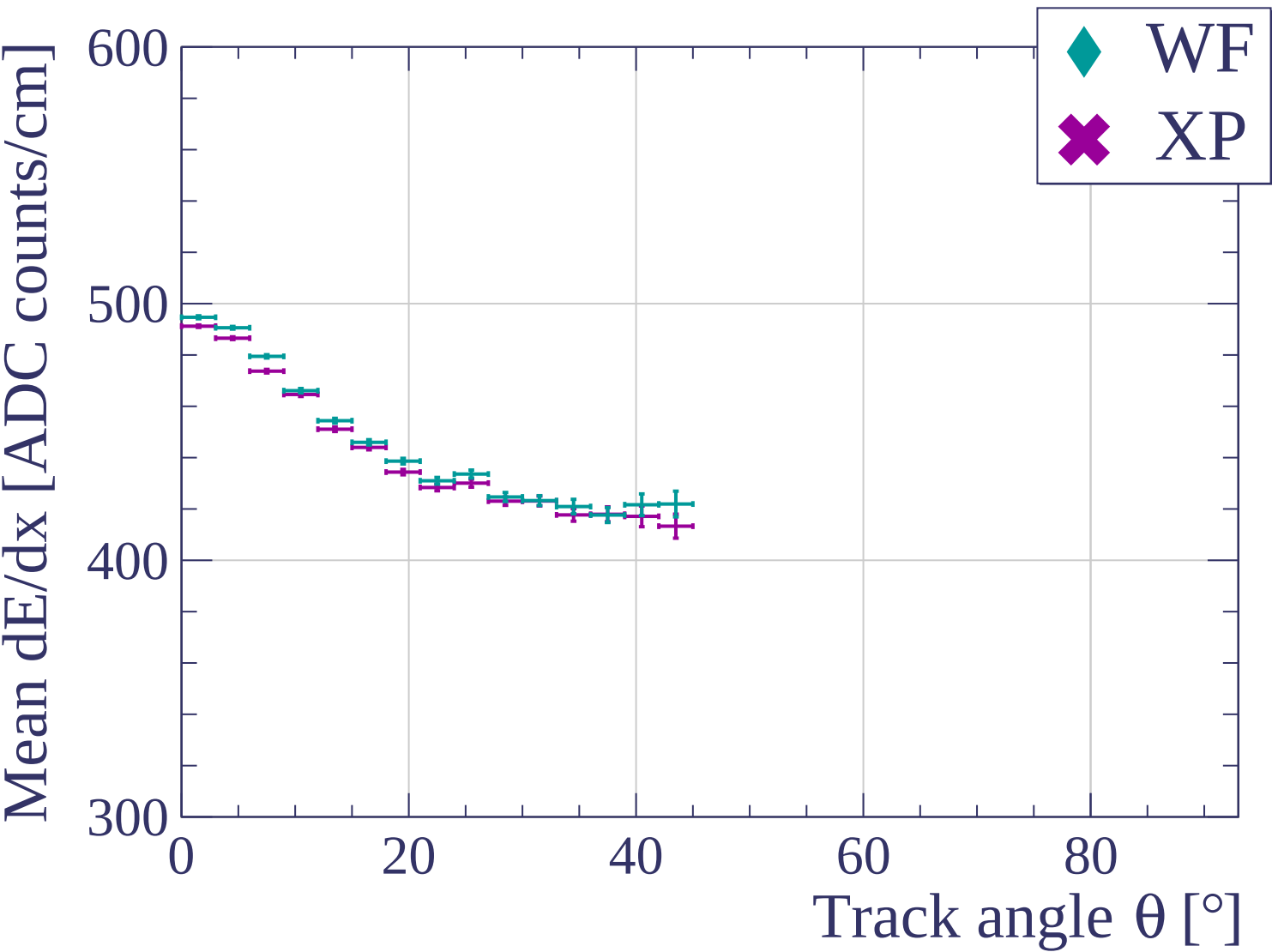




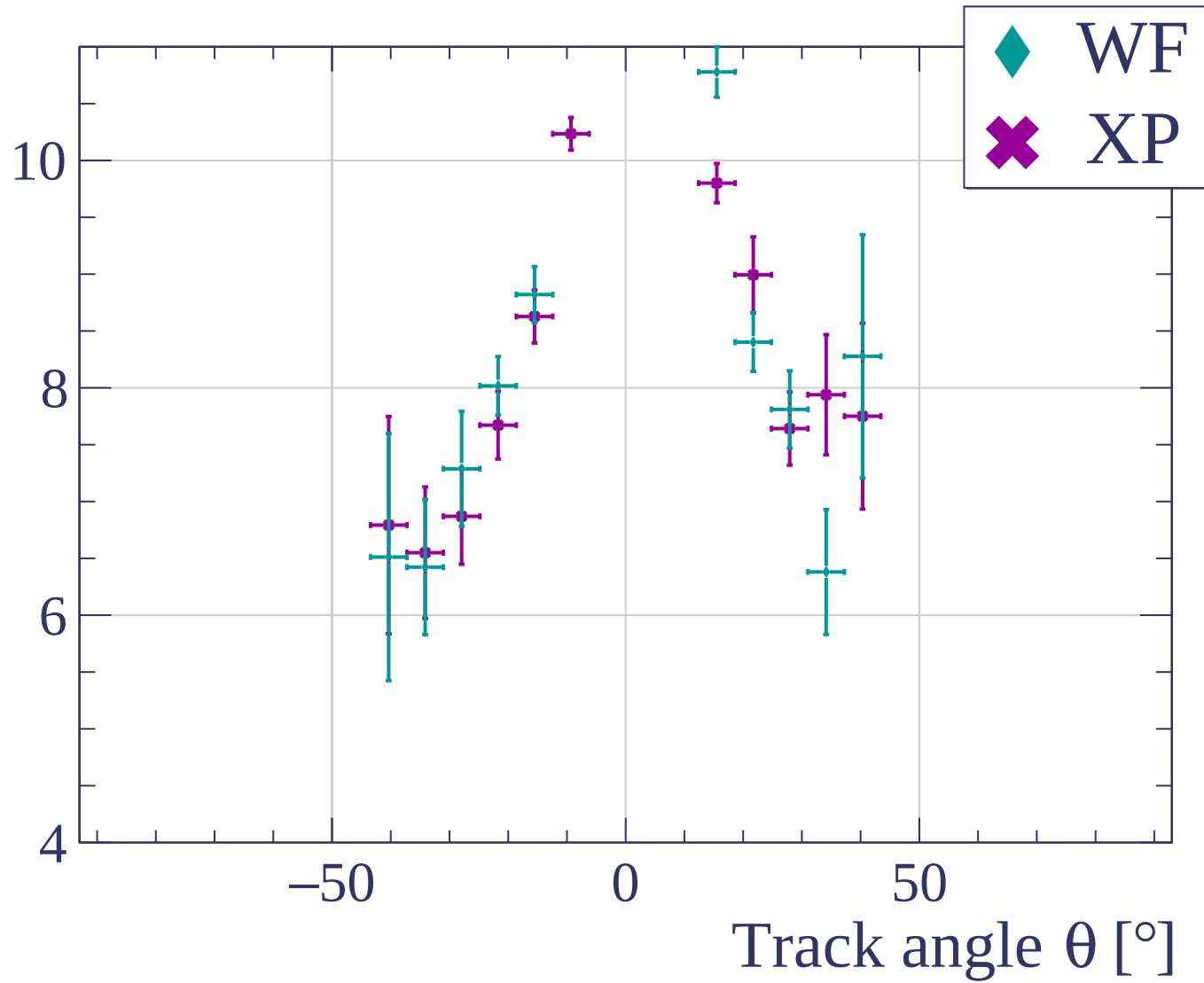


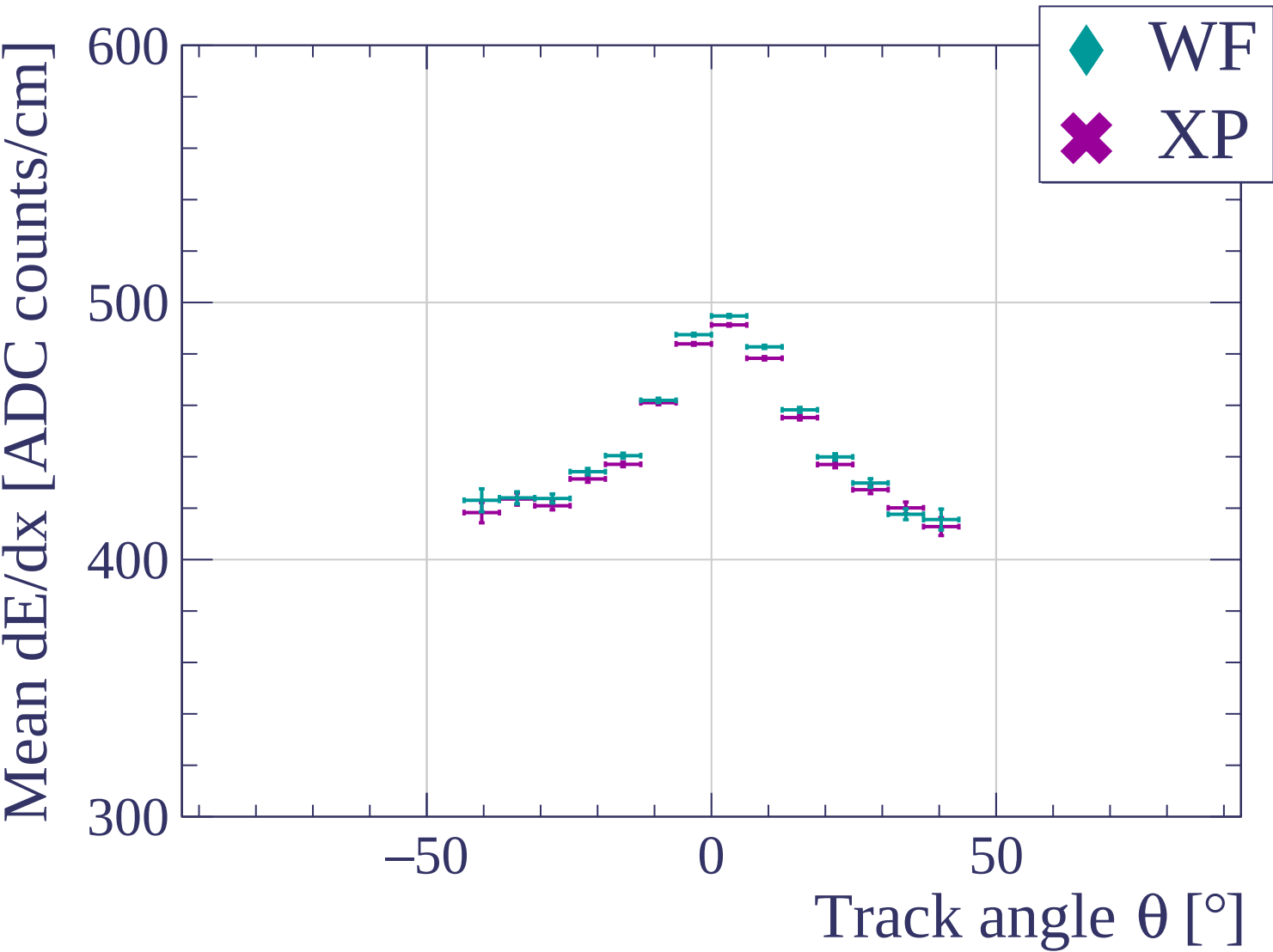


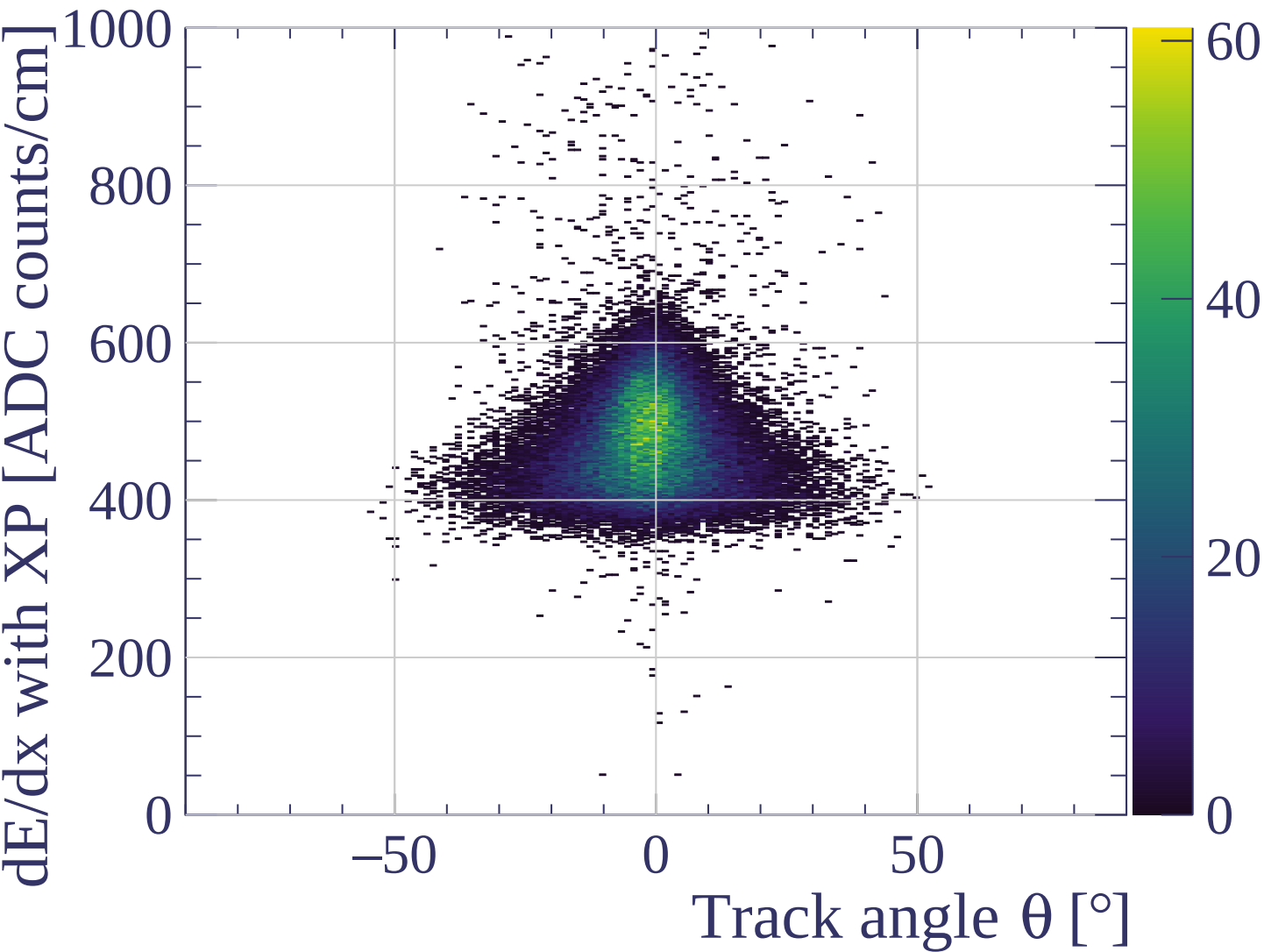


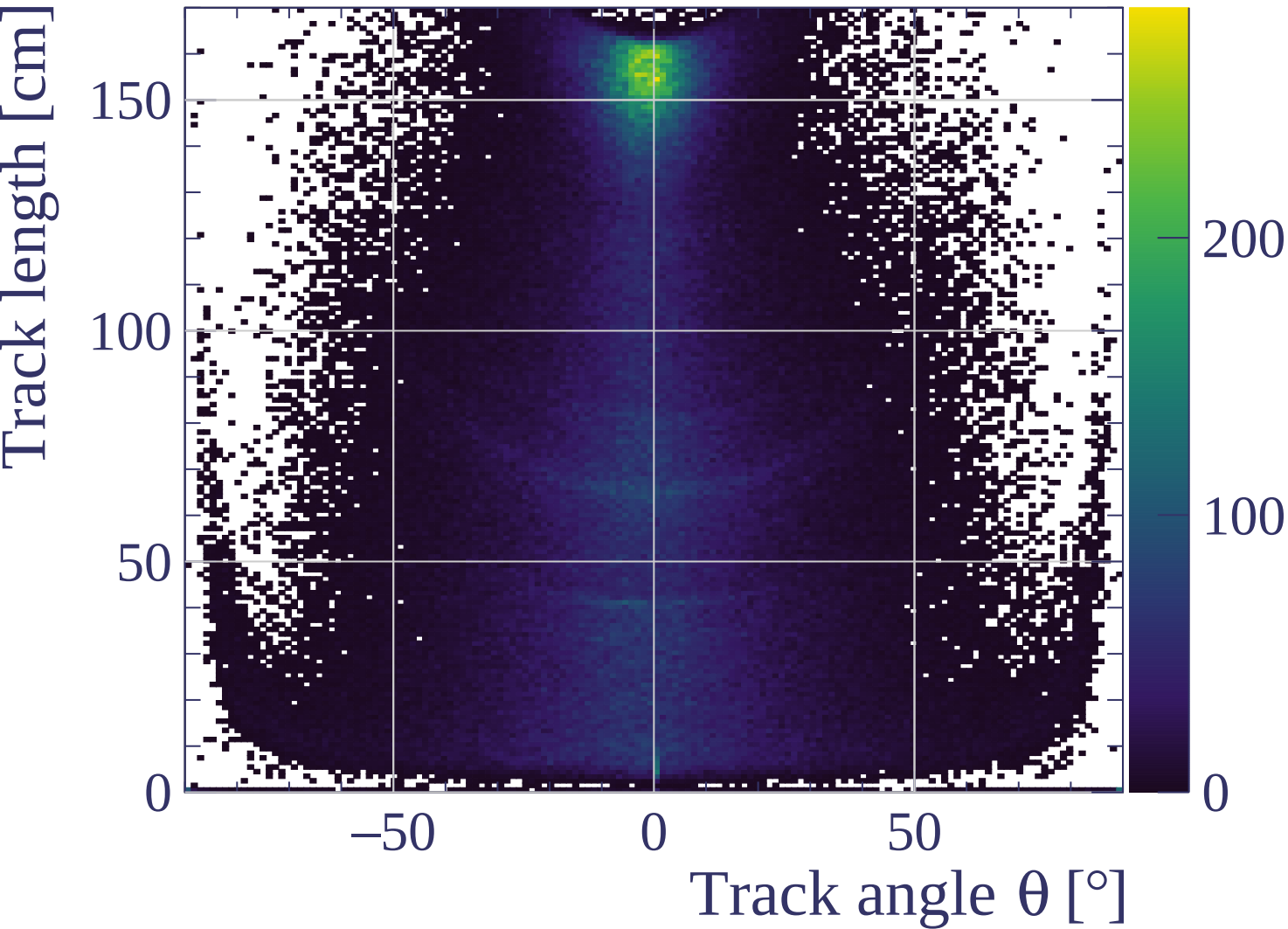


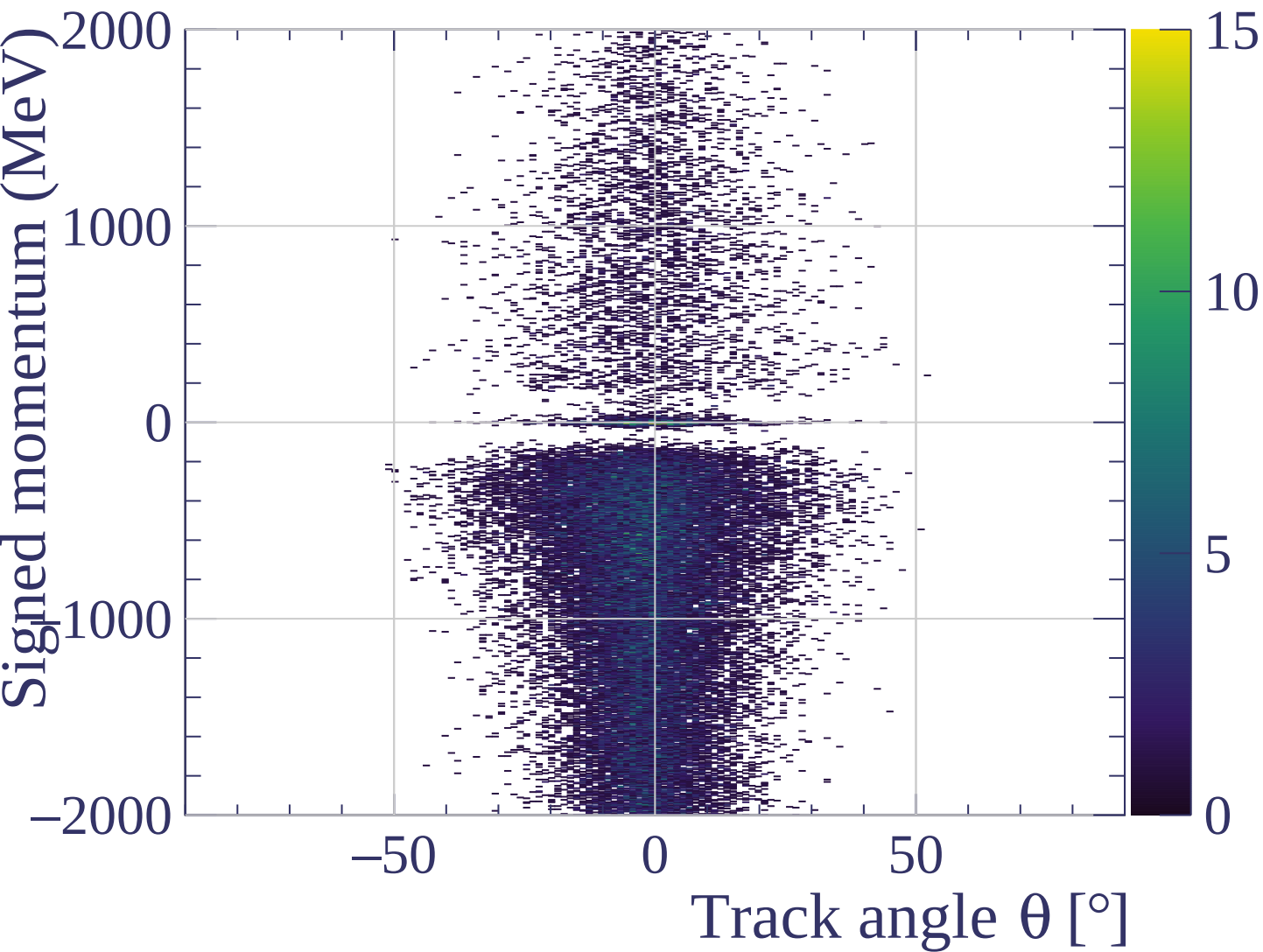
dE/dx resolution [%]



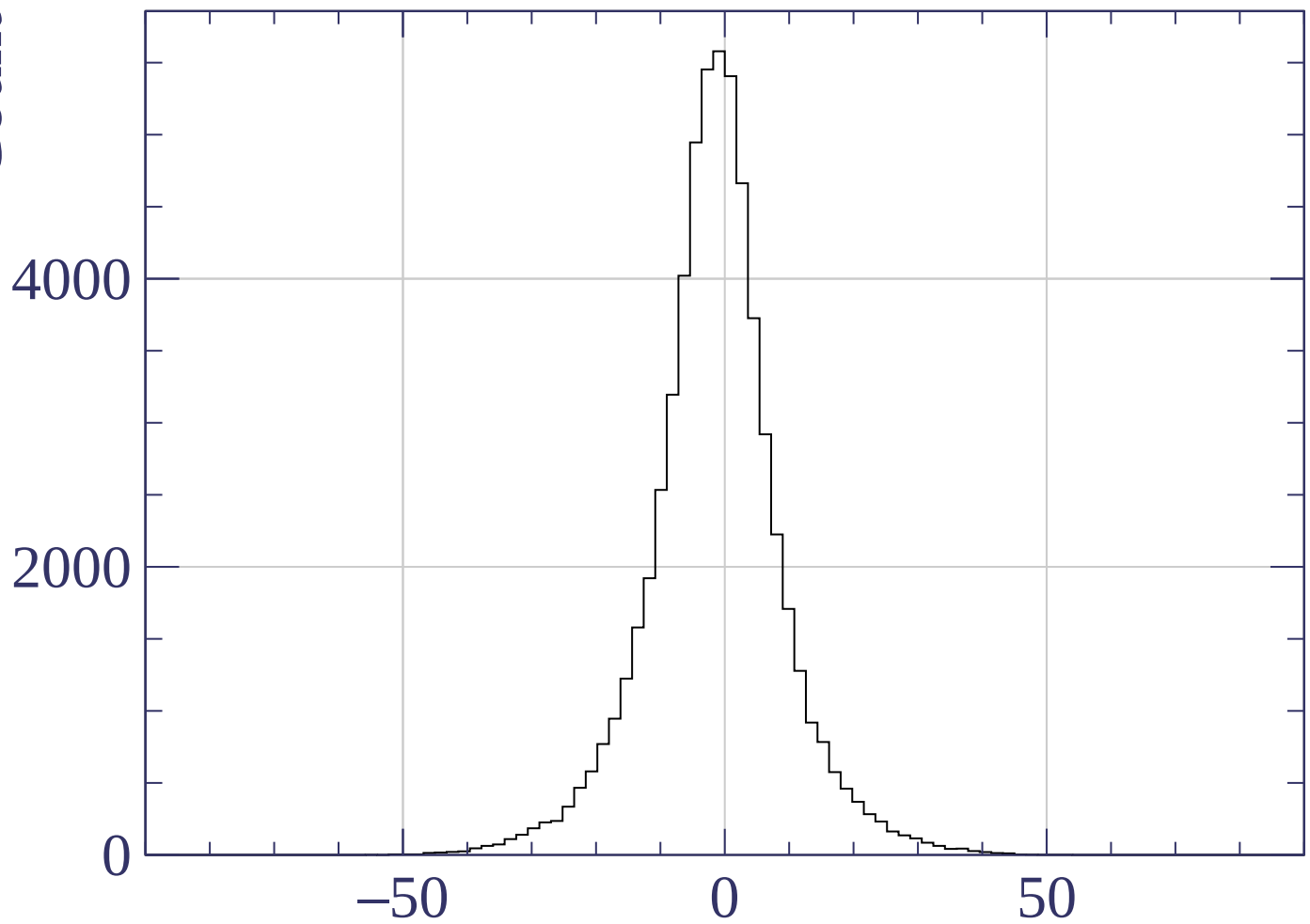






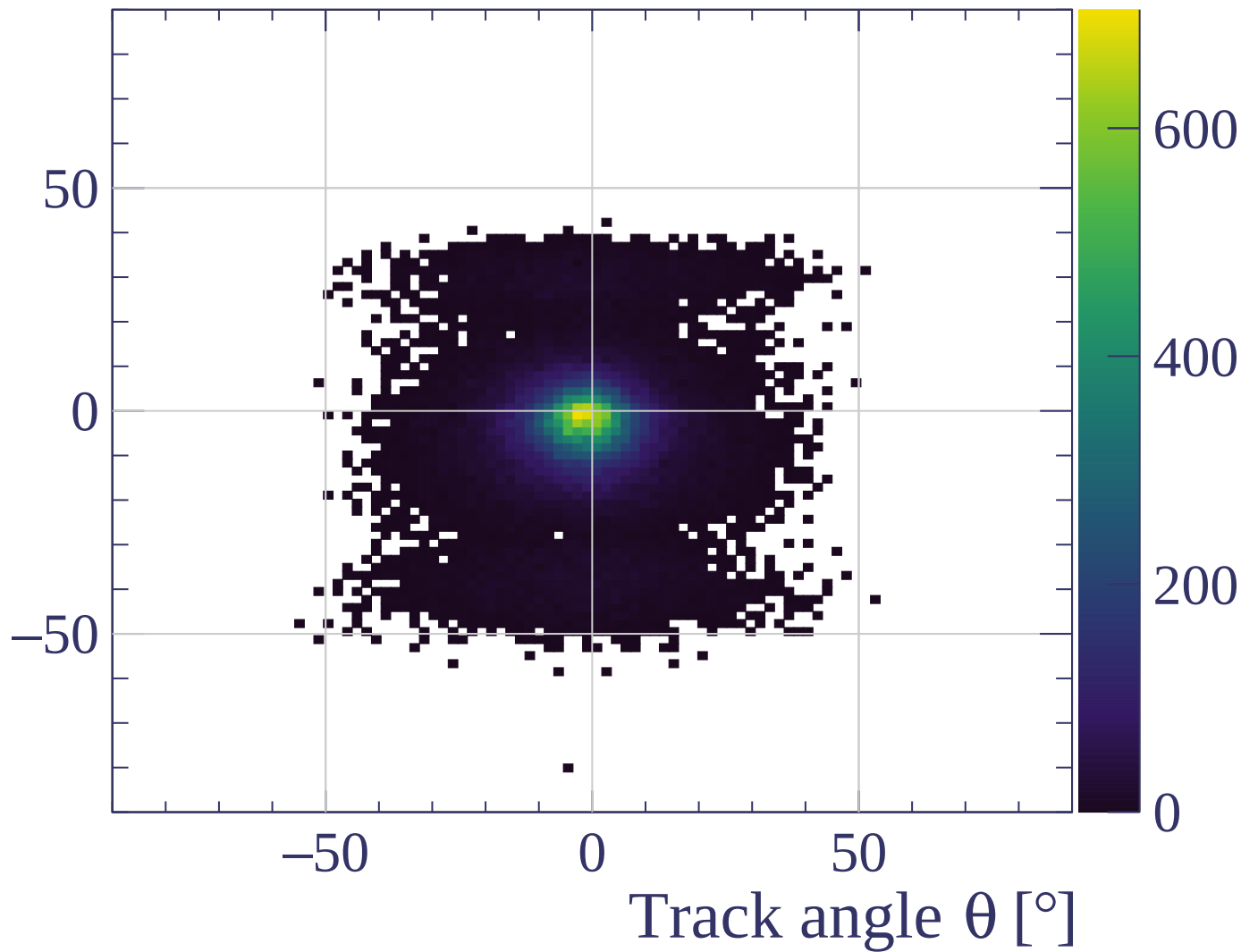


Count

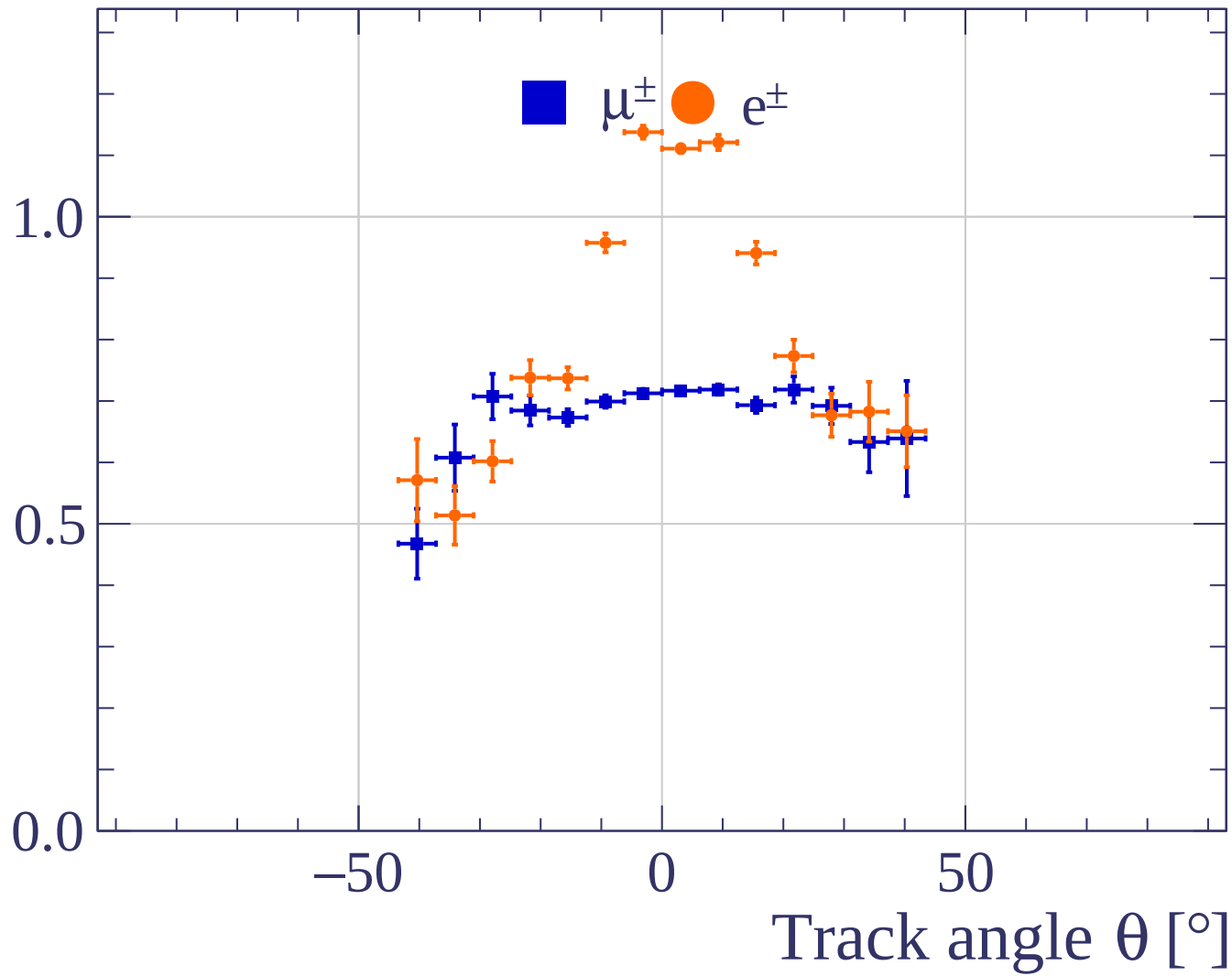


Track angle θ [°]

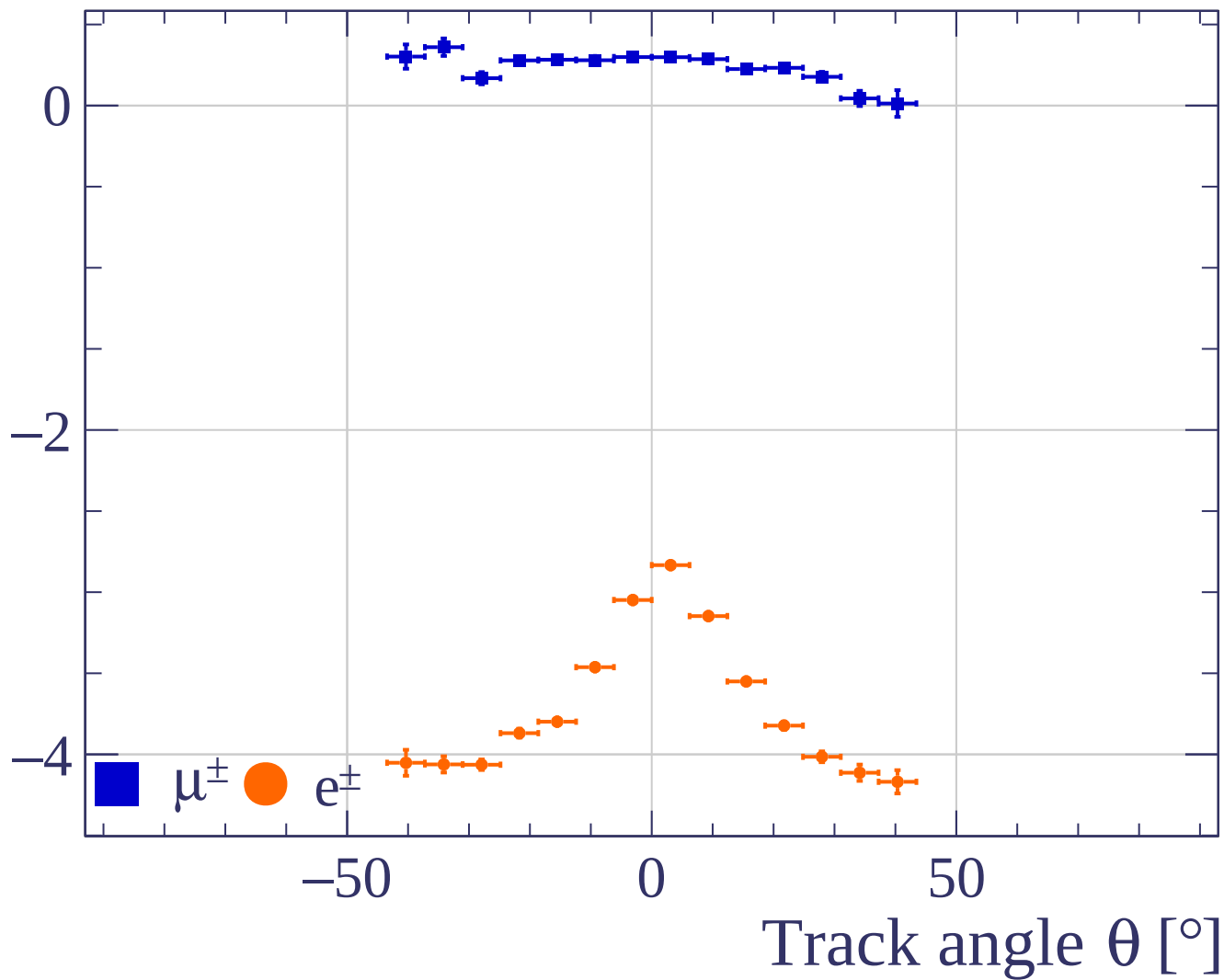
Track angle ϕ [°]



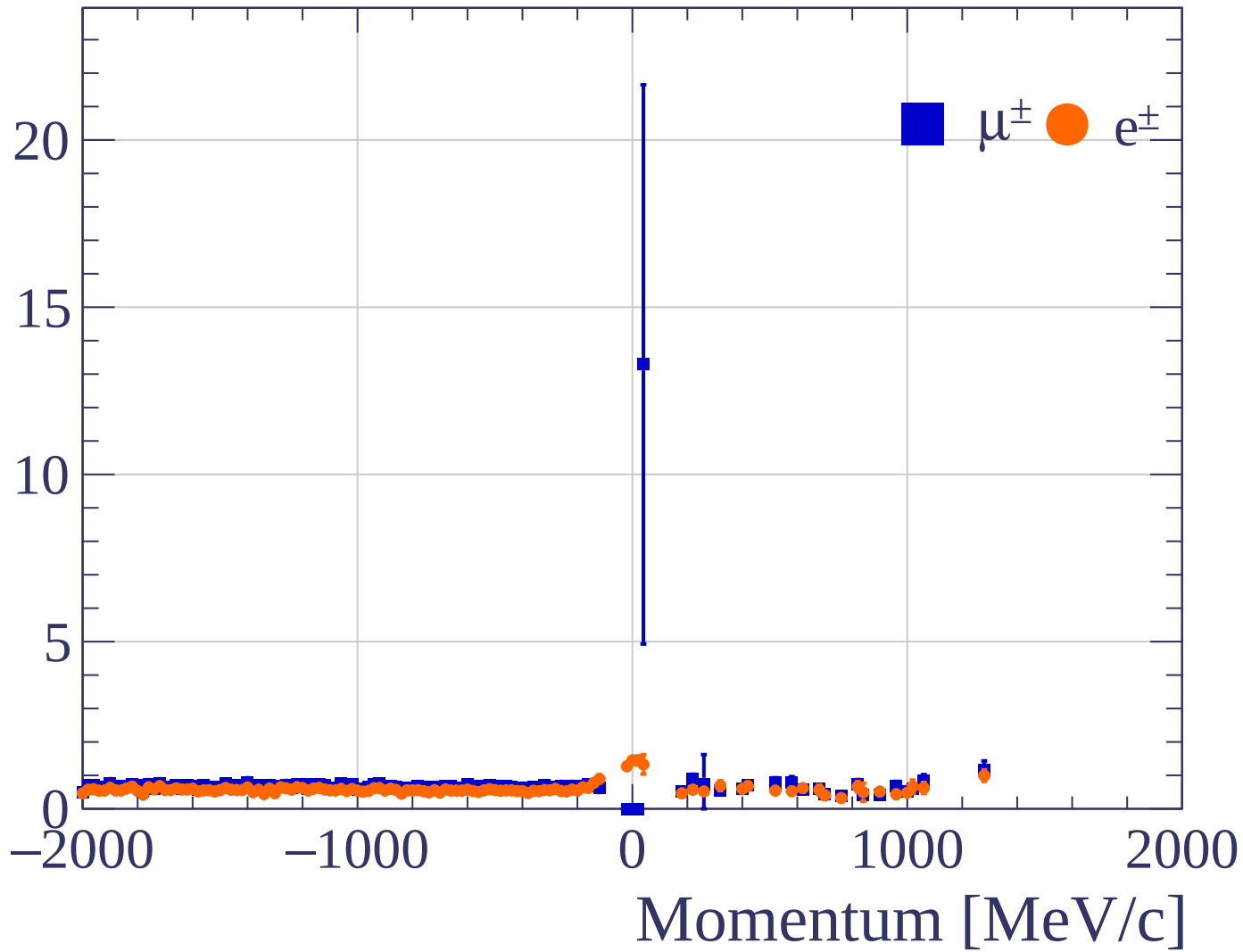
Pulls standard deviation



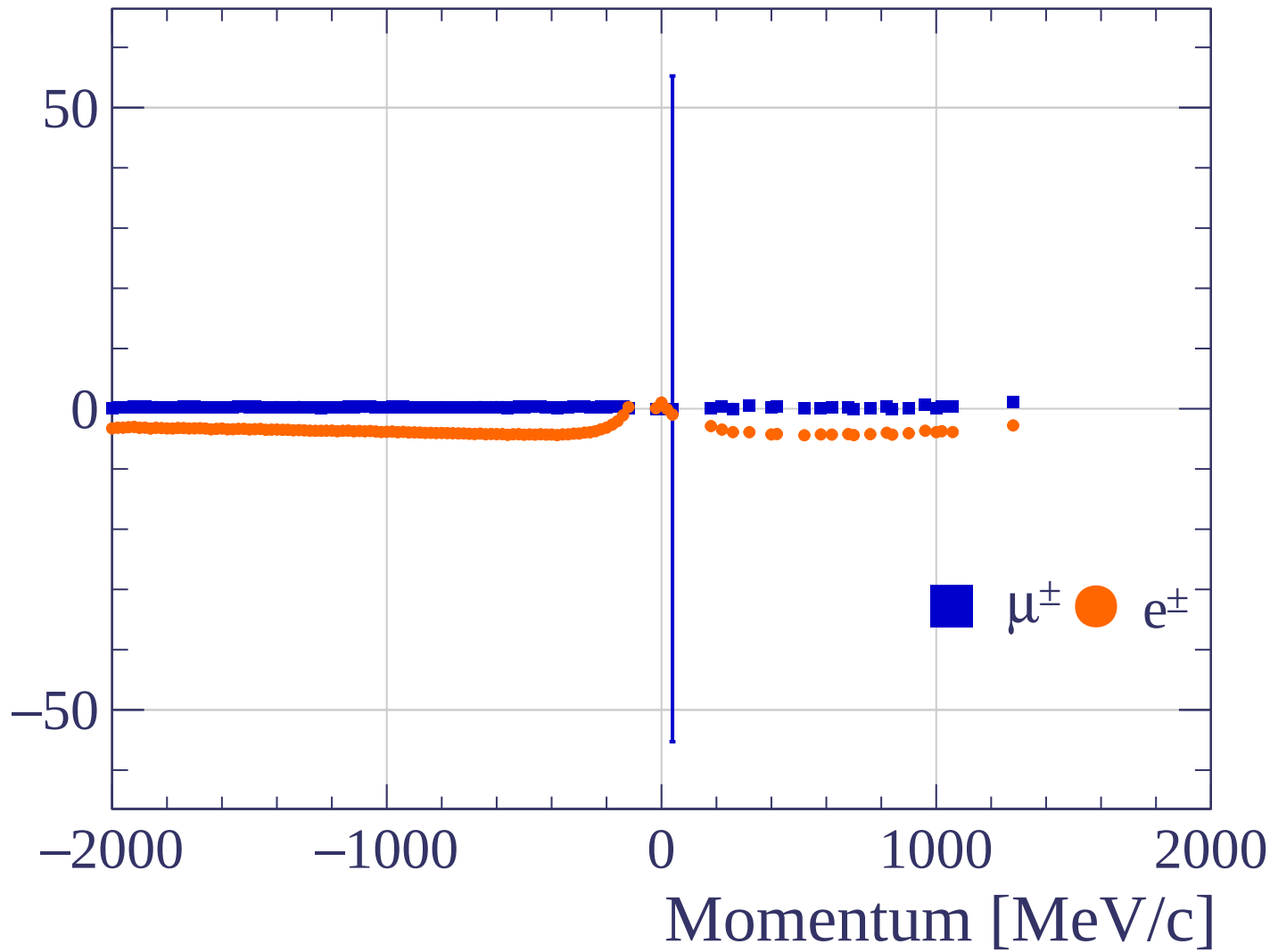
Pulls mean



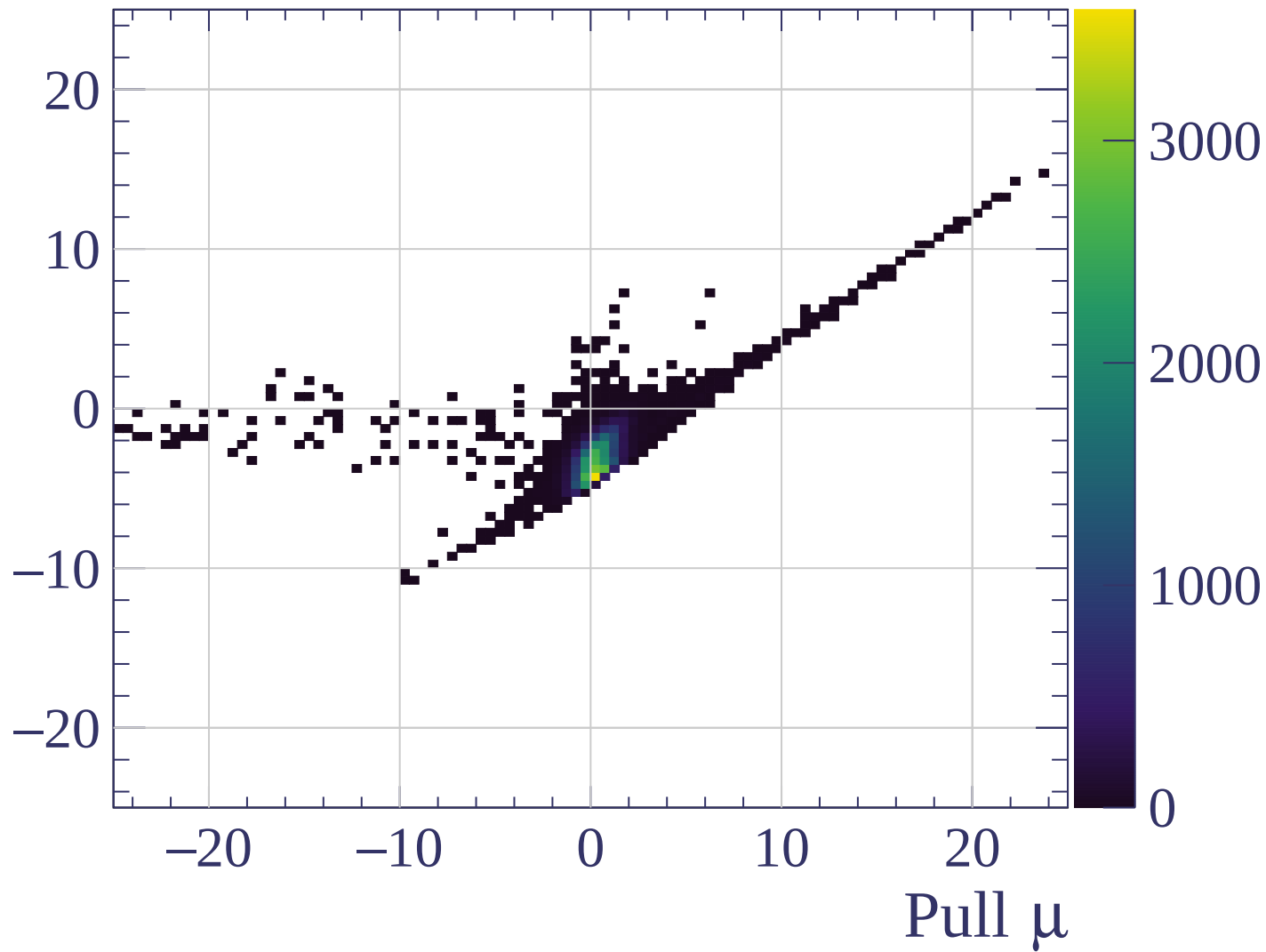
Pulls standard deviation

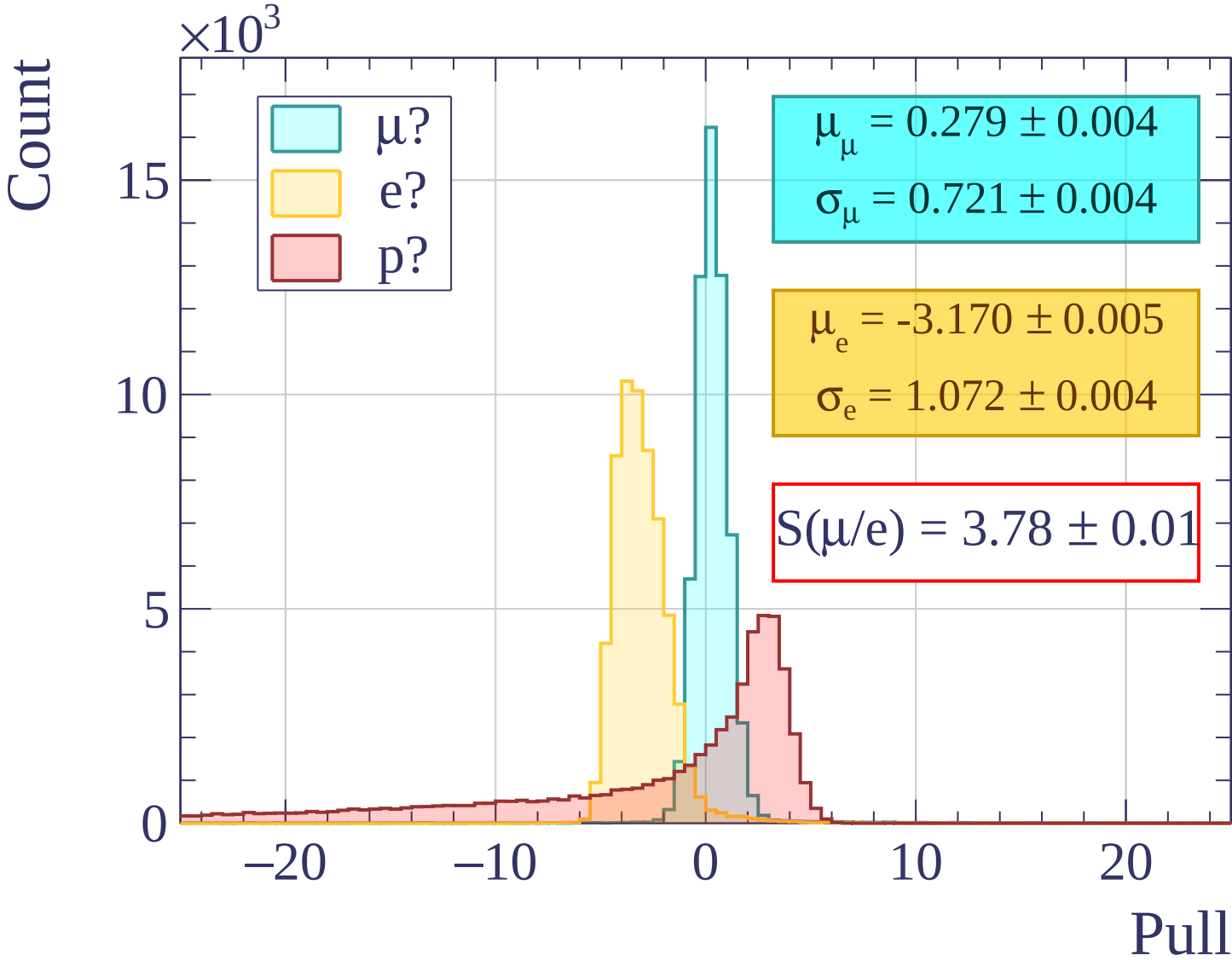


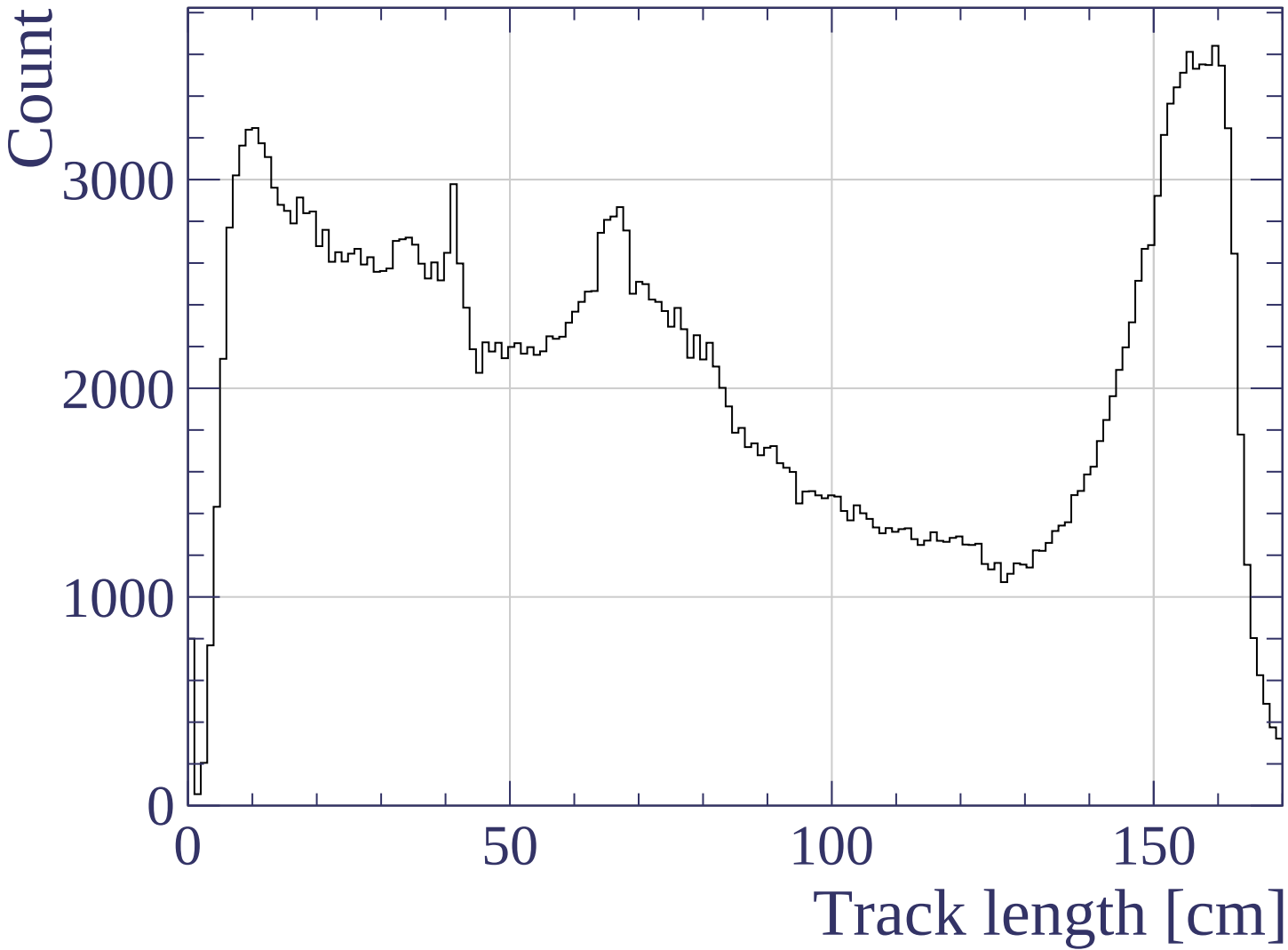
Pulls mean

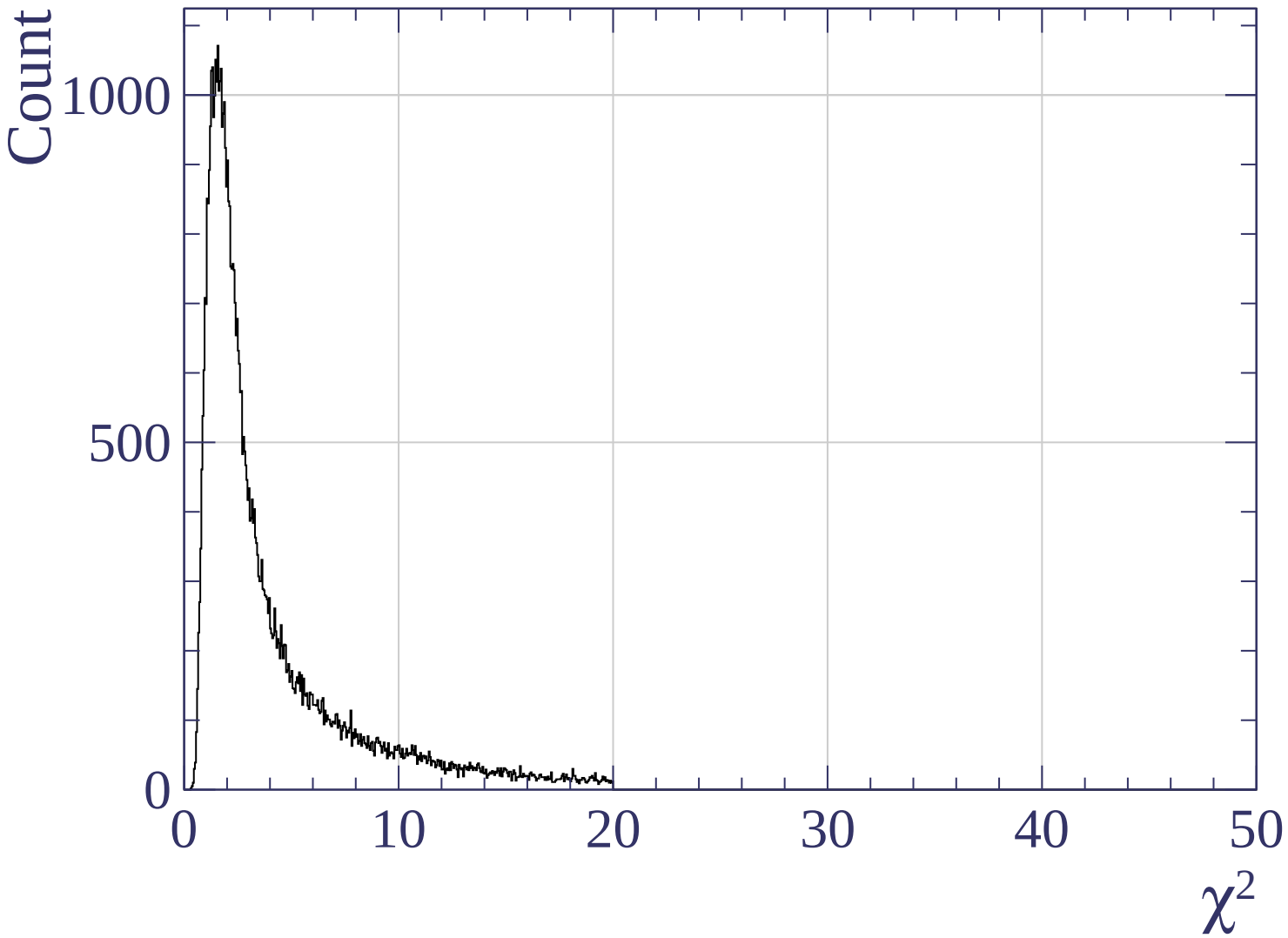


Pull e

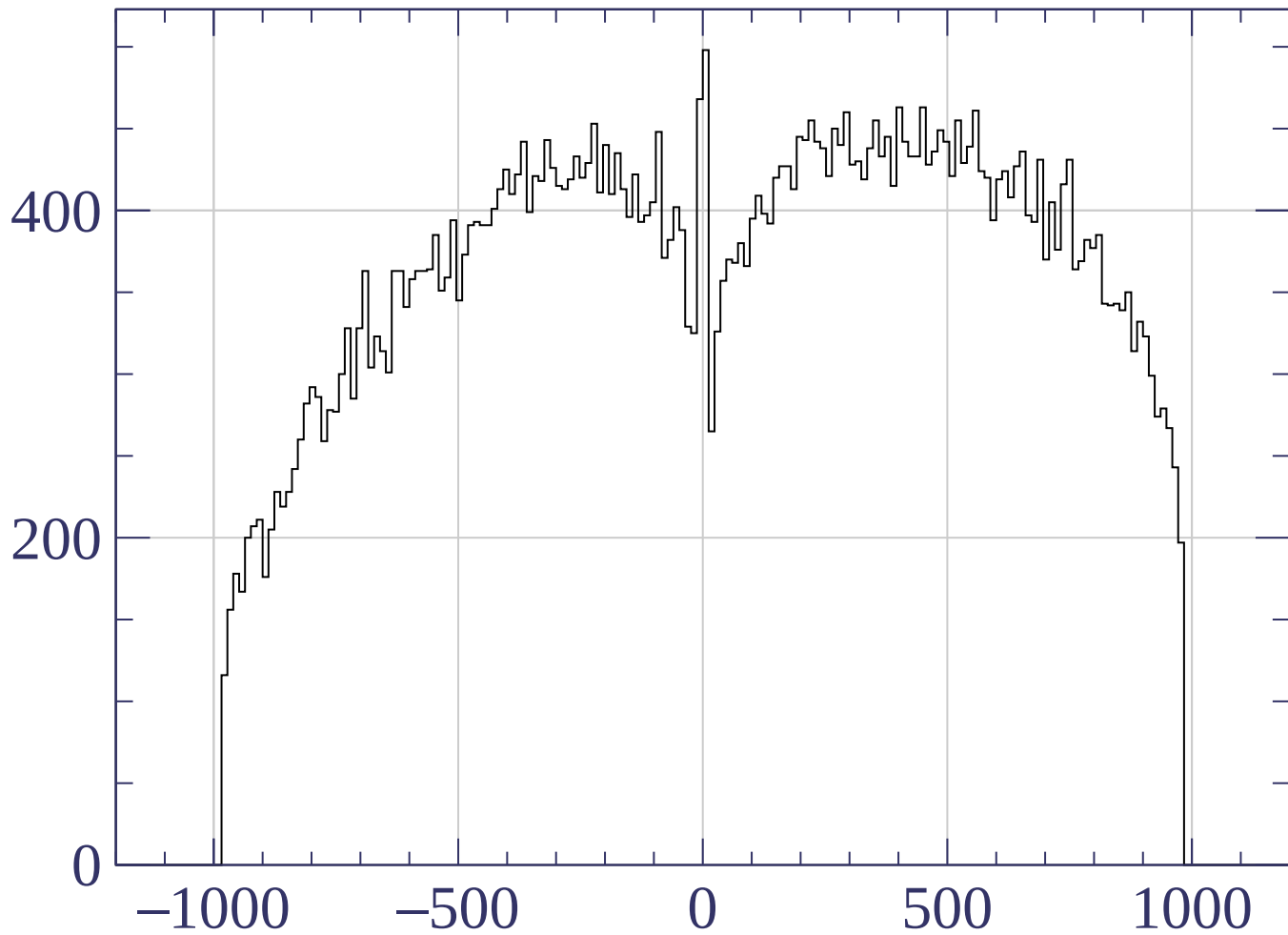




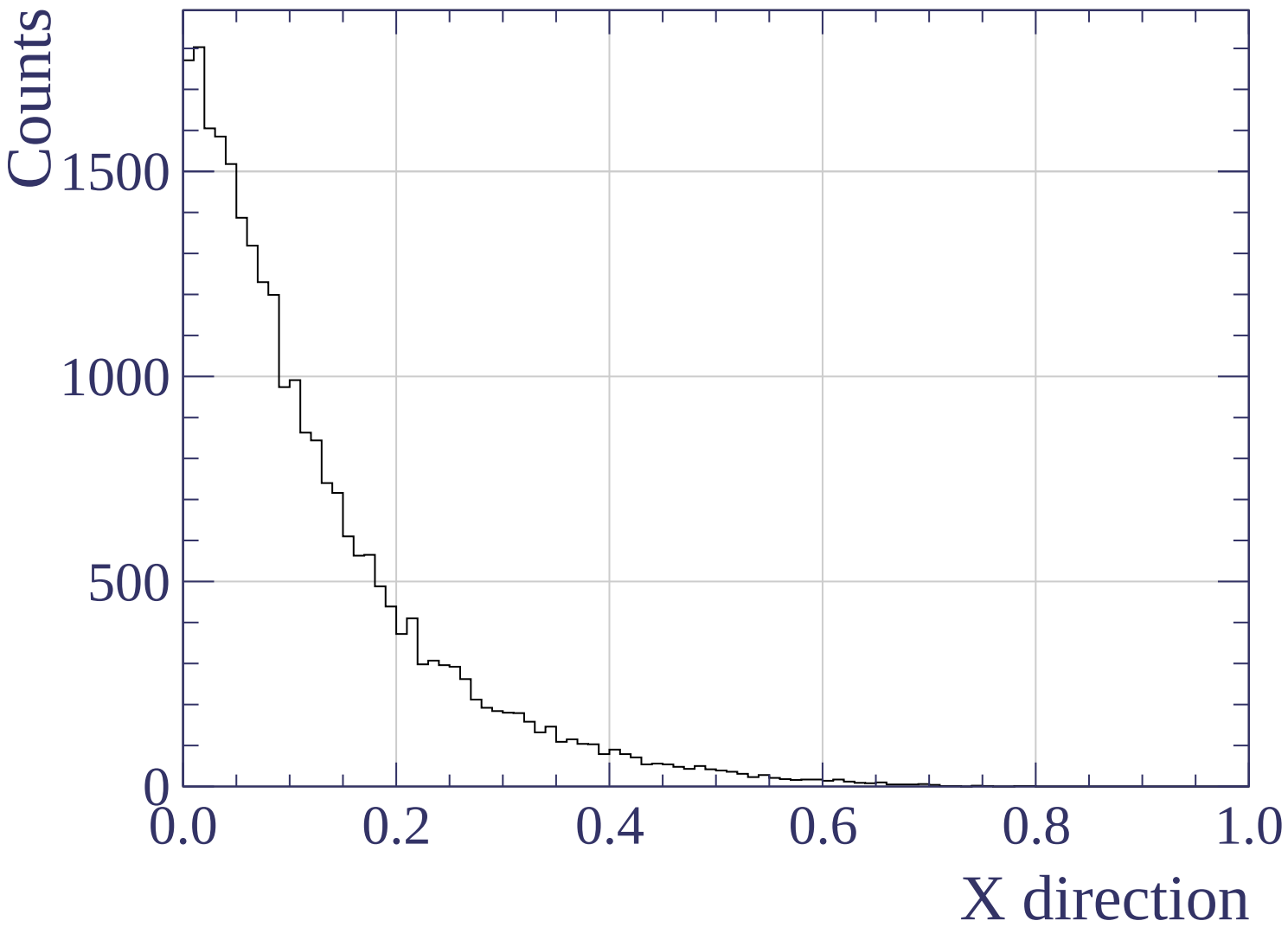


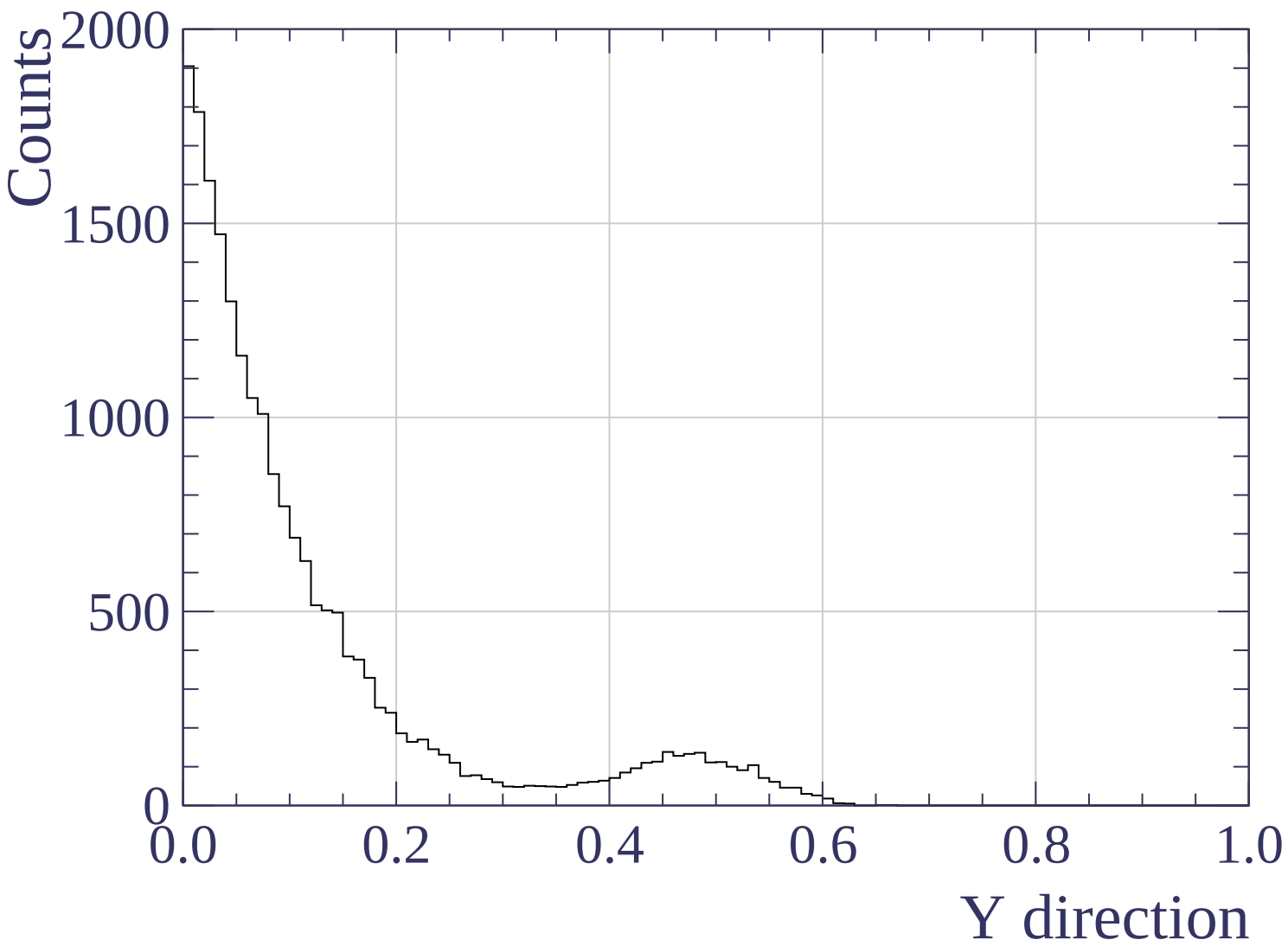


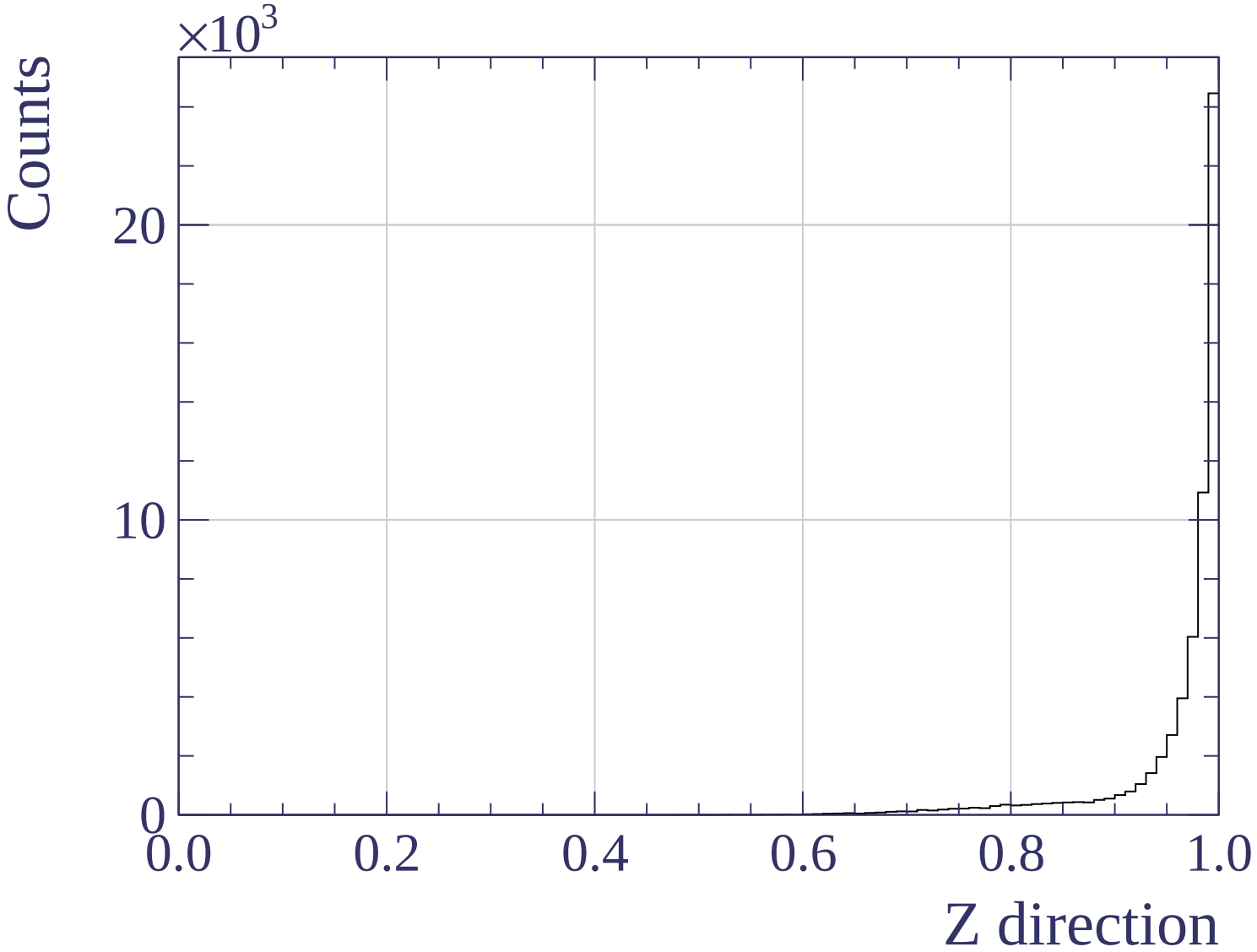
Count

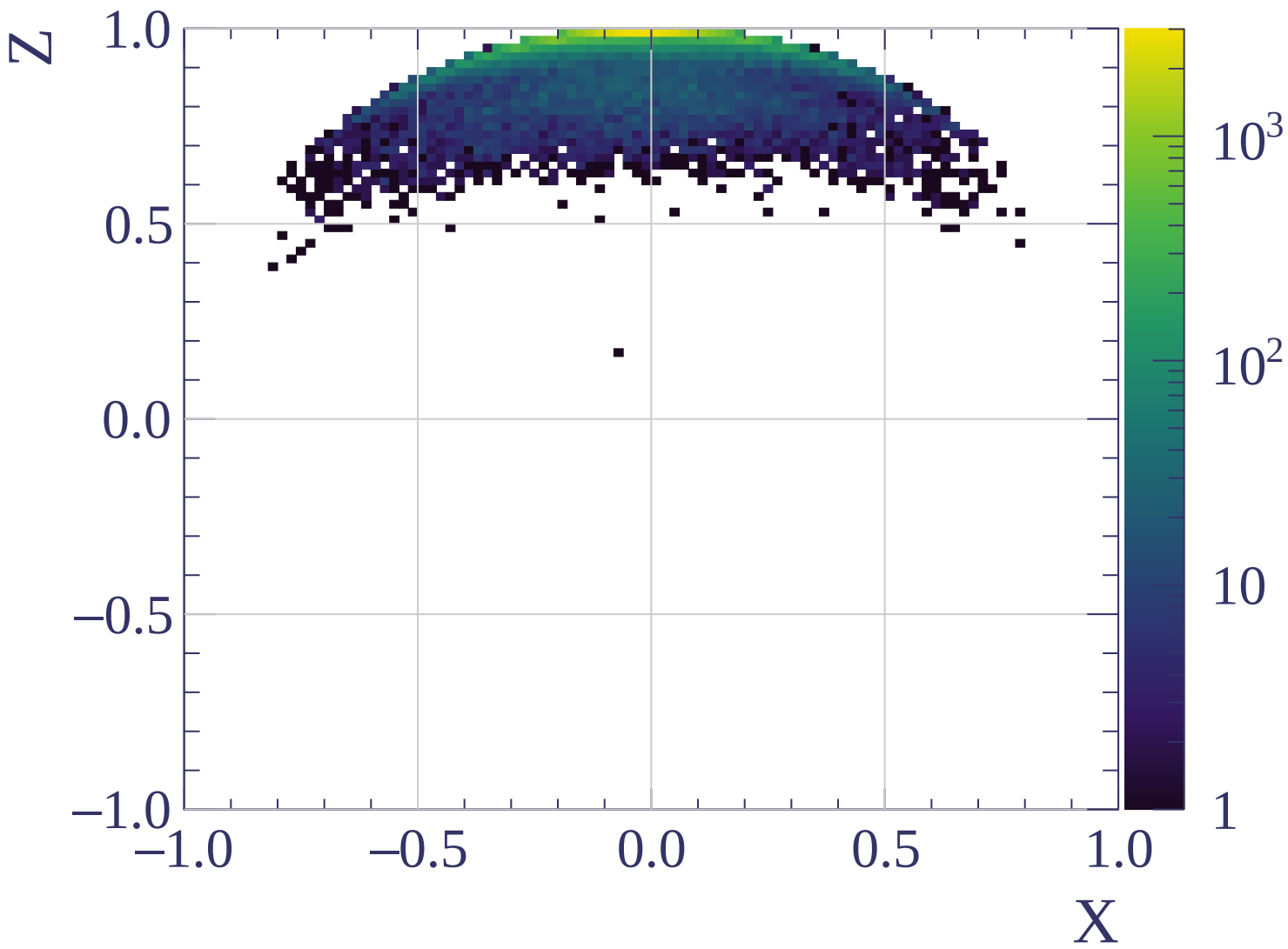


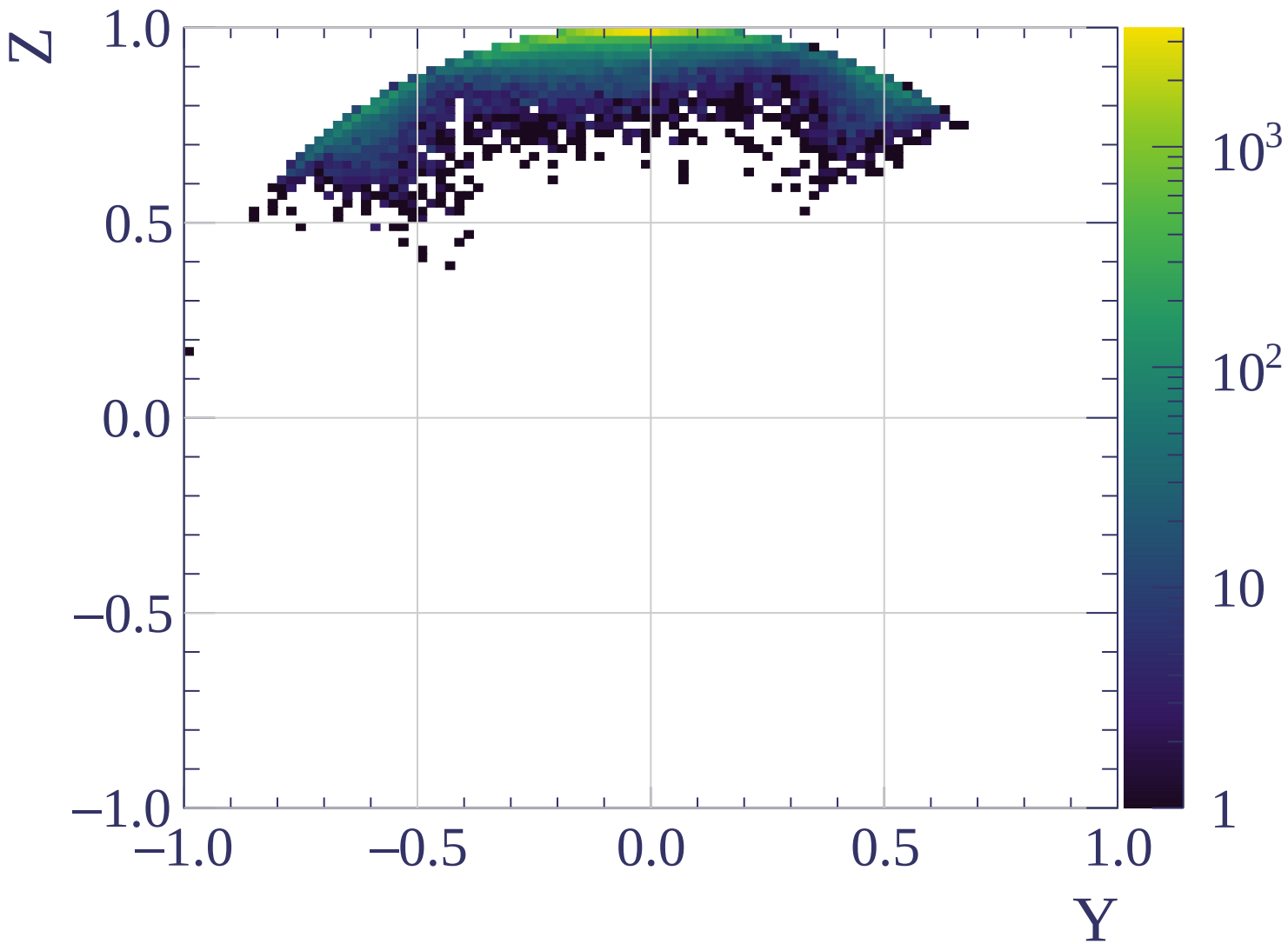
Track X position [mm]

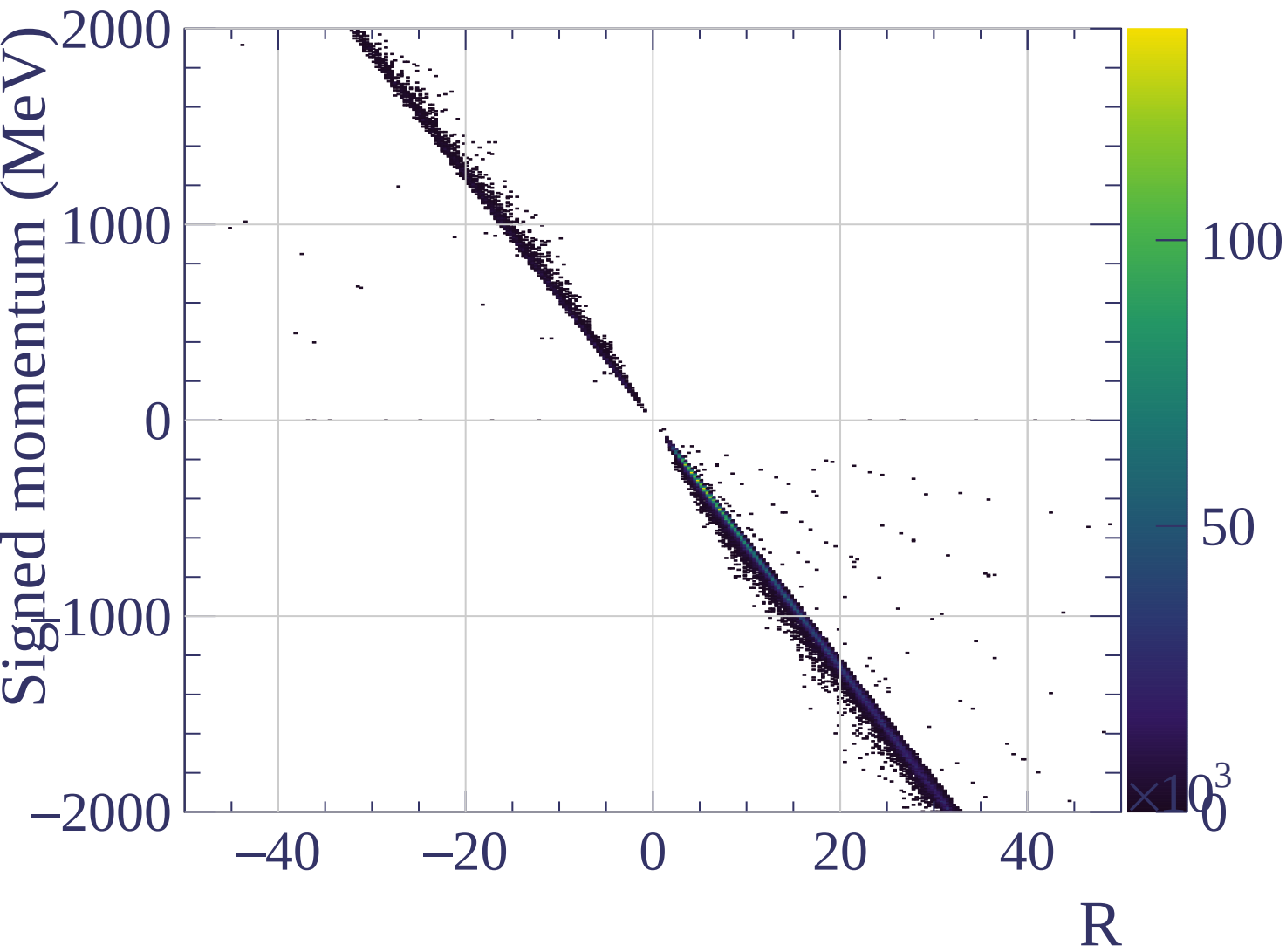


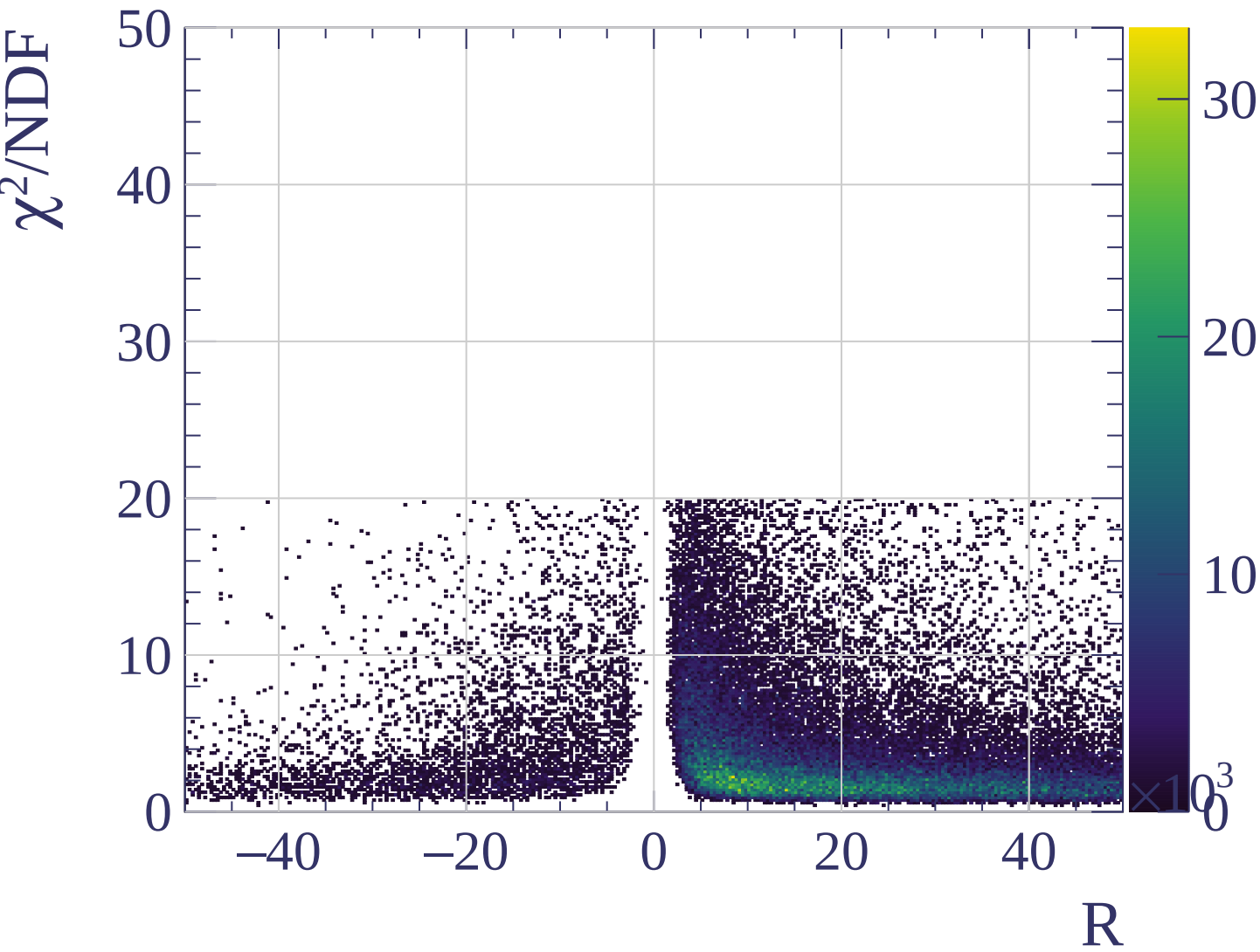


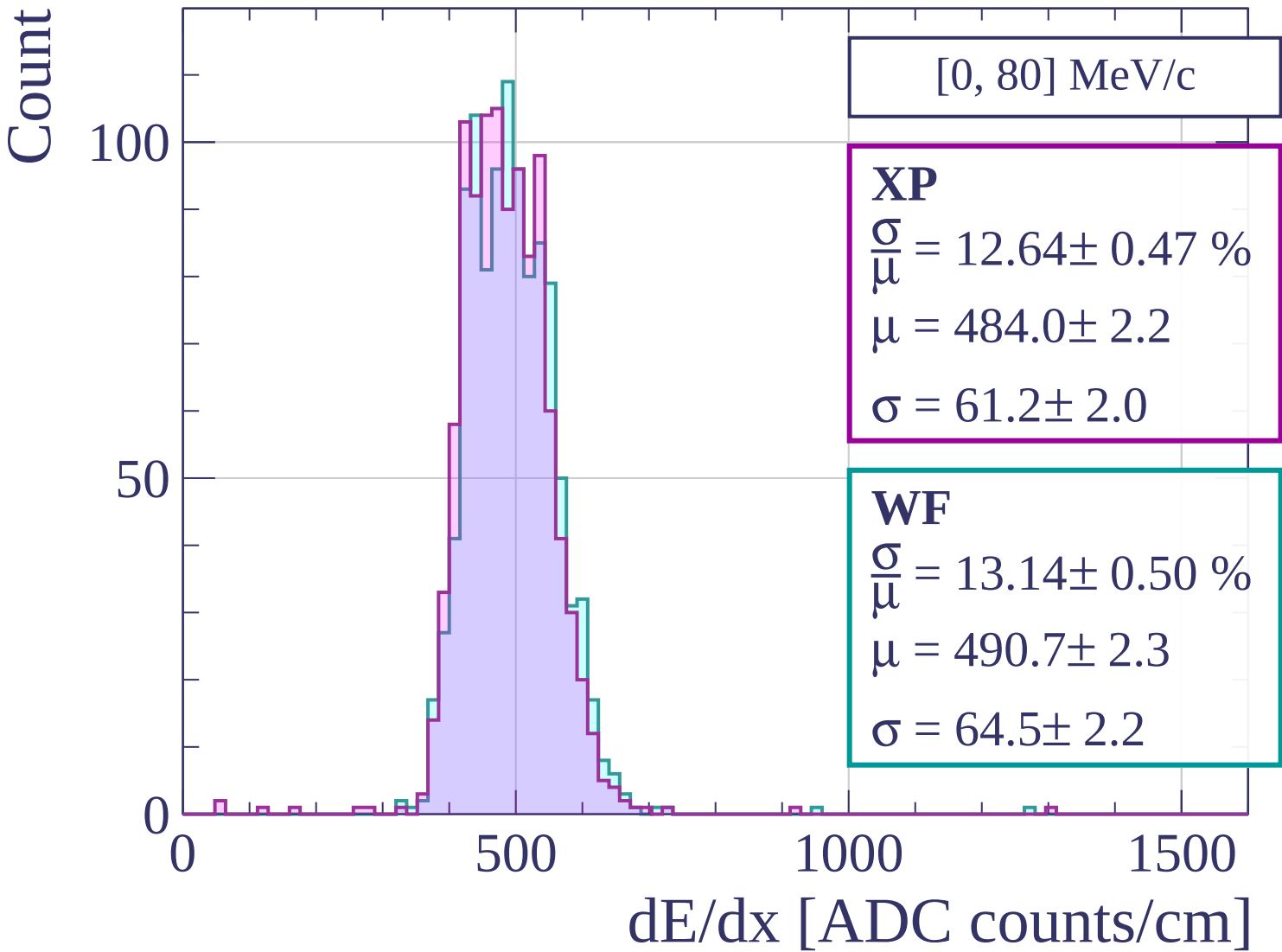




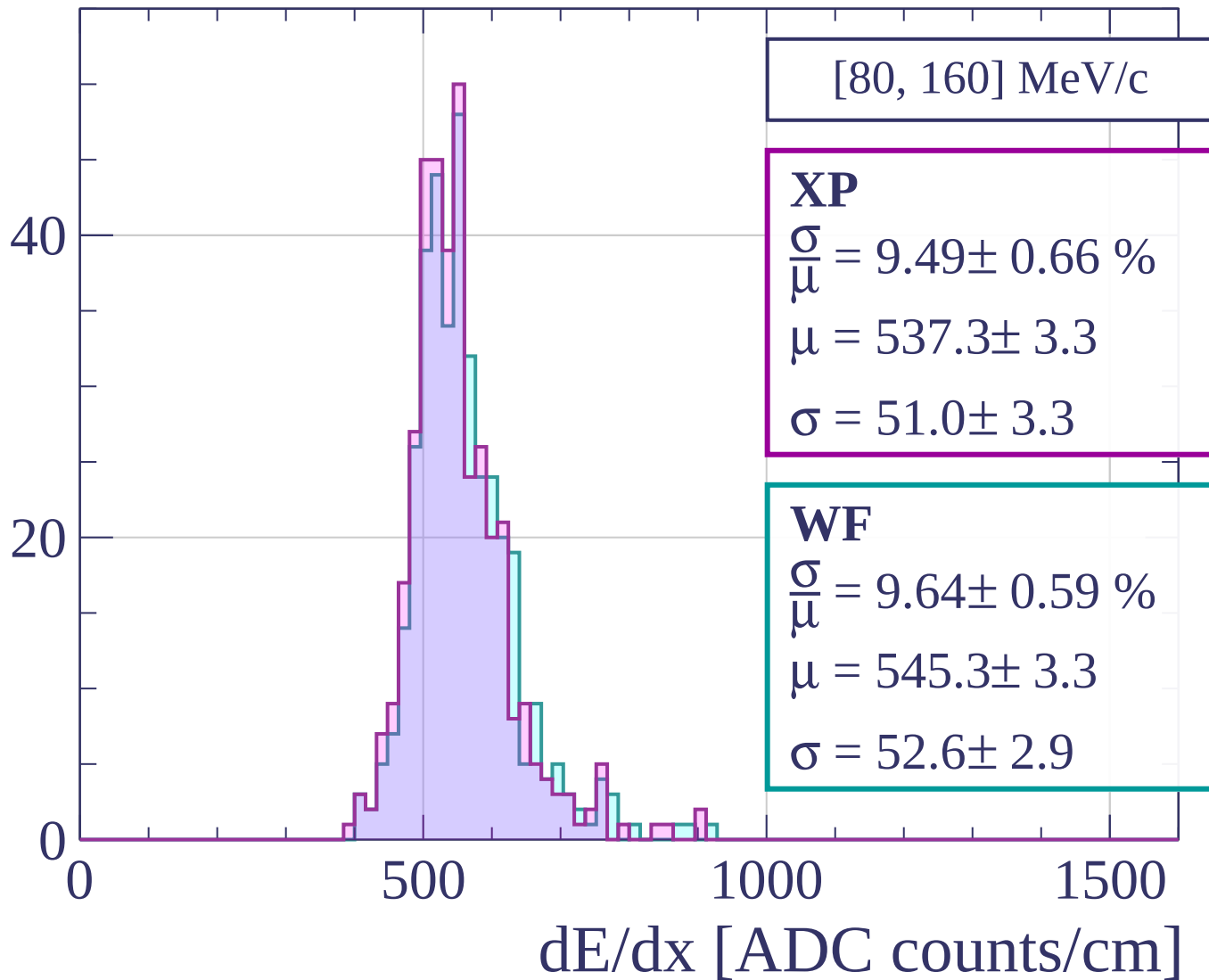




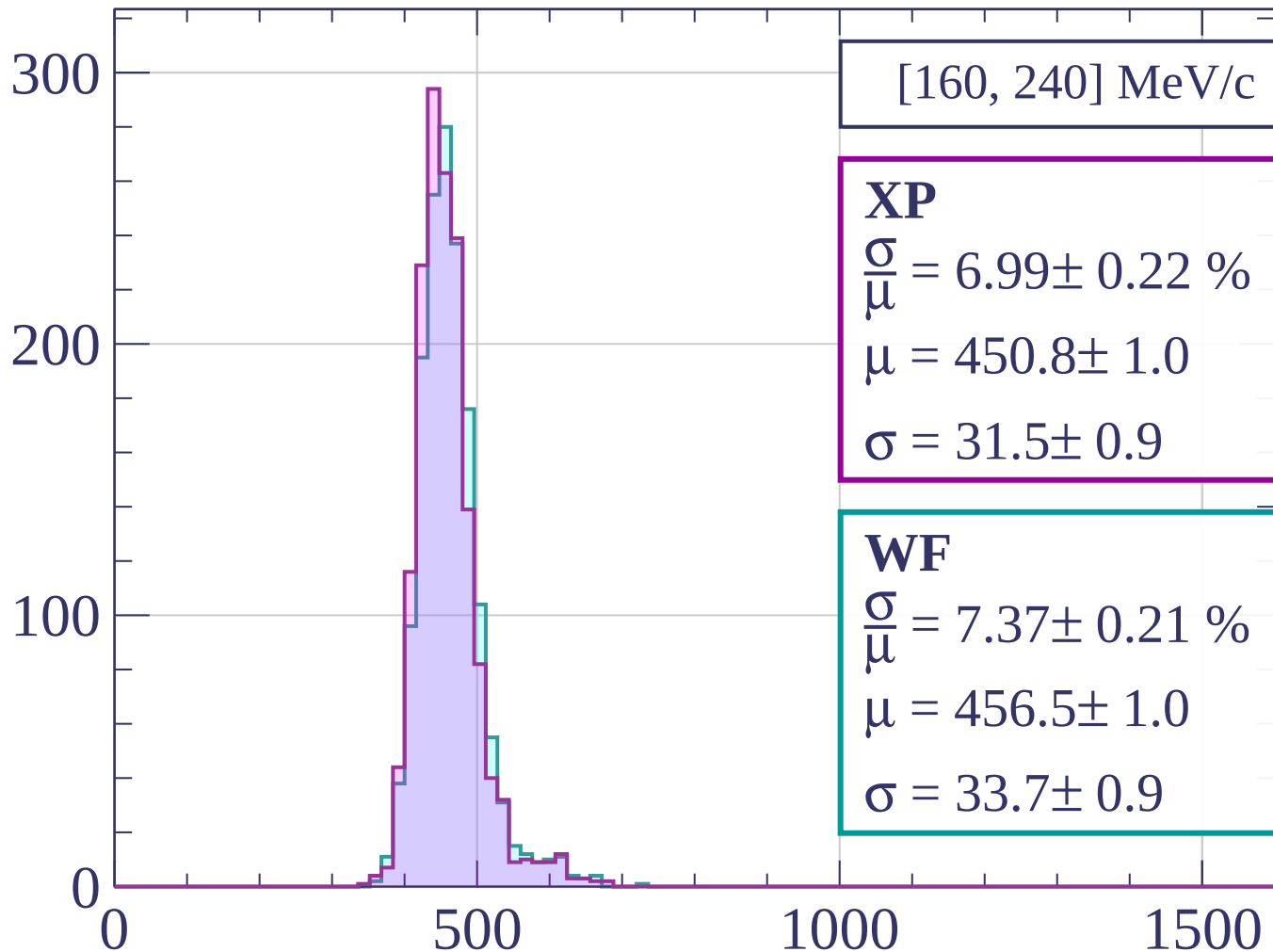




Count



Count



dE/dx [ADC counts/cm]

Count

[240, 320] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.07 \pm 0.17 \%$$

$$\mu = 417.9 \pm 0.7$$

$$\sigma = 25.4 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 6.56 \pm 0.18 \%$$

$$\mu = 422.0 \pm 0.7$$

$$\sigma = 27.7 \pm 0.7$$

400

200

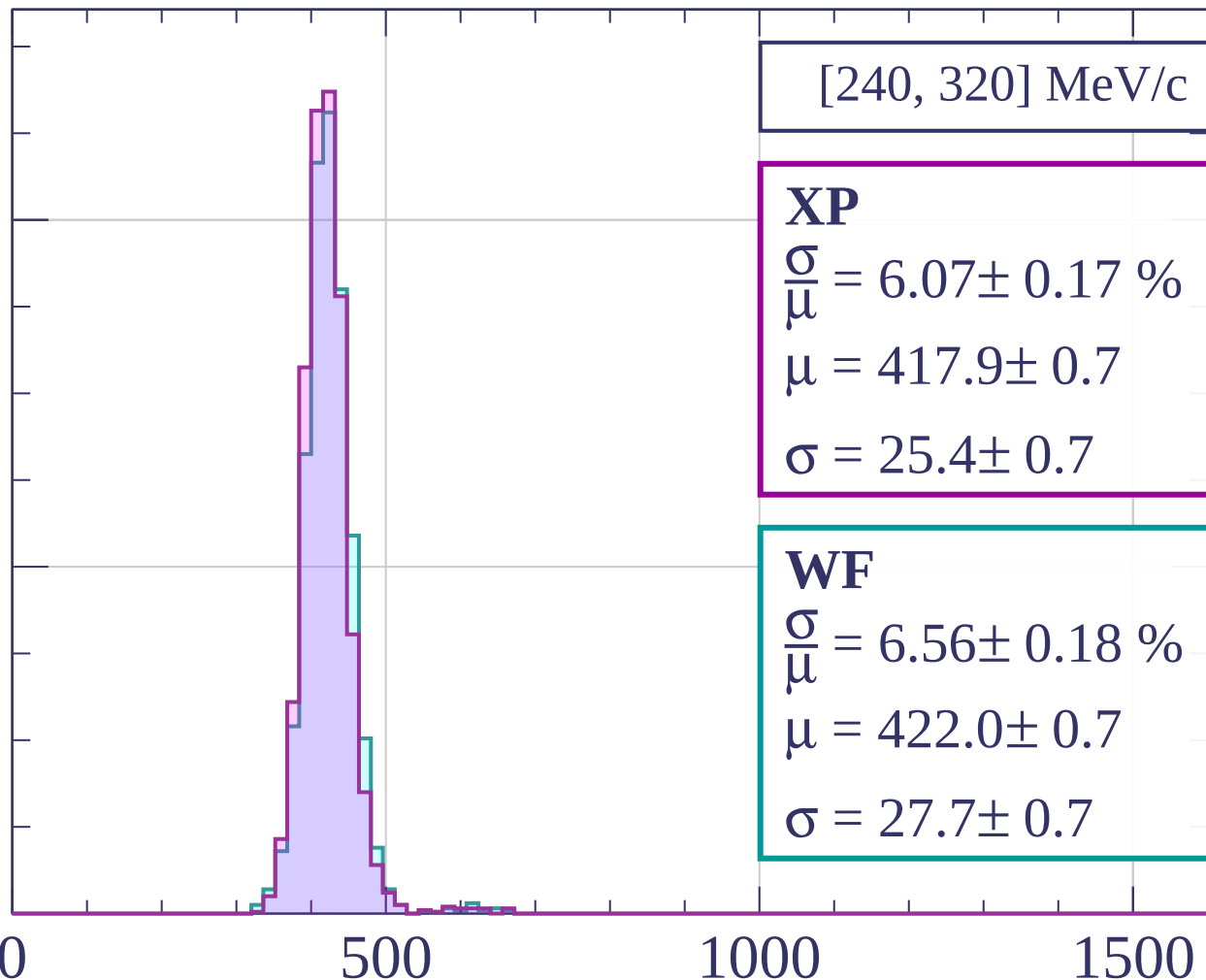
0

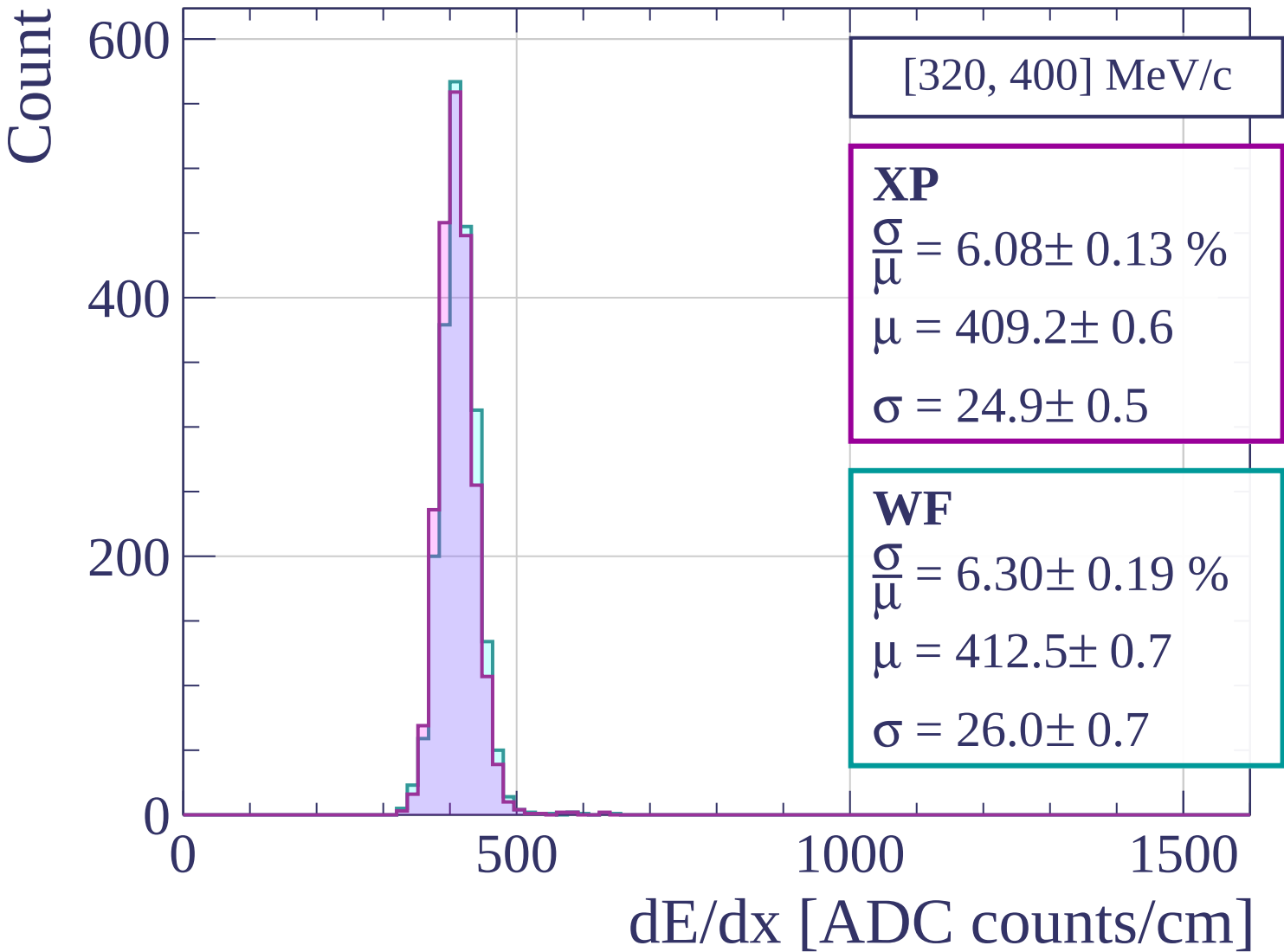
500

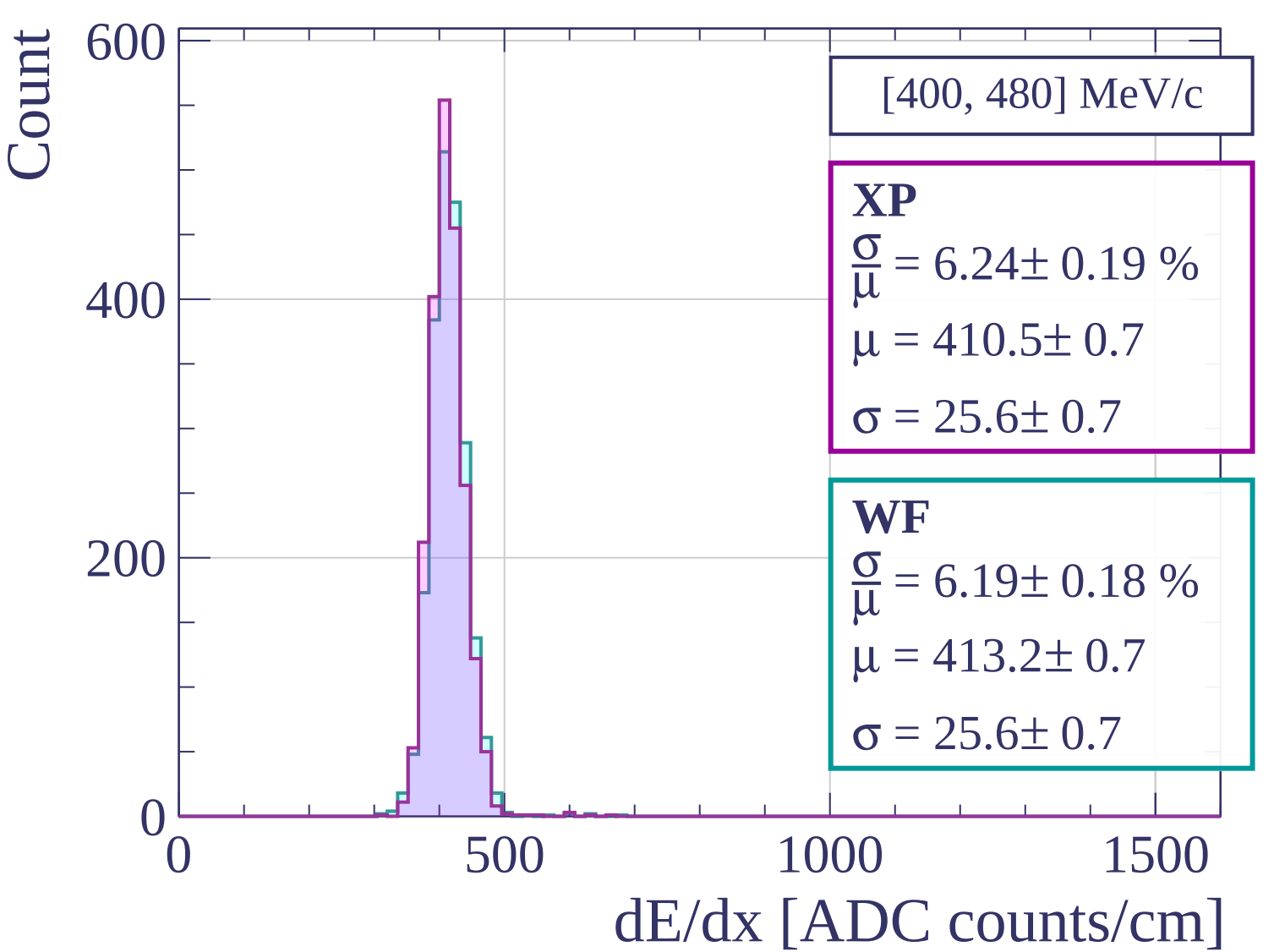
1000

1500

dE/dx [ADC counts/cm]







Count

[480, 560] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.25 \pm 0.19 \%$$

$$\mu = 413.4 \pm 0.7$$

$$\sigma = 25.8 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 6.39 \pm 0.20 \%$$

$$\mu = 417.0 \pm 0.7$$

$$\sigma = 26.6 \pm 0.8$$

400

200

0

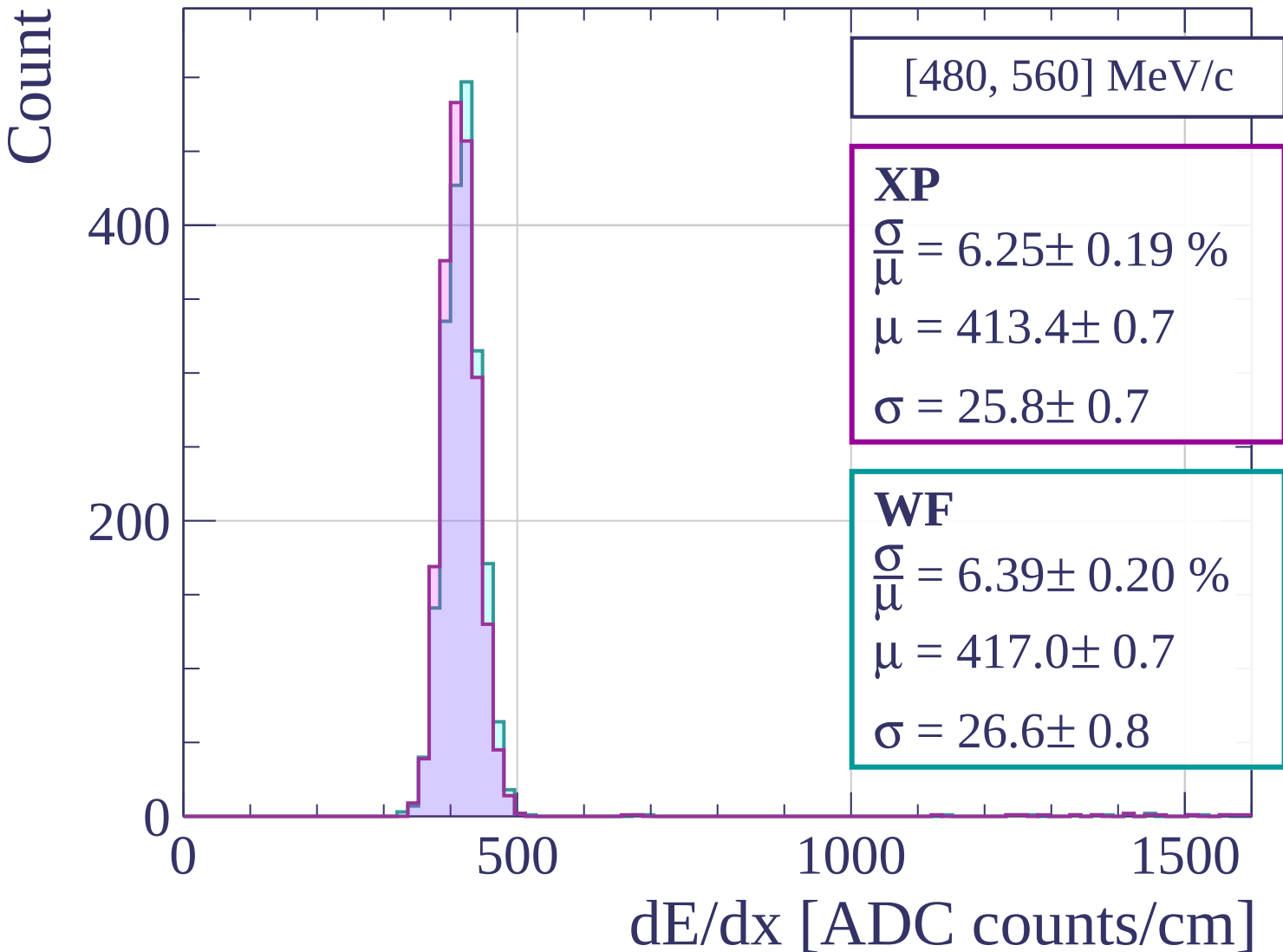
0

500

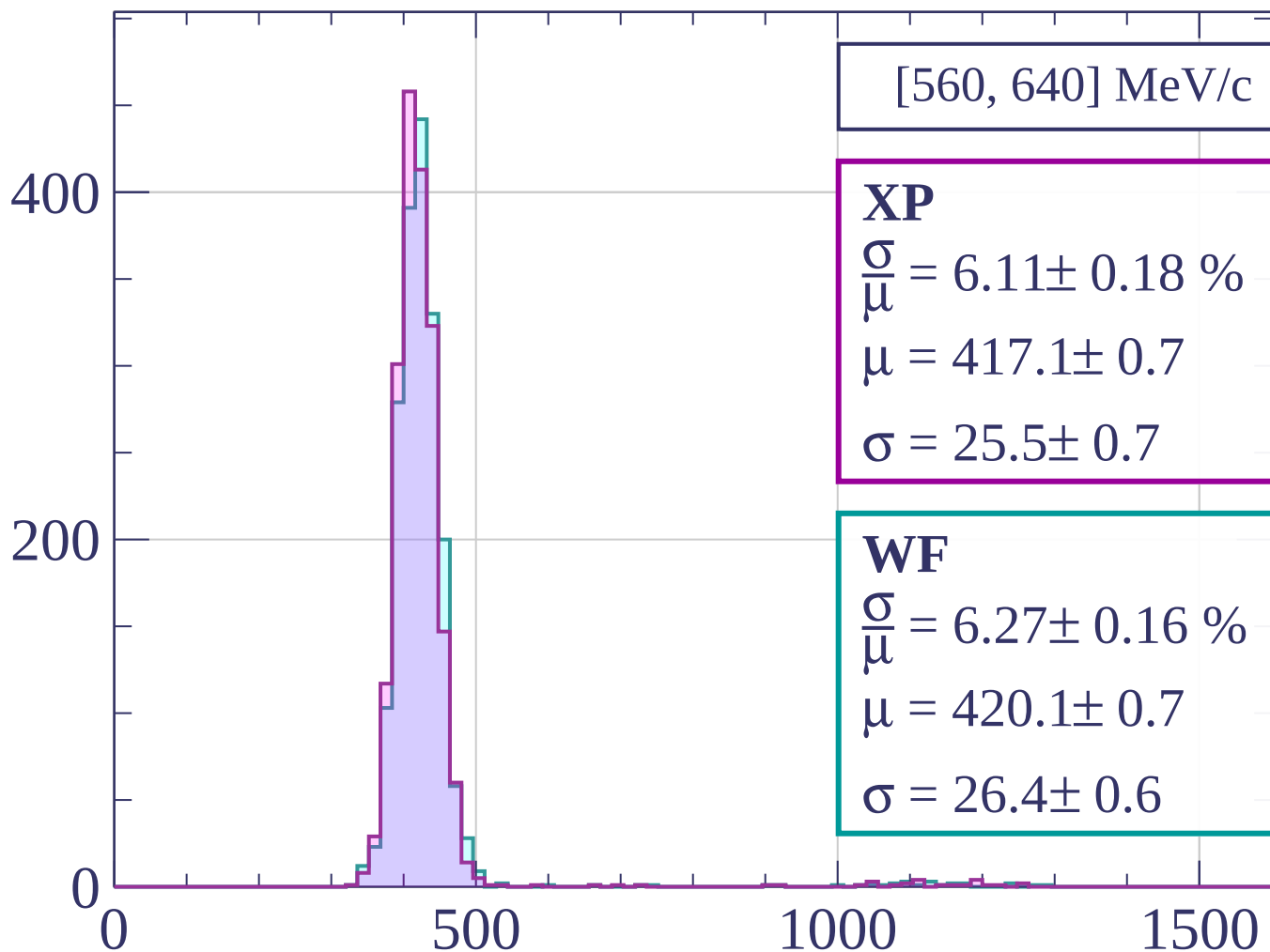
1000

1500

dE/dx [ADC counts/cm]



Count



dE/dx [ADC counts/cm]

Count

[640, 720] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.07 \pm 0.16 \%$$

$$\mu = 422.5 \pm 0.7$$

$$\sigma = 25.7 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 6.22 \pm 0.16 \%$$

$$\mu = 425.7 \pm 0.7$$

$$\sigma = 26.5 \pm 0.7$$

400

300

200

100

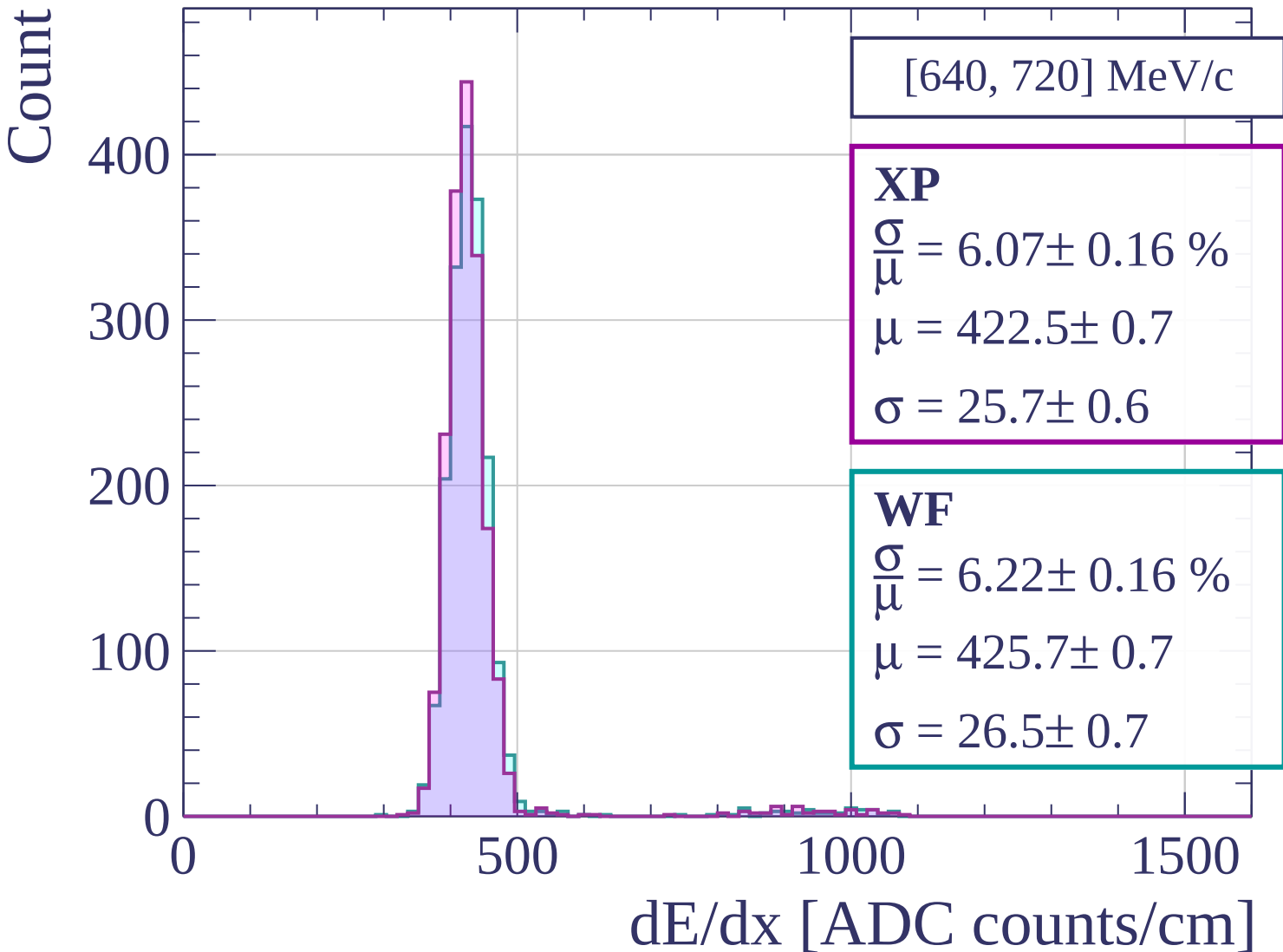
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[720, 800] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.64 \pm 0.17 \%$$

$$\mu = 428.3 \pm 0.7$$

$$\sigma = 24.2 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 5.87 \pm 0.20 \%$$

$$\mu = 430.9 \pm 0.8$$

$$\sigma = 25.3 \pm 0.8$$

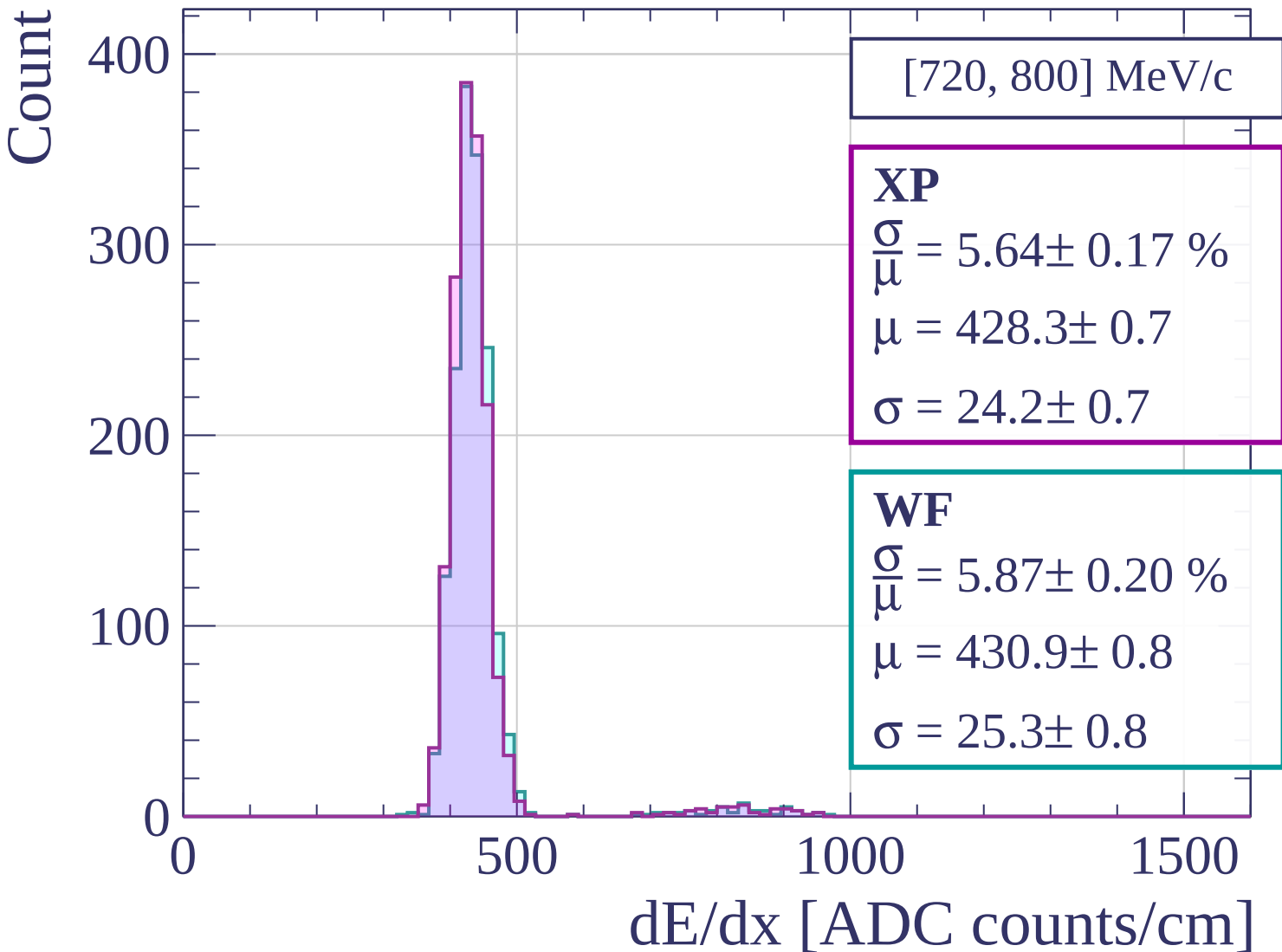
400
300
200
100
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[800, 880] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.00 \pm 0.22 \%$$

$$\mu = 431.7 \pm 0.8$$

$$\sigma = 25.9 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 6.01 \pm 0.18 \%$$

$$\mu = 435.2 \pm 0.8$$

$$\sigma = 26.1 \pm 0.7$$

300

200

100

0

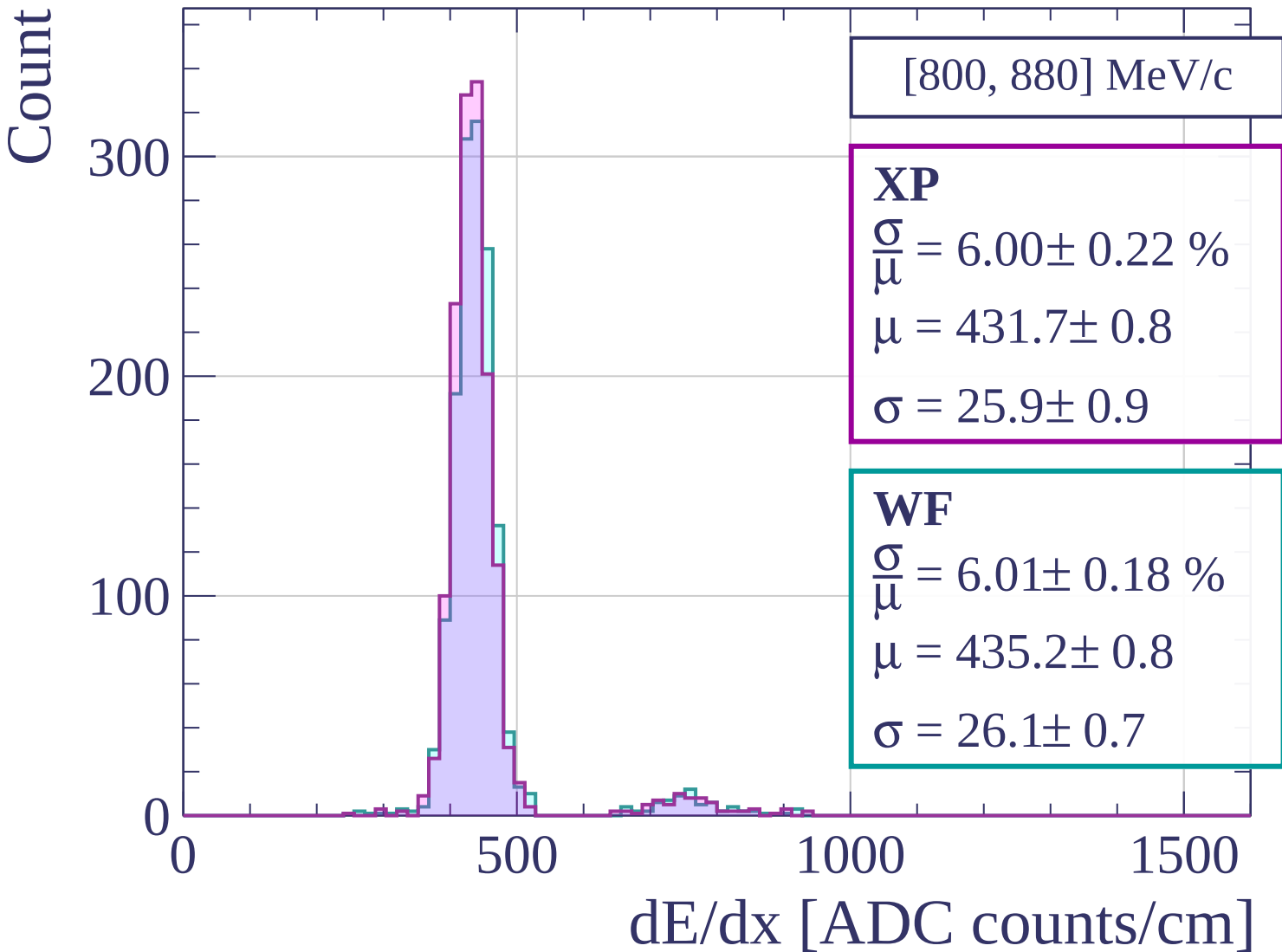
0

500

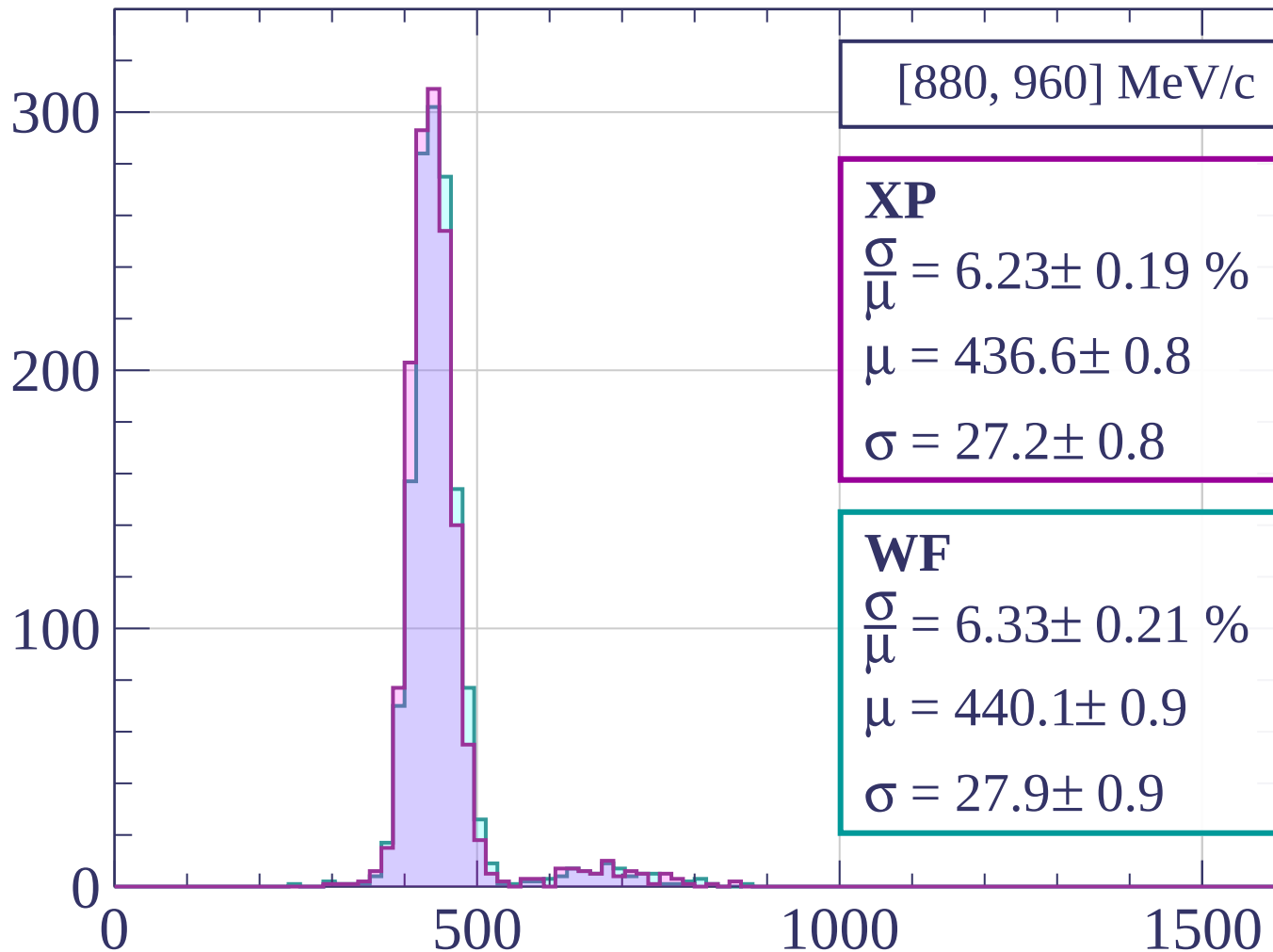
1000

1500

dE/dx [ADC counts/cm]



Count



dE/dx [ADC counts/cm]

Count

[960, 1040] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.74 \pm 0.21 \%$$

$$\mu = 442.0 \pm 0.9$$

$$\sigma = 25.4 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 5.86 \pm 0.23 \%$$

$$\mu = 445.5 \pm 0.9$$

$$\sigma = 26.1 \pm 1.0$$

300

200

100

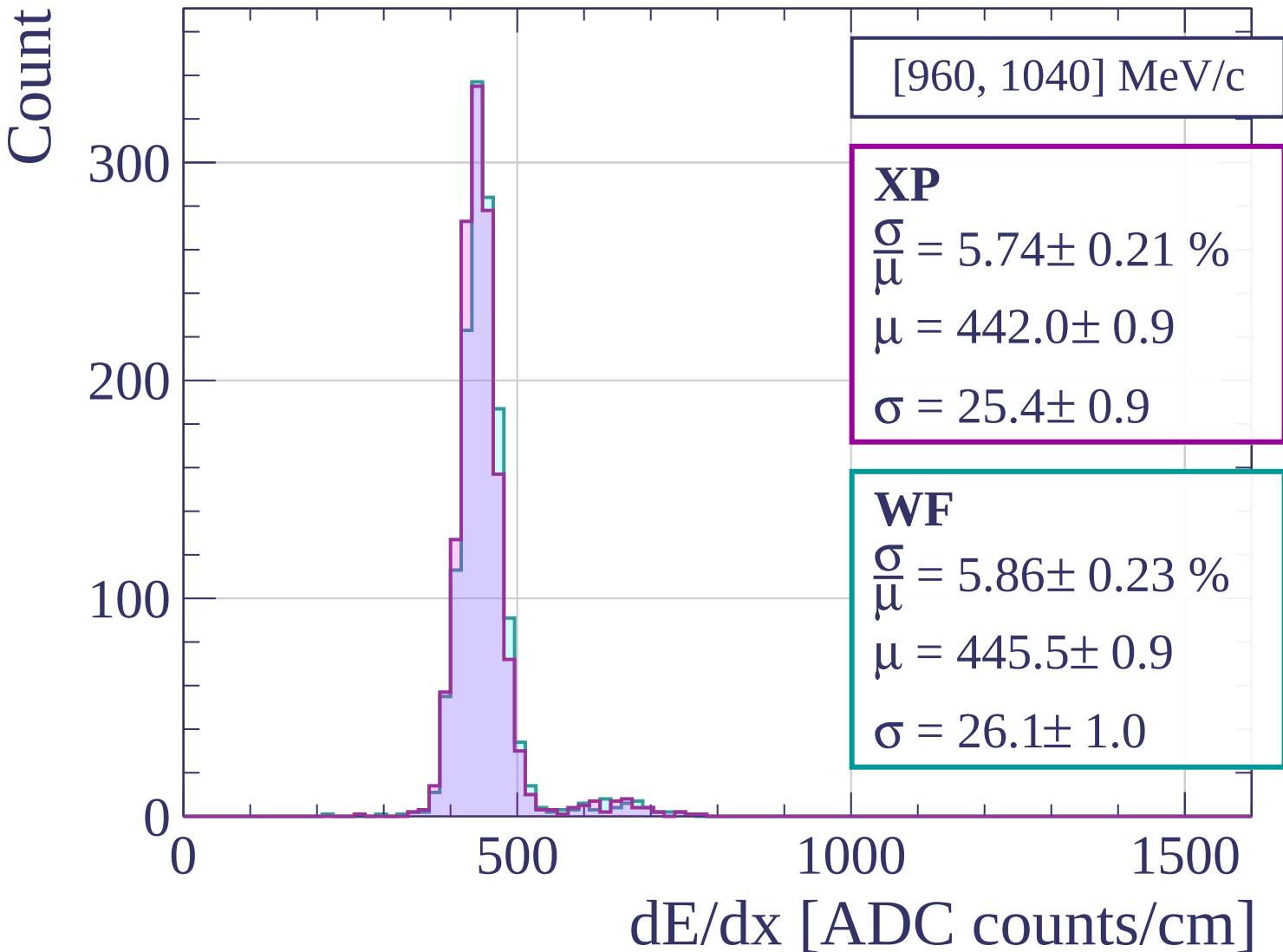
0

500

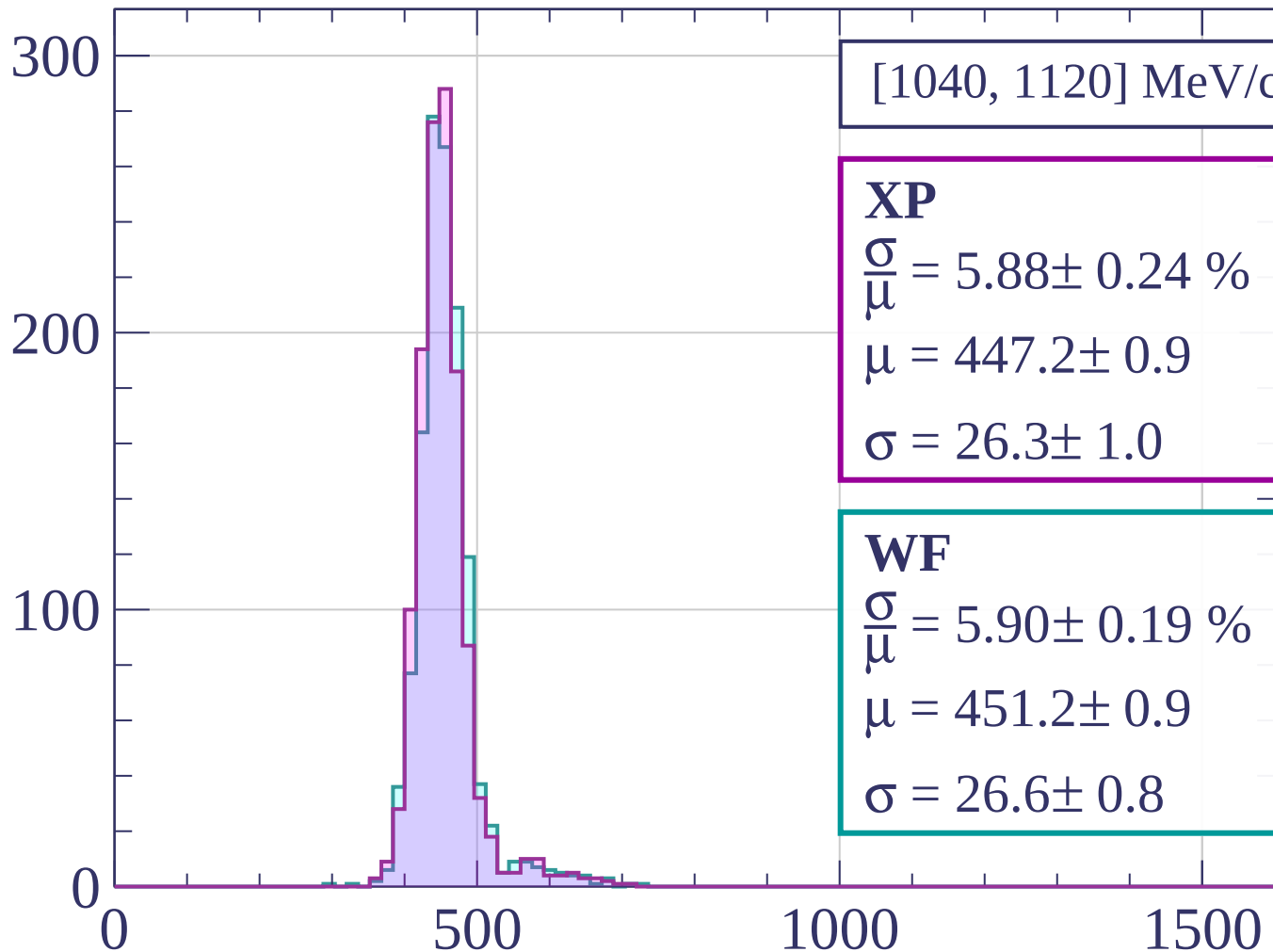
1000

1500

dE/dx [ADC counts/cm]

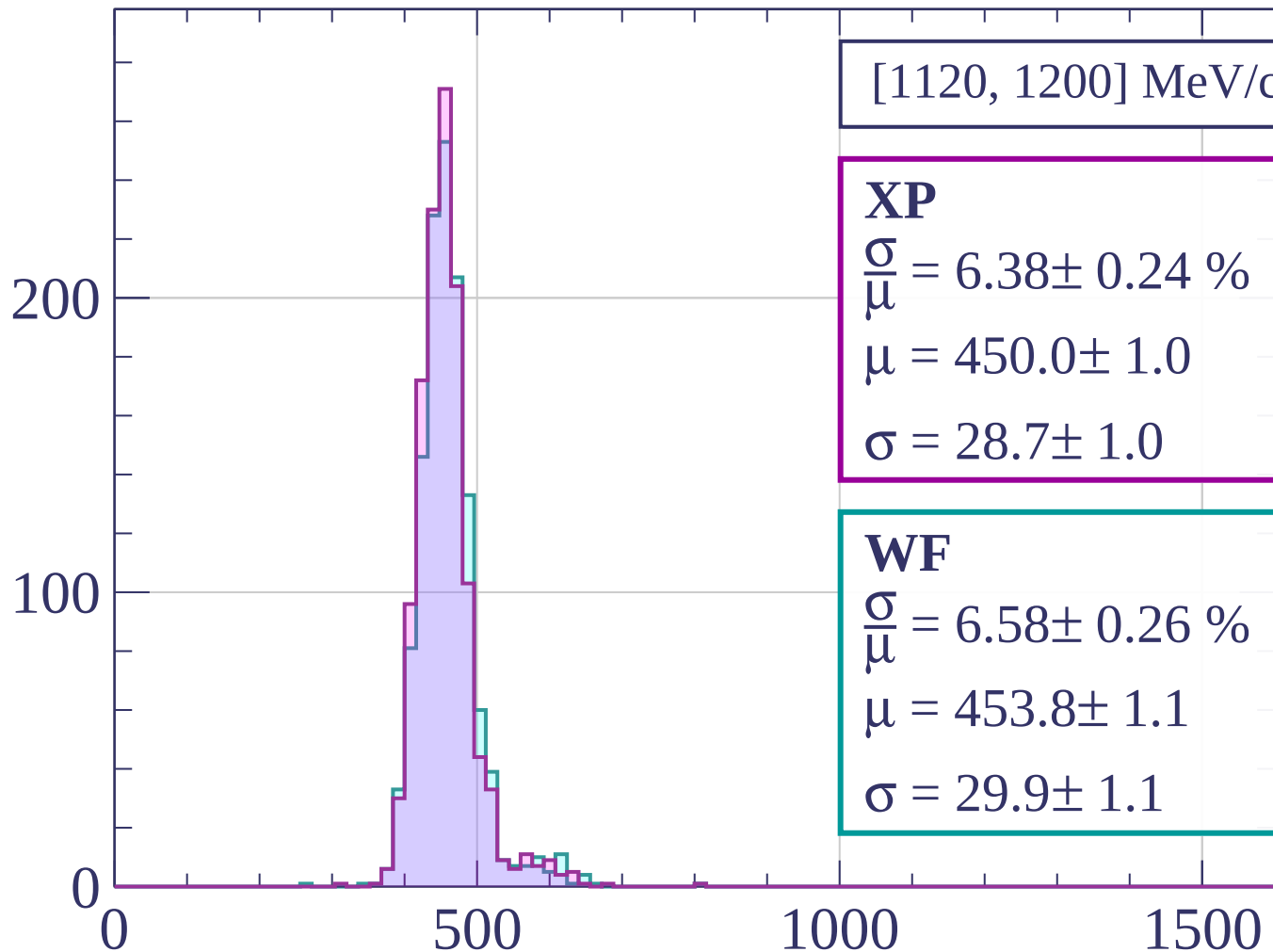


Count



dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count

[1200, 1280] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.55 \pm 0.25 \%$$

$$\mu = 453.2 \pm 1.0$$

$$\sigma = 29.7 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 6.37 \pm 0.24 \%$$

$$\mu = 456.3 \pm 1.0$$

$$\sigma = 29.1 \pm 1.0$$

200

100

0

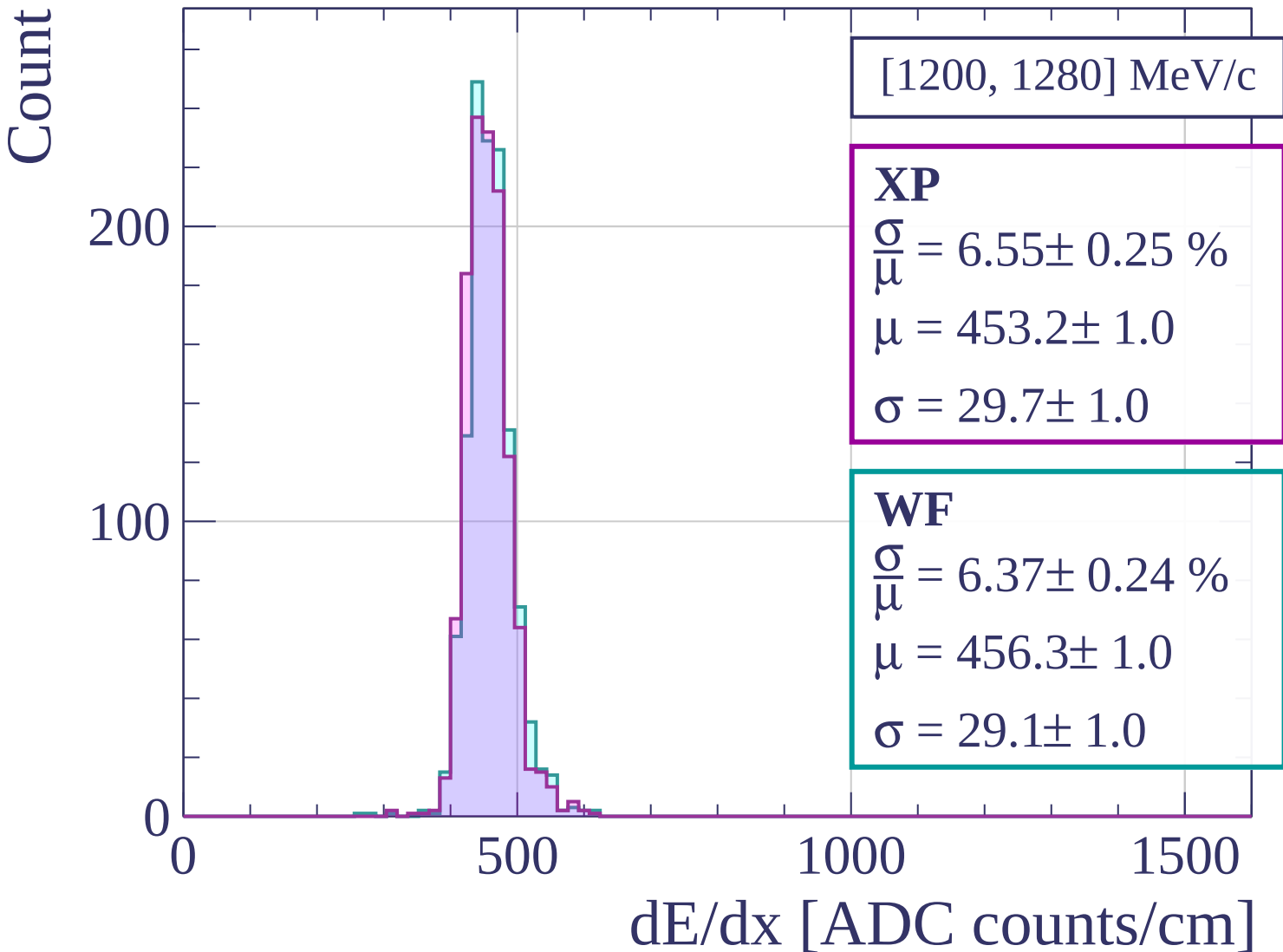
0

500

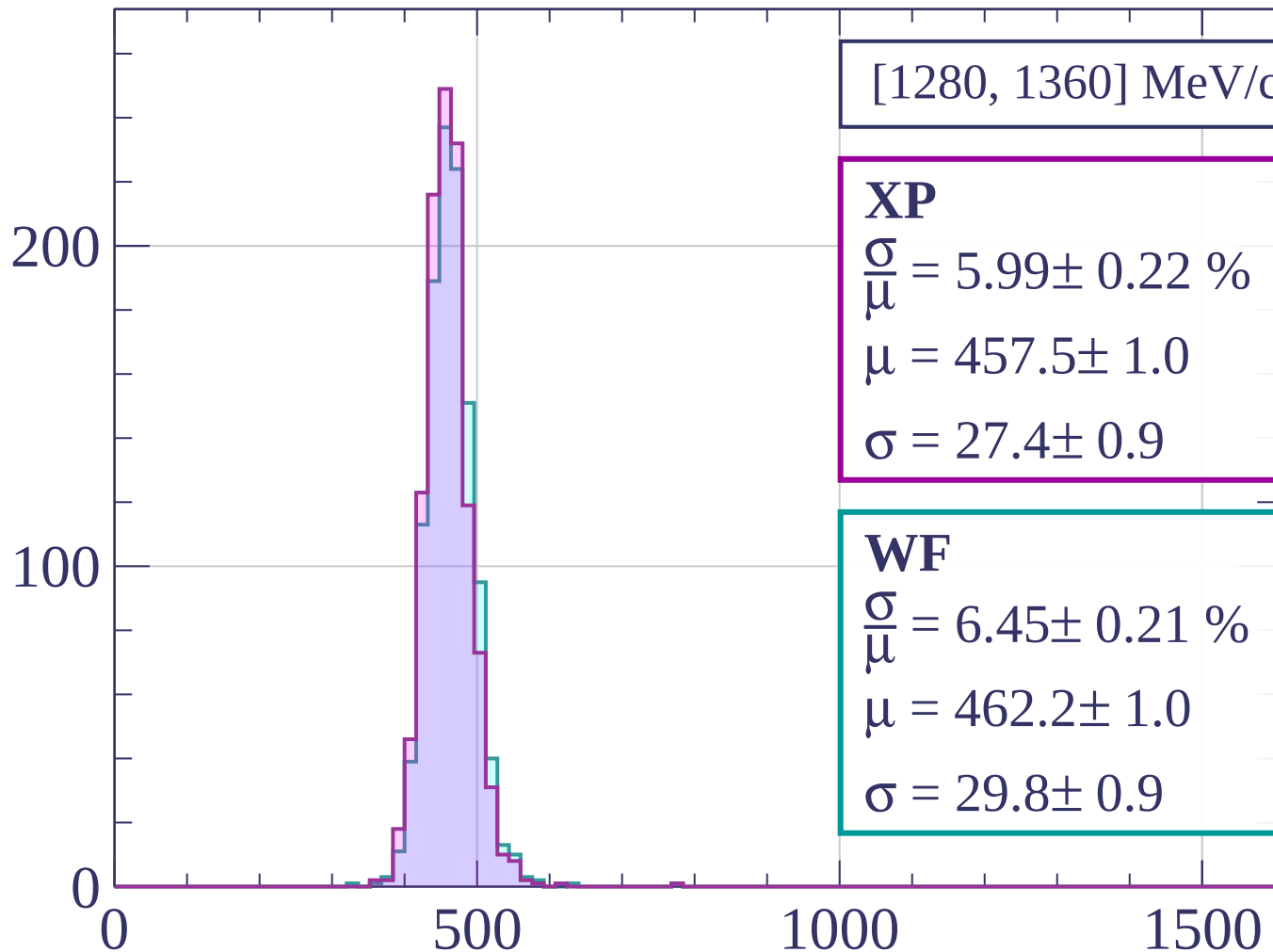
1000

1500

dE/dx [ADC counts/cm]

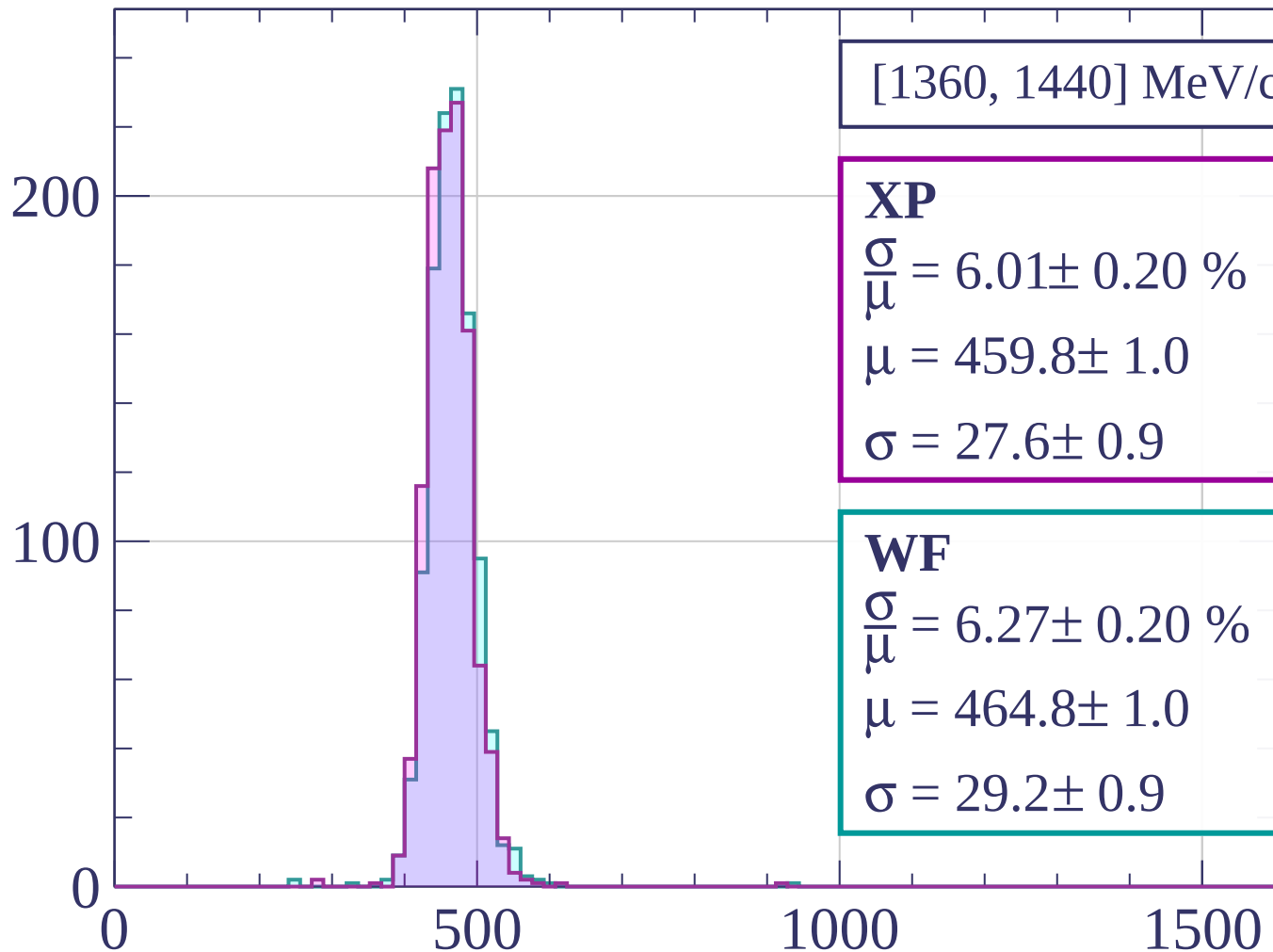


Count



dE/dx [ADC counts/cm]

Count



[1360, 1440] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.01 \pm 0.20 \%$$

$$\mu = 459.8 \pm 1.0$$

$$\sigma = 27.6 \pm 0.9$$

WF

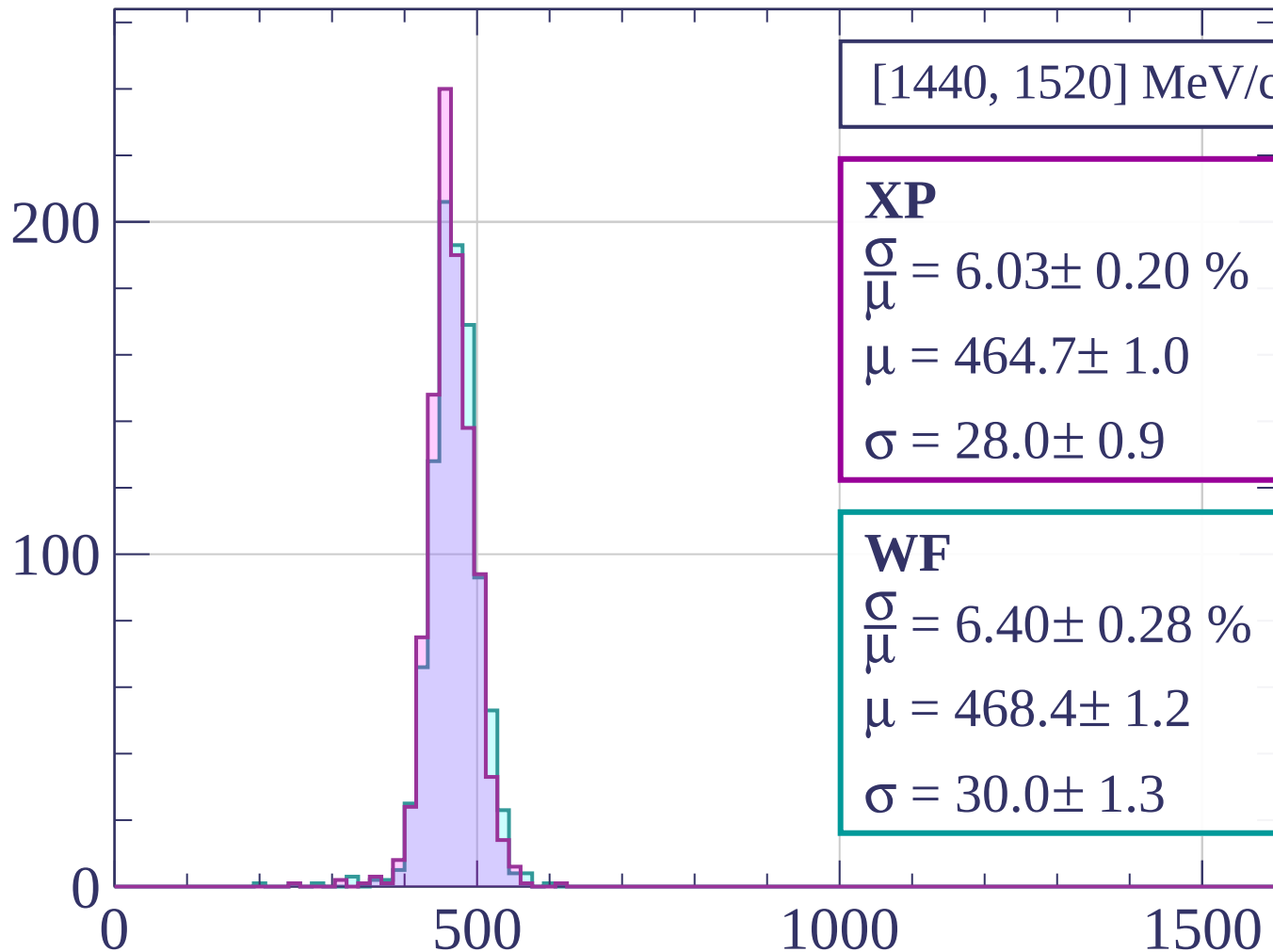
$$\frac{\sigma}{\mu} = 6.27 \pm 0.20 \%$$

$$\mu = 464.8 \pm 1.0$$

$$\sigma = 29.2 \pm 0.9$$

dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count

[1520, 1600] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.92 \pm 0.23 \%$$

$$\mu = 467.4 \pm 1.0$$

$$\sigma = 27.7 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 6.11 \pm 0.25 \%$$

$$\mu = 471.1 \pm 1.1$$

$$\sigma = 28.8 \pm 1.1$$

200

150

100

50

0

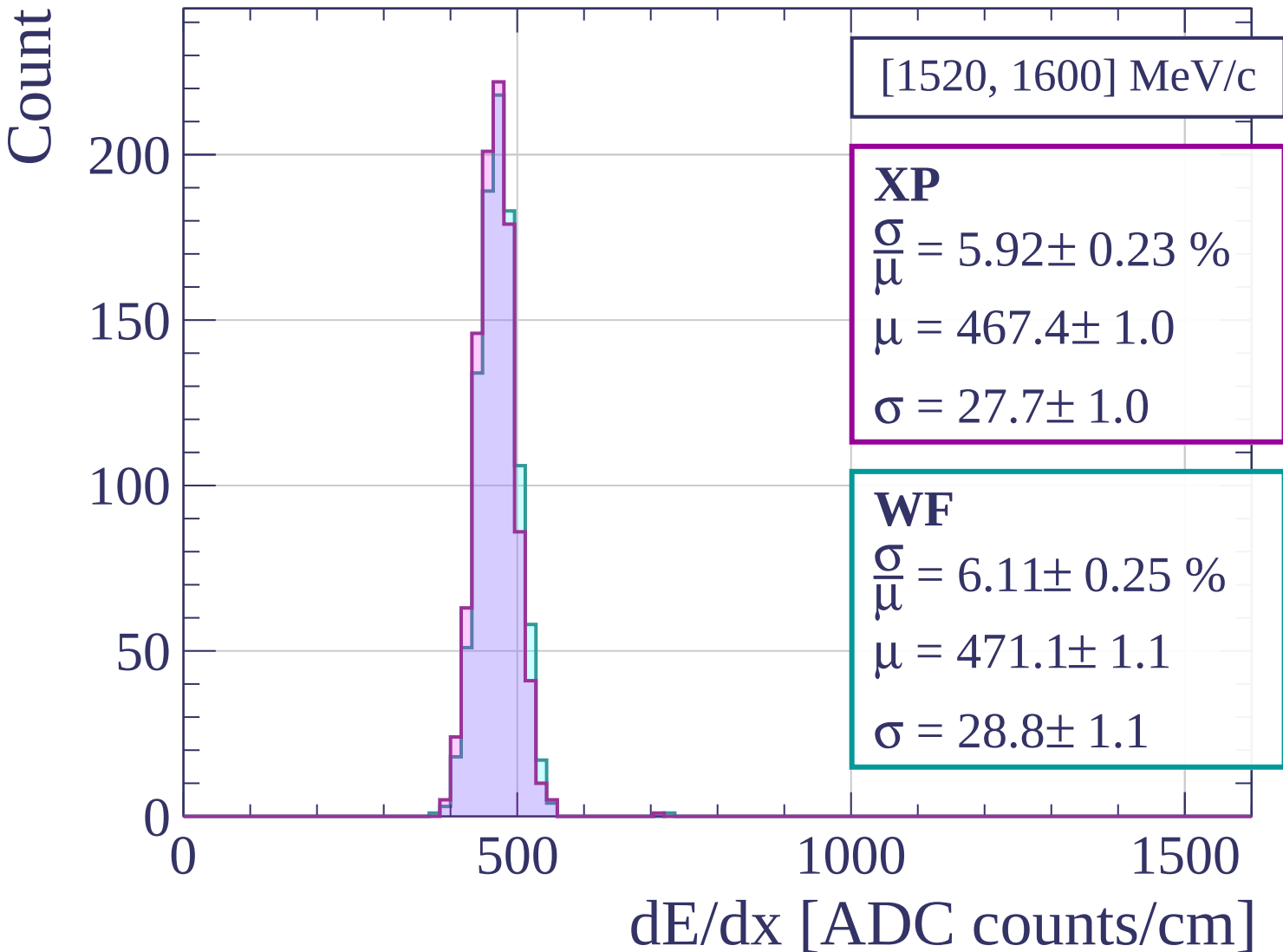
0

500

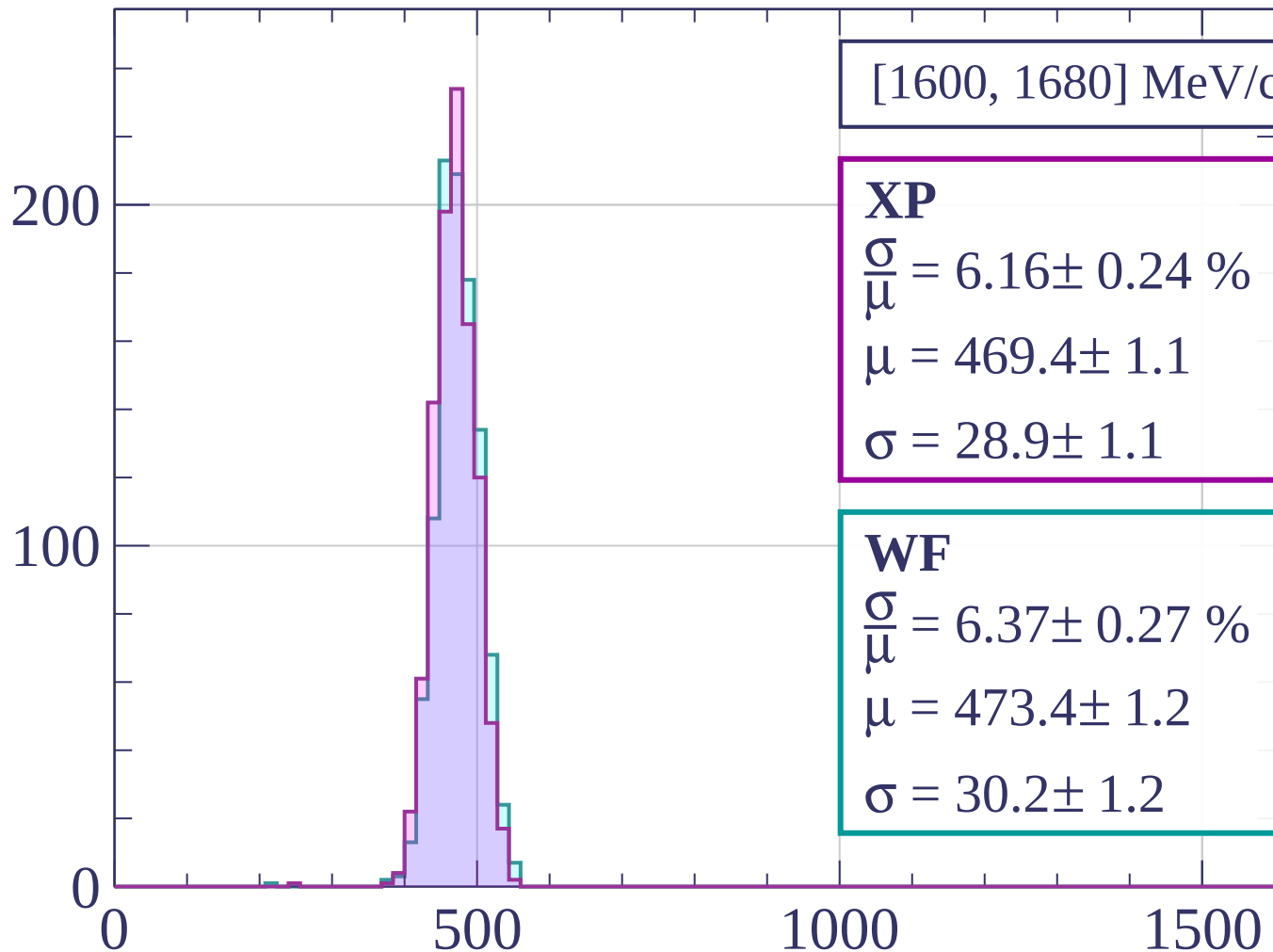
1000

1500

dE/dx [ADC counts/cm]



Count



[1600, 1680] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.16 \pm 0.24 \%$$

$$\mu = 469.4 \pm 1.1$$

$$\sigma = 28.9 \pm 1.1$$

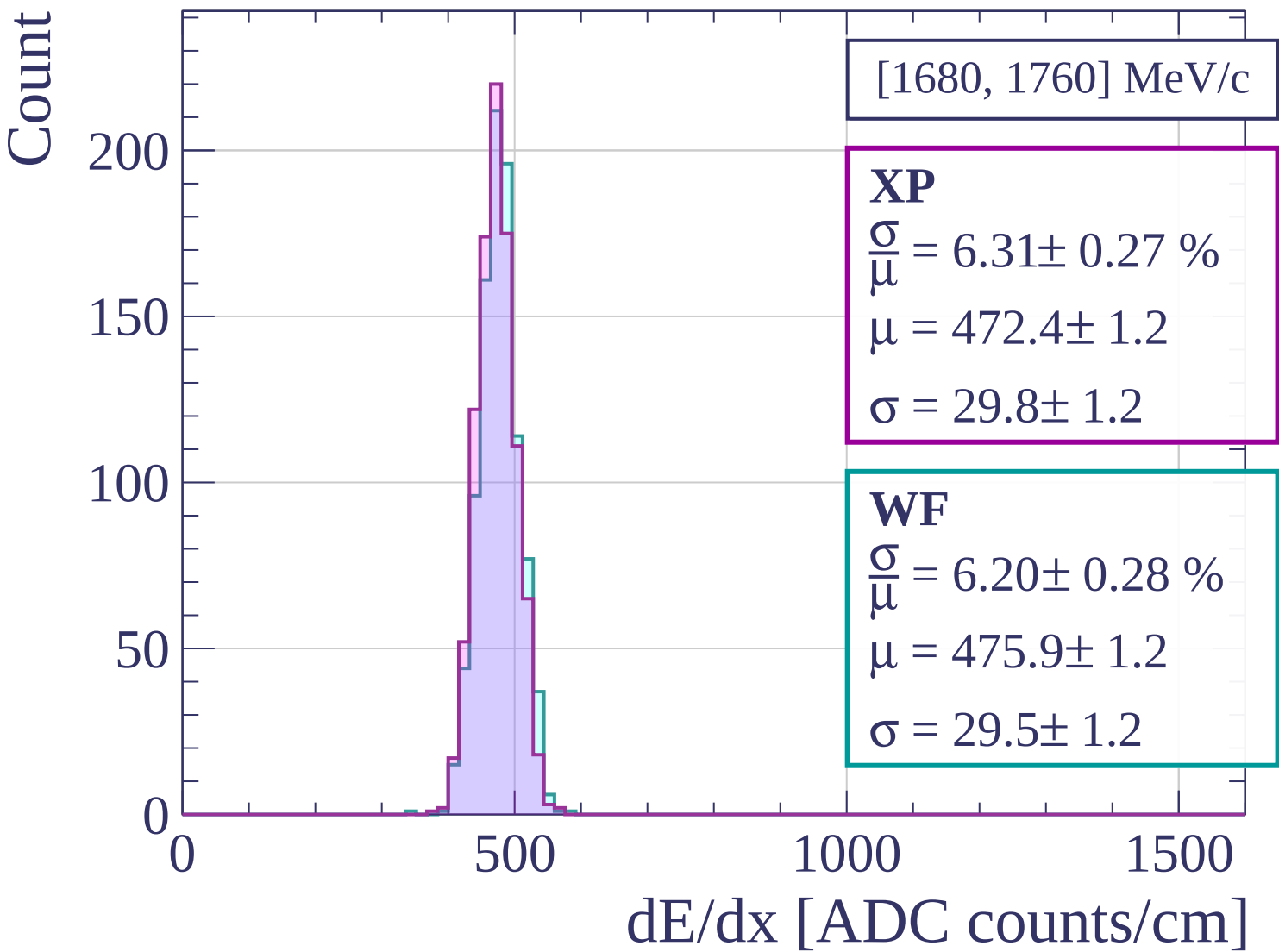
WF

$$\frac{\sigma}{\mu} = 6.37 \pm 0.27 \%$$

$$\mu = 473.4 \pm 1.2$$

$$\sigma = 30.2 \pm 1.2$$

dE/dx [ADC counts/cm]



Count

[1760, 1840] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.17 \pm 0.29 \%$$

$$\mu = 473.7 \pm 1.2$$

$$\sigma = 29.3 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 6.15 \pm 0.29 \%$$

$$\mu = 477.8 \pm 1.3$$

$$\sigma = 29.4 \pm 1.3$$

200

150

100

50

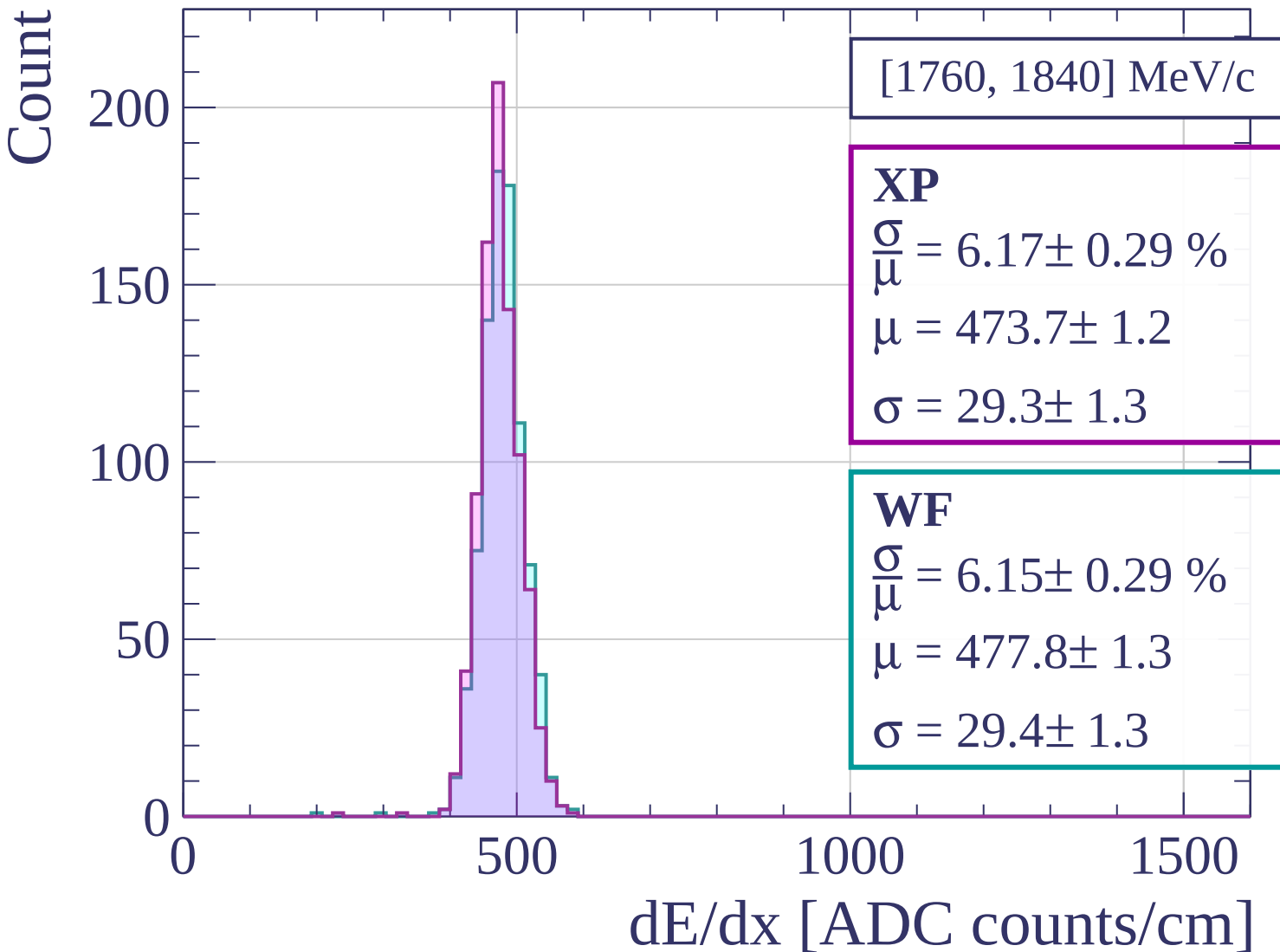
0

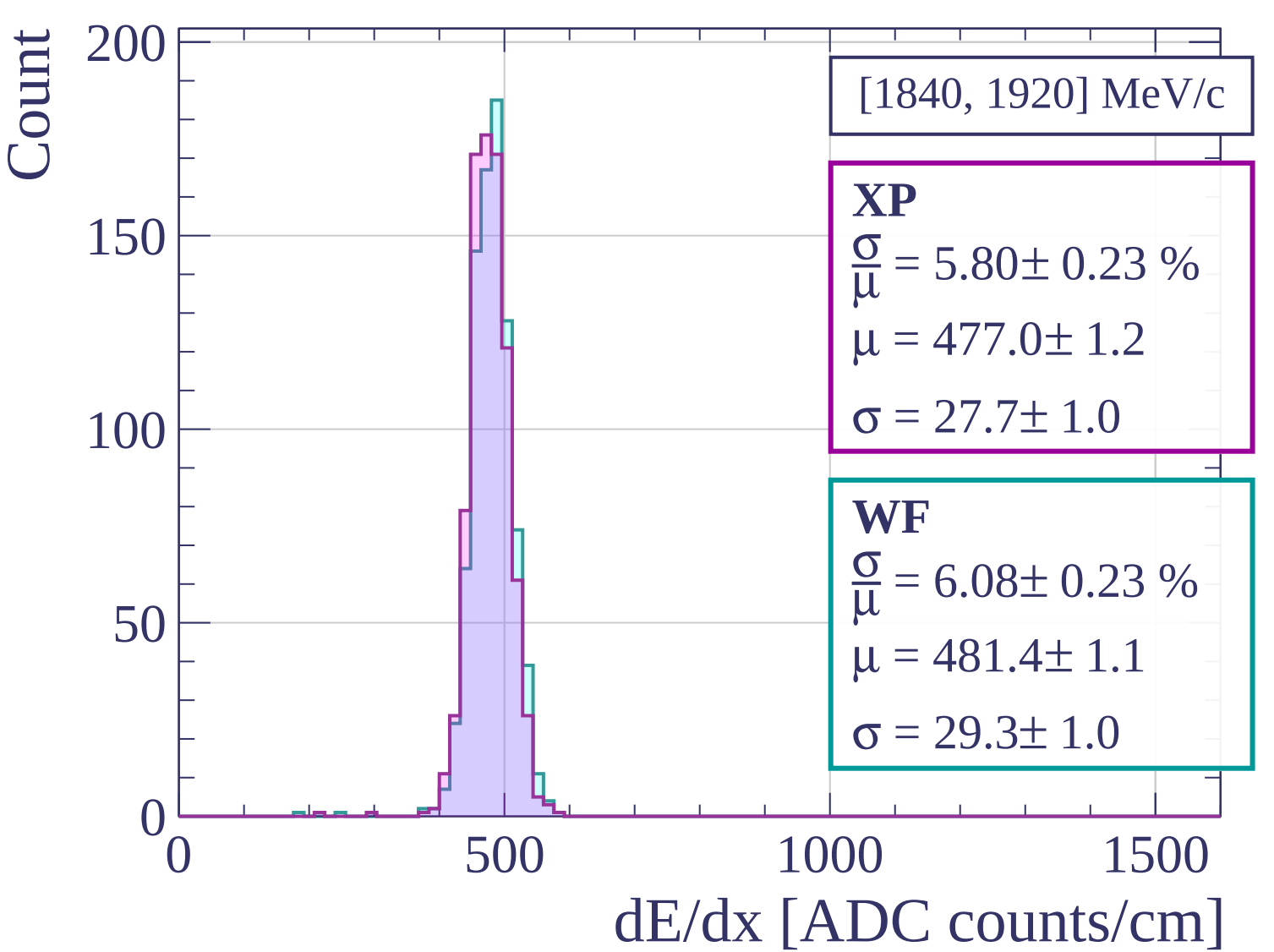
500

1000

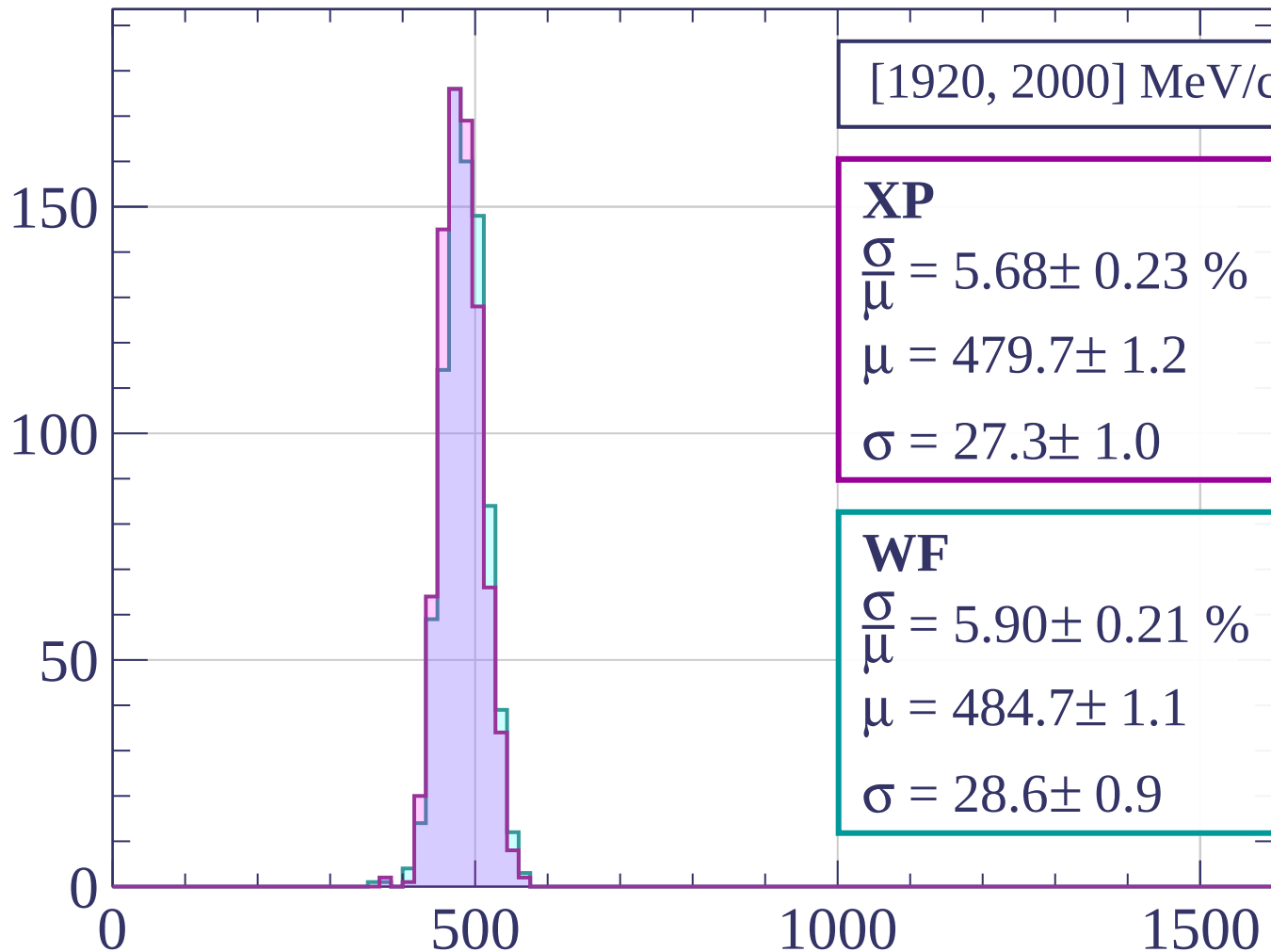
1500

dE/dx [ADC counts/cm]



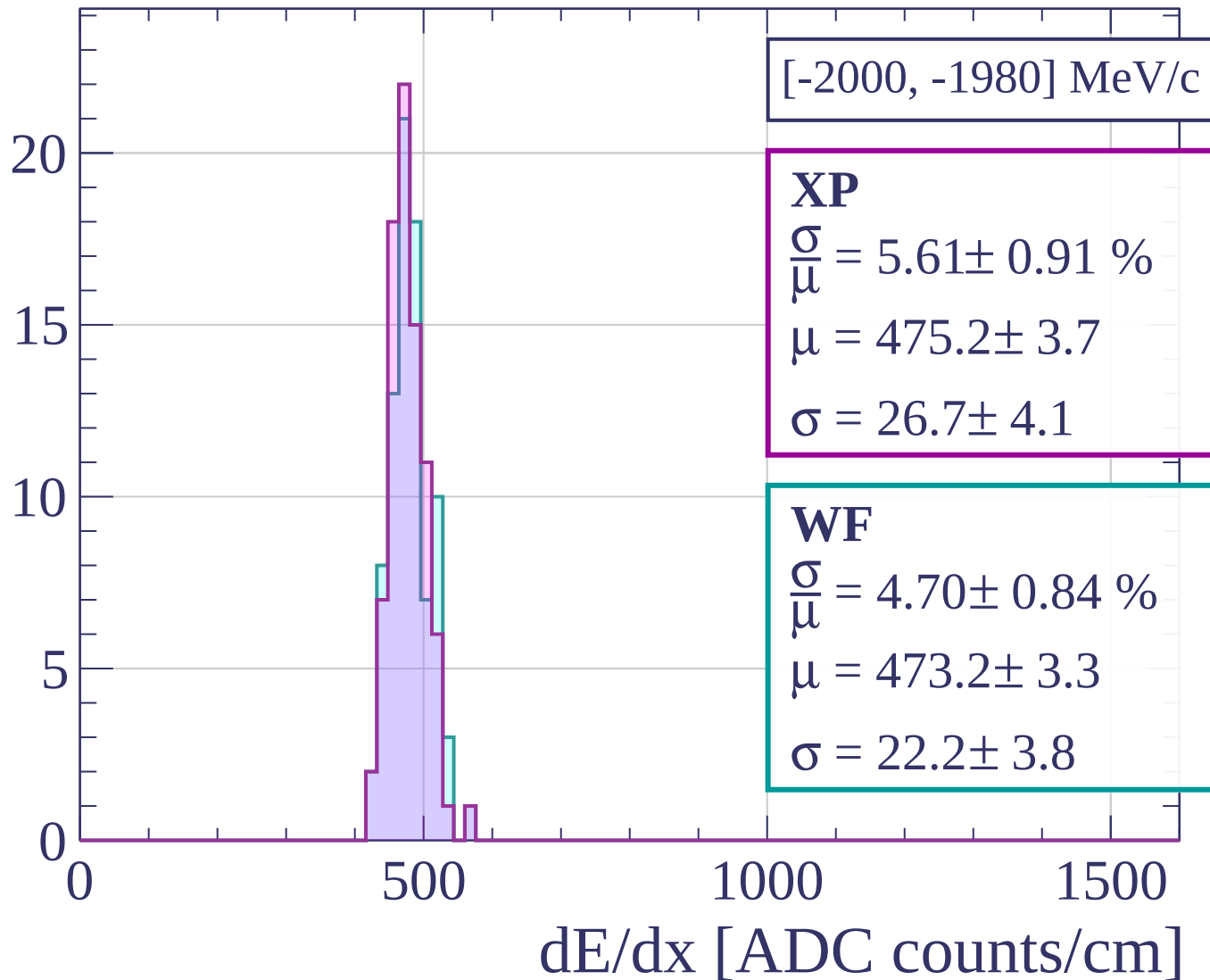


Count

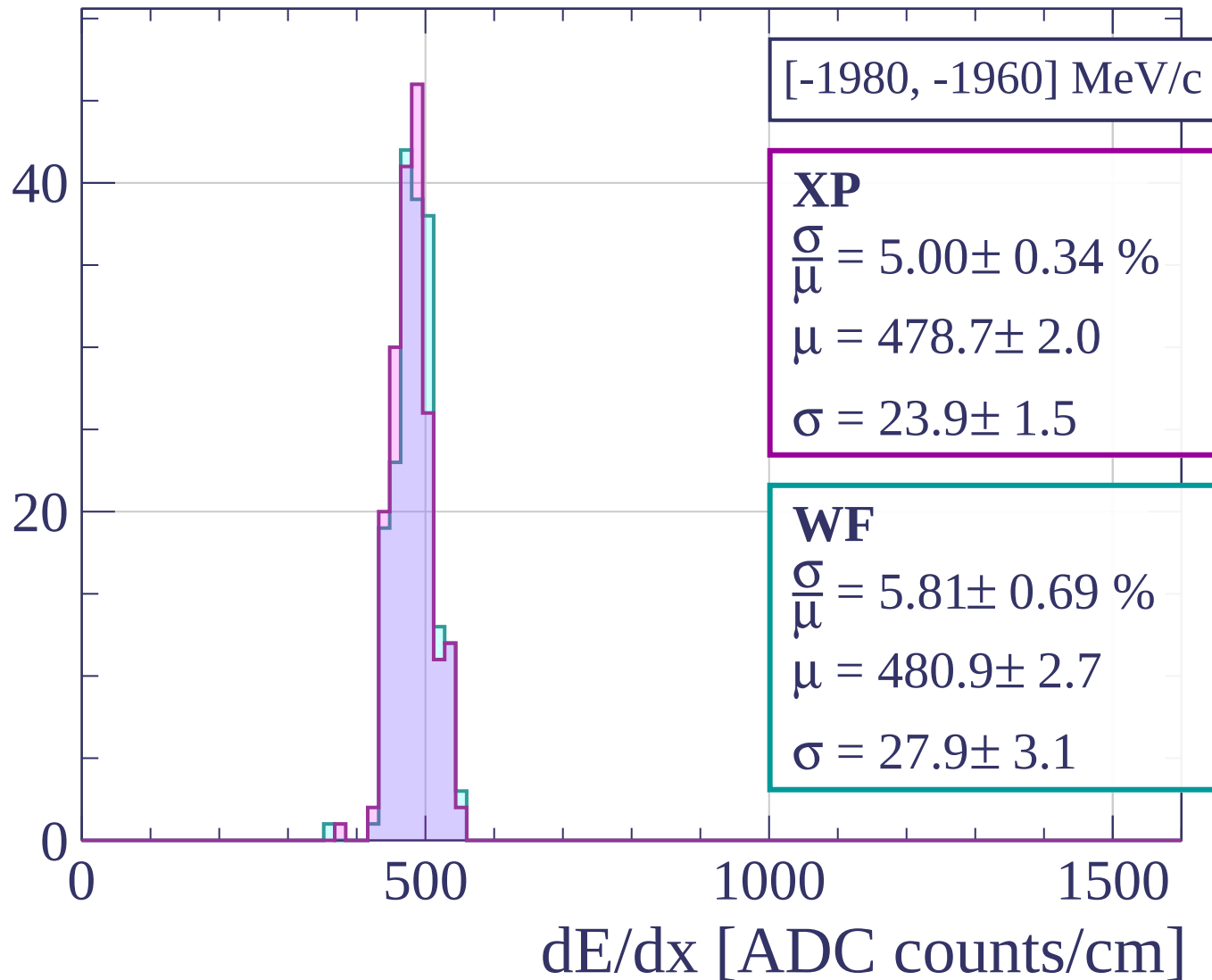


dE/dx [ADC counts/cm]

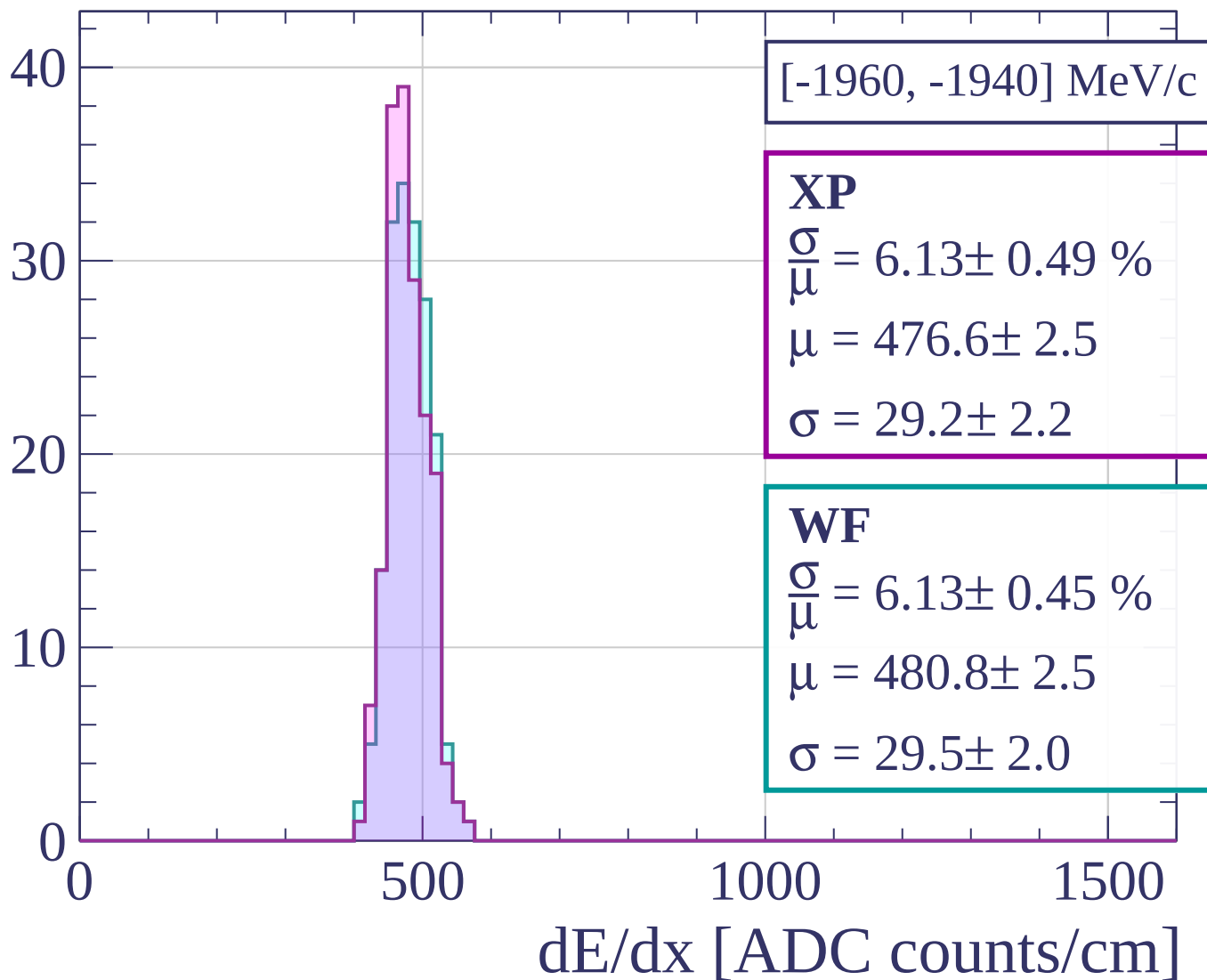
Count



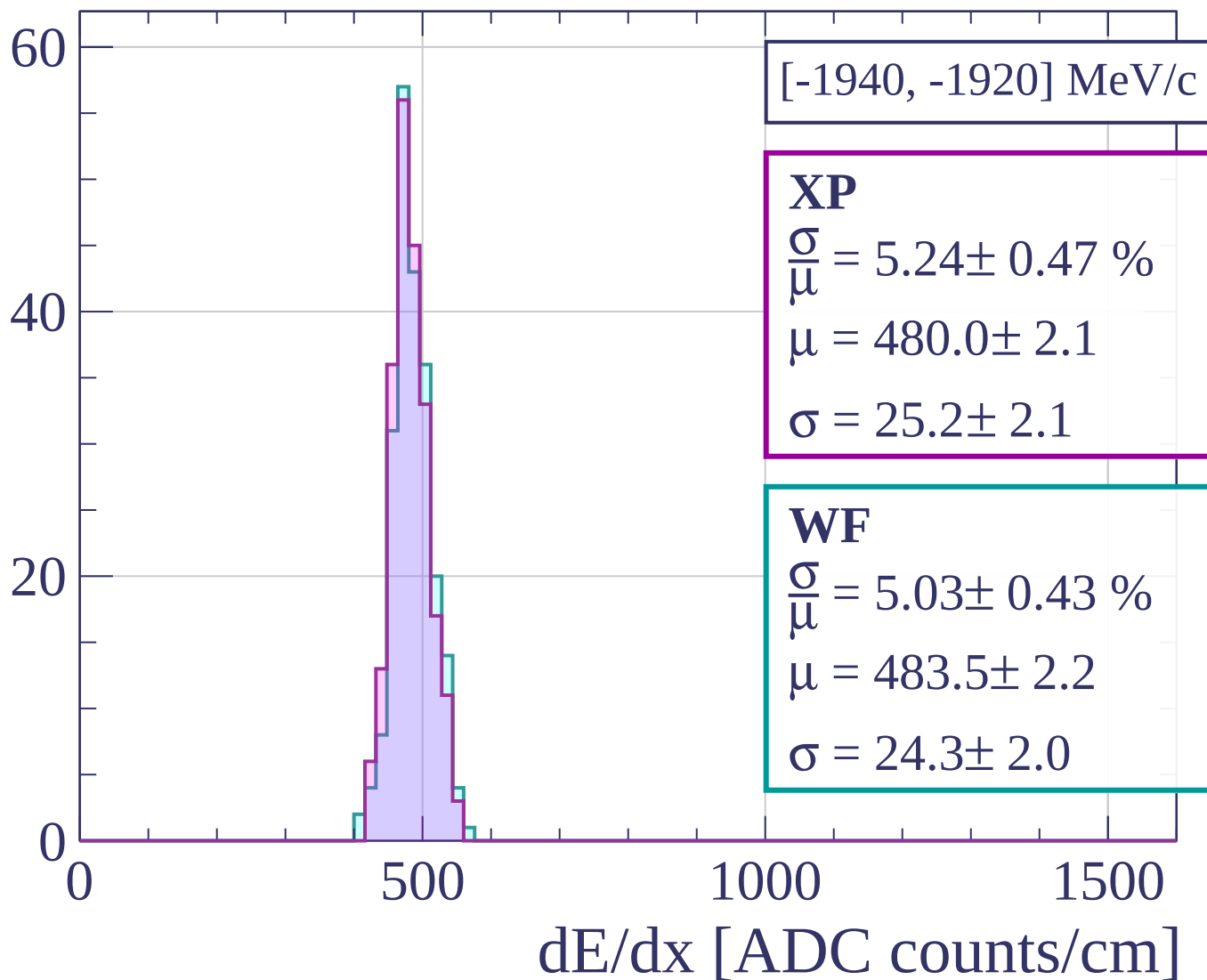
Count



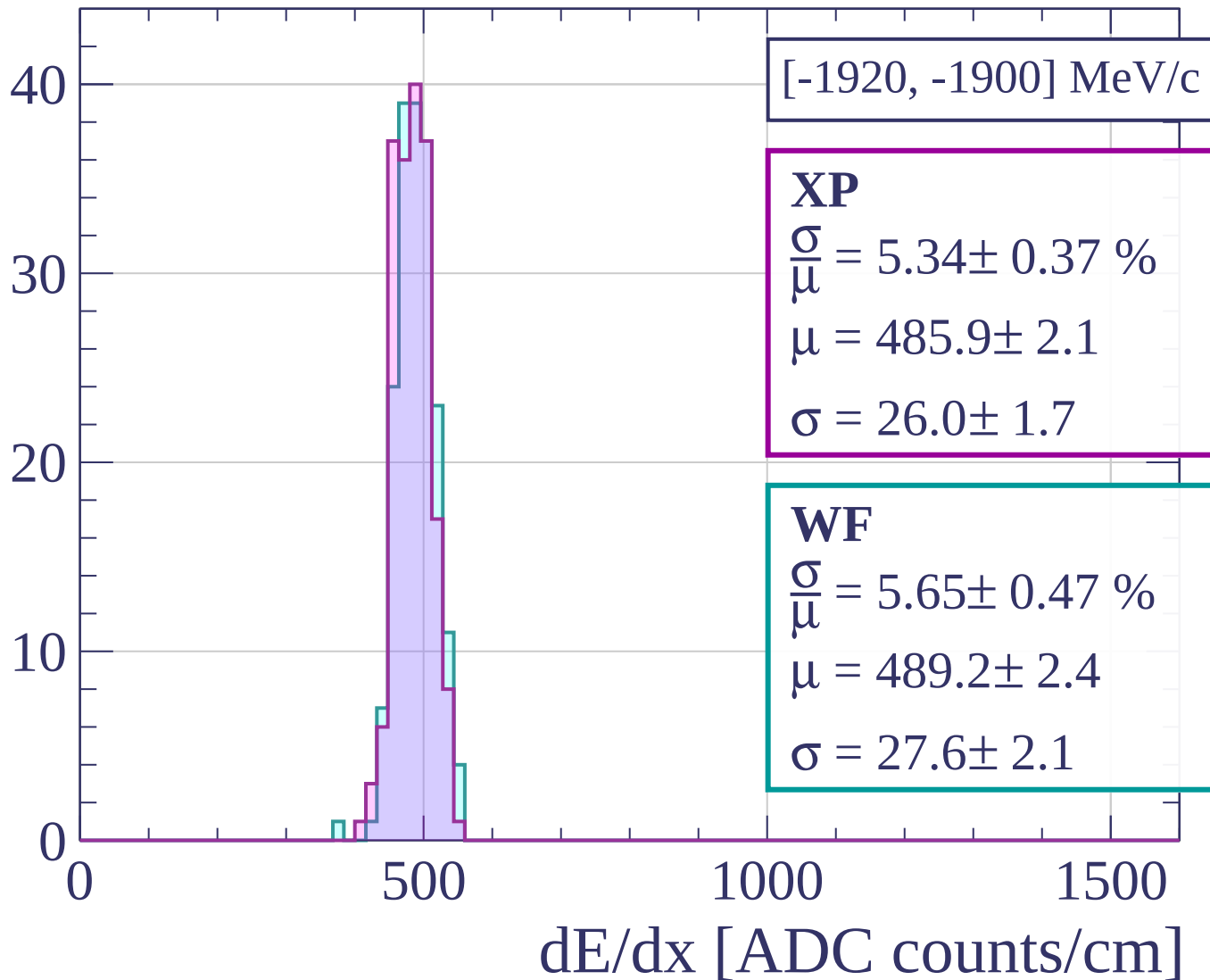
Count



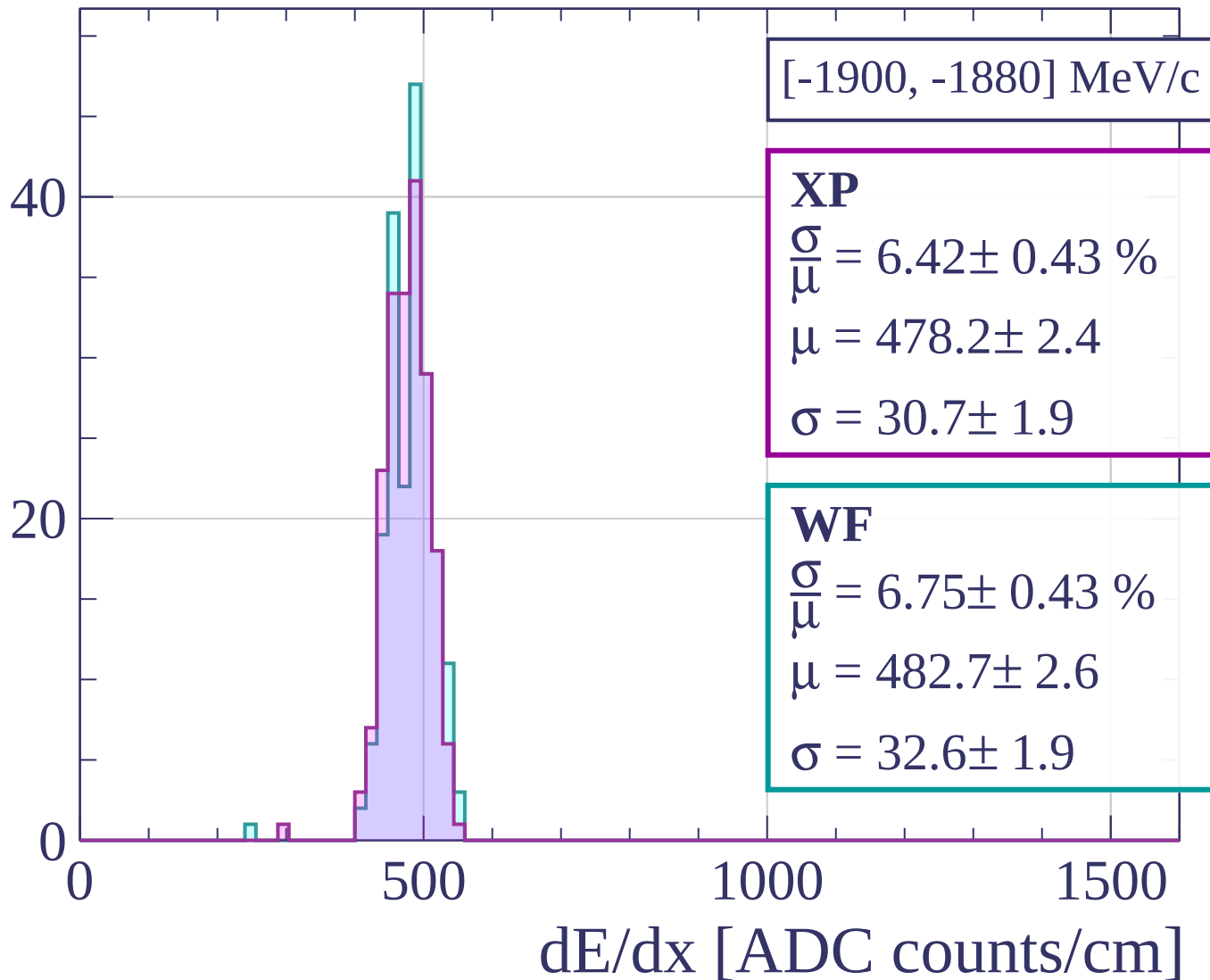
Count



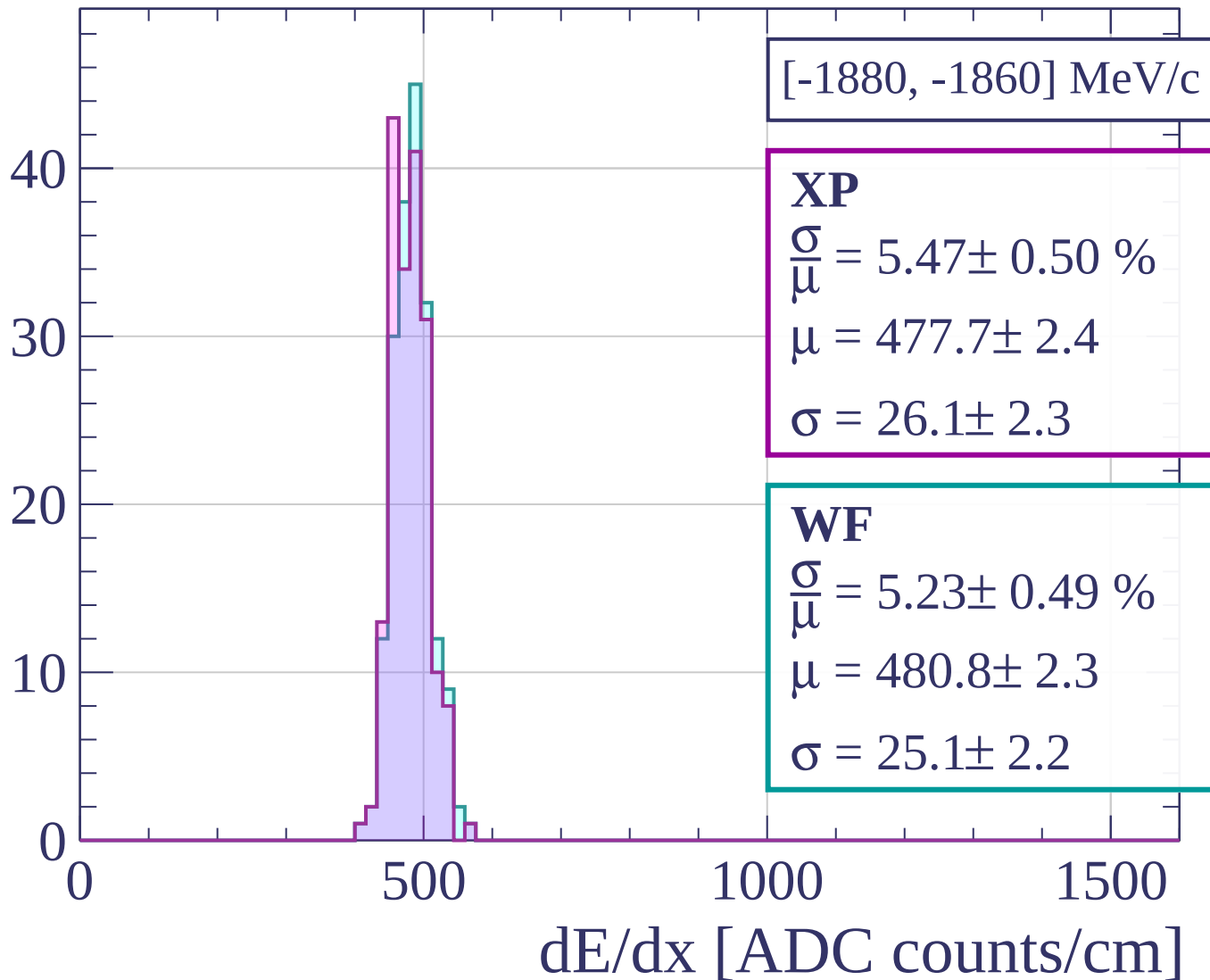
Count



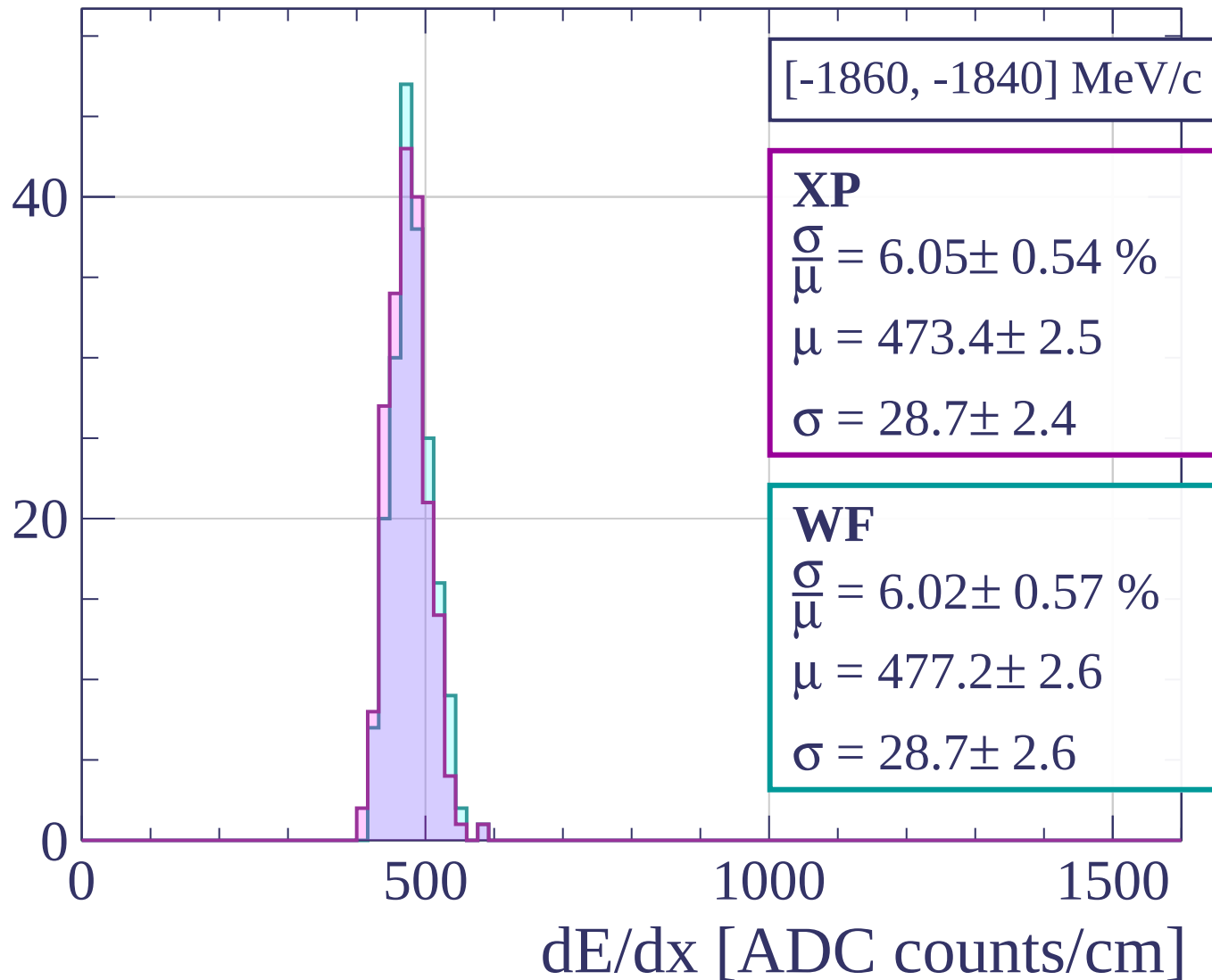
Count



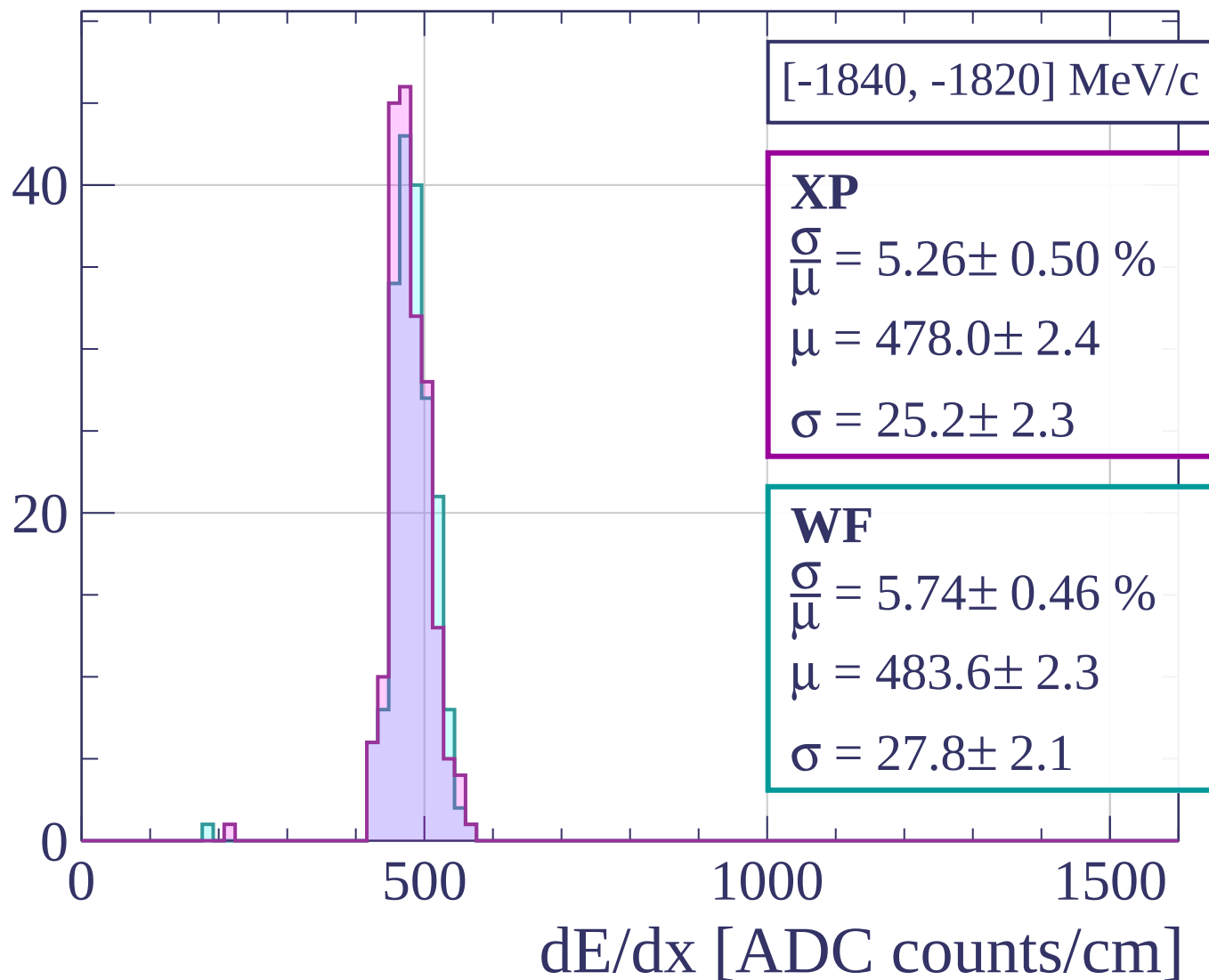
Count



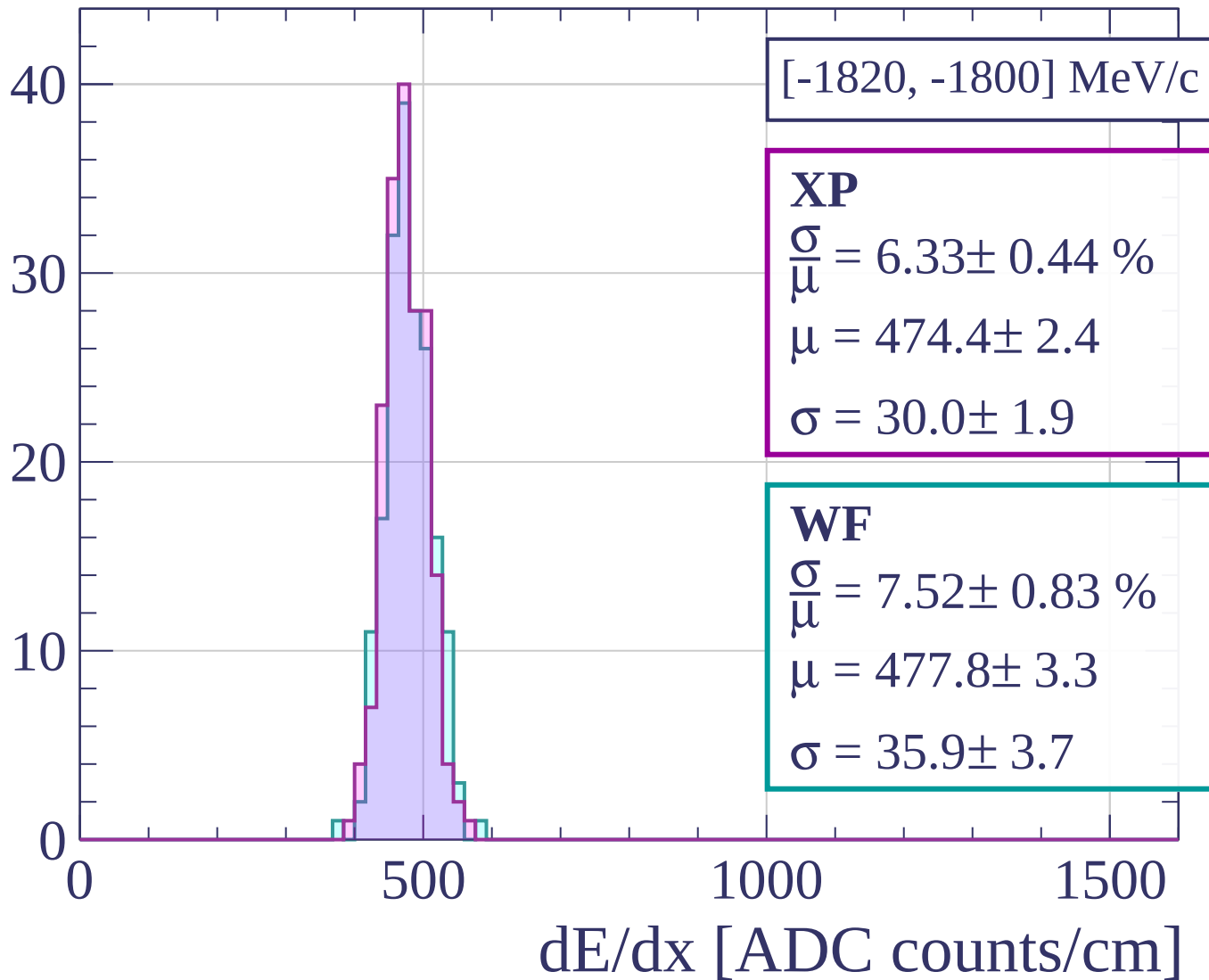
Count



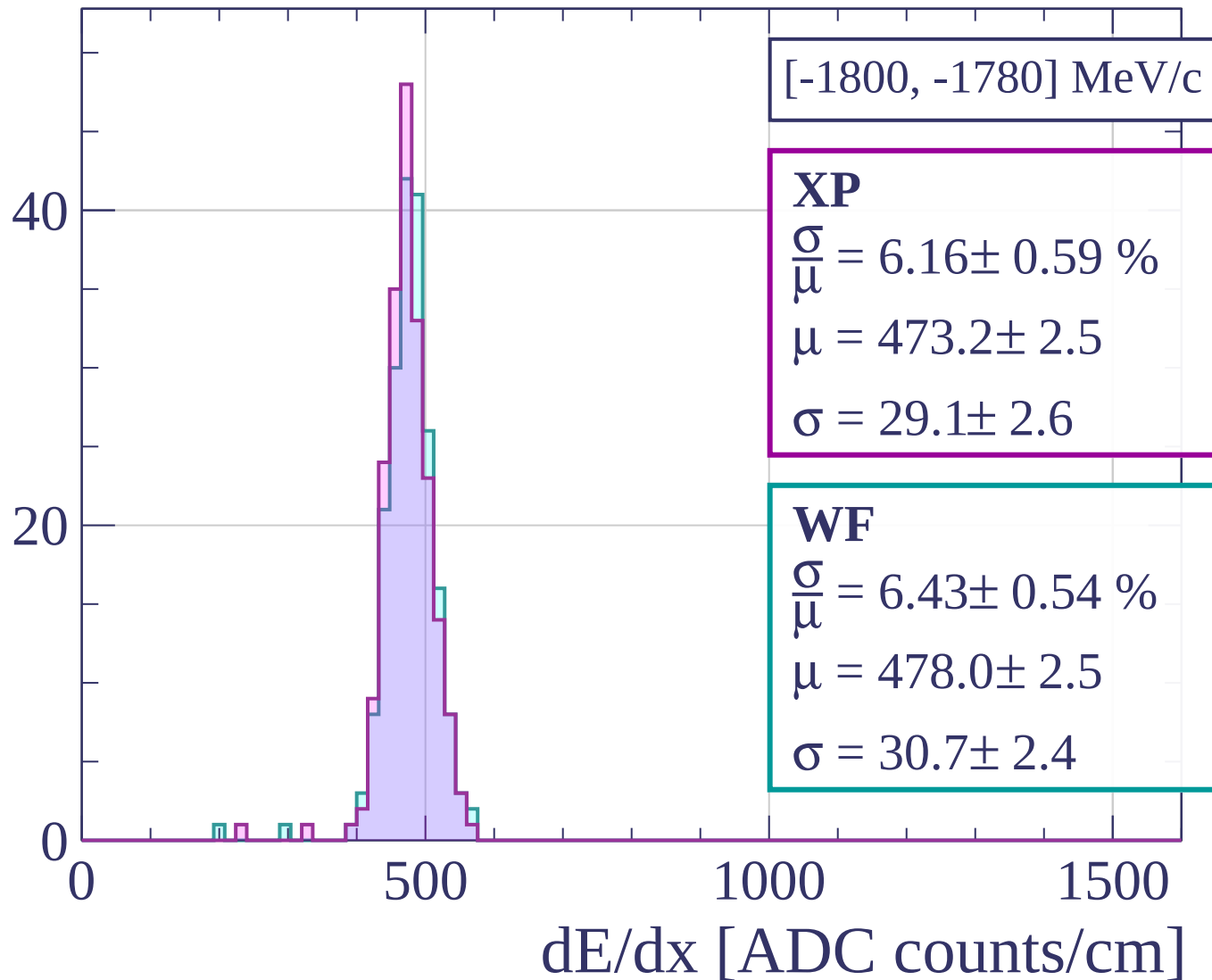
Count



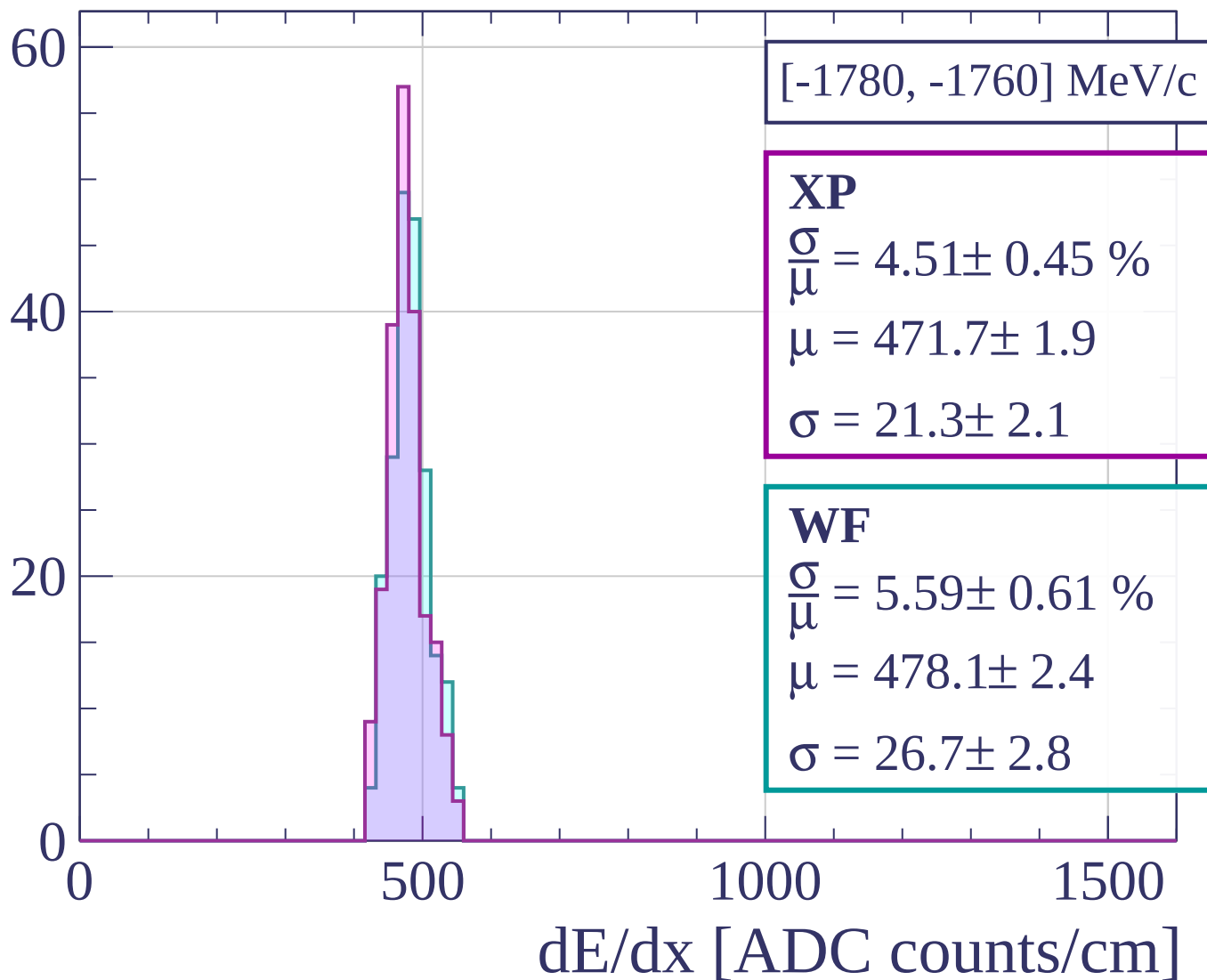
Count



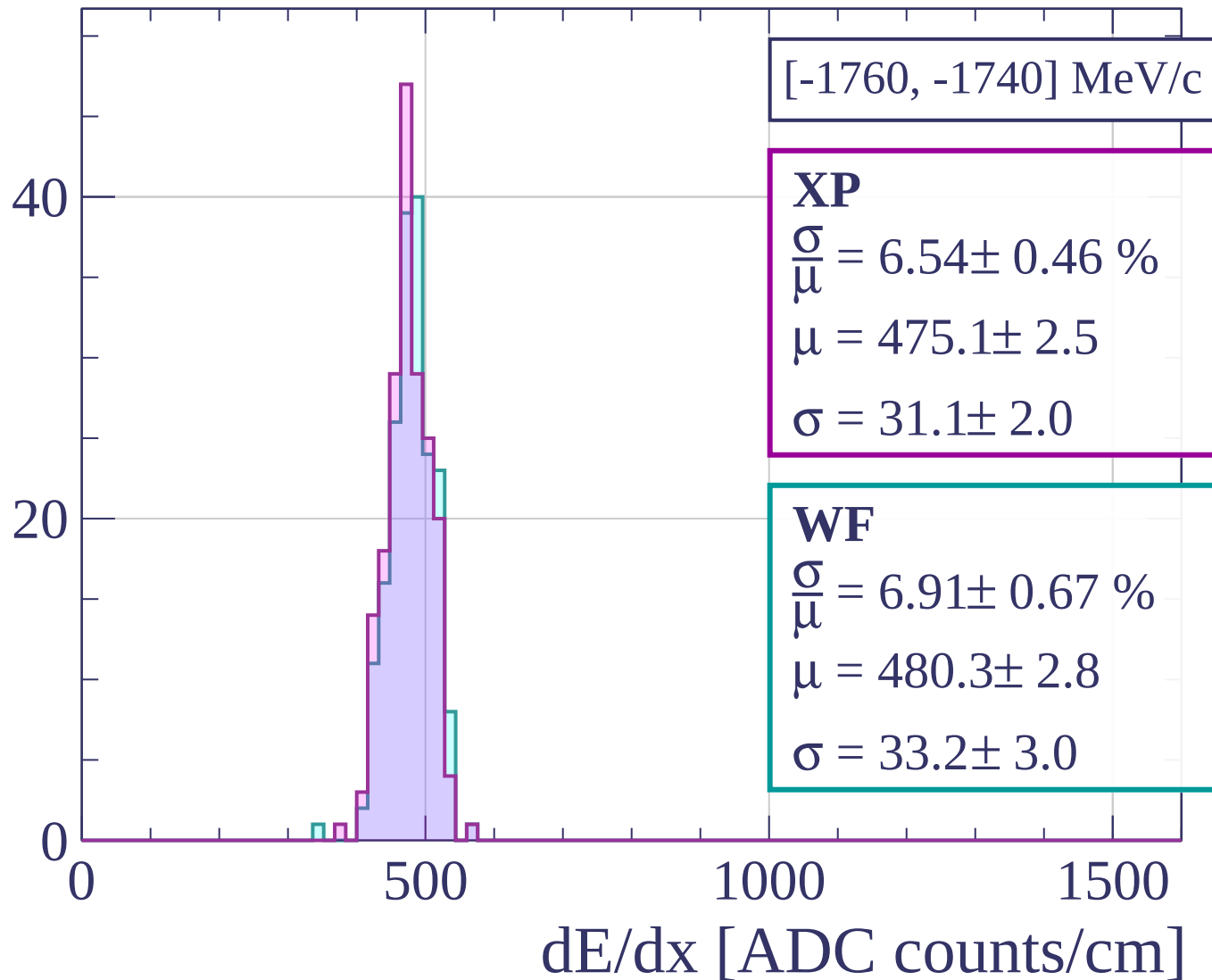
Count



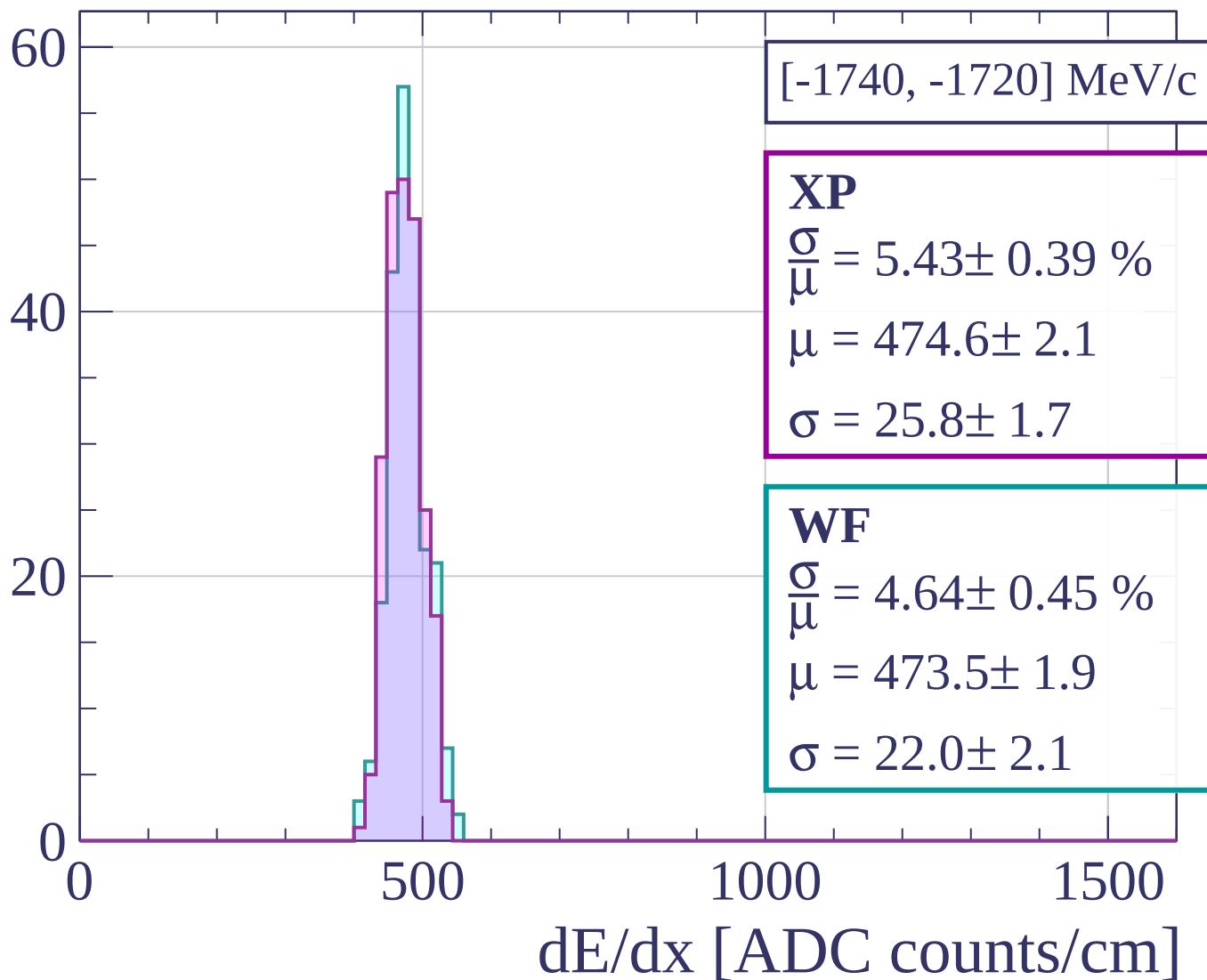
Count



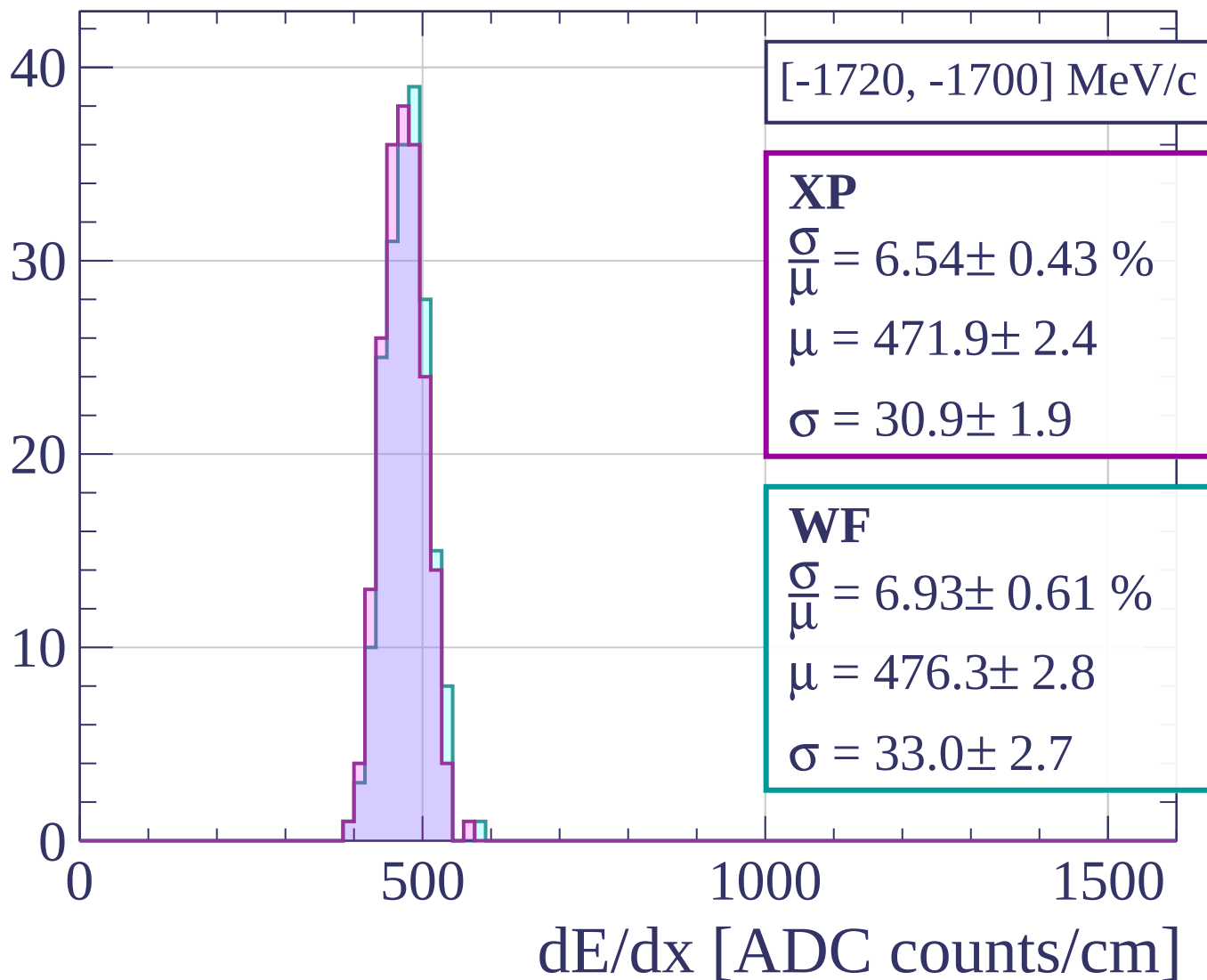
Count



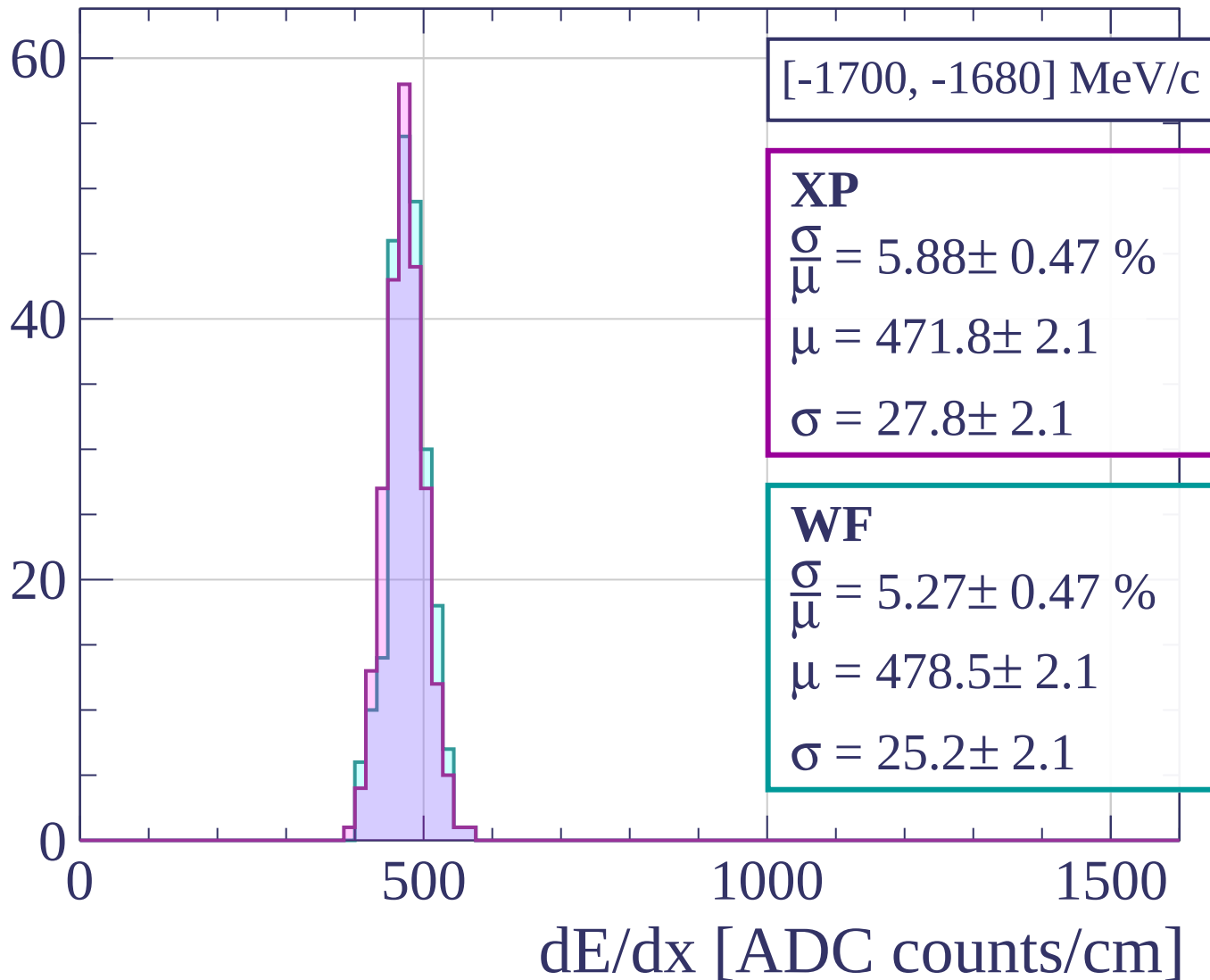
Count



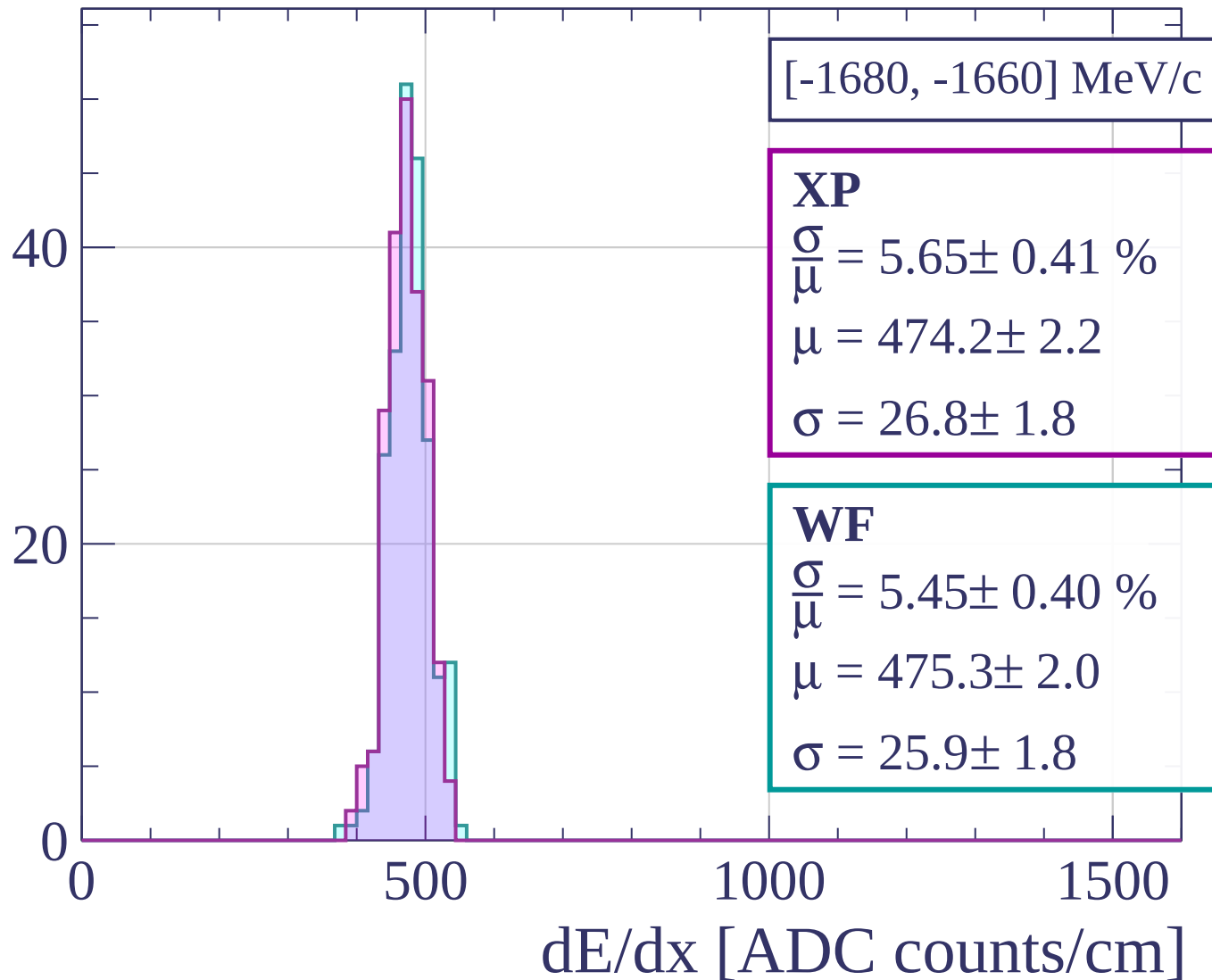
Count



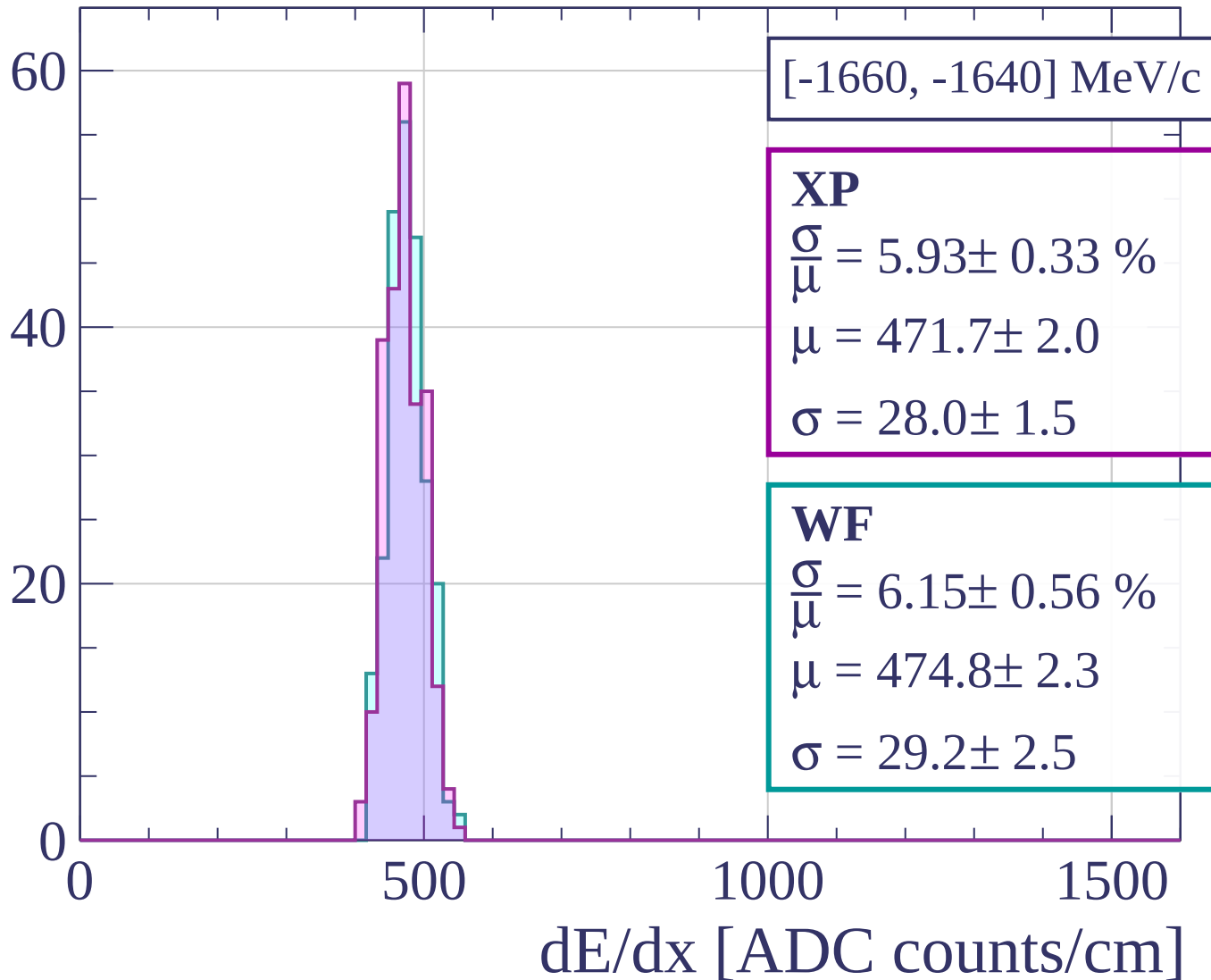
Count



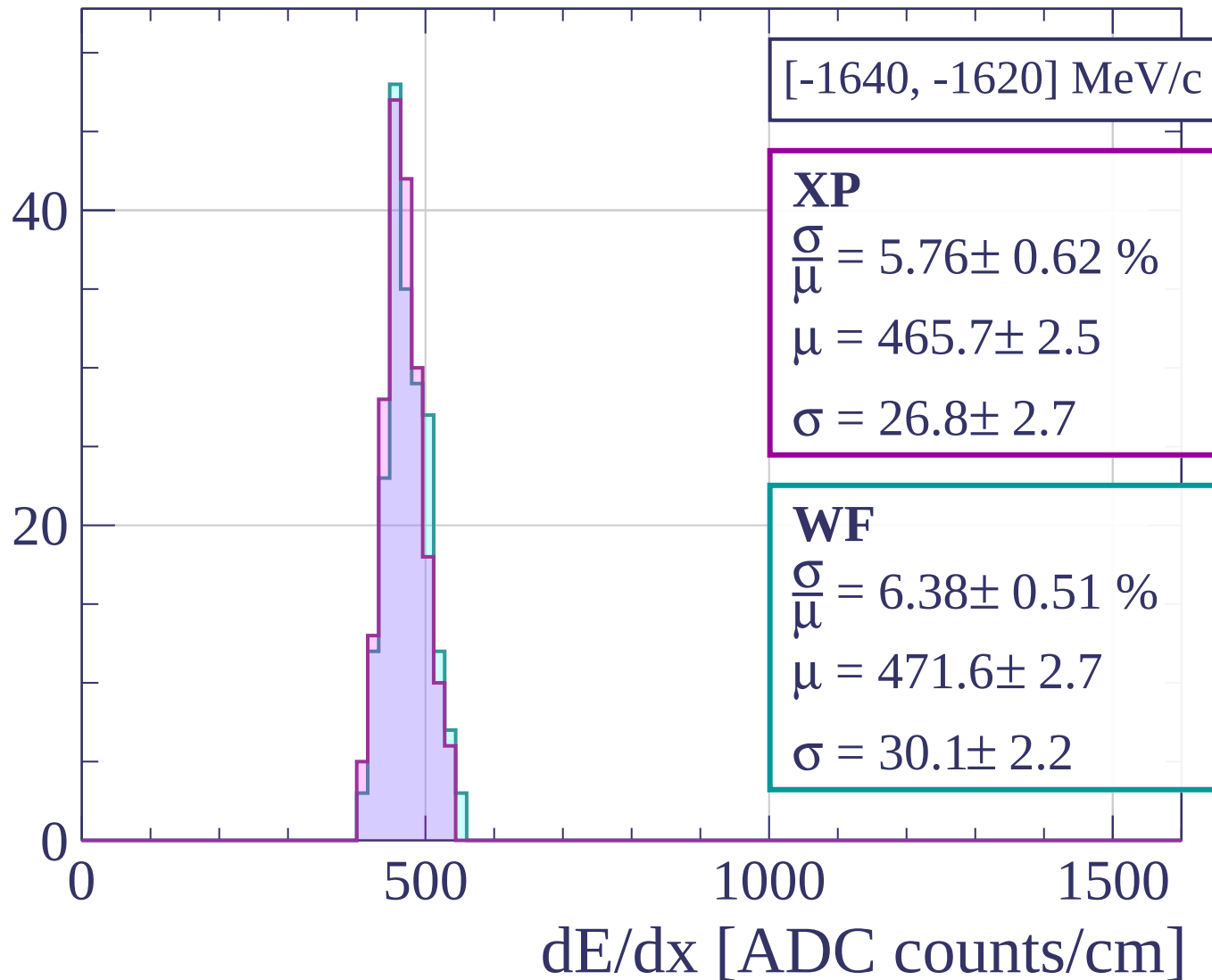
Count

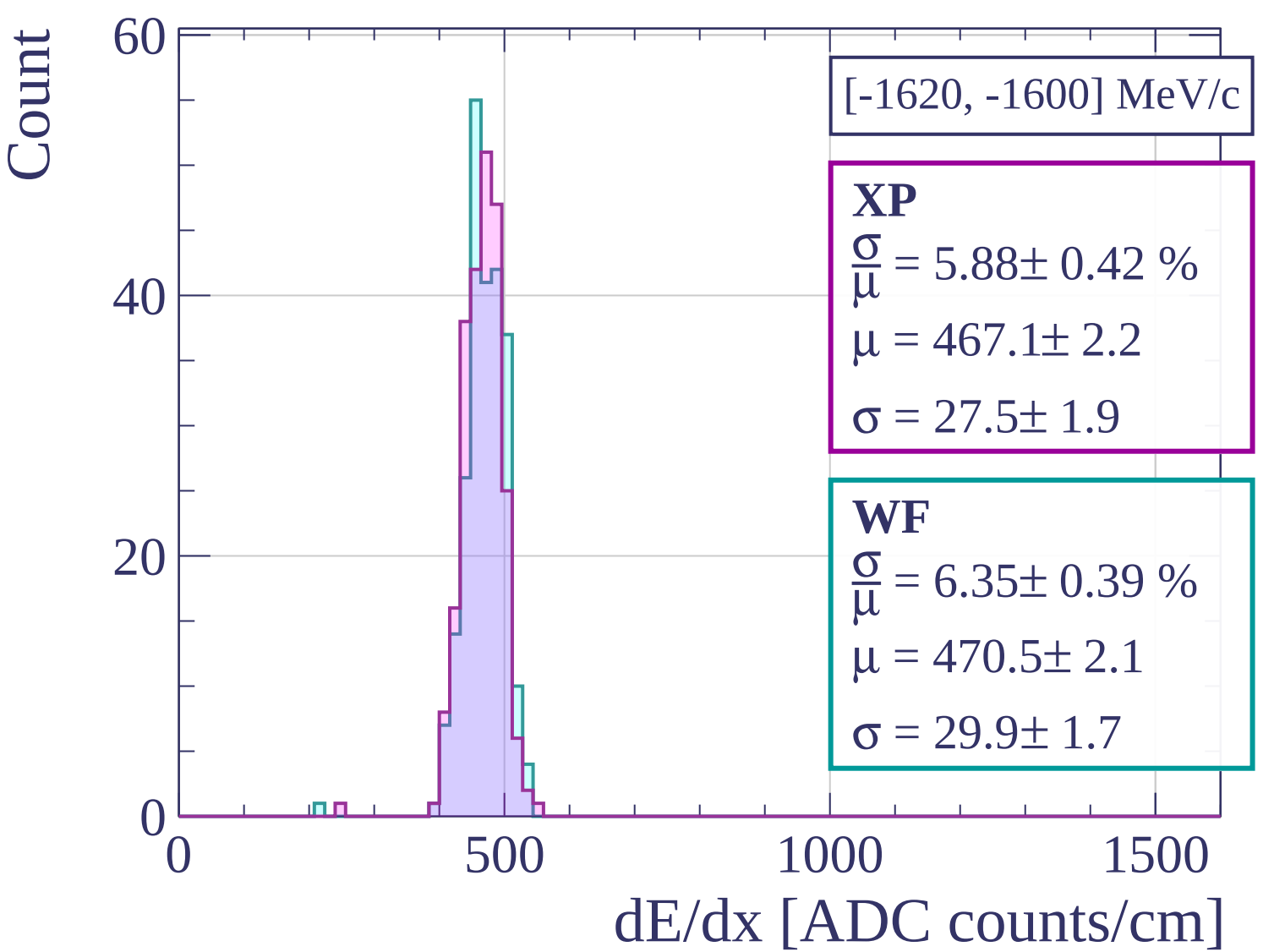


Count

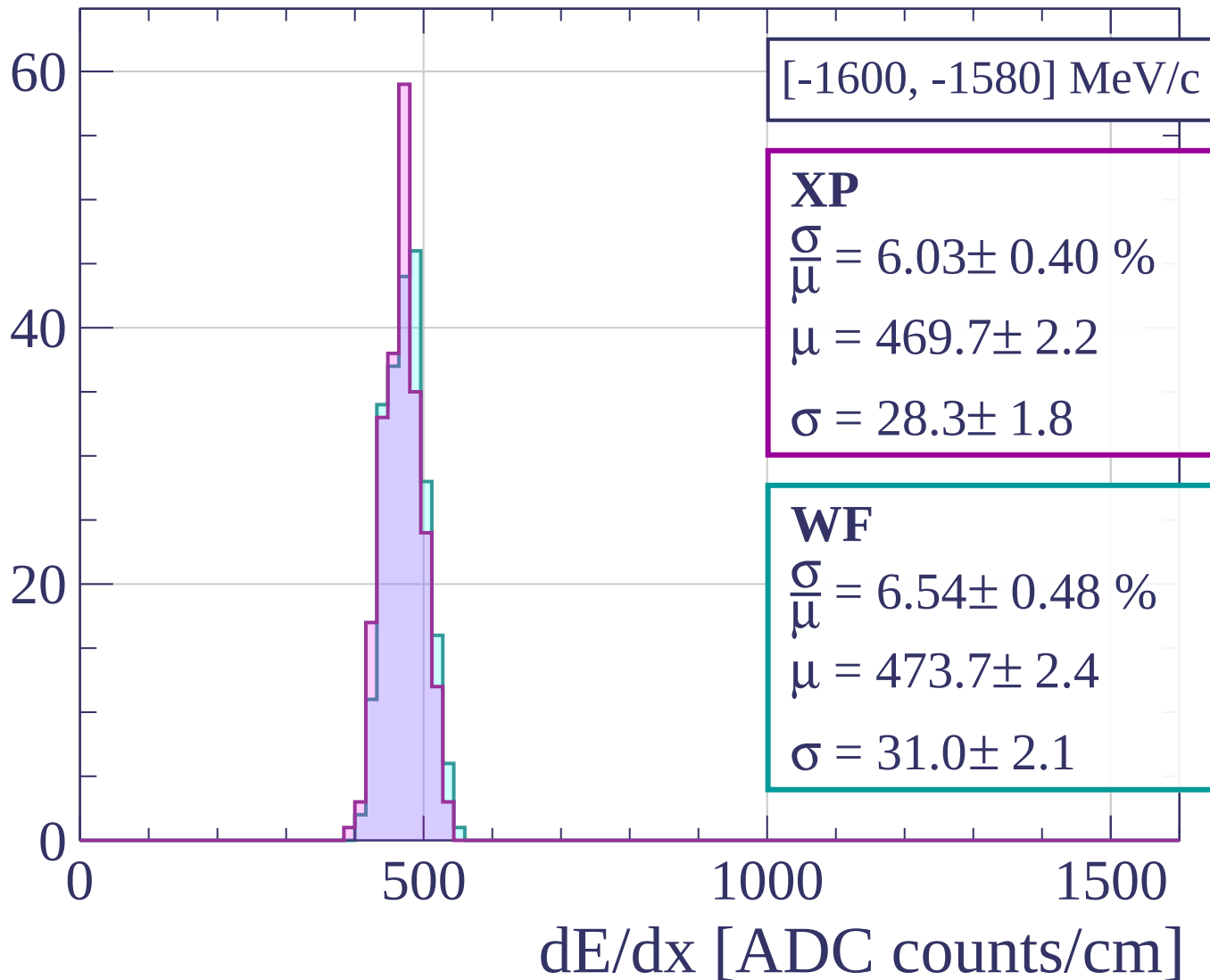


Count

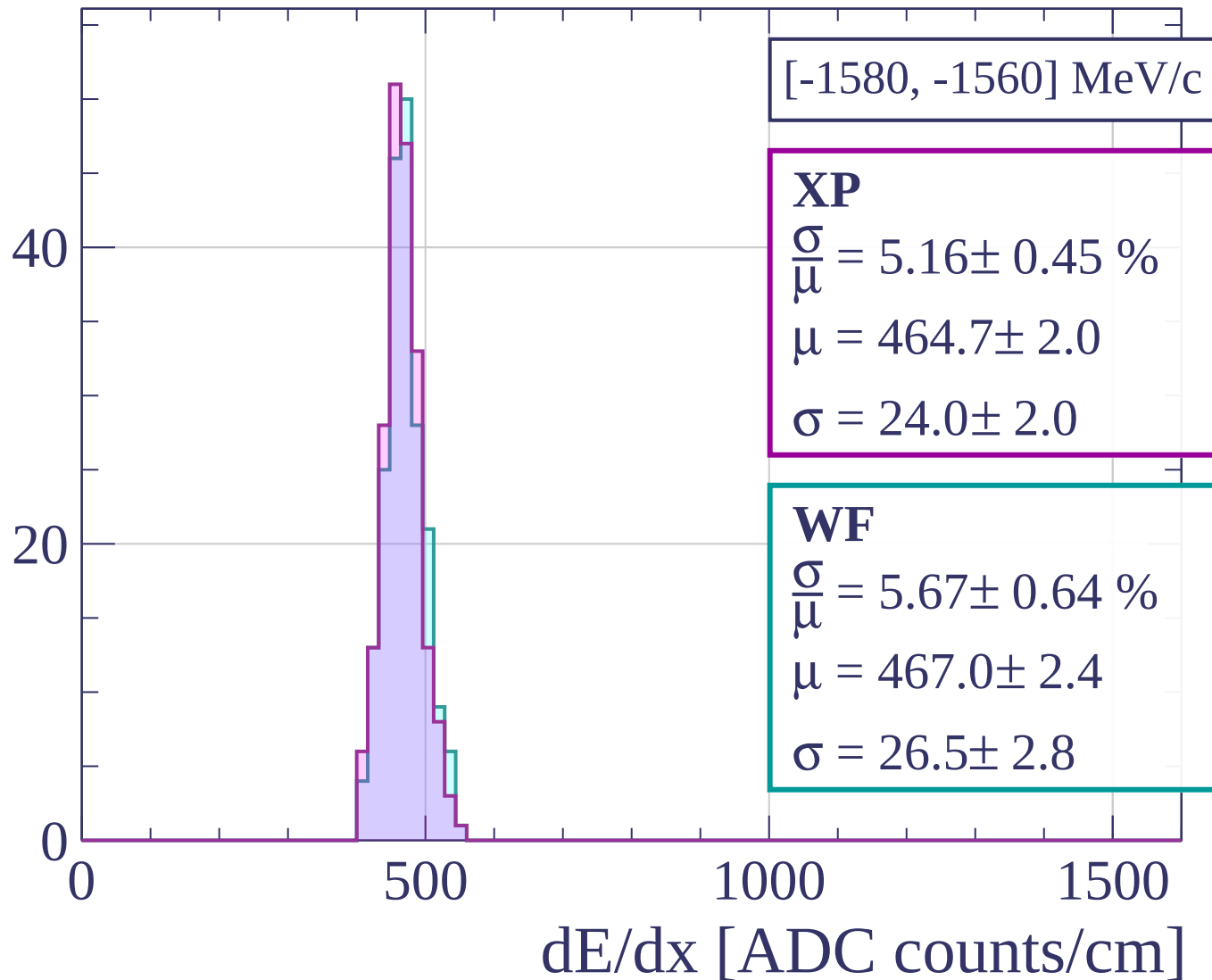




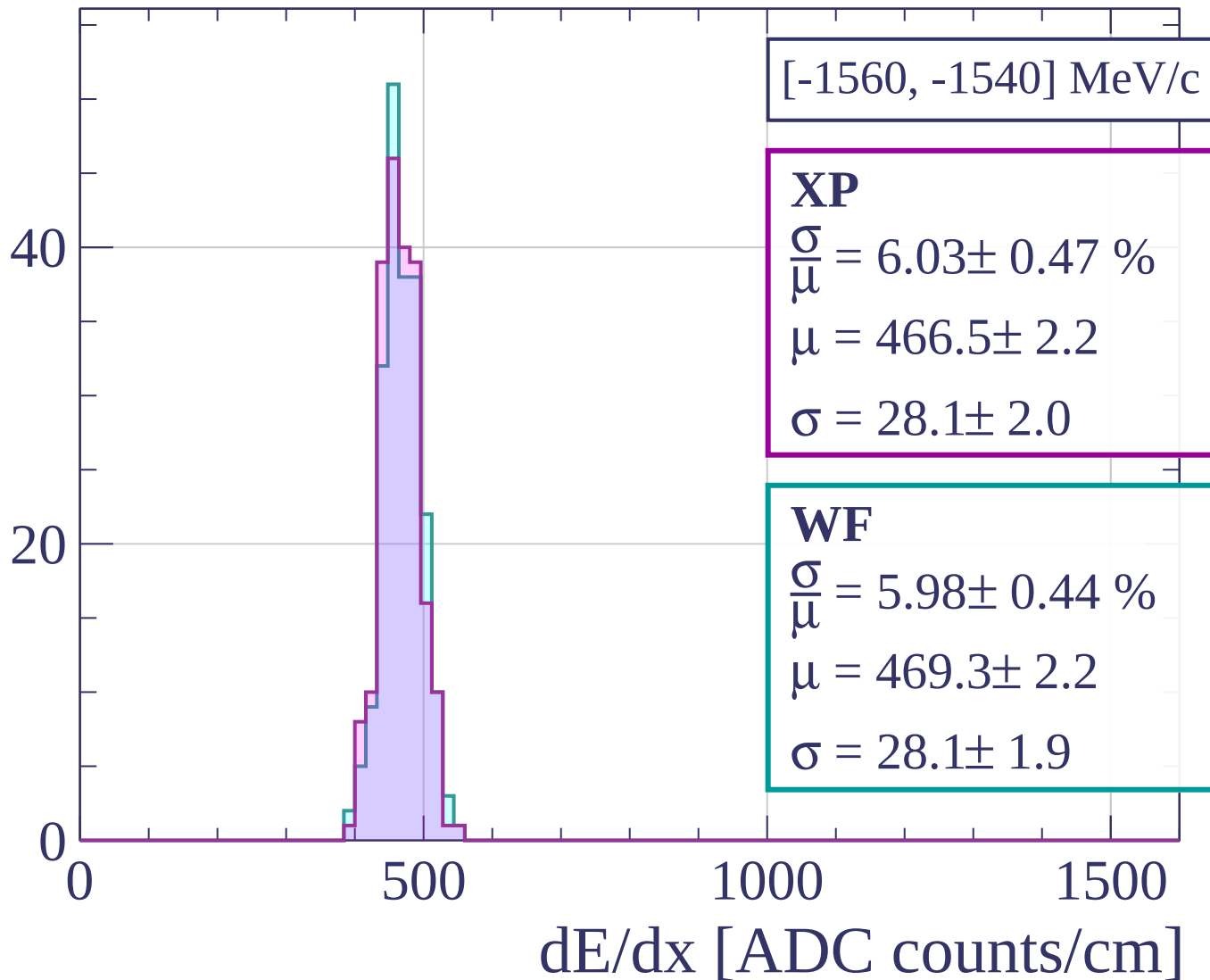
Count



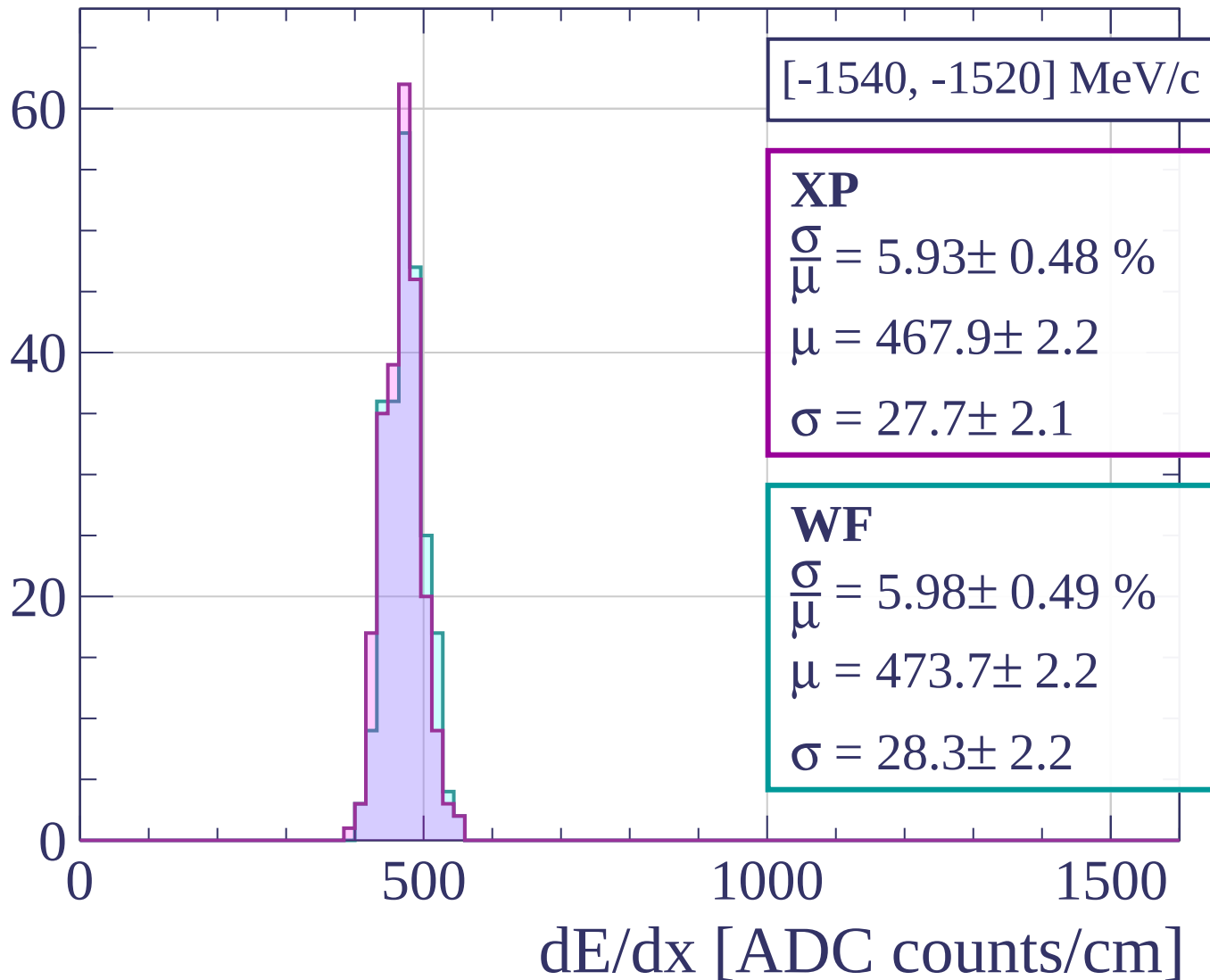
Count



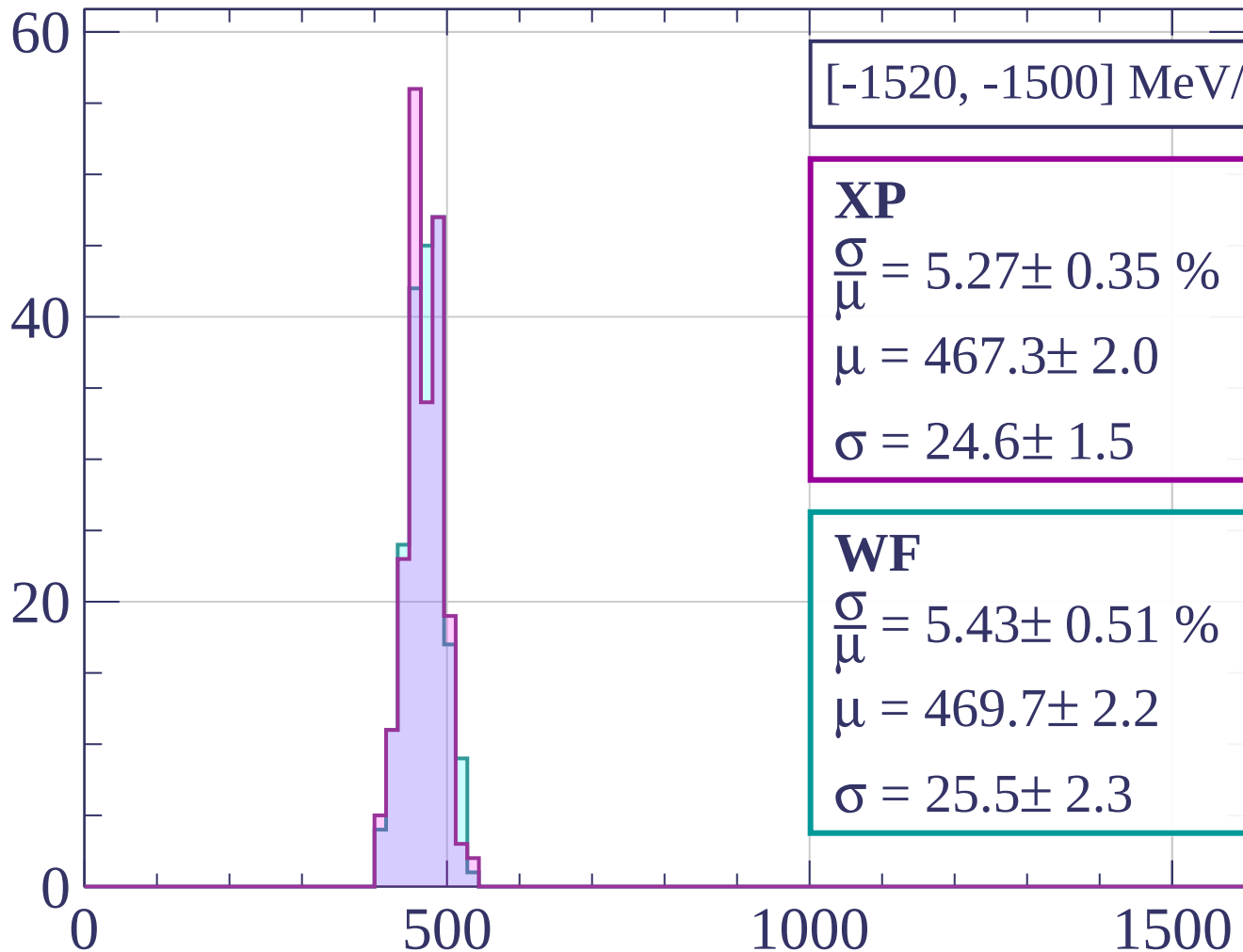
Count



Count



Count



XP

$$\frac{\sigma}{\mu} = 5.27 \pm 0.35 \%$$

$$\mu = 467.3 \pm 2.0$$

$$\sigma = 24.6 \pm 1.5$$

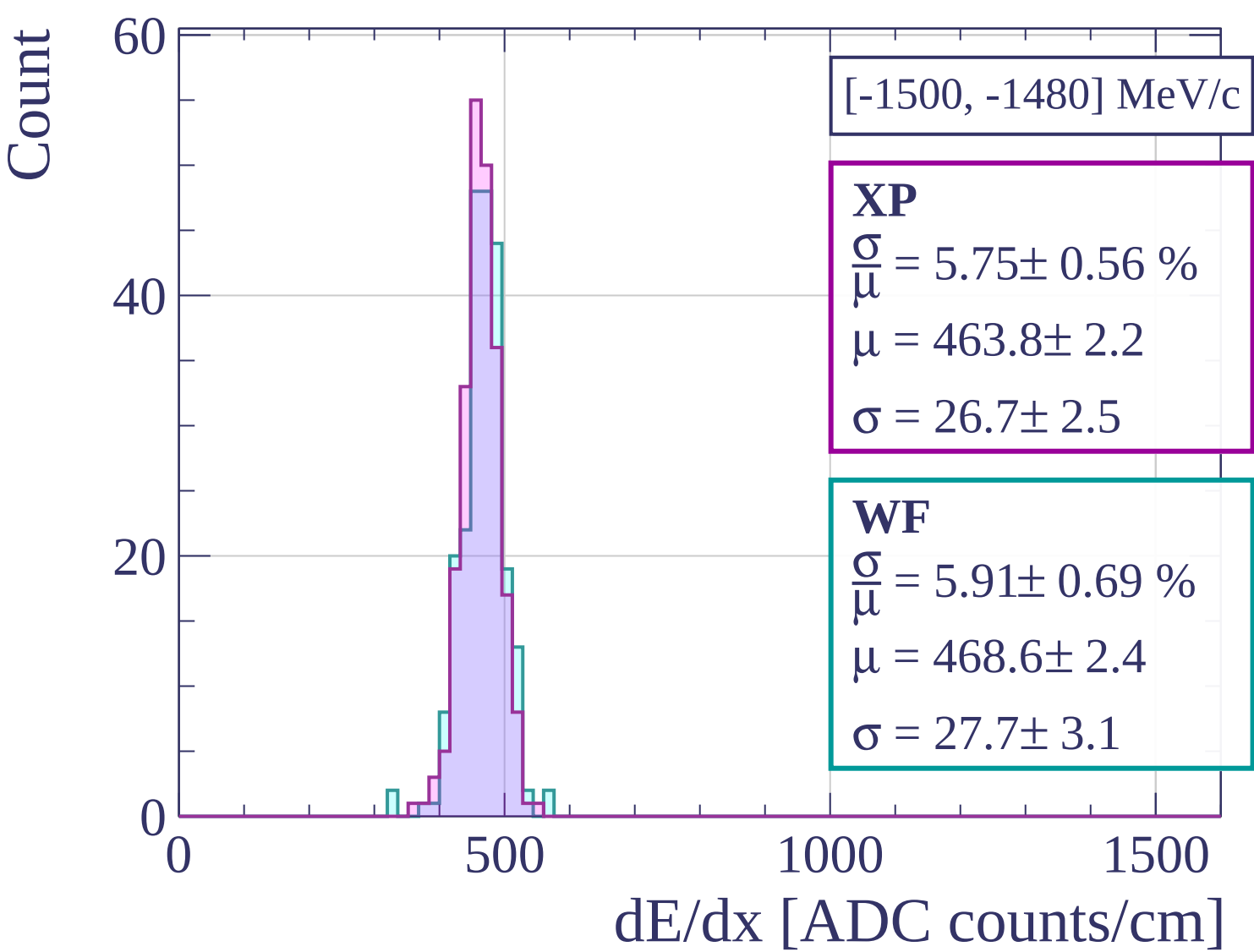
WF

$$\frac{\sigma}{\mu} = 5.43 \pm 0.51 \%$$

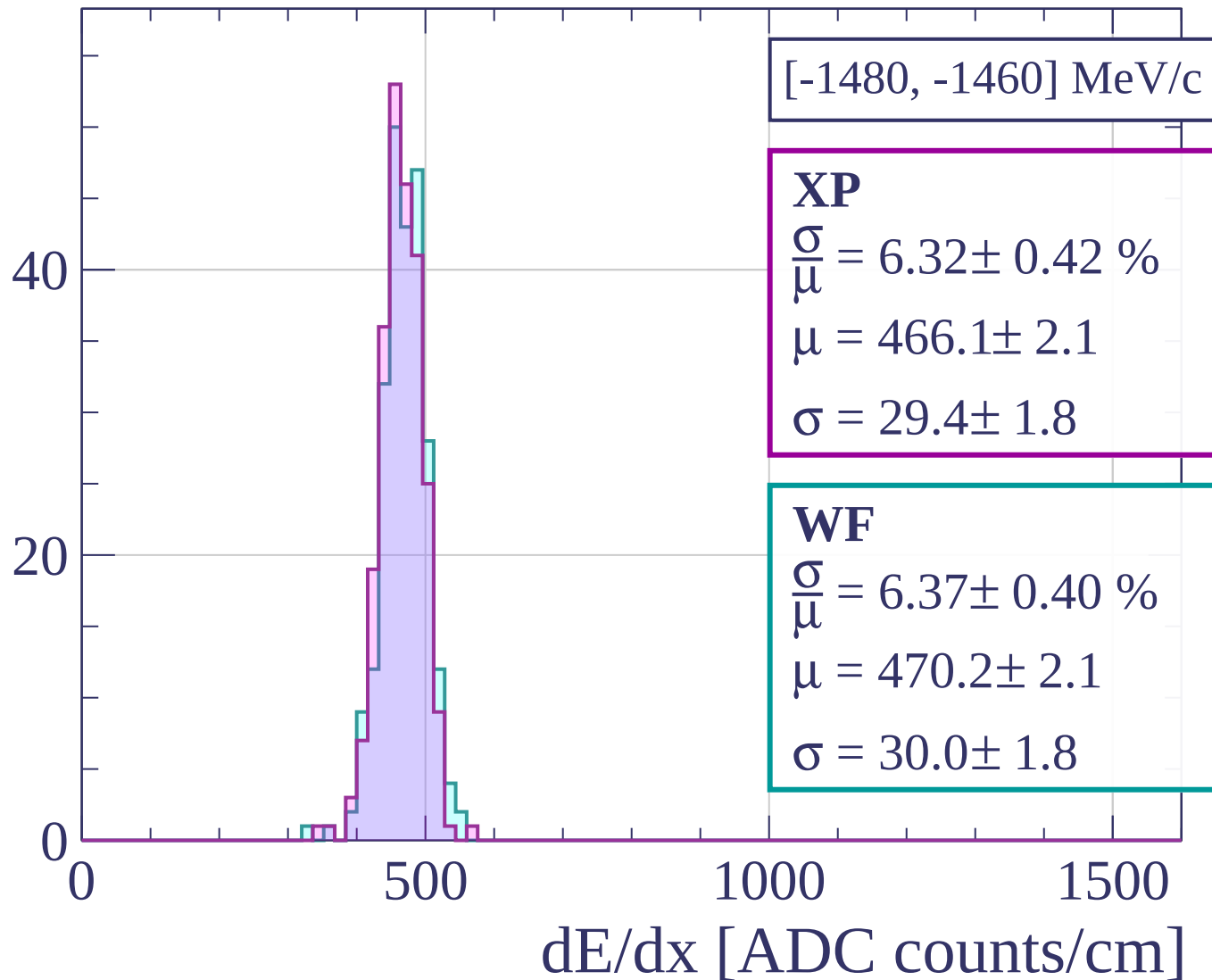
$$\mu = 469.7 \pm 2.2$$

$$\sigma = 25.5 \pm 2.3$$

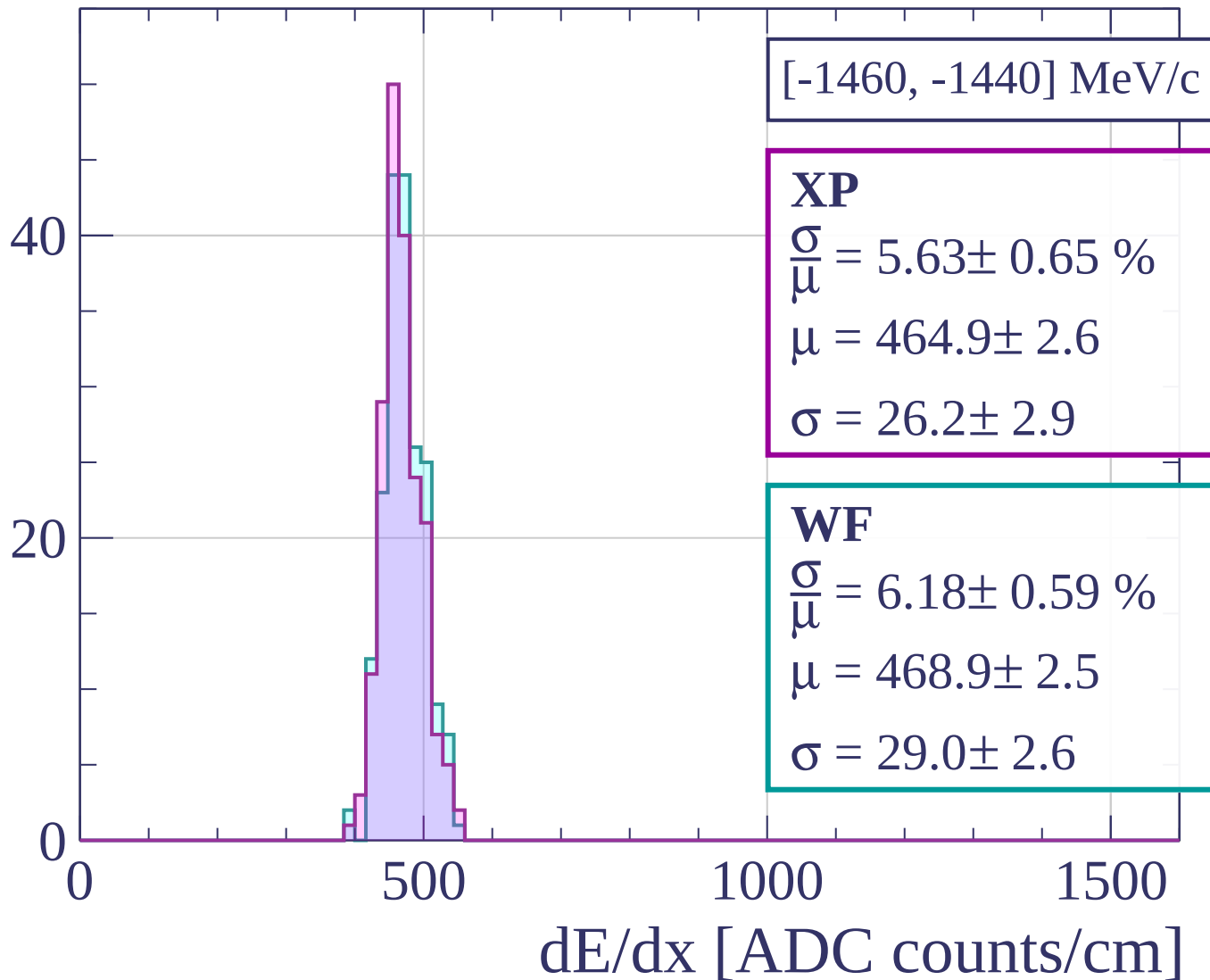
dE/dx [ADC counts/cm]



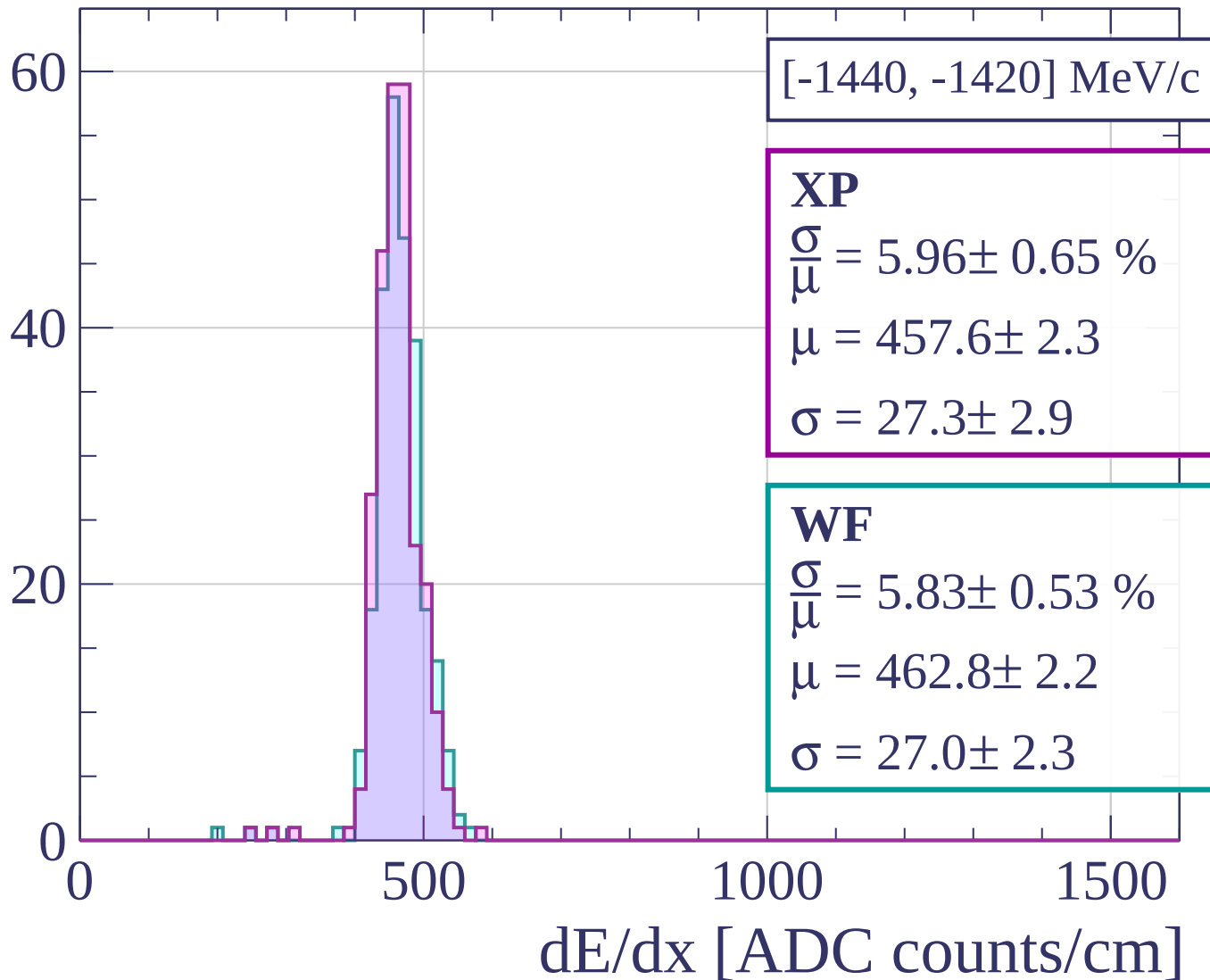
Count

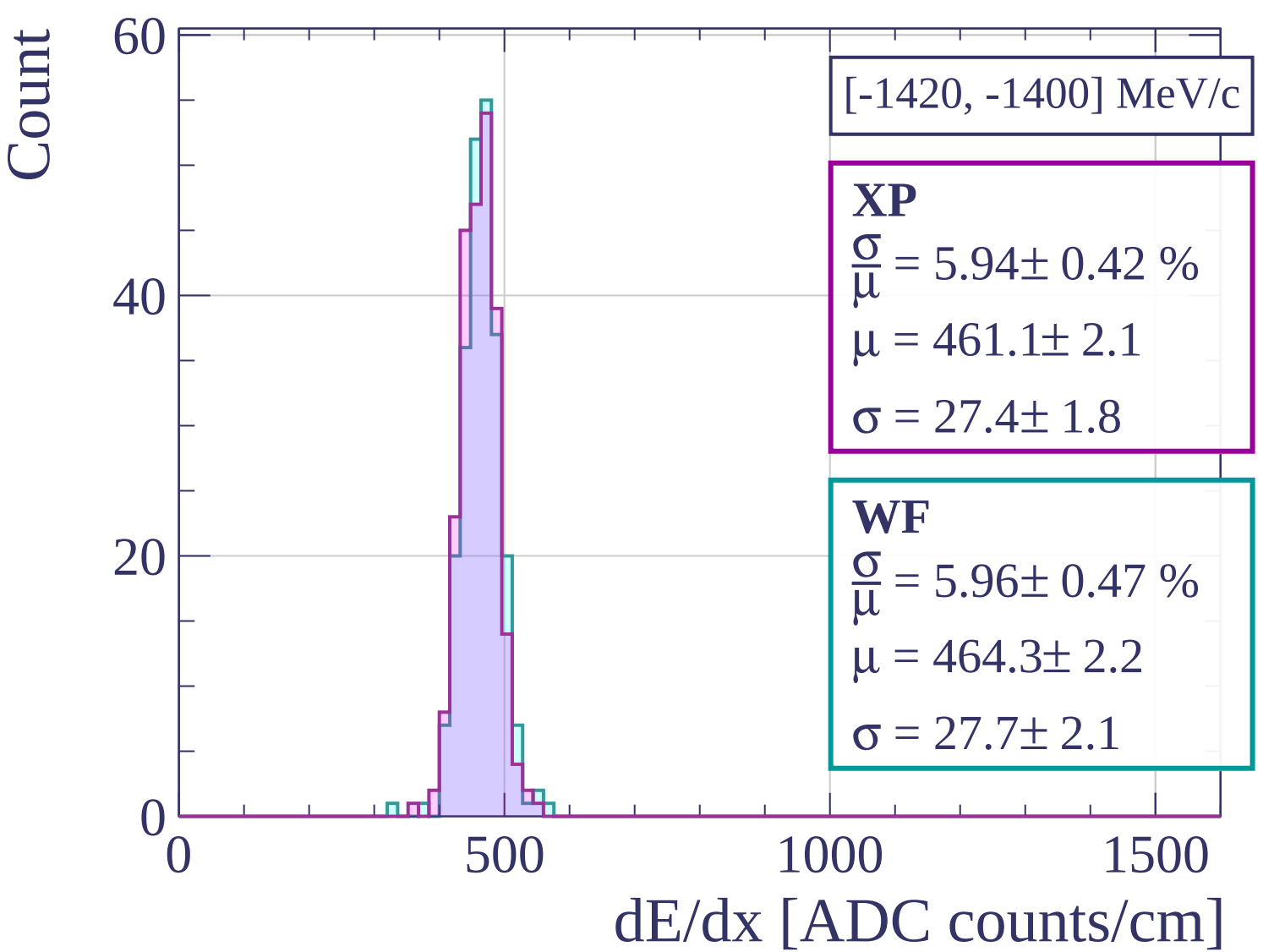


Count

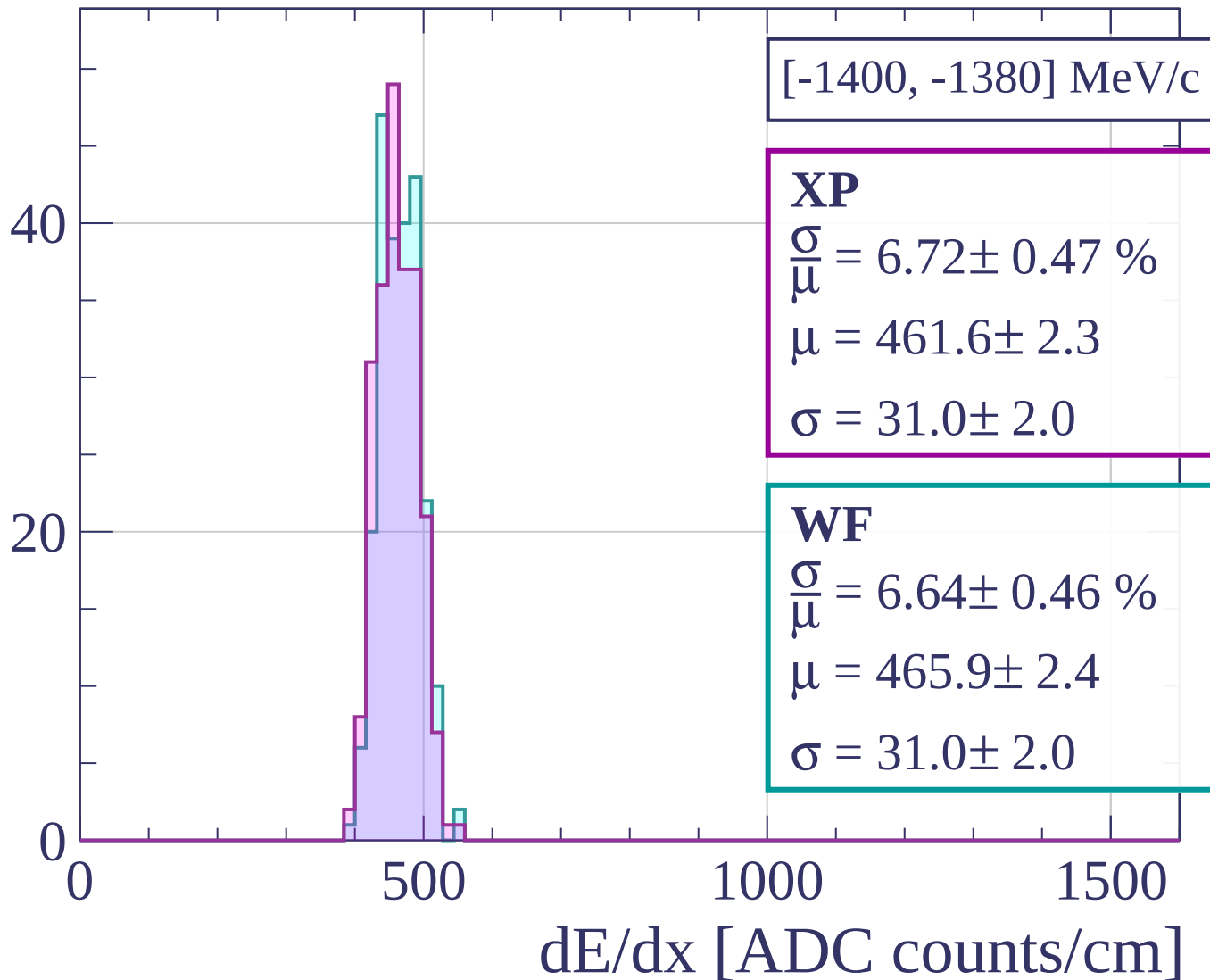


Count

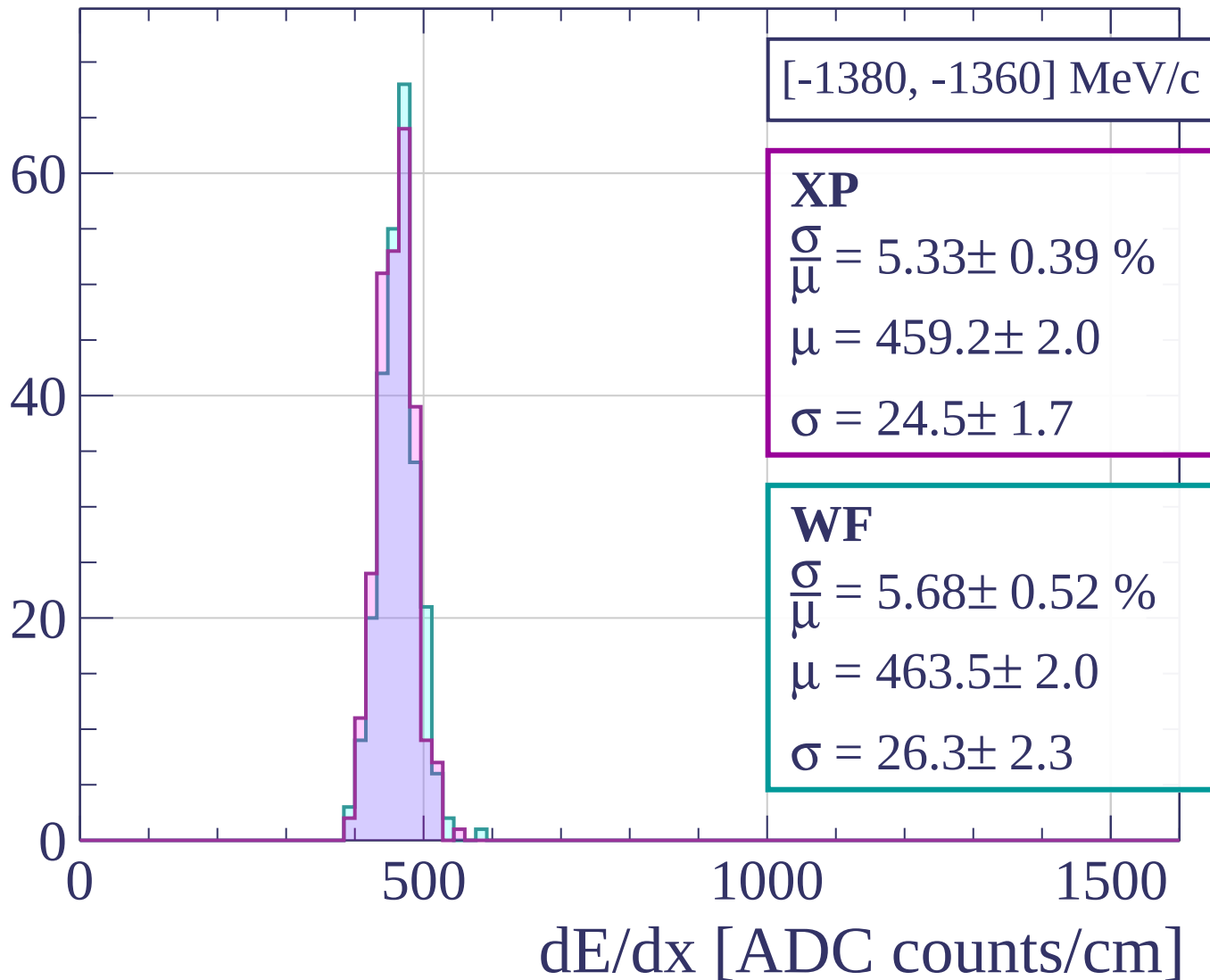




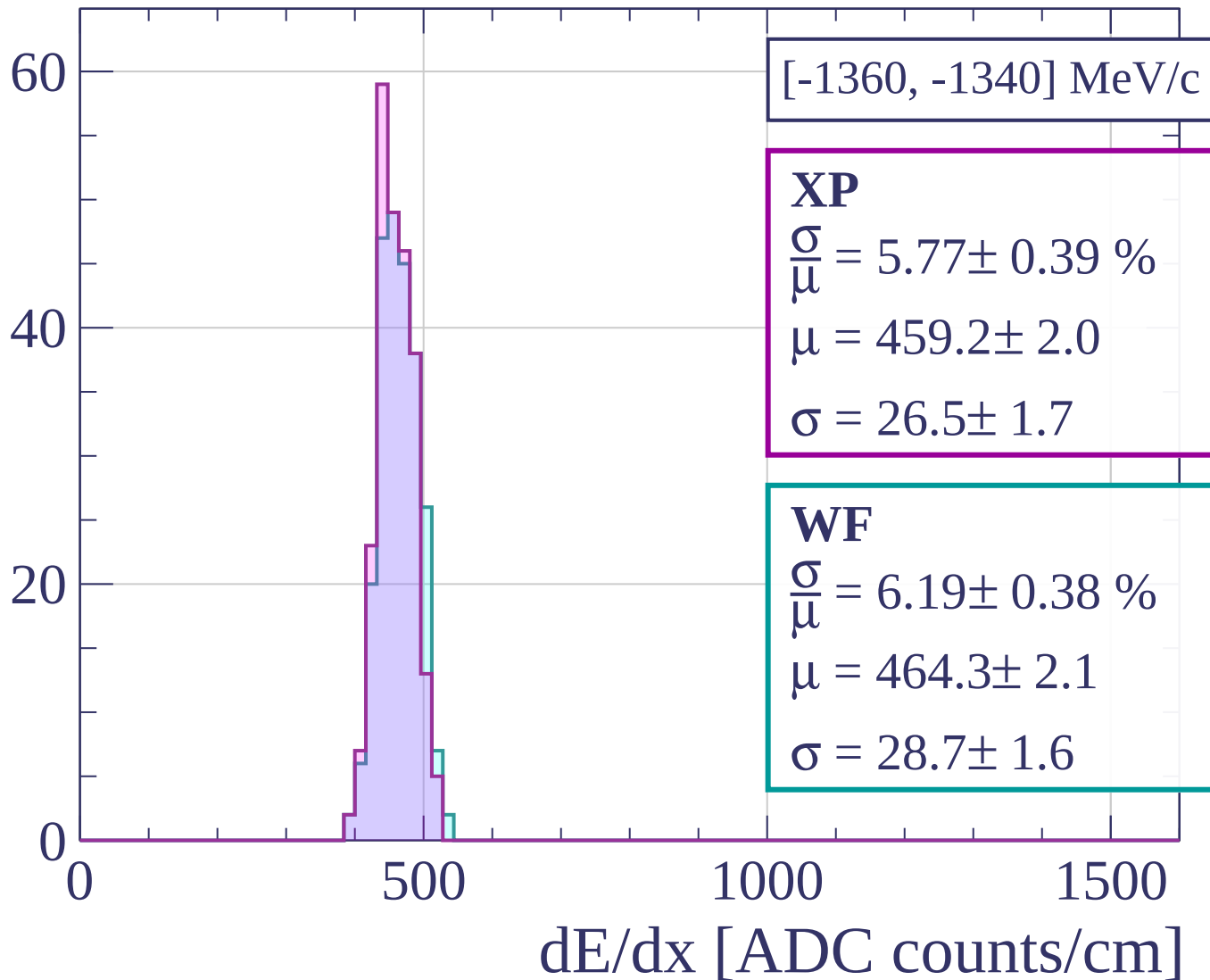
Count



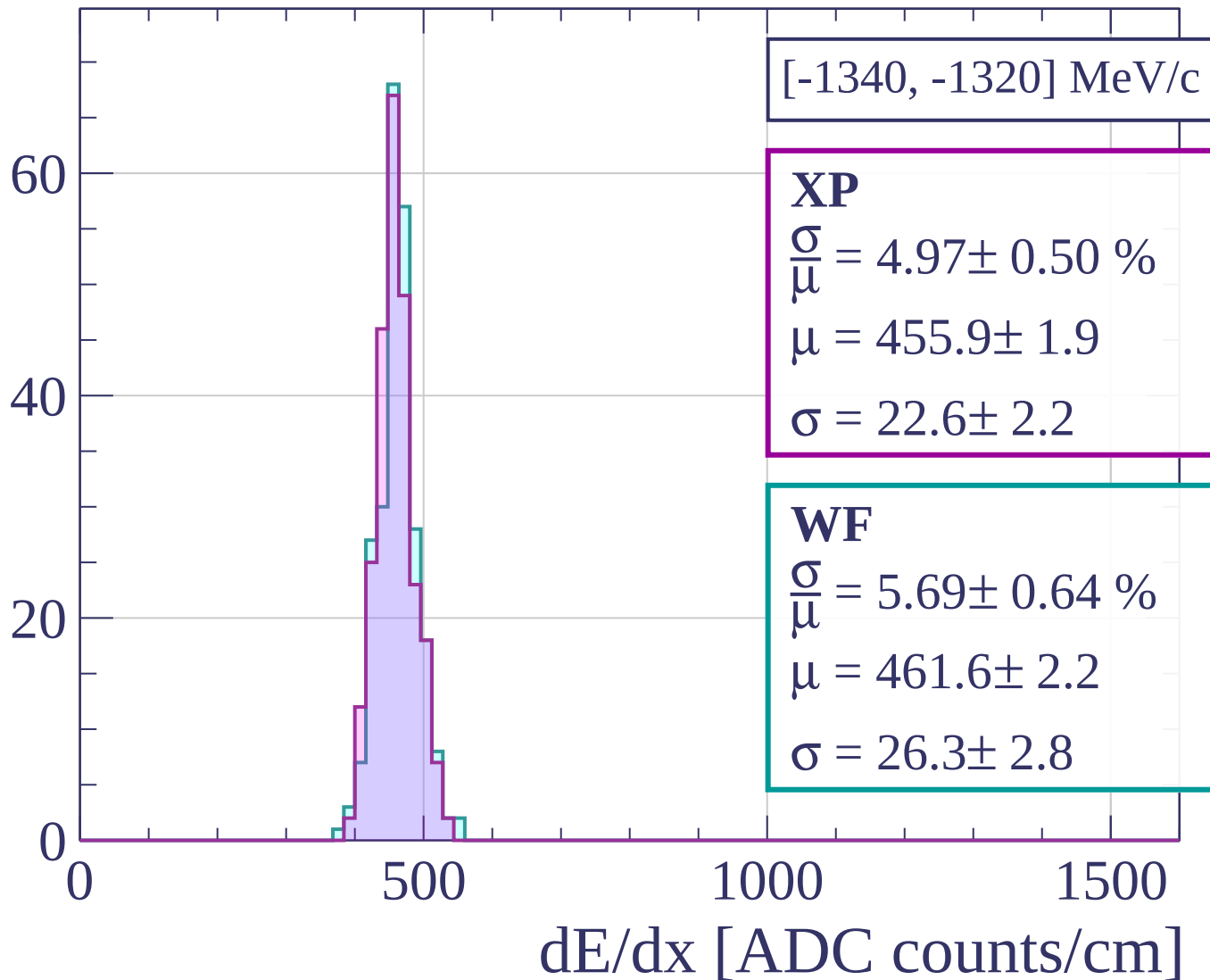
Count



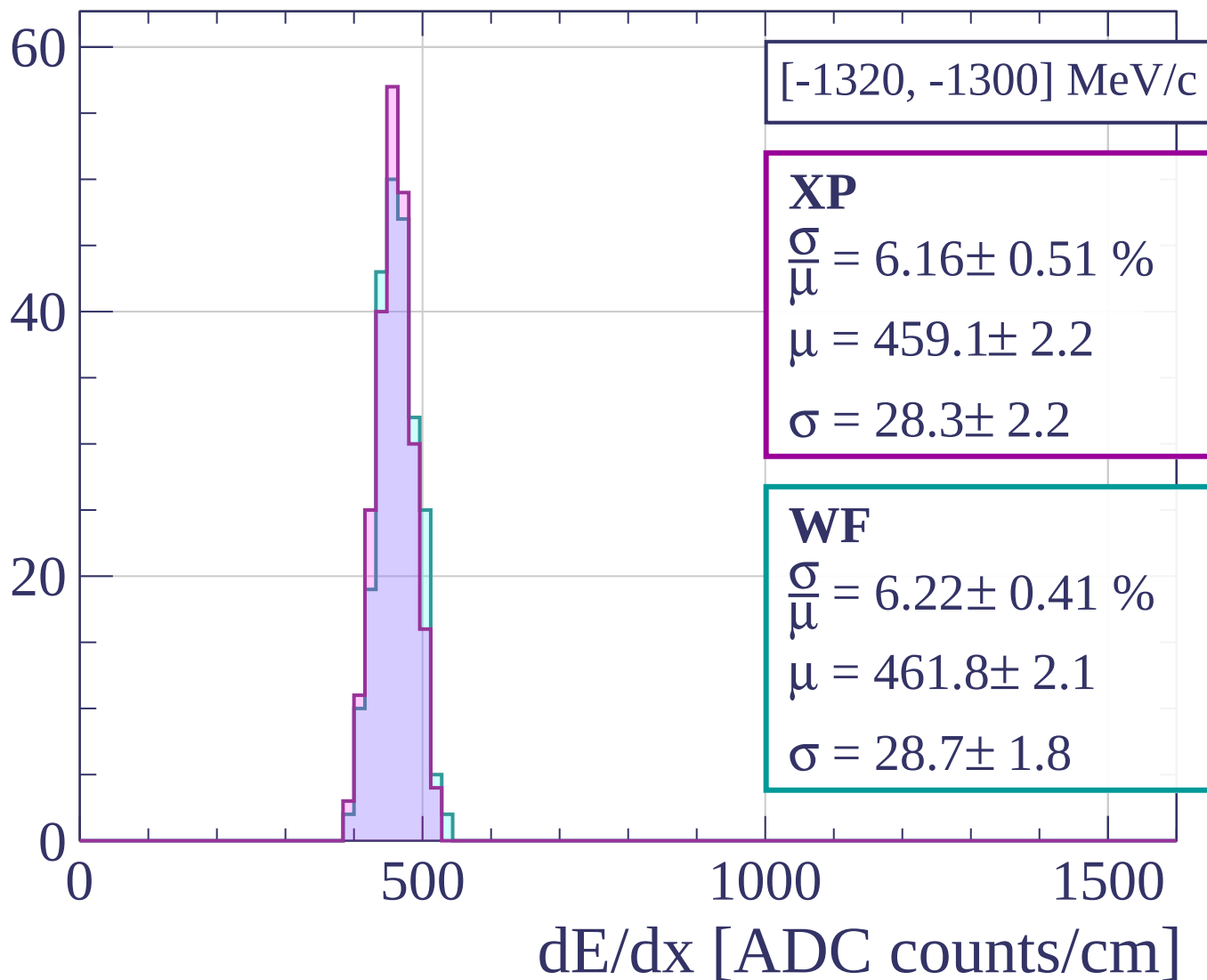
Count



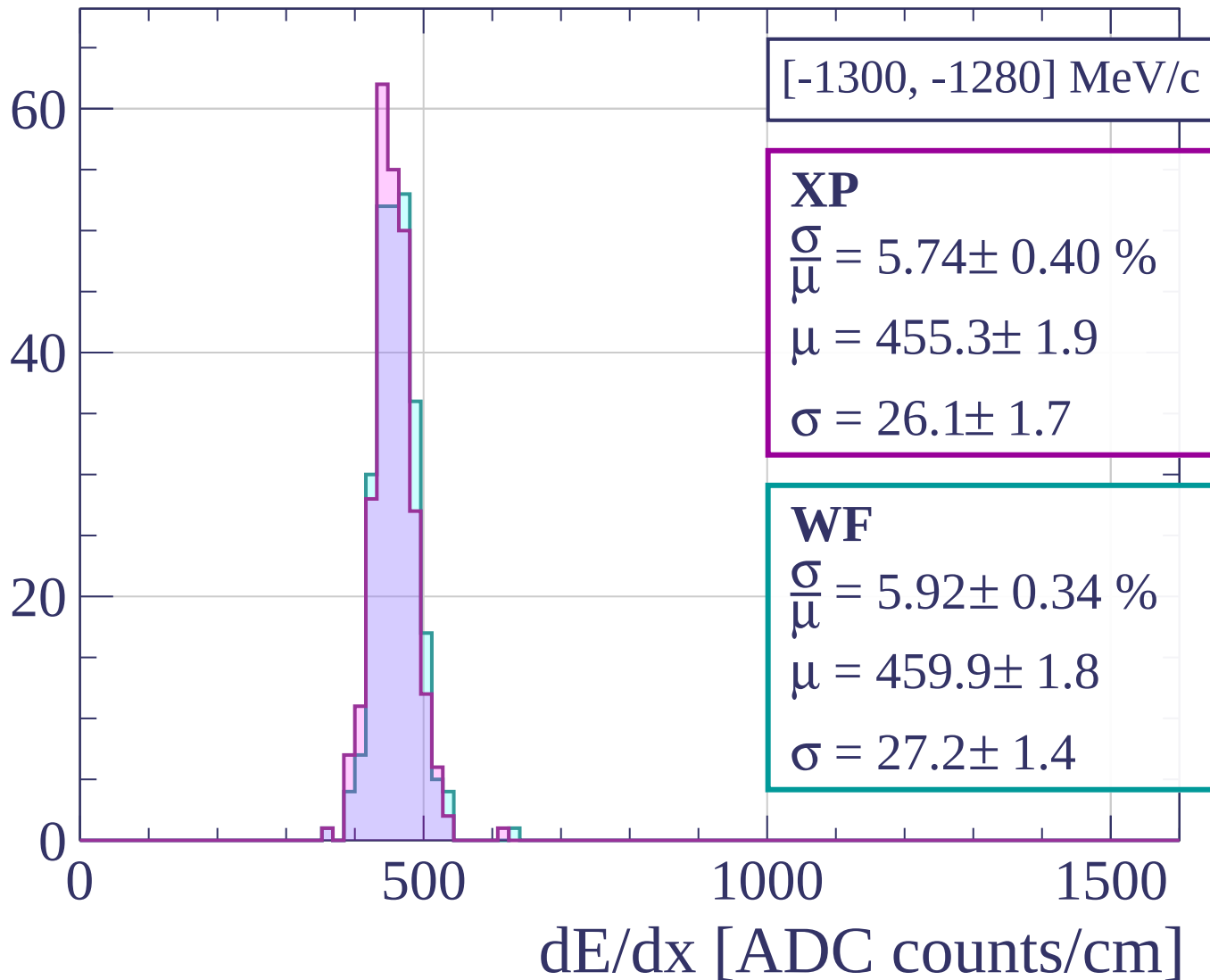
Count



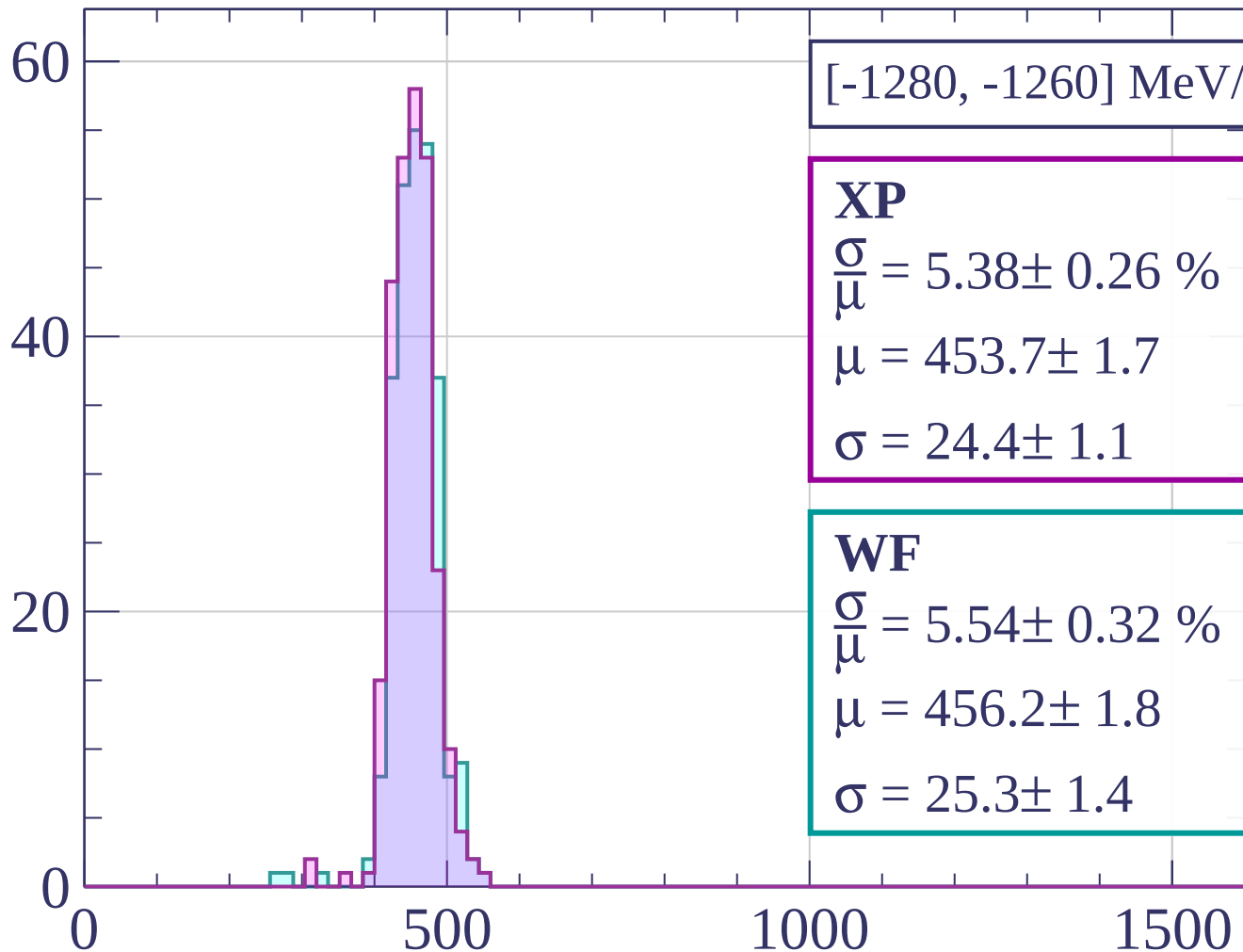
Count



Count



Count



$[-1280, -1260] \text{ MeV/c}$

XP

$$\frac{\sigma}{\mu} = 5.38 \pm 0.26 \%$$

$$\mu = 453.7 \pm 1.7$$

$$\sigma = 24.4 \pm 1.1$$

WF

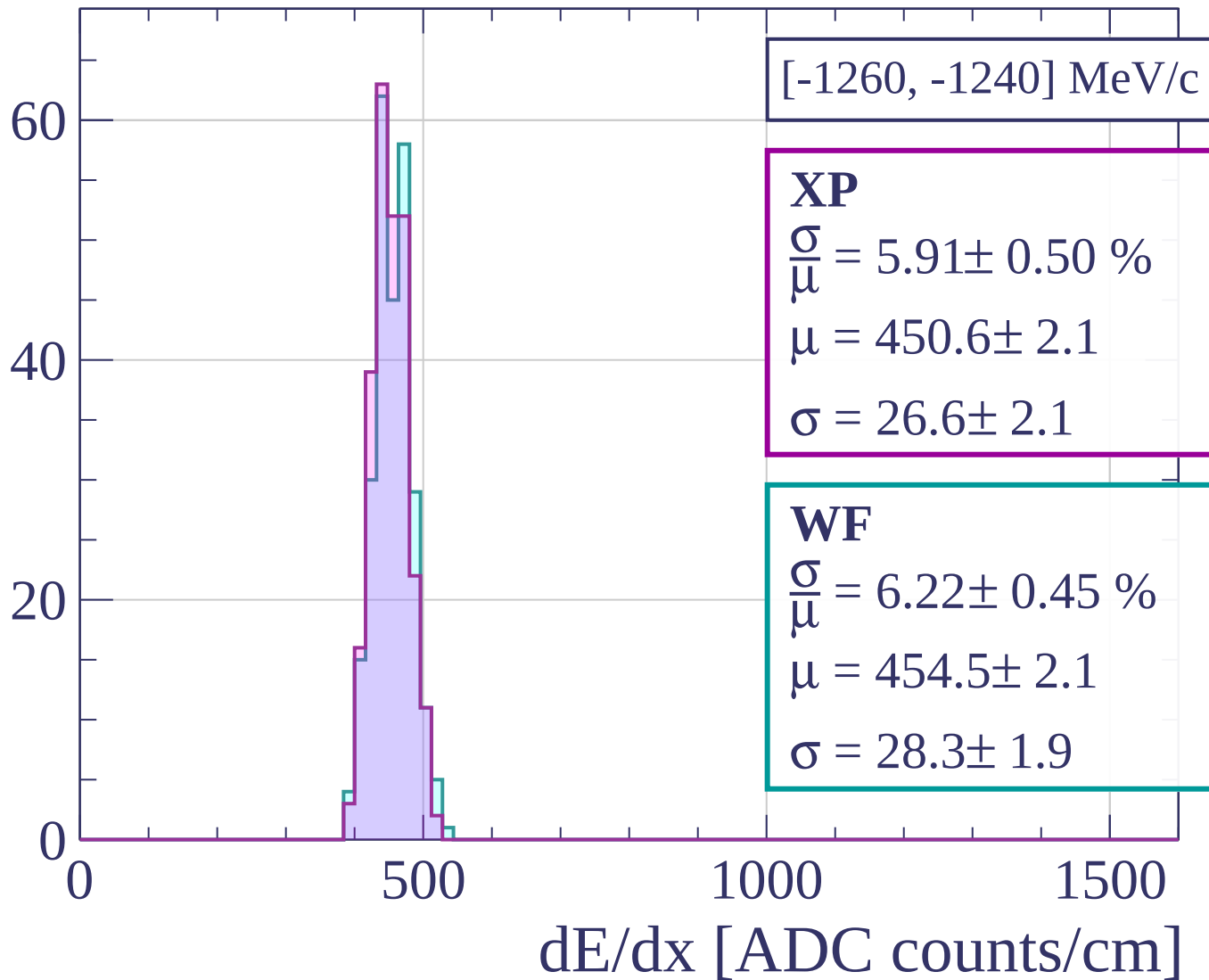
$$\frac{\sigma}{\mu} = 5.54 \pm 0.32 \%$$

$$\mu = 456.2 \pm 1.8$$

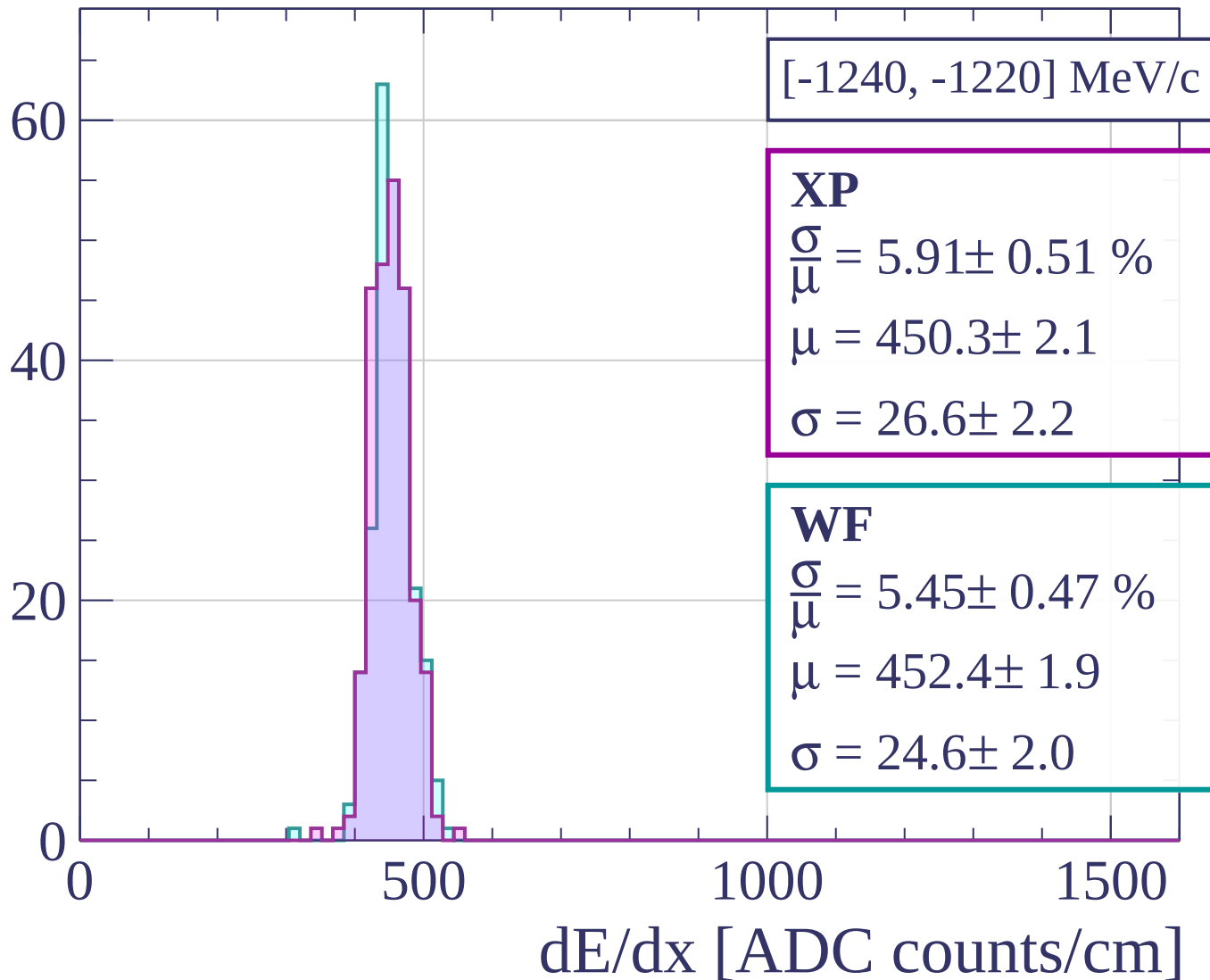
$$\sigma = 25.3 \pm 1.4$$

dE/dx [ADC counts/cm]

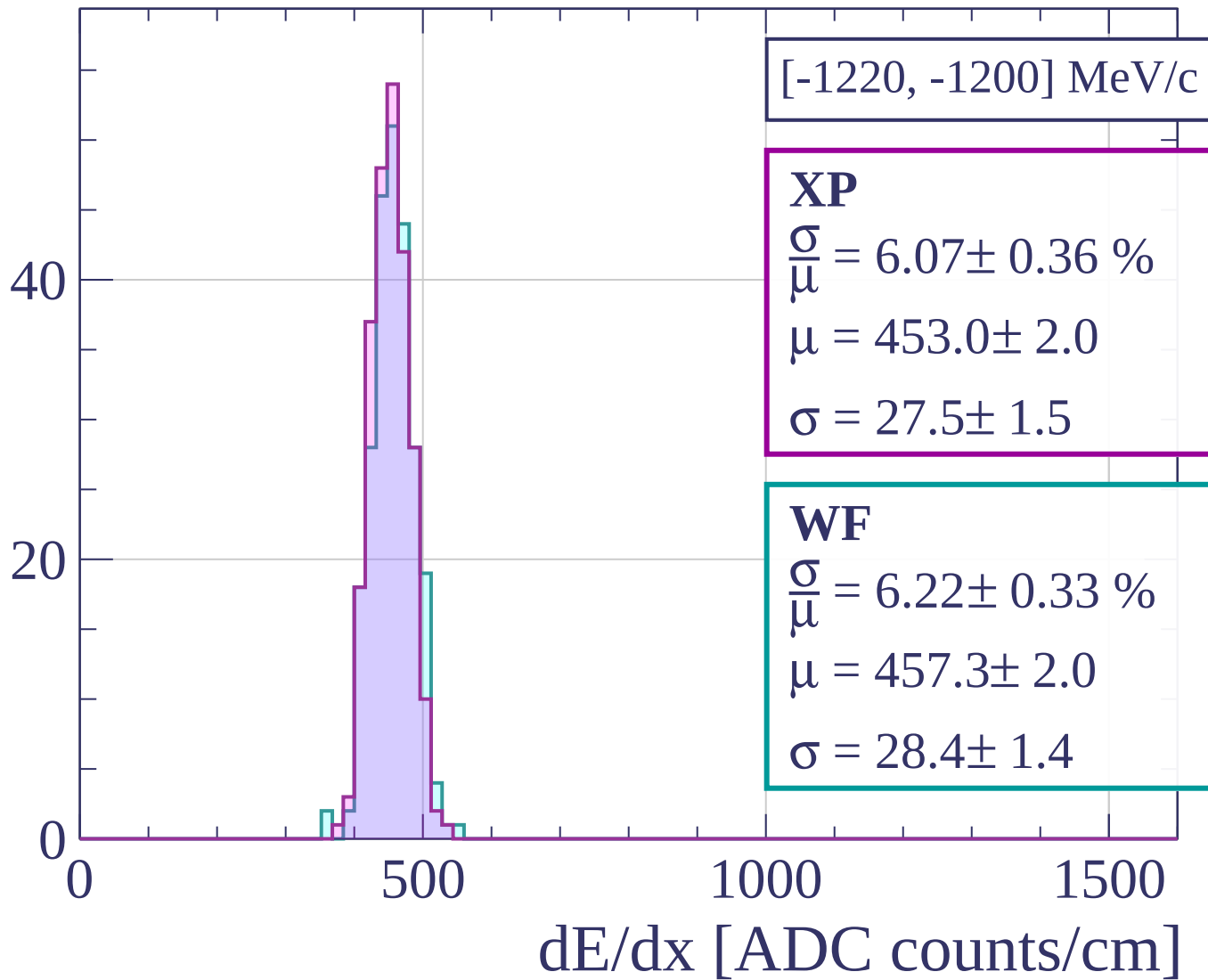
Count



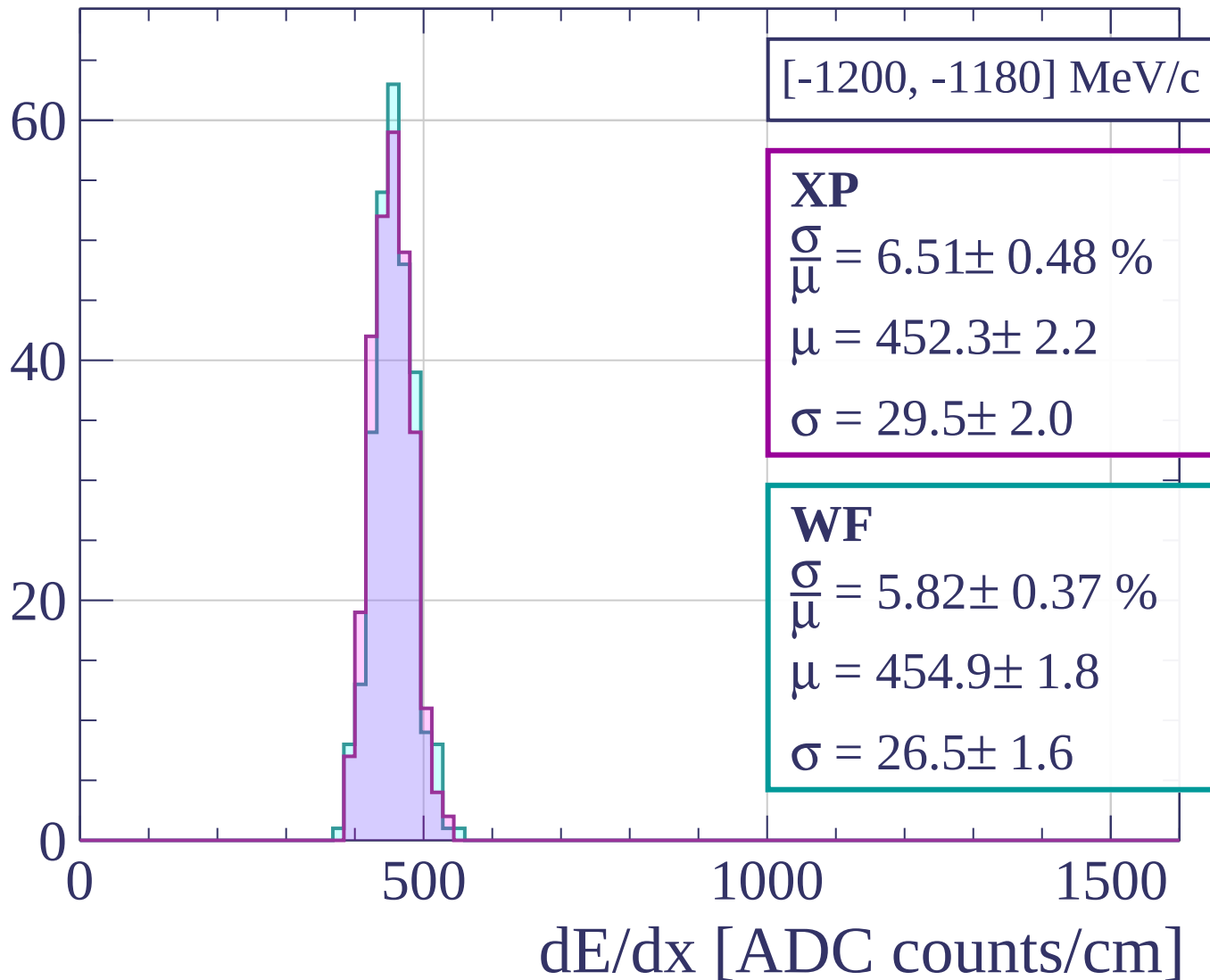
Count



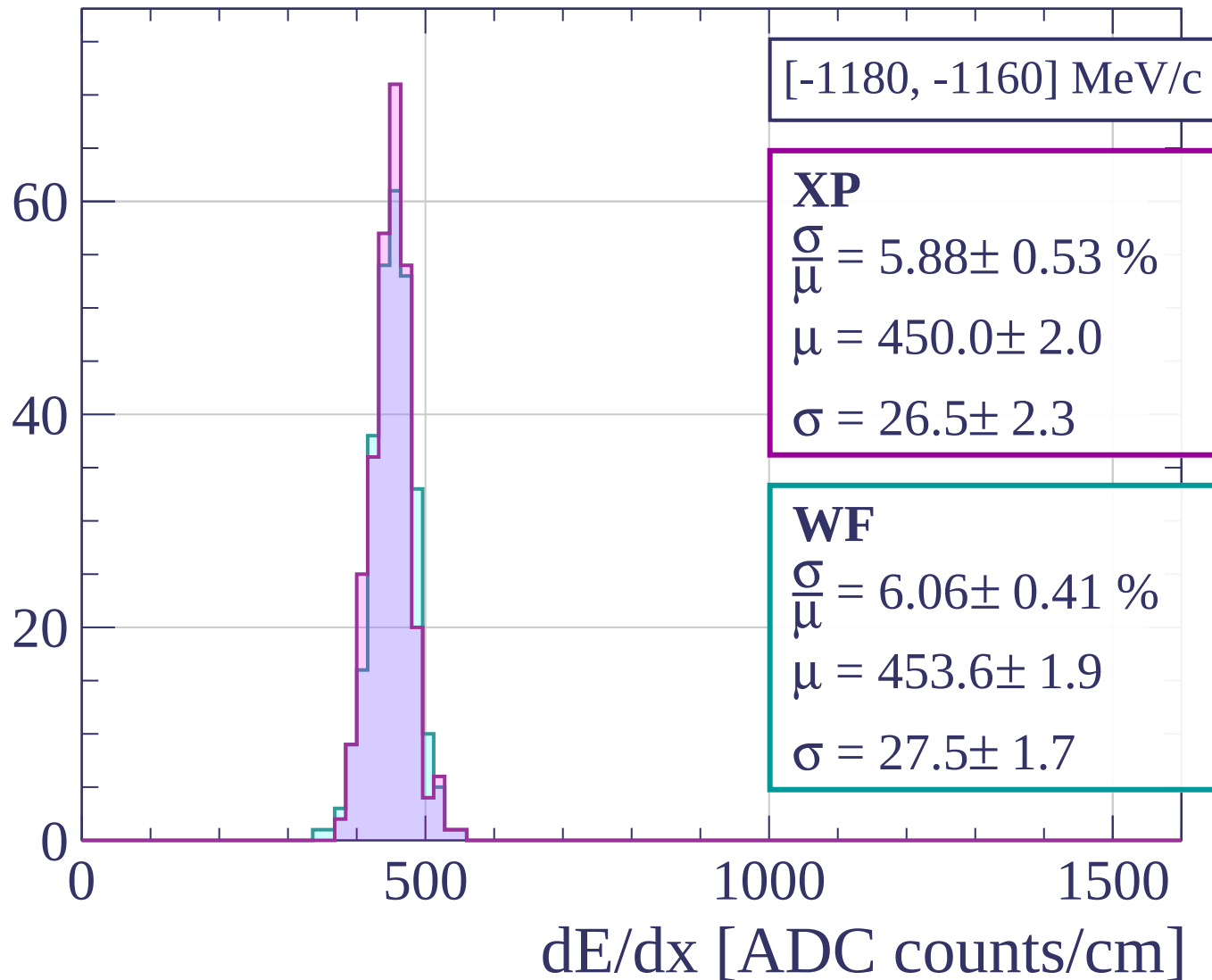
Count



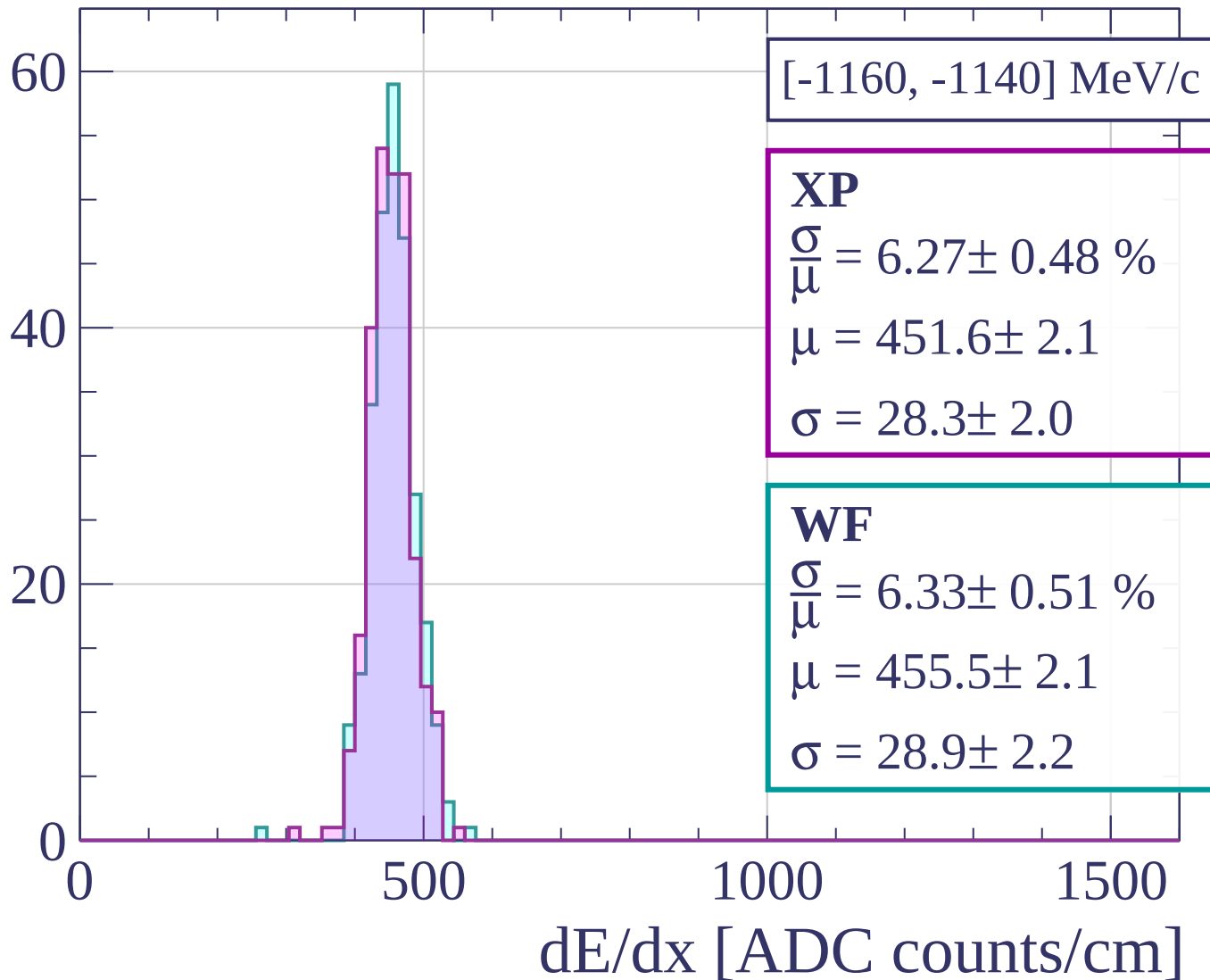
Count



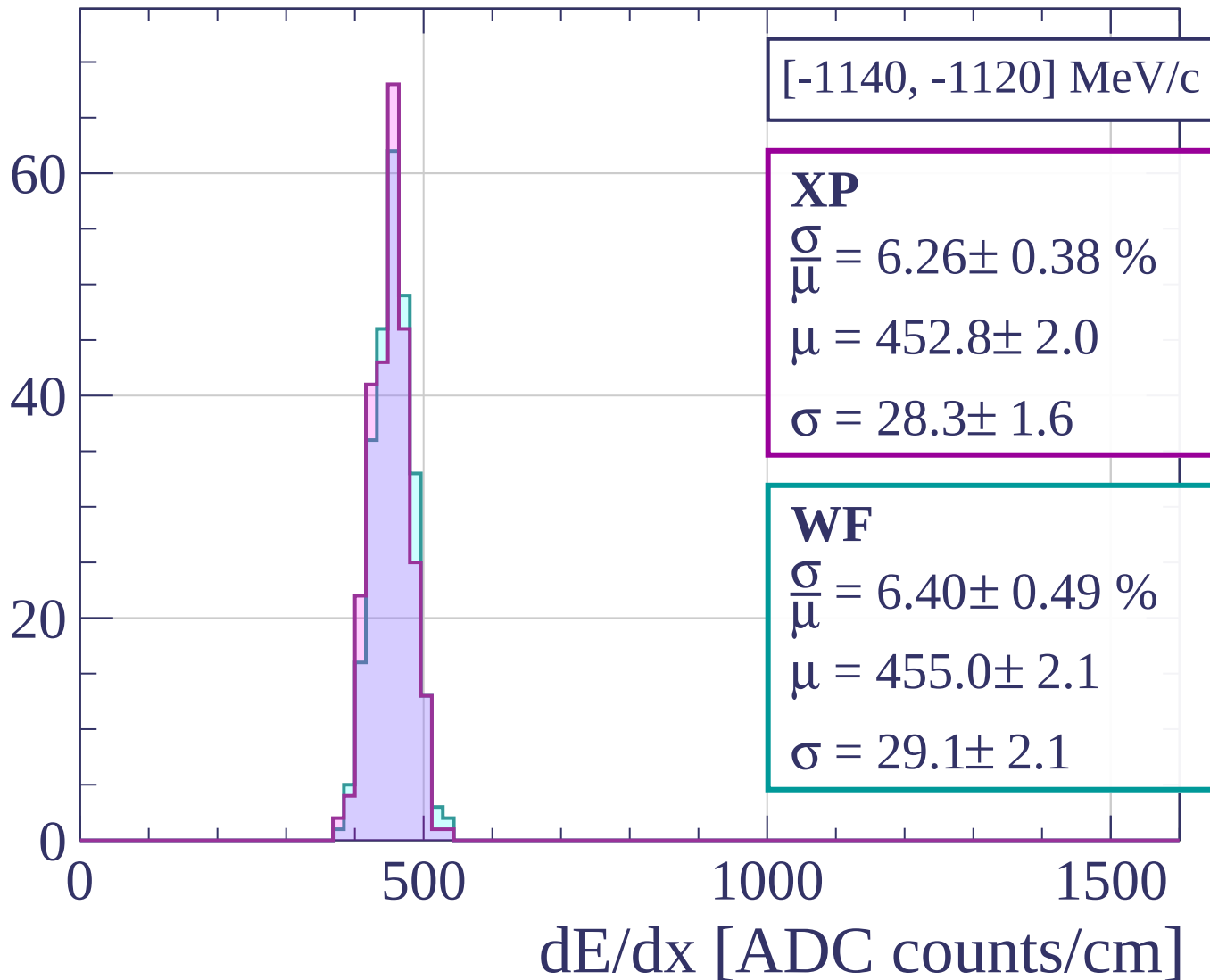
Count



Count



Count



[-1140, -1120] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.26 \pm 0.38 \%$$

$$\mu = 452.8 \pm 2.0$$

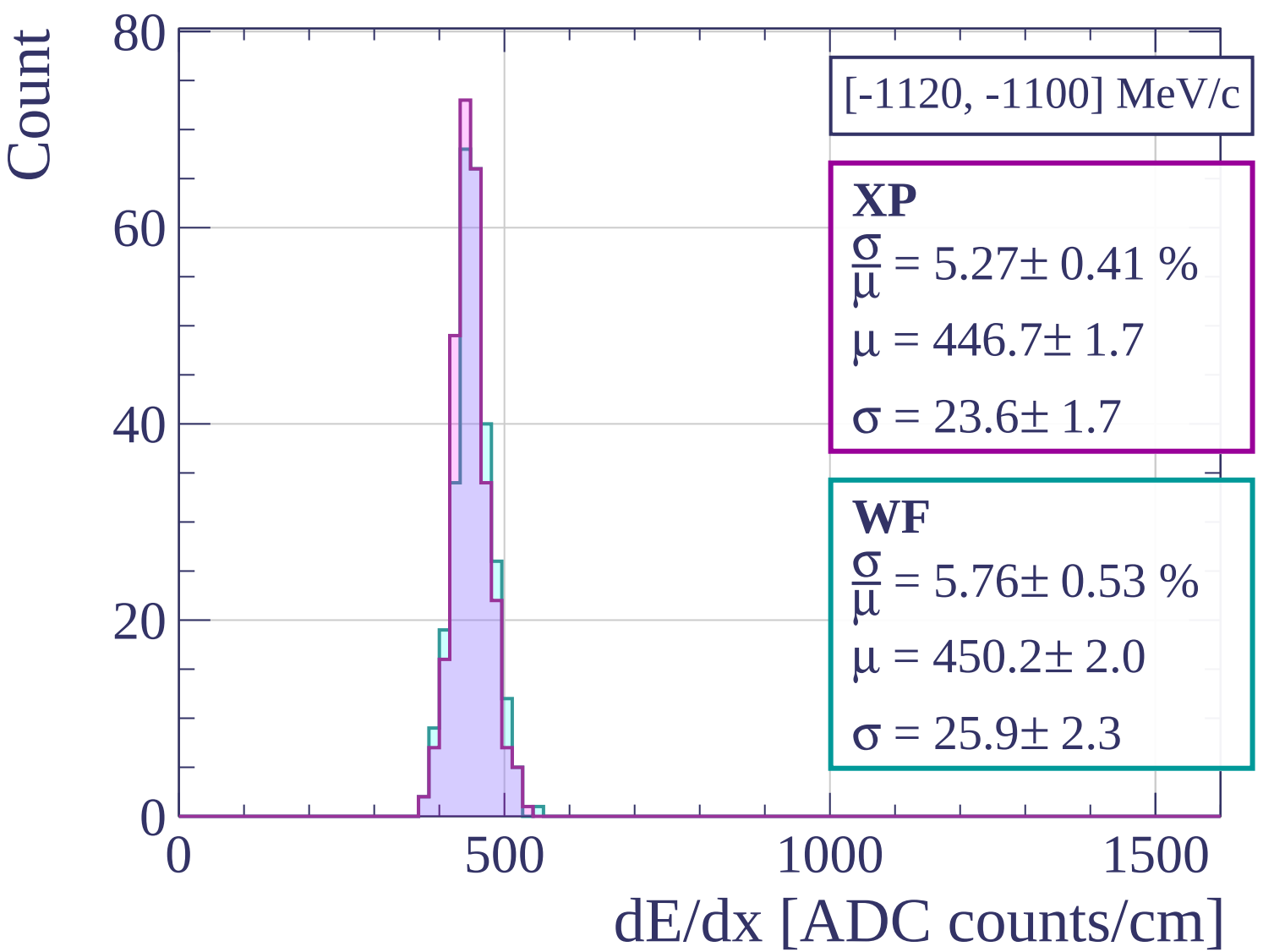
$$\sigma = 28.3 \pm 1.6$$

WF

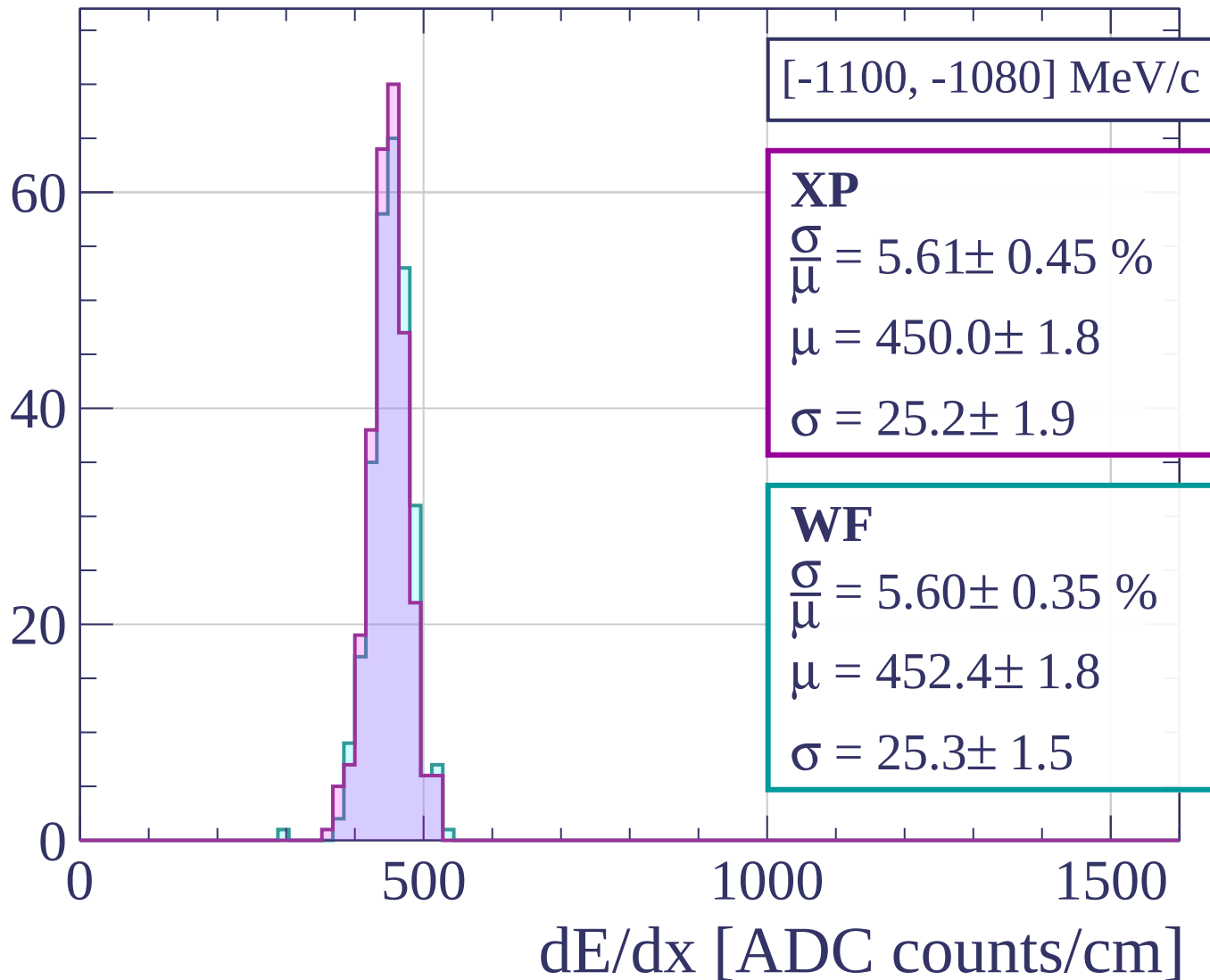
$$\frac{\sigma}{\mu} = 6.40 \pm 0.49 \%$$

$$\mu = 455.0 \pm 2.1$$

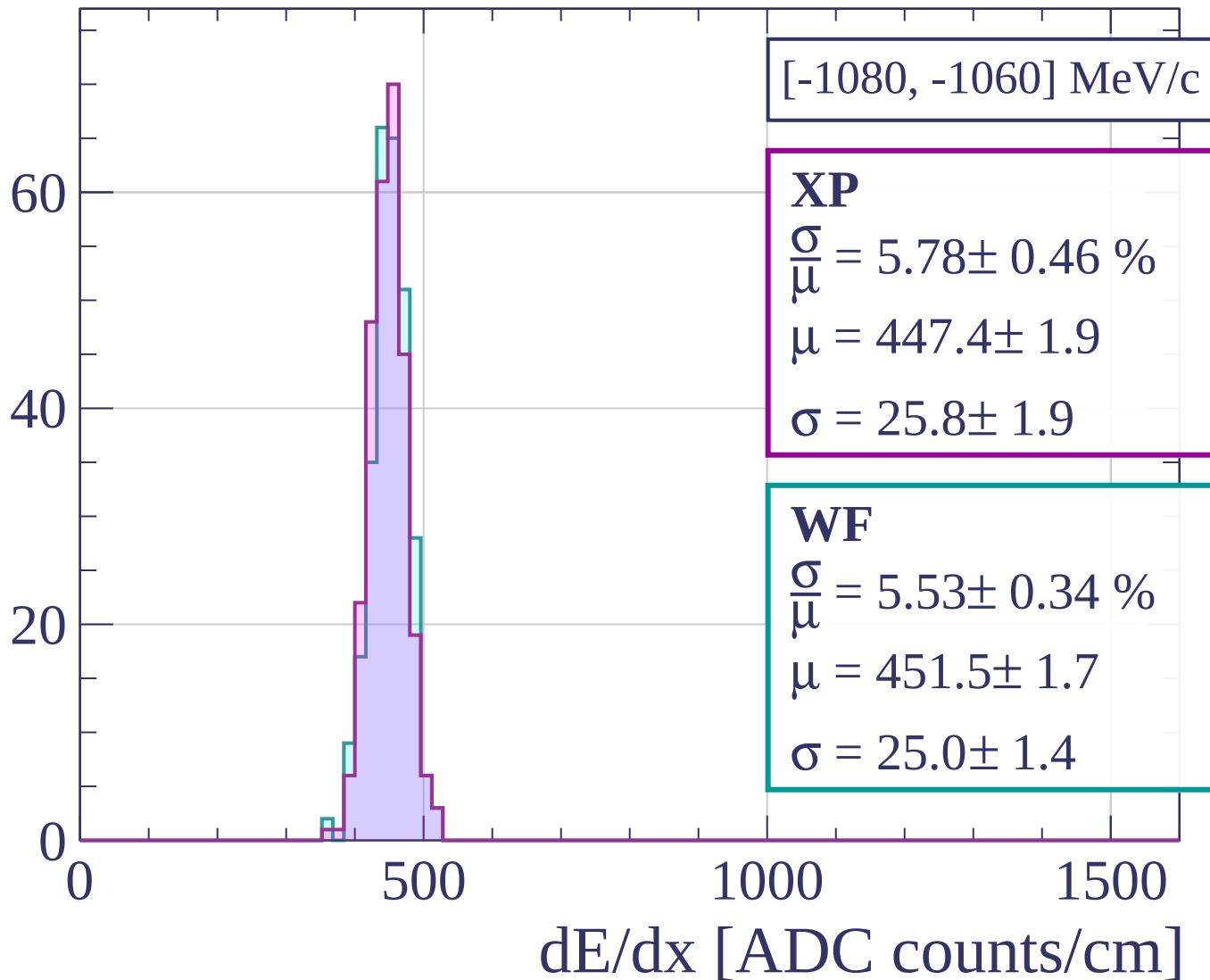
$$\sigma = 29.1 \pm 2.1$$



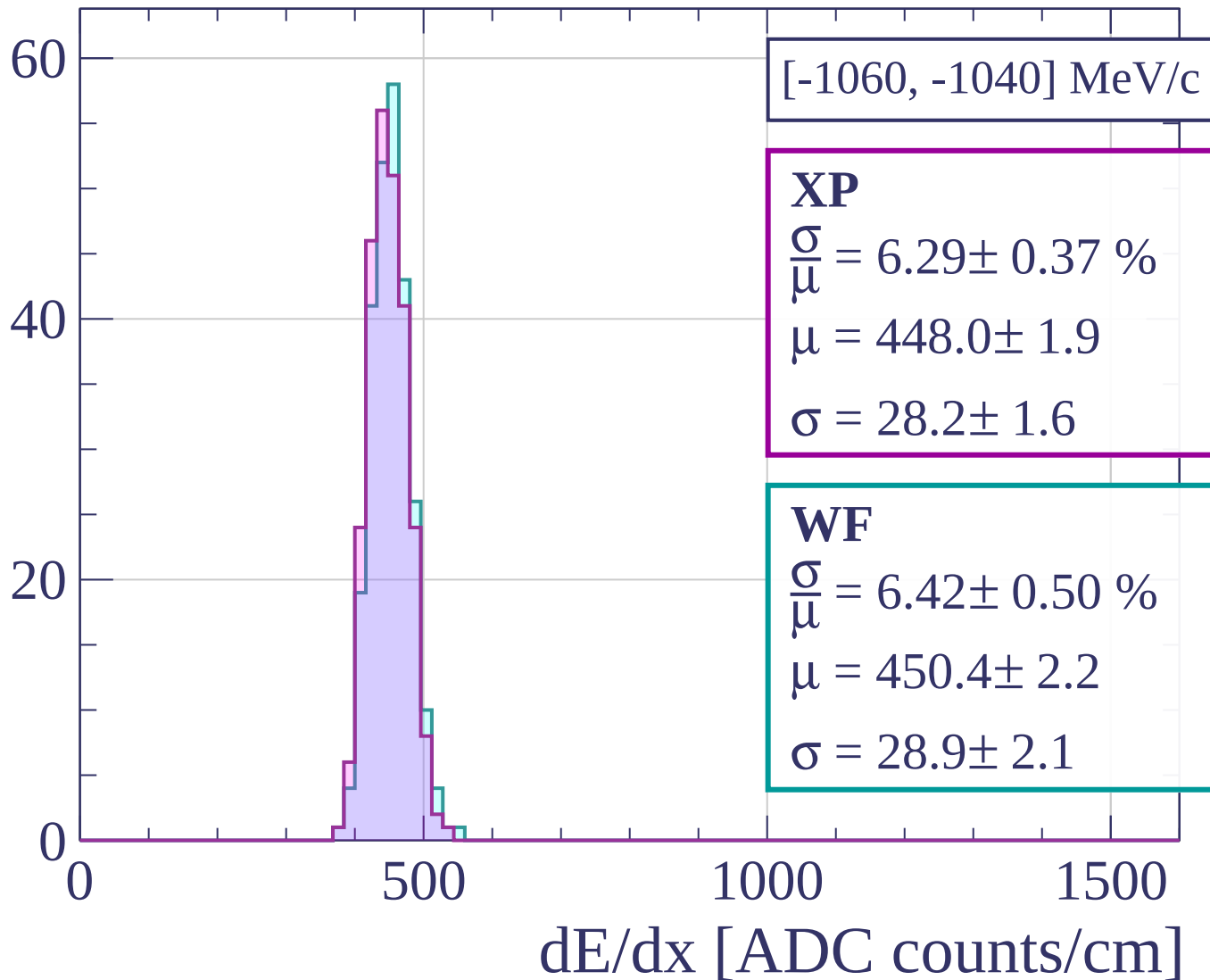
Count



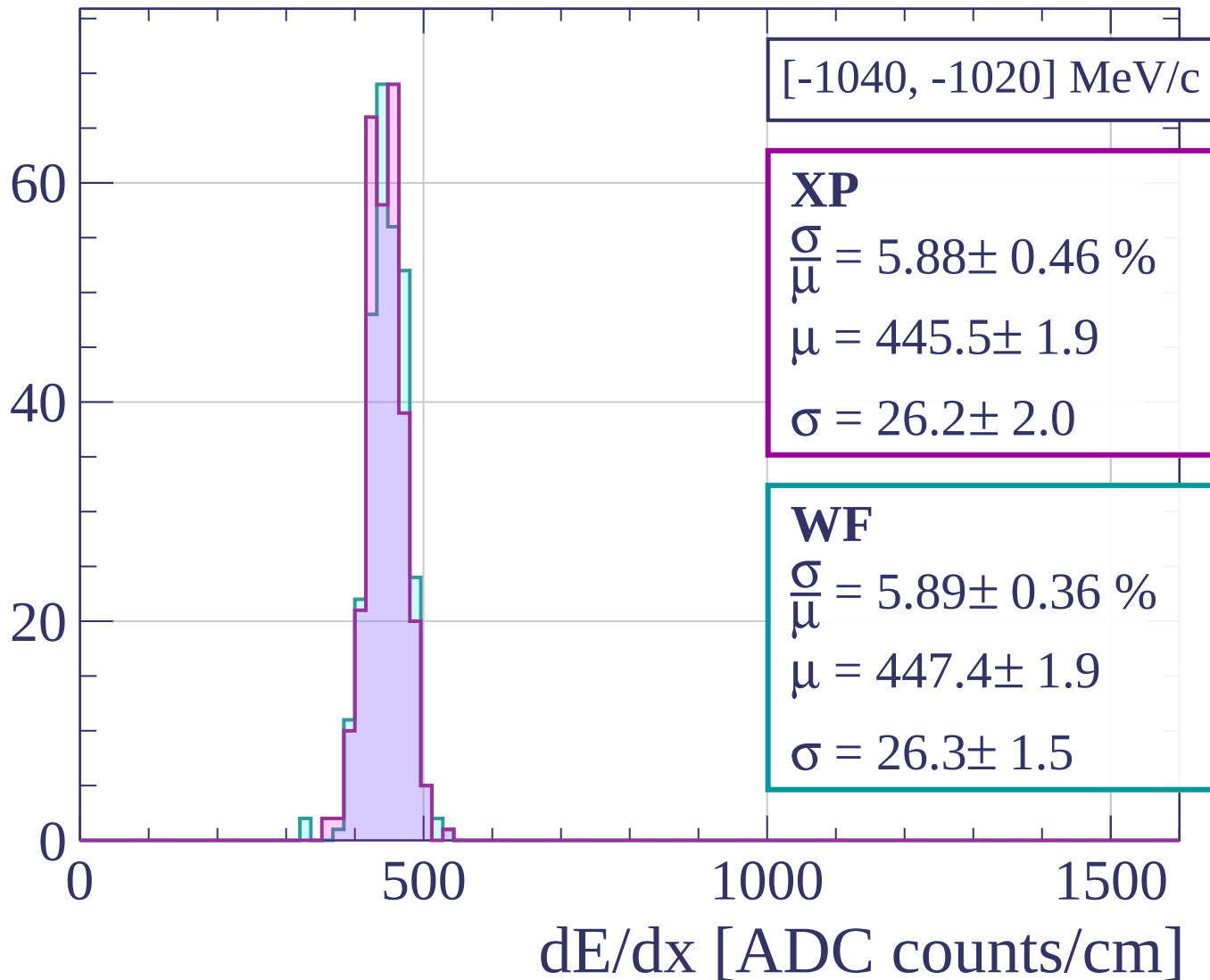
Count



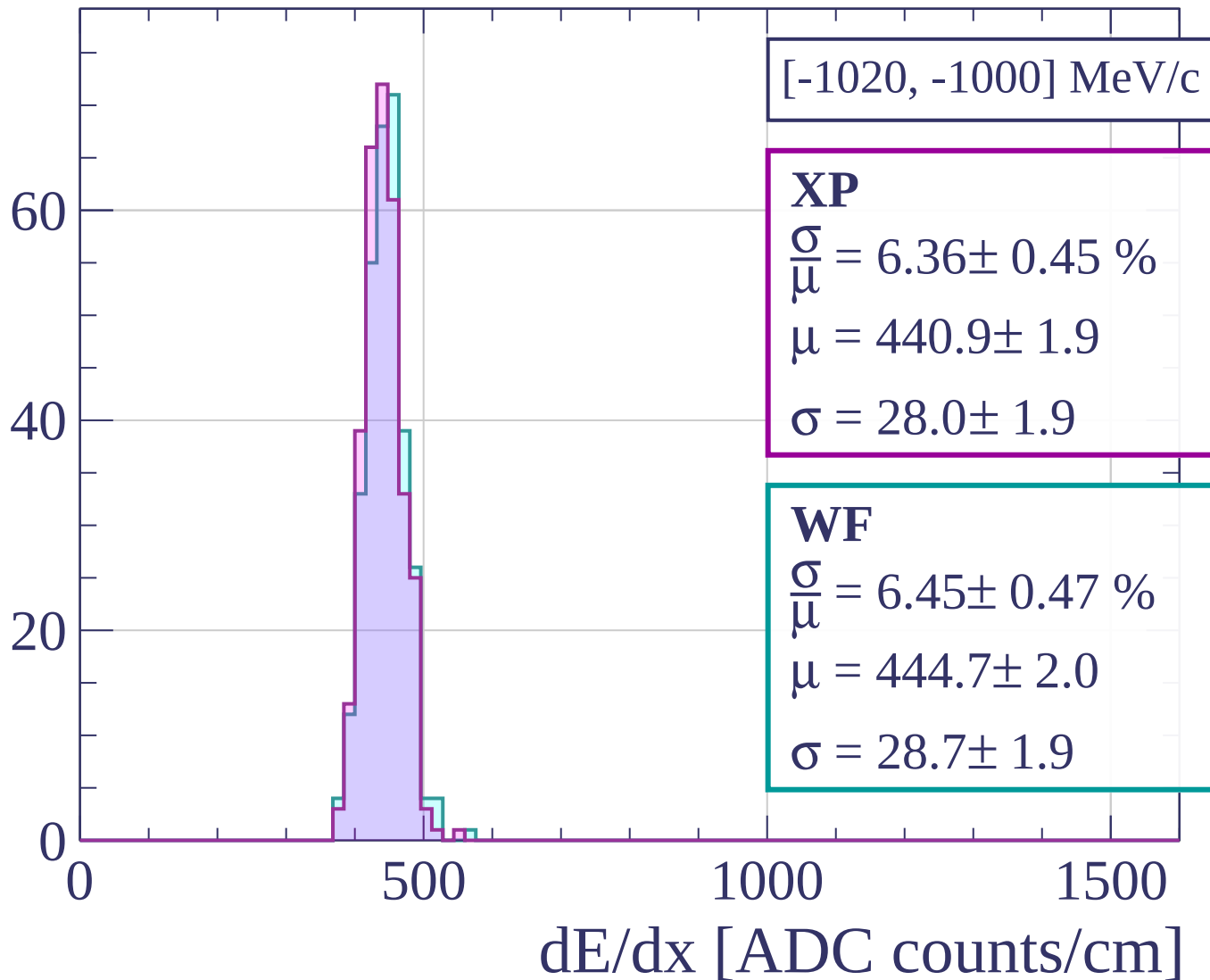
Count



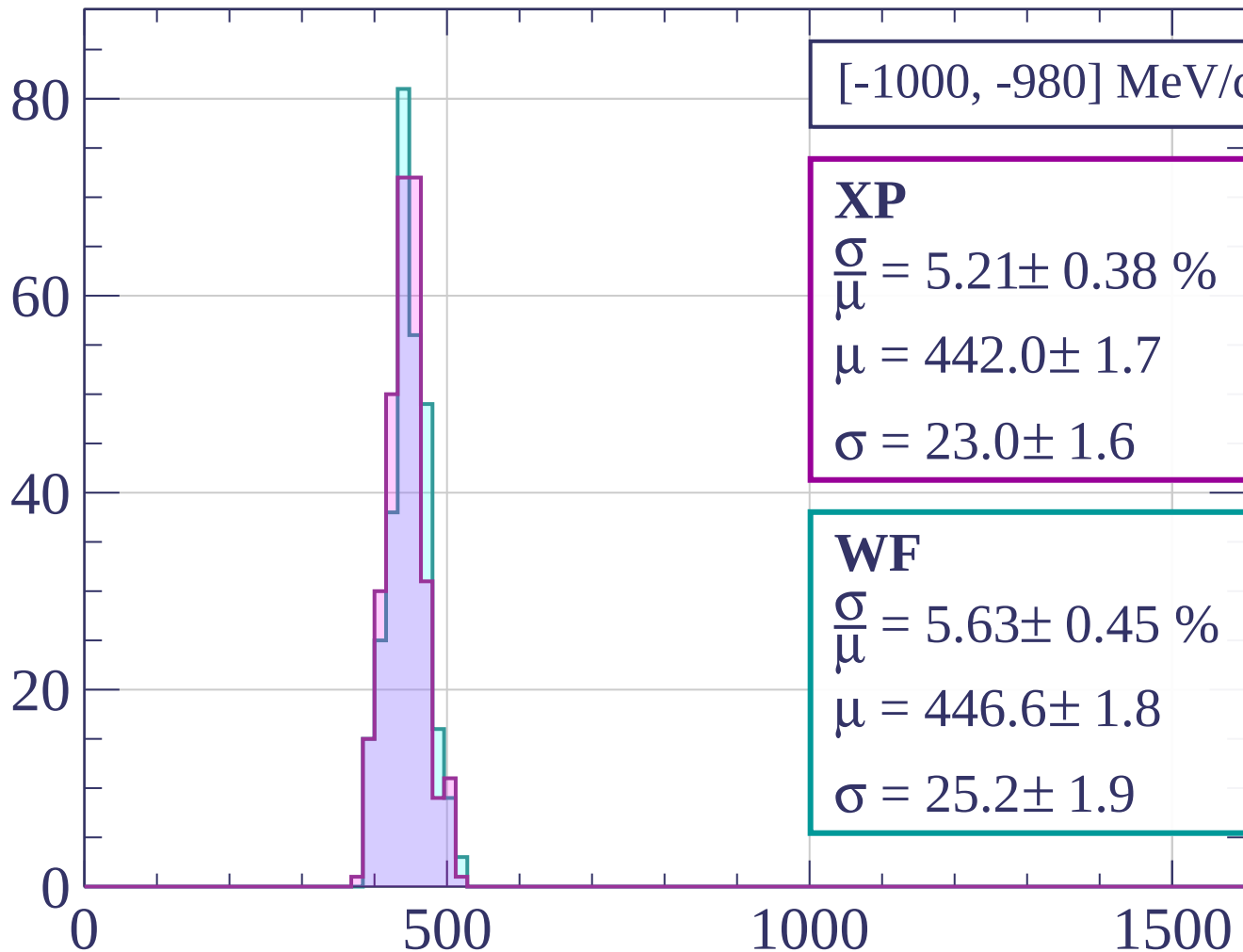
Count



Count

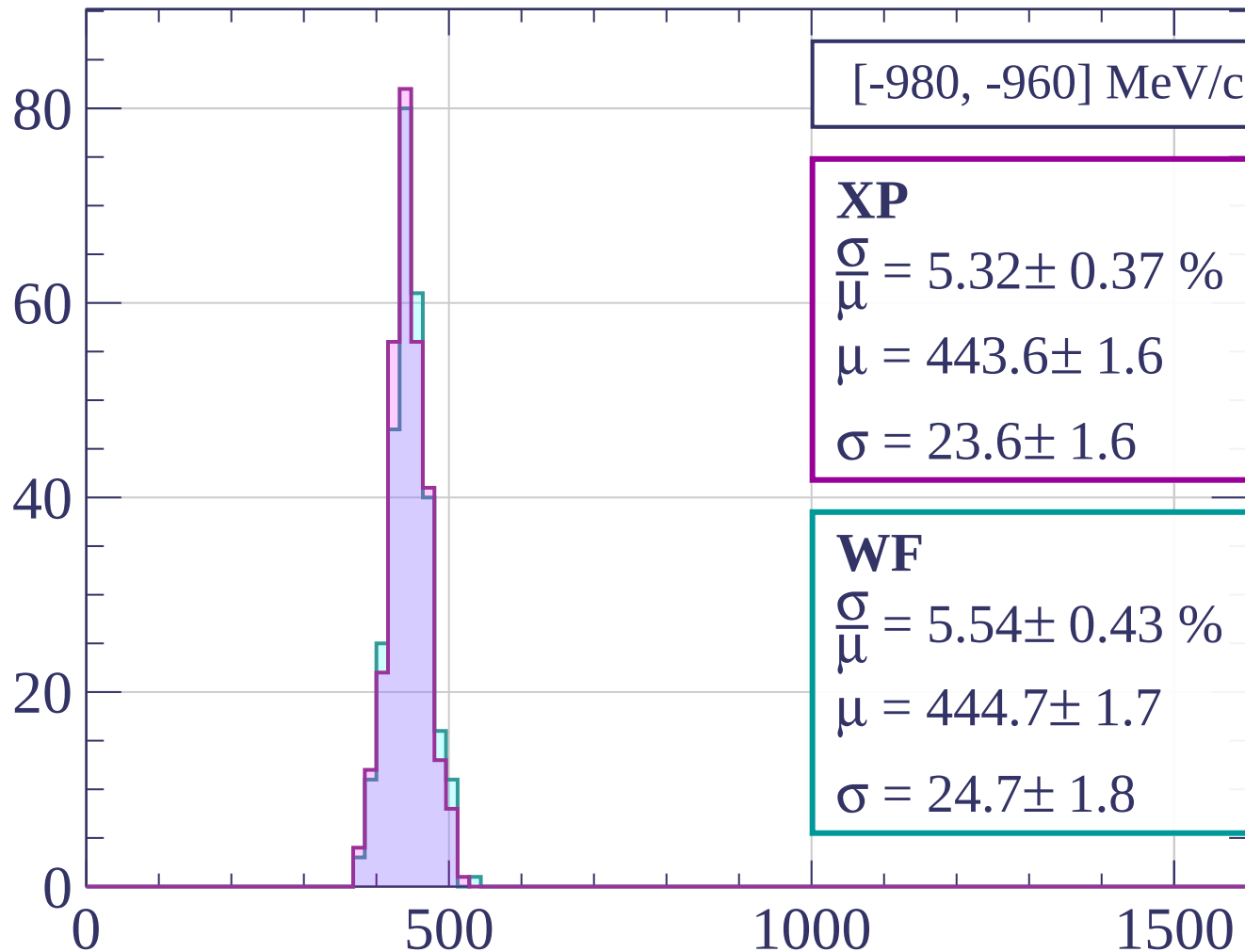


Count



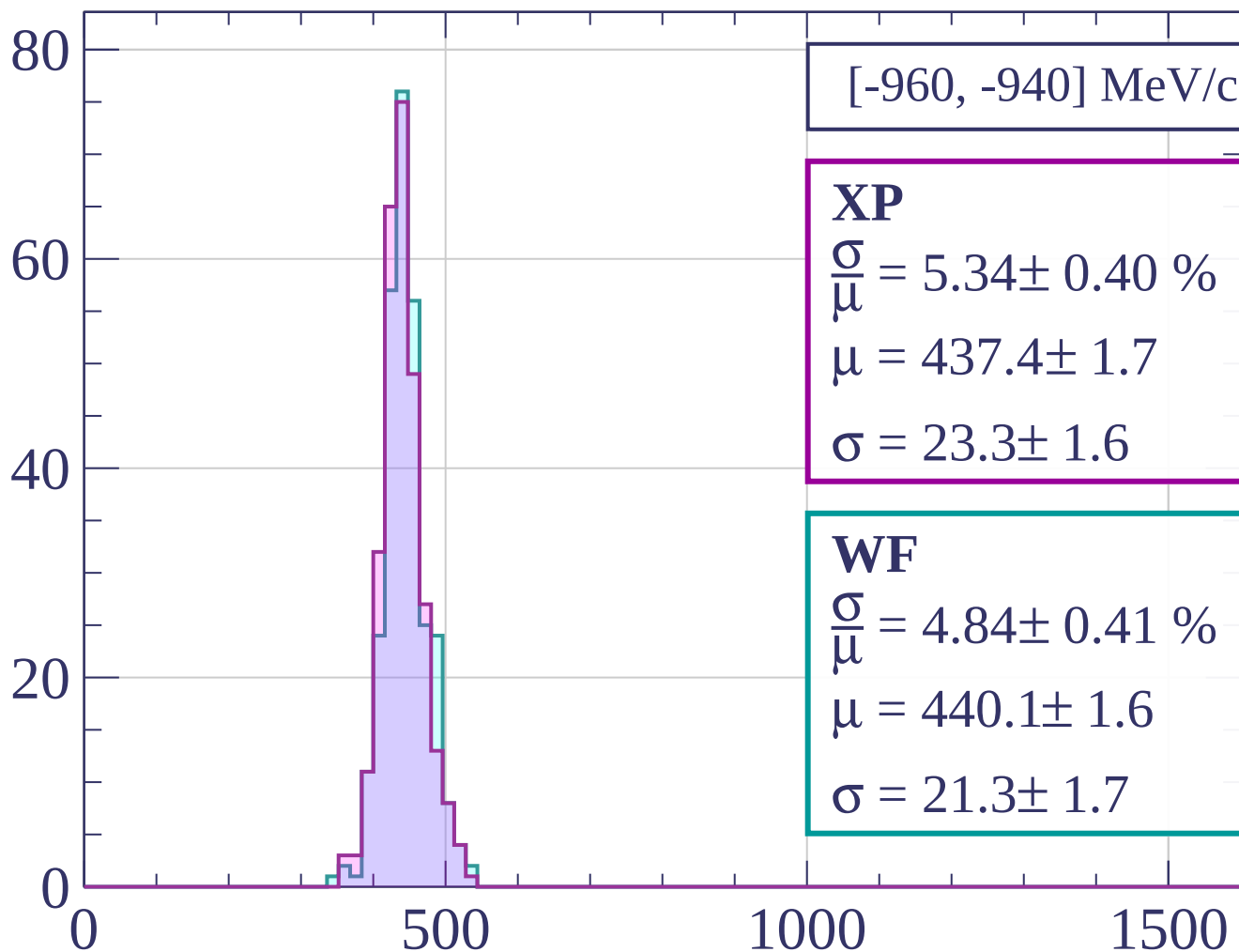
dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count



[-960, -940] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.34 \pm 0.40 \%$$

$$\mu = 437.4 \pm 1.7$$

$$\sigma = 23.3 \pm 1.6$$

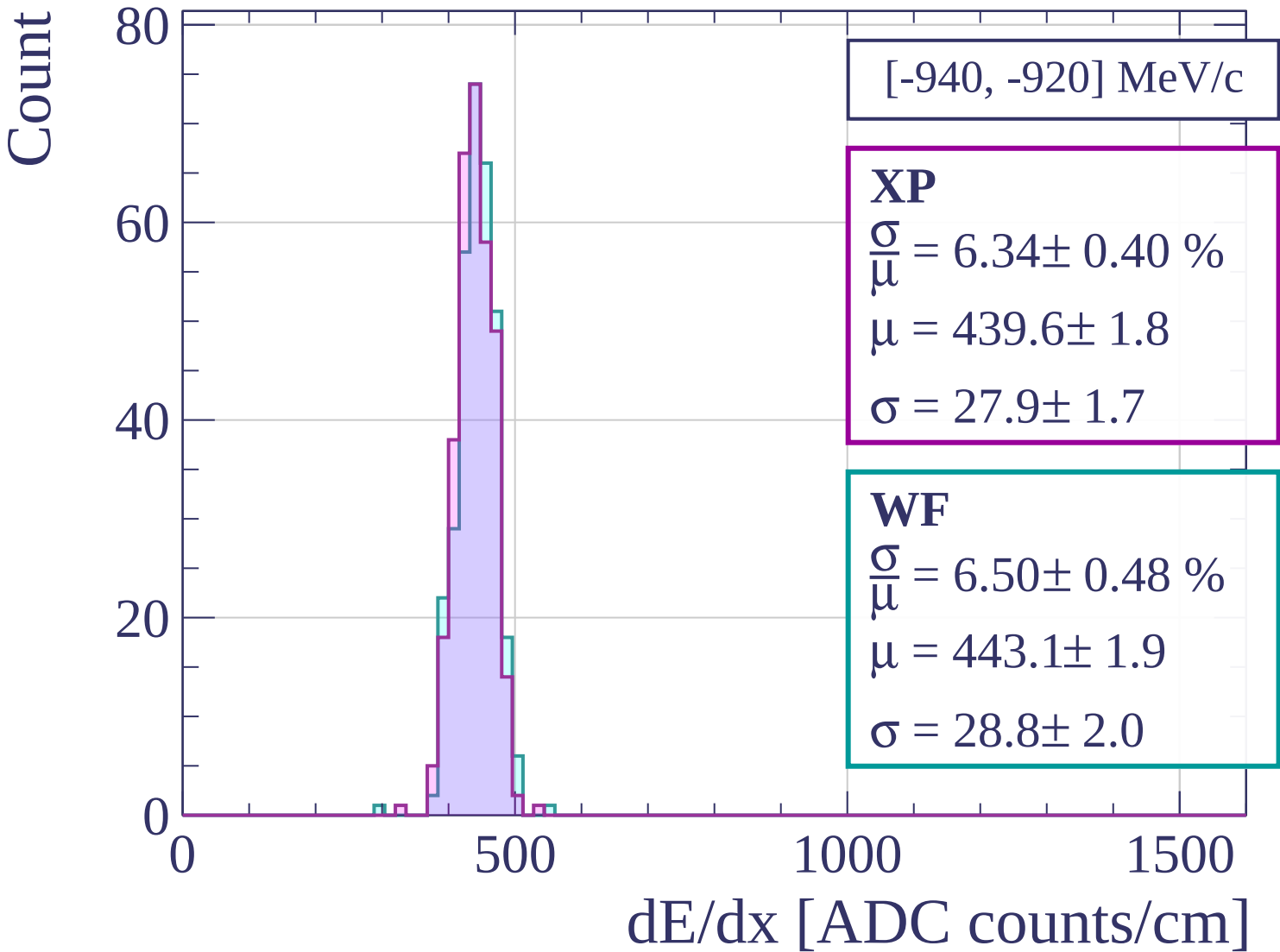
WF

$$\frac{\sigma}{\mu} = 4.84 \pm 0.41 \%$$

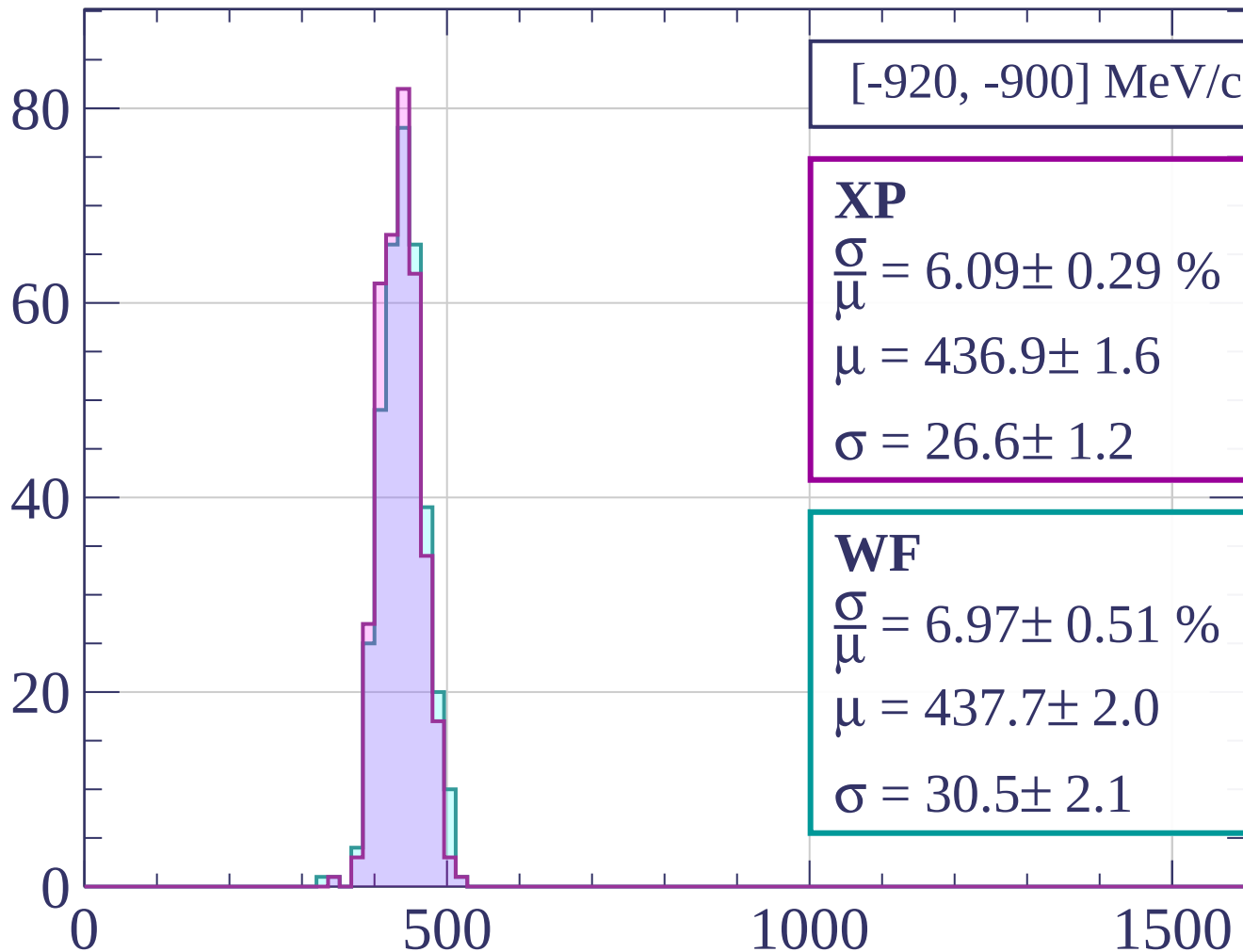
$$\mu = 440.1 \pm 1.6$$

$$\sigma = 21.3 \pm 1.7$$

dE/dx [ADC counts/cm]

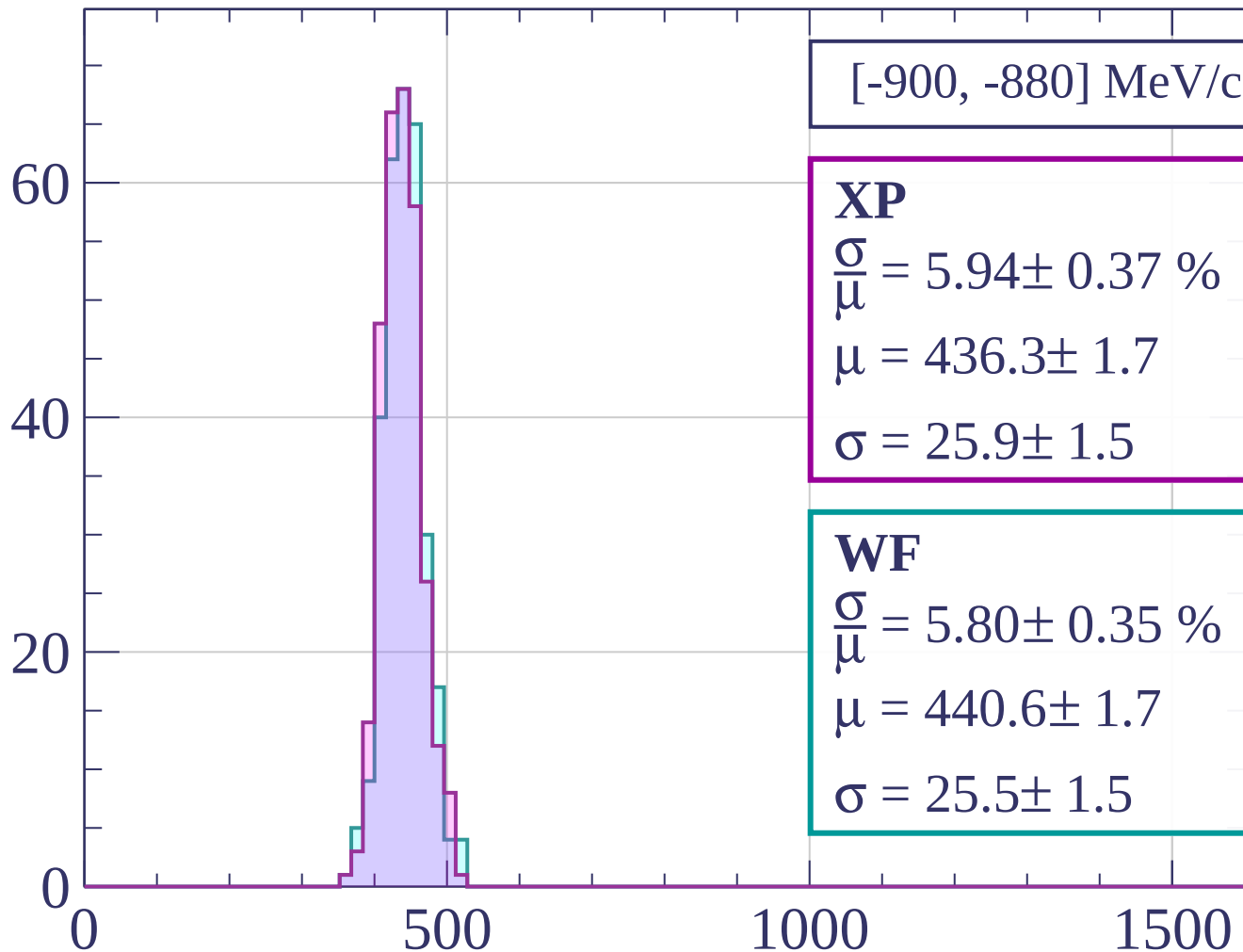


Count



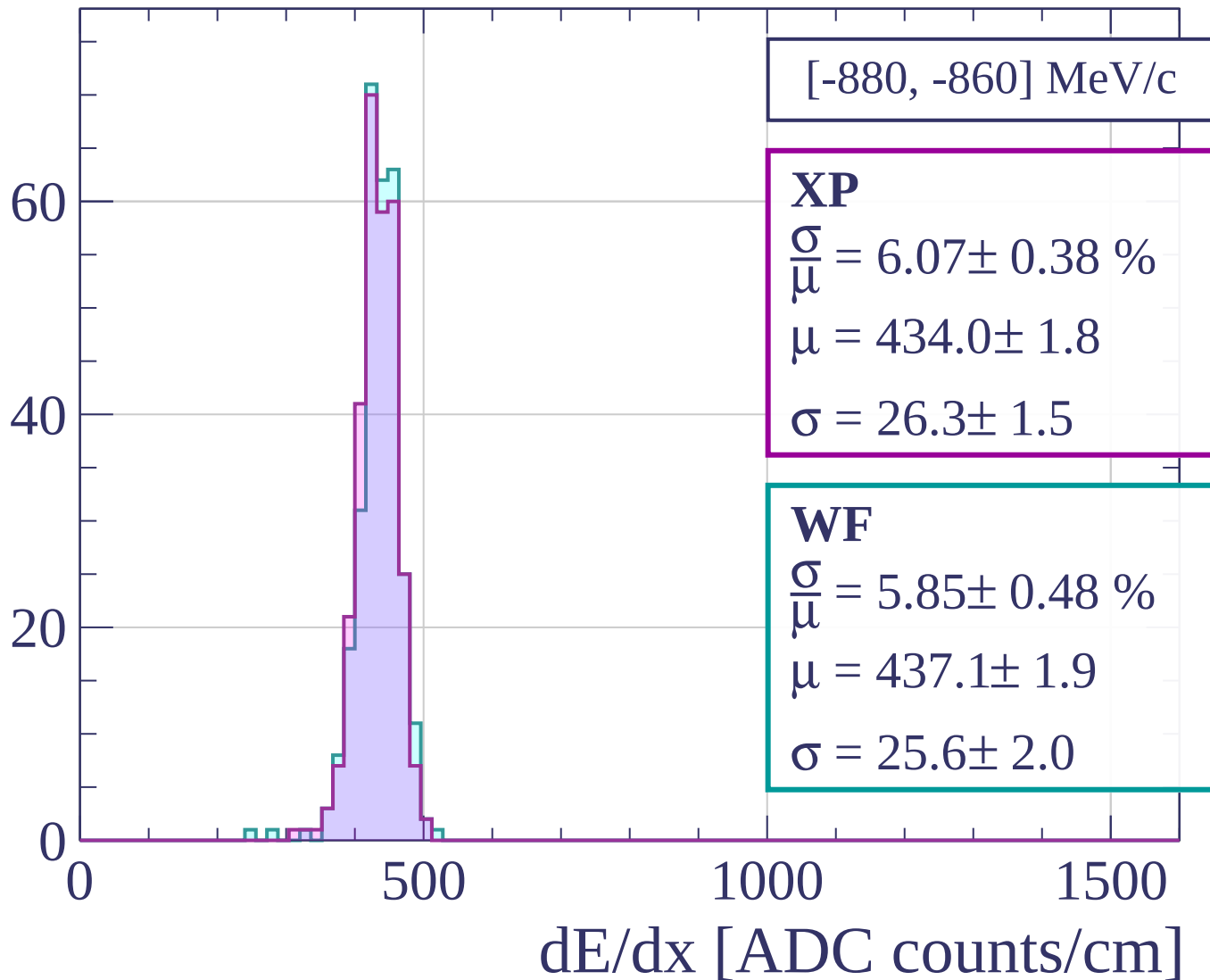
dE/dx [ADC counts/cm]

Count

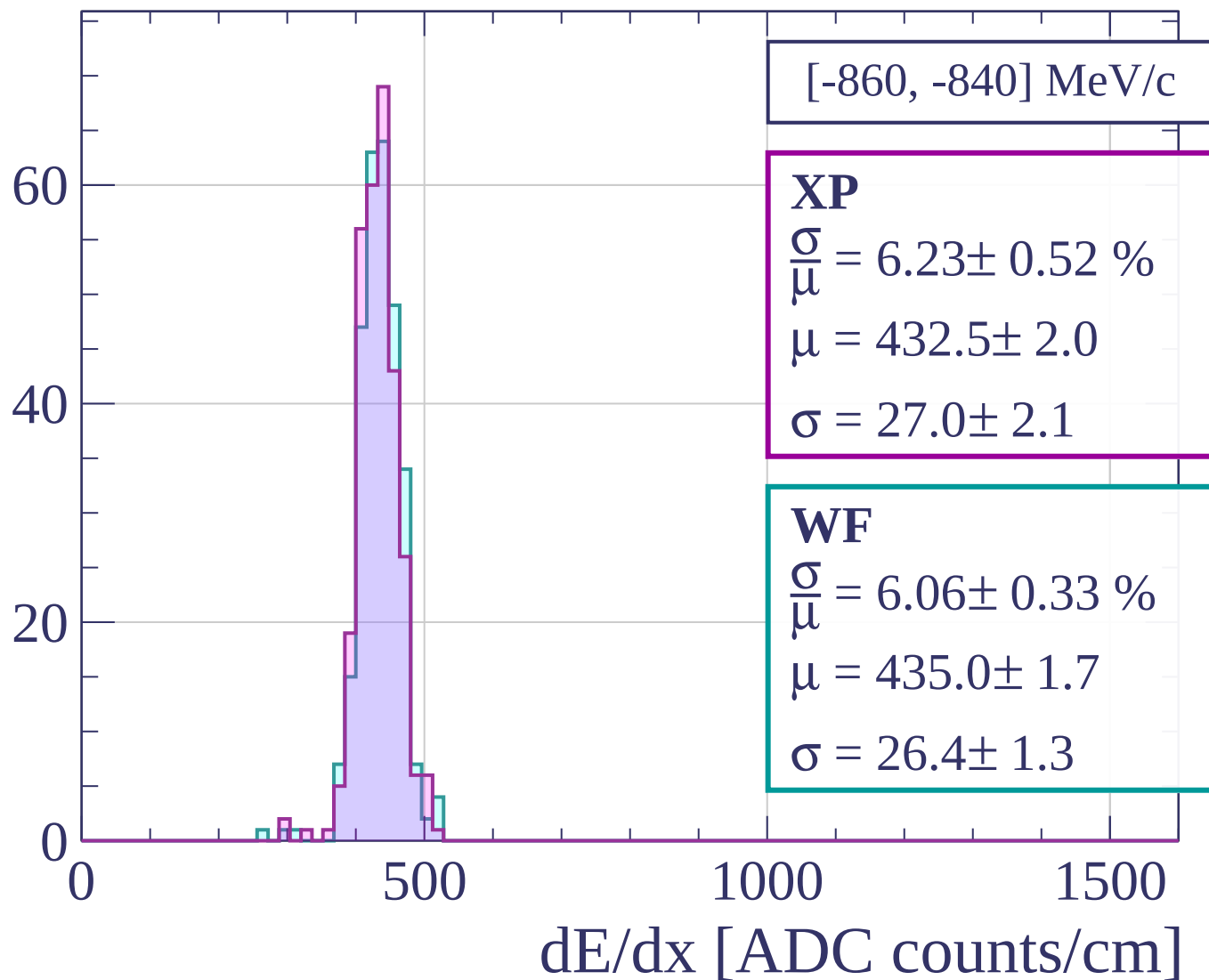


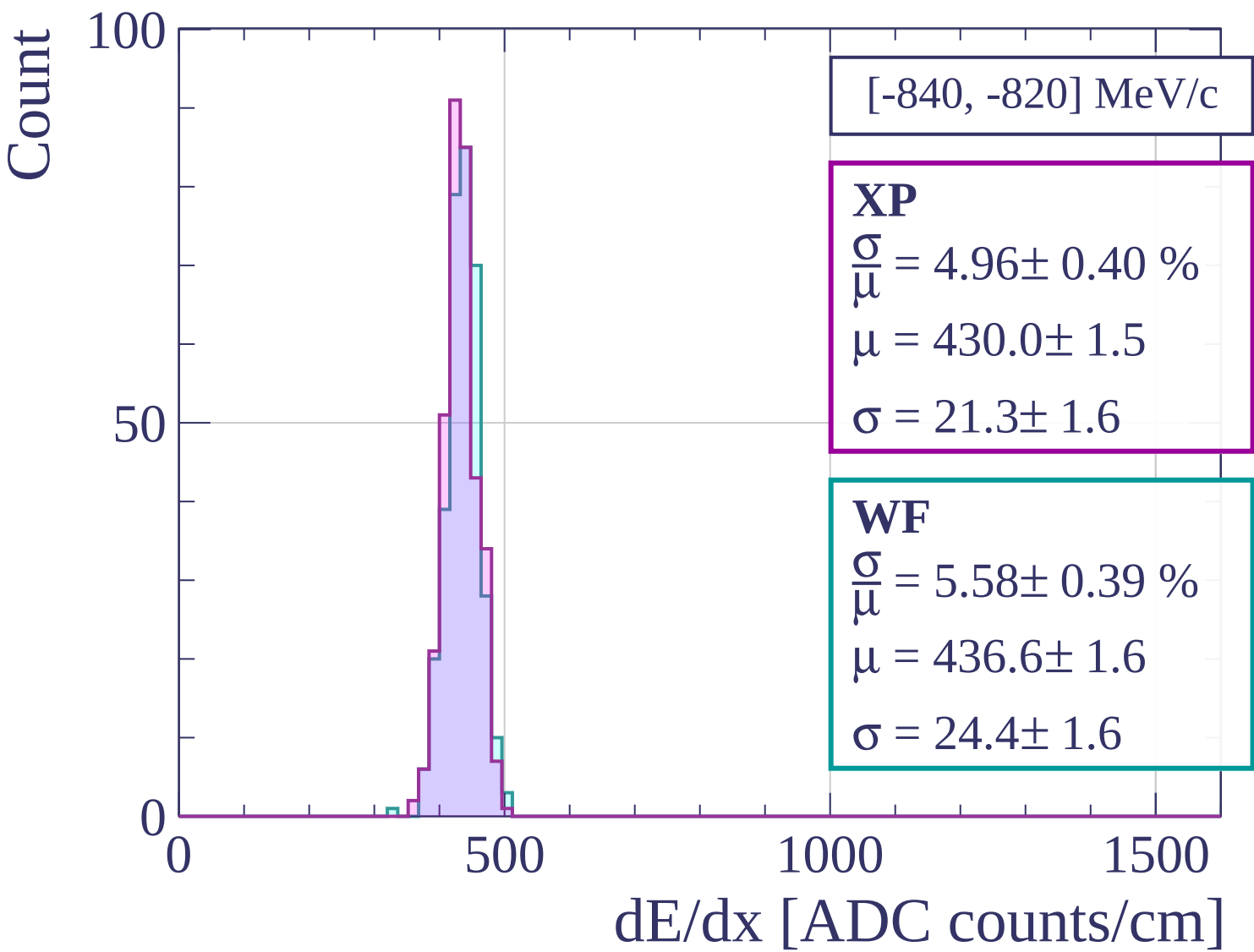
dE/dx [ADC counts/cm]

Count

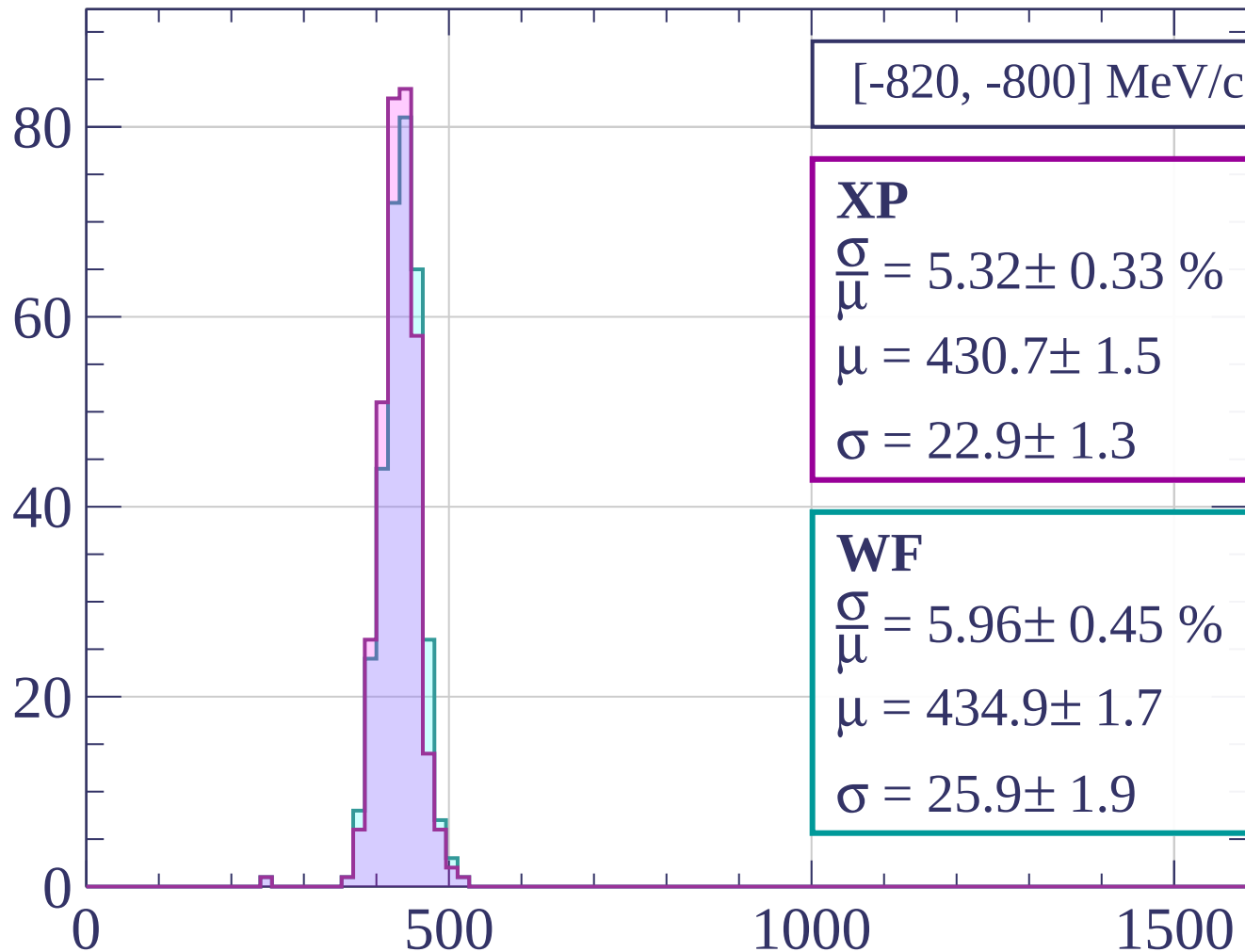


Count



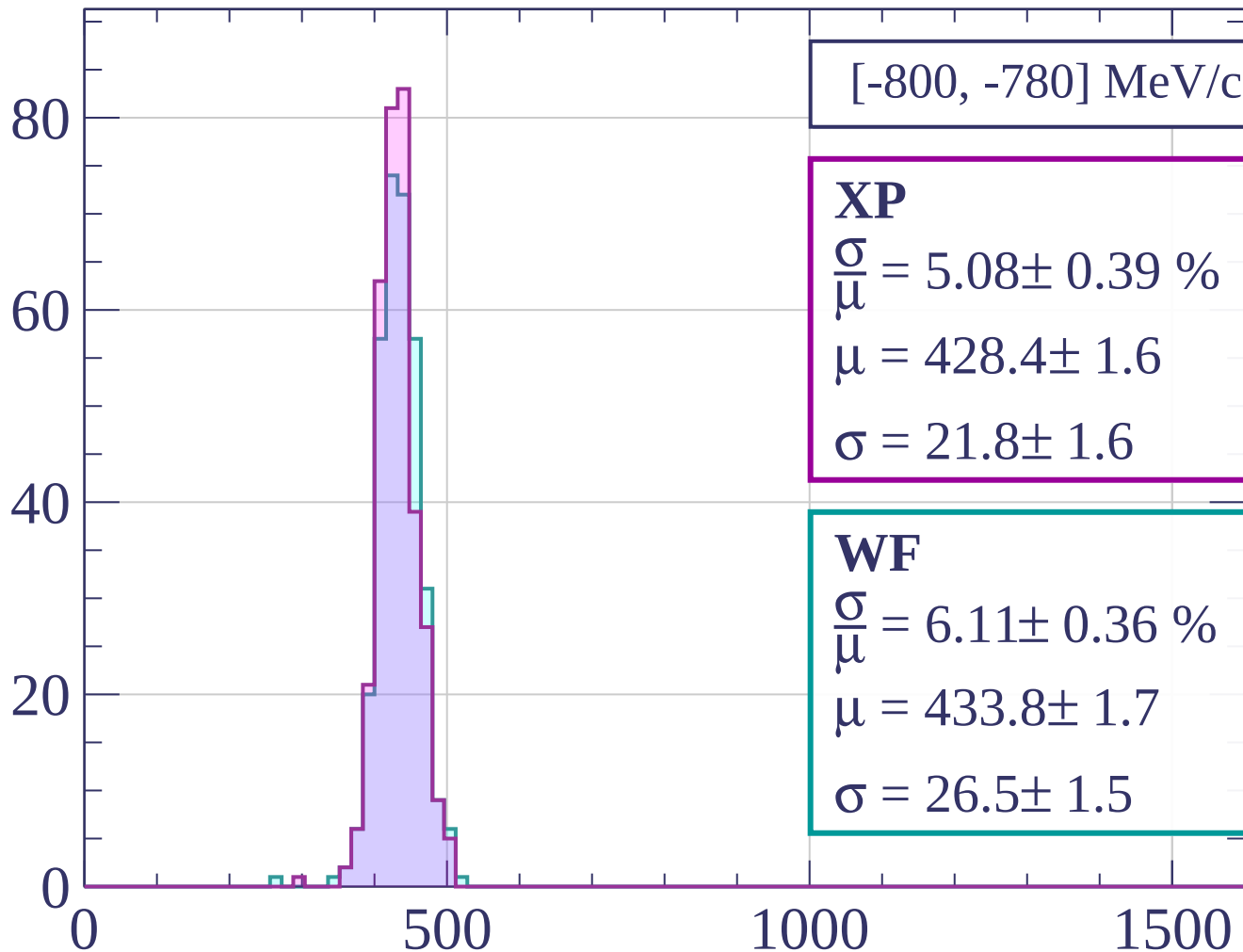


Count



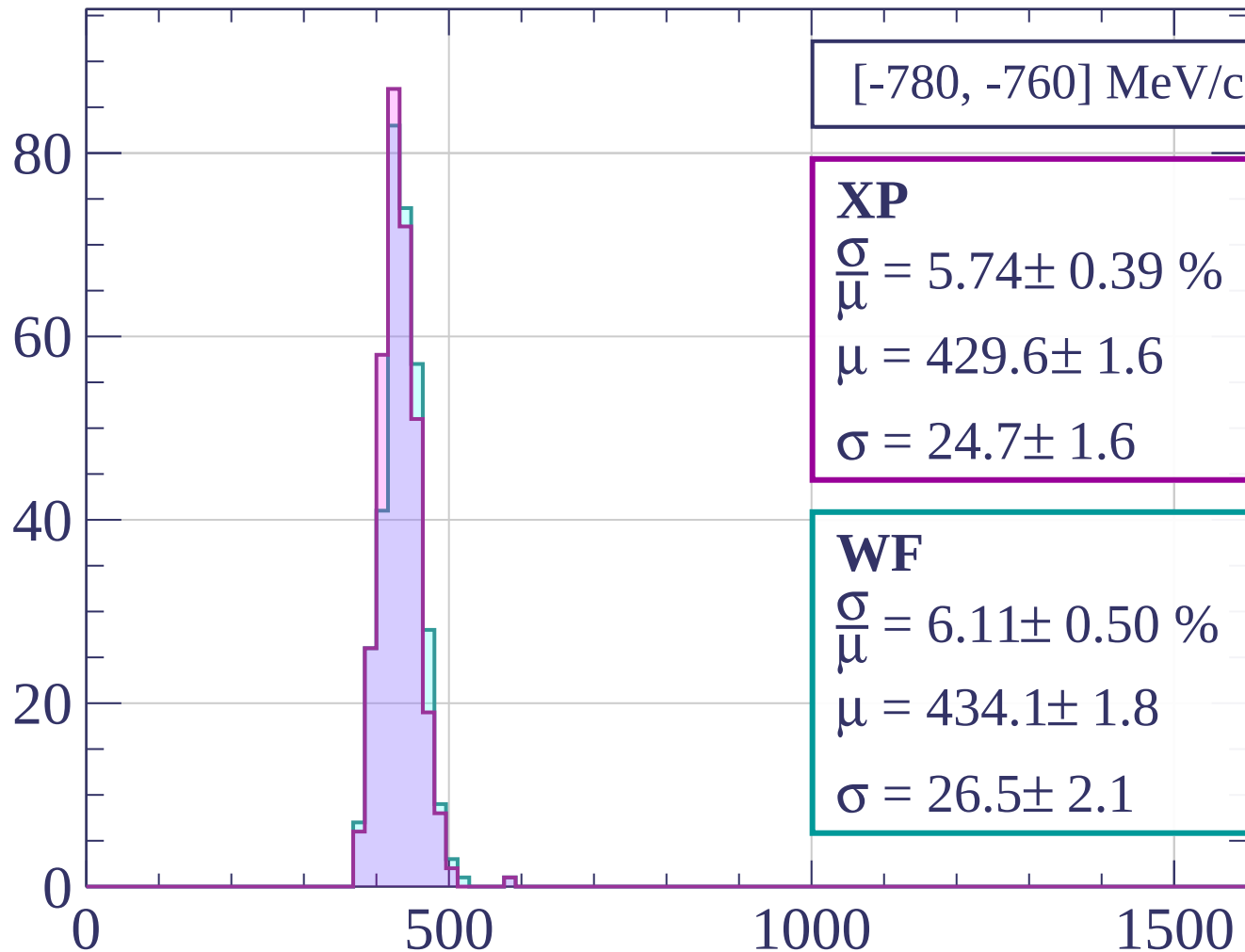
dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count



$[-780, -760]$ MeV/c

XP

$$\frac{\sigma}{\mu} = 5.74 \pm 0.39 \%$$

$$\mu = 429.6 \pm 1.6$$

$$\sigma = 24.7 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 6.11 \pm 0.50 \%$$

$$\mu = 434.1 \pm 1.8$$

$$\sigma = 26.5 \pm 2.1$$

dE/dx [ADC counts/cm]

Count

[-760, -740] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.79 \pm 0.41 \%$$

$$\mu = 428.1 \pm 1.8$$

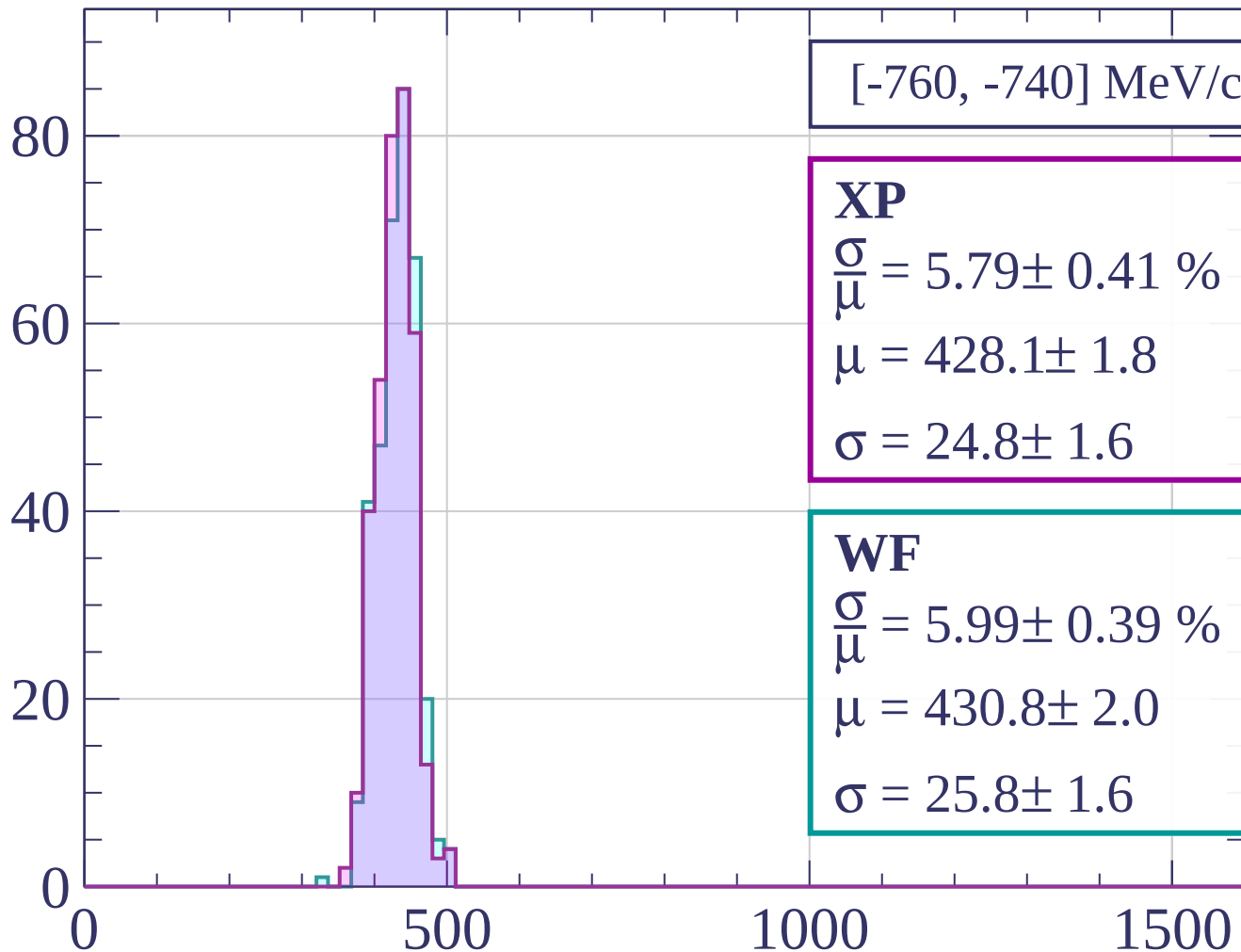
$$\sigma = 24.8 \pm 1.6$$

WF

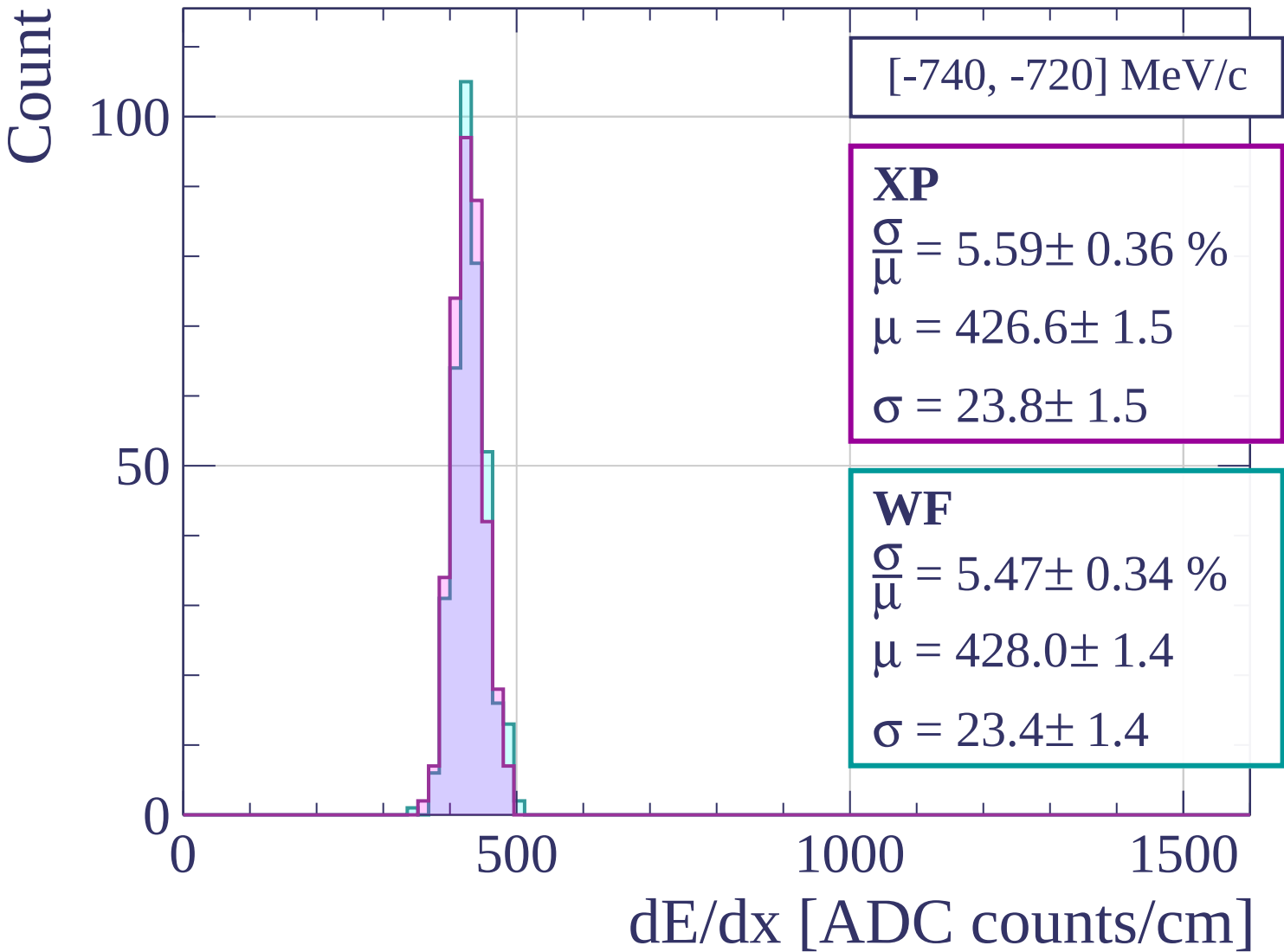
$$\frac{\sigma}{\mu} = 5.99 \pm 0.39 \%$$

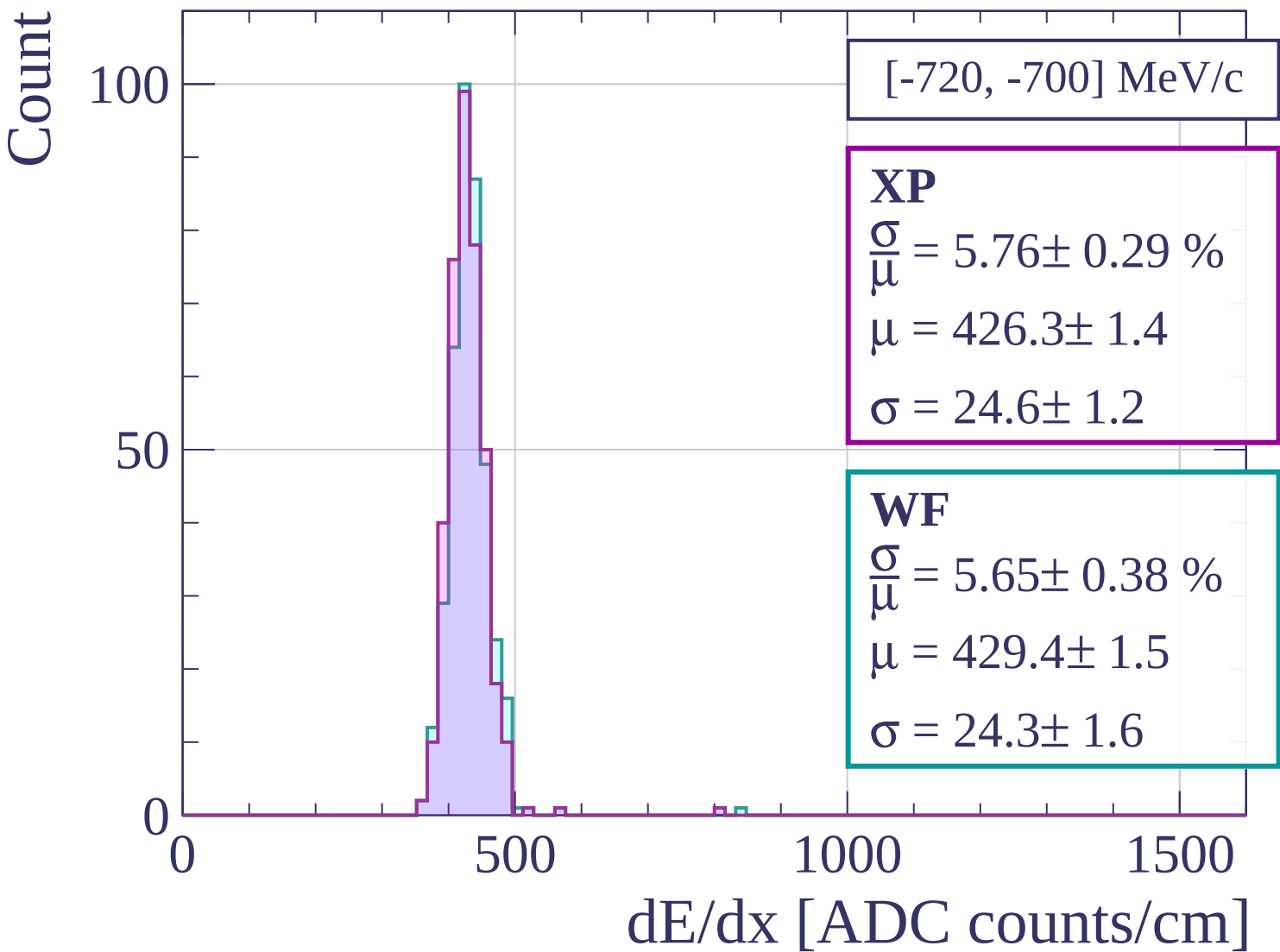
$$\mu = 430.8 \pm 2.0$$

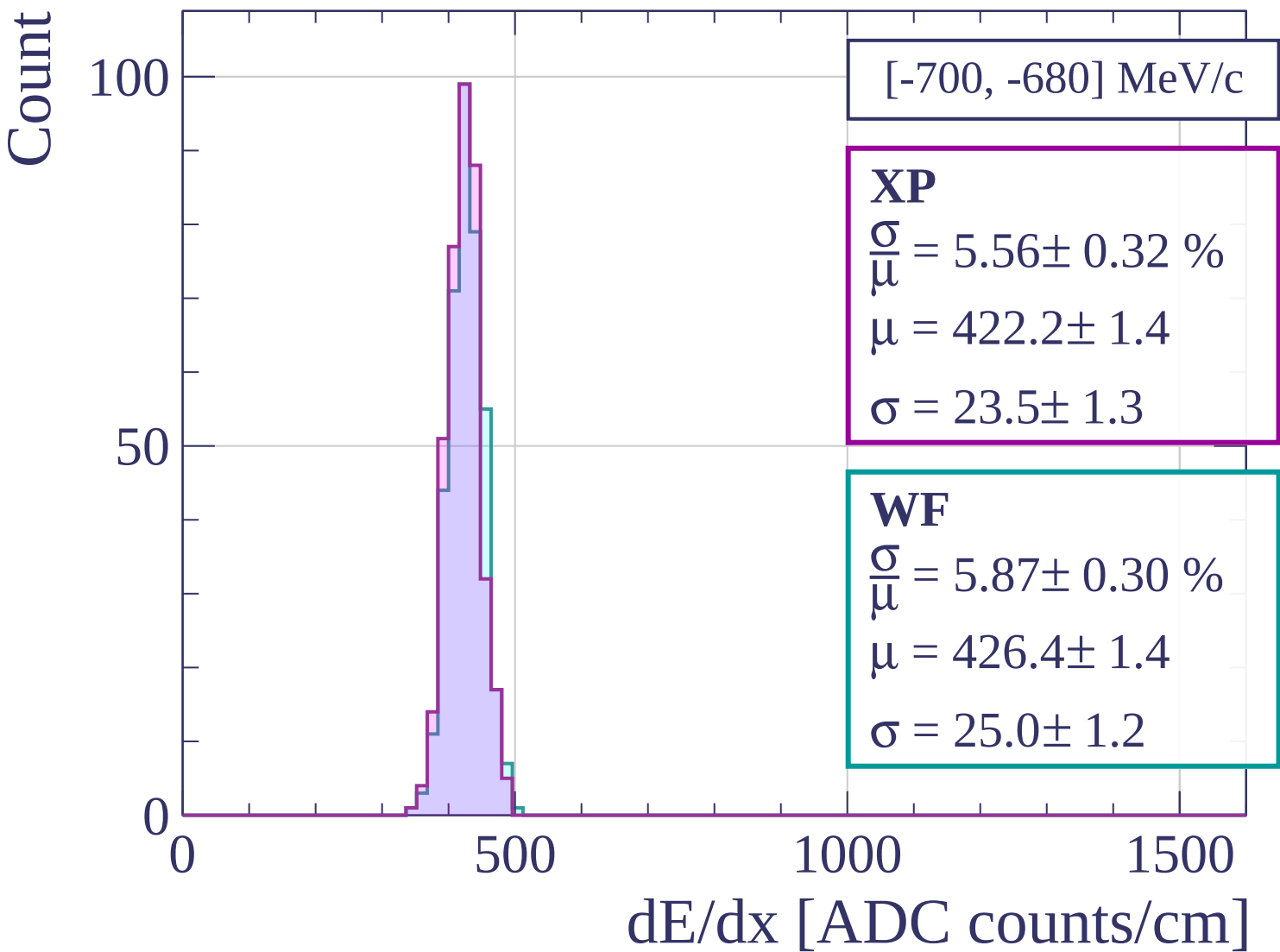
$$\sigma = 25.8 \pm 1.6$$



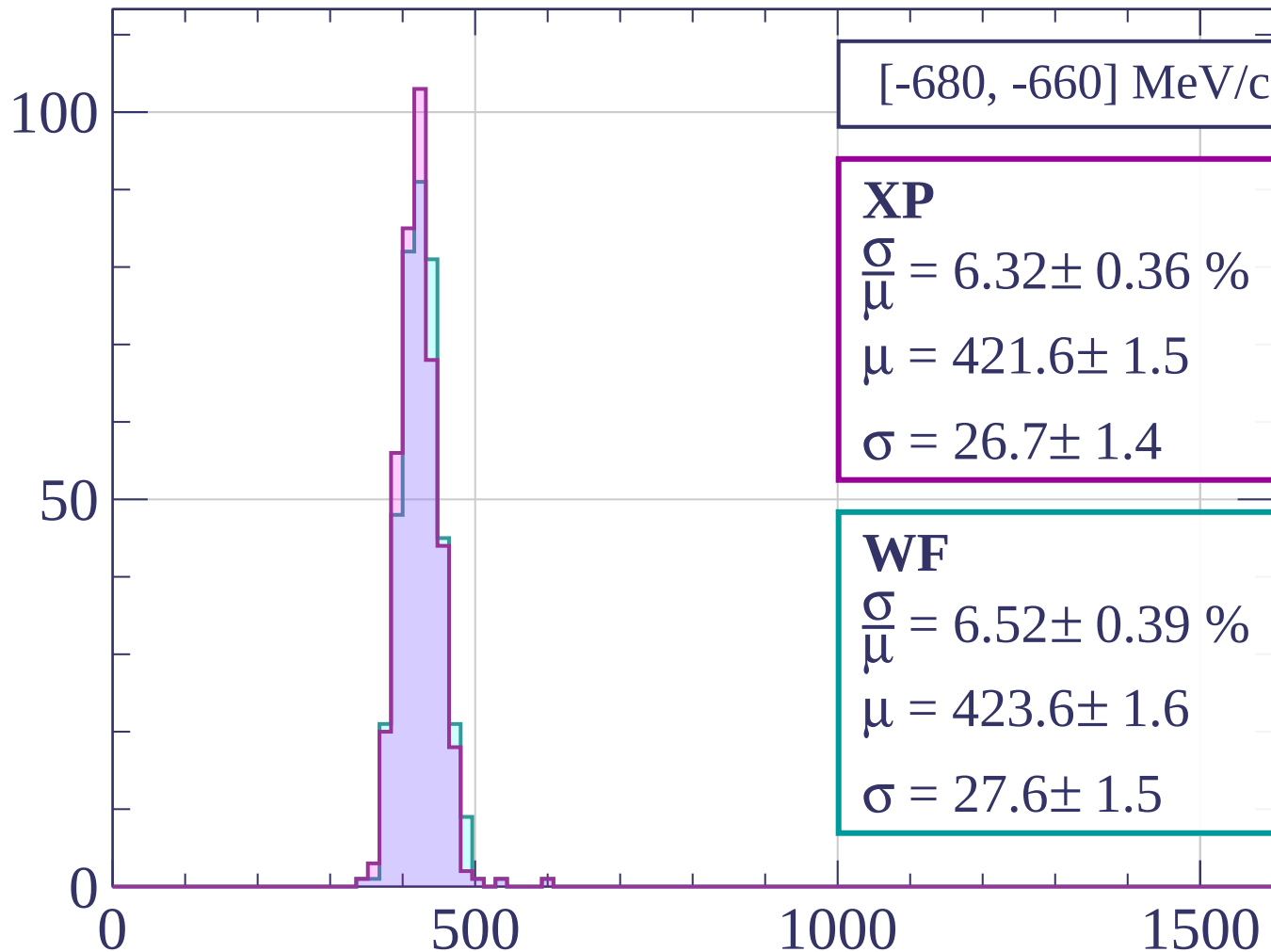
dE/dx [ADC counts/cm]







Count



[-680, -660] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.32 \pm 0.36 \%$$

$$\mu = 421.6 \pm 1.5$$

$$\sigma = 26.7 \pm 1.4$$

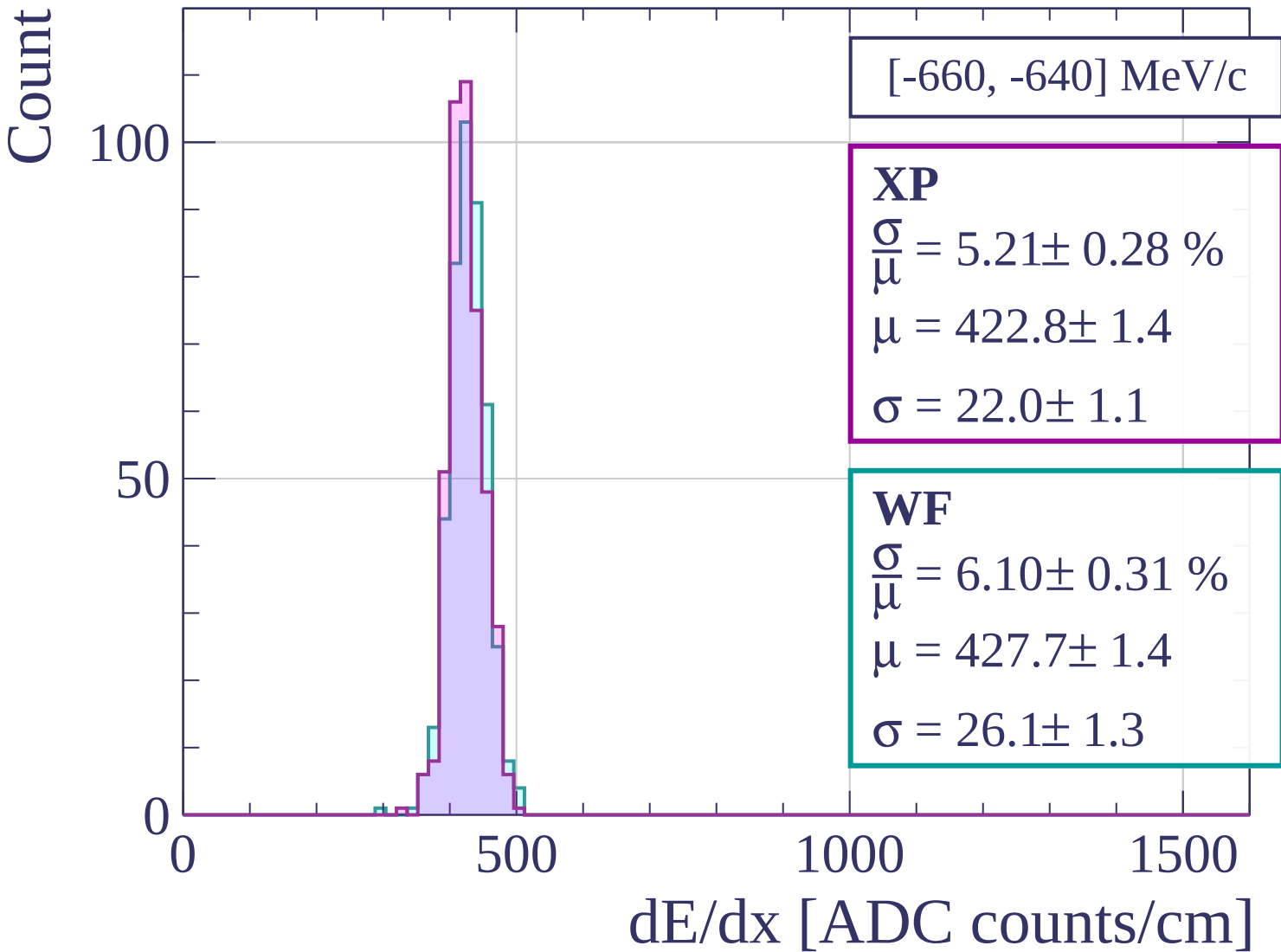
WF

$$\frac{\sigma}{\mu} = 6.52 \pm 0.39 \%$$

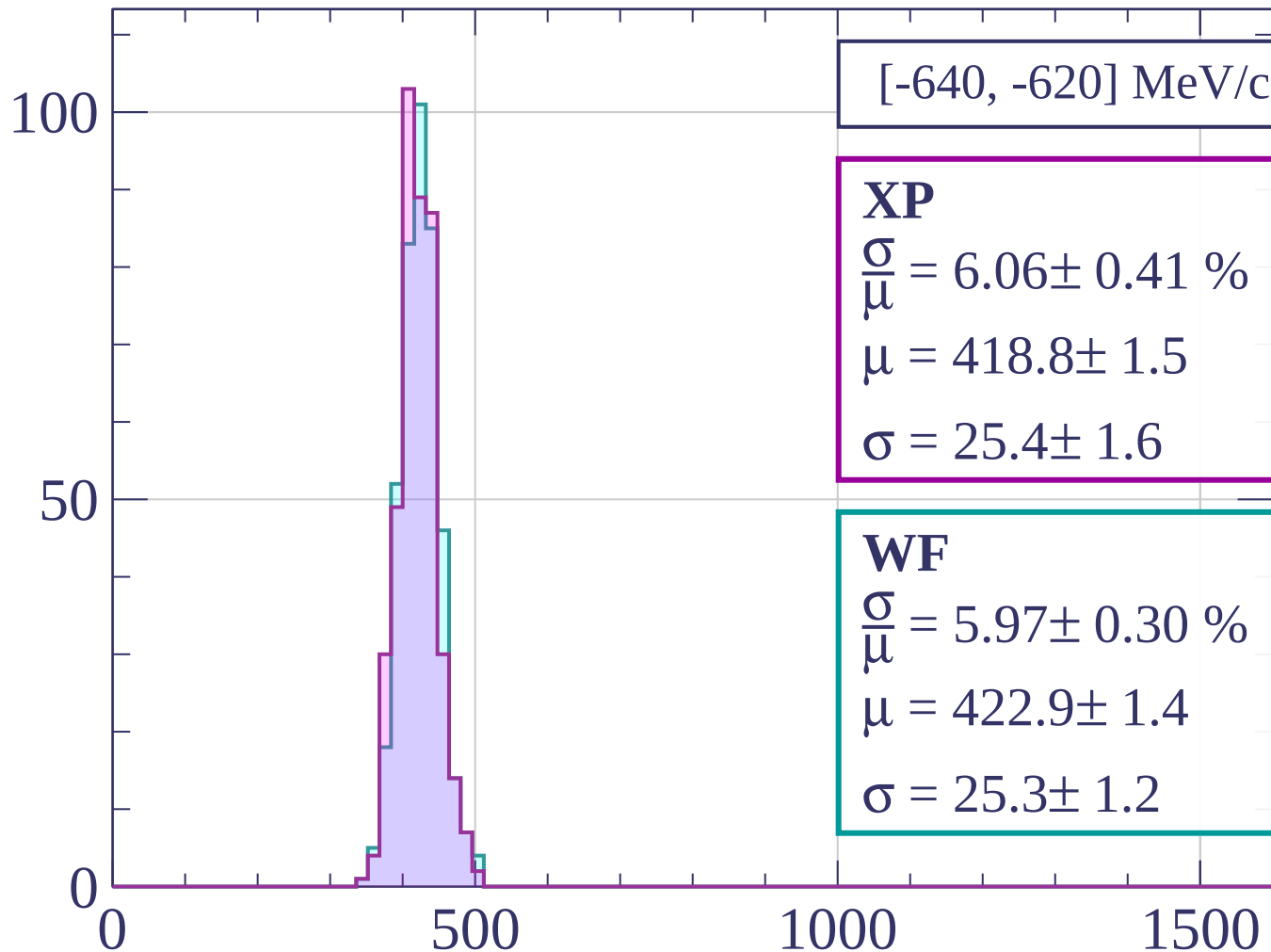
$$\mu = 423.6 \pm 1.6$$

$$\sigma = 27.6 \pm 1.5$$

dE/dx [ADC counts/cm]



Count



[-640, -620] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.06 \pm 0.41 \%$$

$$\mu = 418.8 \pm 1.5$$

$$\sigma = 25.4 \pm 1.6$$

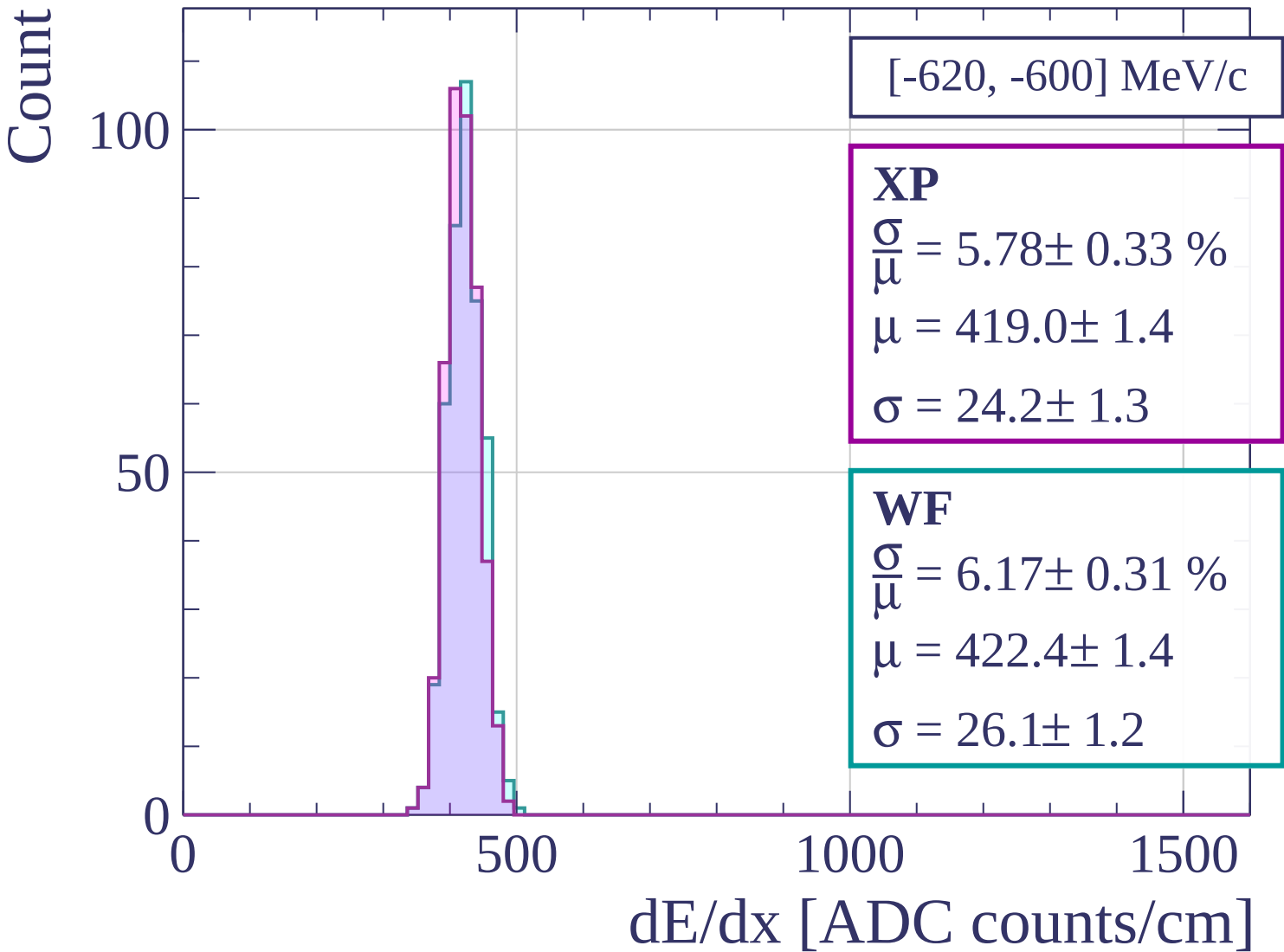
WF

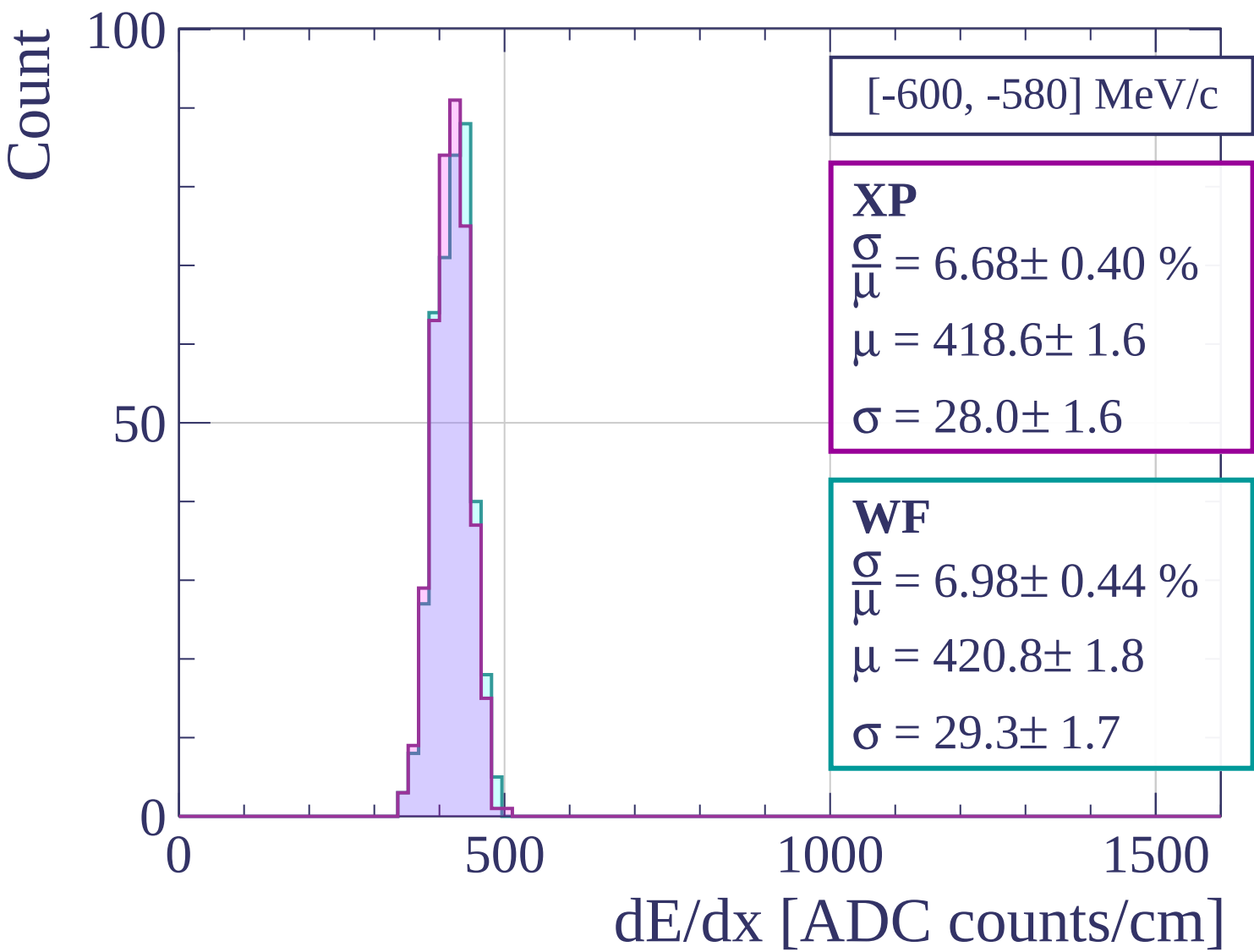
$$\frac{\sigma}{\mu} = 5.97 \pm 0.30 \%$$

$$\mu = 422.9 \pm 1.4$$

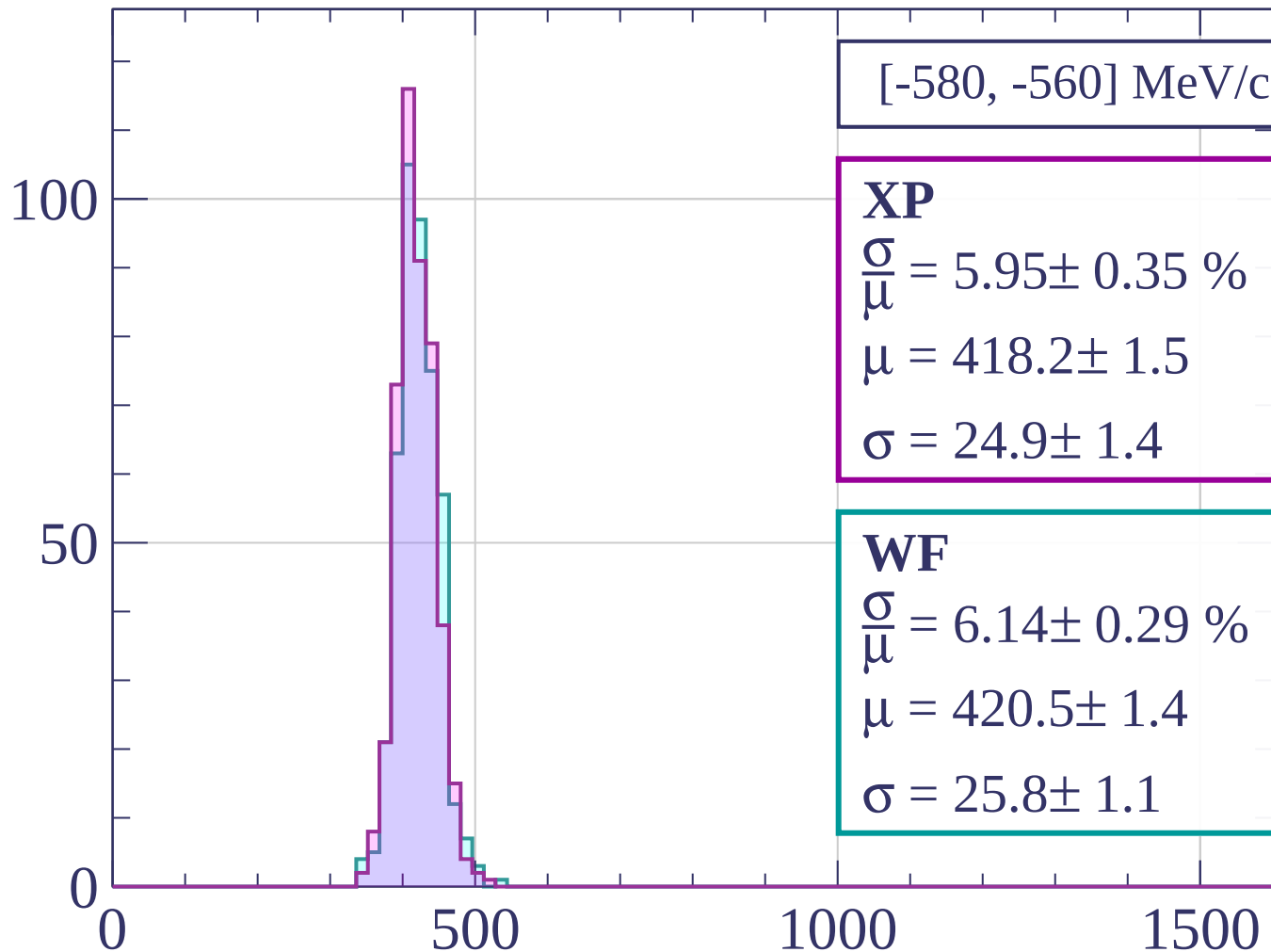
$$\sigma = 25.3 \pm 1.2$$

dE/dx [ADC counts/cm]





Count



[-580, -560] MeV/c

XP

$$\frac{\sigma}{\bar{\mu}} = 5.95 \pm 0.35 \%$$

$$\mu = 418.2 \pm 1.5$$

$$\sigma = 24.9 \pm 1.4$$

WF

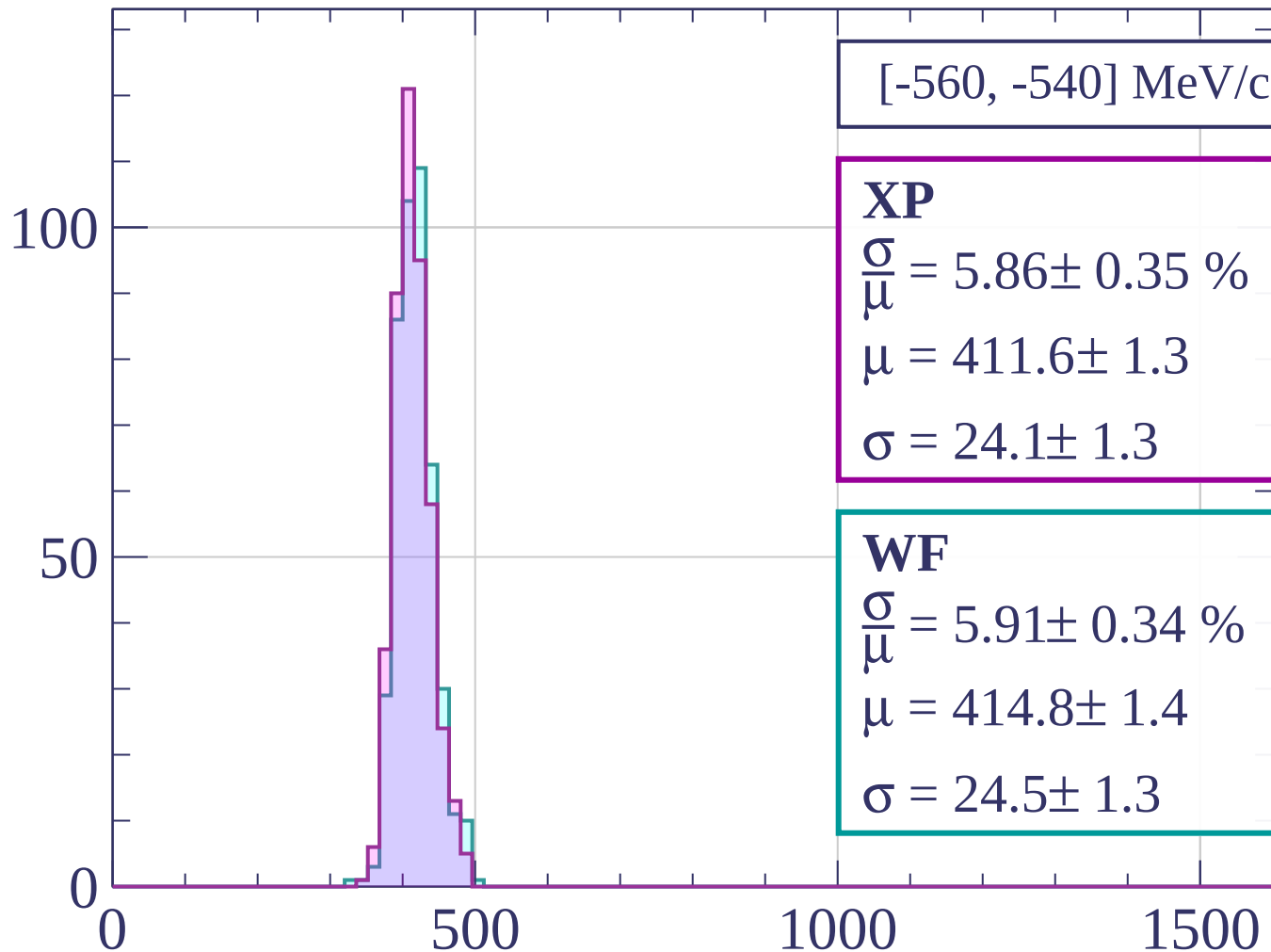
$$\frac{\sigma}{\bar{\mu}} = 6.14 \pm 0.29 \%$$

$$\mu = 420.5 \pm 1.4$$

$$\sigma = 25.8 \pm 1.1$$

dE/dx [ADC counts/cm]

Count



[-560, -540] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.86 \pm 0.35 \%$$

$$\mu = 411.6 \pm 1.3$$

$$\sigma = 24.1 \pm 1.3$$

WF

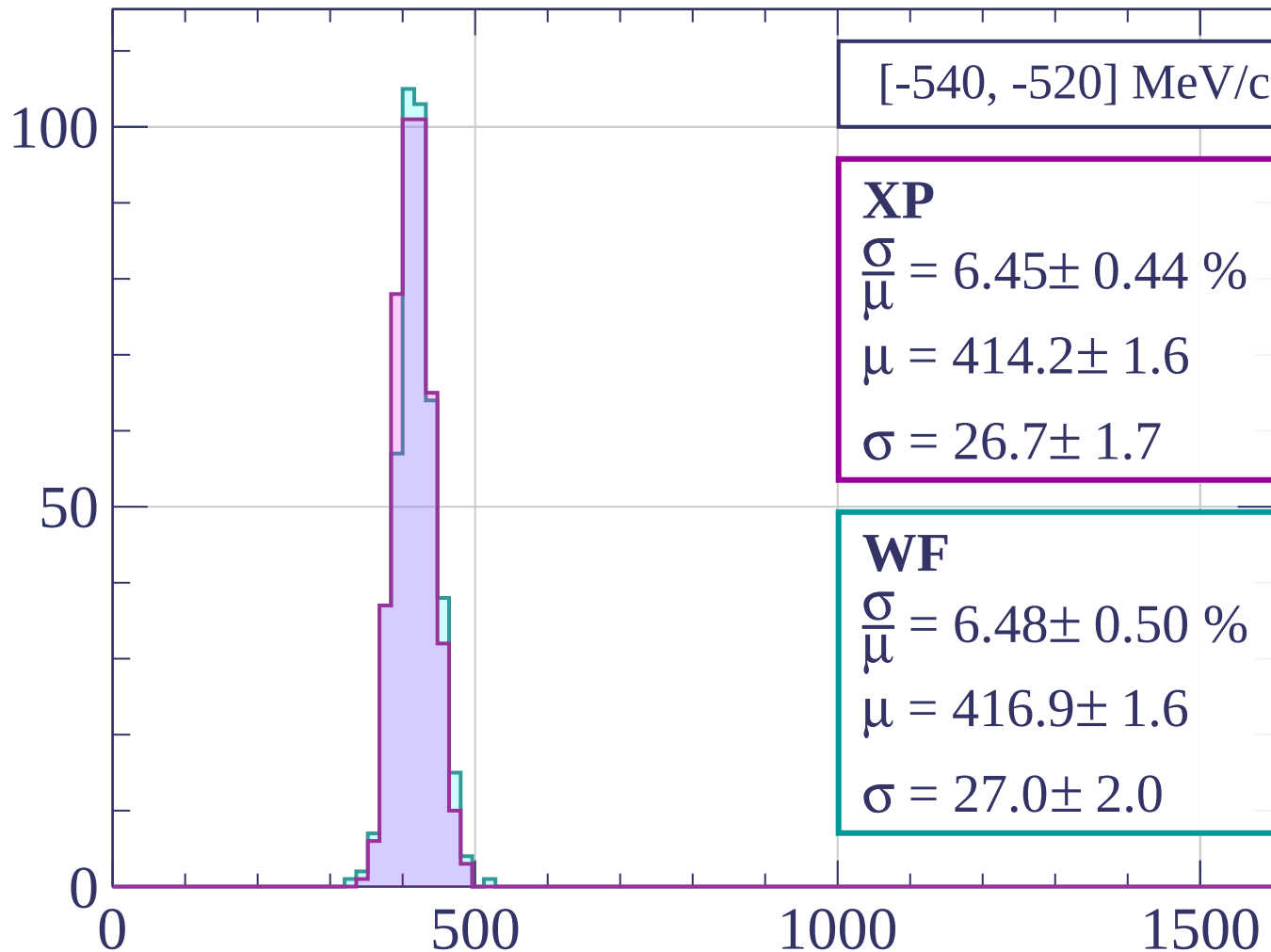
$$\frac{\sigma}{\mu} = 5.91 \pm 0.34 \%$$

$$\mu = 414.8 \pm 1.4$$

$$\sigma = 24.5 \pm 1.3$$

dE/dx [ADC counts/cm]

Count



$[-540, -520]$ MeV/c

XP

$$\frac{\sigma}{\mu} = 6.45 \pm 0.44 \%$$

$$\mu = 414.2 \pm 1.6$$

$$\sigma = 26.7 \pm 1.7$$

WF

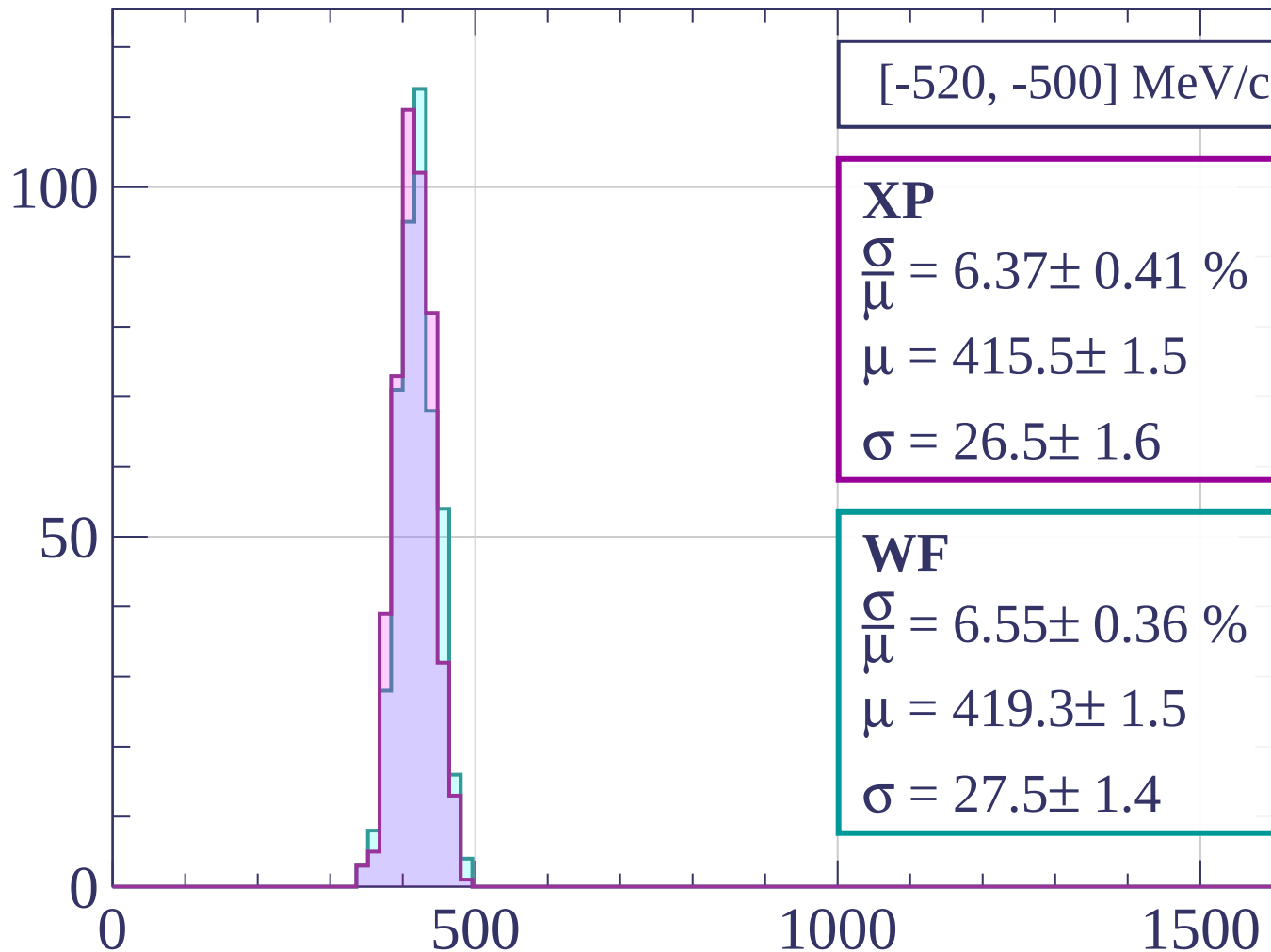
$$\frac{\sigma}{\mu} = 6.48 \pm 0.50 \%$$

$$\mu = 416.9 \pm 1.6$$

$$\sigma = 27.0 \pm 2.0$$

dE/dx [ADC counts/cm]

Count



[-520, -500] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.37 \pm 0.41 \%$$

$$\mu = 415.5 \pm 1.5$$

$$\sigma = 26.5 \pm 1.6$$

WF

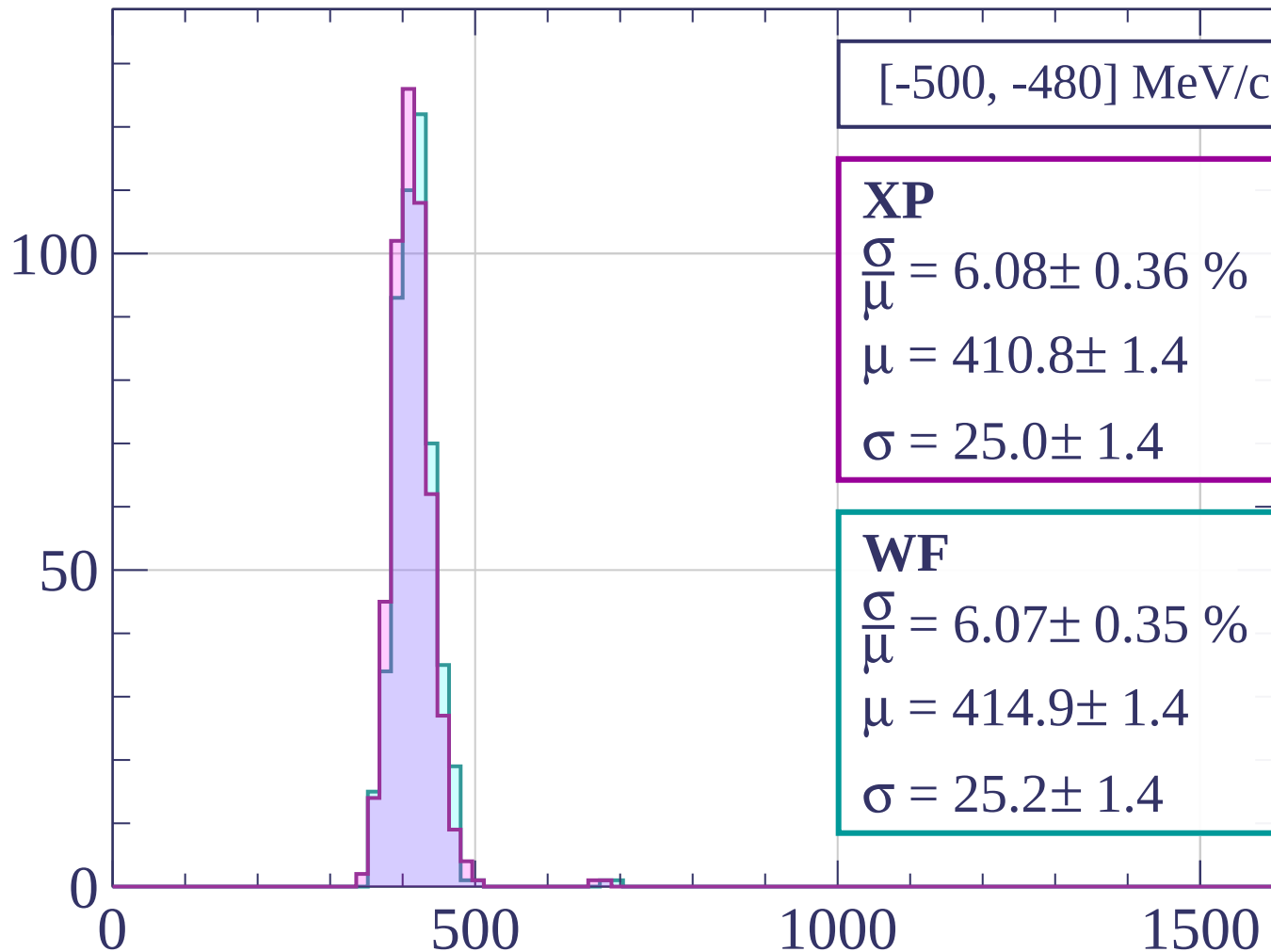
$$\frac{\sigma}{\mu} = 6.55 \pm 0.36 \%$$

$$\mu = 419.3 \pm 1.5$$

$$\sigma = 27.5 \pm 1.4$$

dE/dx [ADC counts/cm]

Count



[-500, -480] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.08 \pm 0.36 \%$$

$$\mu = 410.8 \pm 1.4$$

$$\sigma = 25.0 \pm 1.4$$

WF

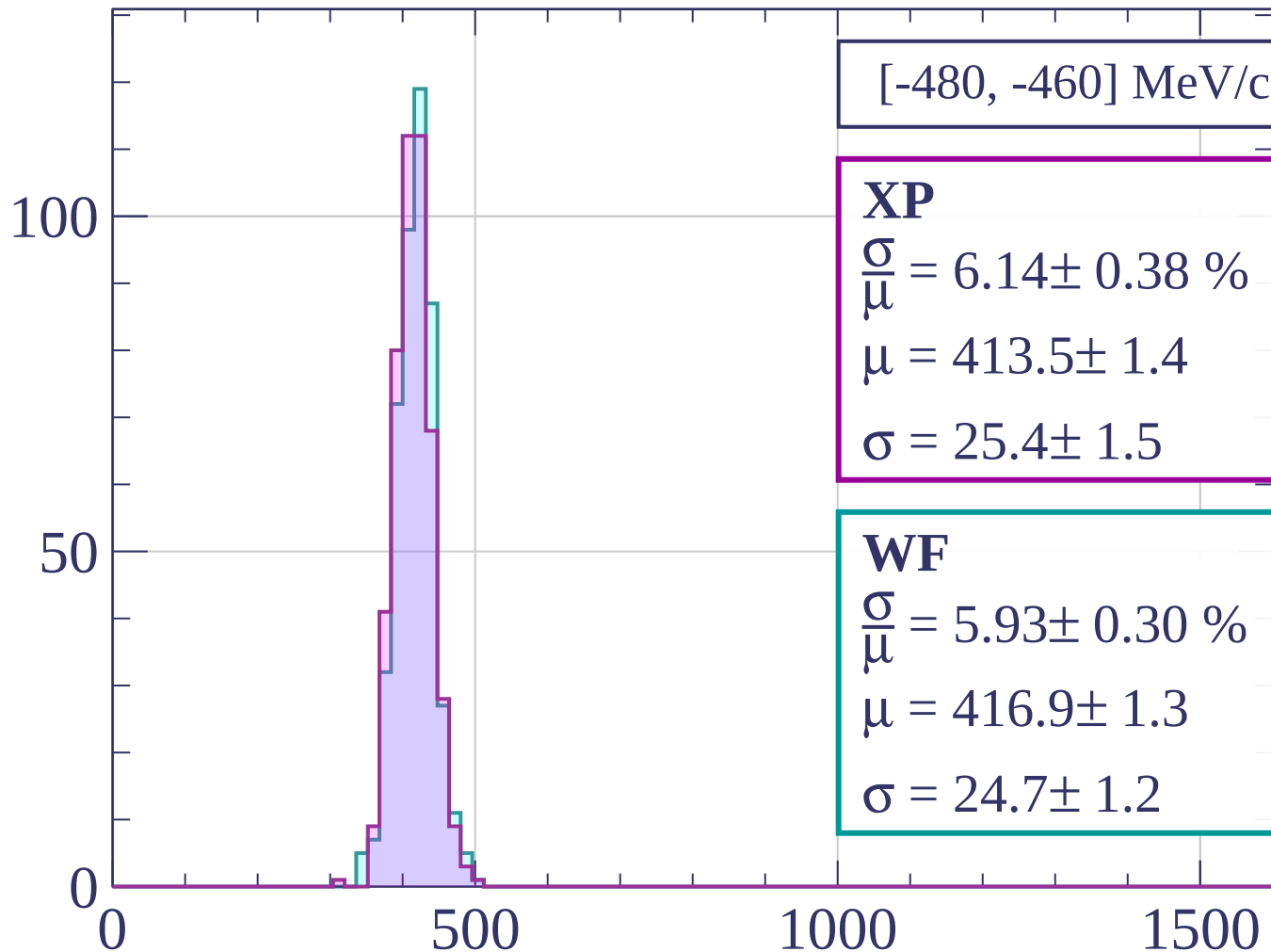
$$\frac{\sigma}{\mu} = 6.07 \pm 0.35 \%$$

$$\mu = 414.9 \pm 1.4$$

$$\sigma = 25.2 \pm 1.4$$

dE/dx [ADC counts/cm]

Count



[-480, -460] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.14 \pm 0.38 \%$$

$$\mu = 413.5 \pm 1.4$$

$$\sigma = 25.4 \pm 1.5$$

WF

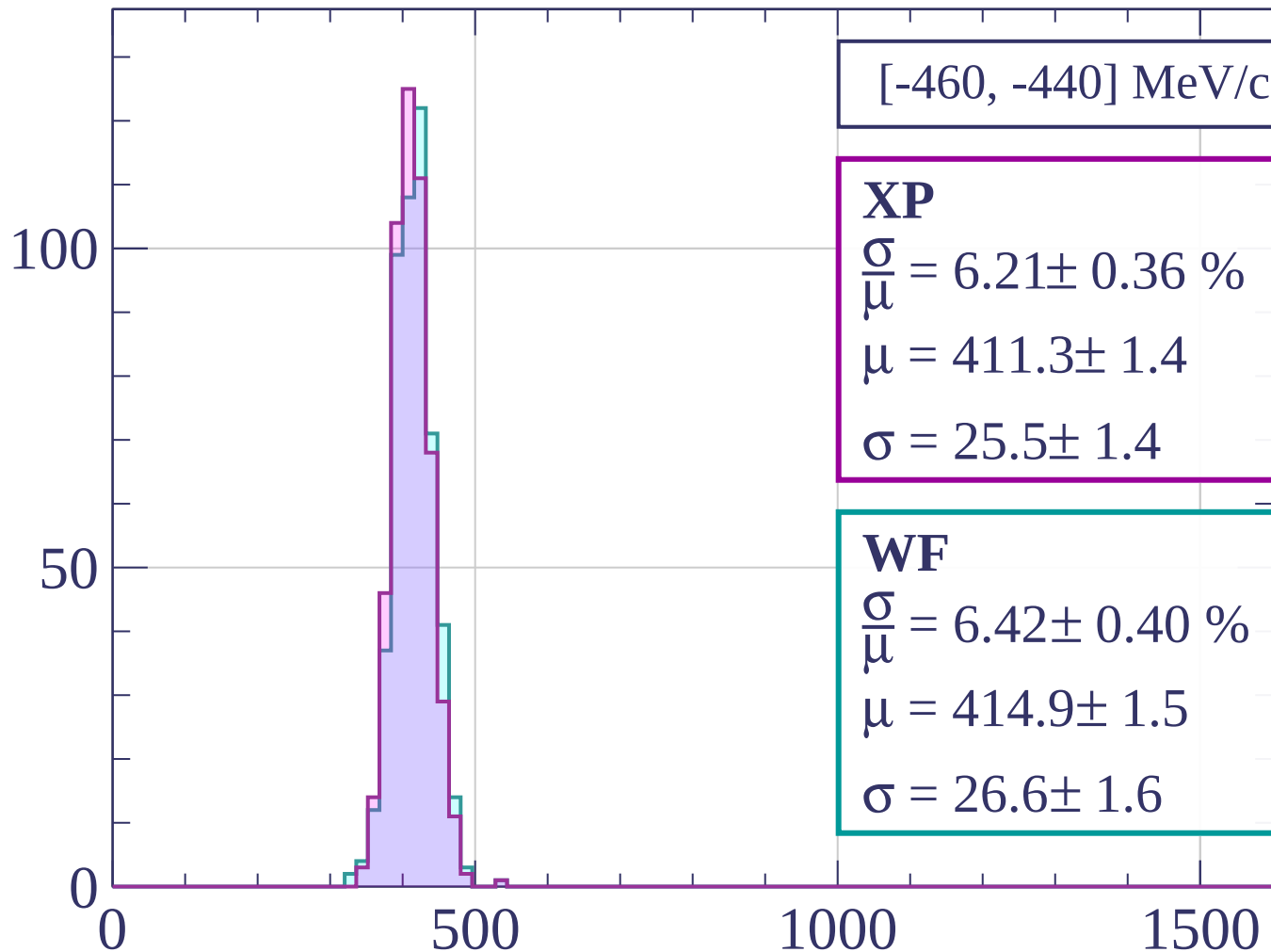
$$\frac{\sigma}{\mu} = 5.93 \pm 0.30 \%$$

$$\mu = 416.9 \pm 1.3$$

$$\sigma = 24.7 \pm 1.2$$

dE/dx [ADC counts/cm]

Count



[-460, -440] MeV/c

XP

$$\frac{\sigma}{\bar{\mu}} = 6.21 \pm 0.36 \%$$

$$\mu = 411.3 \pm 1.4$$

$$\sigma = 25.5 \pm 1.4$$

WF

$$\frac{\sigma}{\bar{\mu}} = 6.42 \pm 0.40 \%$$

$$\mu = 414.9 \pm 1.5$$

$$\sigma = 26.6 \pm 1.6$$

dE/dx [ADC counts/cm]

Count

[-440, -420] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.55 \pm 0.41 \%$$

$$\mu = 410.3 \pm 1.4$$

$$\sigma = 22.8 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 6.14 \pm 0.37 \%$$

$$\mu = 415.0 \pm 1.4$$

$$\sigma = 25.5 \pm 1.5$$

100

50

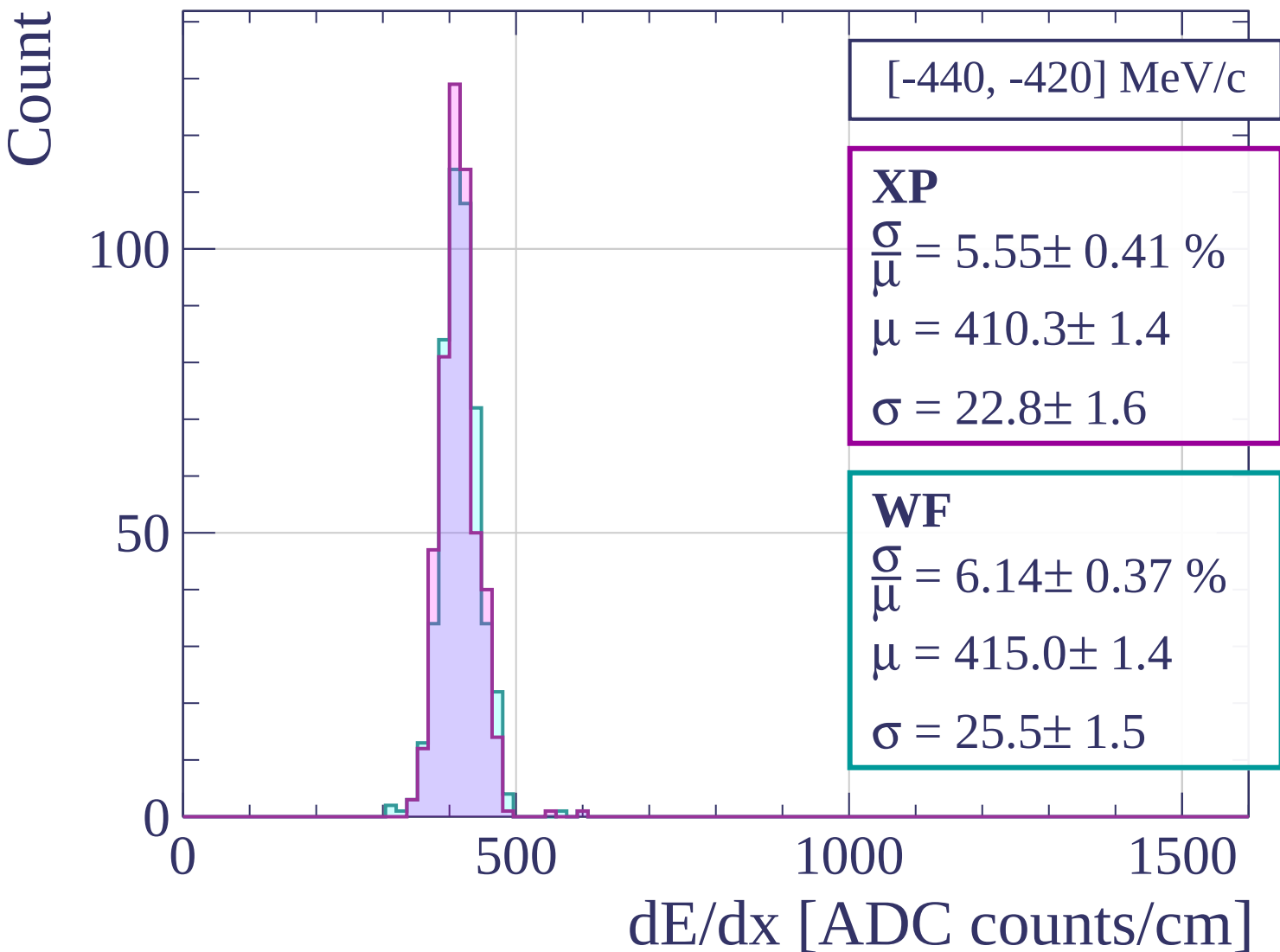
0

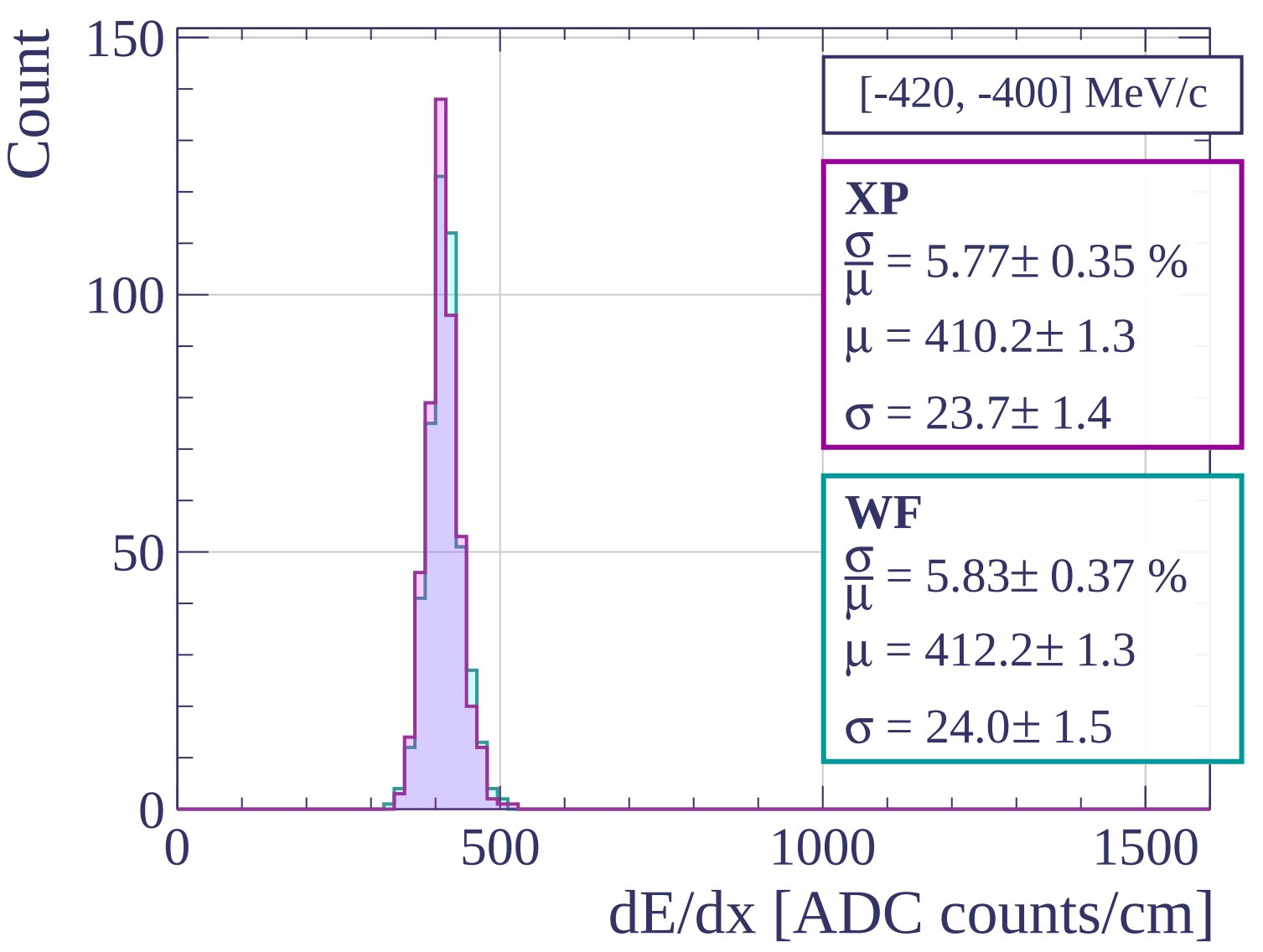
500

1000

1500

dE/dx [ADC counts/cm]





Count

[-400, -380] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.79 \pm 0.25 \%$$

$$\mu = 409.5 \pm 1.2$$

$$\sigma = 23.7 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 6.15 \pm 0.37 \%$$

$$\mu = 411.7 \pm 1.4$$

$$\sigma = 25.3 \pm 1.4$$

100

50

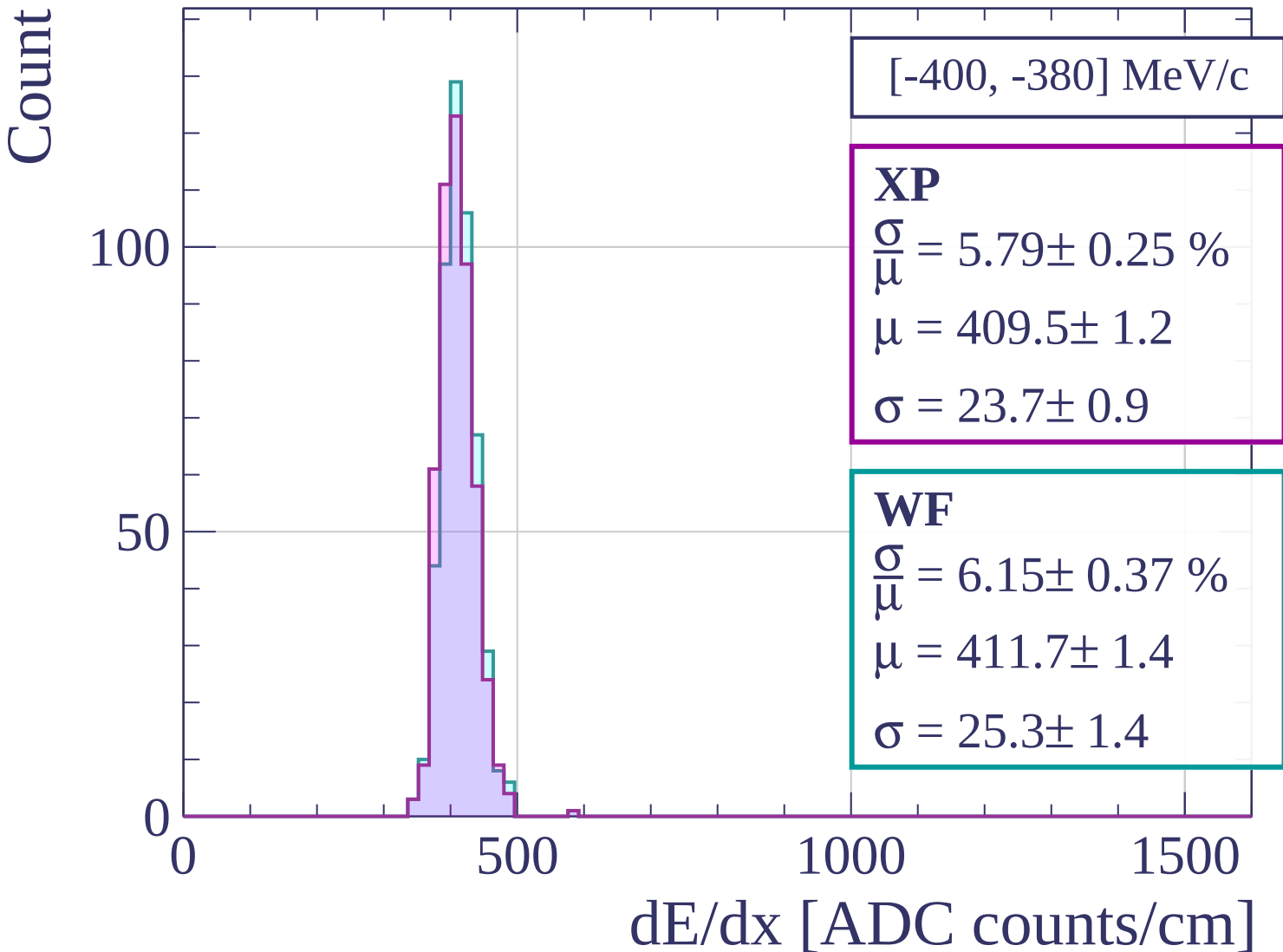
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-380, -360] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.58 \pm 0.29 \%$$

$$\mu = 404.4 \pm 1.2$$

$$\sigma = 22.6 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 6.01 \pm 0.27 \%$$

$$\mu = 409.1 \pm 1.2$$

$$\sigma = 24.6 \pm 1.0$$

100

50

0

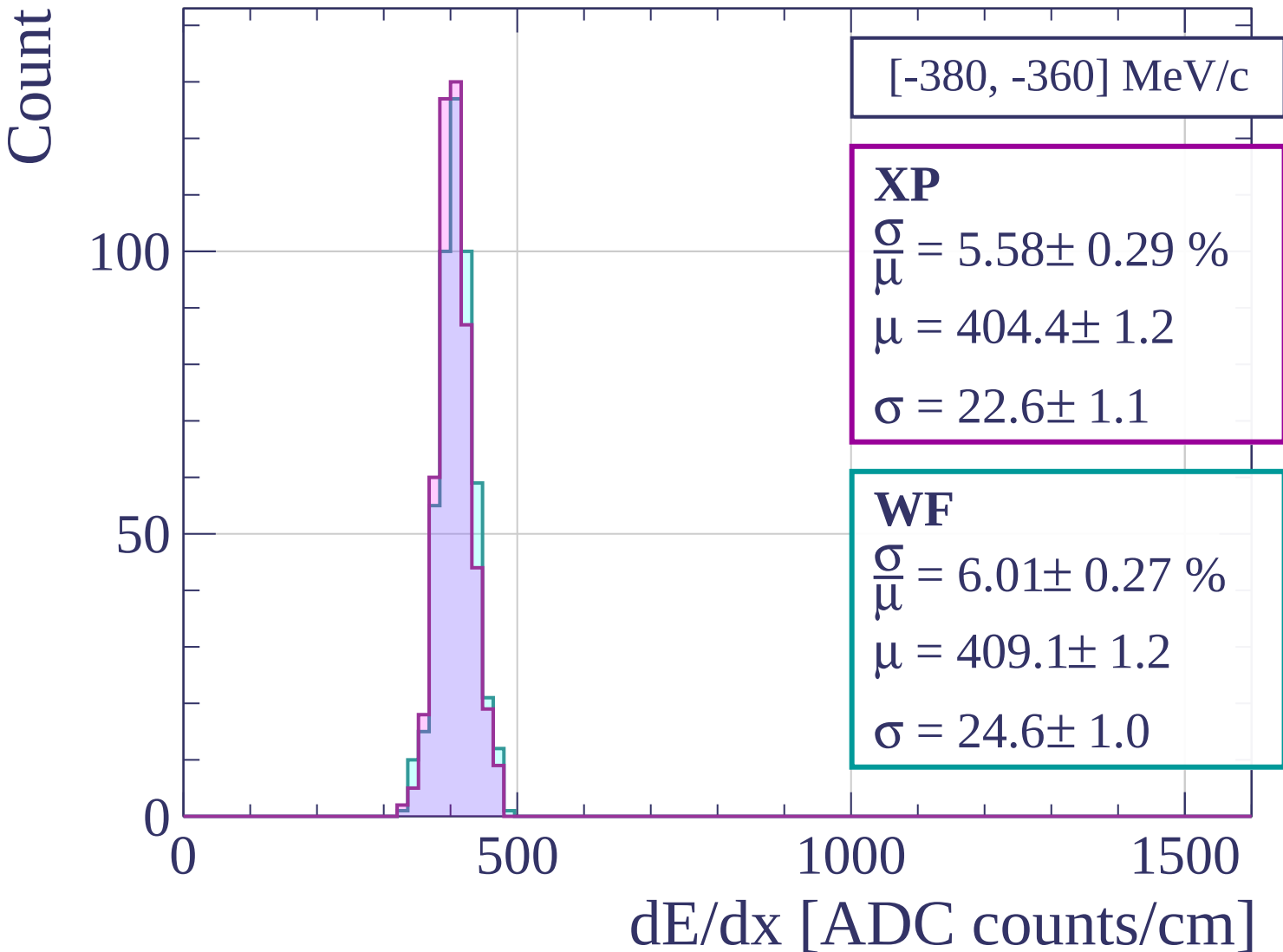
0

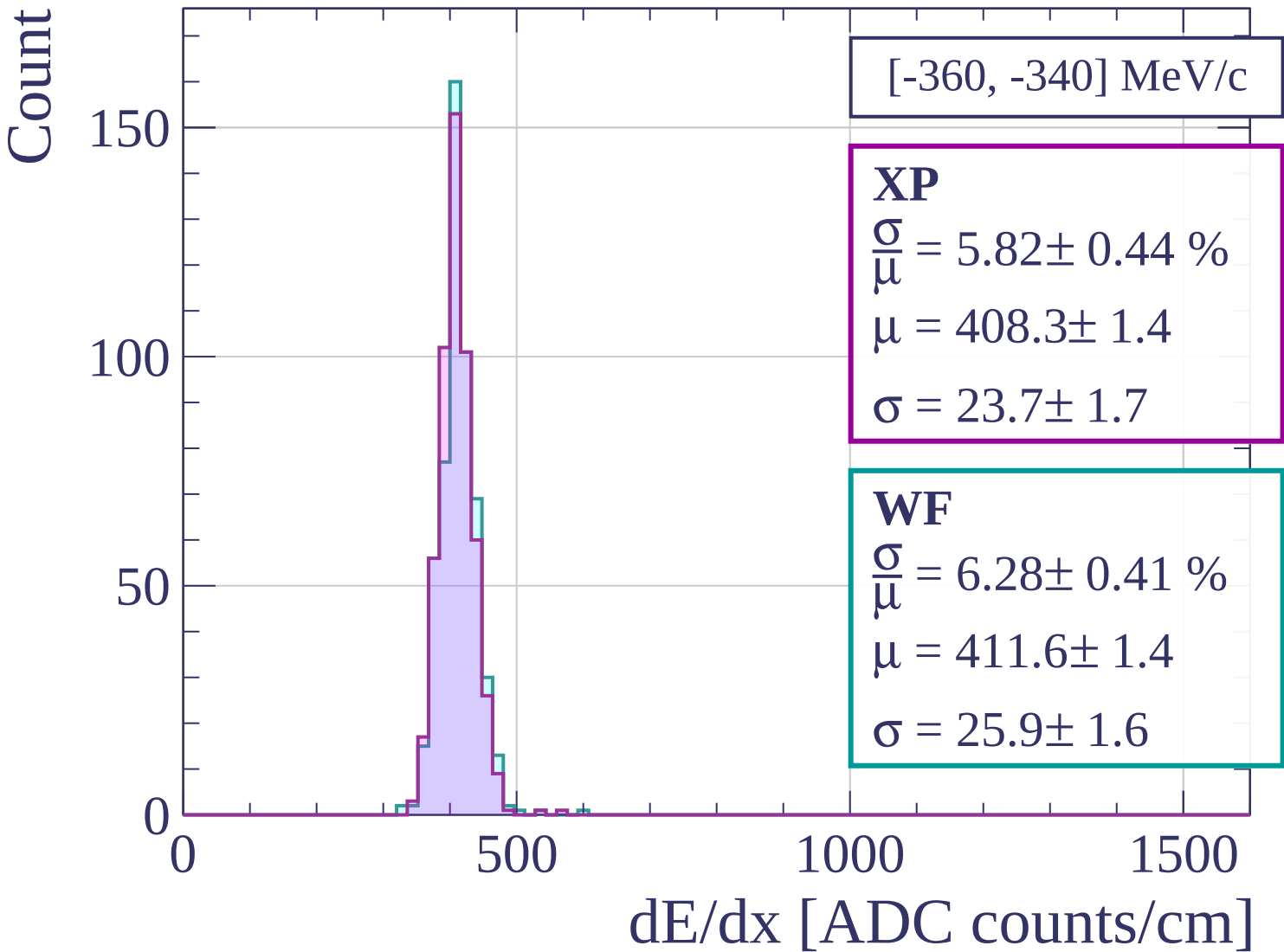
500

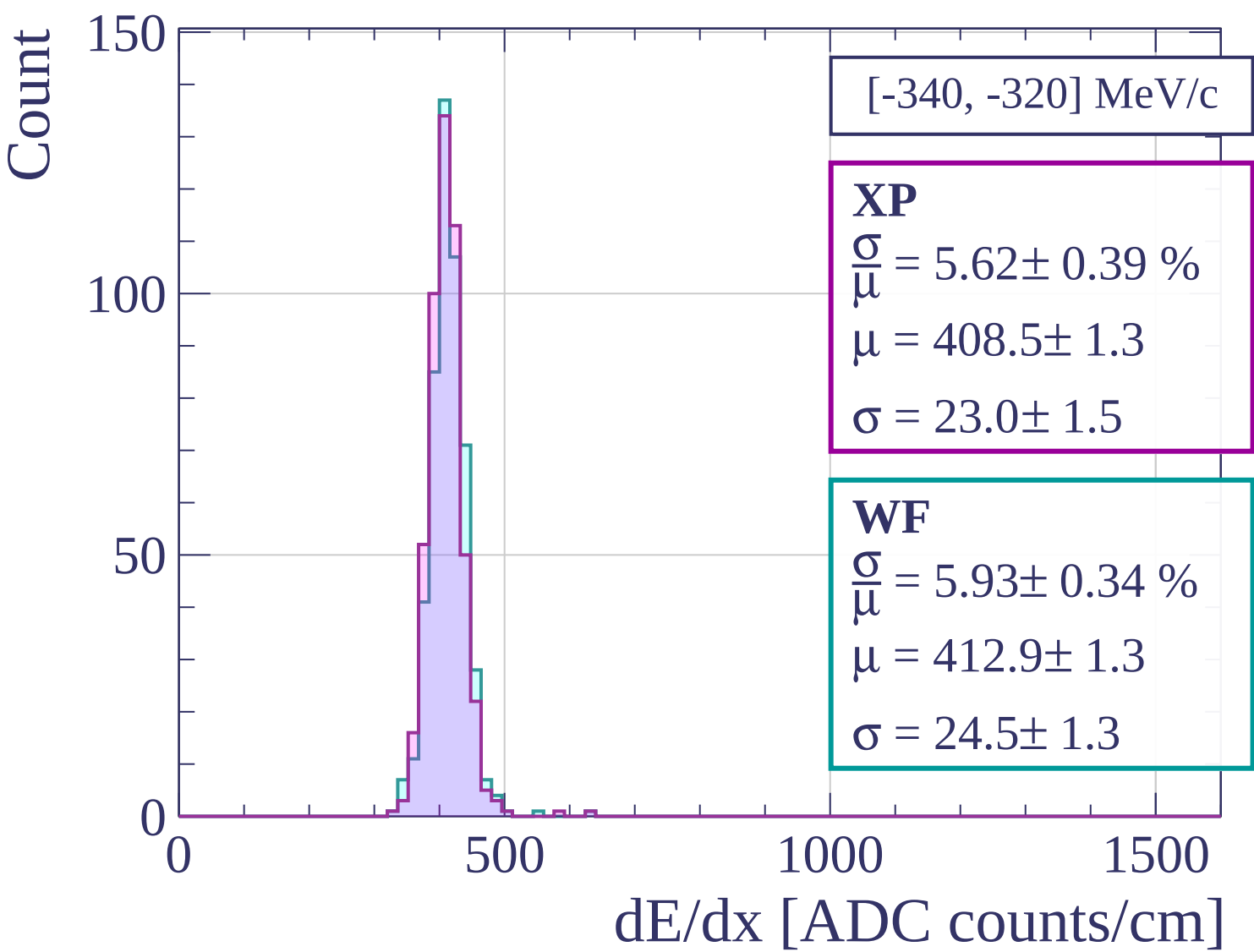
1000

1500

dE/dx [ADC counts/cm]







Count

[-320, -300] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.16 \pm 0.28 \%$$

$$\mu = 413.1 \pm 1.3$$

$$\sigma = 25.4 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 6.64 \pm 0.34 \%$$

$$\mu = 417.2 \pm 1.5$$

$$\sigma = 27.7 \pm 1.3$$

100

50

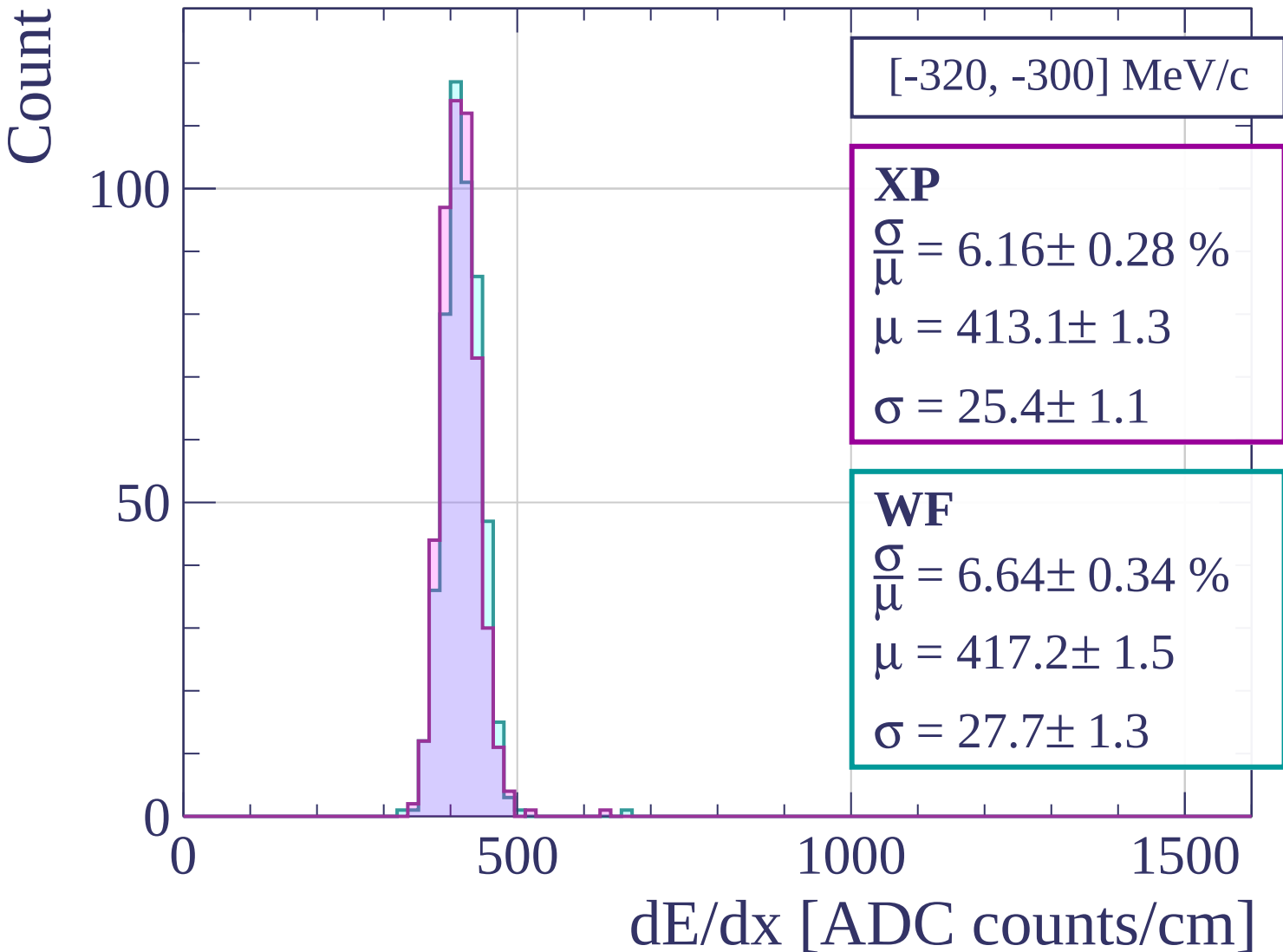
0

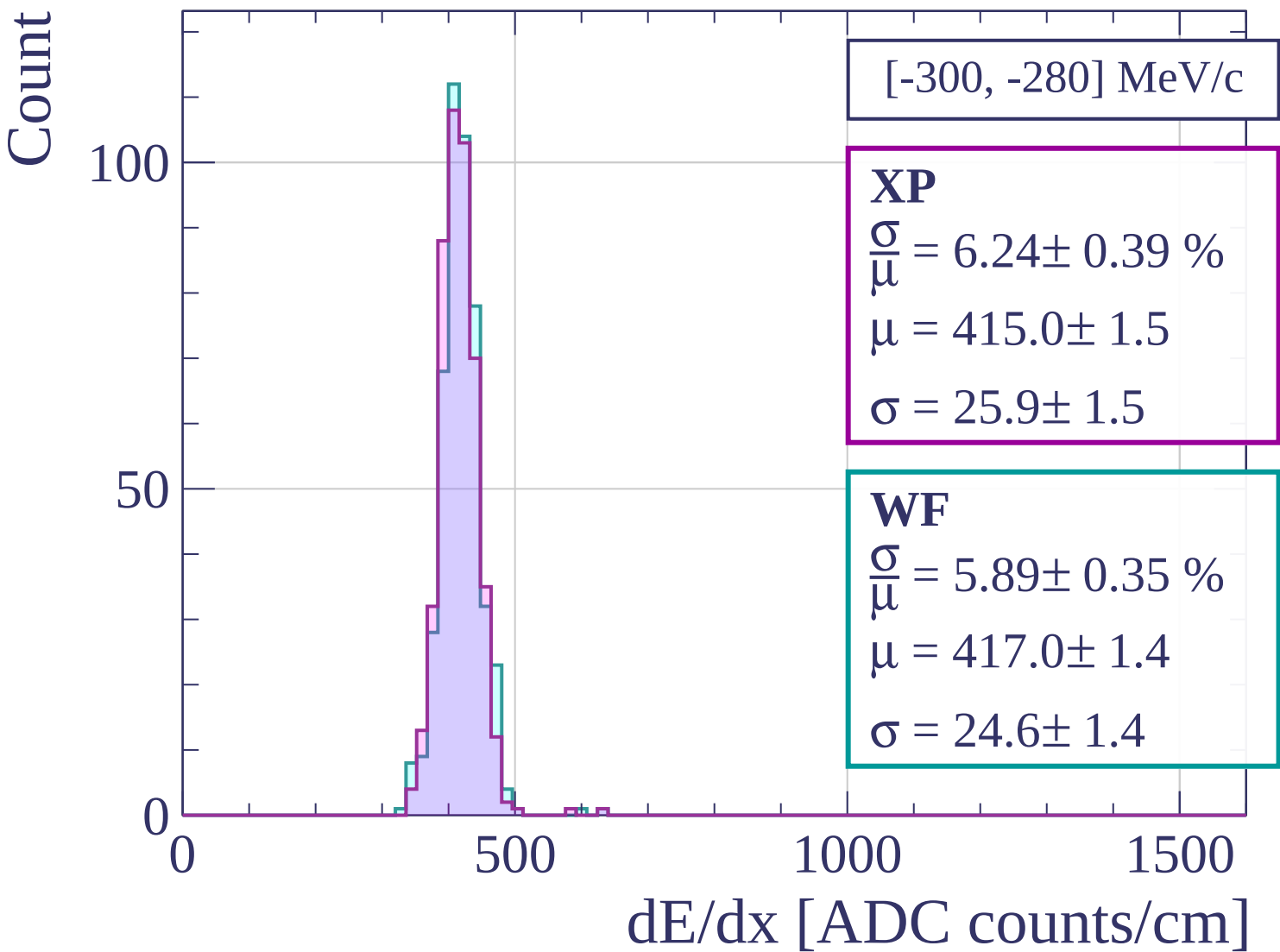
500

1000

1500

dE/dx [ADC counts/cm]





Count

[-280, -260] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.13 \pm 0.40 \%$$

$$\mu = 417.8 \pm 1.5$$

$$\sigma = 25.6 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 7.04 \pm 0.45 \%$$

$$\mu = 421.8 \pm 1.7$$

$$\sigma = 29.7 \pm 1.8$$

100

50

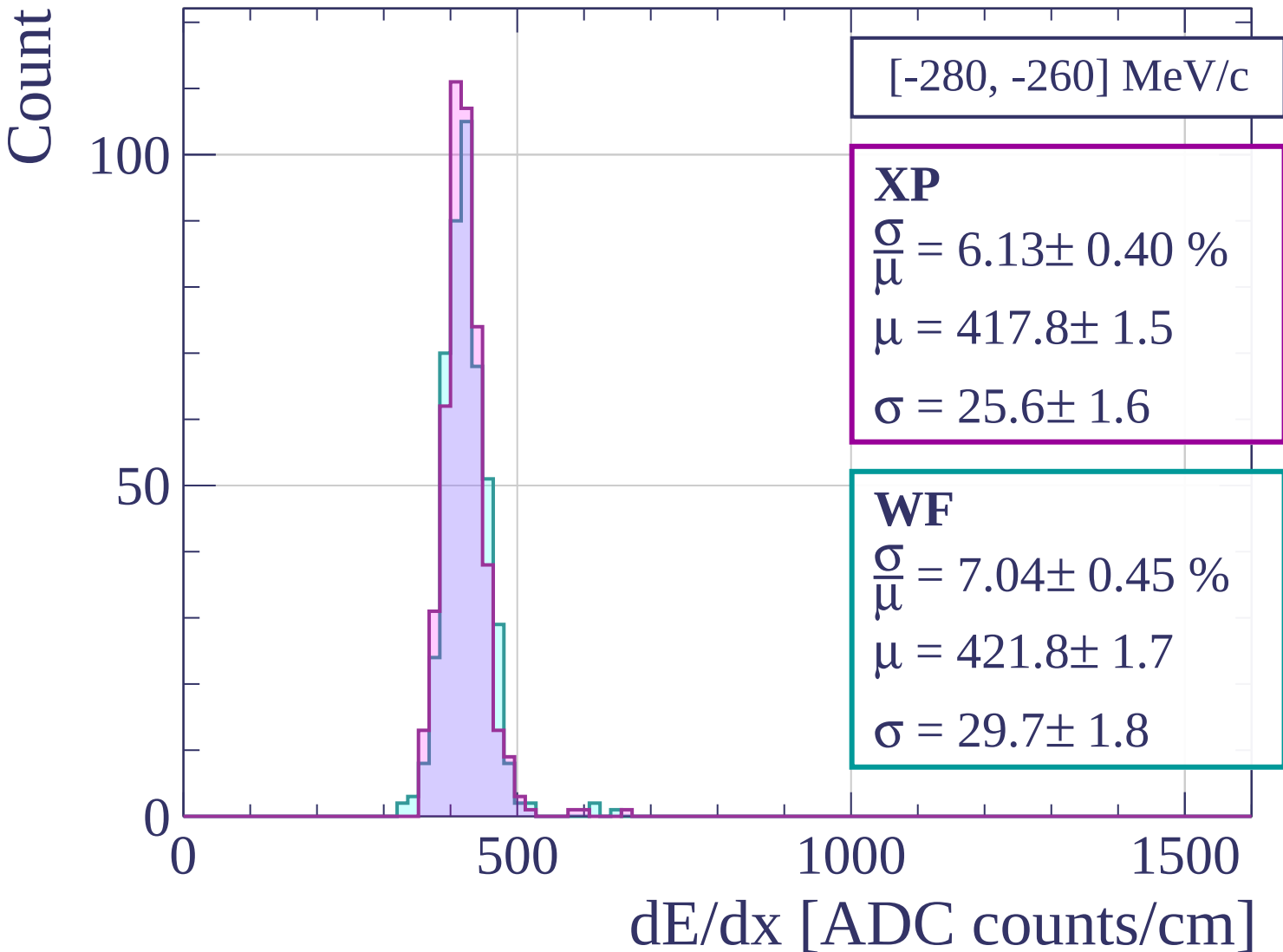
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-260, -240] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.77 \pm 0.32 \%$$

$$\mu = 419.6 \pm 1.3$$

$$\sigma = 24.2 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 5.98 \pm 0.30 \%$$

$$\mu = 424.8 \pm 1.3$$

$$\sigma = 25.4 \pm 1.2$$

100

50

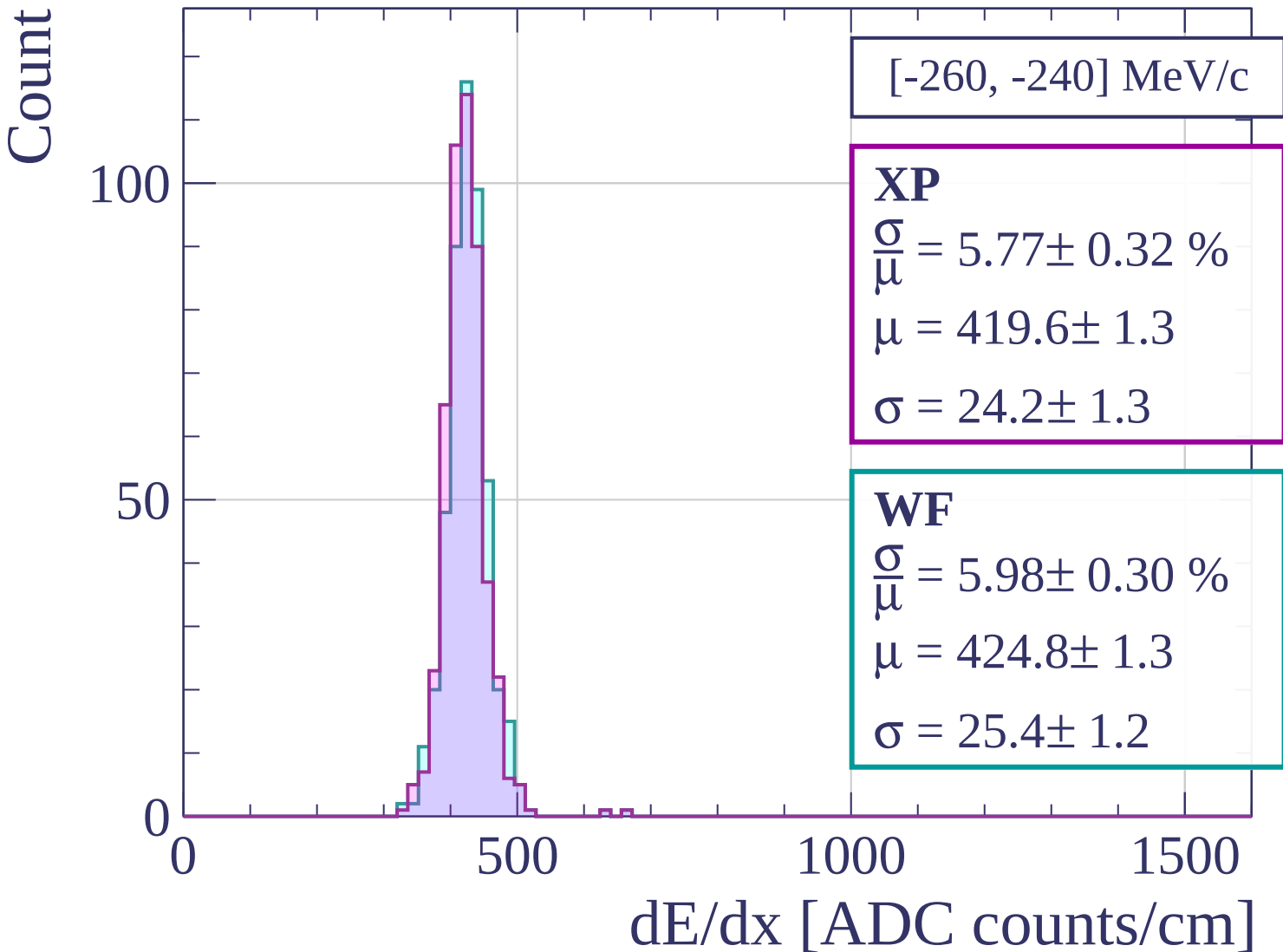
0

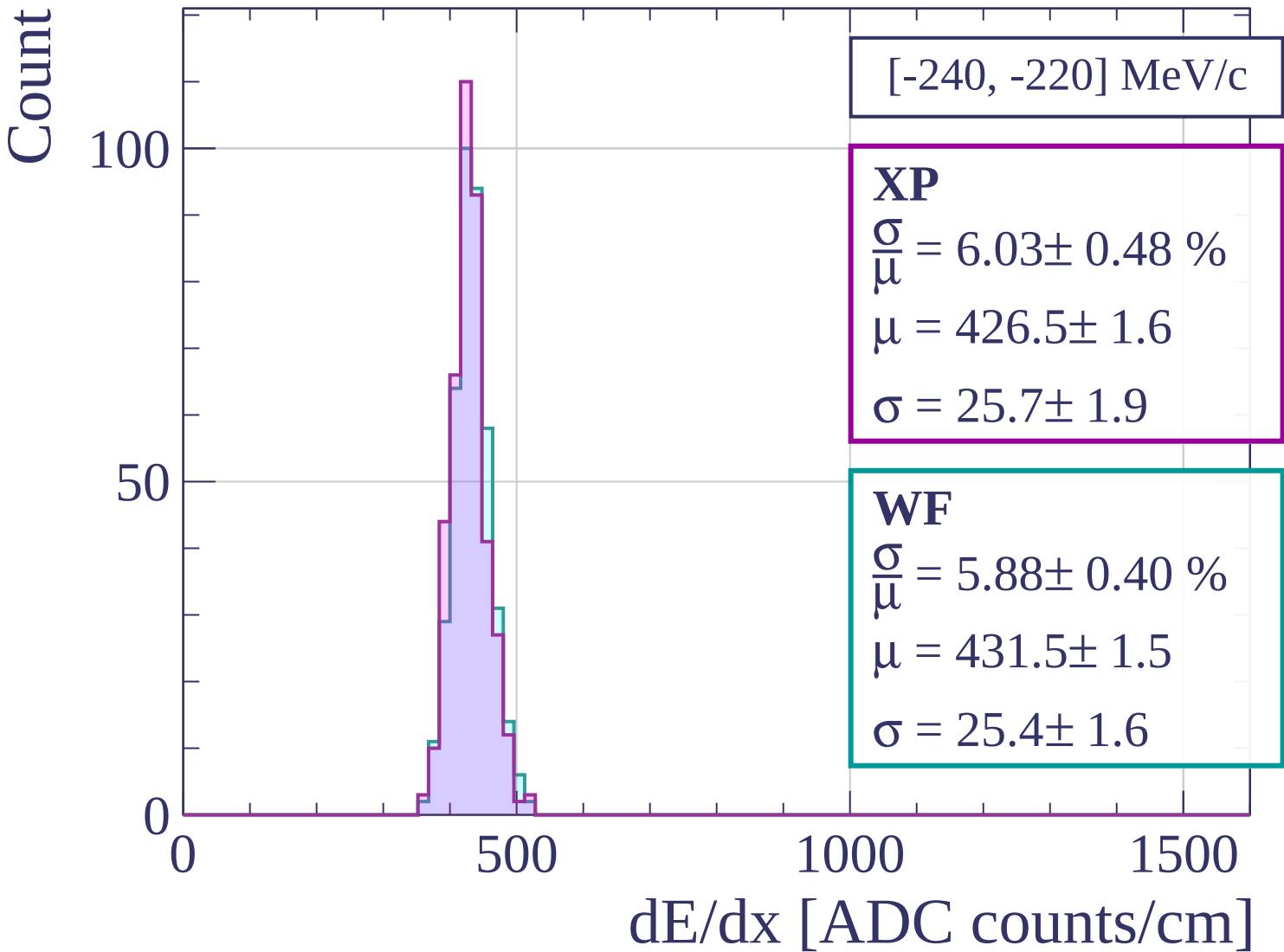
500

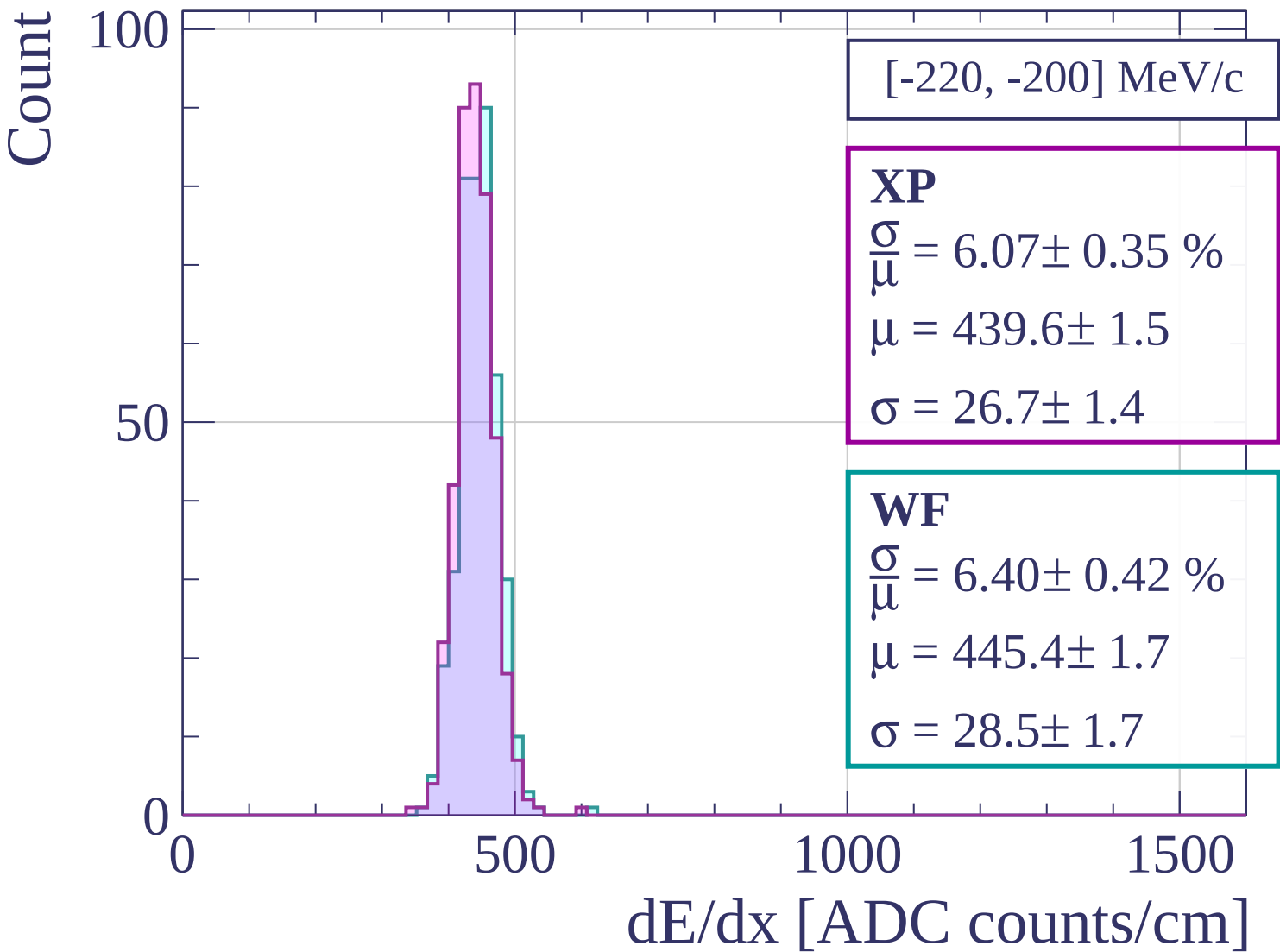
1000

1500

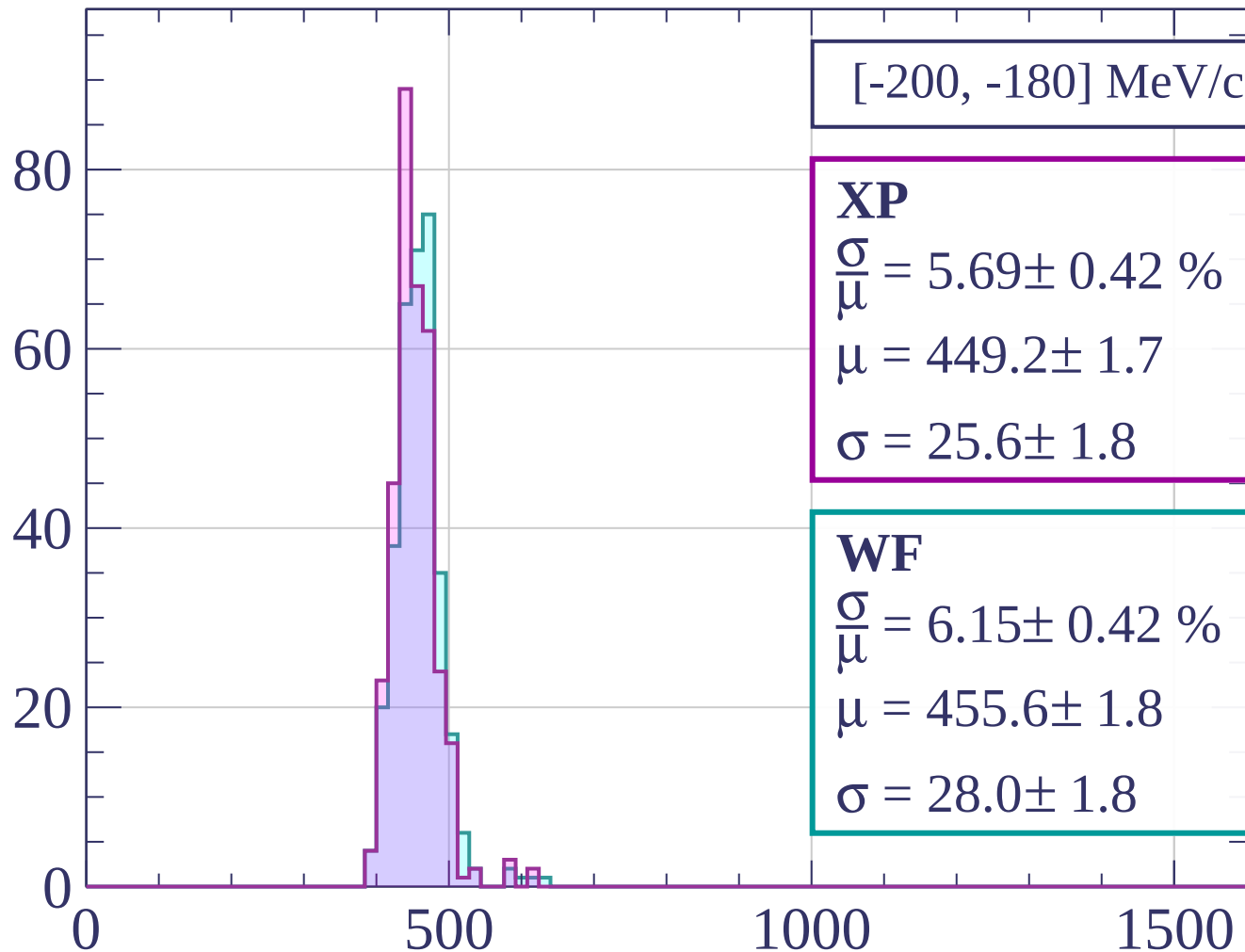
dE/dx [ADC counts/cm]







Count



$[-200, -180]$ MeV/c

XP

$$\frac{\sigma}{\mu} = 5.69 \pm 0.42 \%$$

$$\mu = 449.2 \pm 1.7$$

$$\sigma = 25.6 \pm 1.8$$

WF

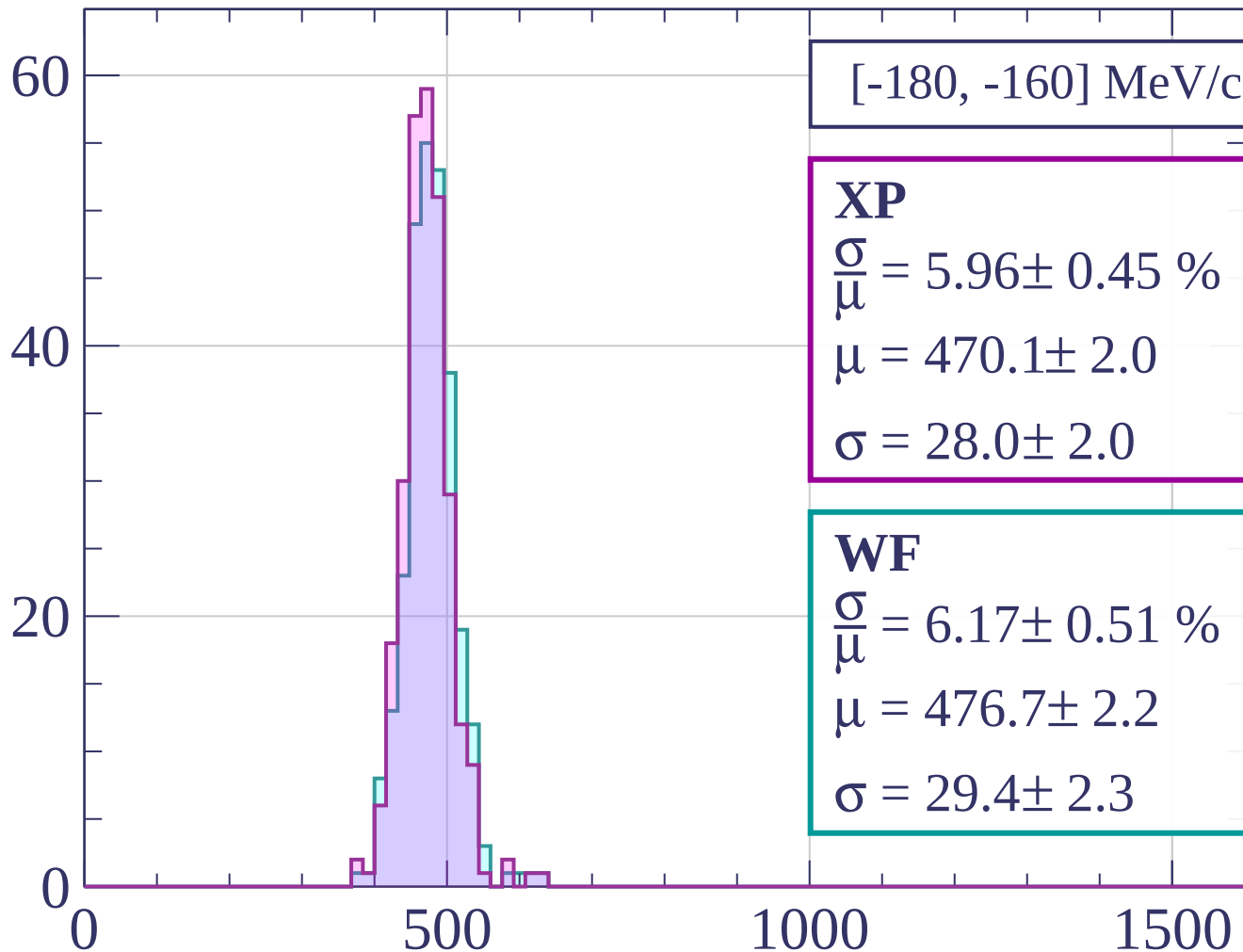
$$\frac{\sigma}{\mu} = 6.15 \pm 0.42 \%$$

$$\mu = 455.6 \pm 1.8$$

$$\sigma = 28.0 \pm 1.8$$

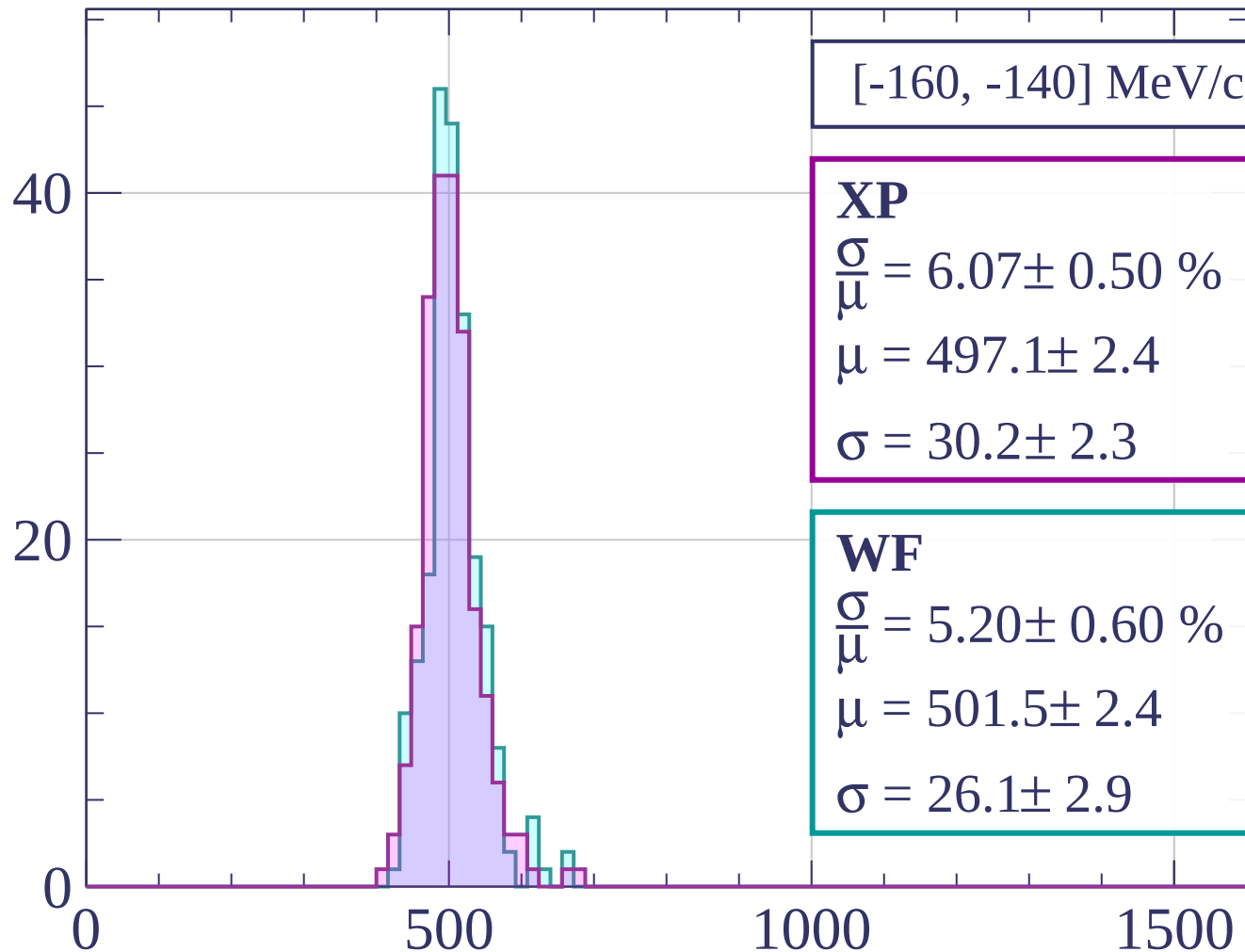
dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count



[-160, -140] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.07 \pm 0.50 \%$$

$$\mu = 497.1 \pm 2.4$$

$$\sigma = 30.2 \pm 2.3$$

WF

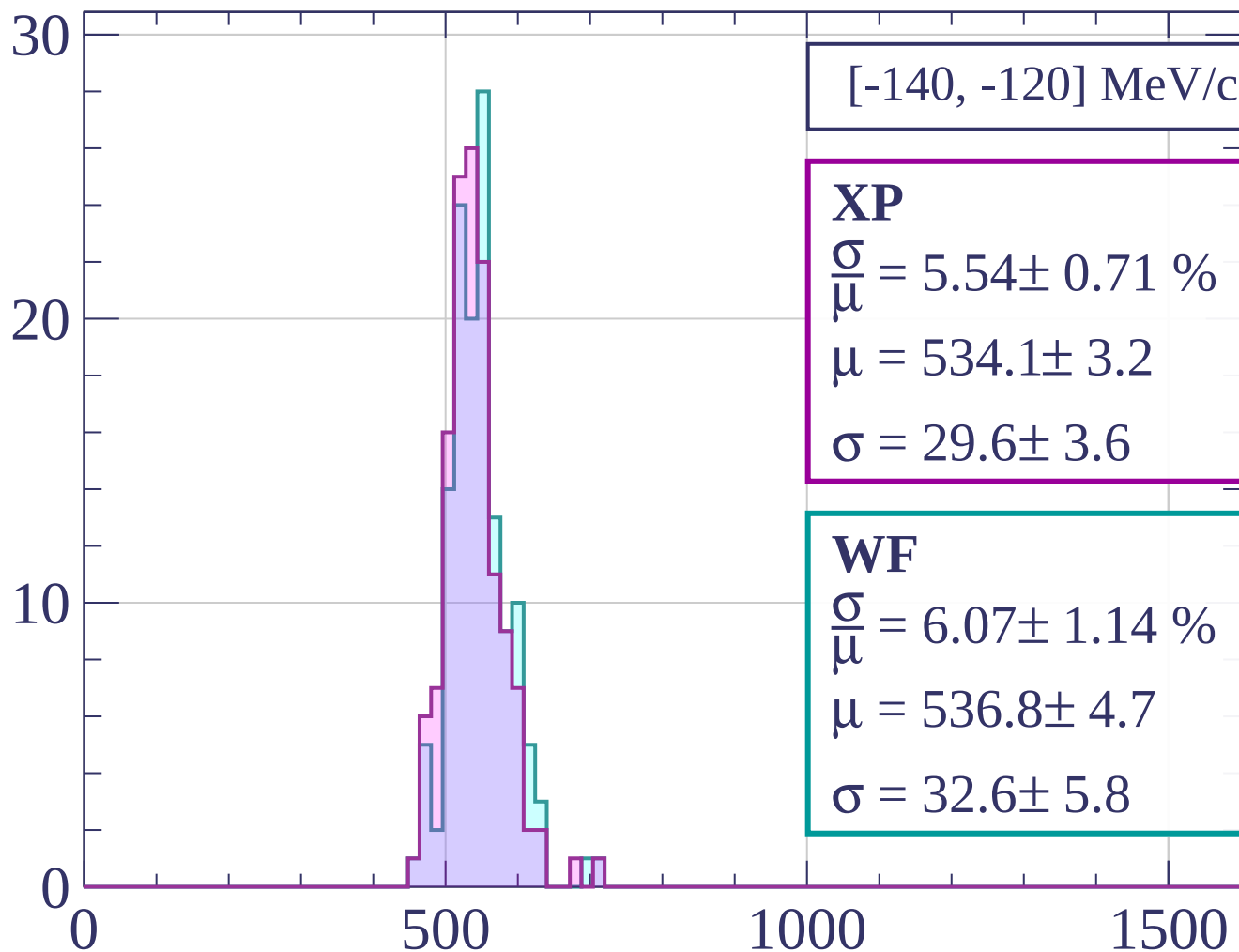
$$\frac{\sigma}{\mu} = 5.20 \pm 0.60 \%$$

$$\mu = 501.5 \pm 2.4$$

$$\sigma = 26.1 \pm 2.9$$

dE/dx [ADC counts/cm]

Count



[-140, -120] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.54 \pm 0.71 \%$$

$$\mu = 534.1 \pm 3.2$$

$$\sigma = 29.6 \pm 3.6$$

WF

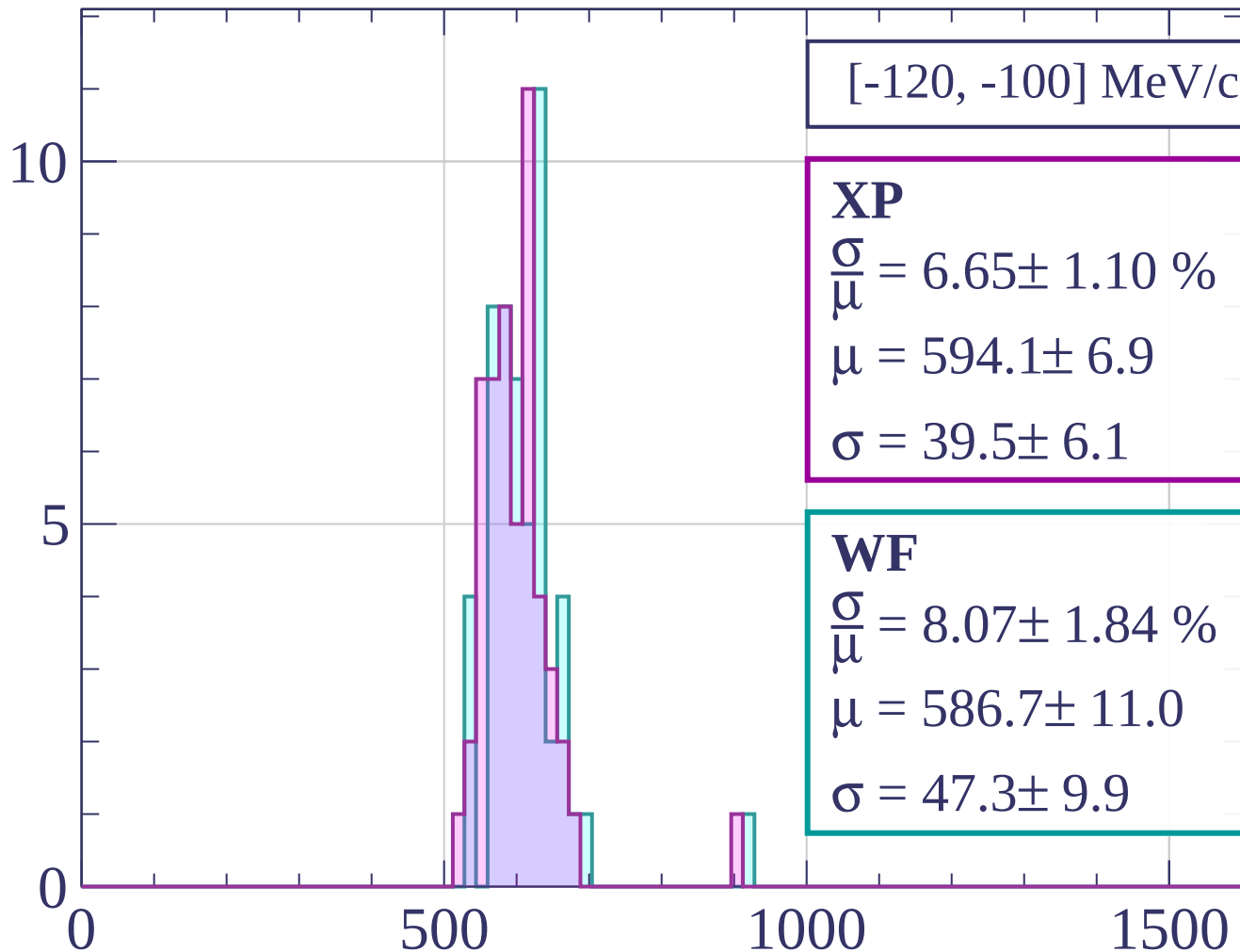
$$\frac{\sigma}{\mu} = 6.07 \pm 1.14 \%$$

$$\mu = 536.8 \pm 4.7$$

$$\sigma = 32.6 \pm 5.8$$

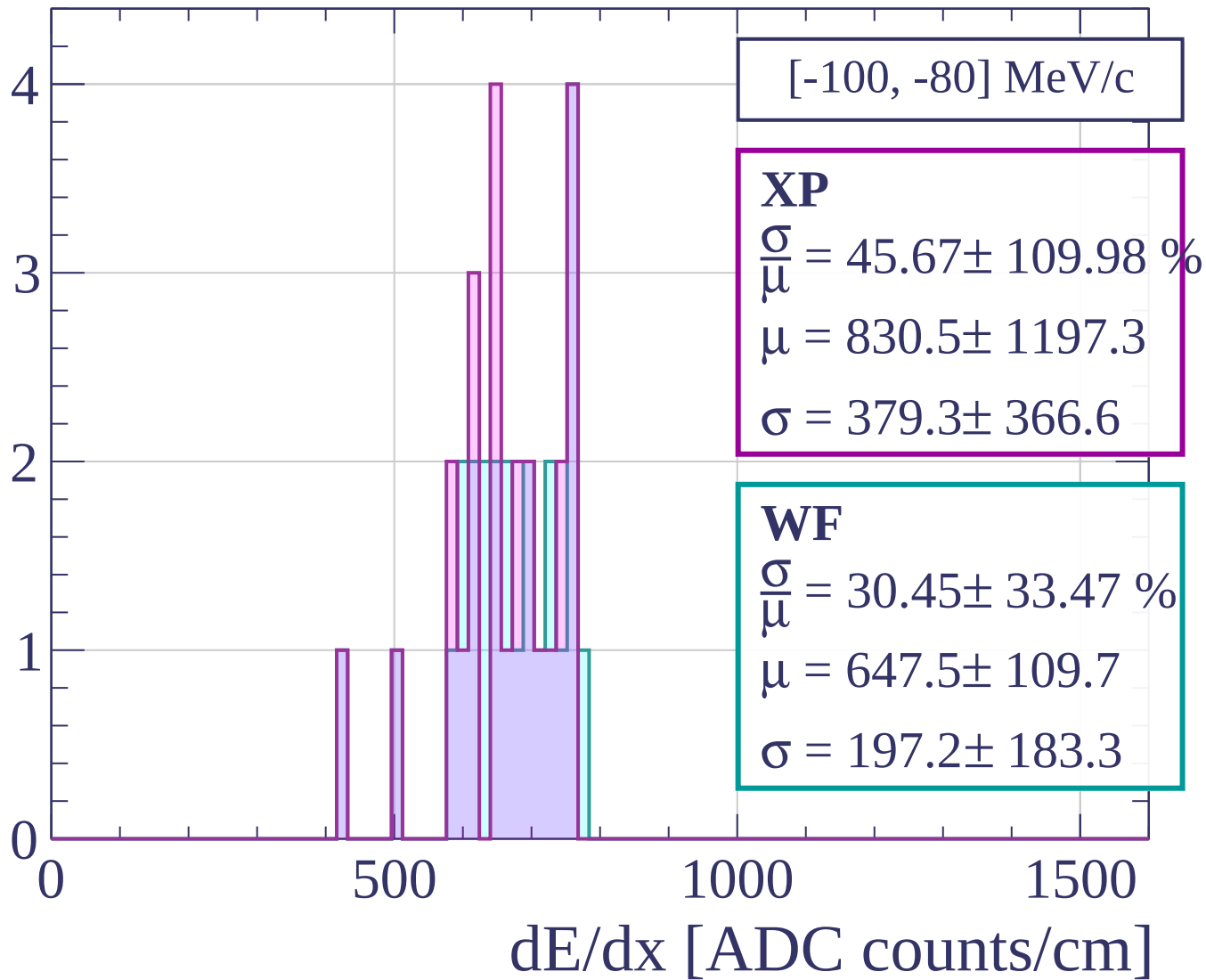
dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count



Count

[-80, -60] MeV/c

XP

$$\frac{\sigma}{\mu} = 47.76 \pm 145.98 \%$$

$$\mu = 535.6 \pm 261.2$$

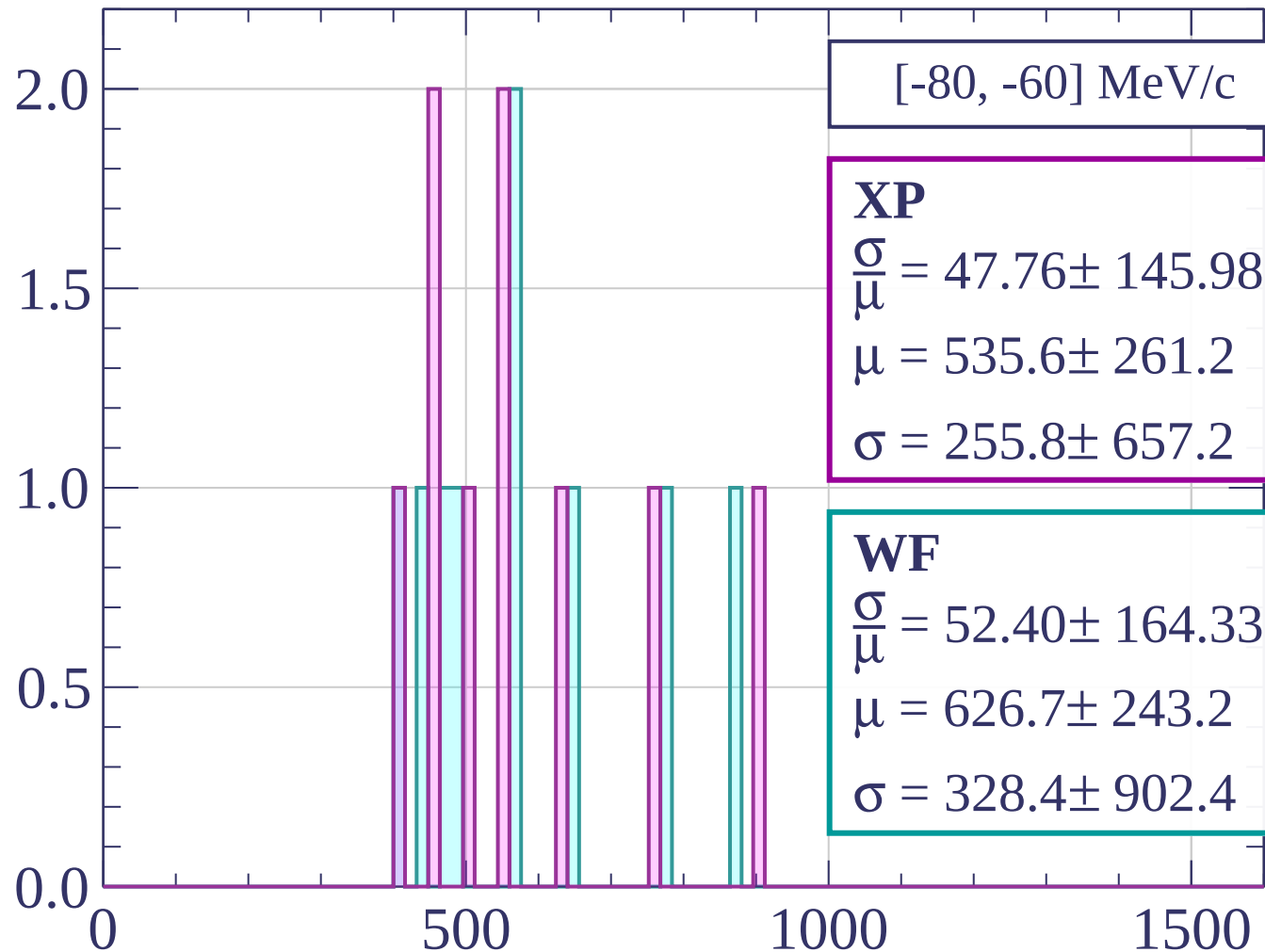
$$\sigma = 255.8 \pm 657.2$$

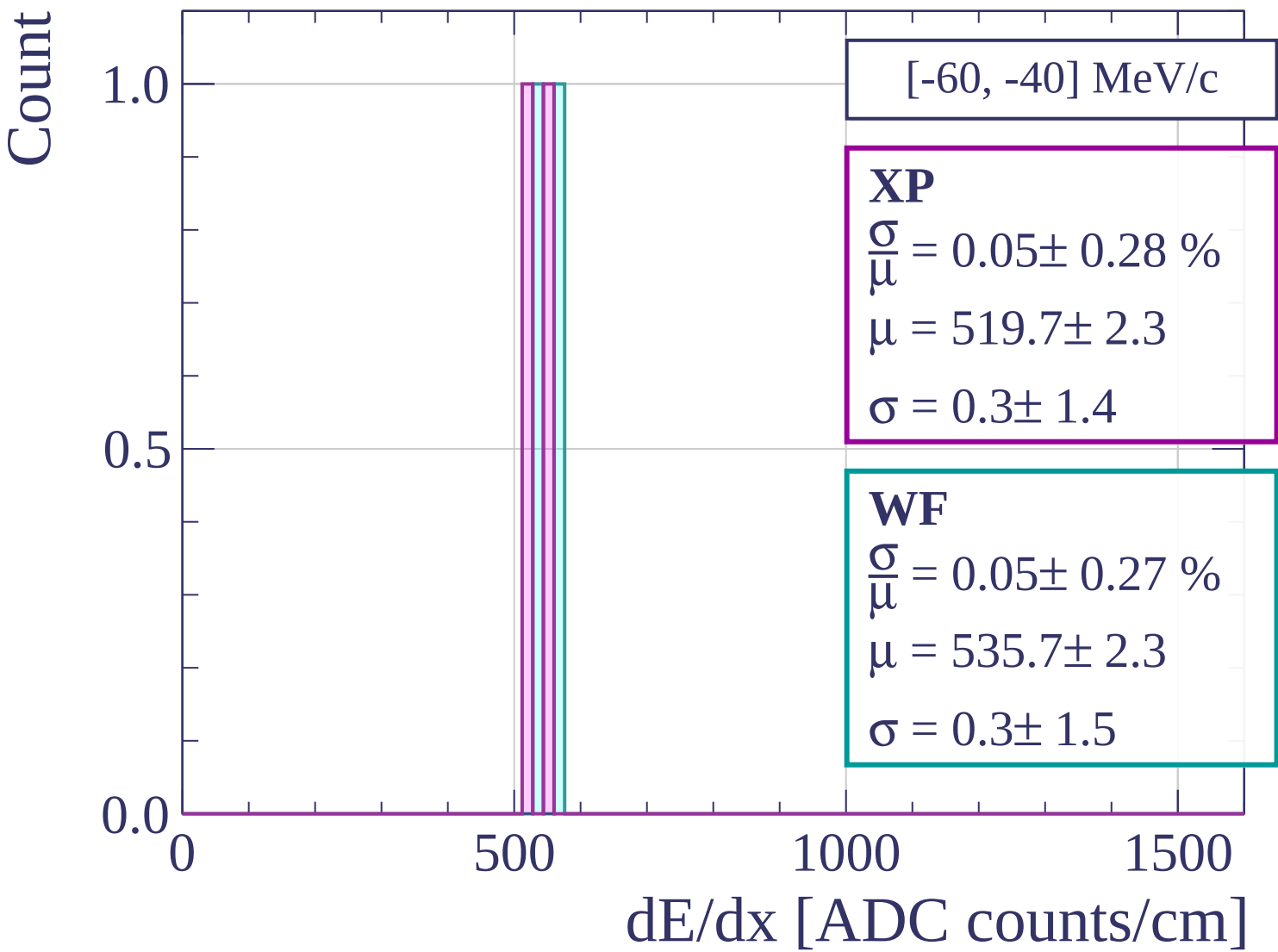
WF

$$\frac{\sigma}{\mu} = 52.40 \pm 164.33 \%$$

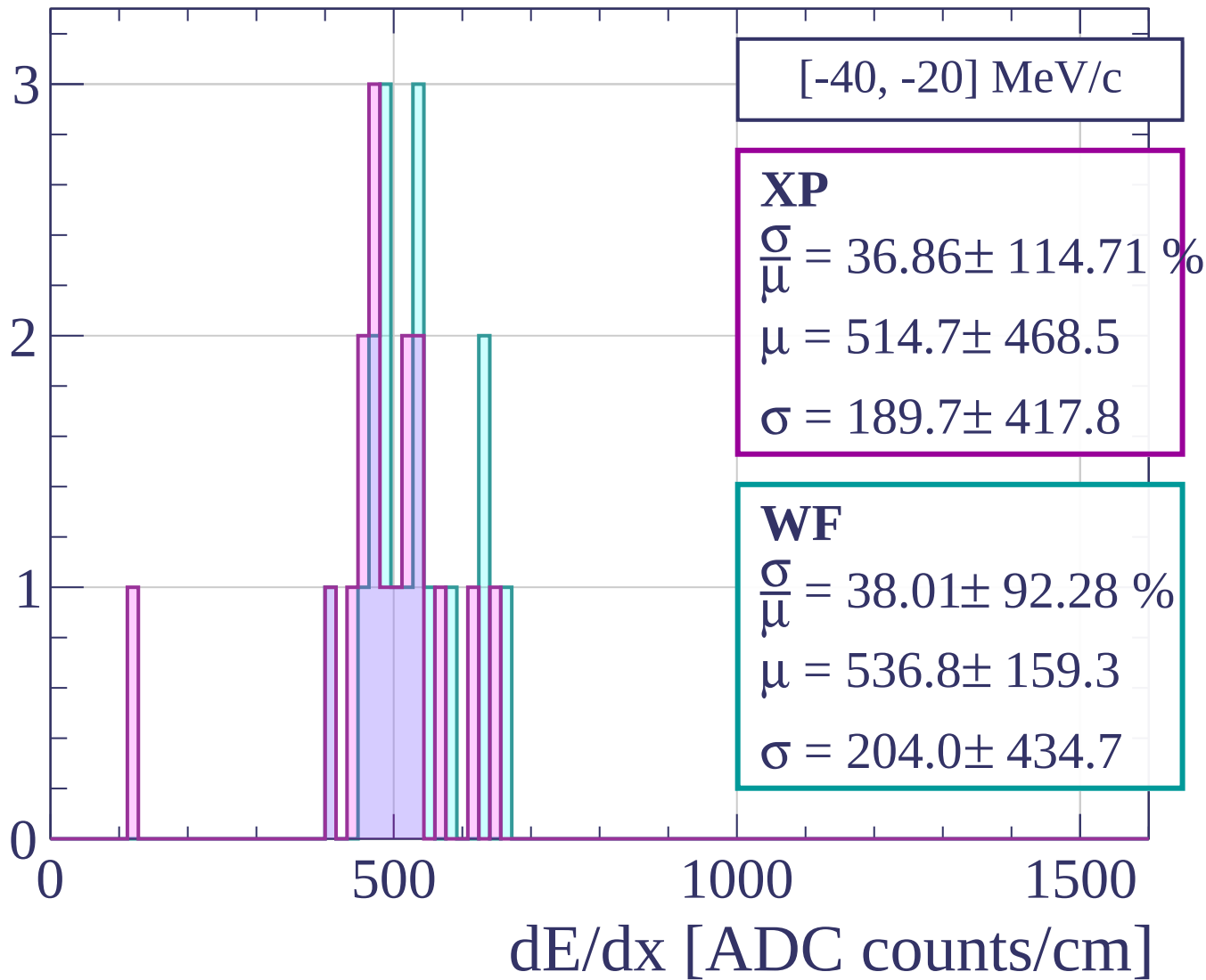
$$\mu = 626.7 \pm 243.2$$

$$\sigma = 328.4 \pm 902.4$$

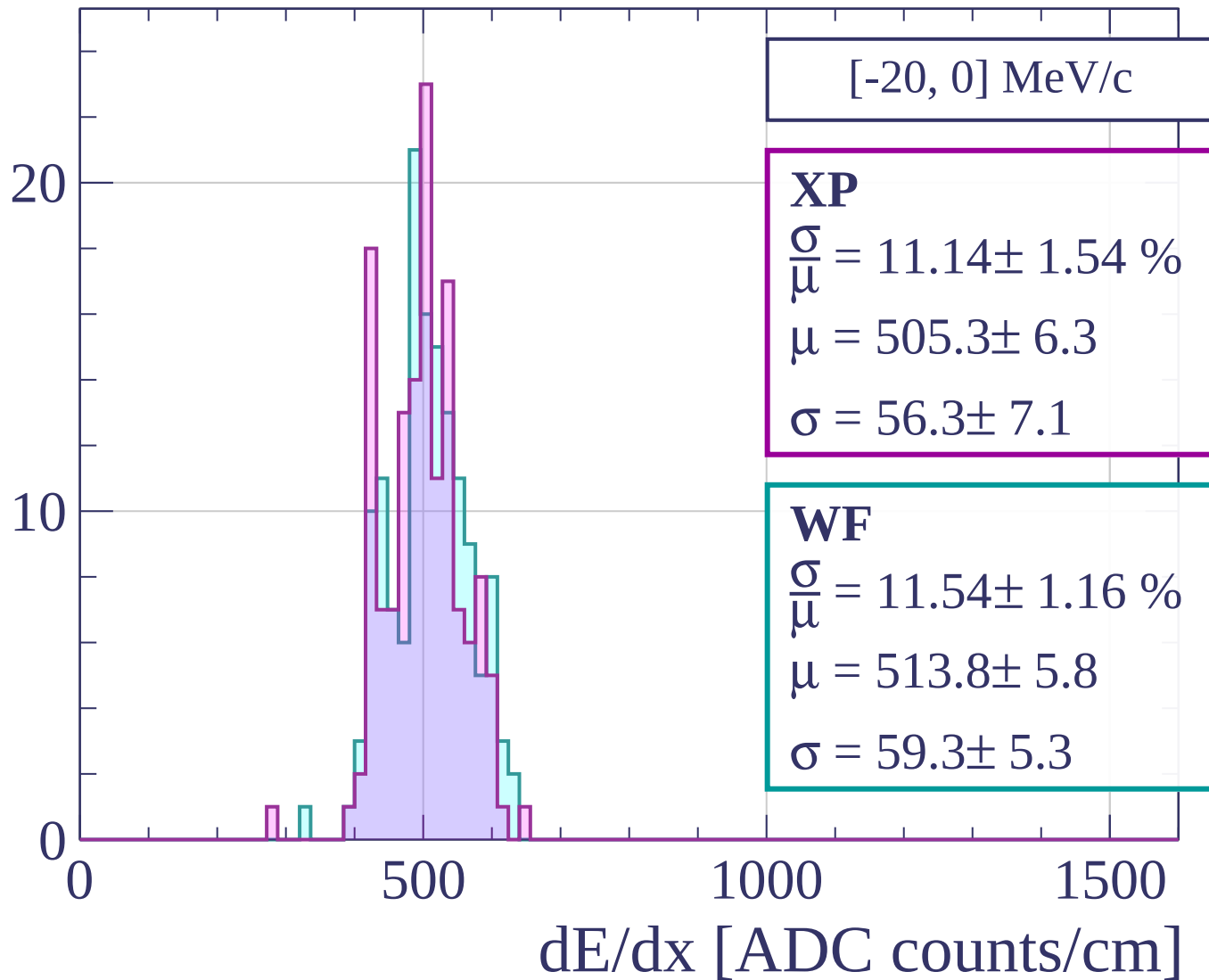




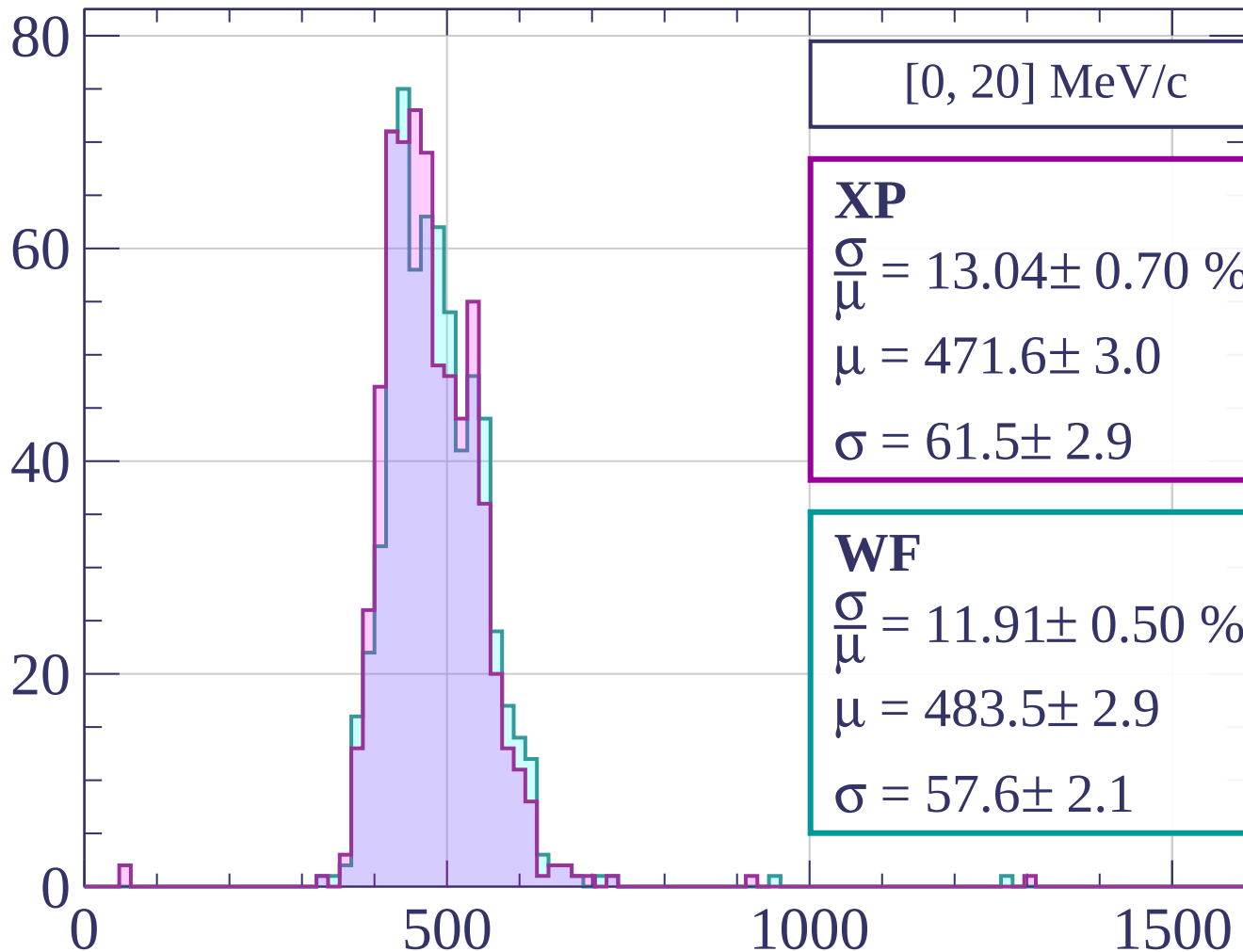
Count



Count



Count



$[0, 20] \text{ MeV/c}$

XP

$$\frac{\sigma}{\mu} = 13.04 \pm 0.70 \%$$

$$\mu = 471.6 \pm 3.0$$

$$\sigma = 61.5 \pm 2.9$$

WF

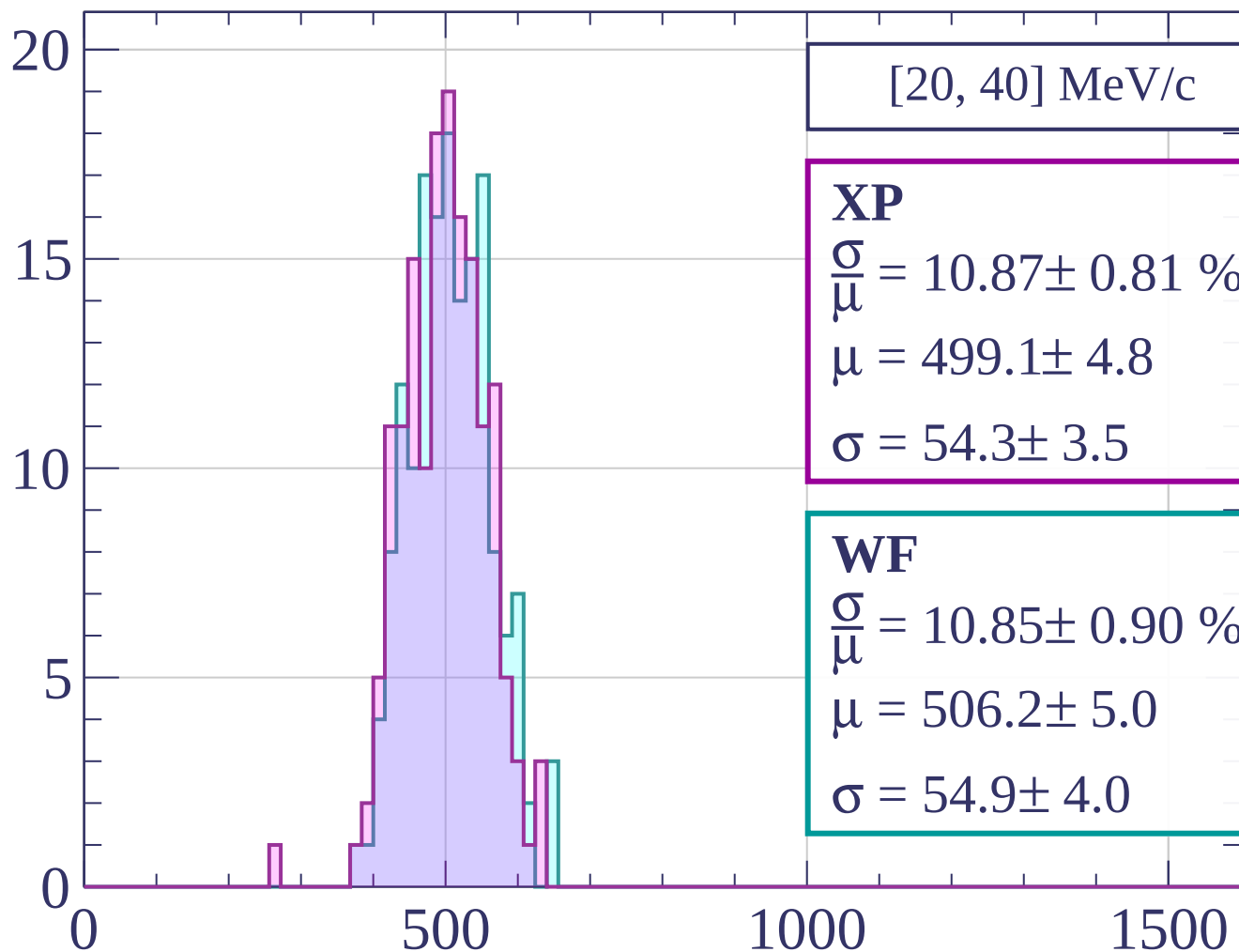
$$\frac{\sigma}{\mu} = 11.91 \pm 0.50 \%$$

$$\mu = 483.5 \pm 2.9$$

$$\sigma = 57.6 \pm 2.1$$

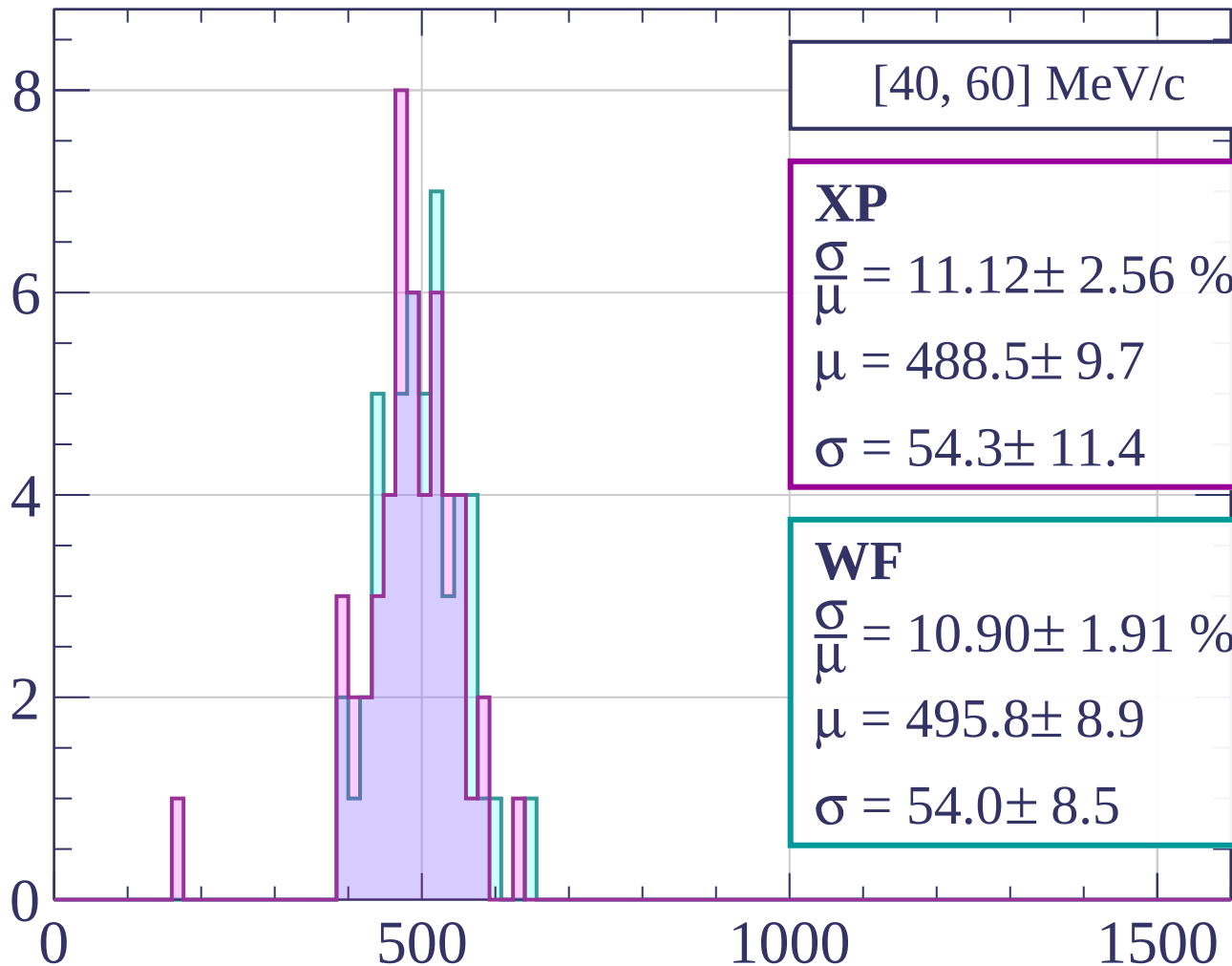
dE/dx [ADC counts/cm]

Count



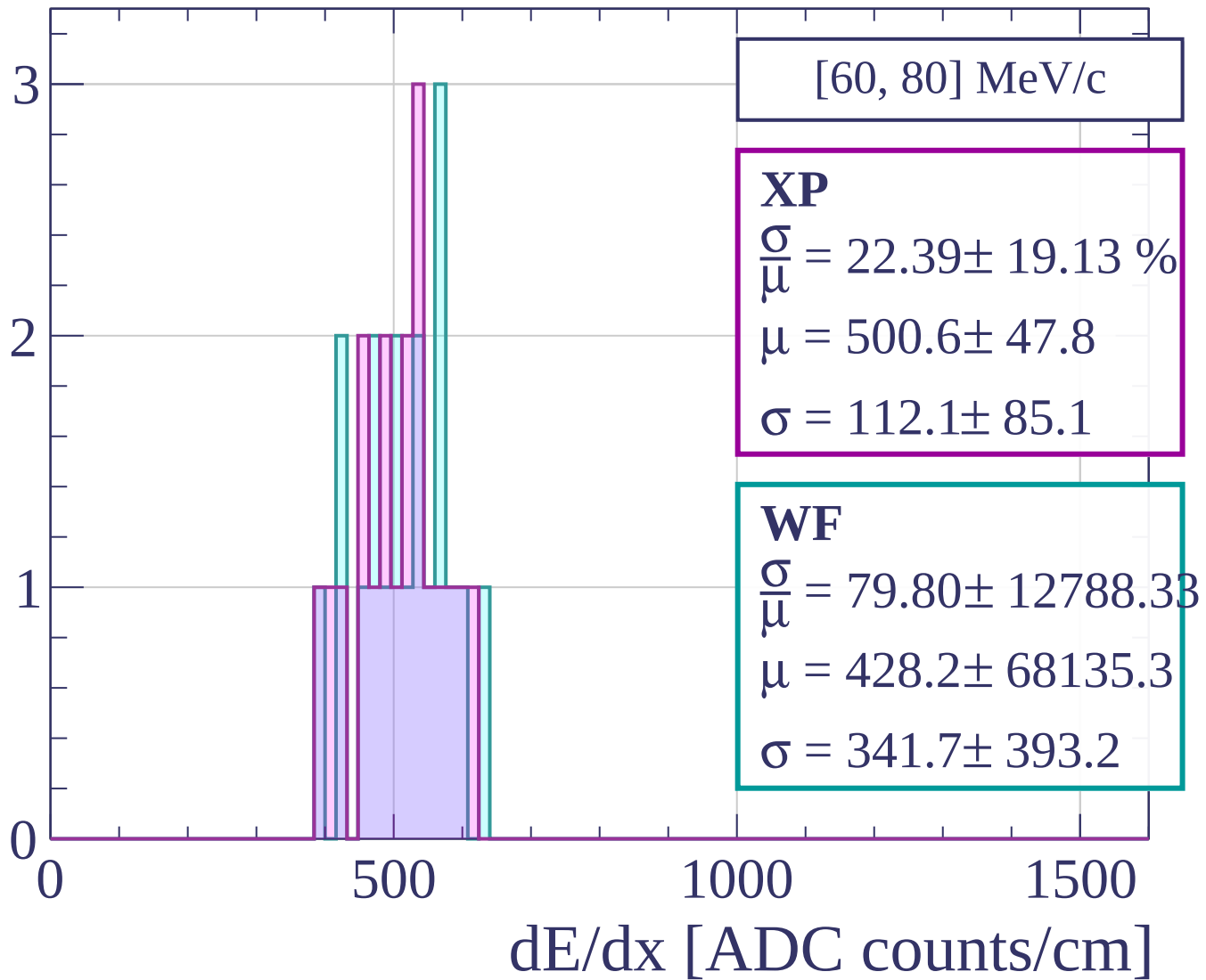
dE/dx [ADC counts/cm]

Count

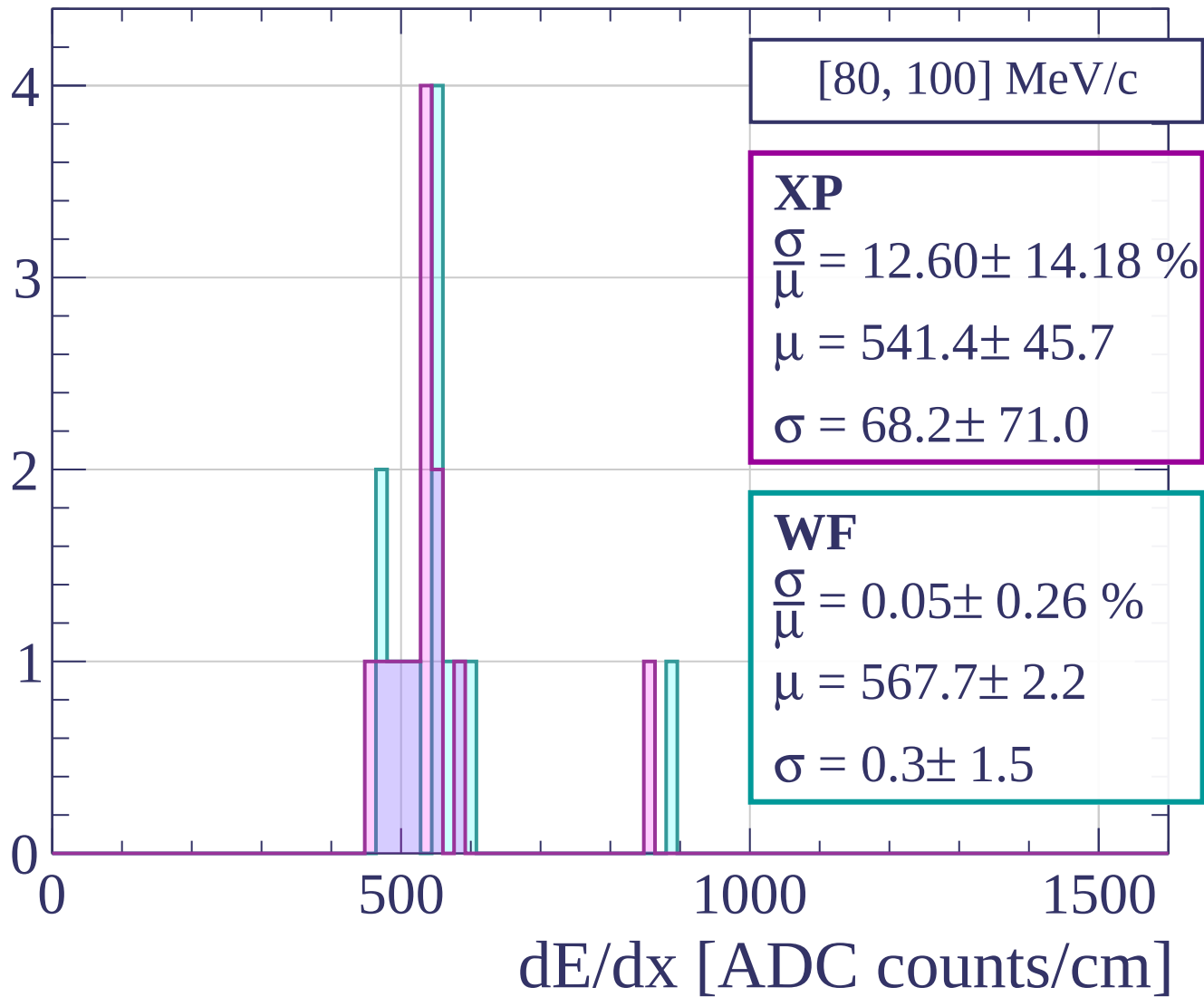


dE/dx [ADC counts/cm]

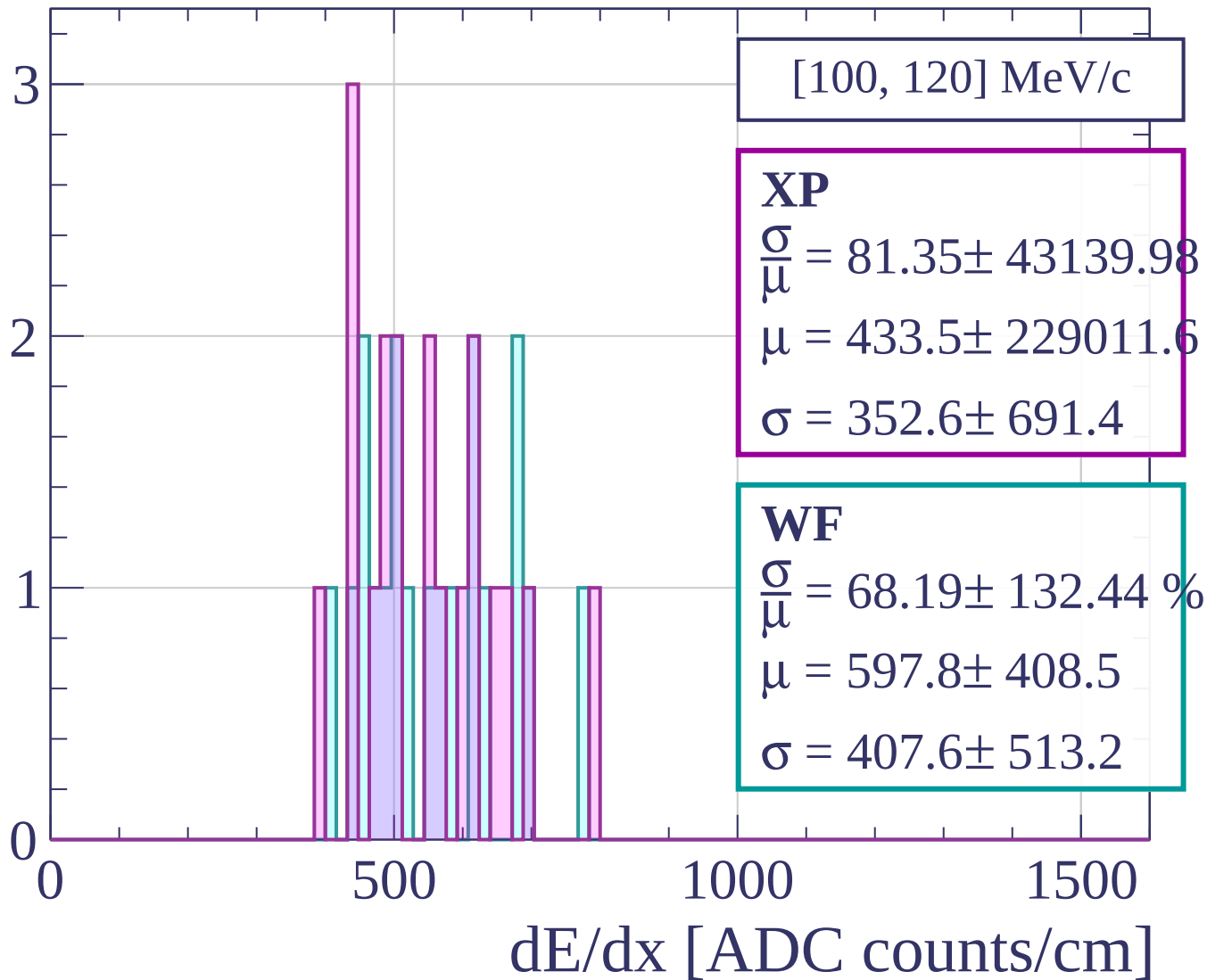
Count



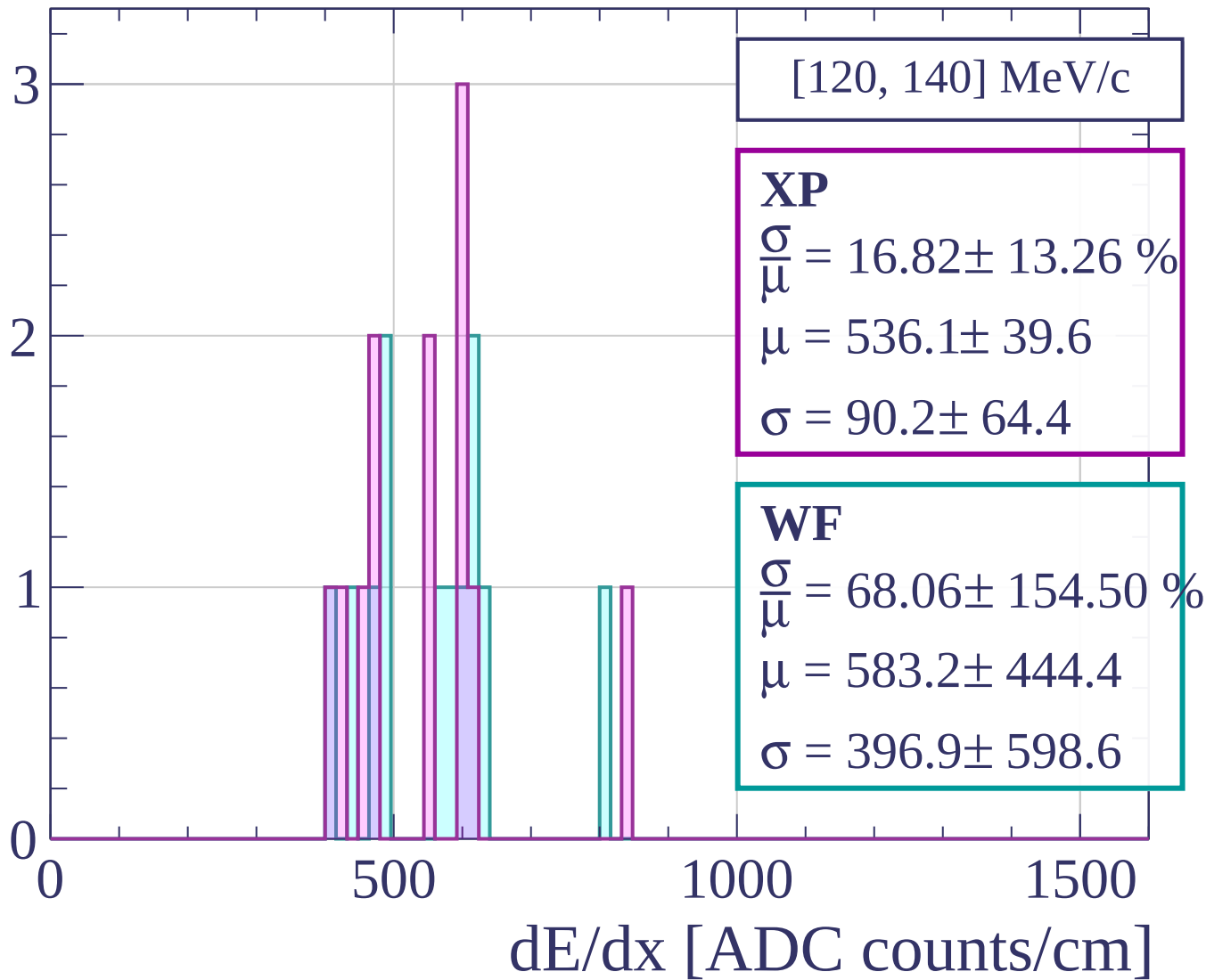
Count



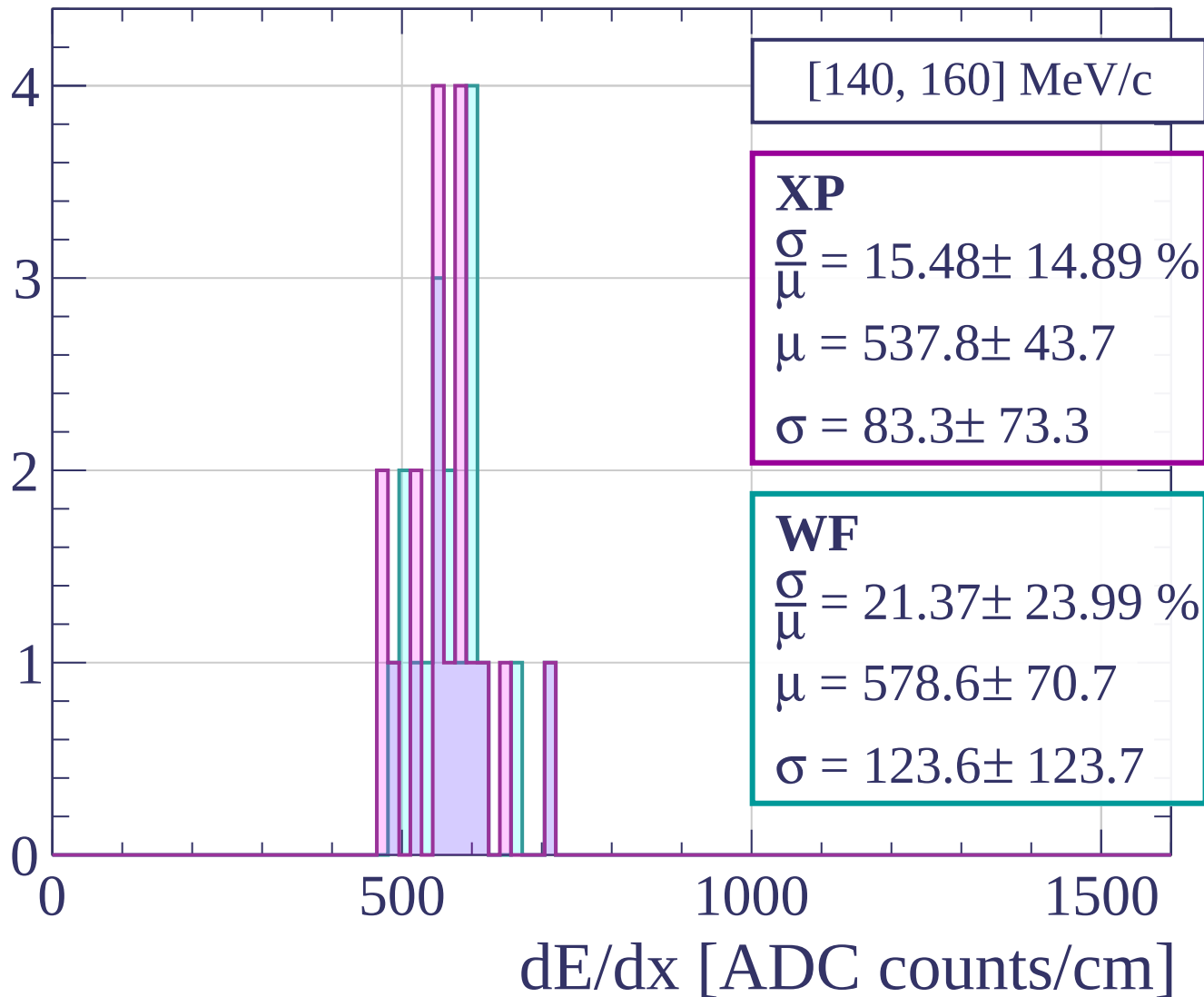
Count



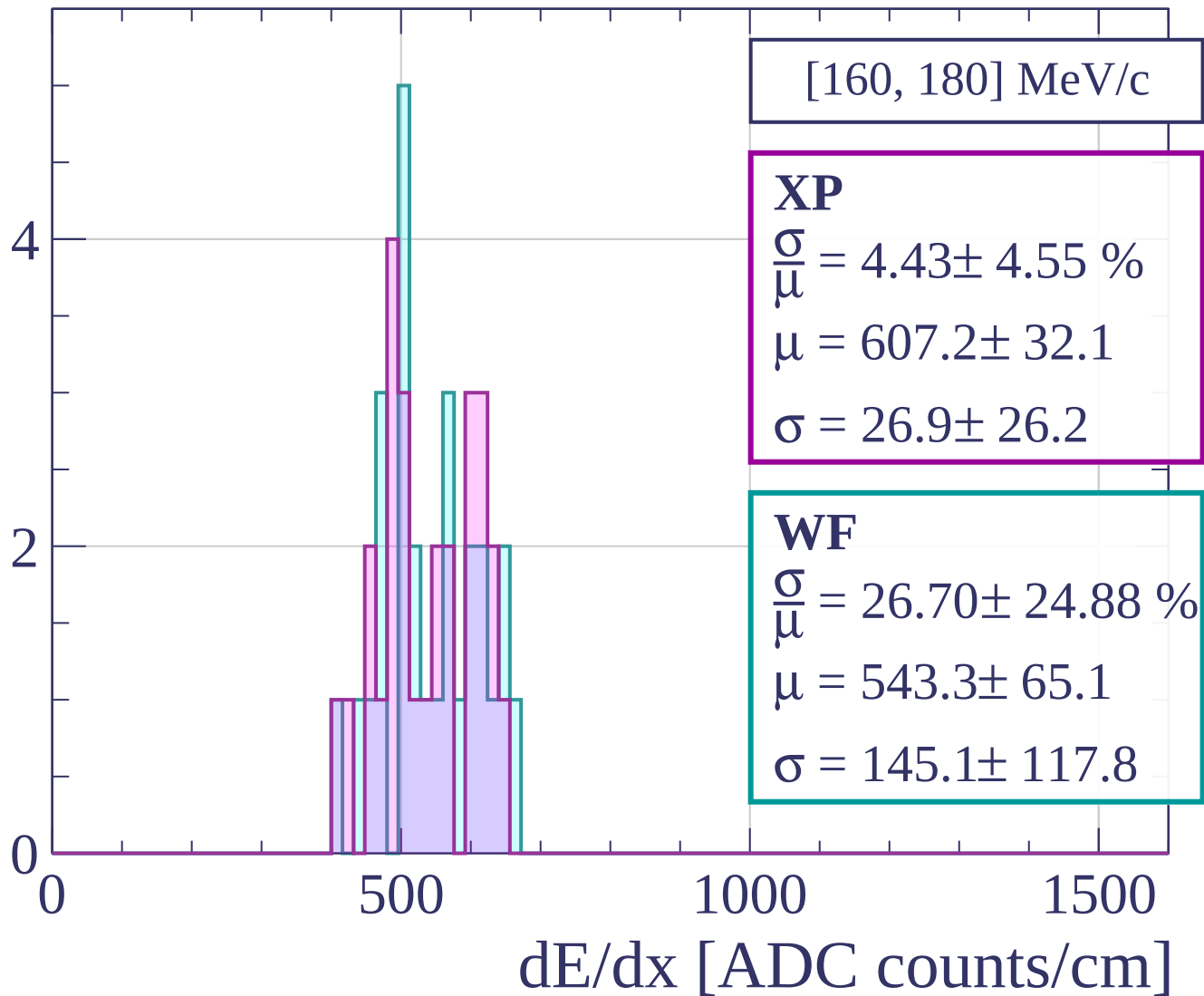
Count



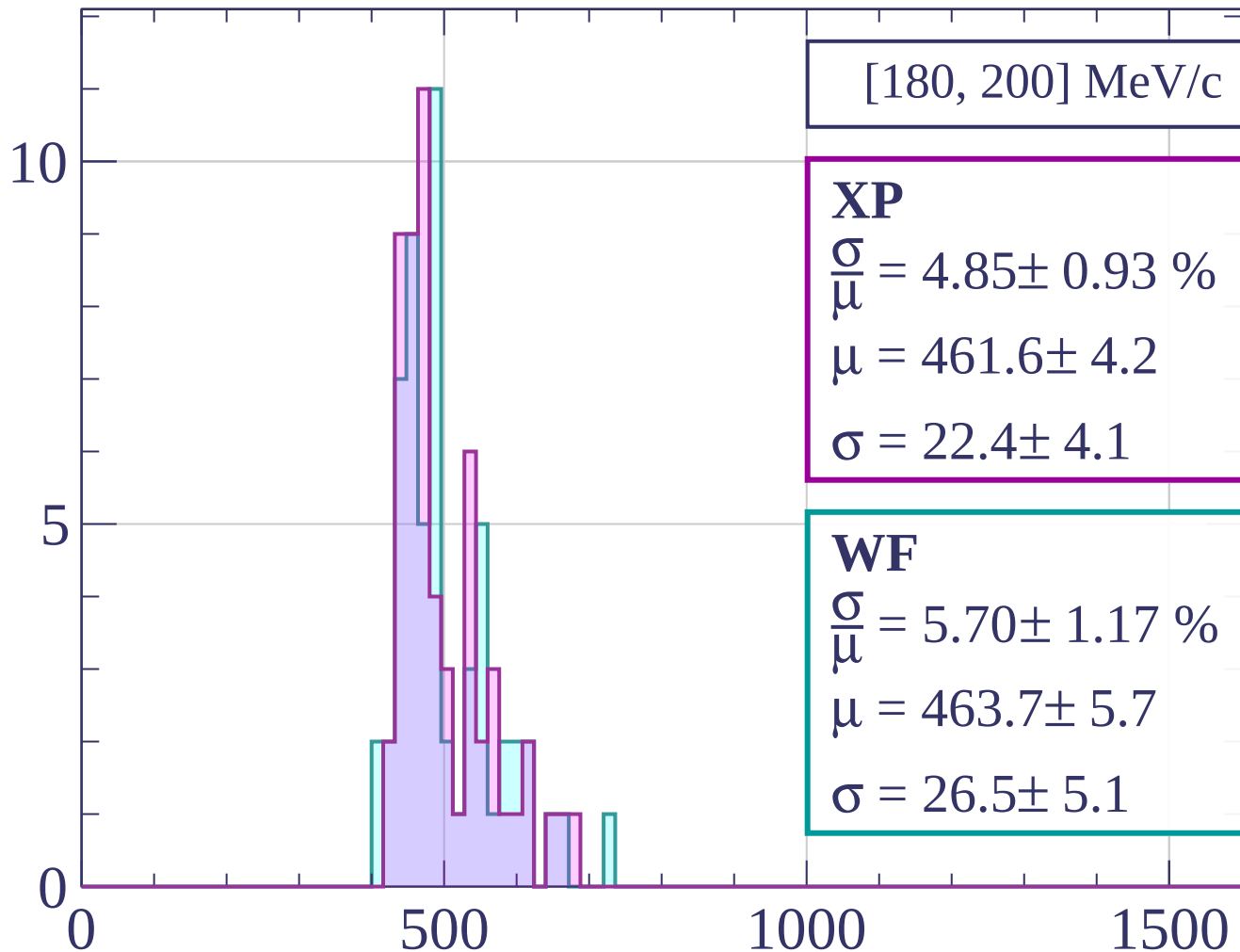
Count



Count



Count



[180, 200] MeV/c

XP

$$\frac{\sigma}{\mu} = 4.85 \pm 0.93 \%$$

$$\mu = 461.6 \pm 4.2$$

$$\sigma = 22.4 \pm 4.1$$

WF

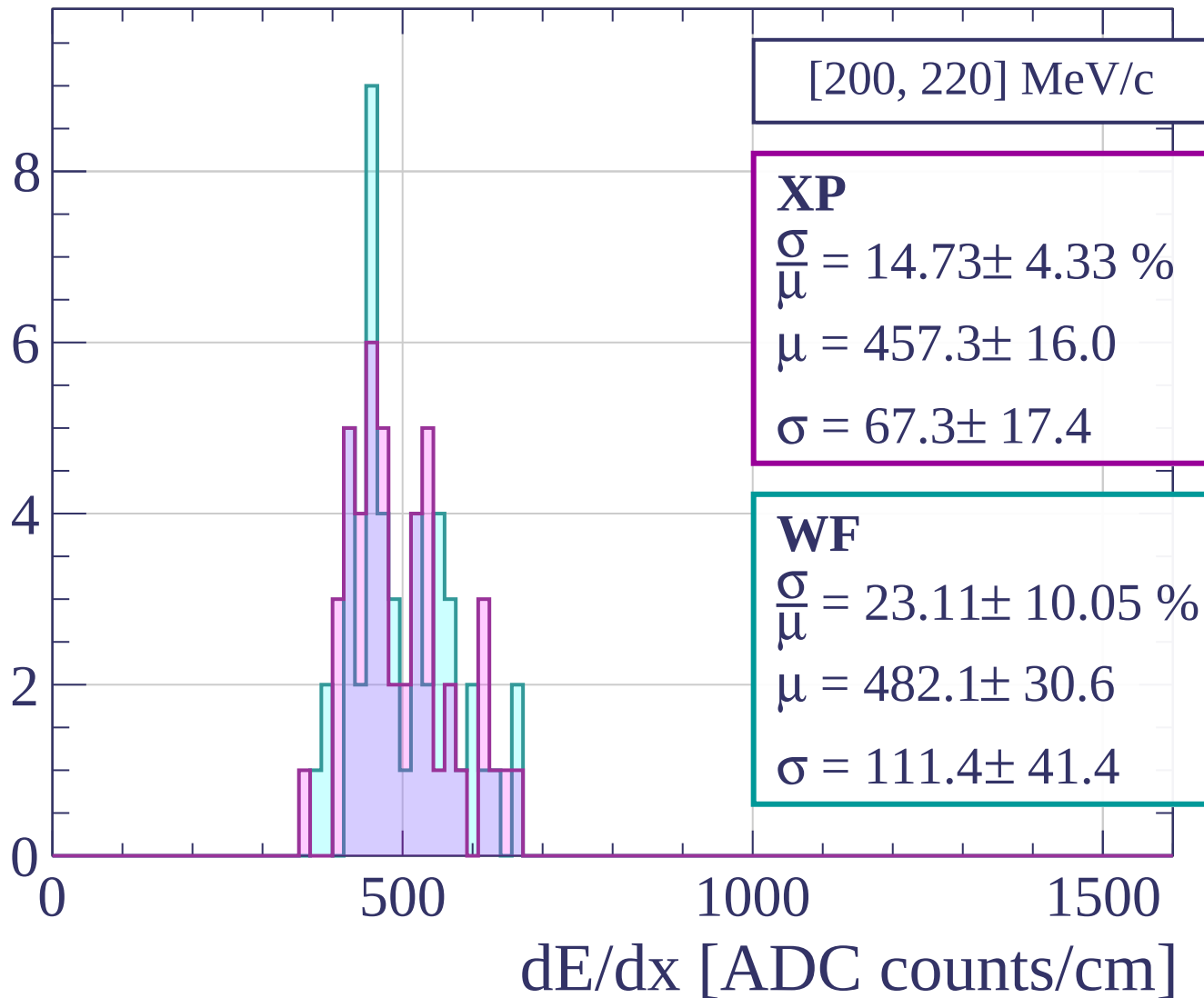
$$\frac{\sigma}{\mu} = 5.70 \pm 1.17 \%$$

$$\mu = 463.7 \pm 5.7$$

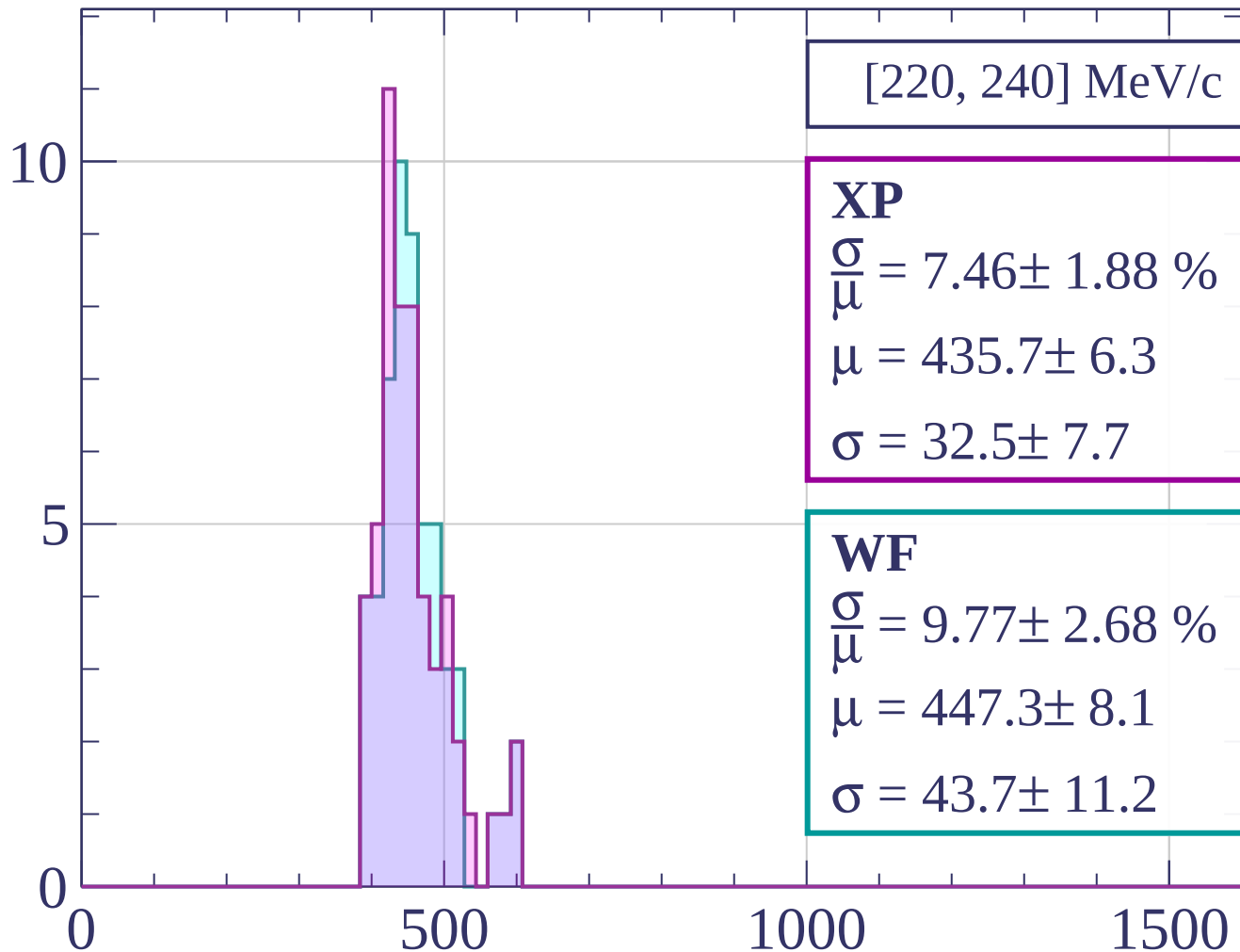
$$\sigma = 26.5 \pm 5.1$$

dE/dx [ADC counts/cm]

Count

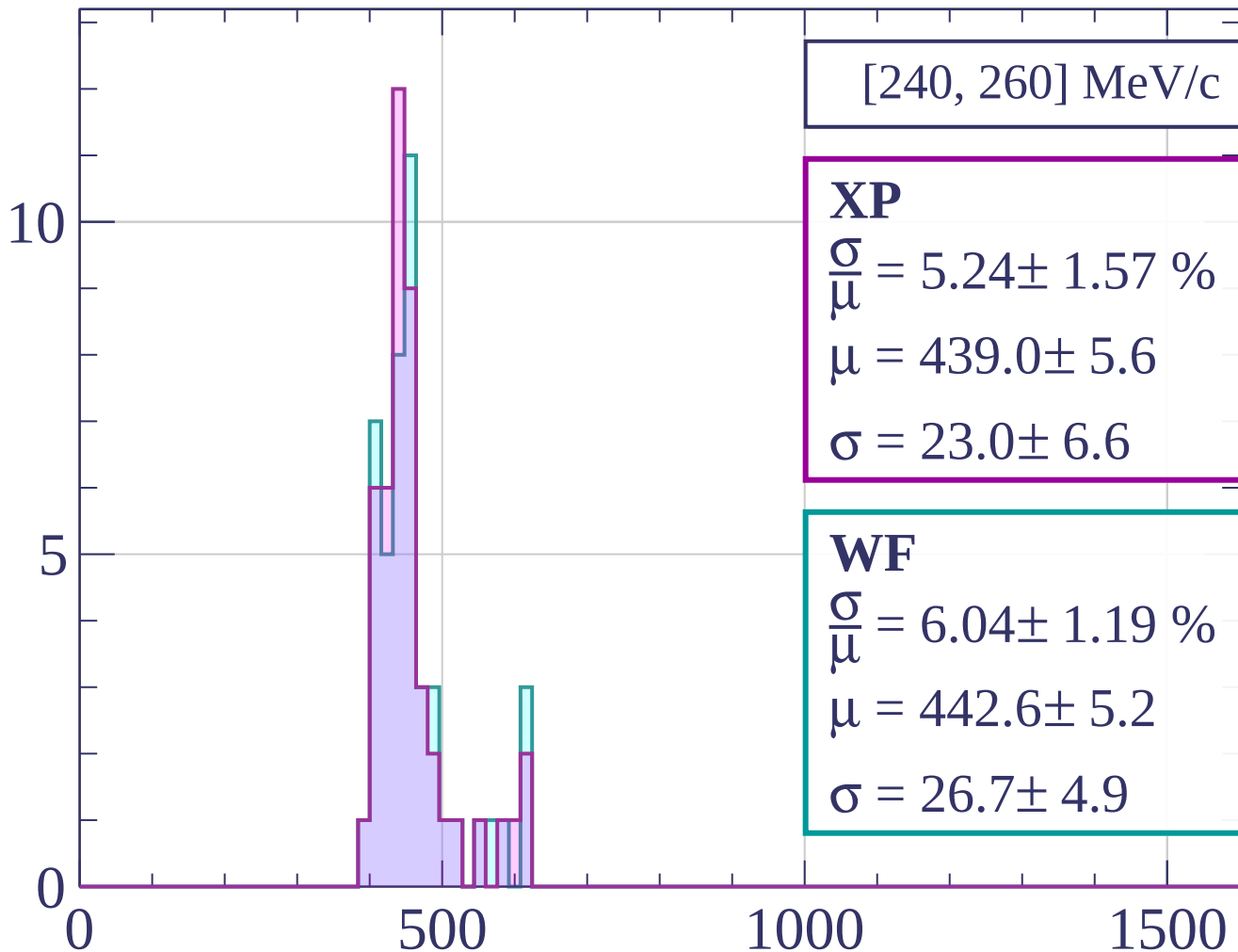


Count



dE/dx [ADC counts/cm]

Count



[240, 260] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.24 \pm 1.57 \%$$

$$\mu = 439.0 \pm 5.6$$

$$\sigma = 23.0 \pm 6.6$$

WF

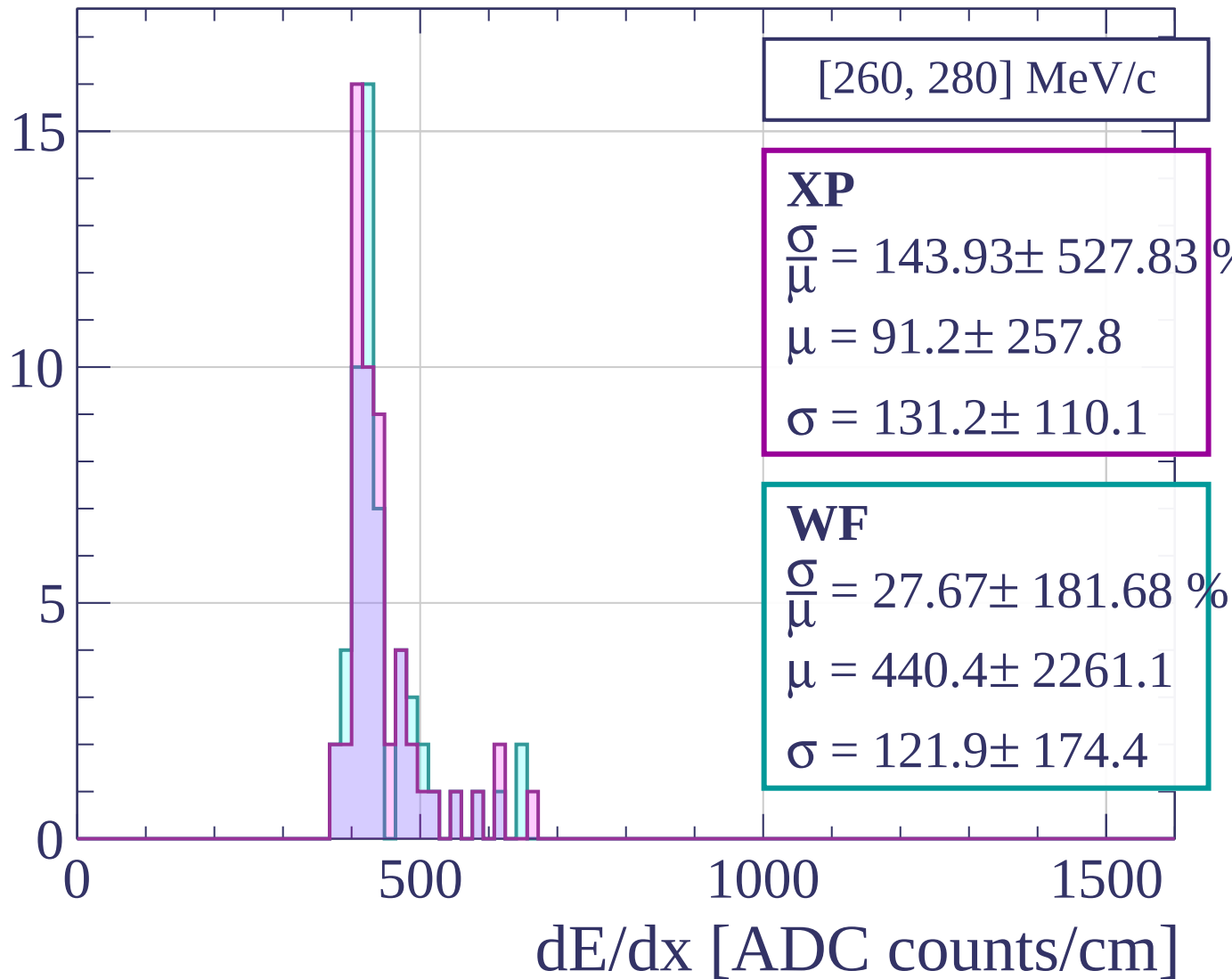
$$\frac{\sigma}{\mu} = 6.04 \pm 1.19 \%$$

$$\mu = 442.6 \pm 5.2$$

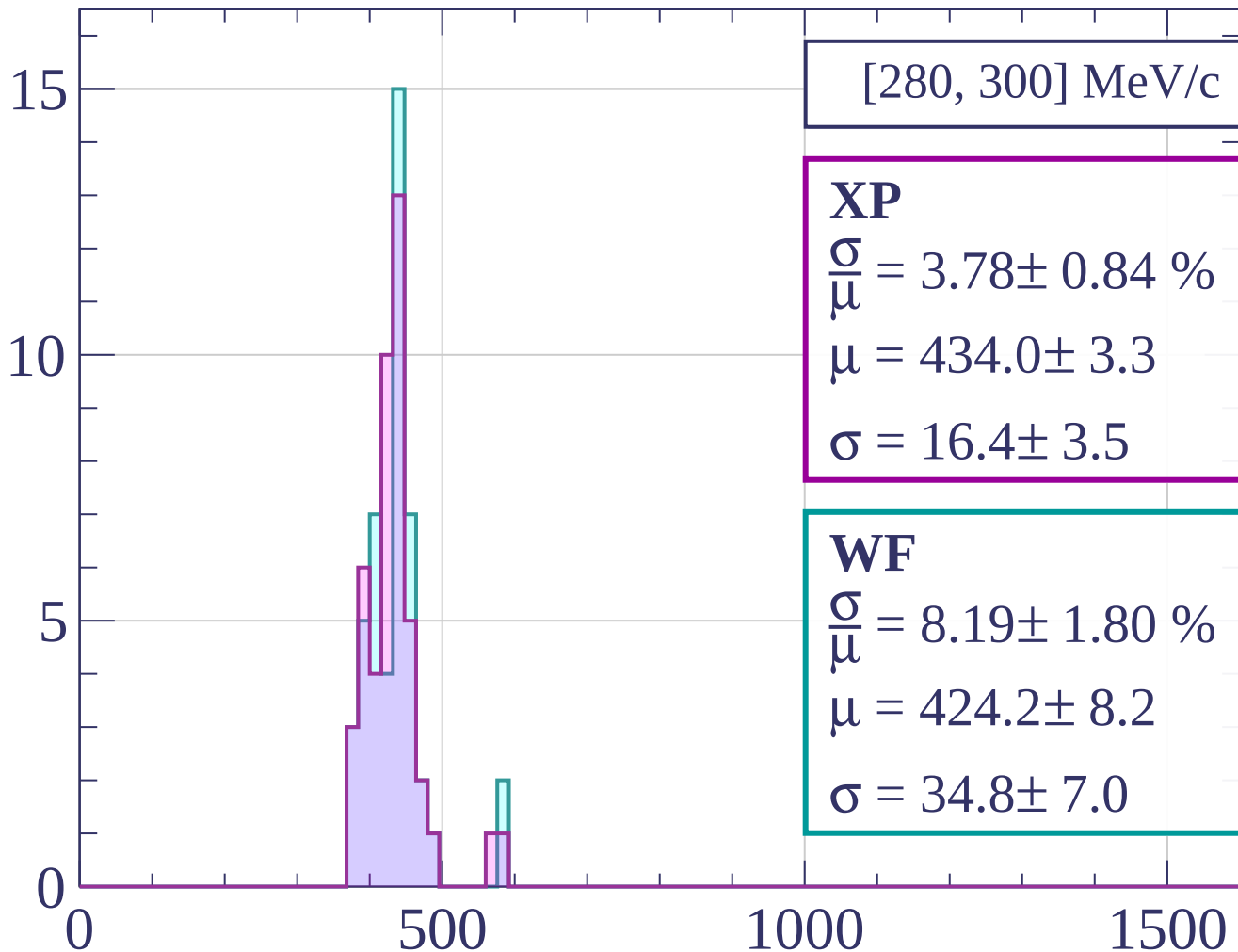
$$\sigma = 26.7 \pm 4.9$$

dE/dx [ADC counts/cm]

Count



Count



[280, 300] MeV/c

XP

$$\frac{\sigma}{\mu} = 3.78 \pm 0.84 \%$$

$$\mu = 434.0 \pm 3.3$$

$$\sigma = 16.4 \pm 3.5$$

WF

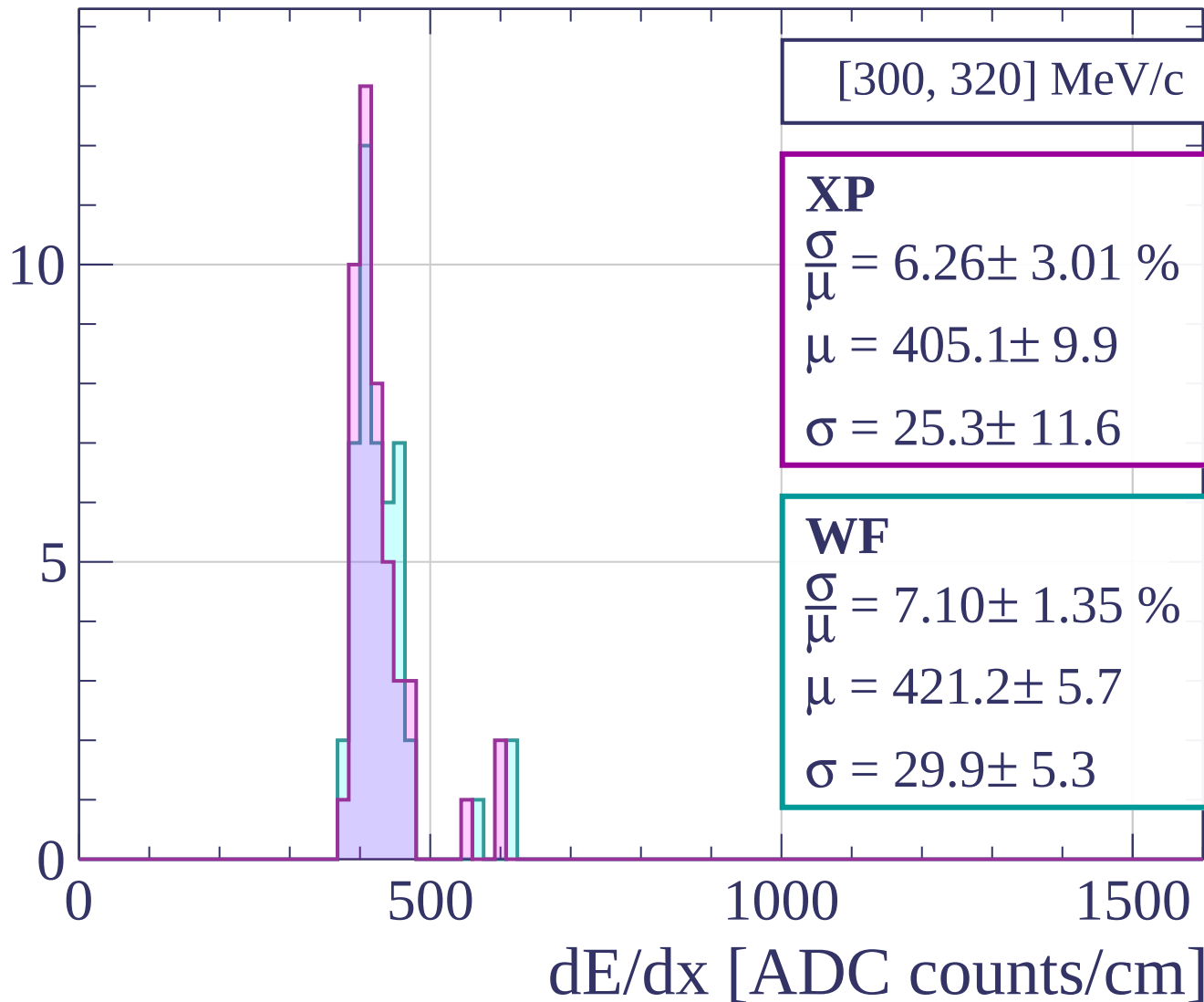
$$\frac{\sigma}{\mu} = 8.19 \pm 1.80 \%$$

$$\mu = 424.2 \pm 8.2$$

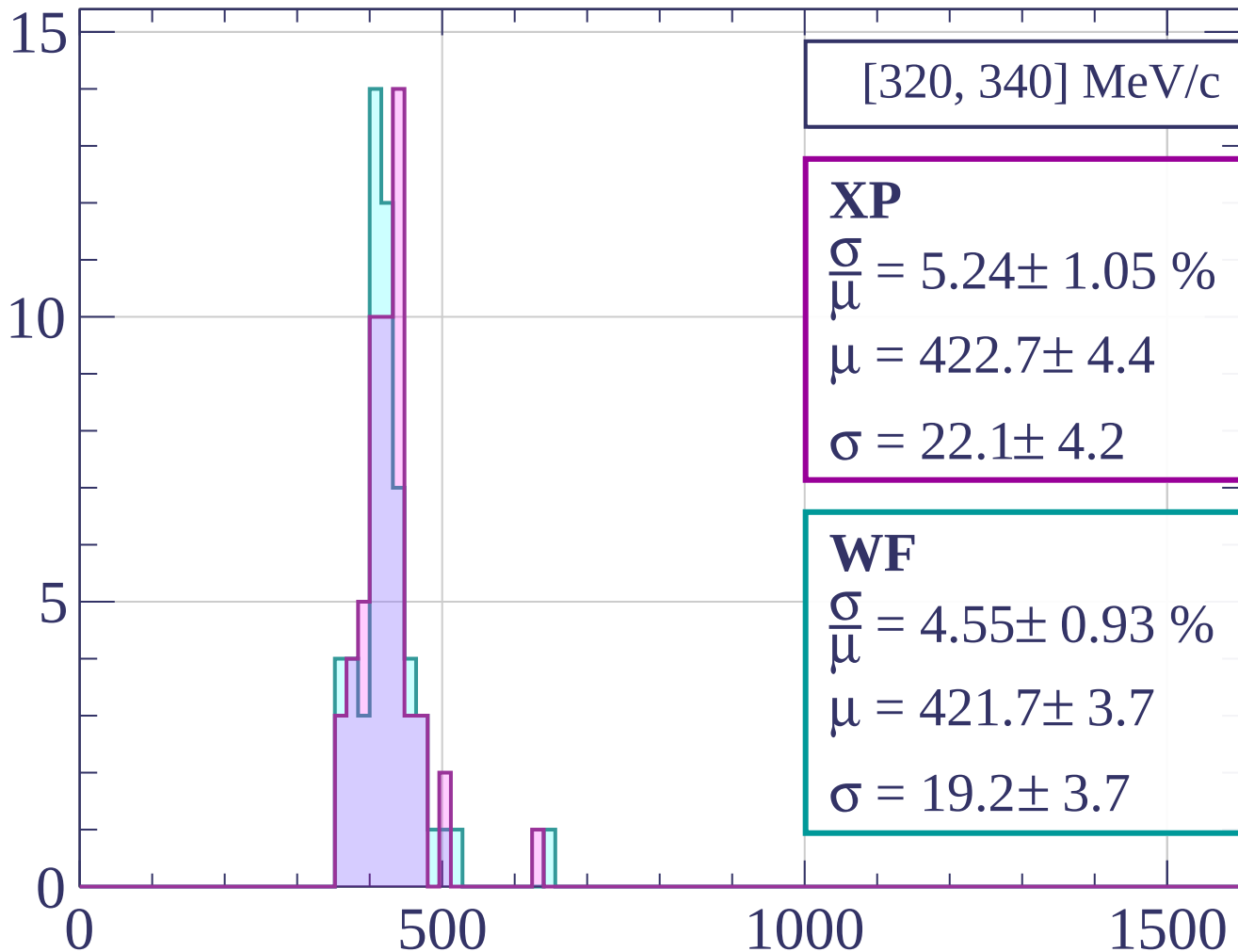
$$\sigma = 34.8 \pm 7.0$$

dE/dx [ADC counts/cm]

Count

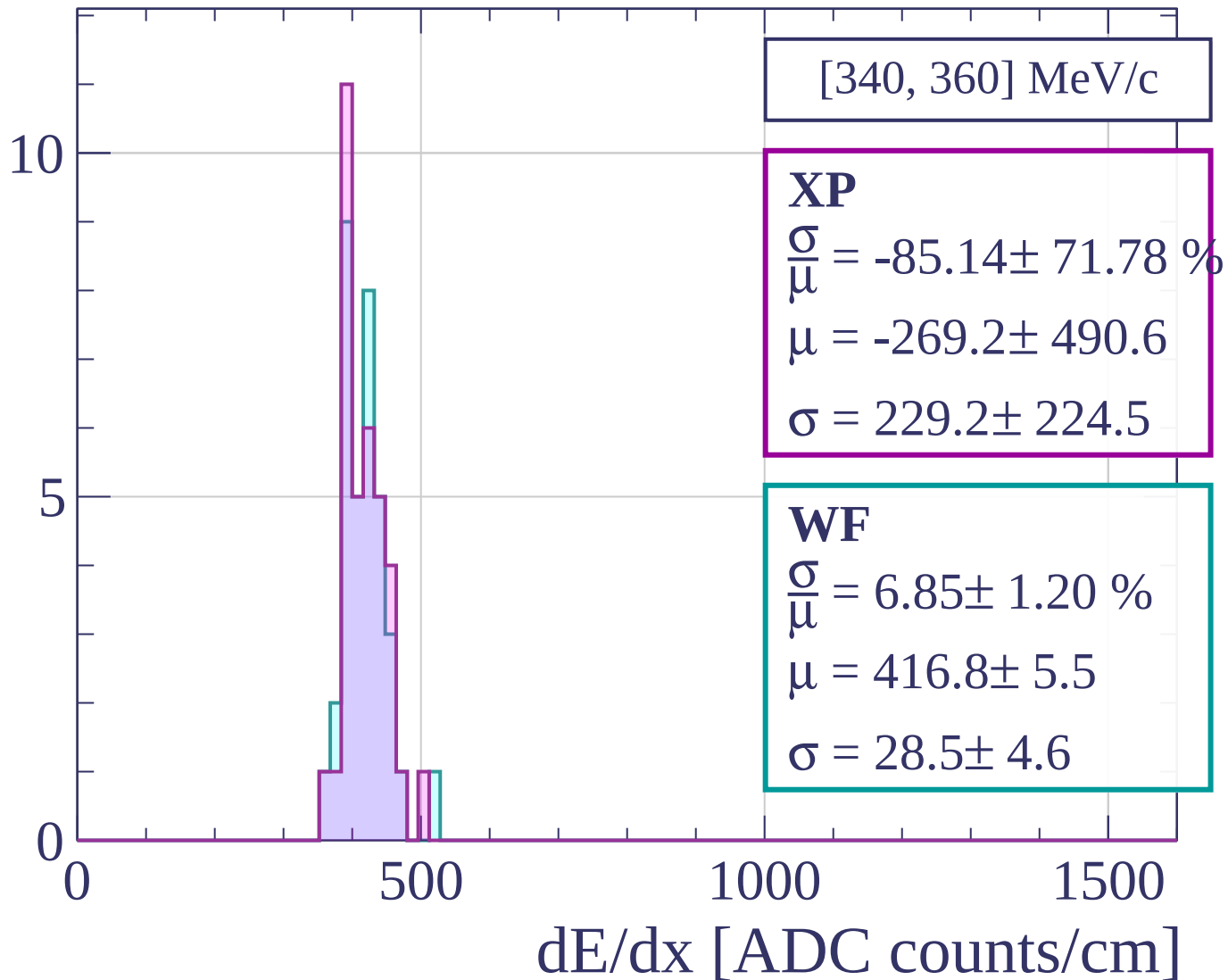


Count

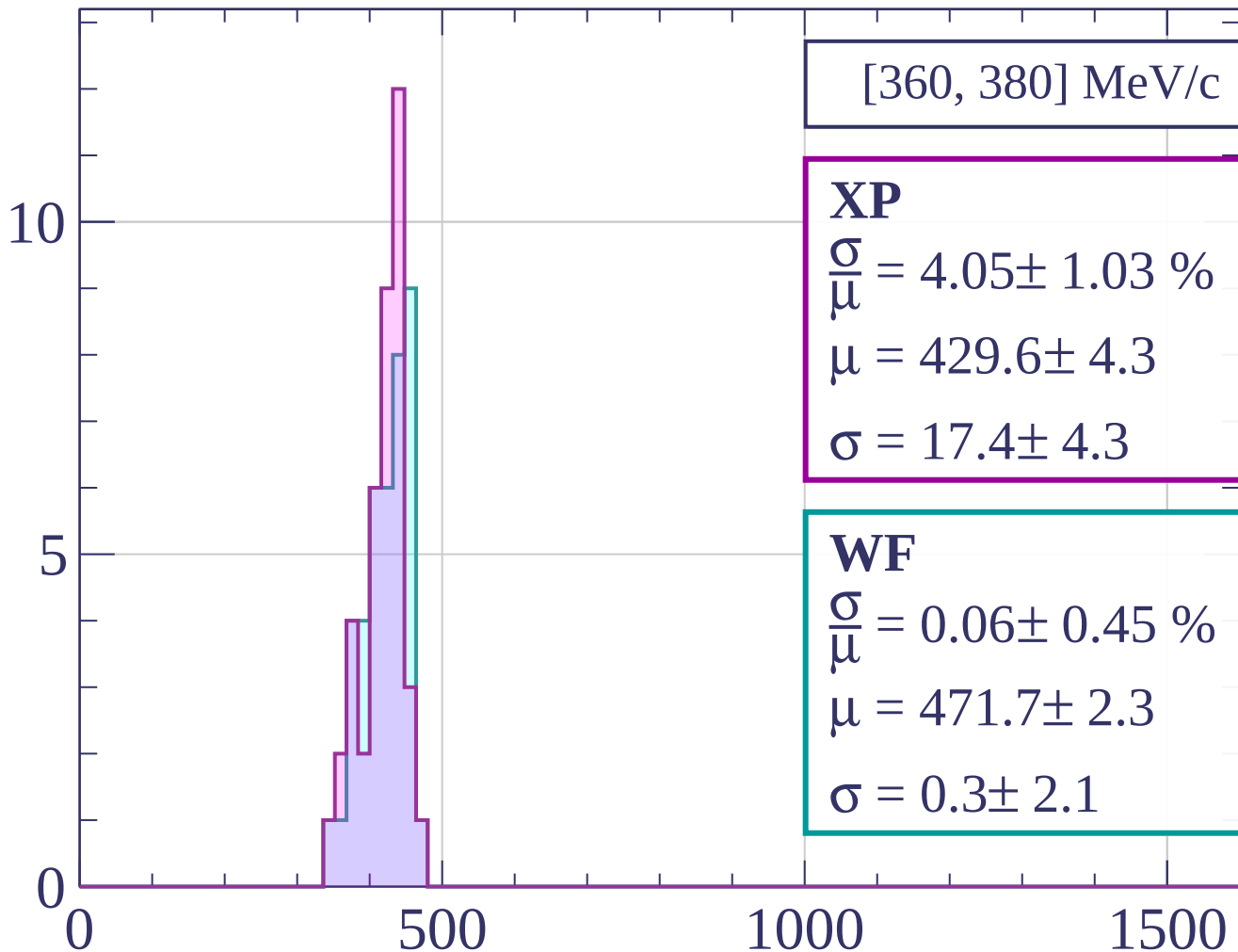


dE/dx [ADC counts/cm]

Count

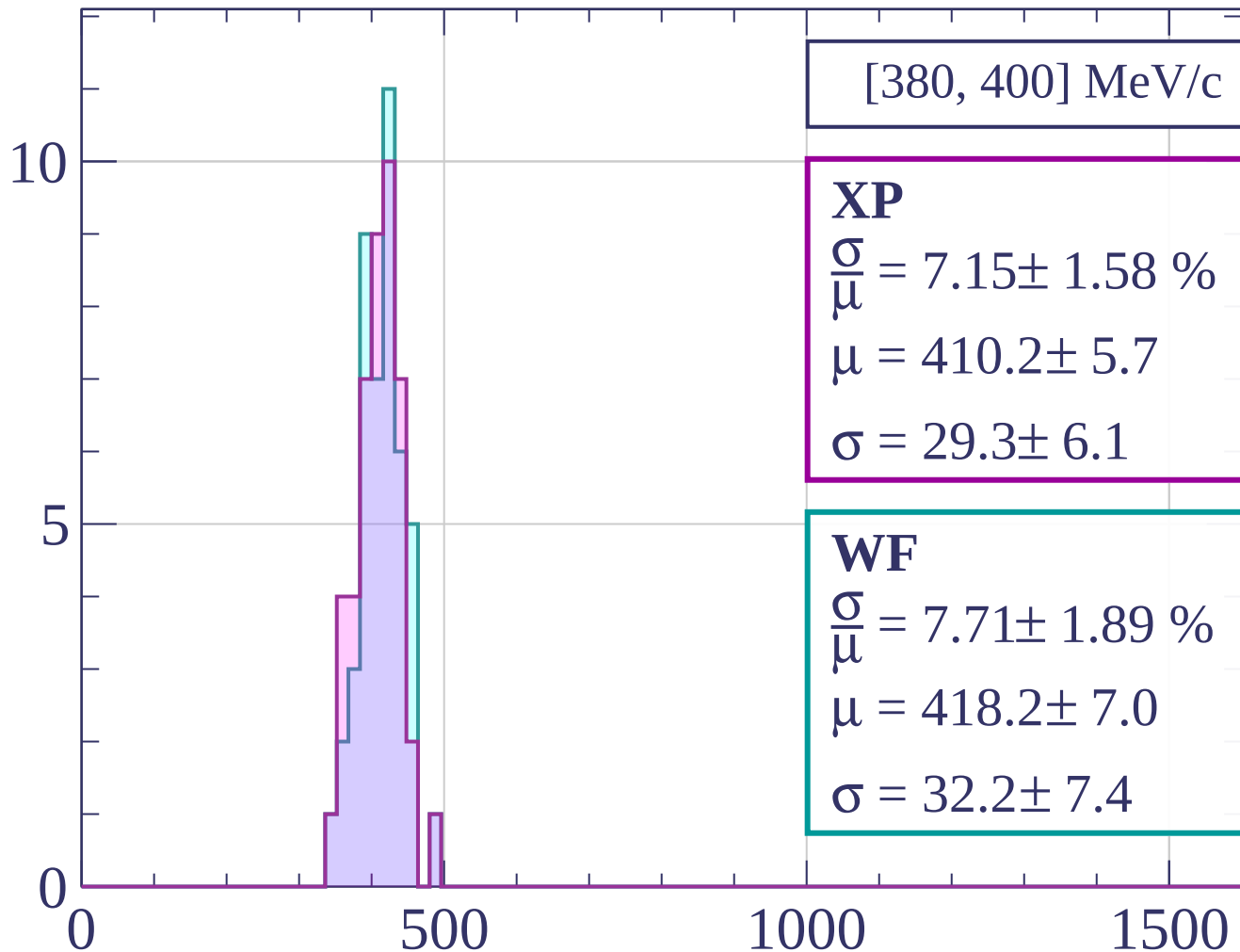


Count



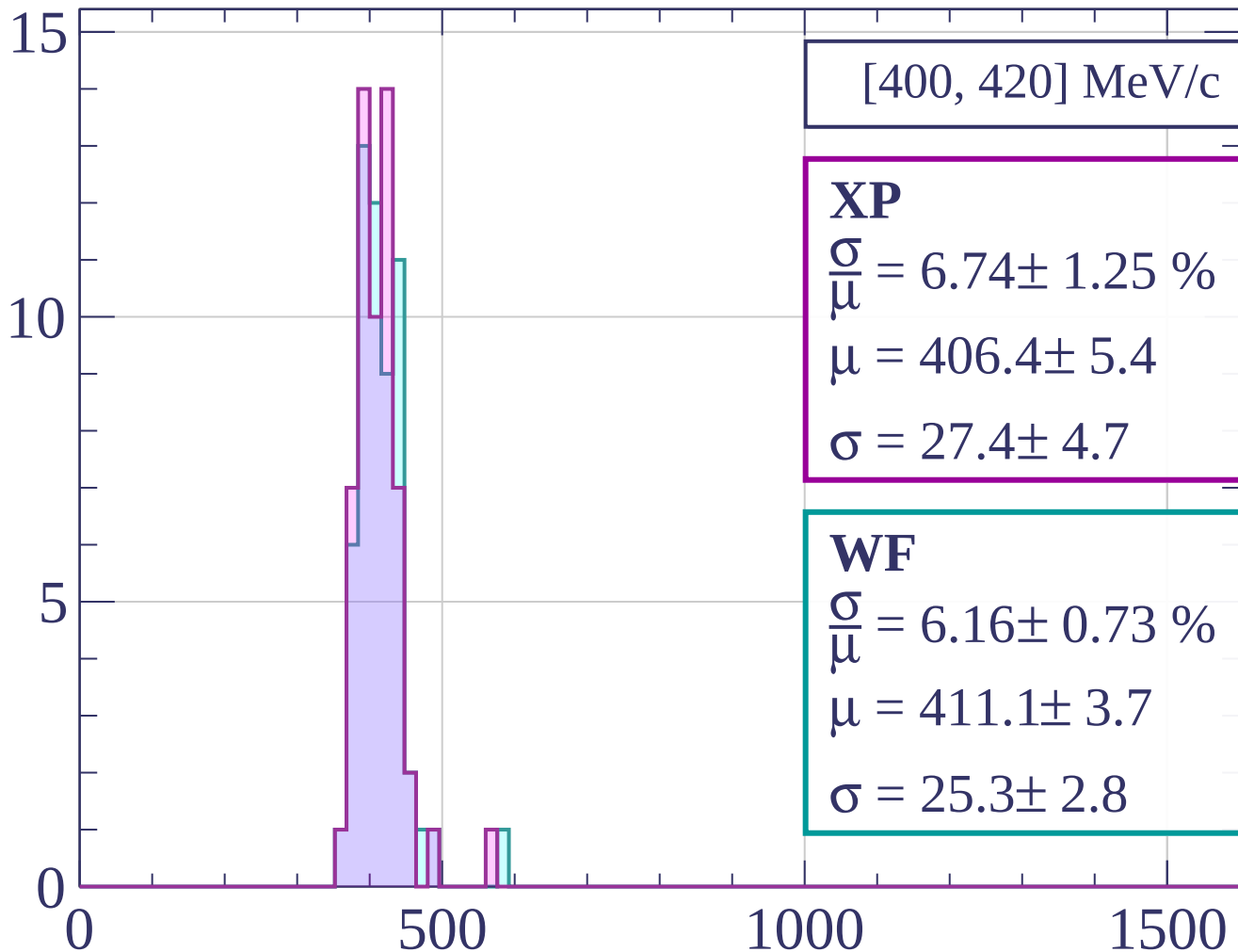
dE/dx [ADC counts/cm]

Count



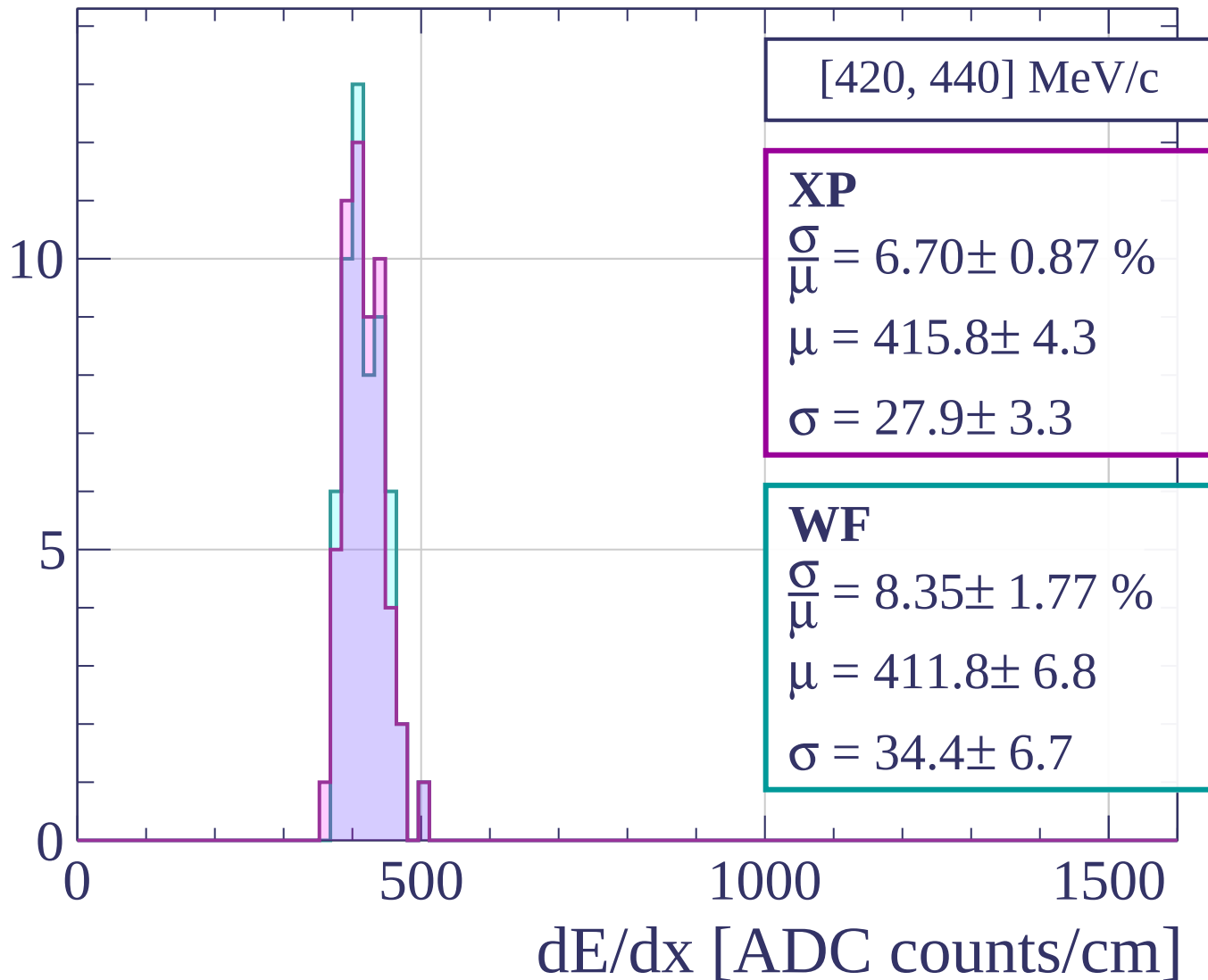
dE/dx [ADC counts/cm]

Count

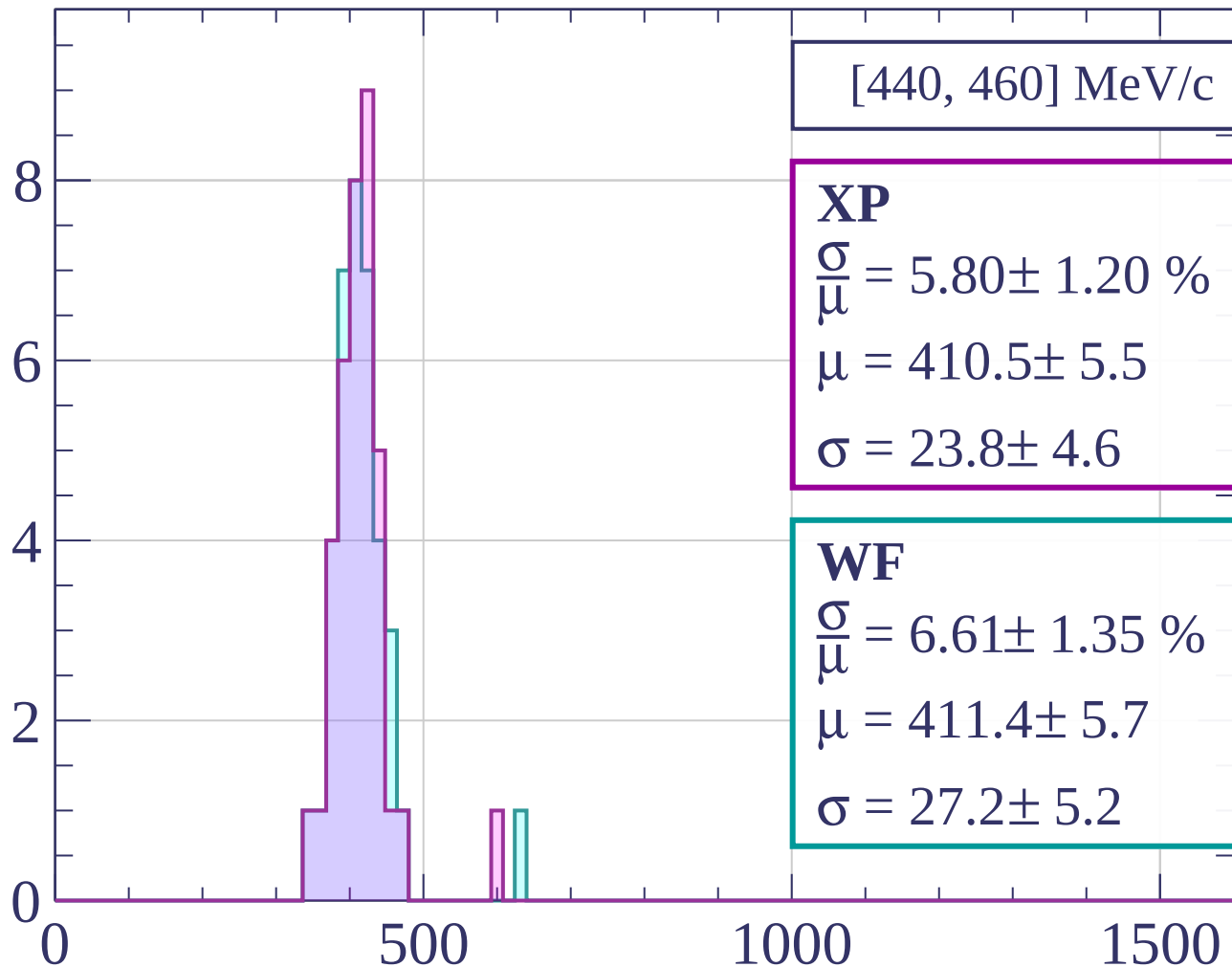


dE/dx [ADC counts/cm]

Count

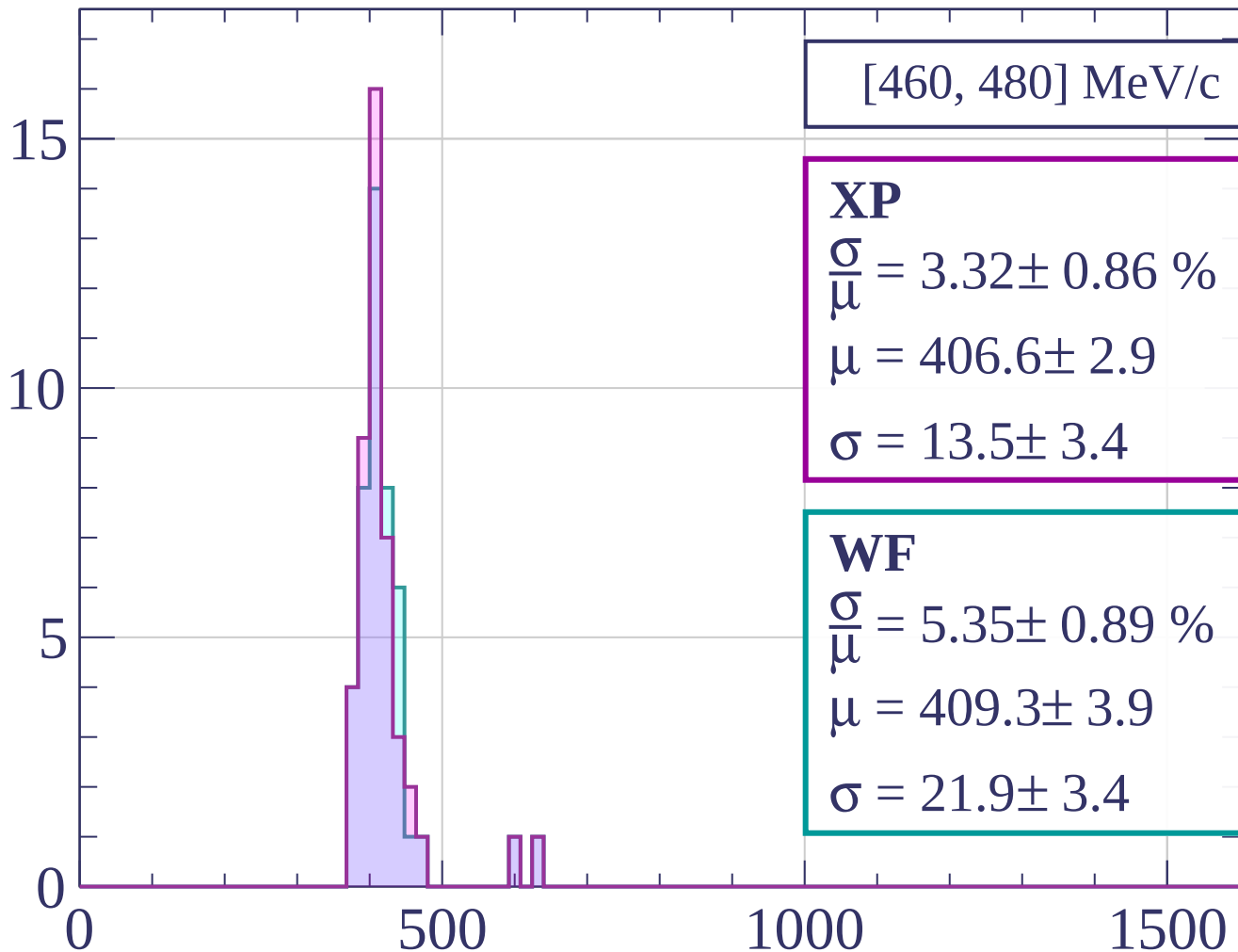


Count



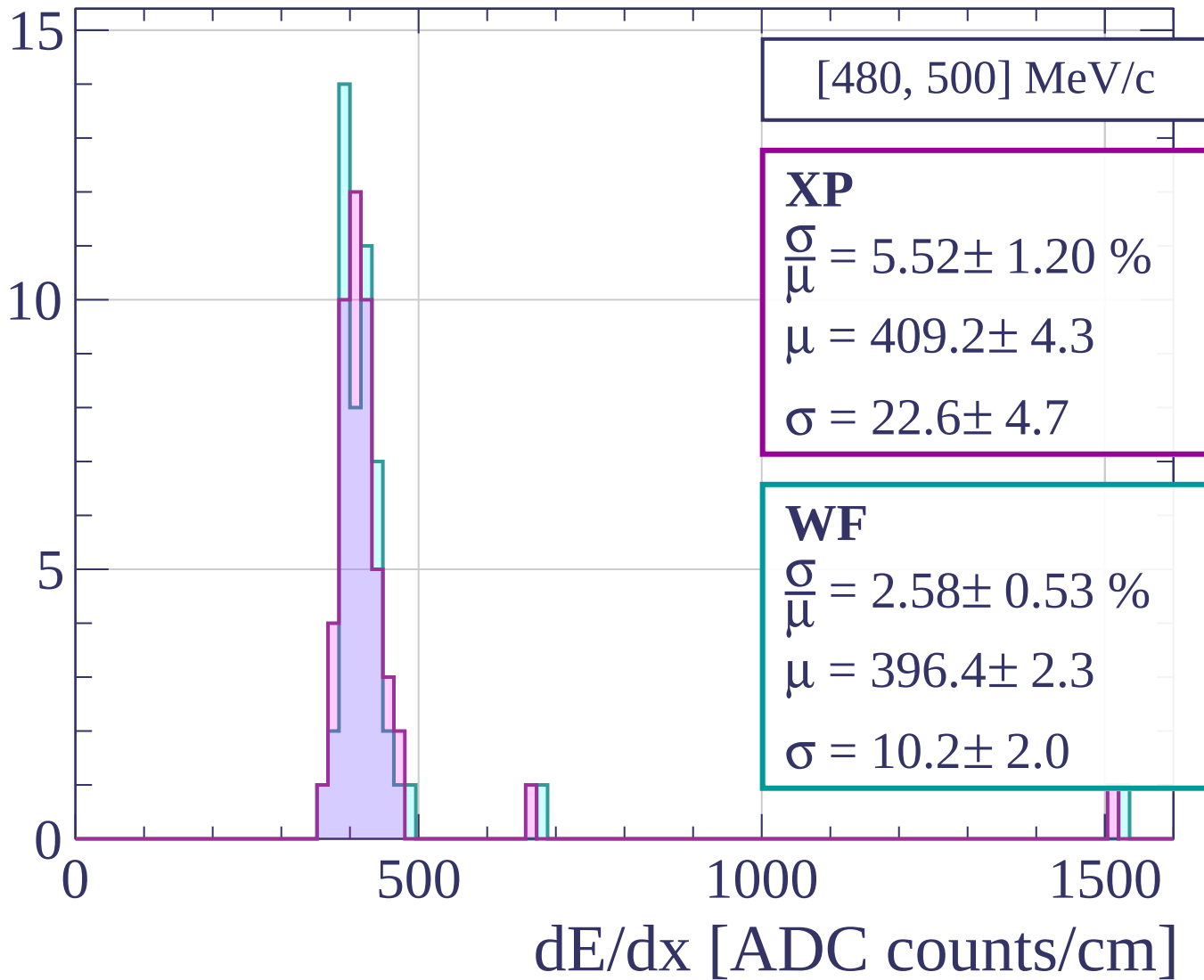
dE/dx [ADC counts/cm]

Count

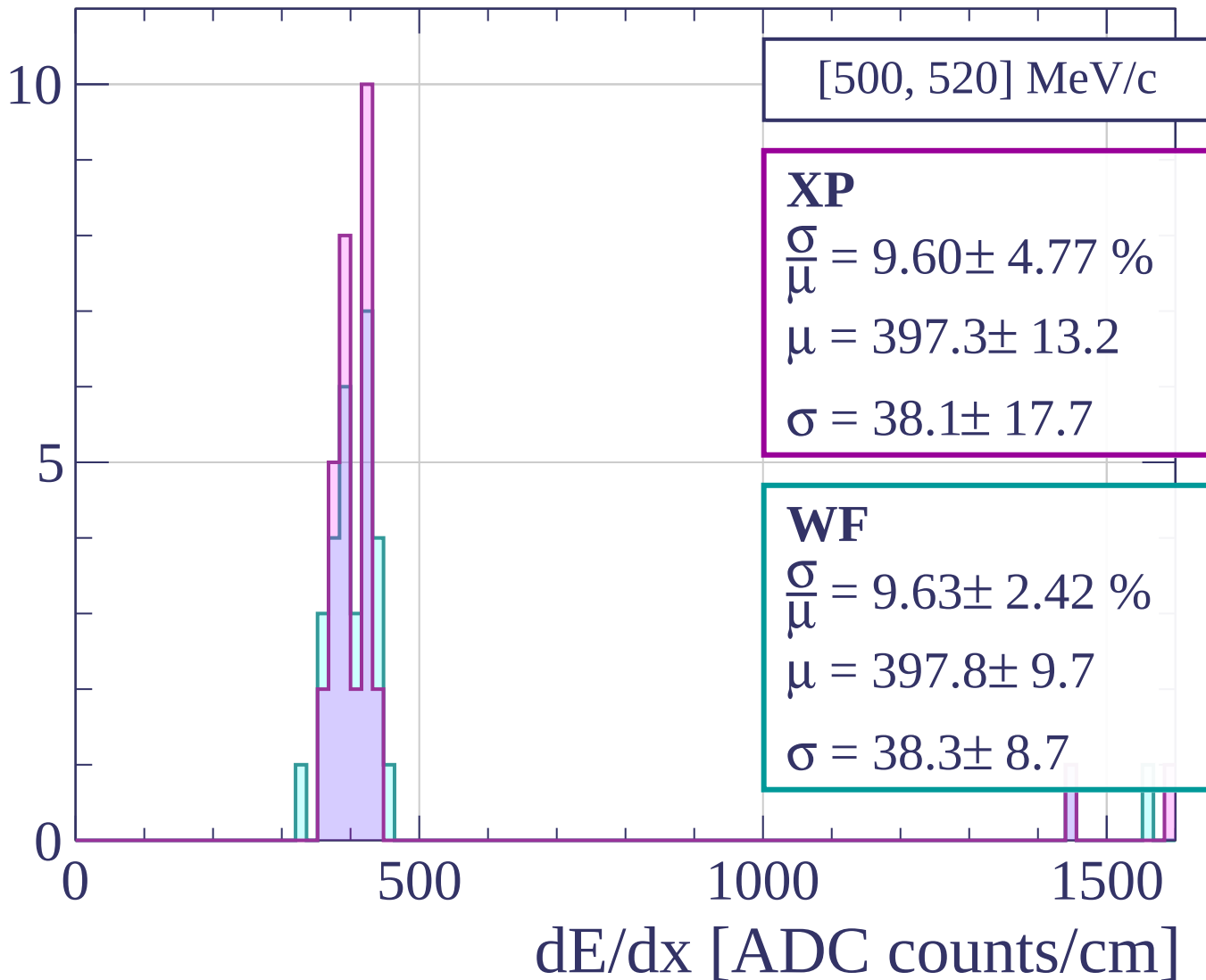


dE/dx [ADC counts/cm]

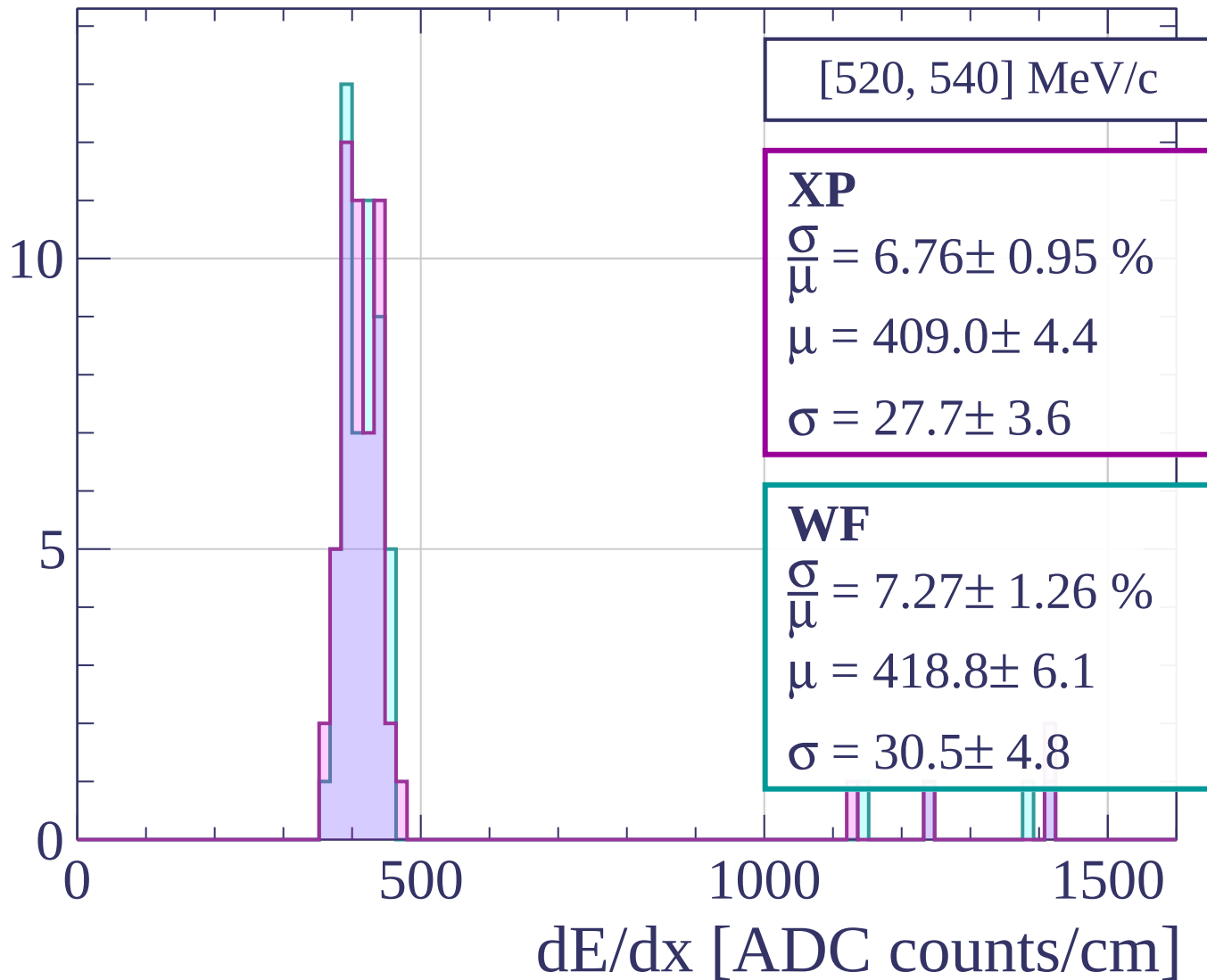
Count



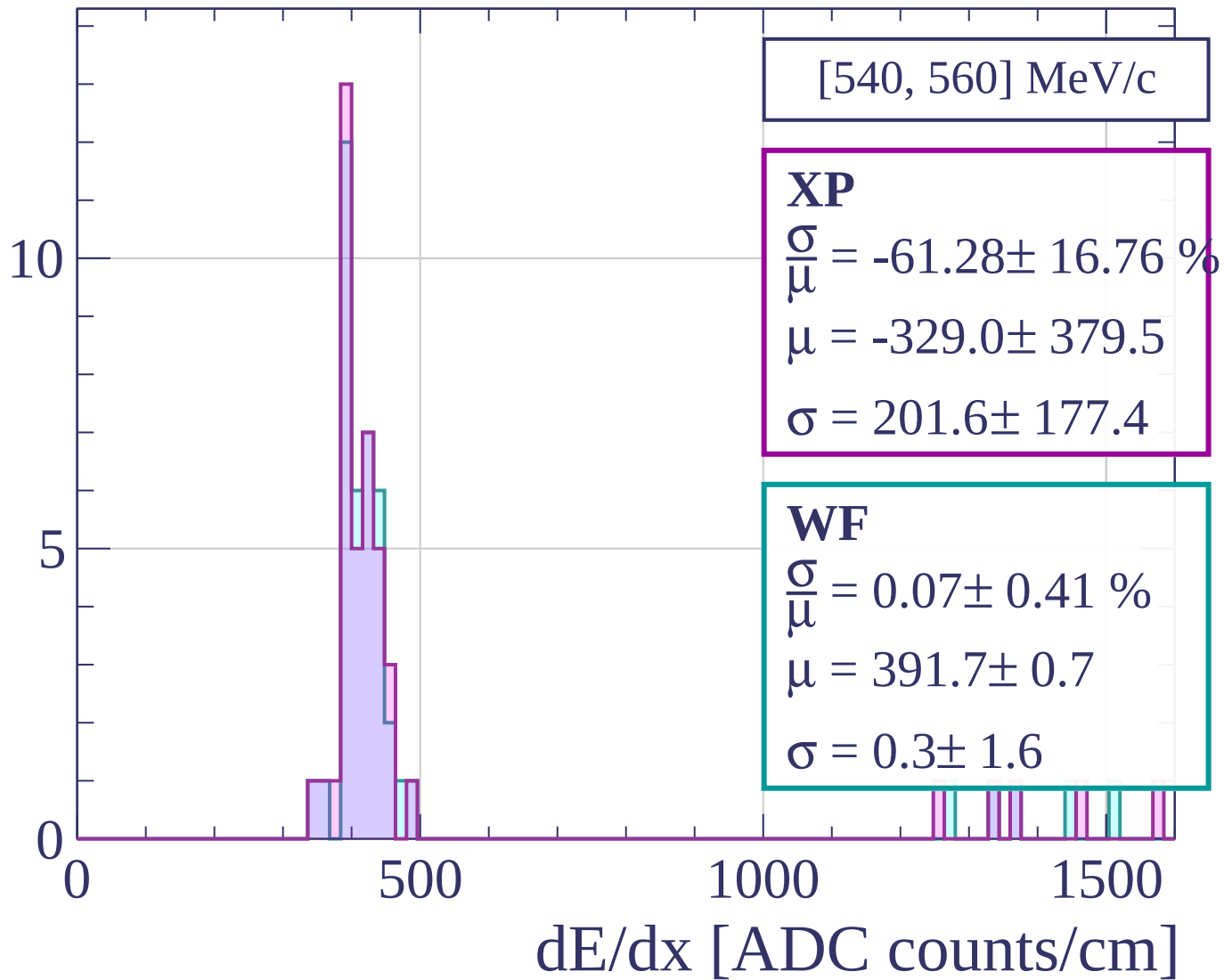
Count



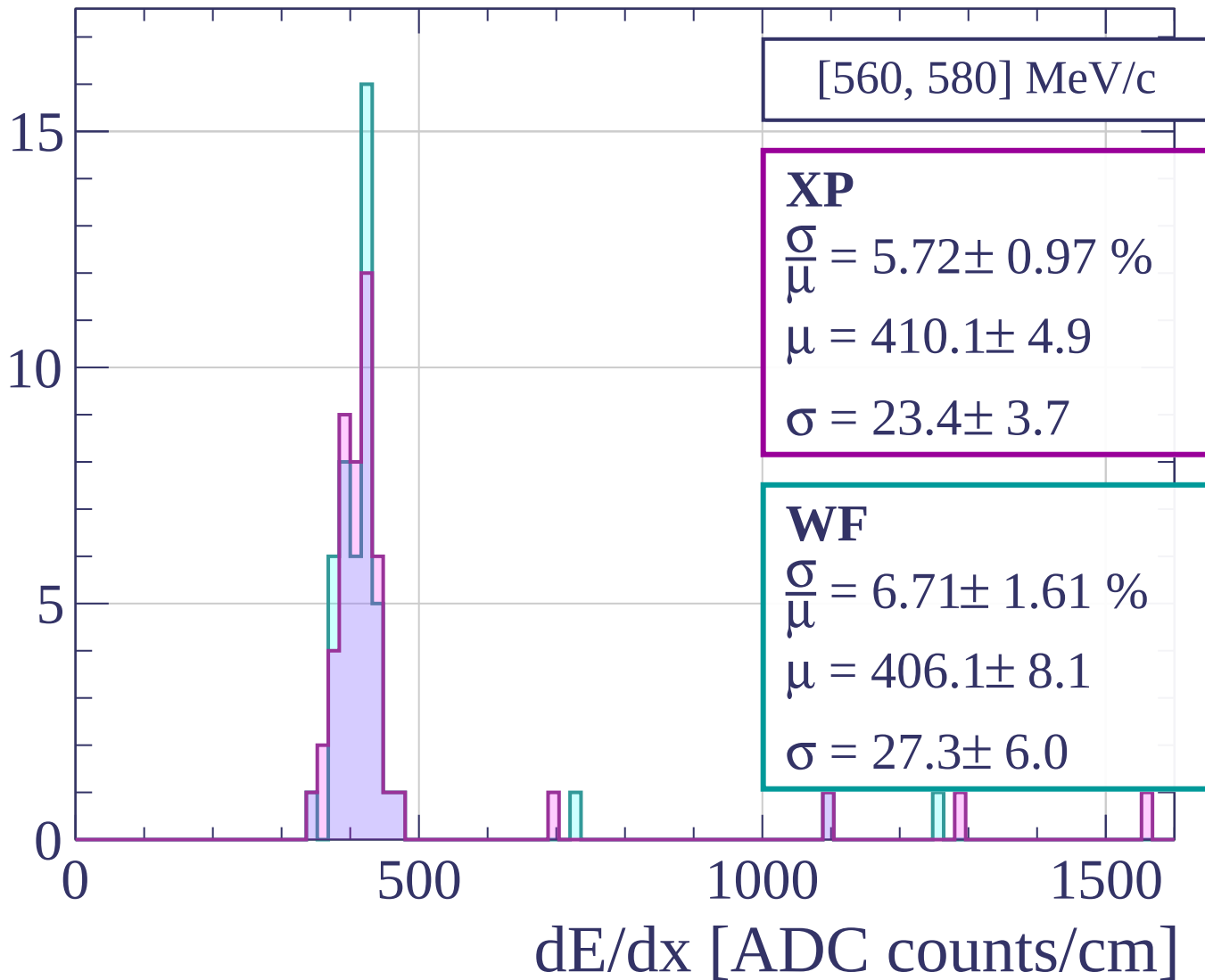
Count



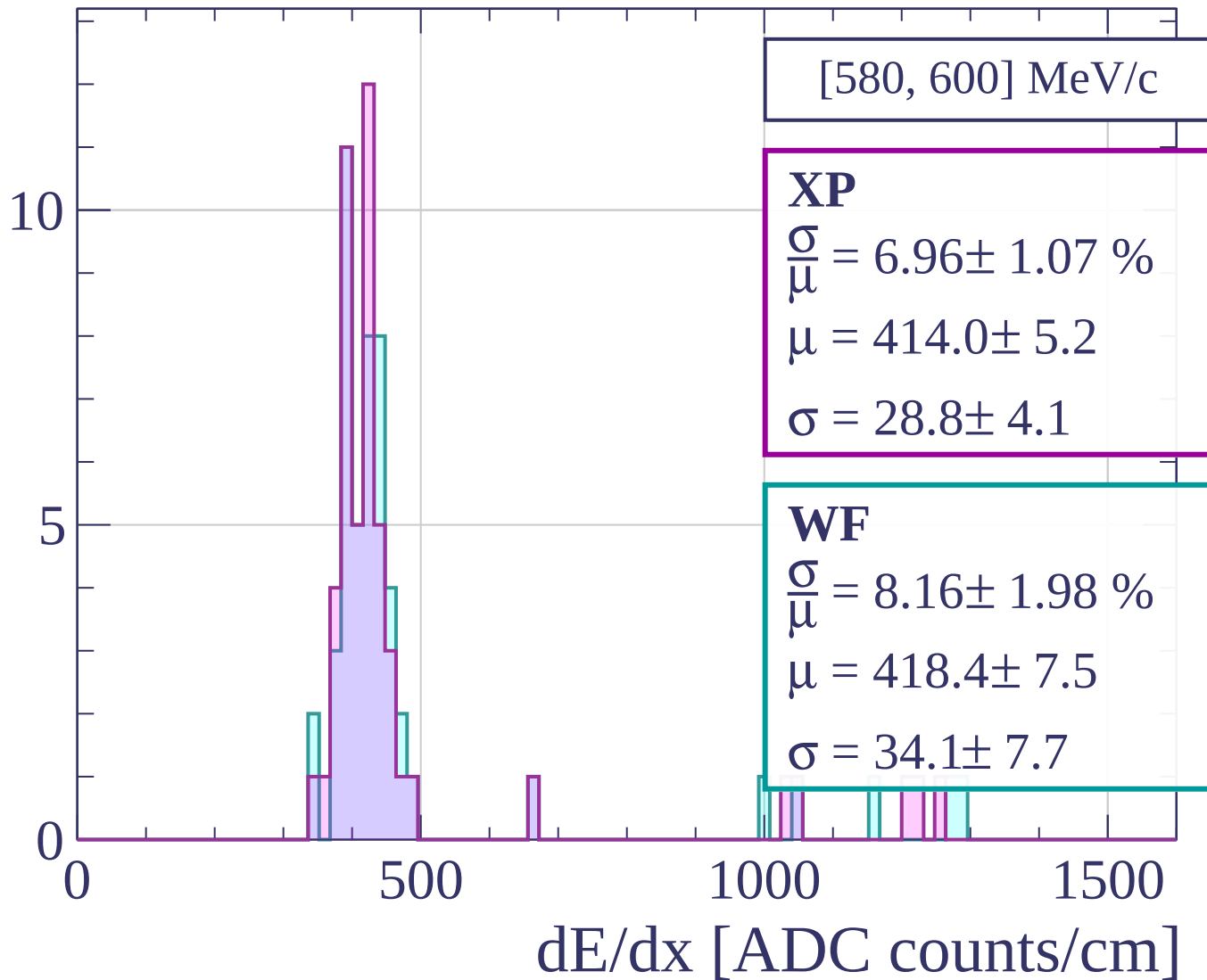
Count



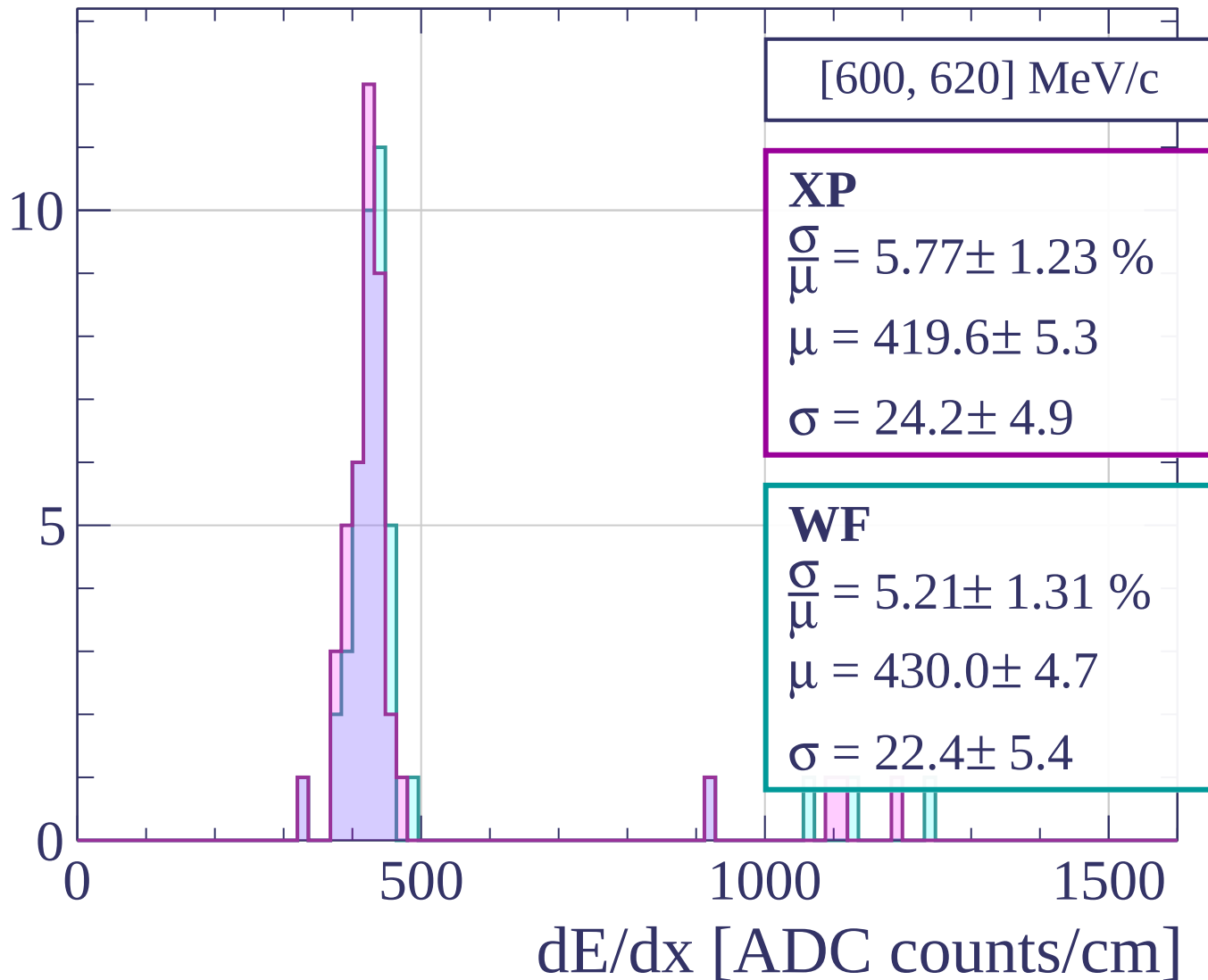
Count



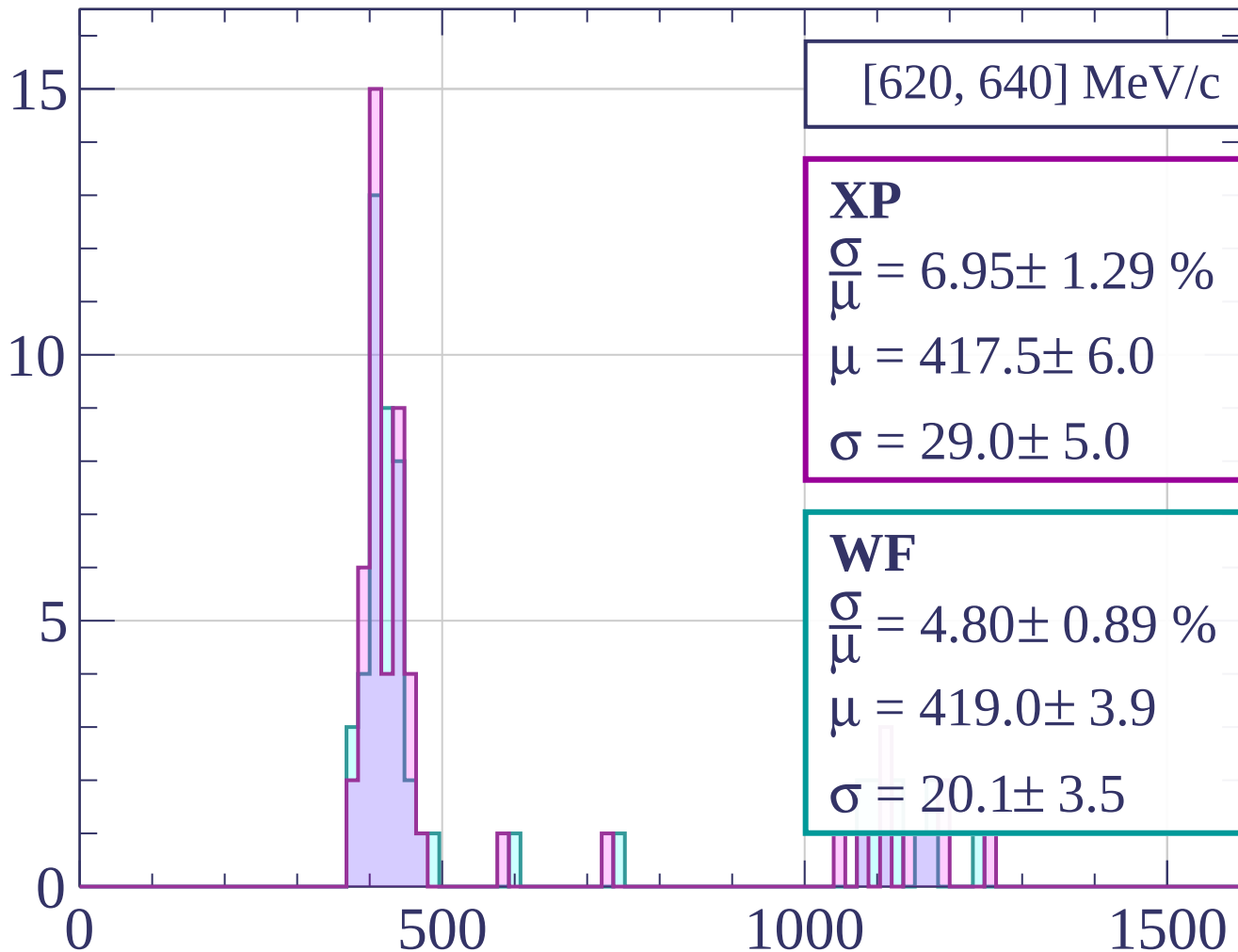
Count



Count

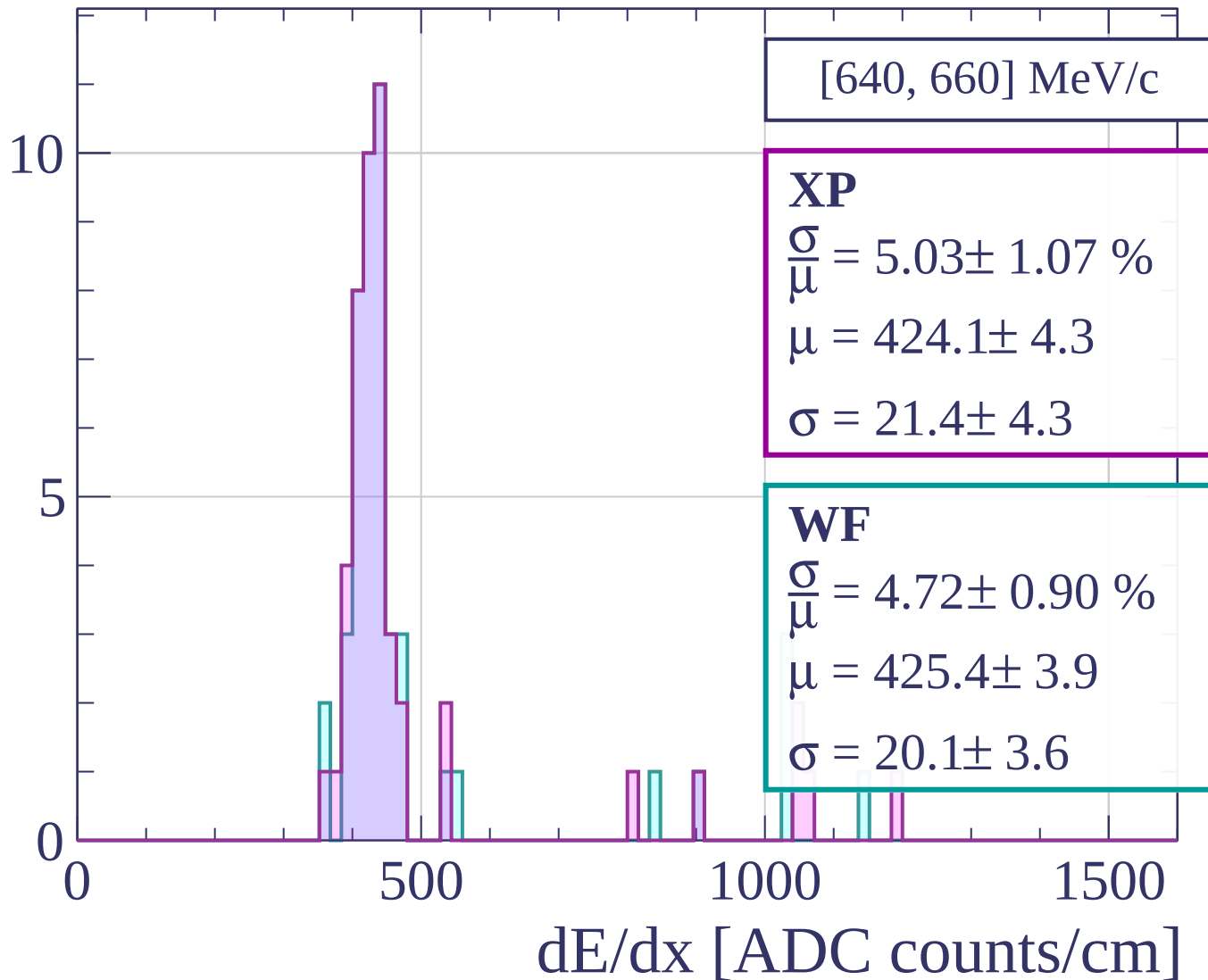


Count

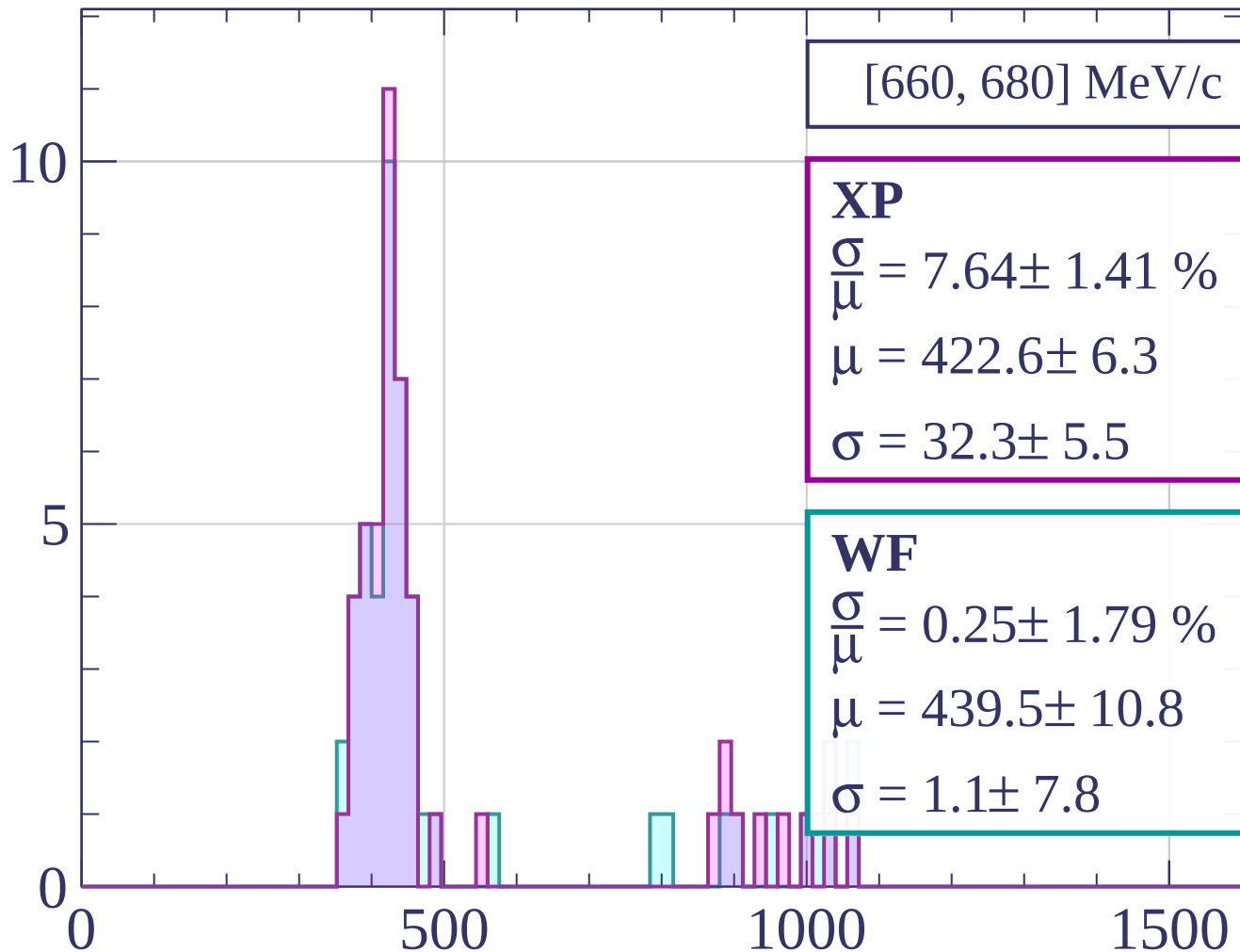


dE/dx [ADC counts/cm]

Count

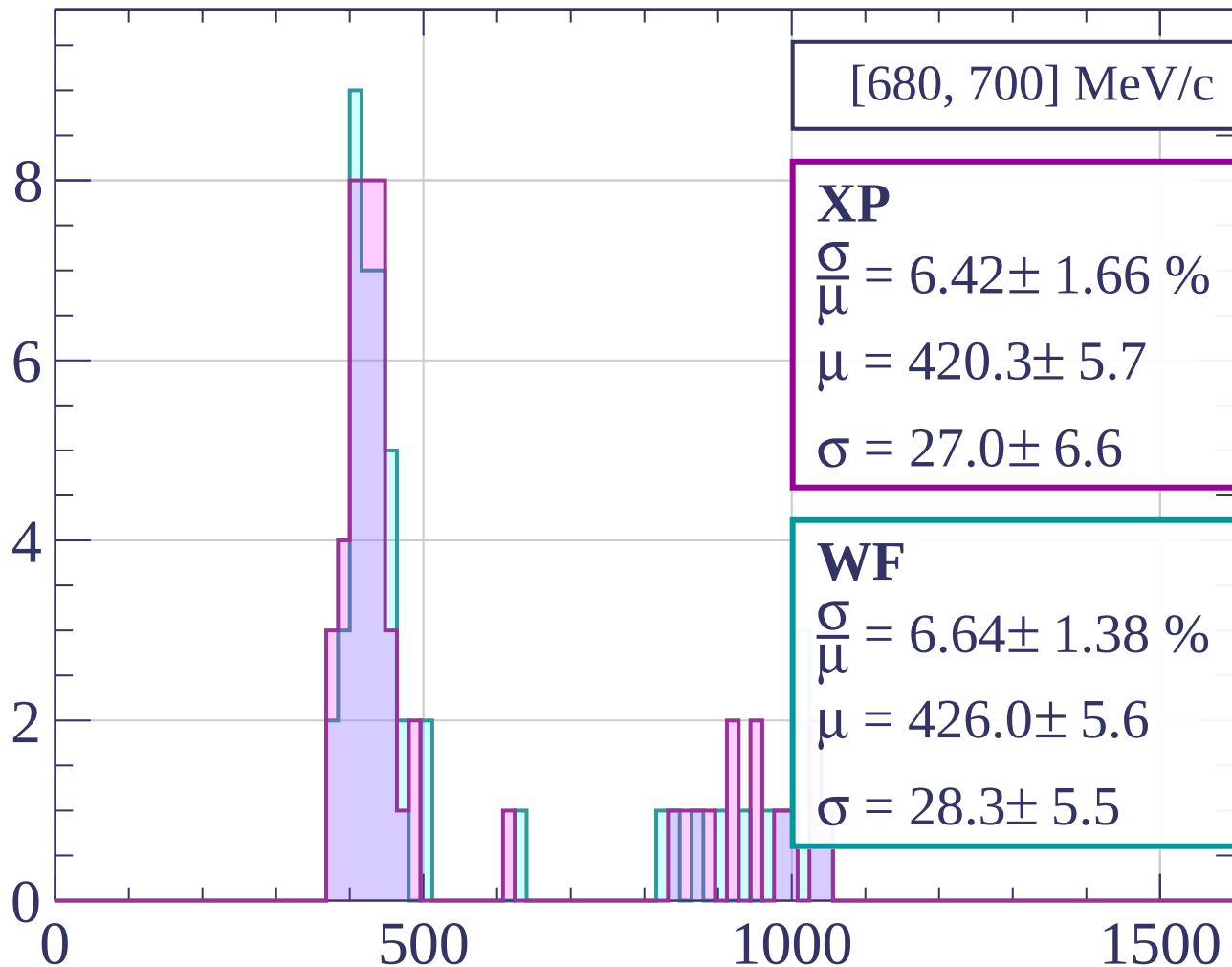


Count



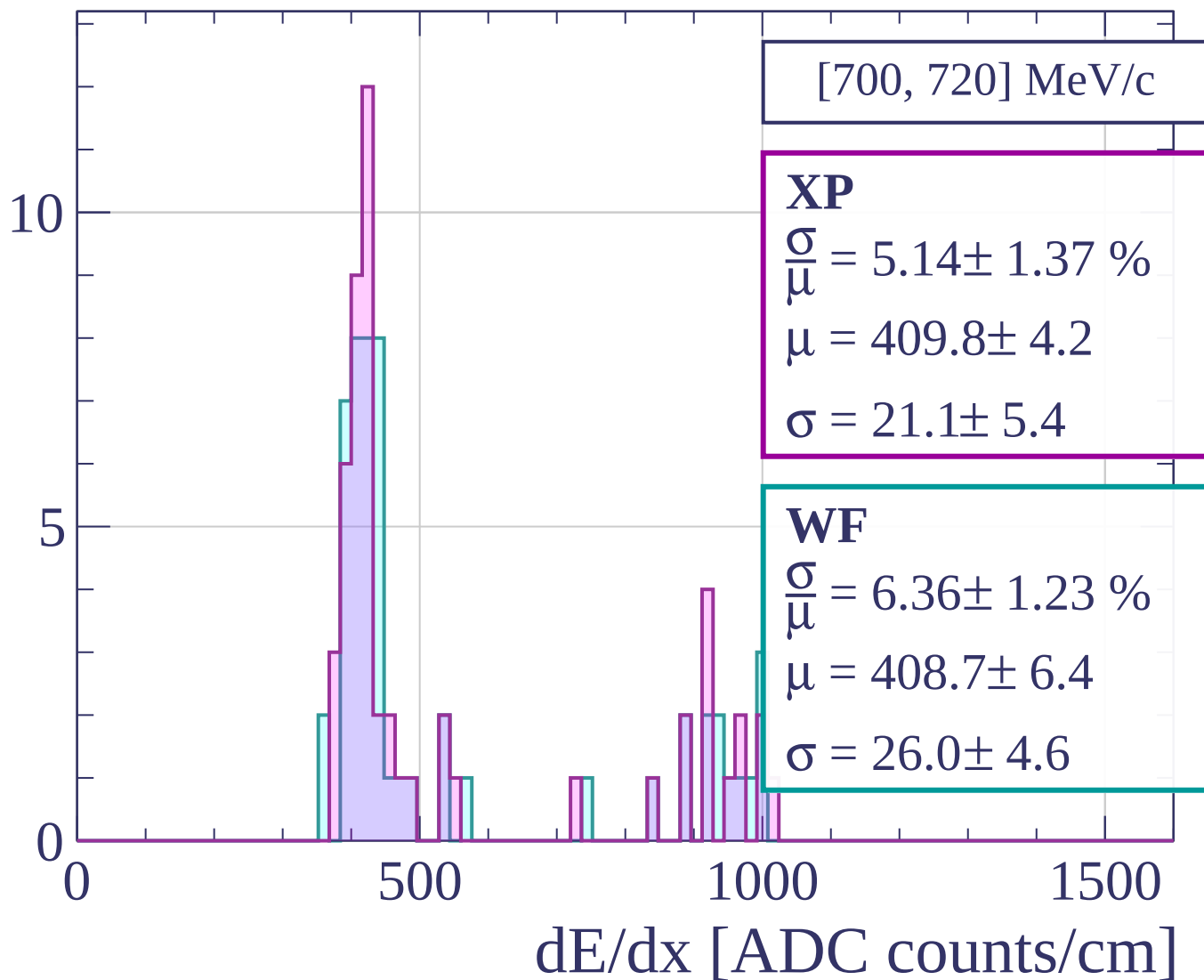
dE/dx [ADC counts/cm]

Count

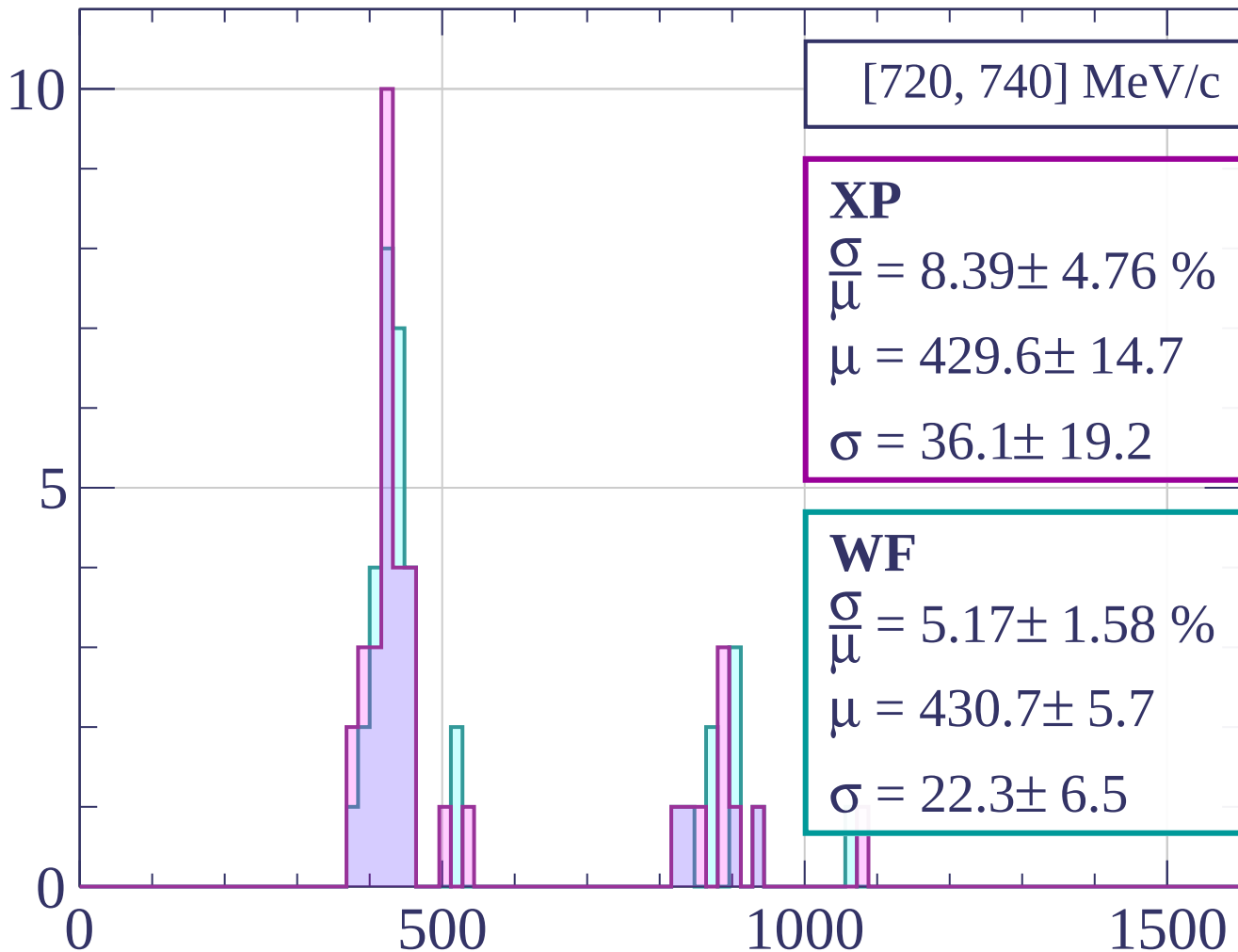


dE/dx [ADC counts/cm]

Count



Count



[720, 740] MeV/c

XP

$$\frac{\sigma}{\mu} = 8.39 \pm 4.76 \%$$

$$\mu = 429.6 \pm 14.7$$

$$\sigma = 36.1 \pm 19.2$$

WF

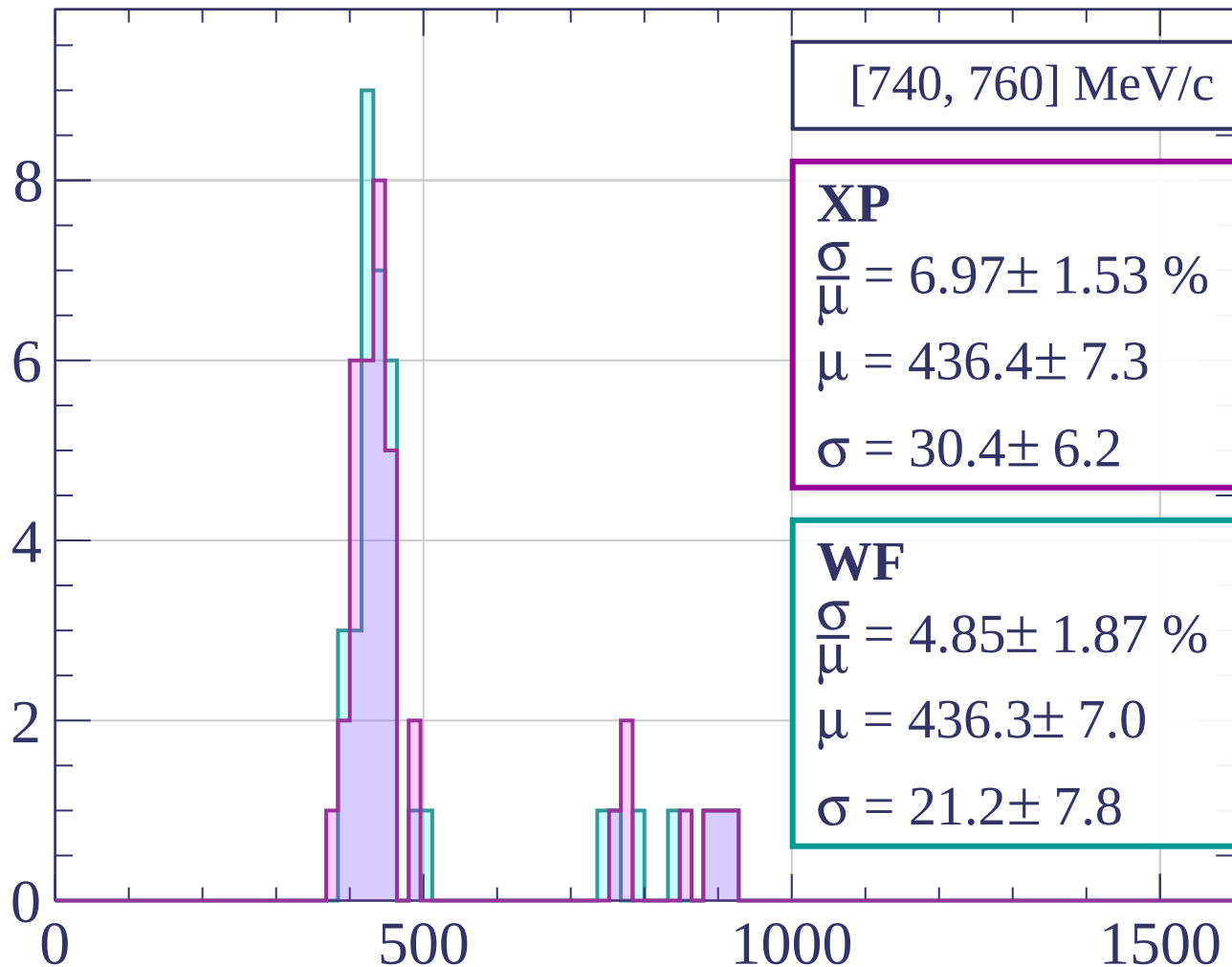
$$\frac{\sigma}{\mu} = 5.17 \pm 1.58 \%$$

$$\mu = 430.7 \pm 5.7$$

$$\sigma = 22.3 \pm 6.5$$

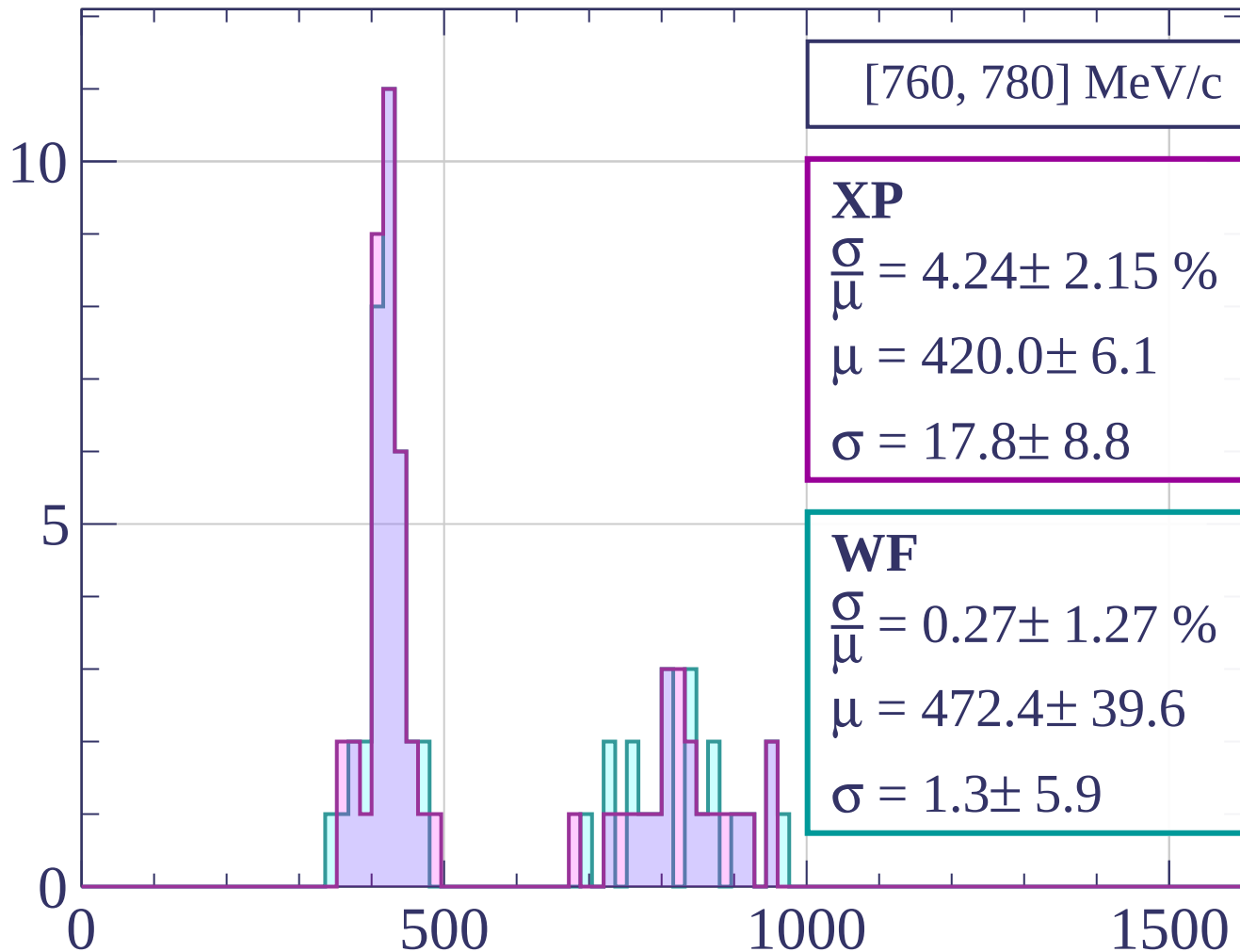
dE/dx [ADC counts/cm]

Count



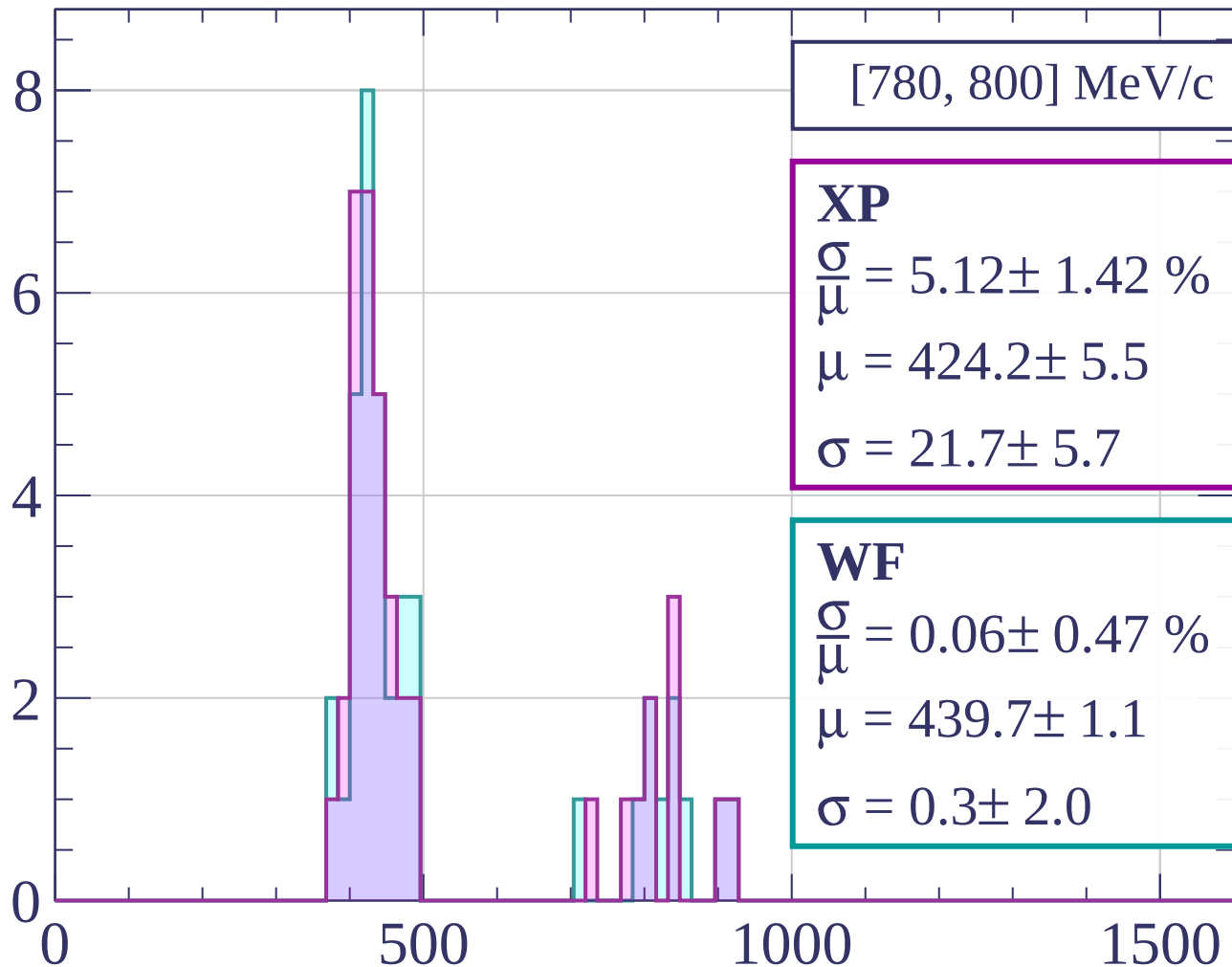
dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count



[780, 800] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.12 \pm 1.42 \%$$

$$\mu = 424.2 \pm 5.5$$

$$\sigma = 21.7 \pm 5.7$$

WF

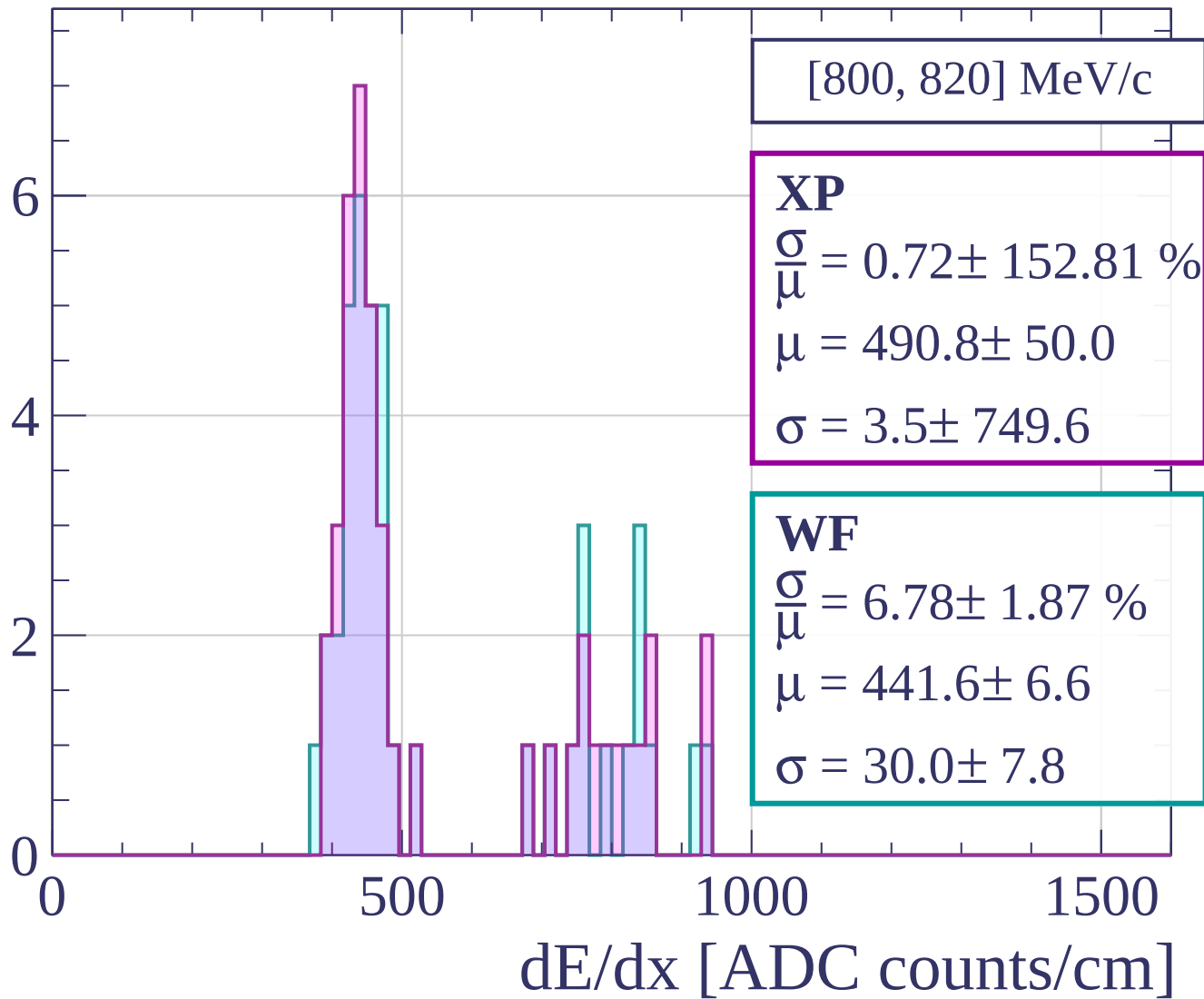
$$\frac{\sigma}{\mu} = 0.06 \pm 0.47 \%$$

$$\mu = 439.7 \pm 1.1$$

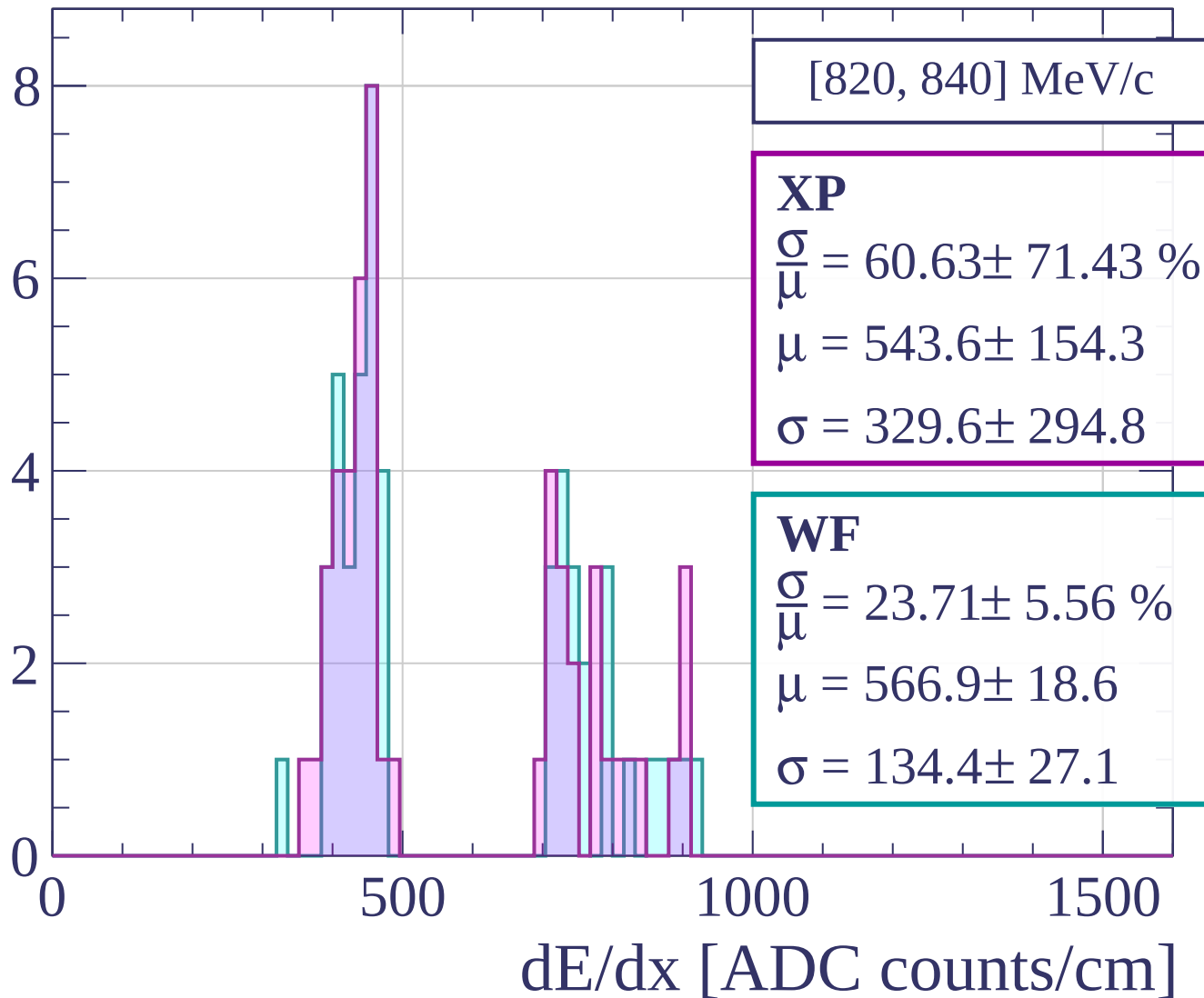
$$\sigma = 0.3 \pm 2.0$$

dE/dx [ADC counts/cm]

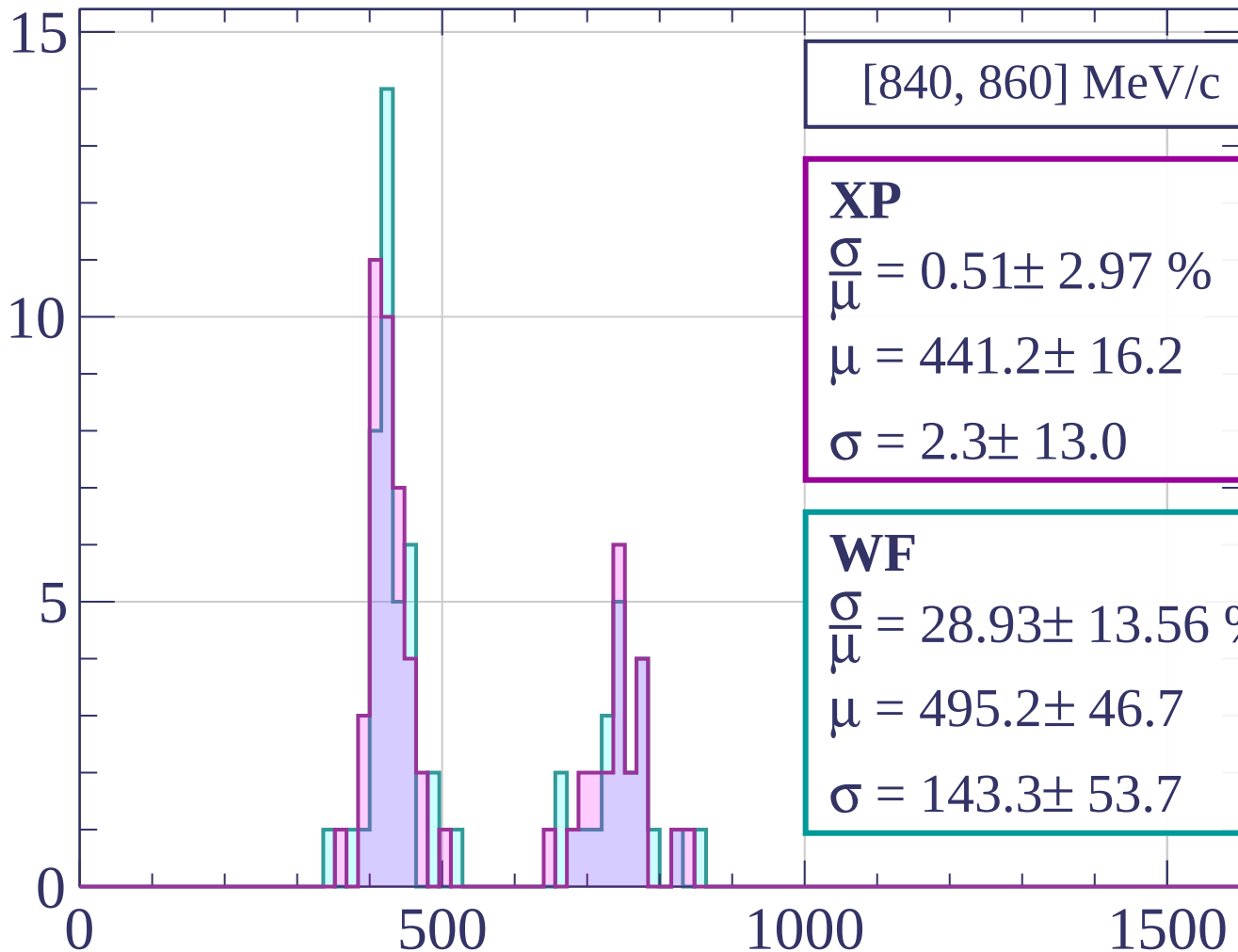
Count



Count

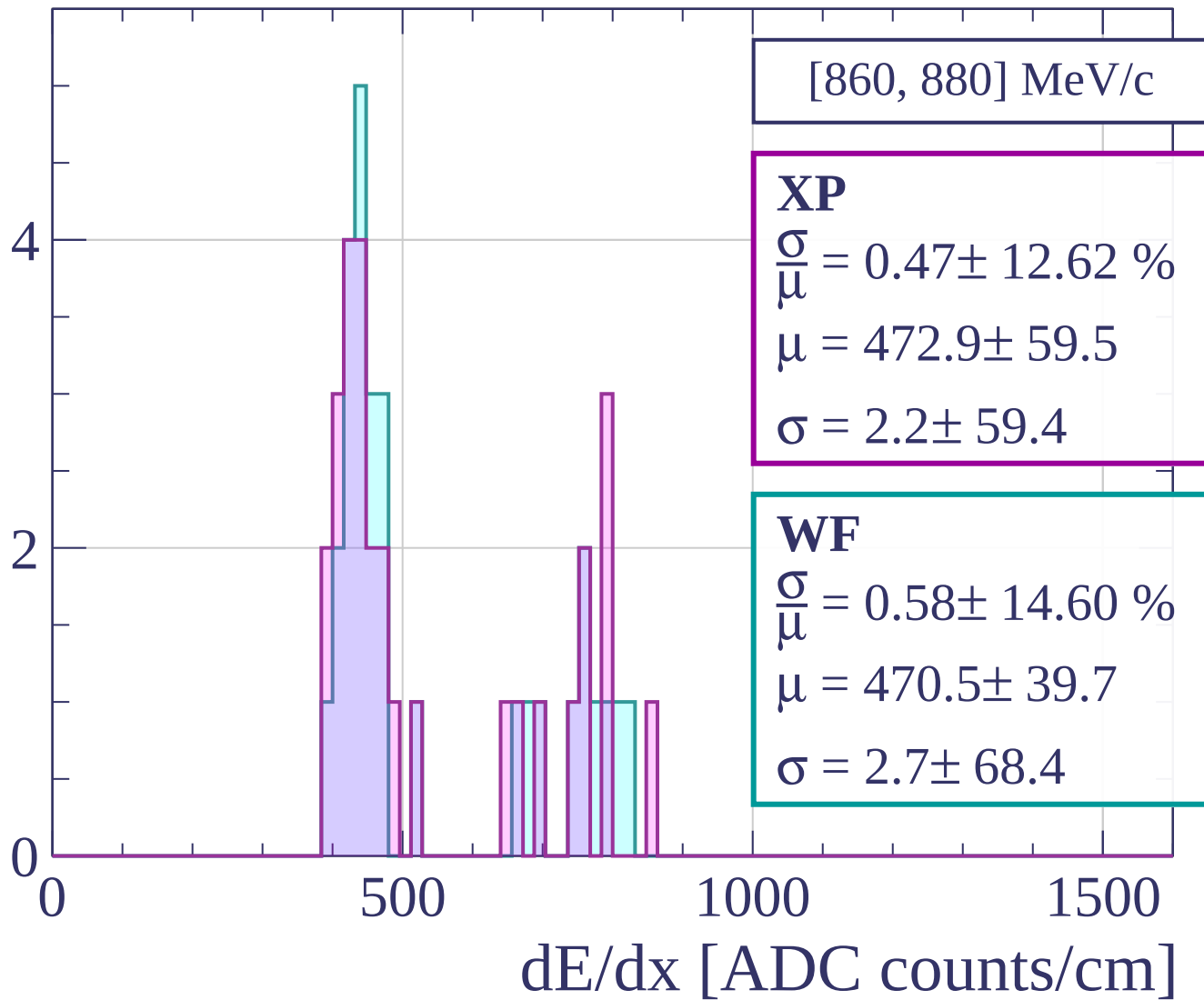


Count

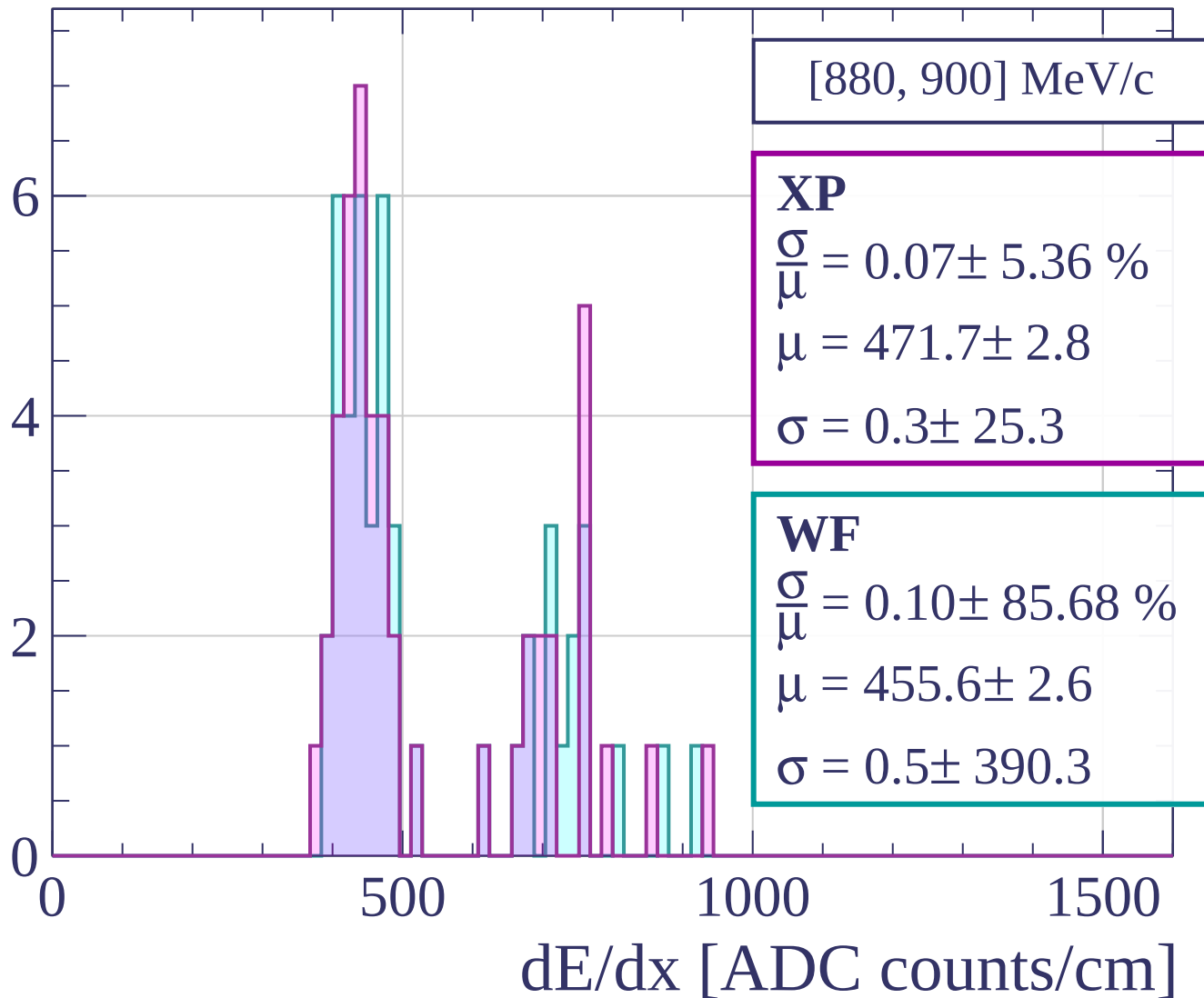


dE/dx [ADC counts/cm]

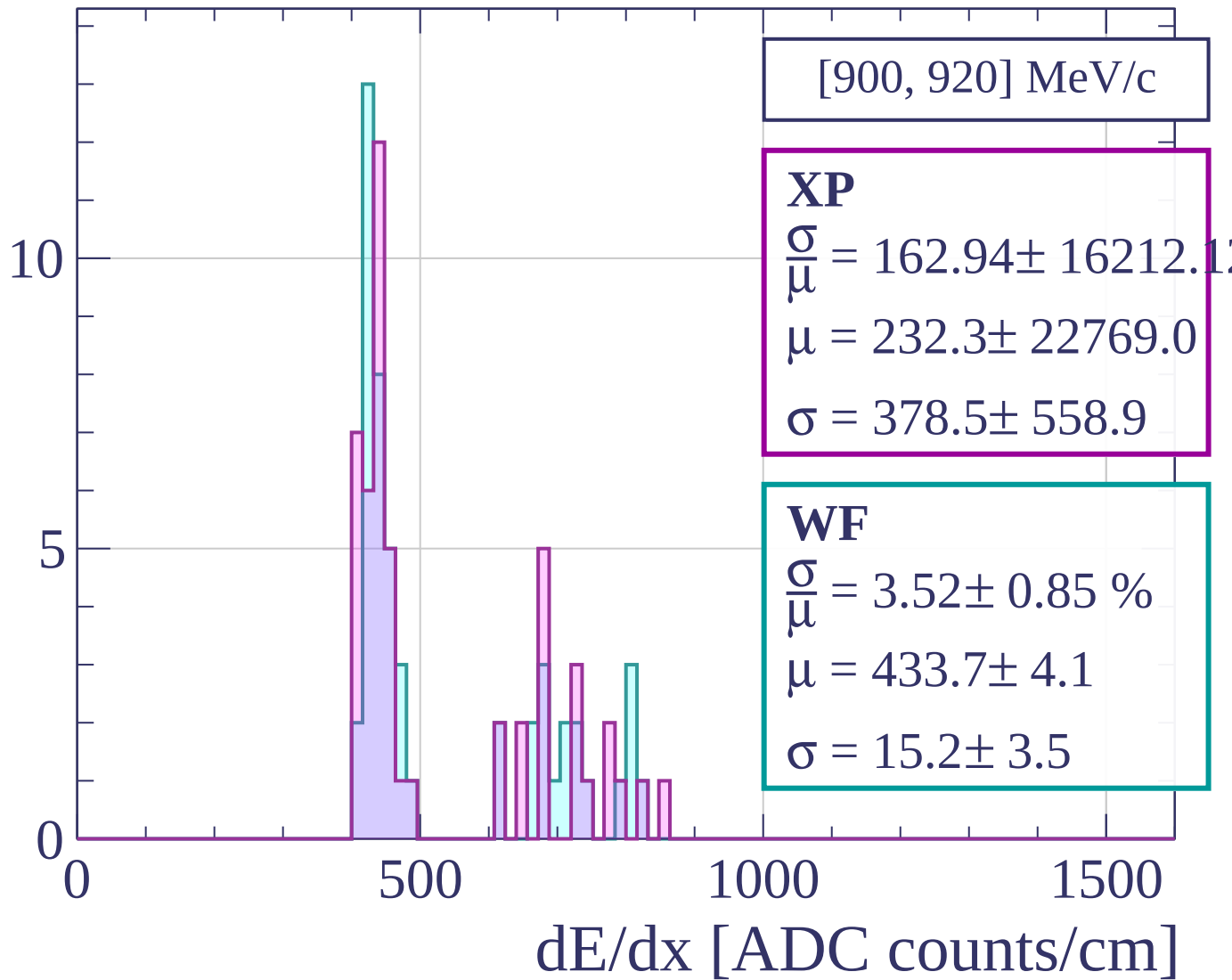
Count



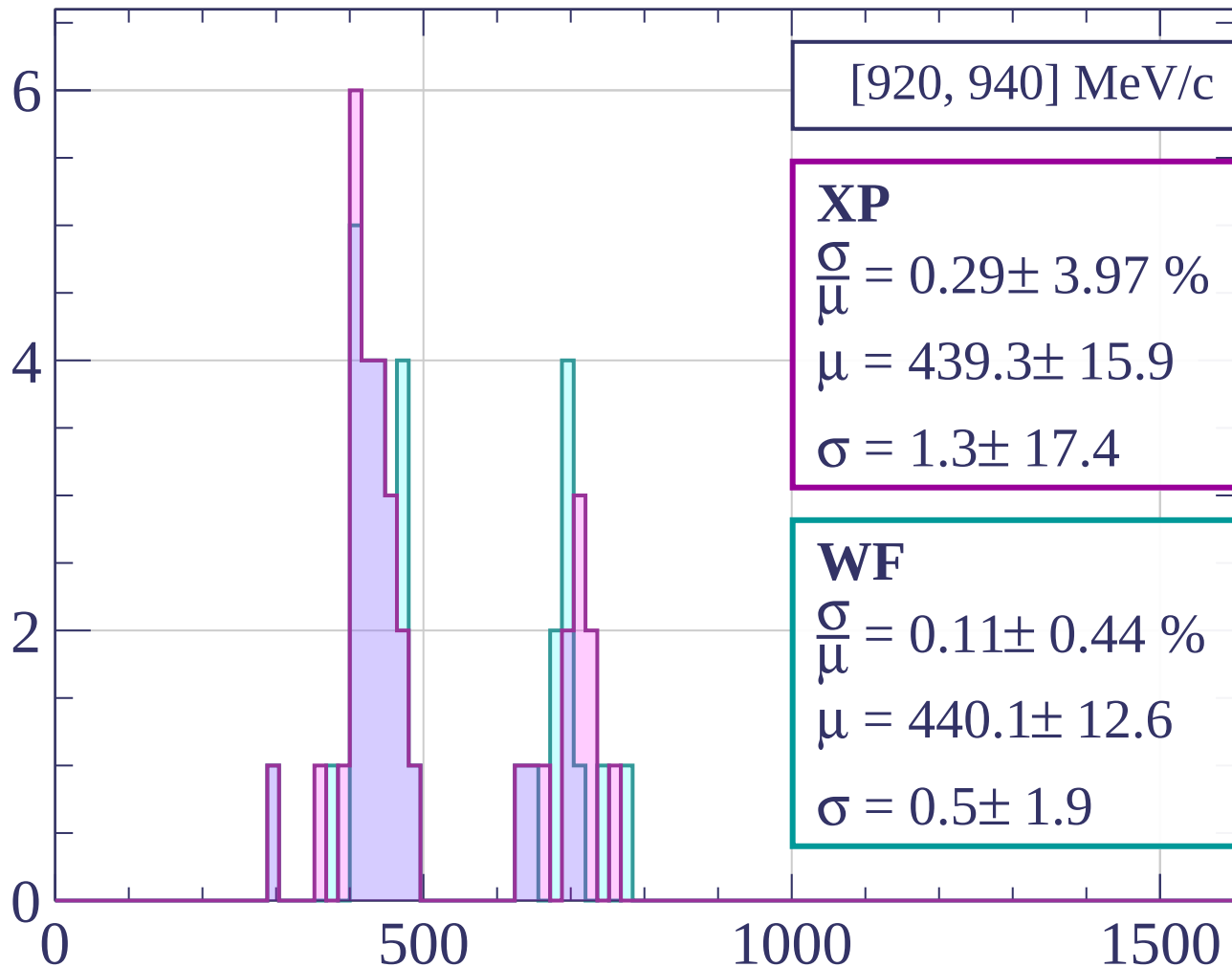
Count



Count

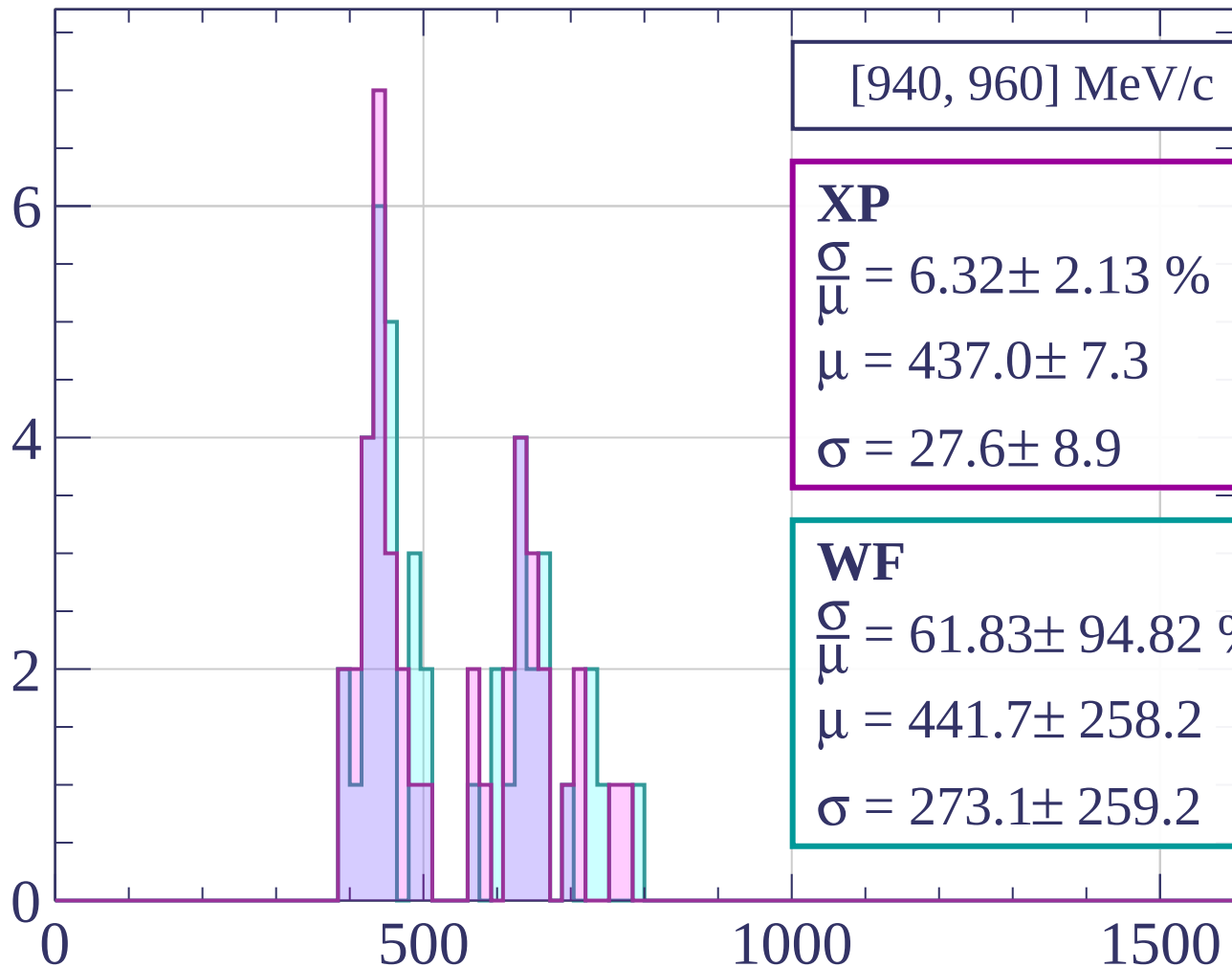


Count



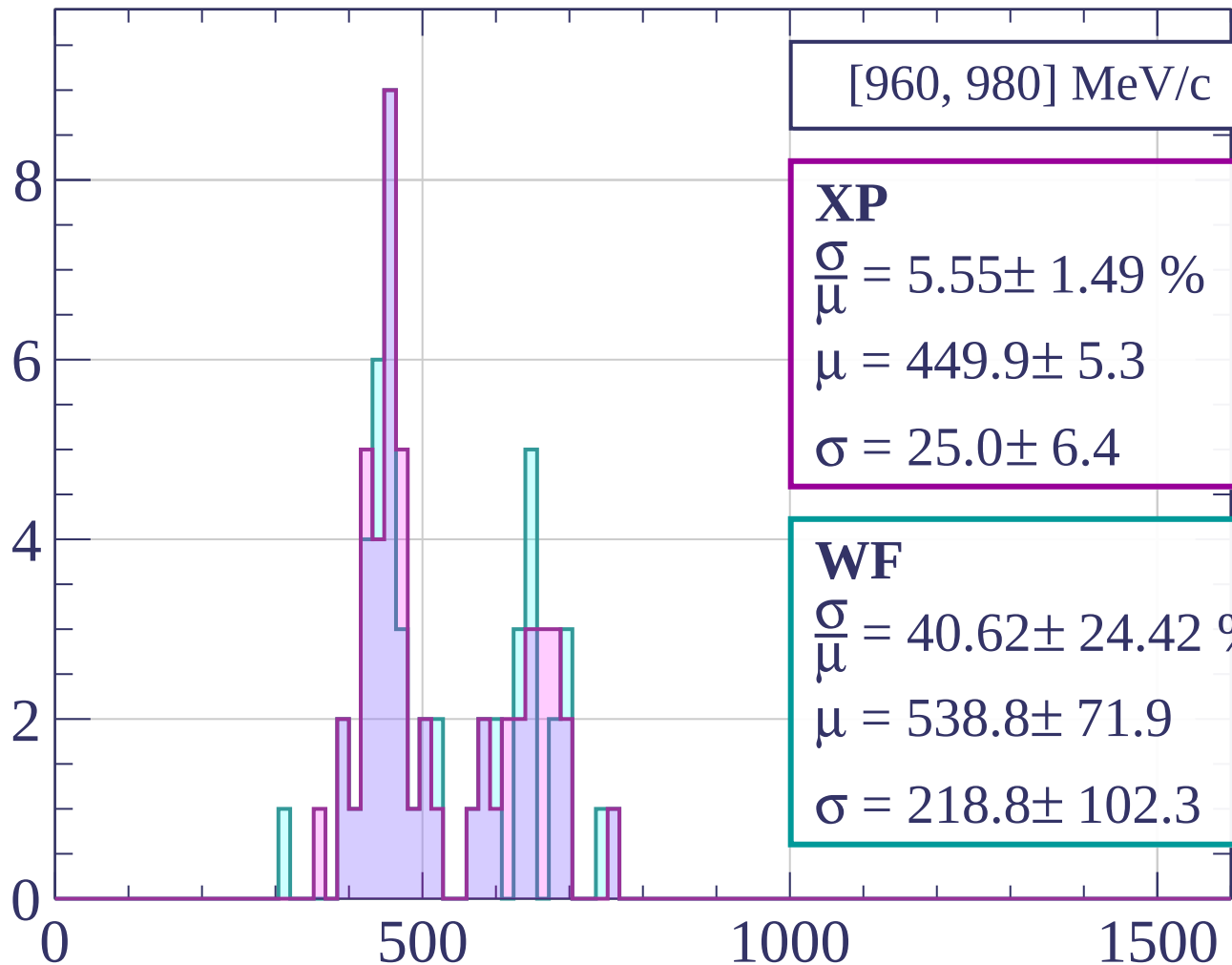
dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count



[960, 980] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.55 \pm 1.49 \%$$

$$\mu = 449.9 \pm 5.3$$

$$\sigma = 25.0 \pm 6.4$$

WF

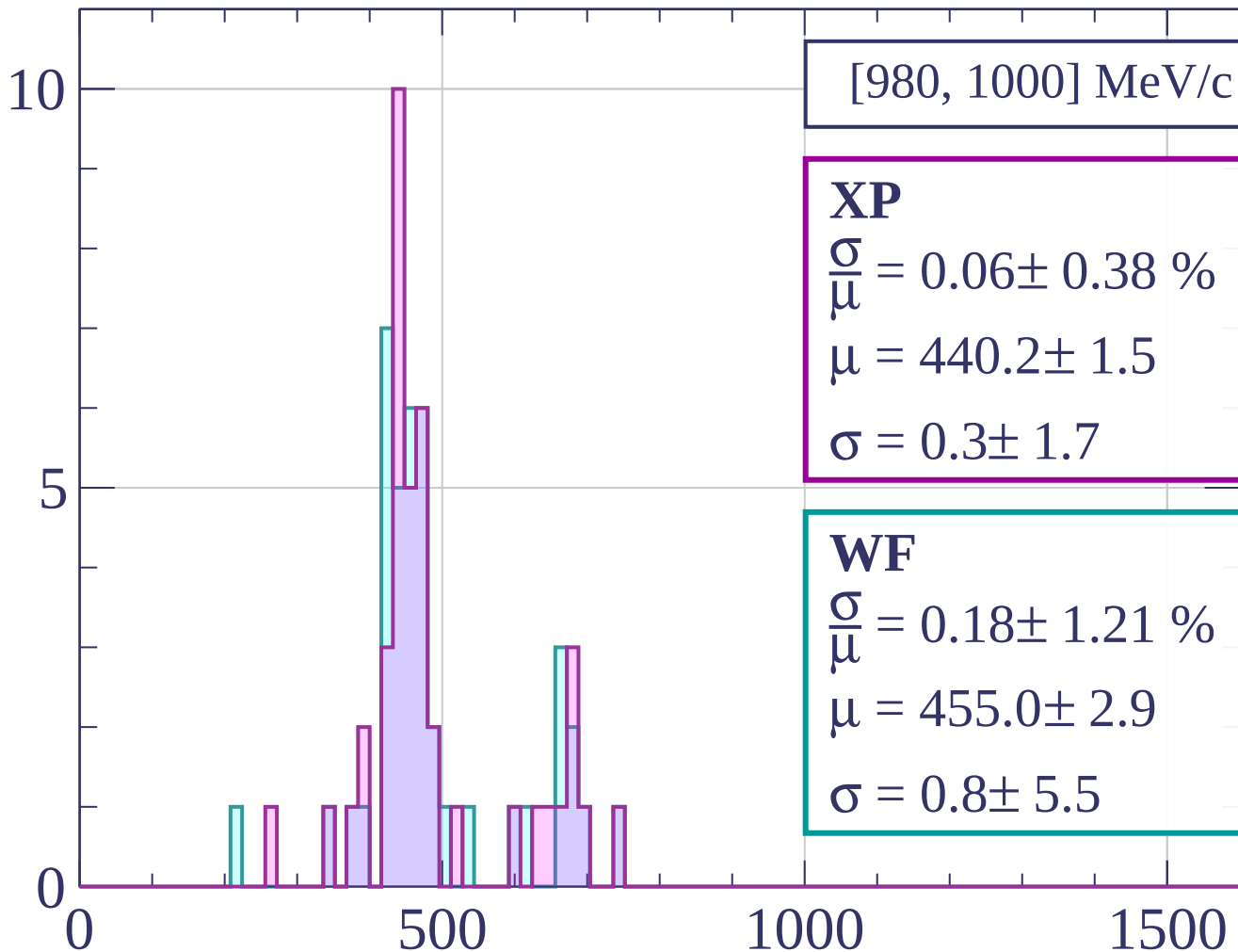
$$\frac{\sigma}{\mu} = 40.62 \pm 24.42 \%$$

$$\mu = 538.8 \pm 71.9$$

$$\sigma = 218.8 \pm 102.3$$

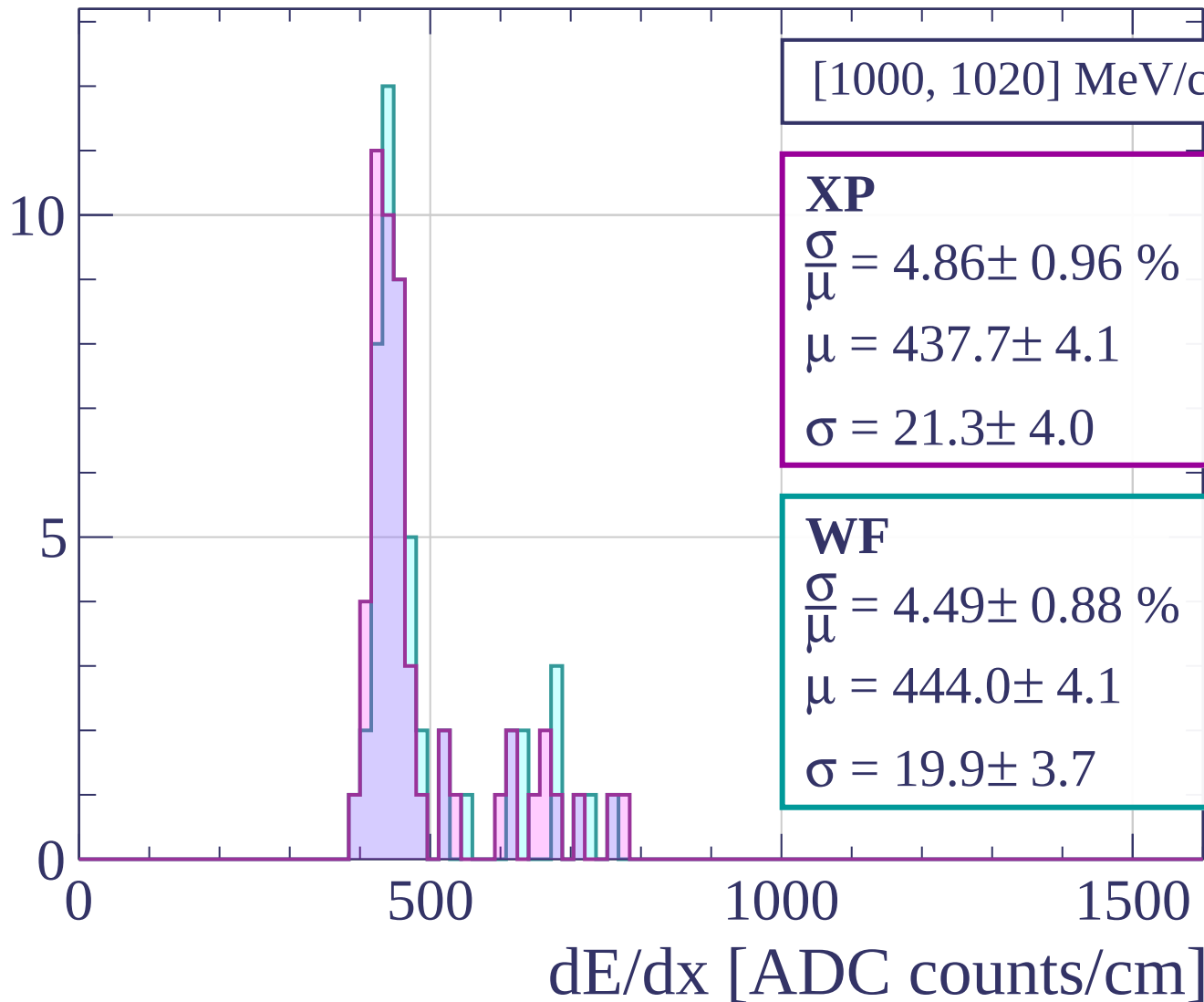
dE/dx [ADC counts/cm]

Count

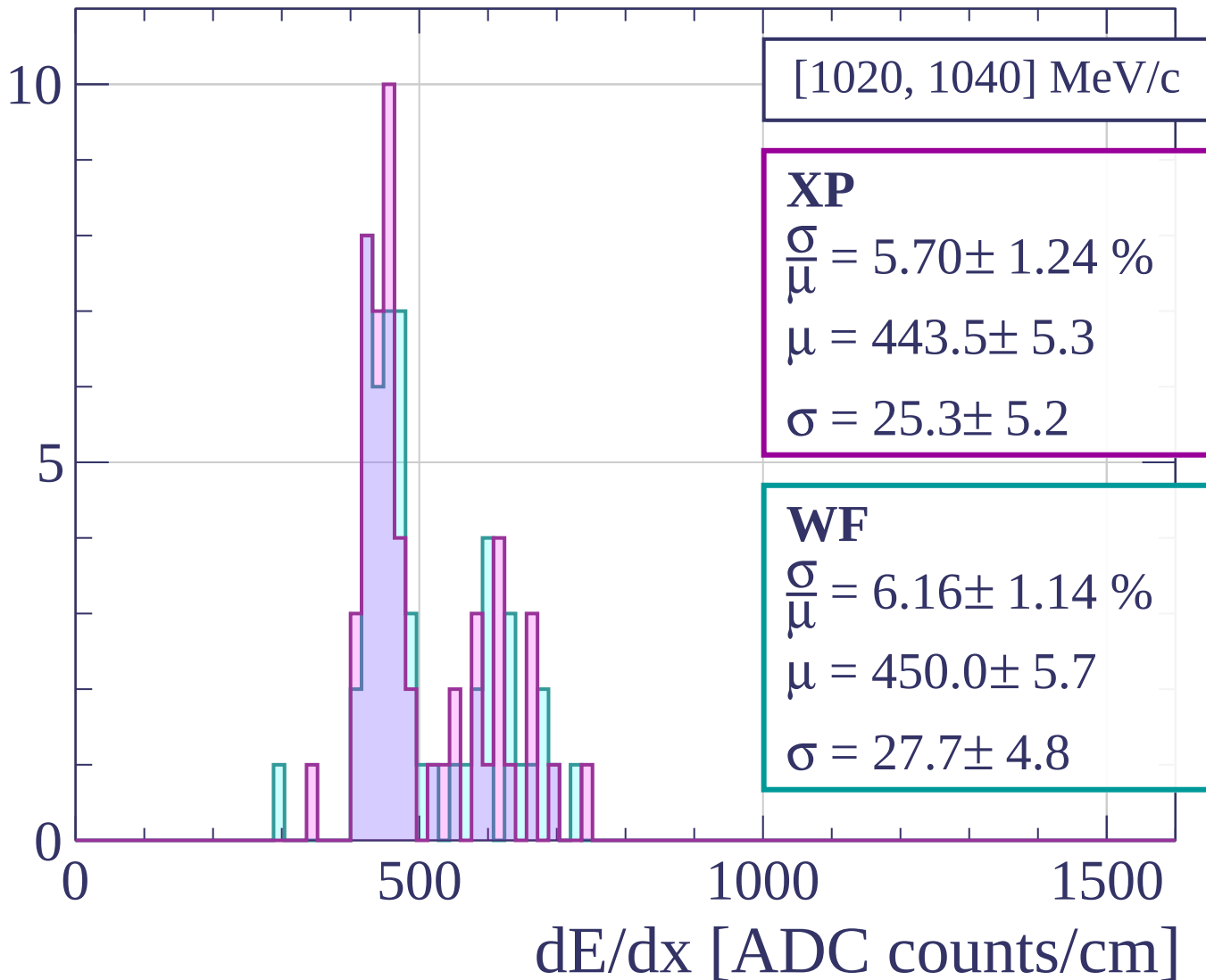


dE/dx [ADC counts/cm]

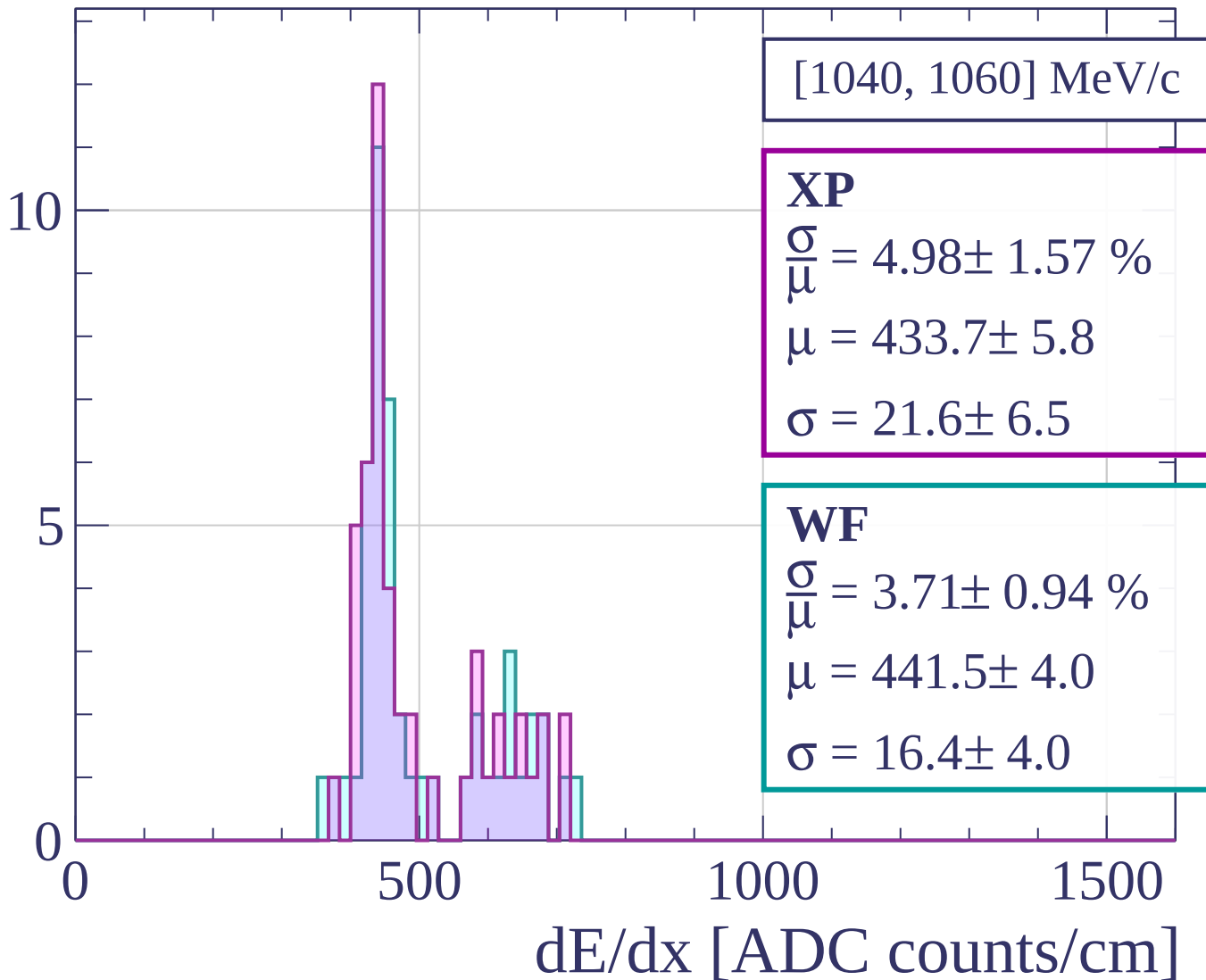
Count



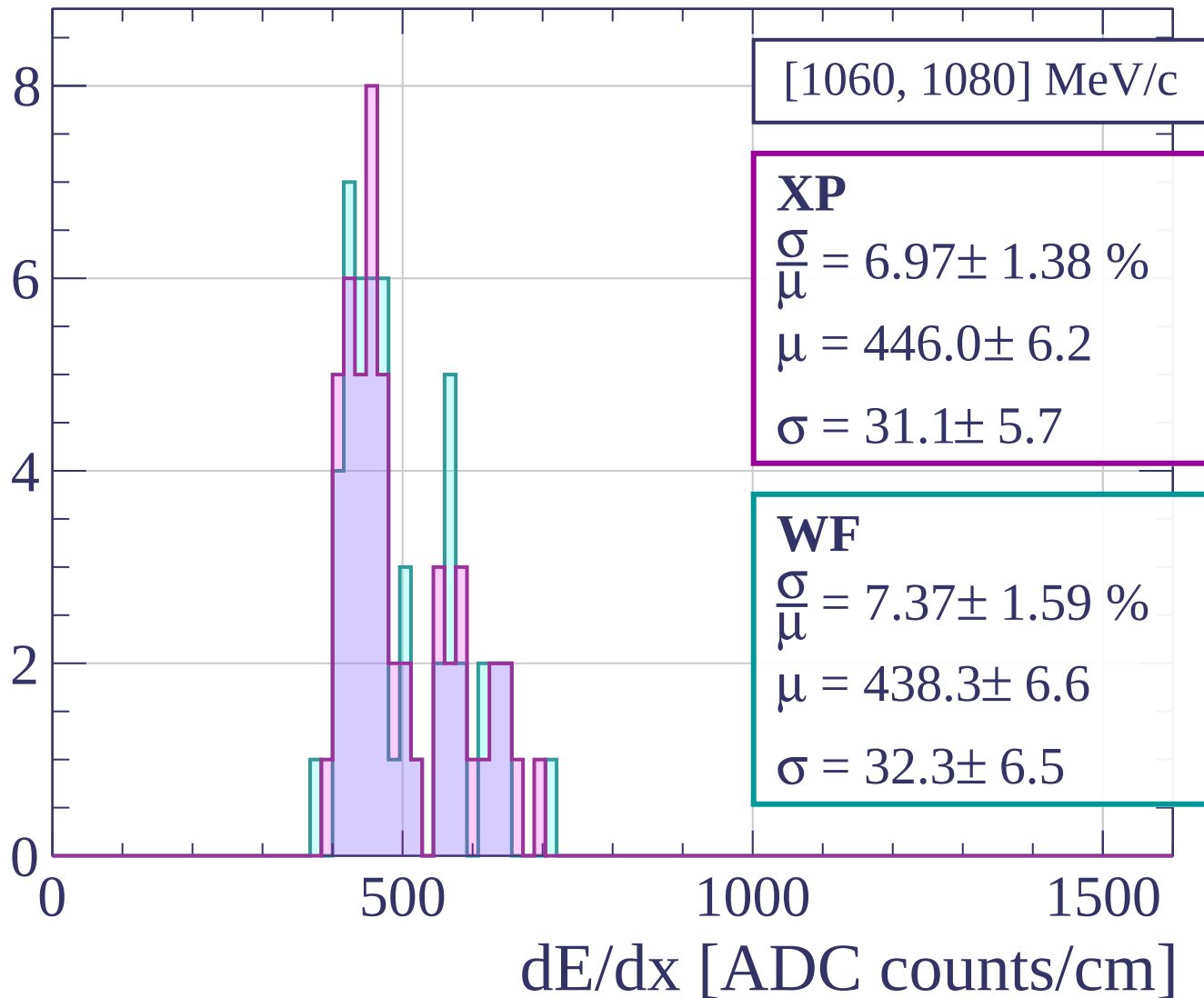
Count



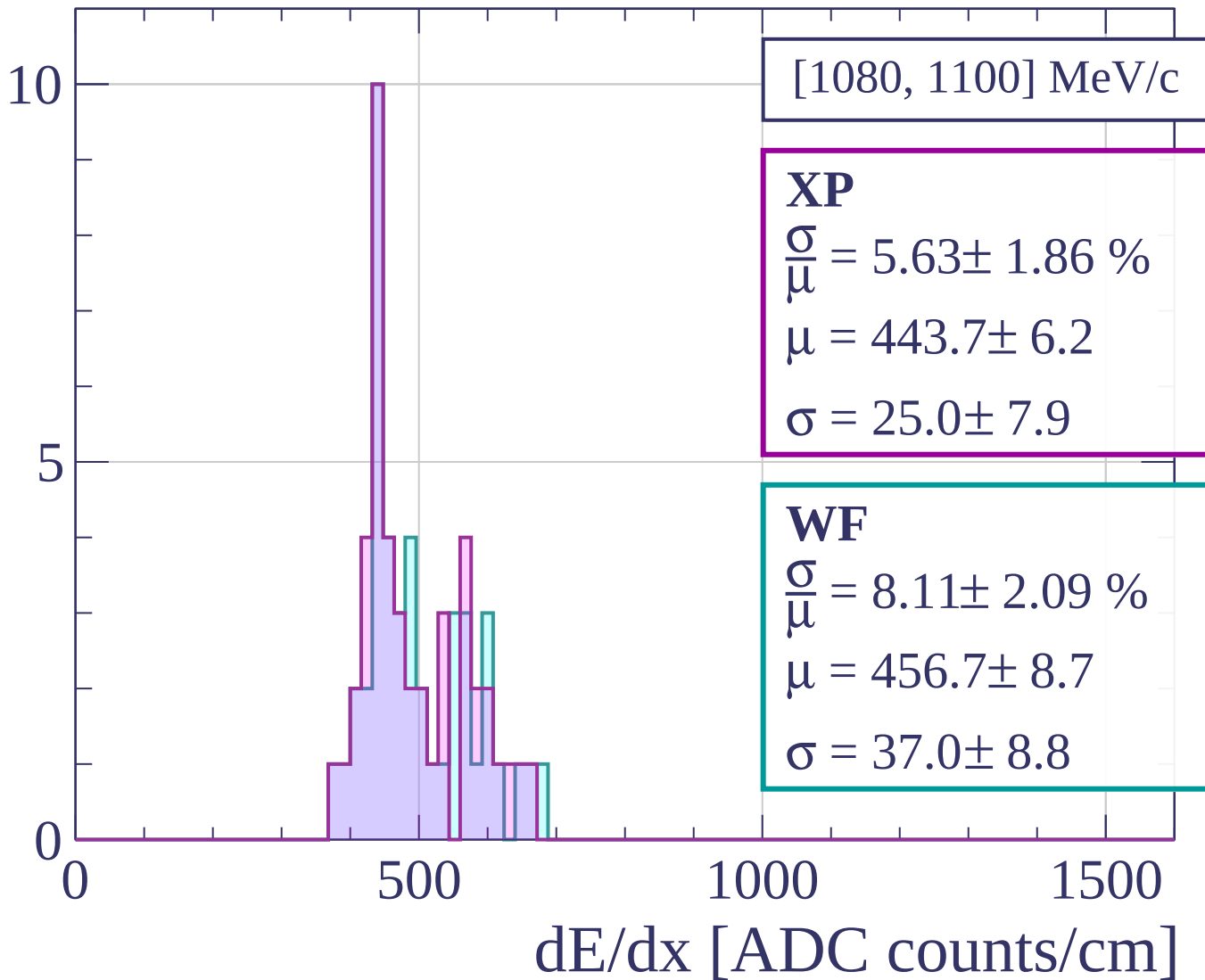
Count



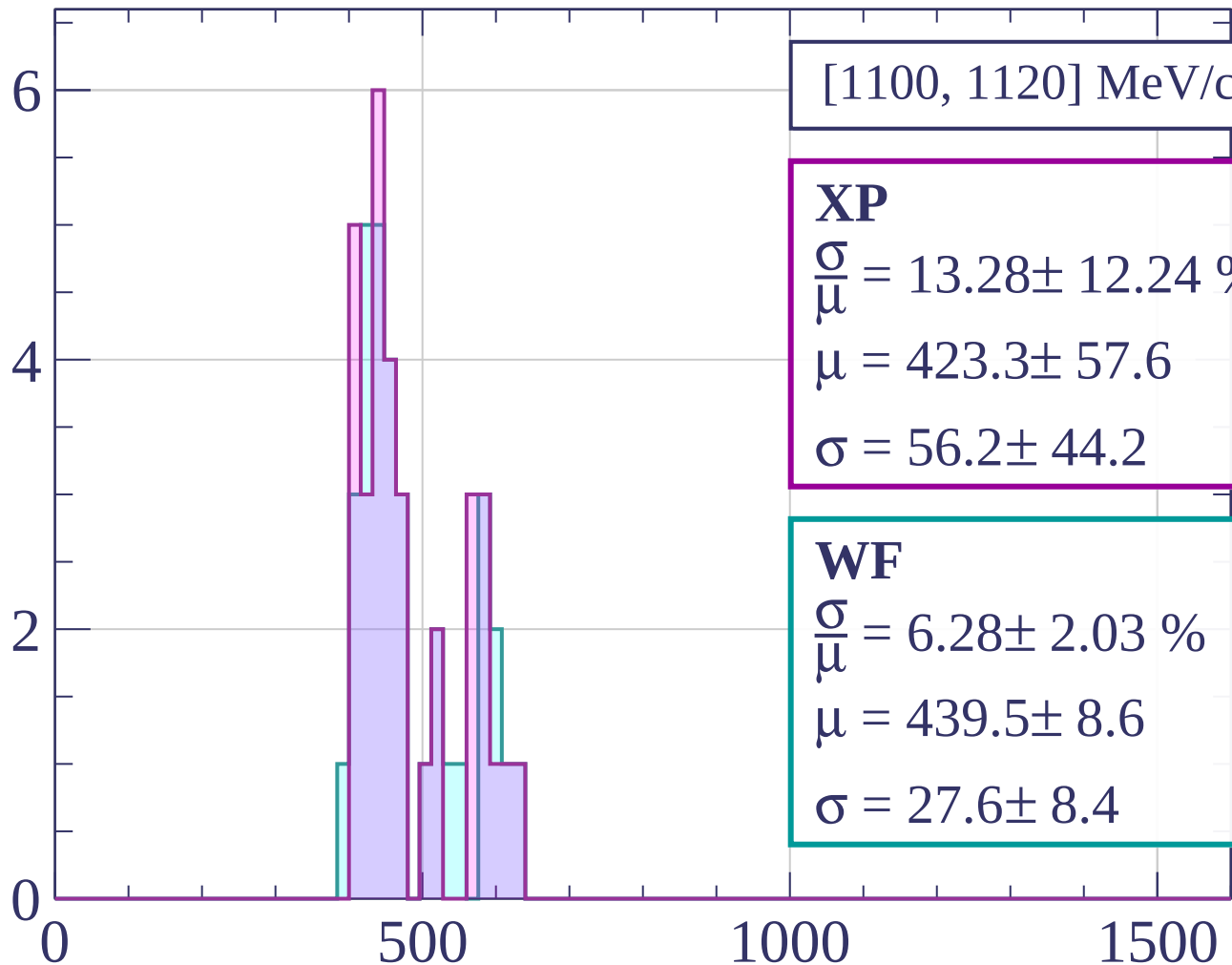
Count



Count

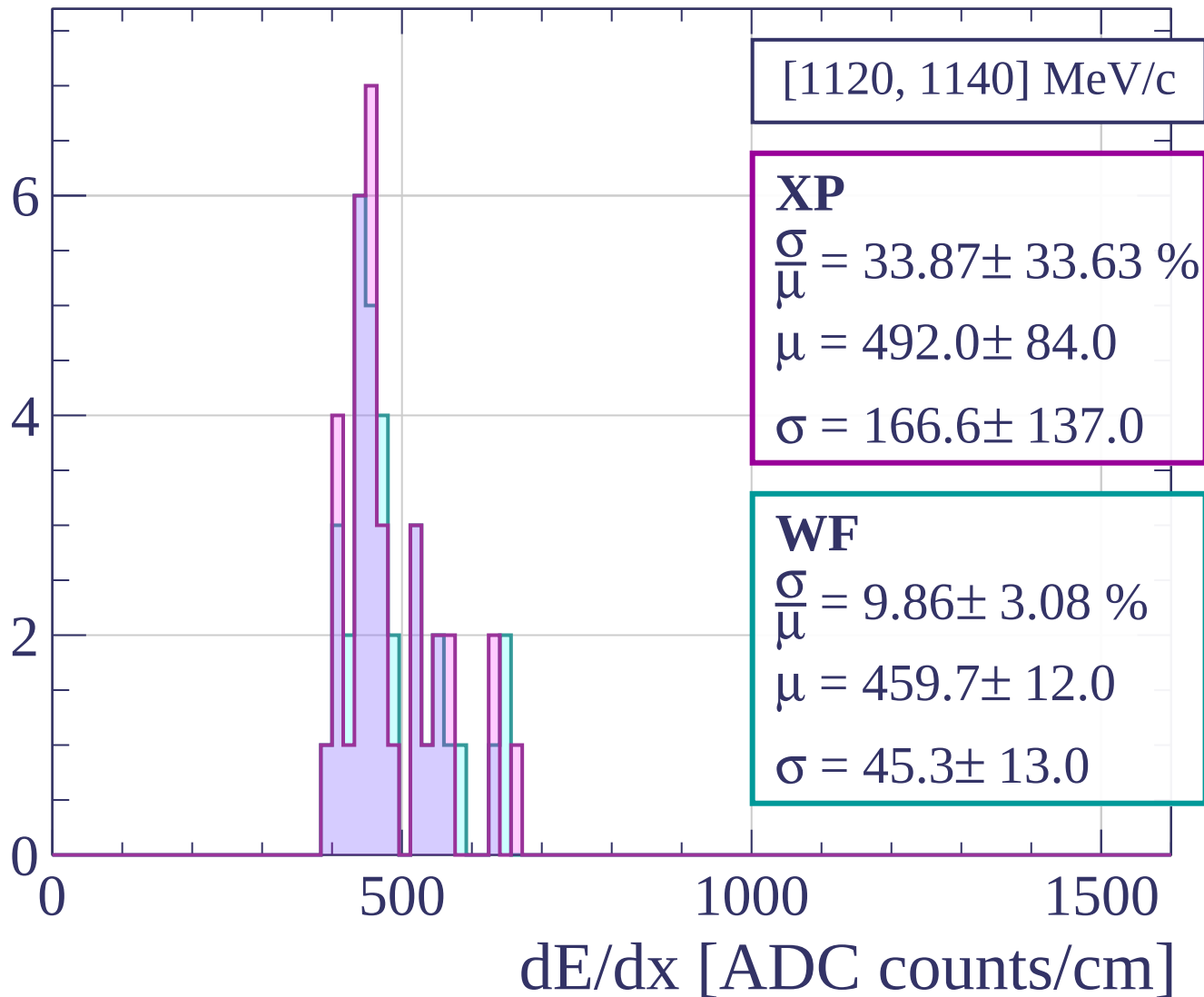


Count

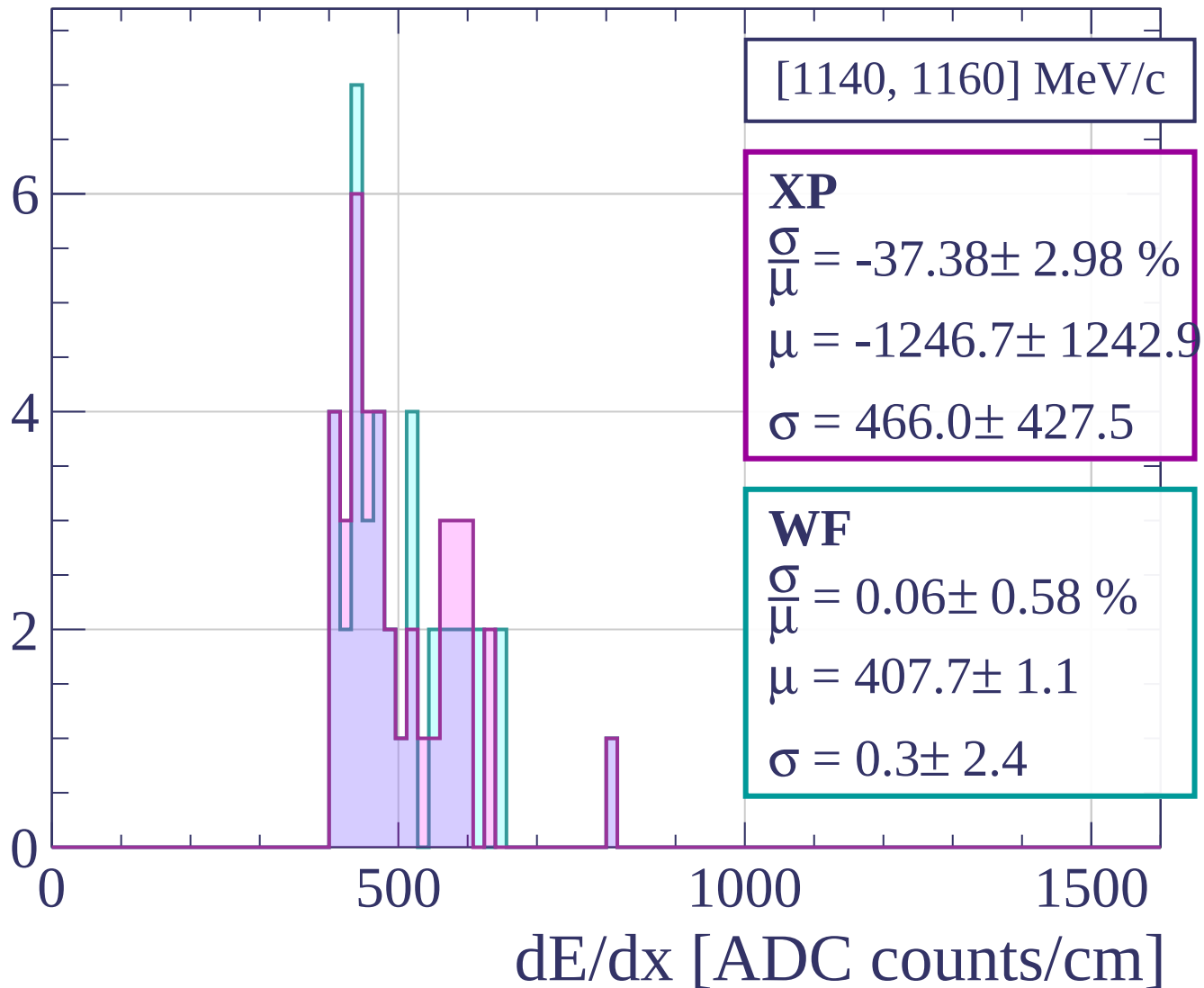


dE/dx [ADC counts/cm]

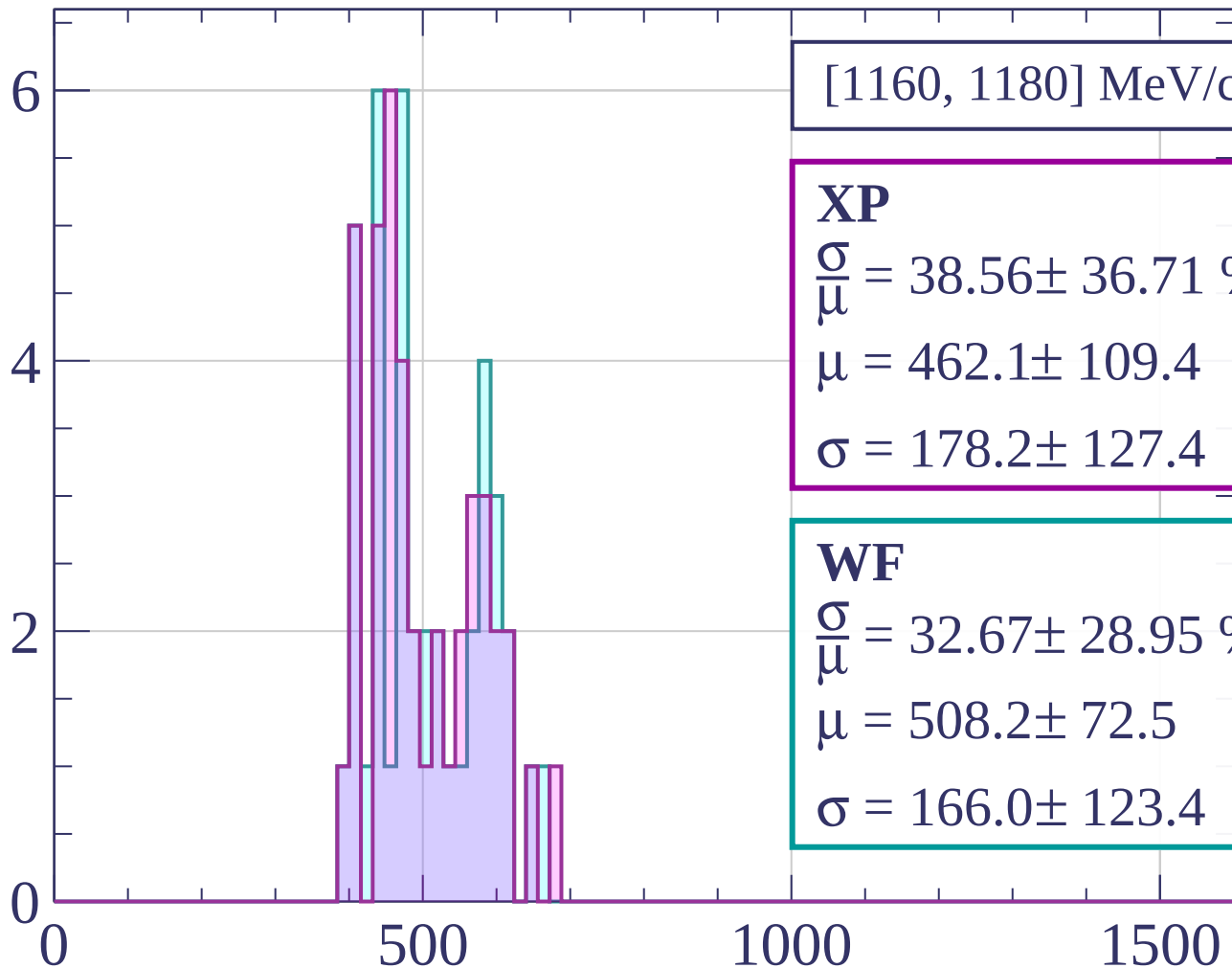
Count



Count



Count



[1160, 1180] MeV/c

XP

$$\frac{\sigma}{\mu} = 38.56 \pm 36.71 \%$$

$$\mu = 462.1 \pm 109.4$$

$$\sigma = 178.2 \pm 127.4$$

WF

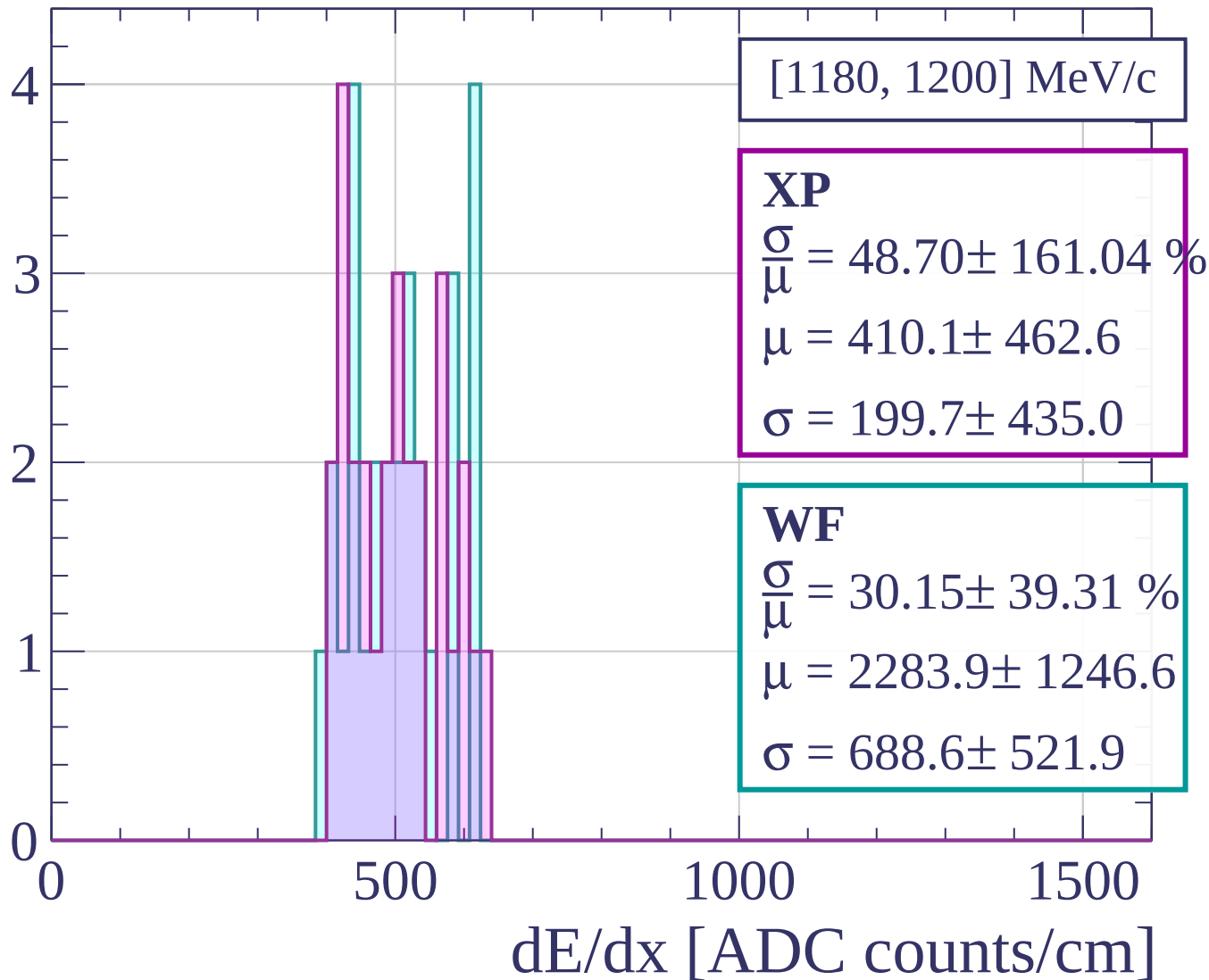
$$\frac{\sigma}{\mu} = 32.67 \pm 28.95 \%$$

$$\mu = 508.2 \pm 72.5$$

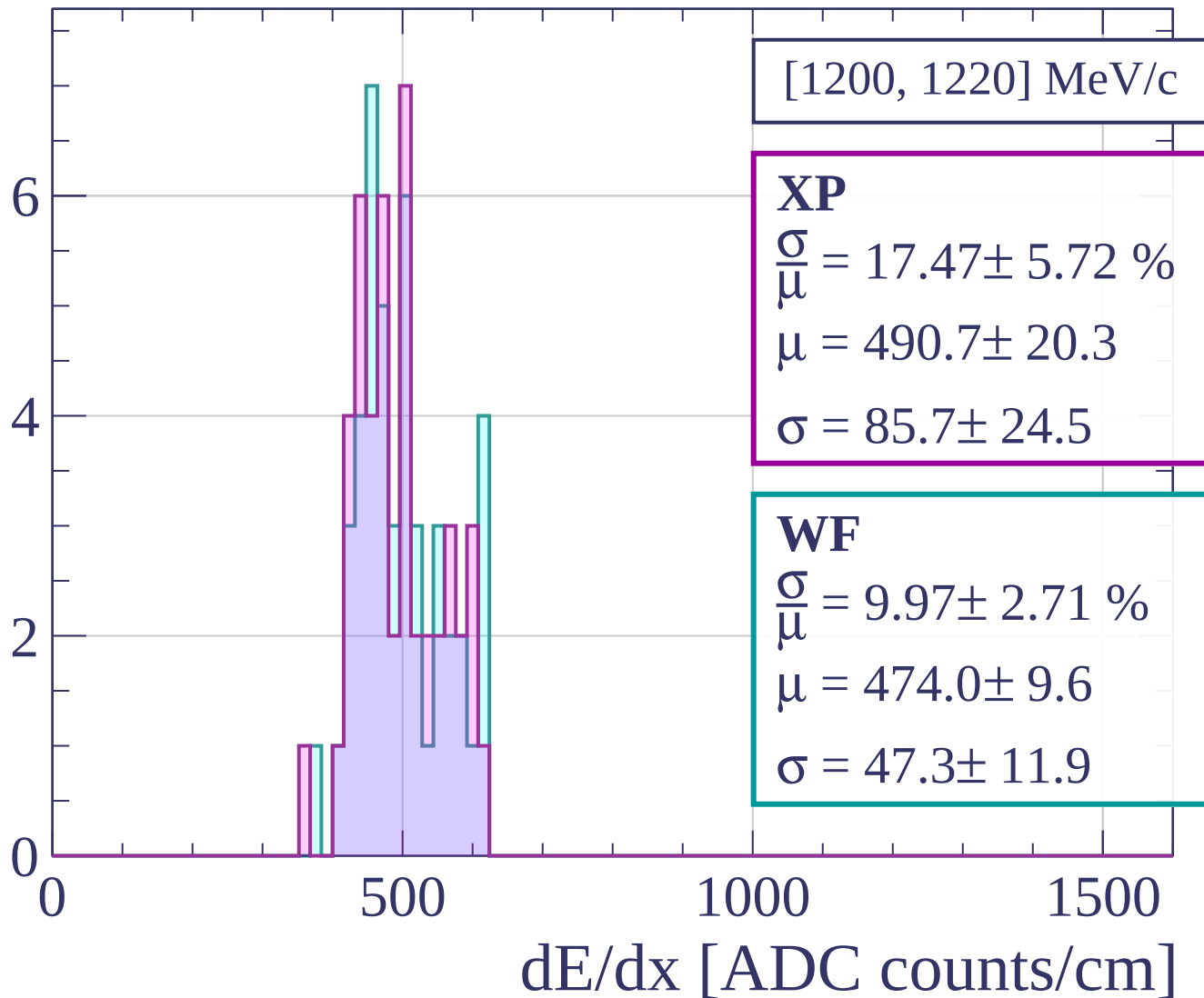
$$\sigma = 166.0 \pm 123.4$$

dE/dx [ADC counts/cm]

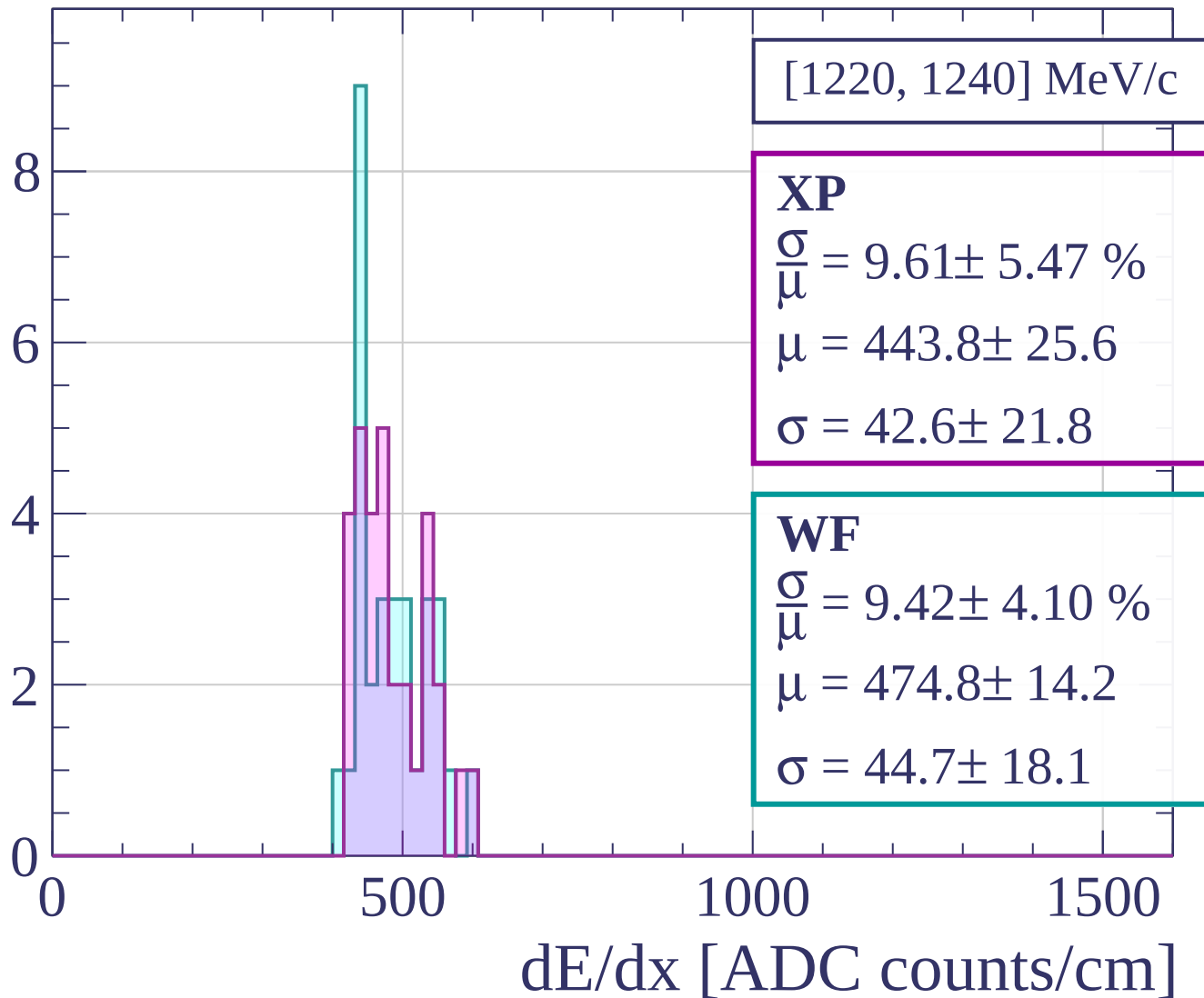
Count



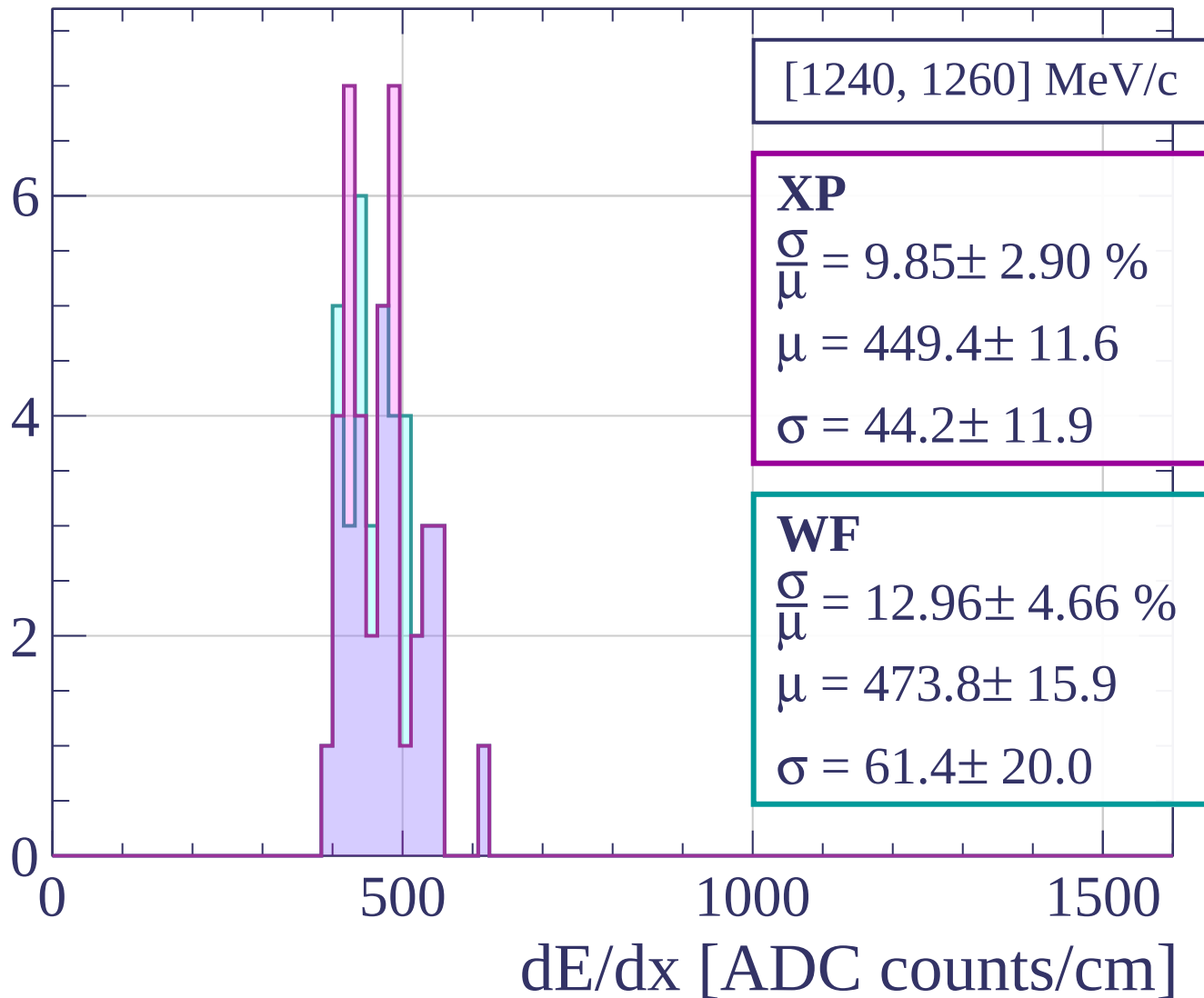
Count



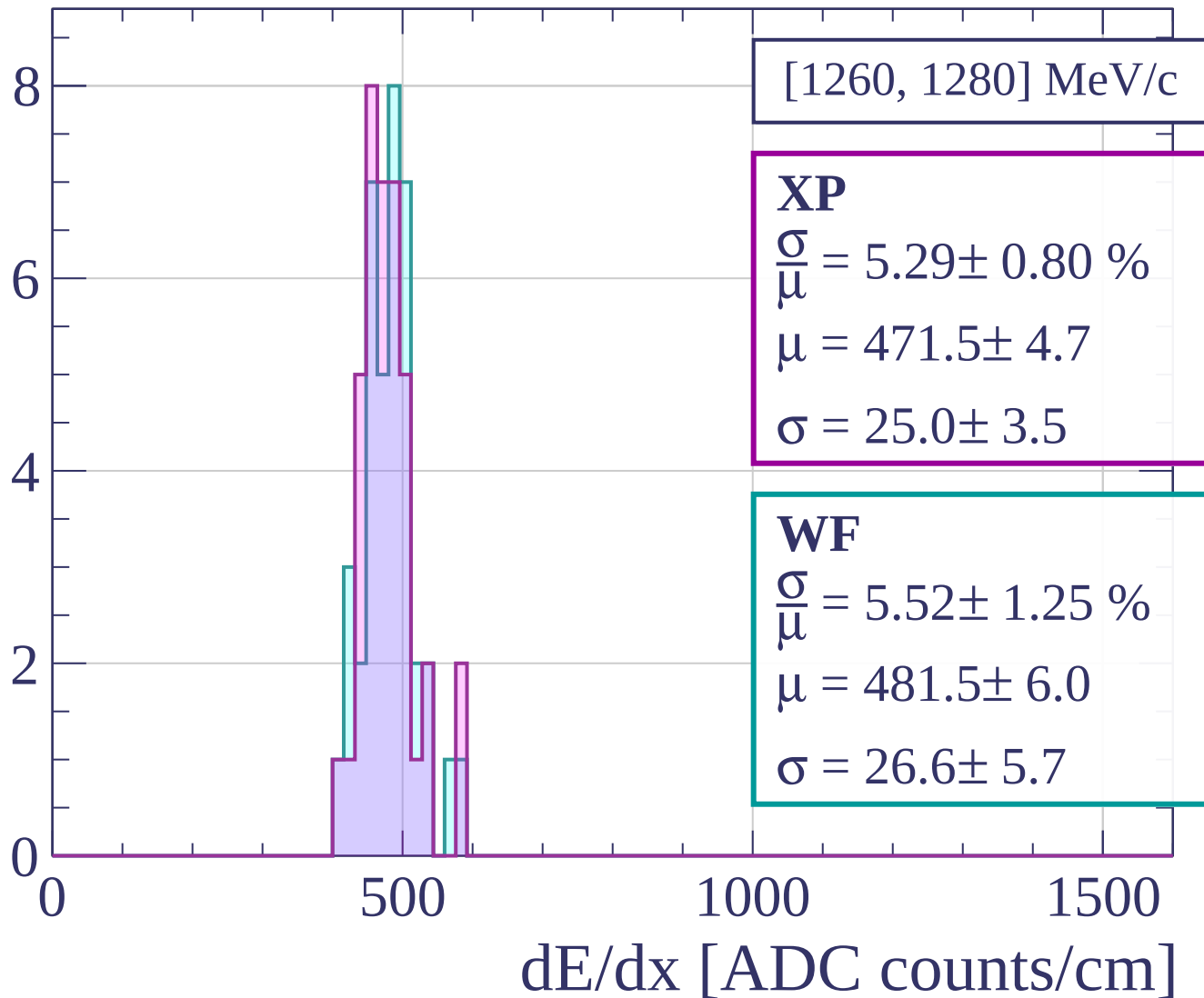
Count



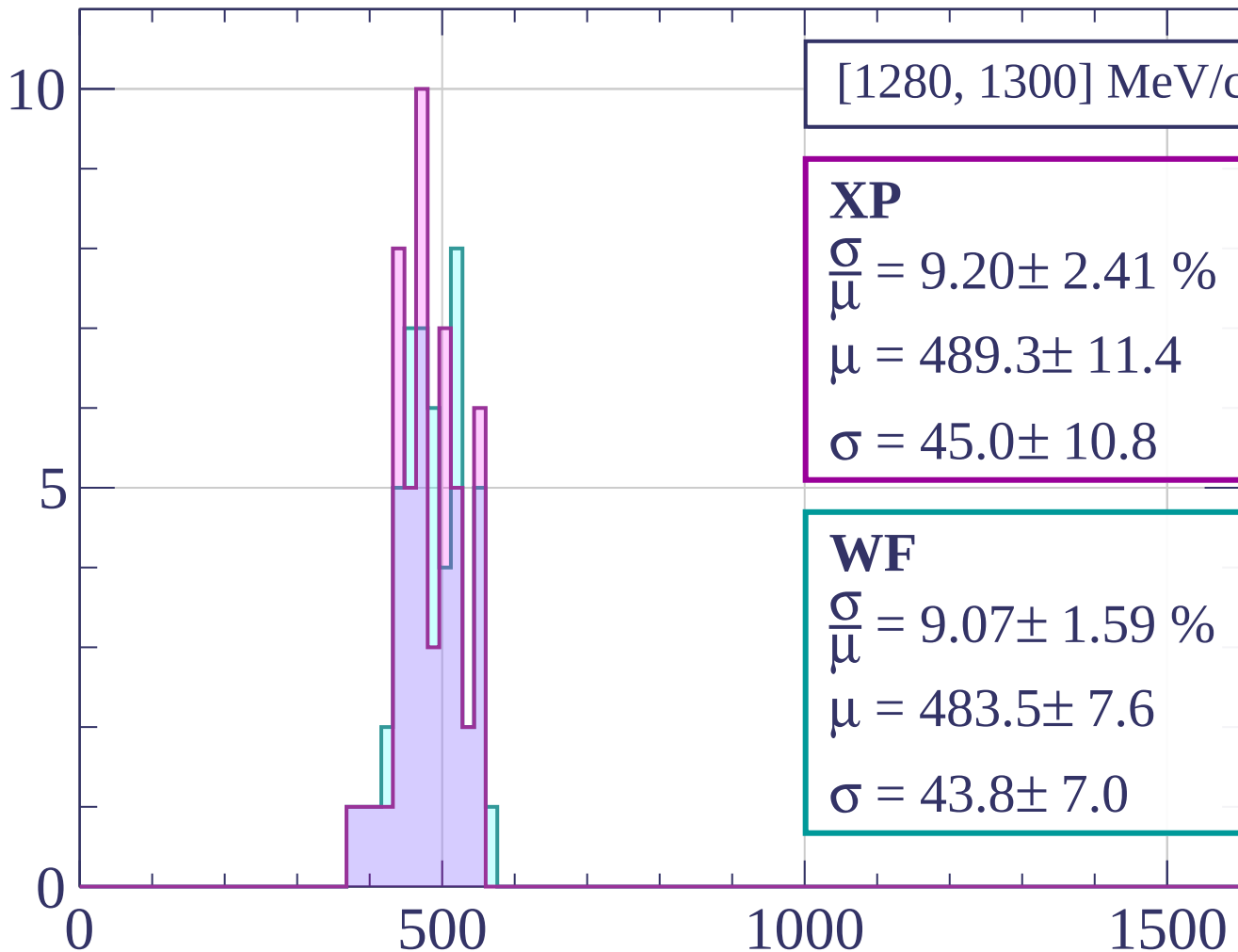
Count



Count

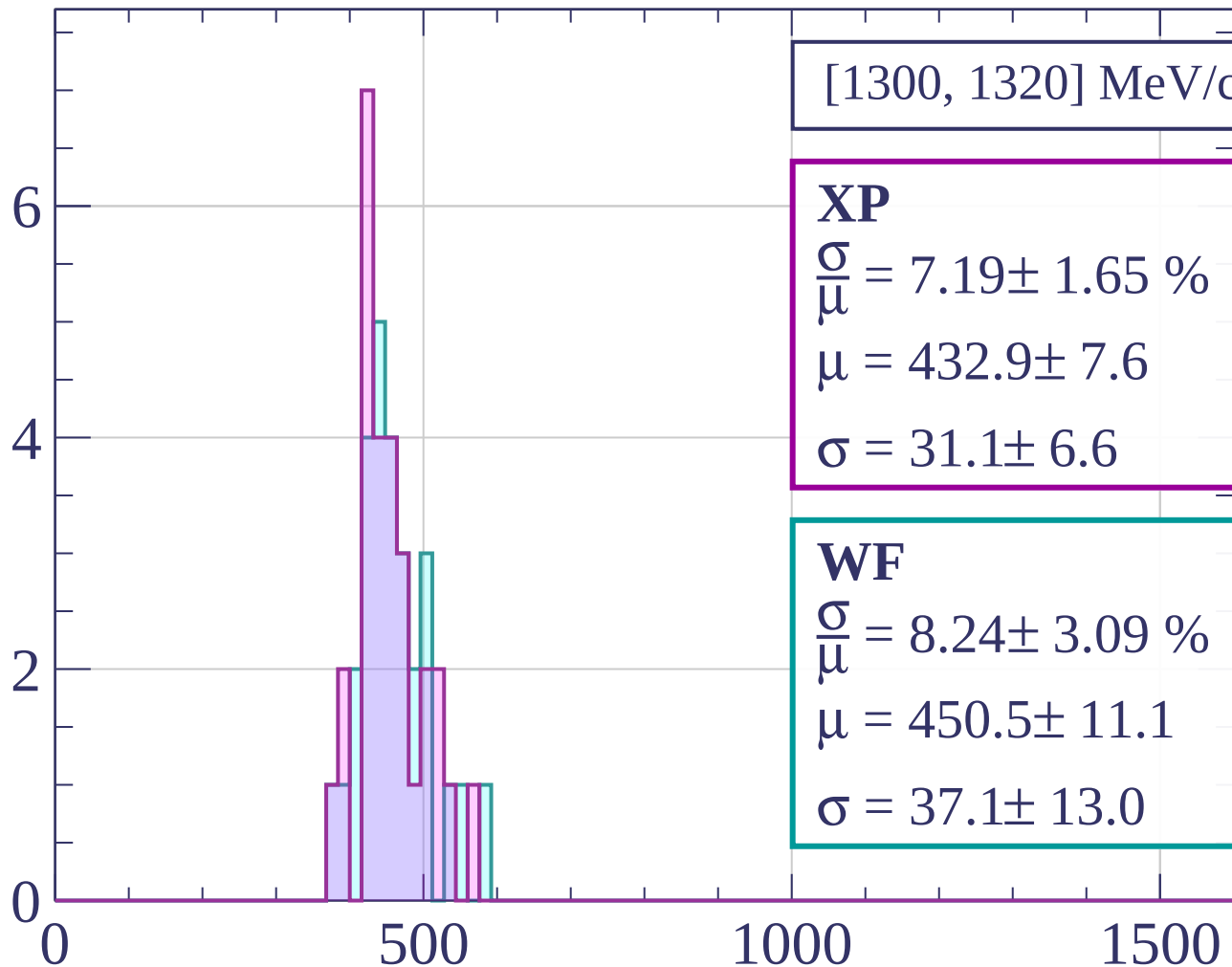


Count



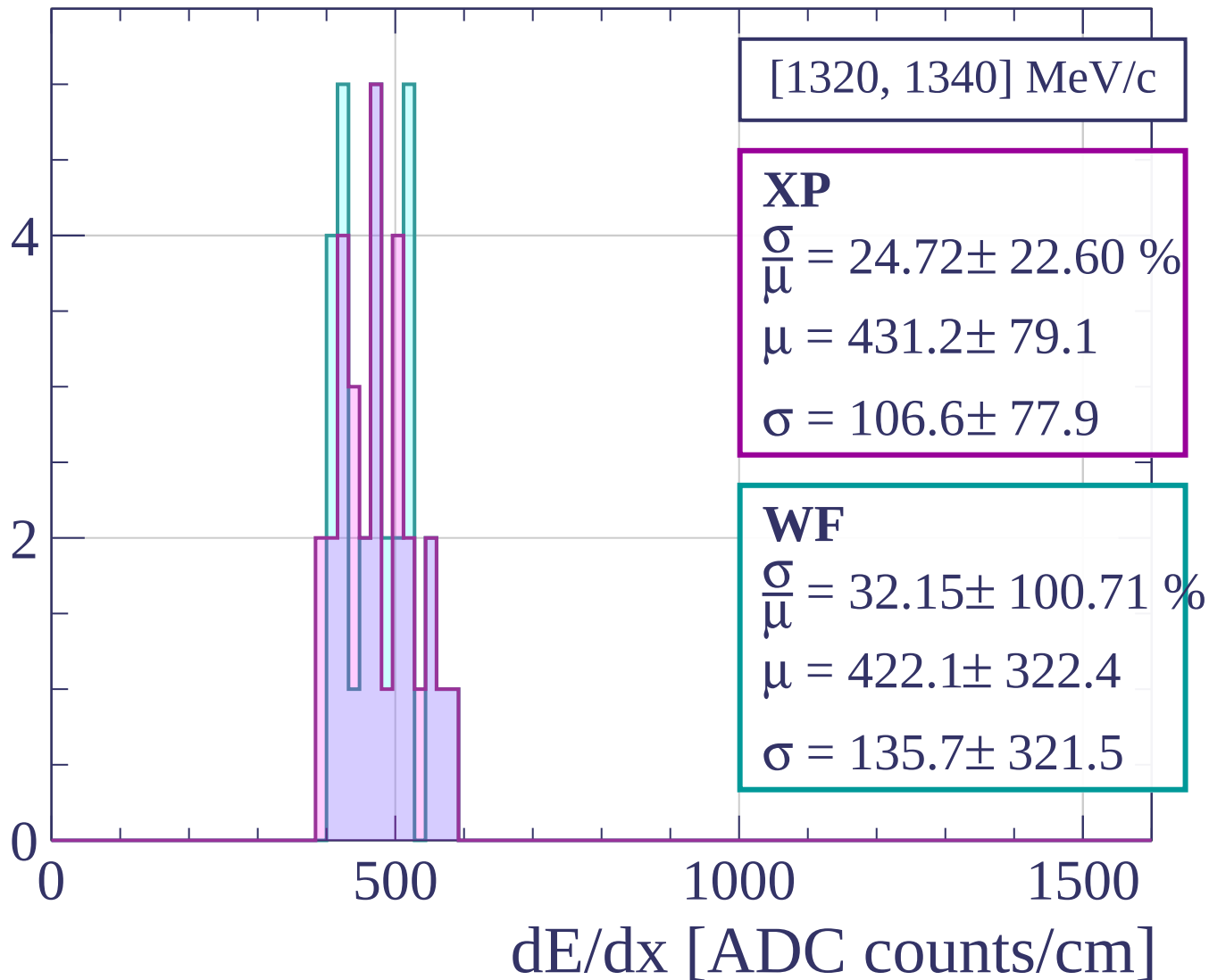
dE/dx [ADC counts/cm]

Count

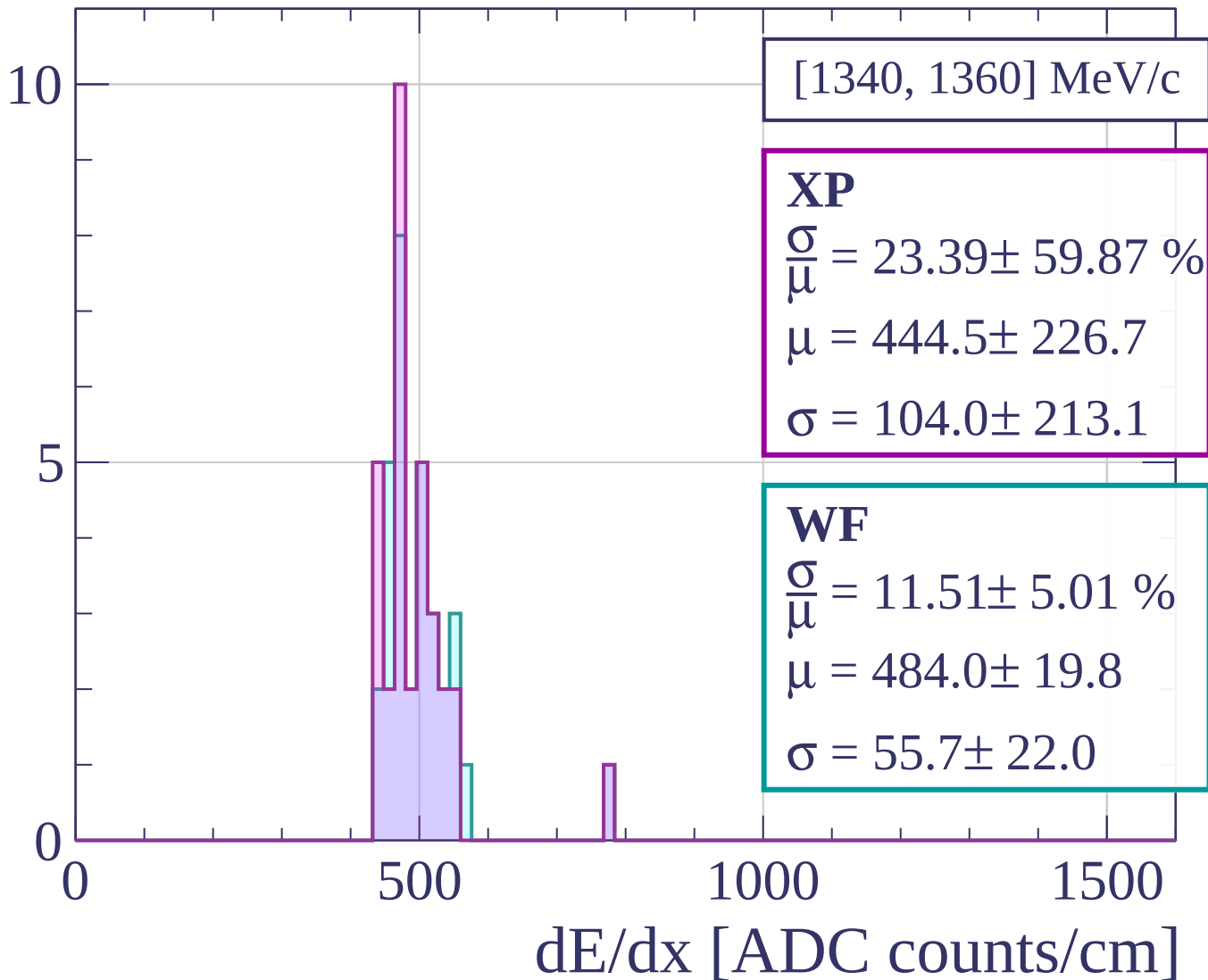


dE/dx [ADC counts/cm]

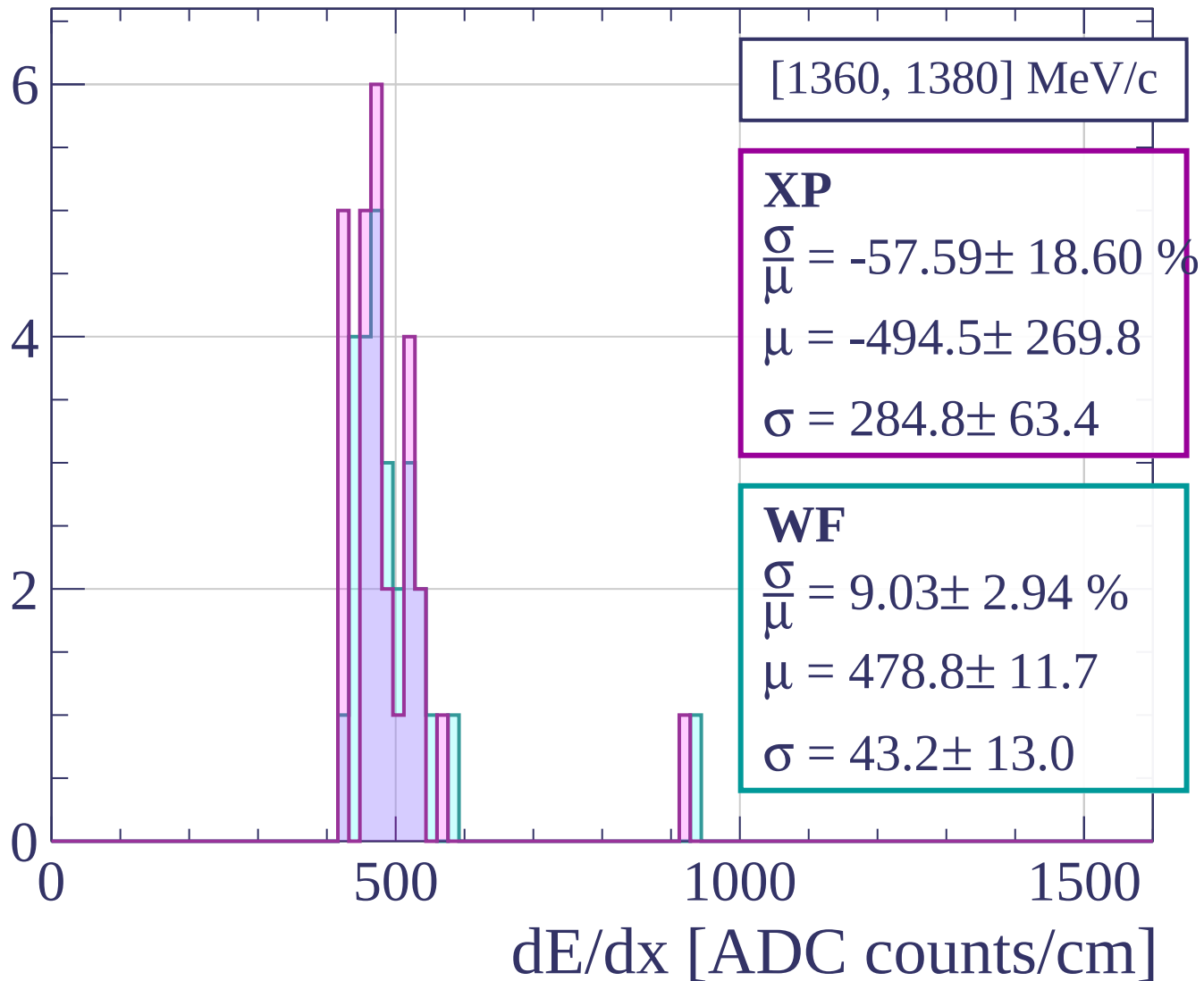
Count



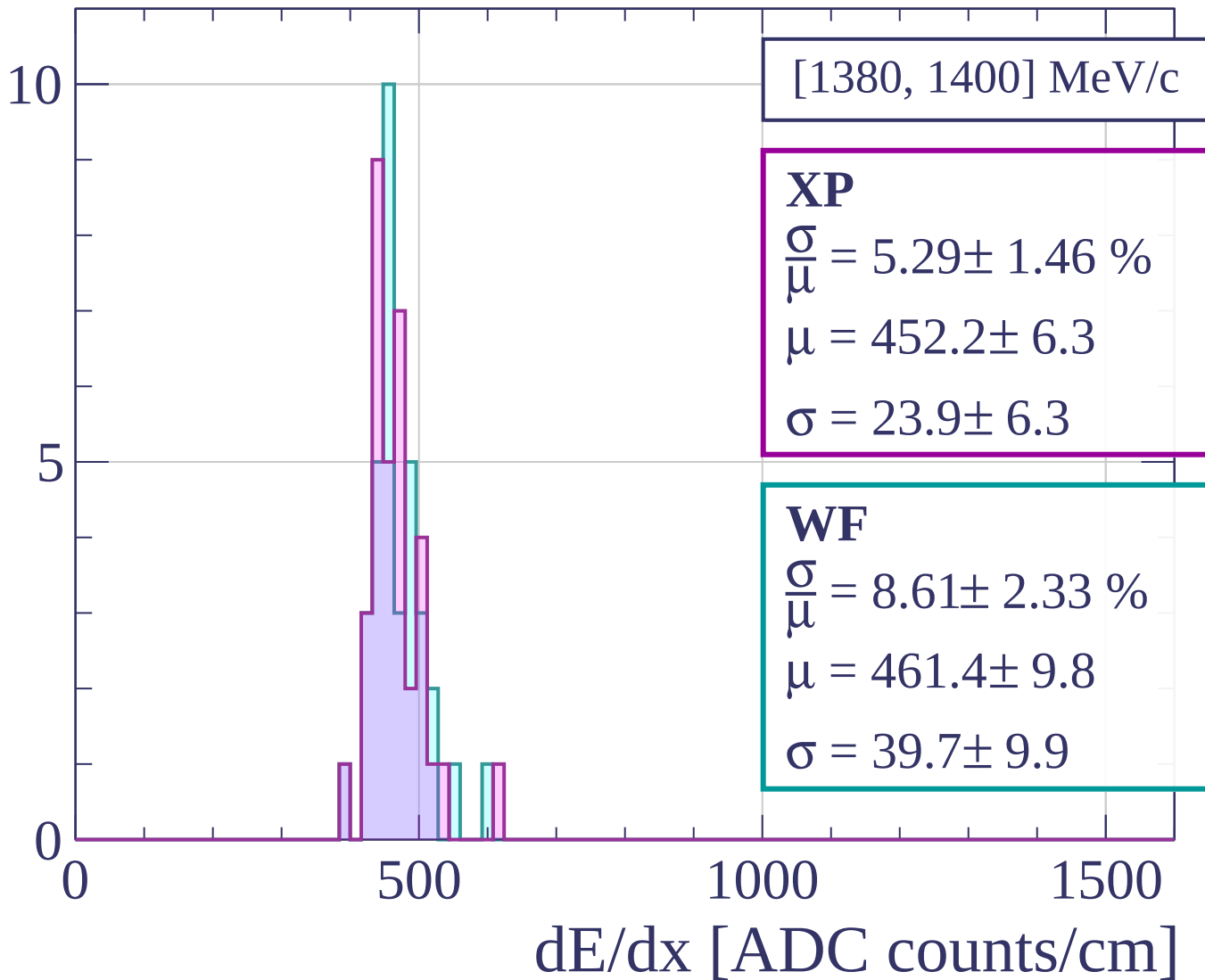
Count



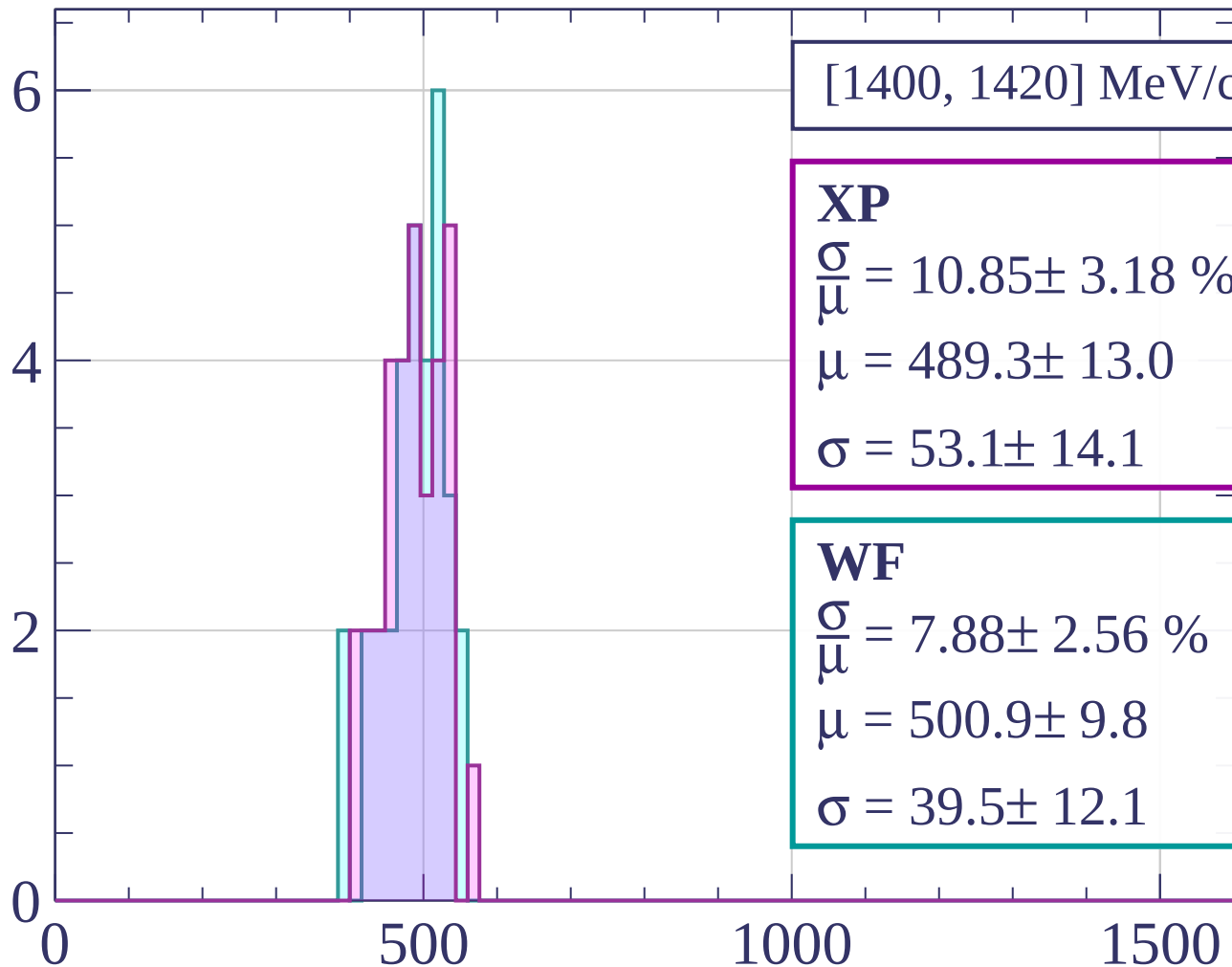
Count



Count

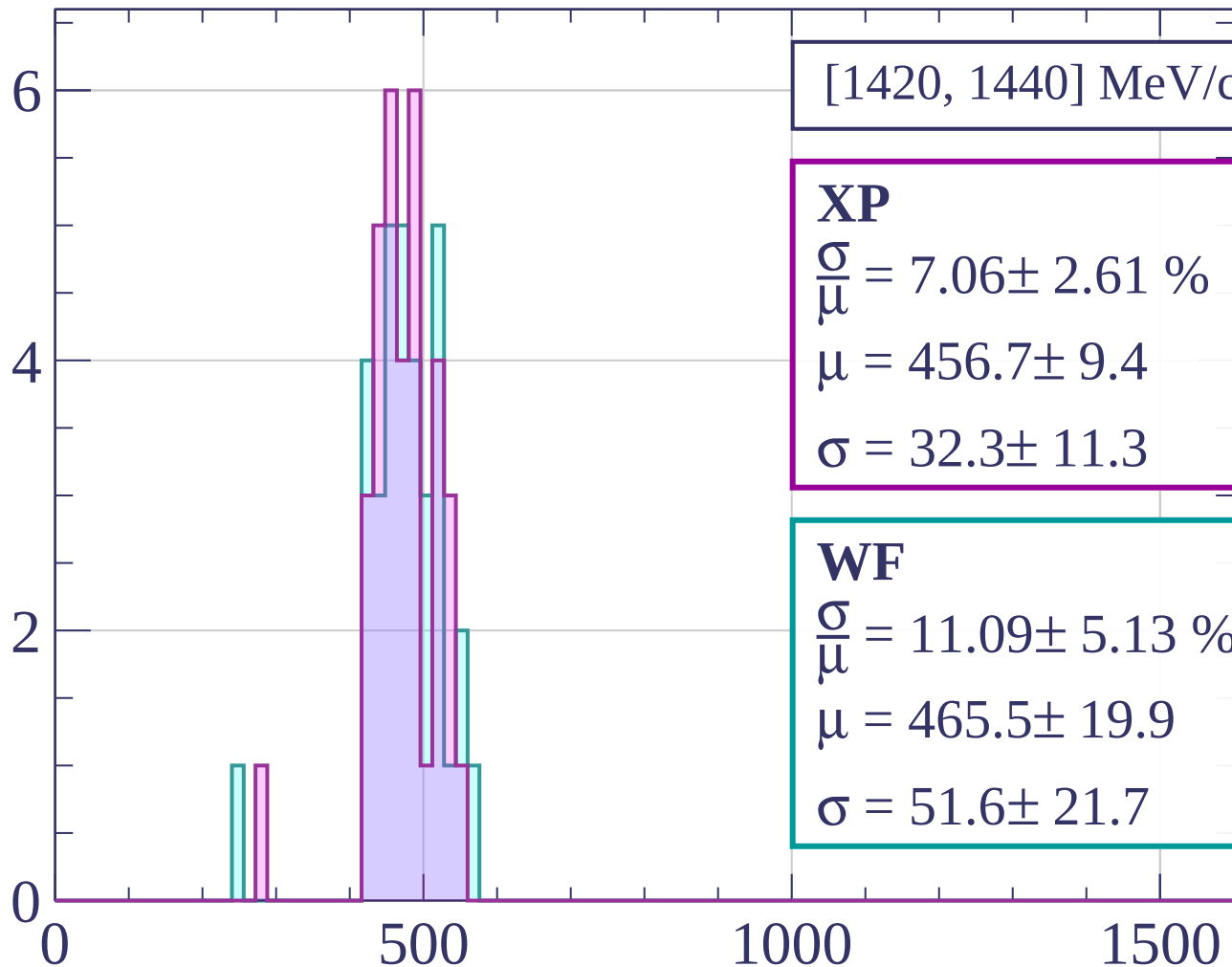


Count



dE/dx [ADC counts/cm]

Count



[1420, 1440] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.06 \pm 2.61 \%$$

$$\mu = 456.7 \pm 9.4$$

$$\sigma = 32.3 \pm 11.3$$

WF

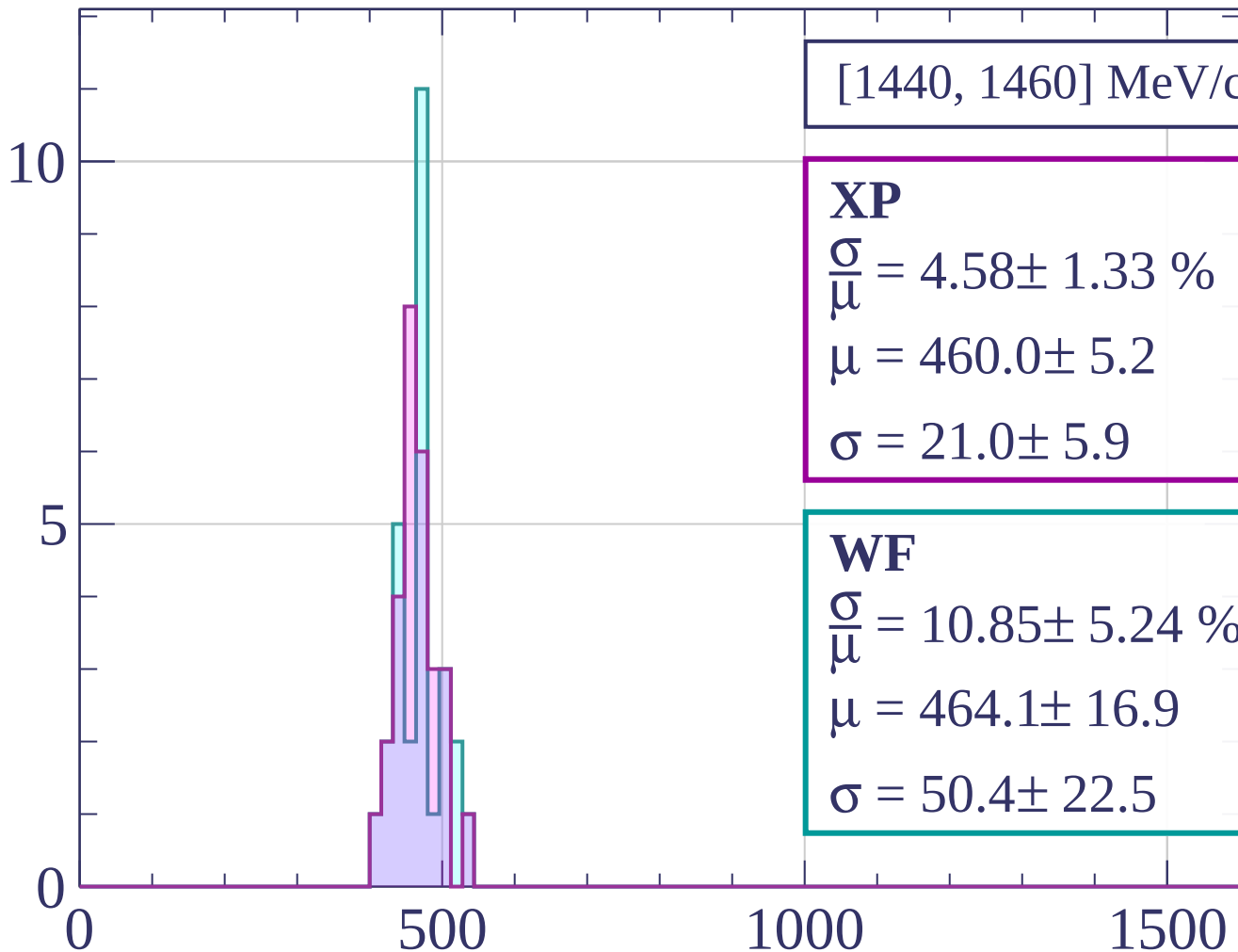
$$\frac{\sigma}{\mu} = 11.09 \pm 5.13 \%$$

$$\mu = 465.5 \pm 19.9$$

$$\sigma = 51.6 \pm 21.7$$

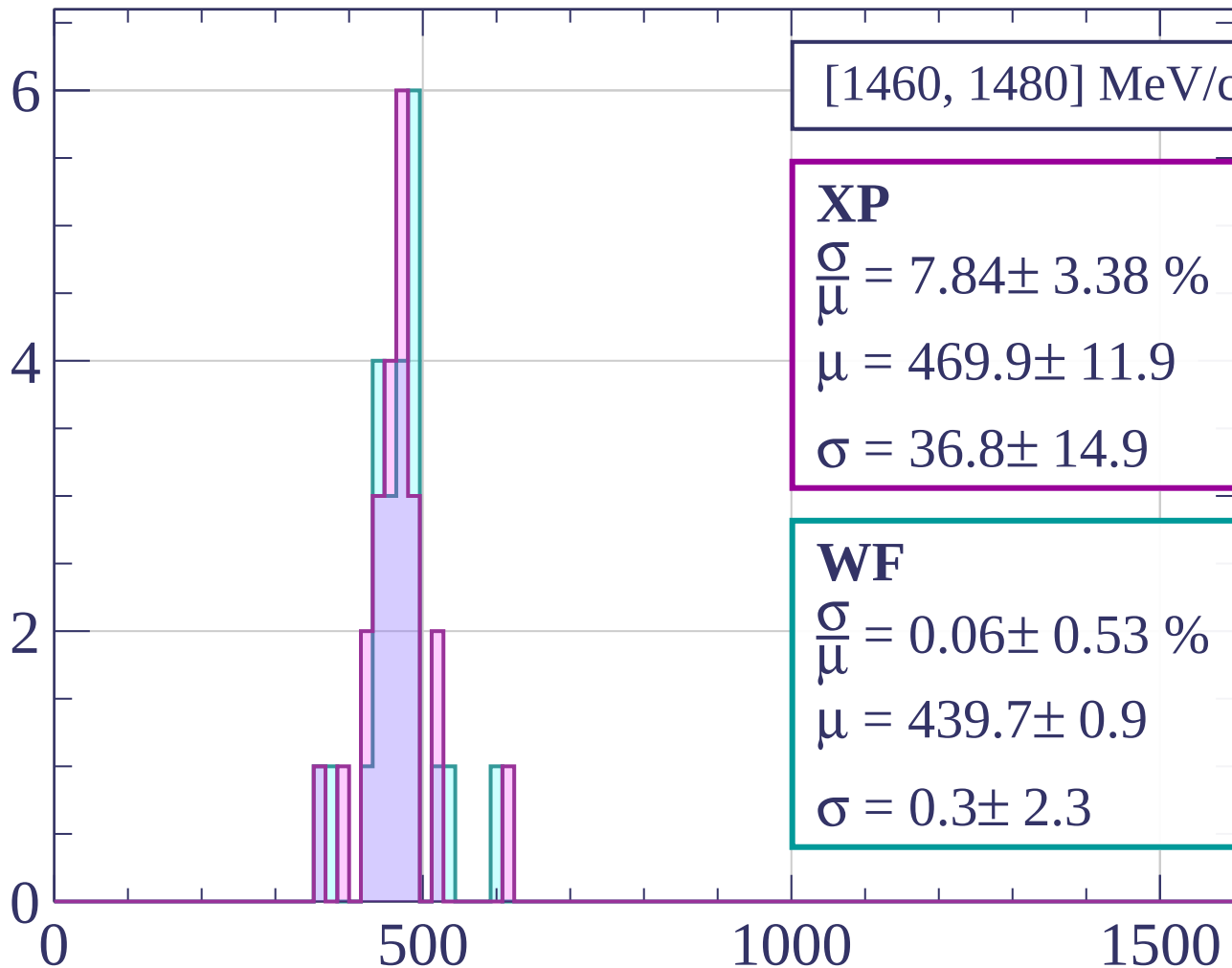
dE/dx [ADC counts/cm]

Count



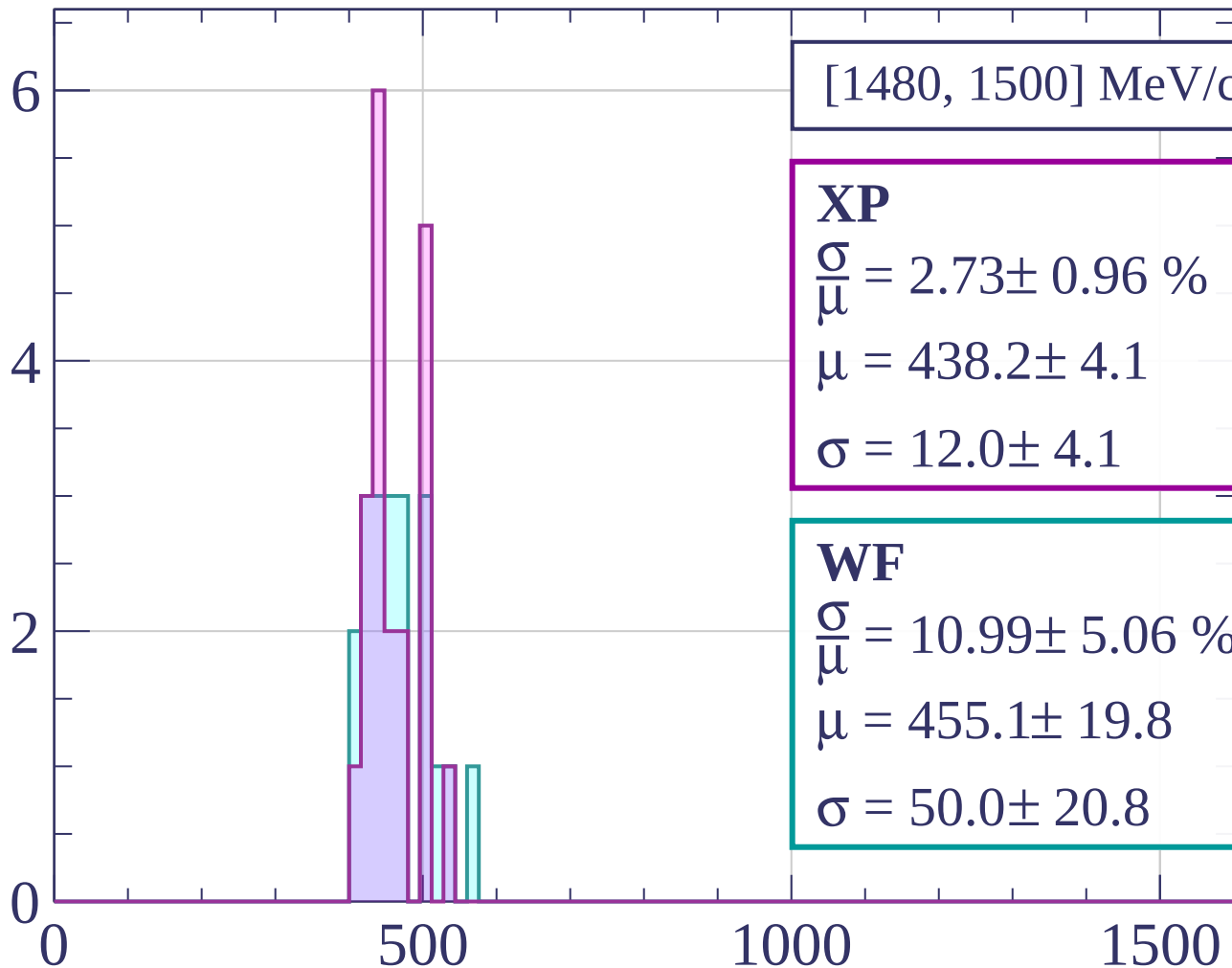
dE/dx [ADC counts/cm]

Count



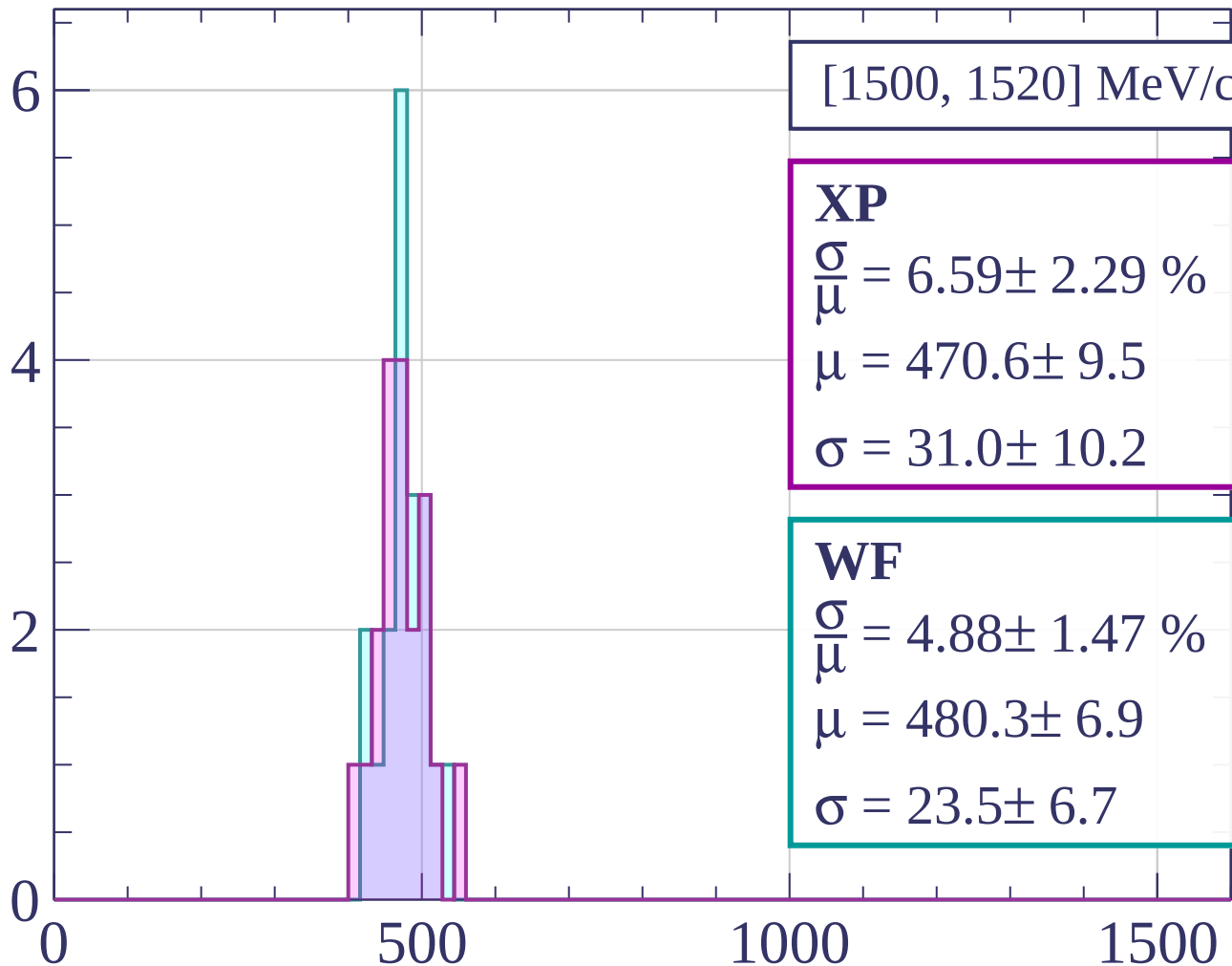
dE/dx [ADC counts/cm]

Count



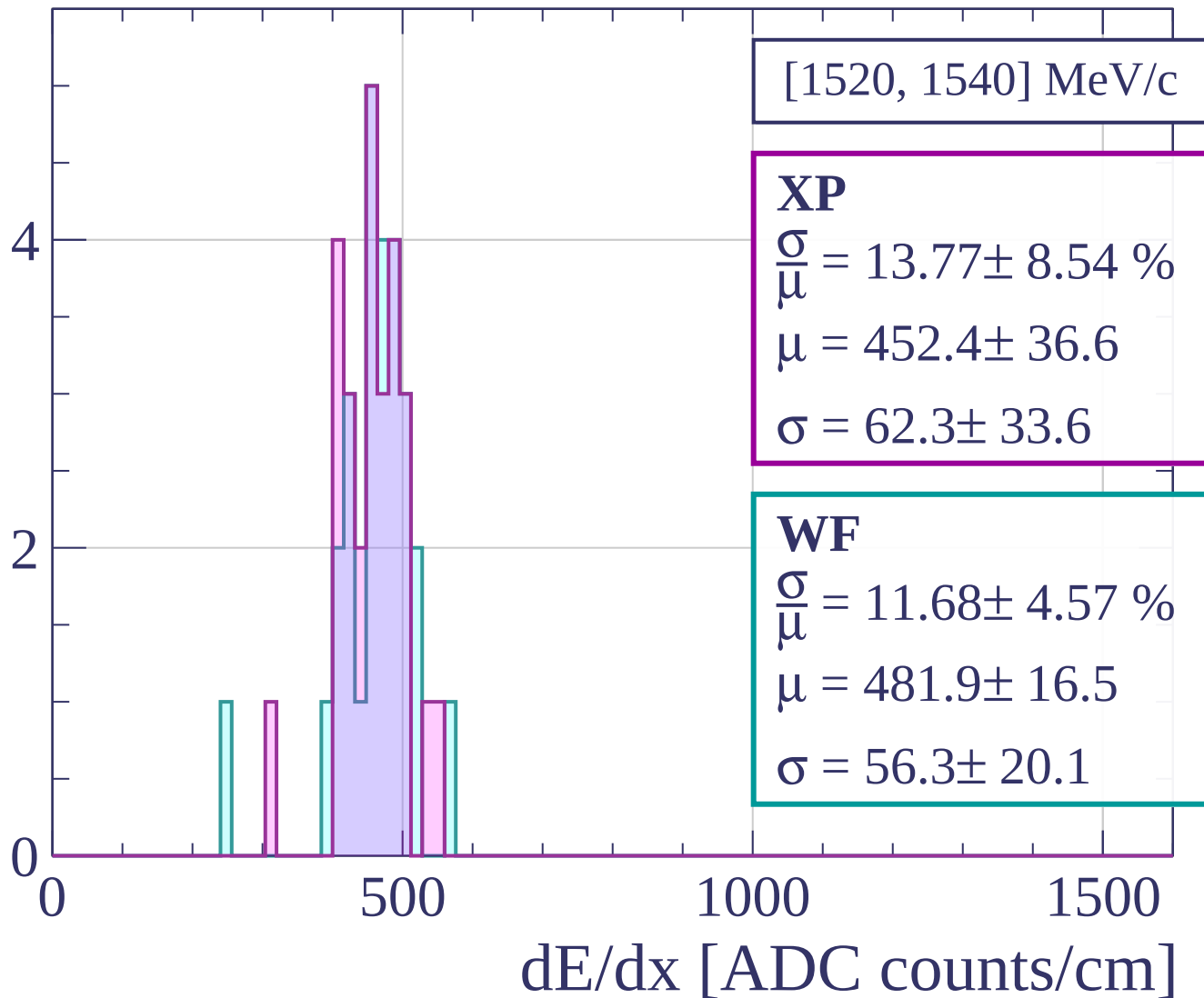
dE/dx [ADC counts/cm]

Count

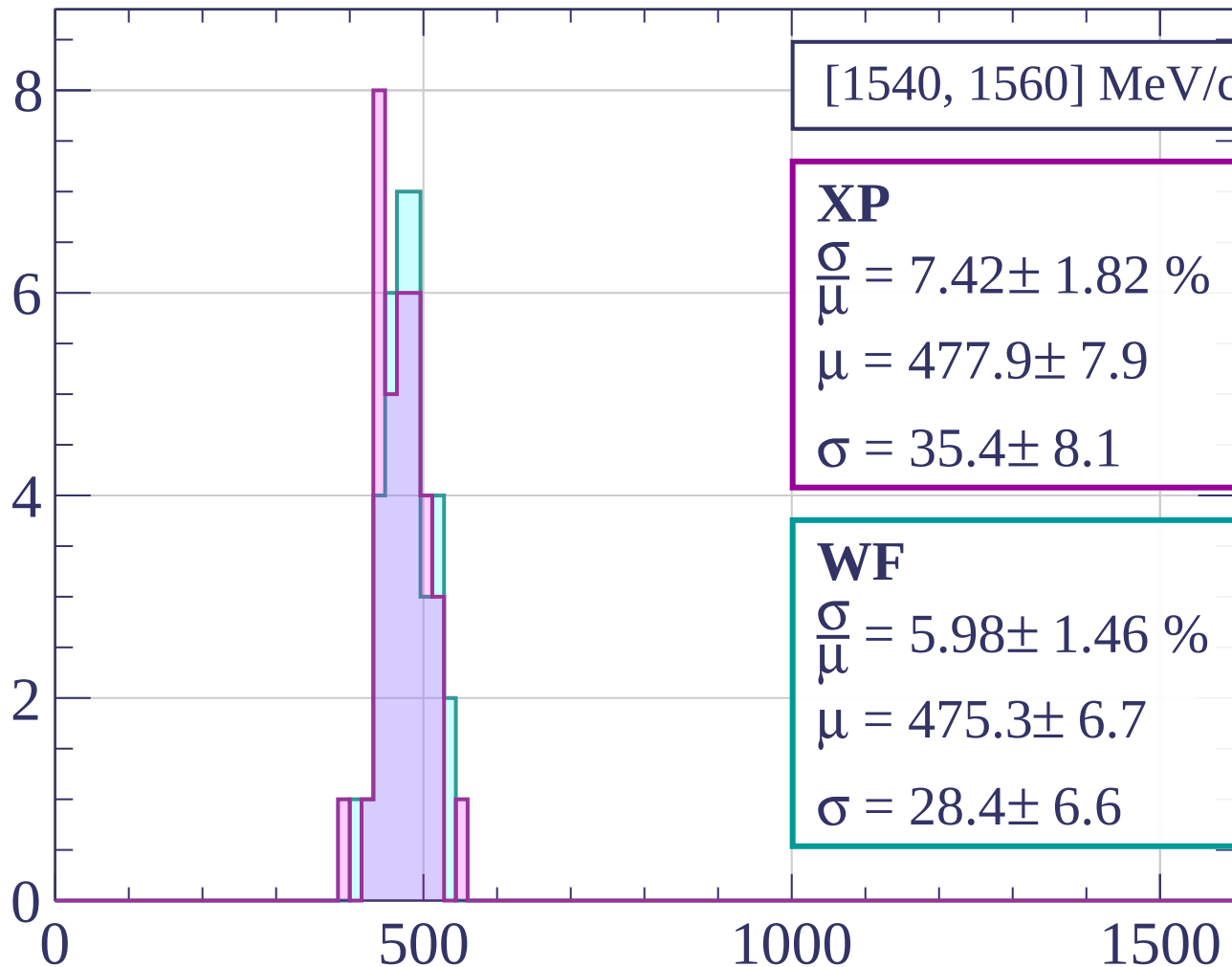


dE/dx [ADC counts/cm]

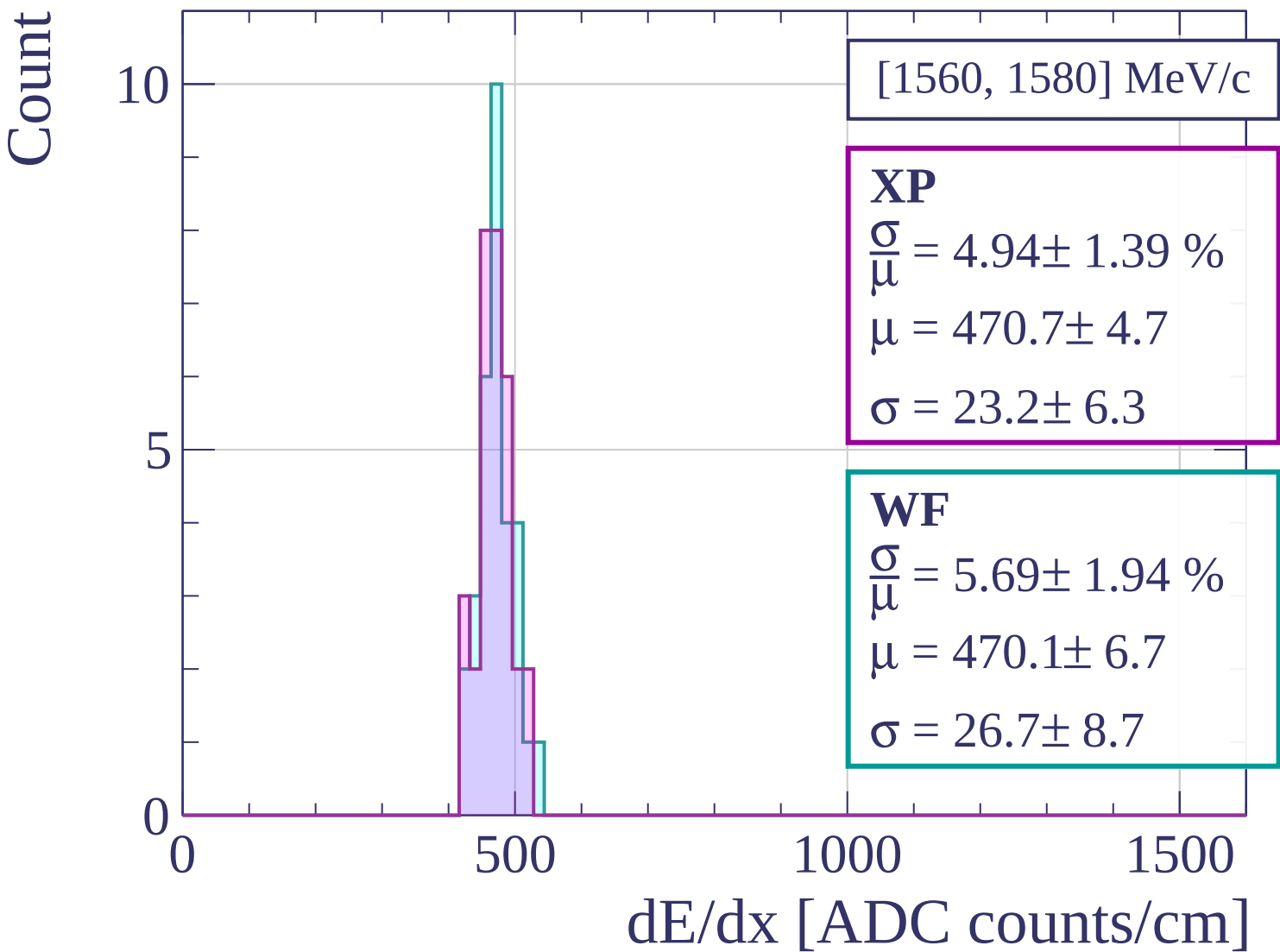
Count



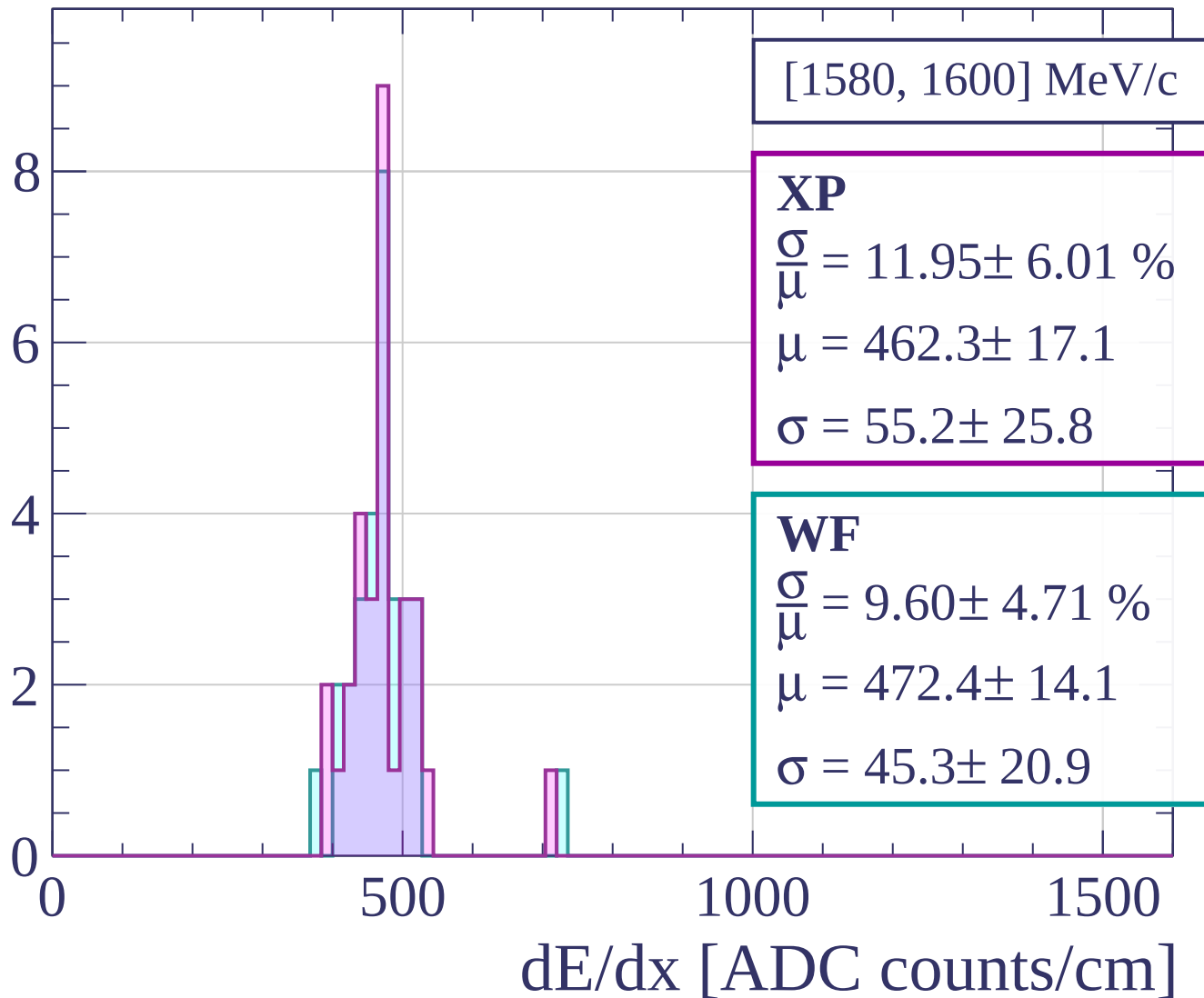
Count



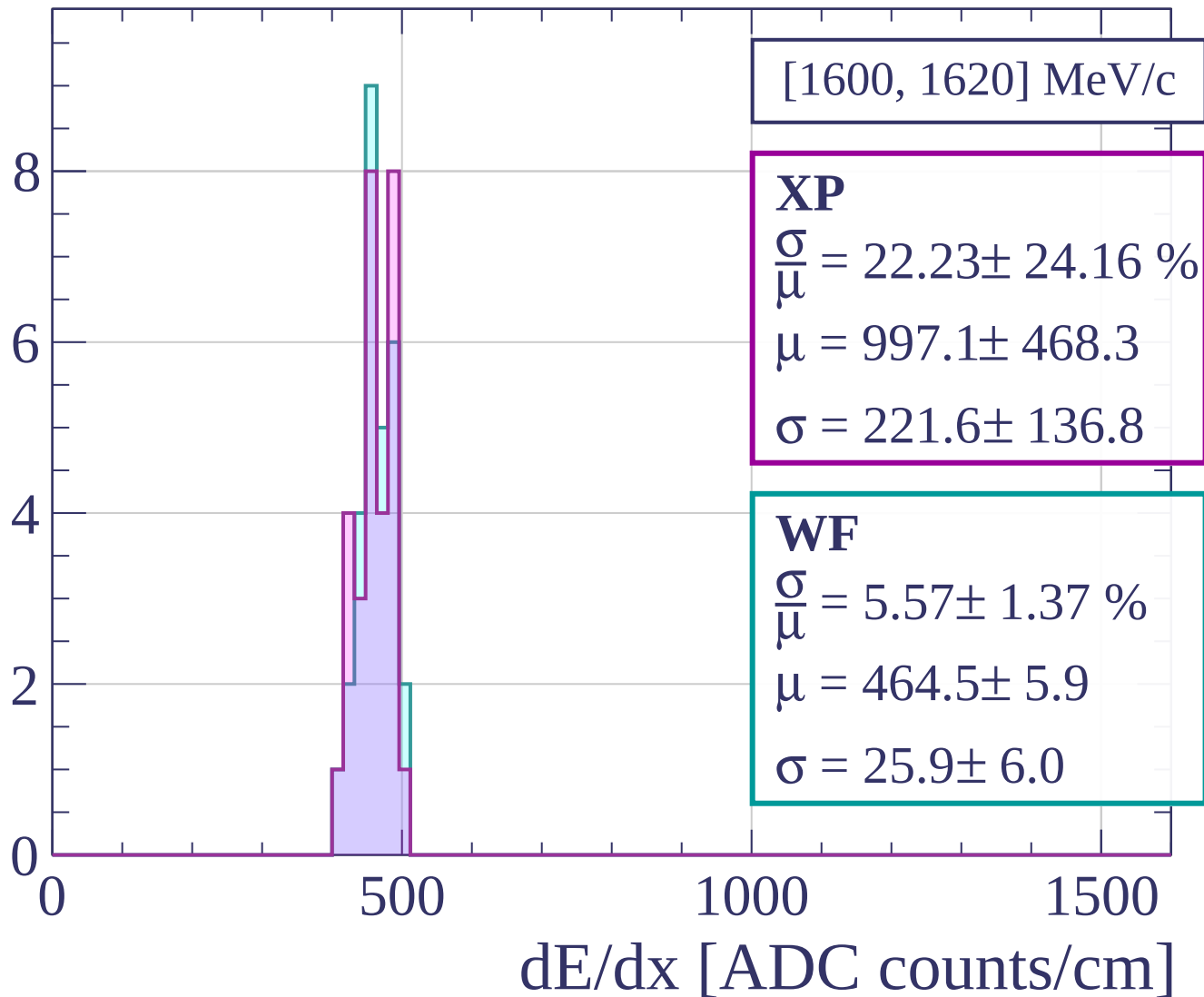
dE/dx [ADC counts/cm]



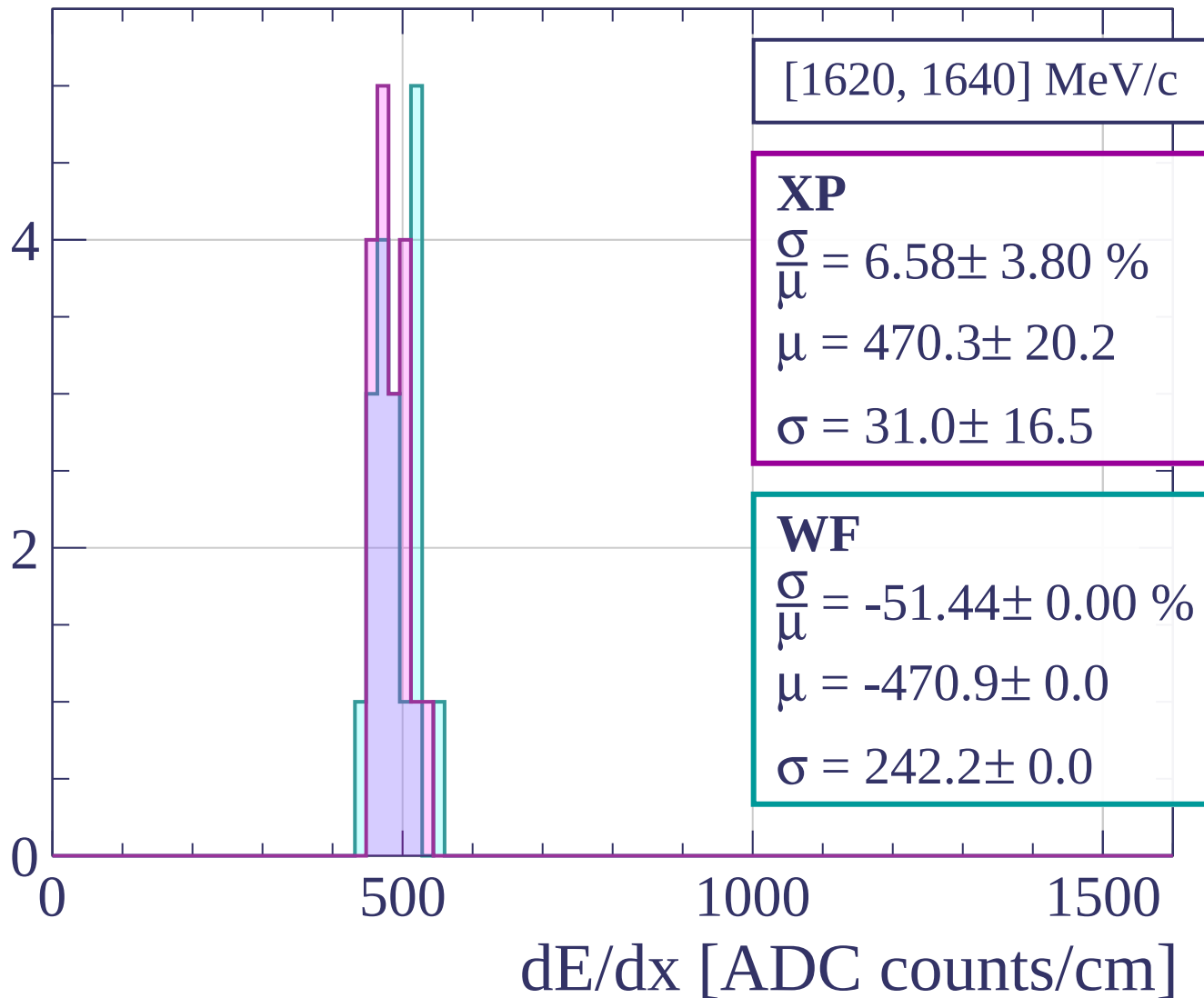
Count



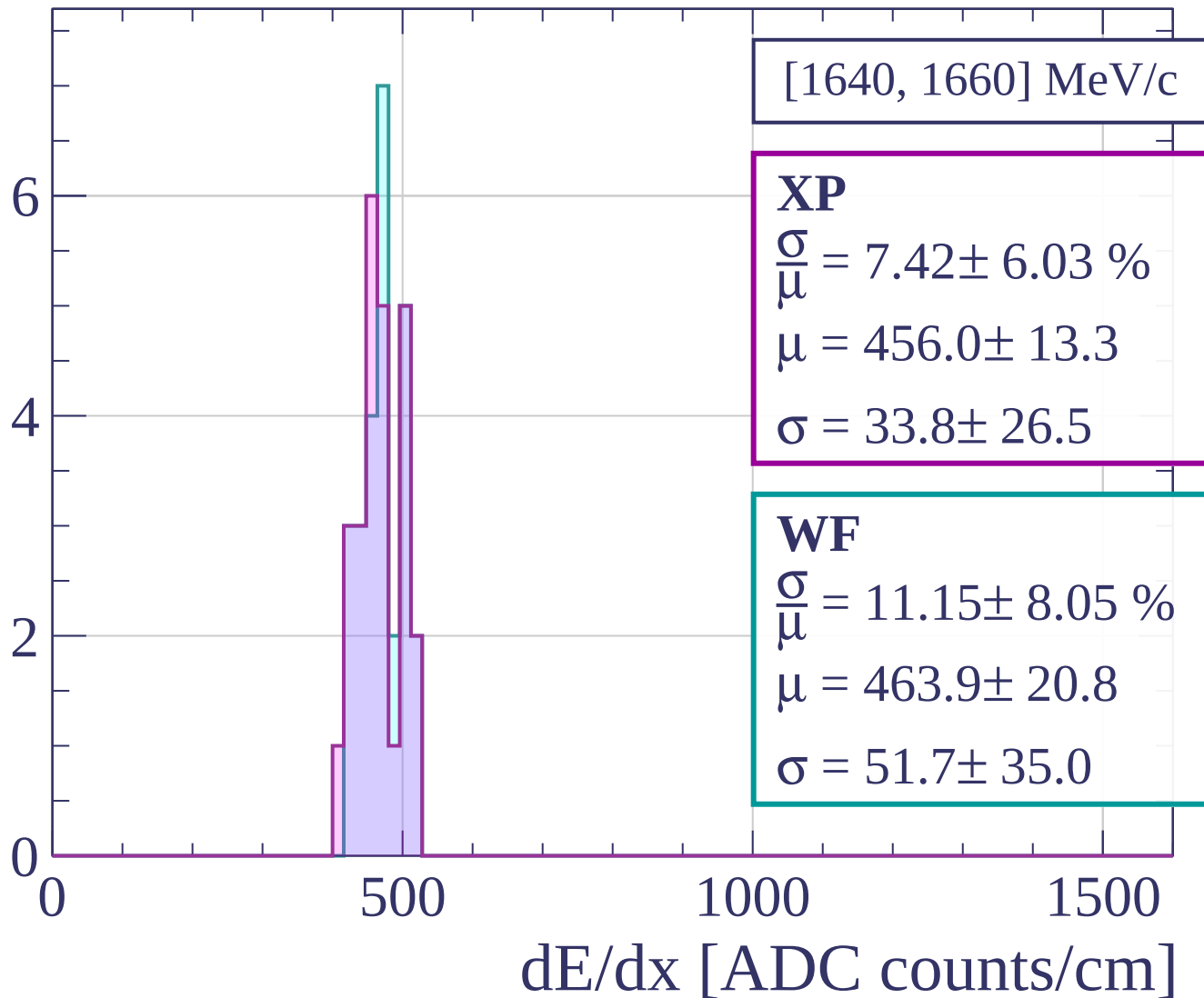
Count



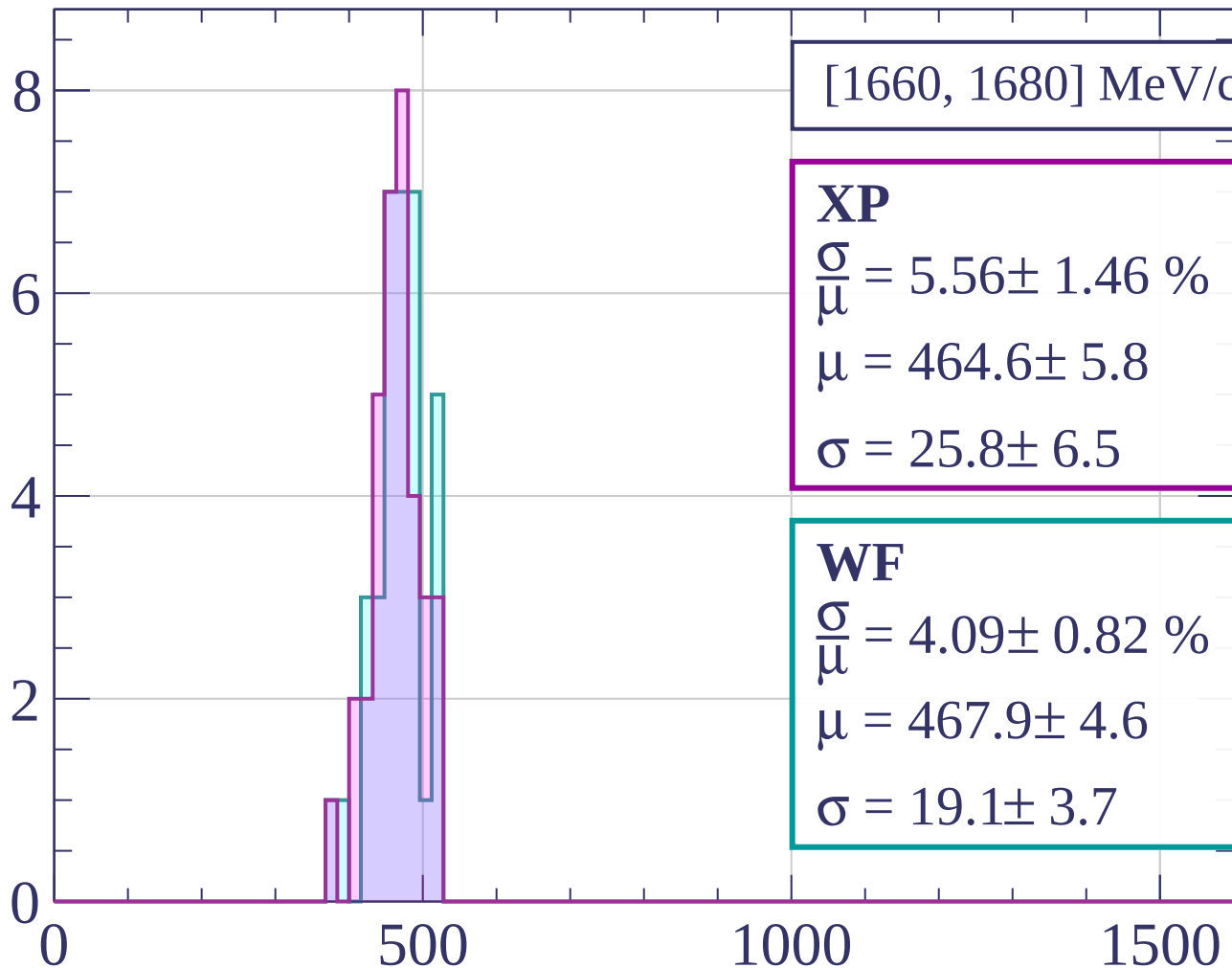
Count



Count

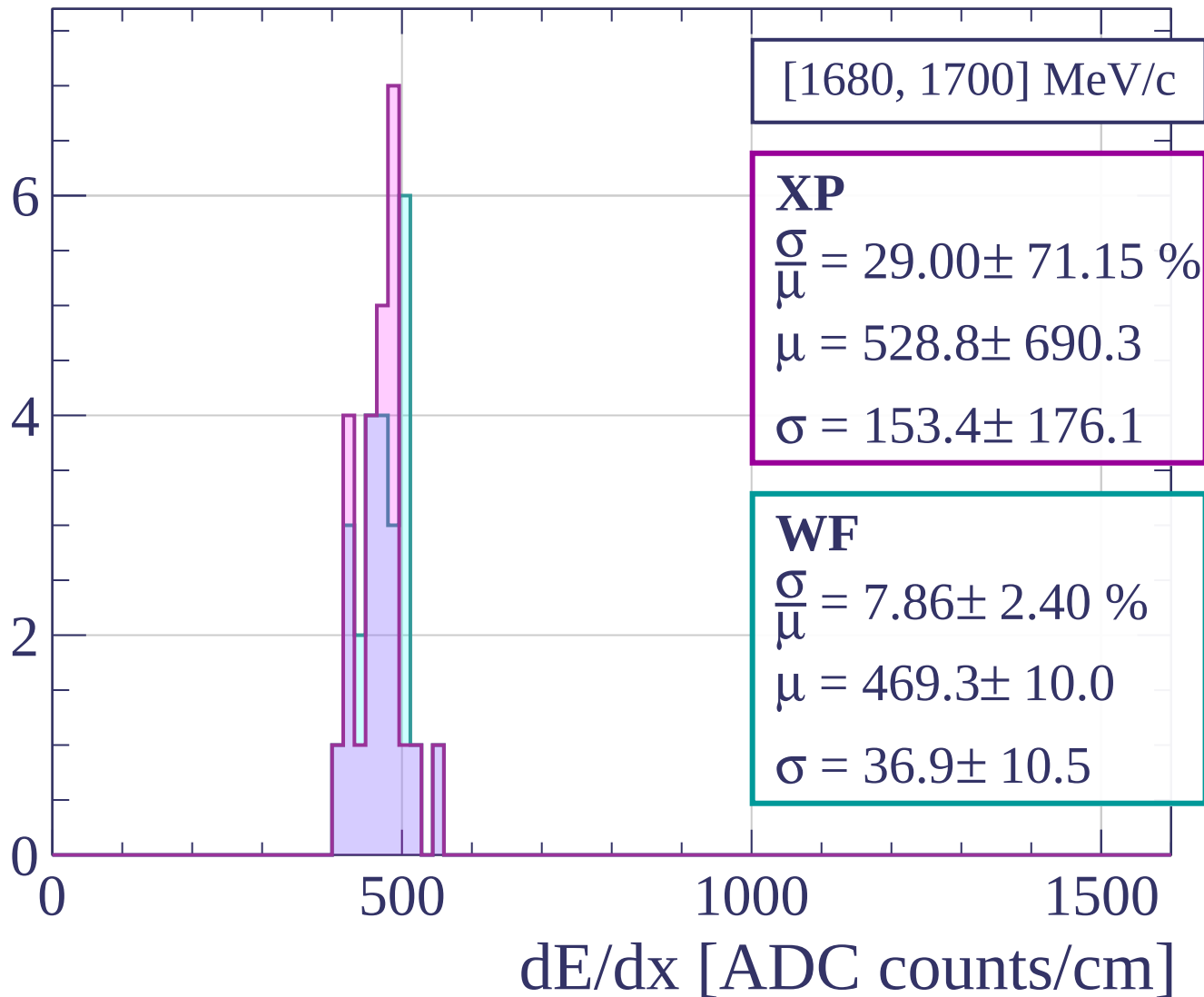


Count

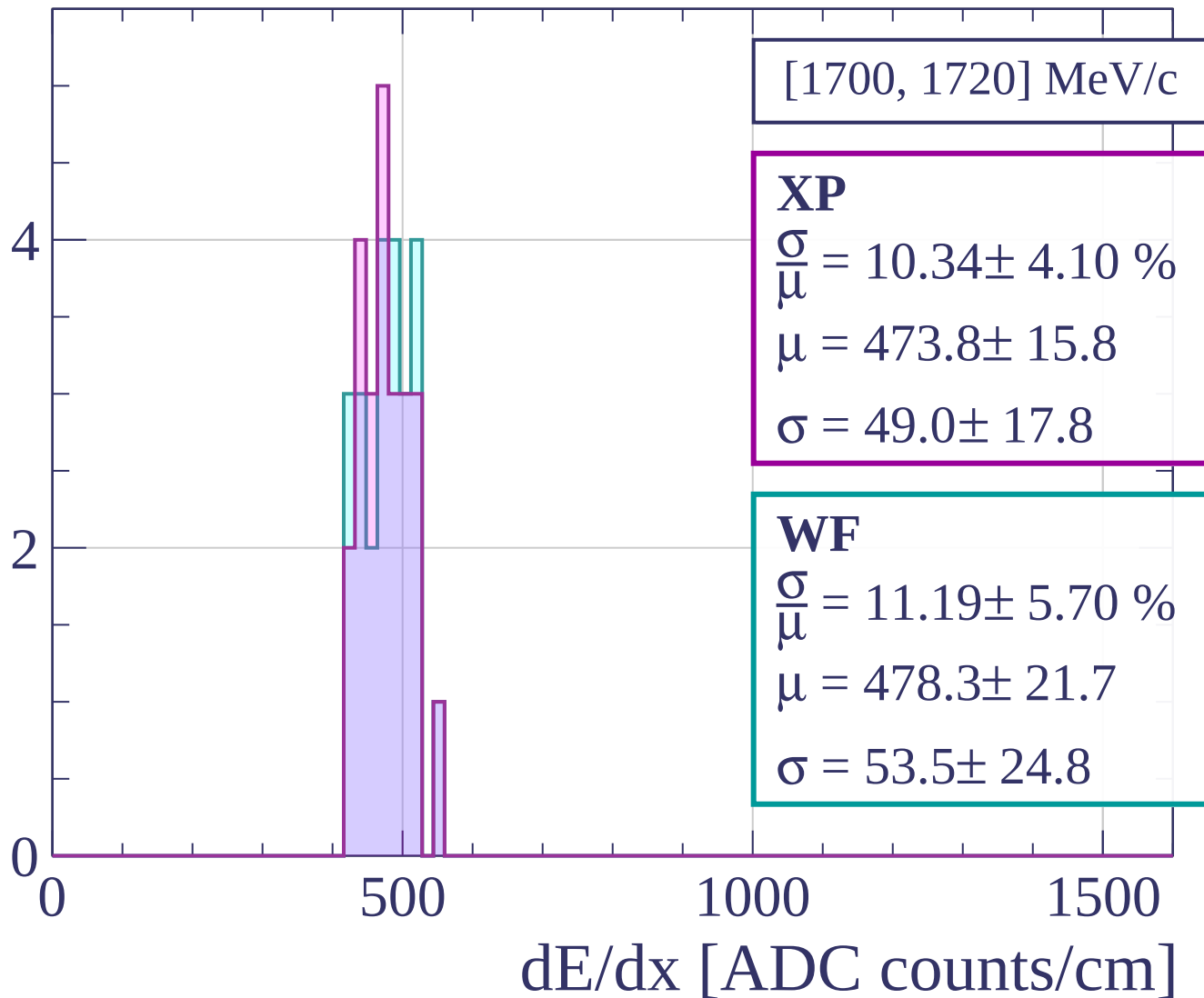


dE/dx [ADC counts/cm]

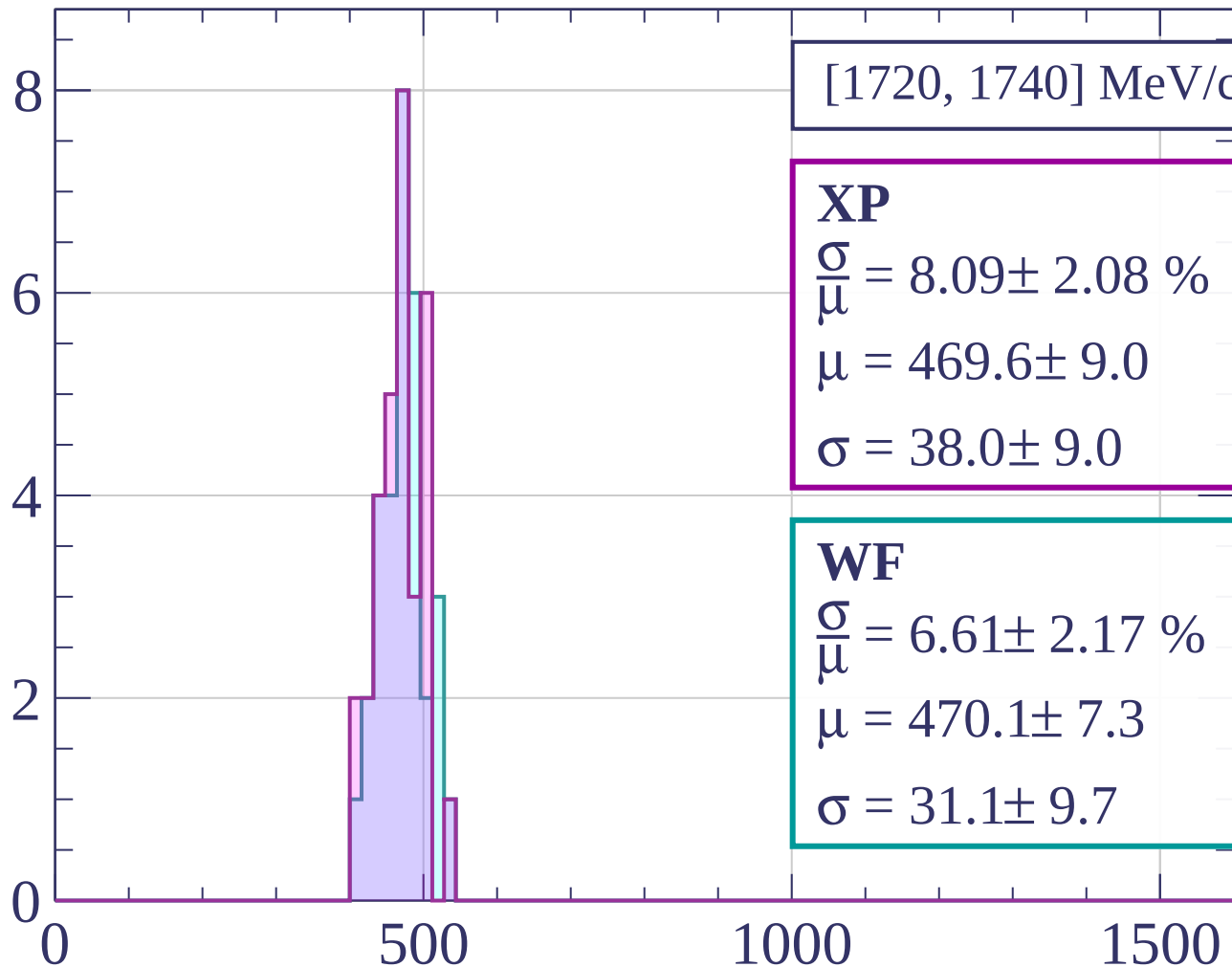
Count



Count

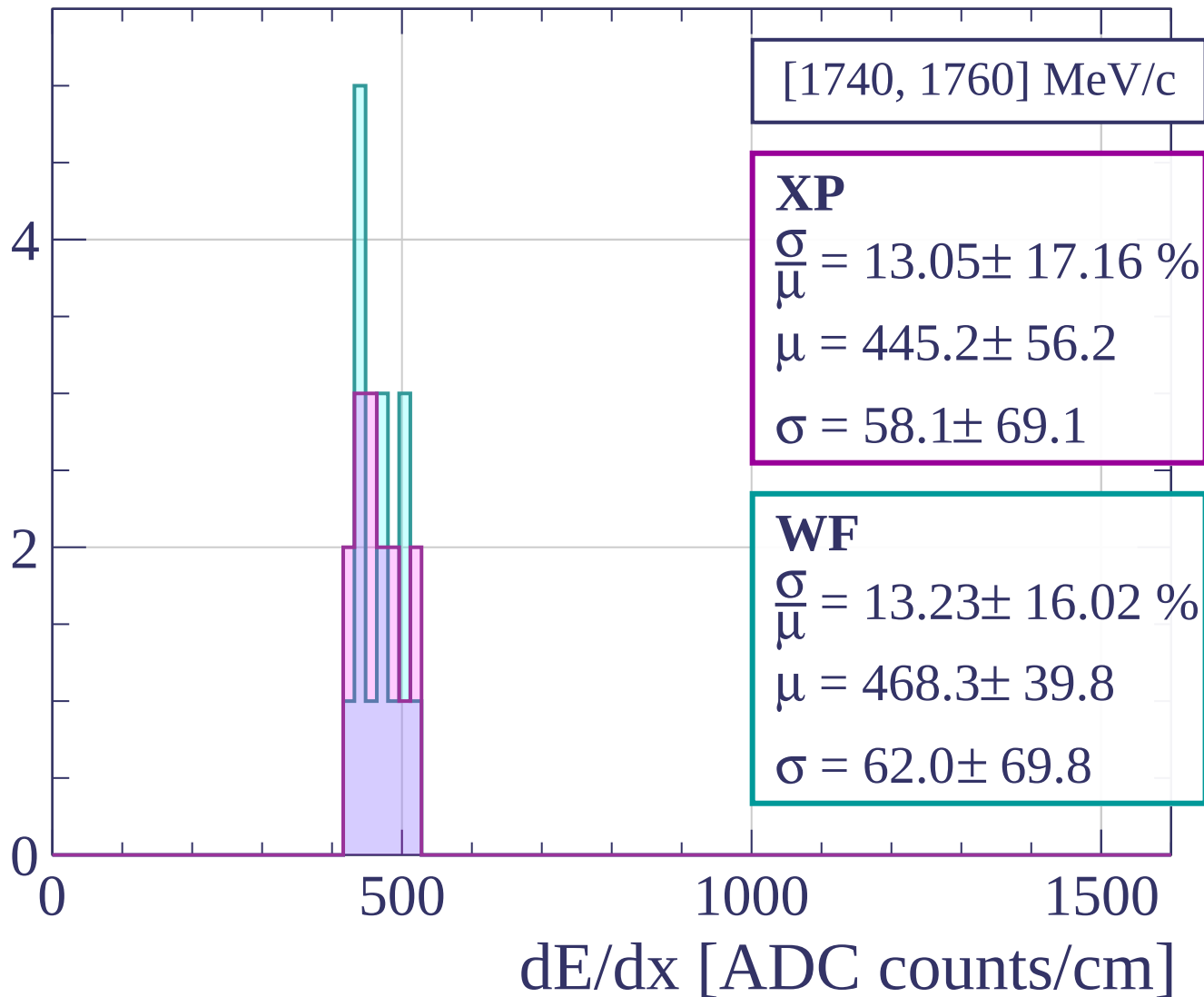


Count

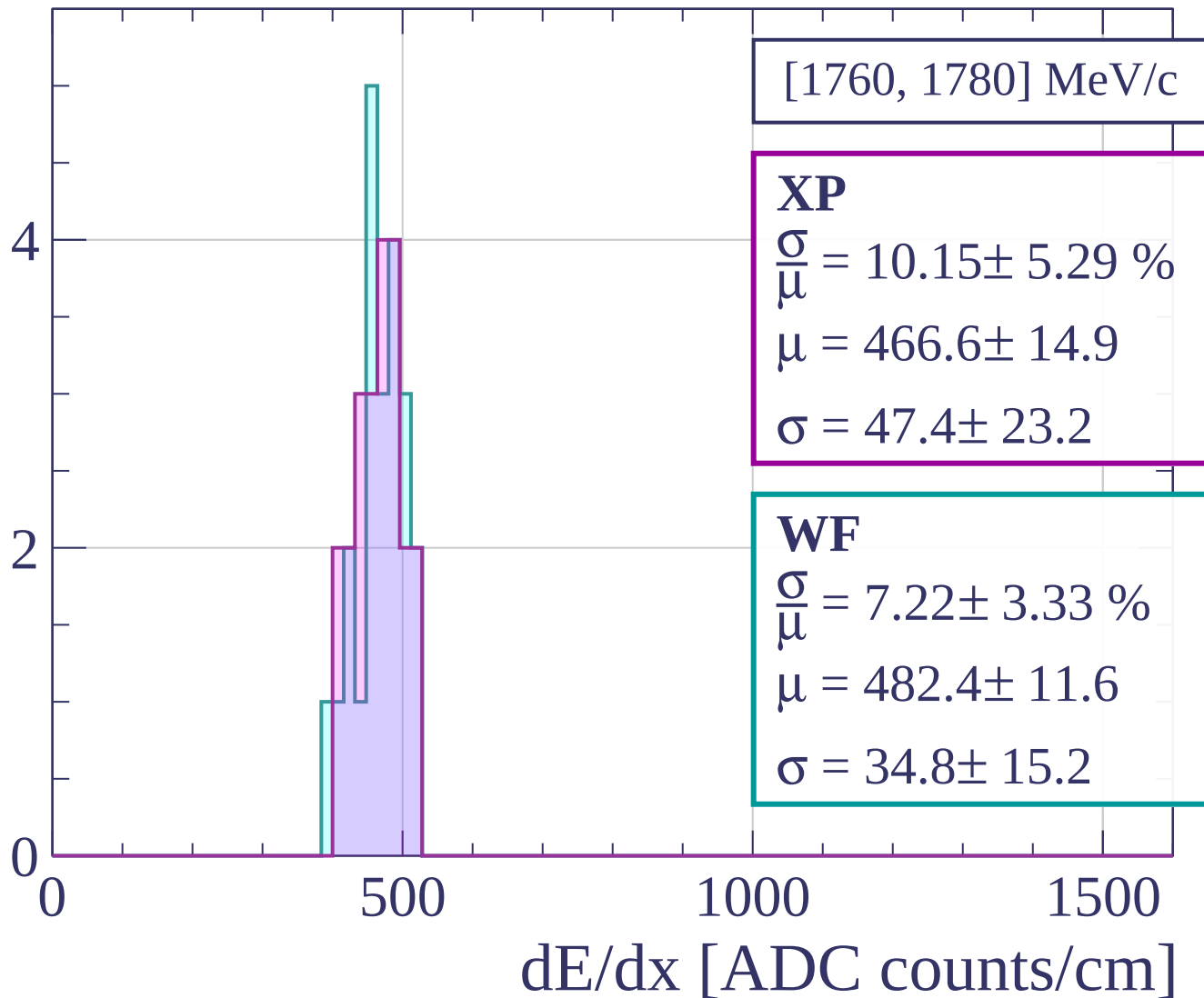


dE/dx [ADC counts/cm]

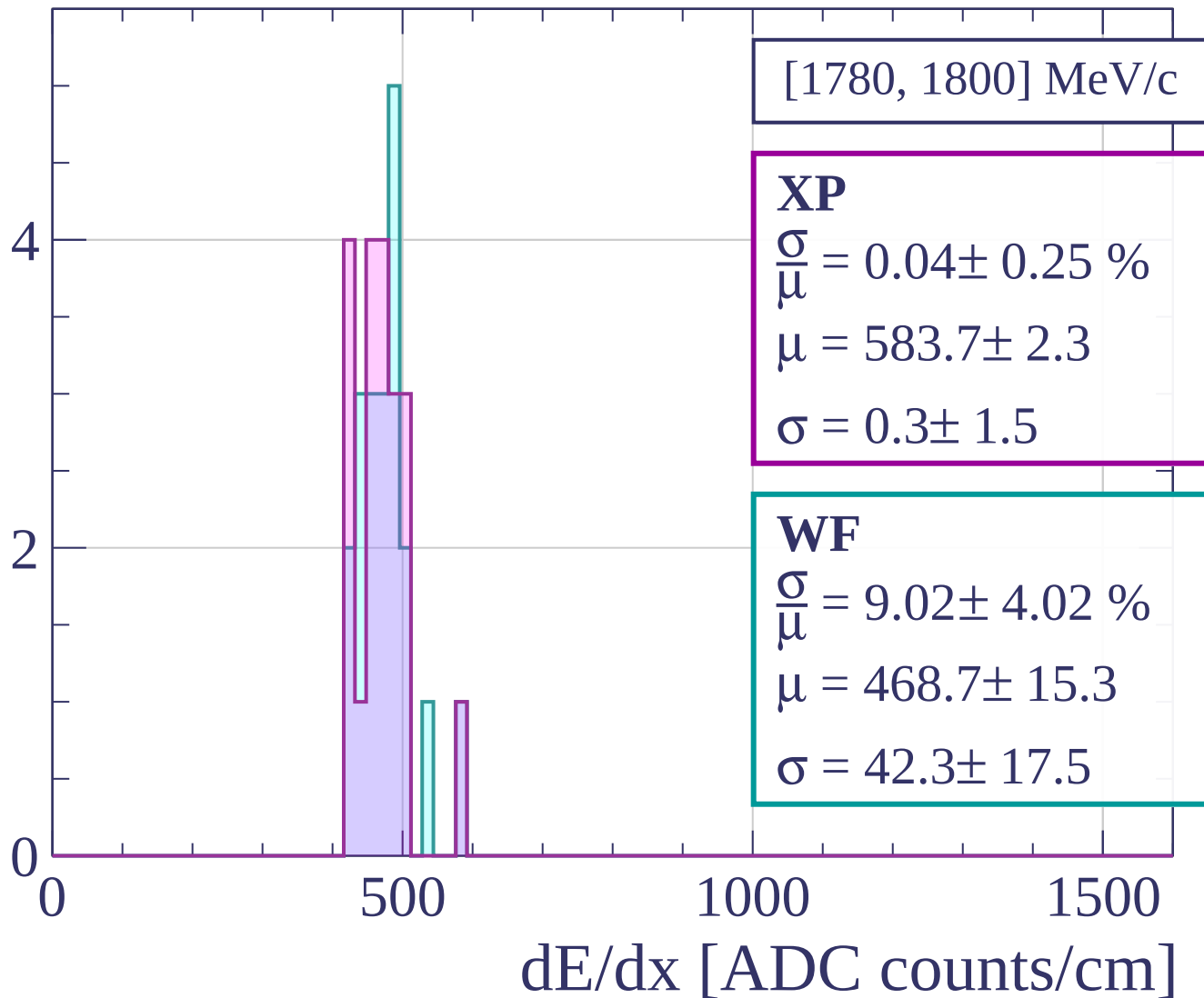
Count



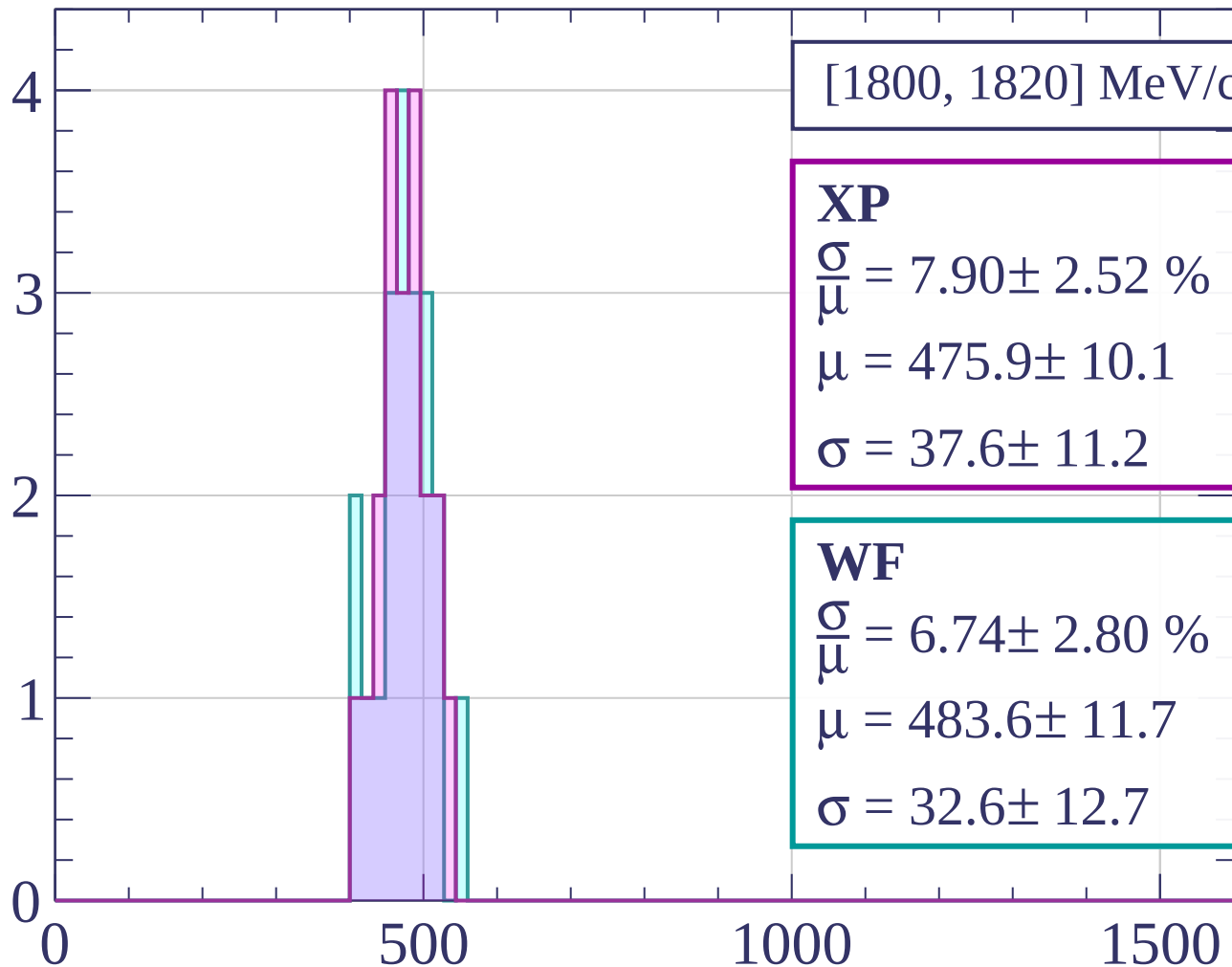
Count



Count

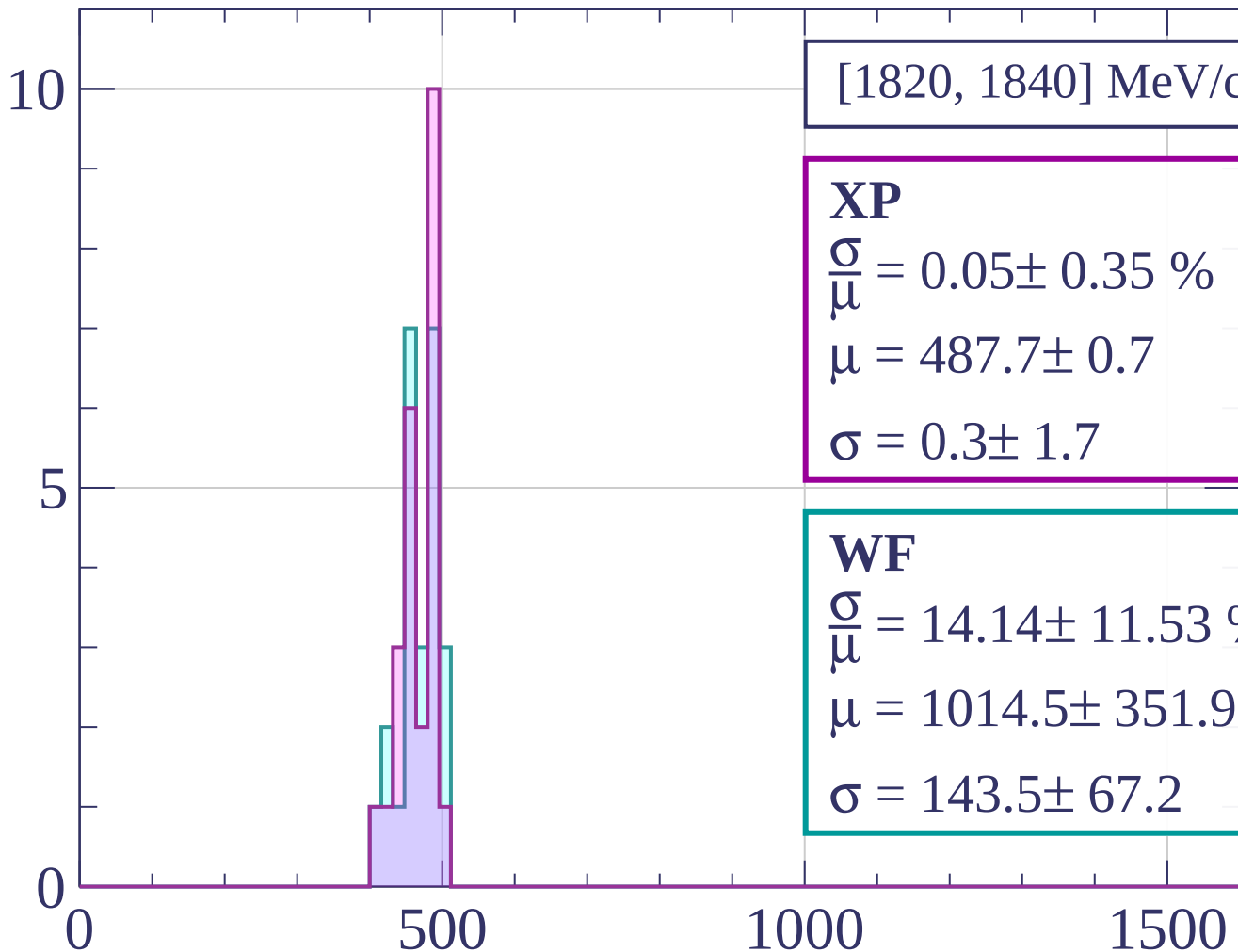


Count



dE/dx [ADC counts/cm]

Count



[1820, 1840] MeV/c

XP

$$\frac{\sigma}{\mu} = 0.05 \pm 0.35 \%$$

$$\mu = 487.7 \pm 0.7$$

$$\sigma = 0.3 \pm 1.7$$

WF

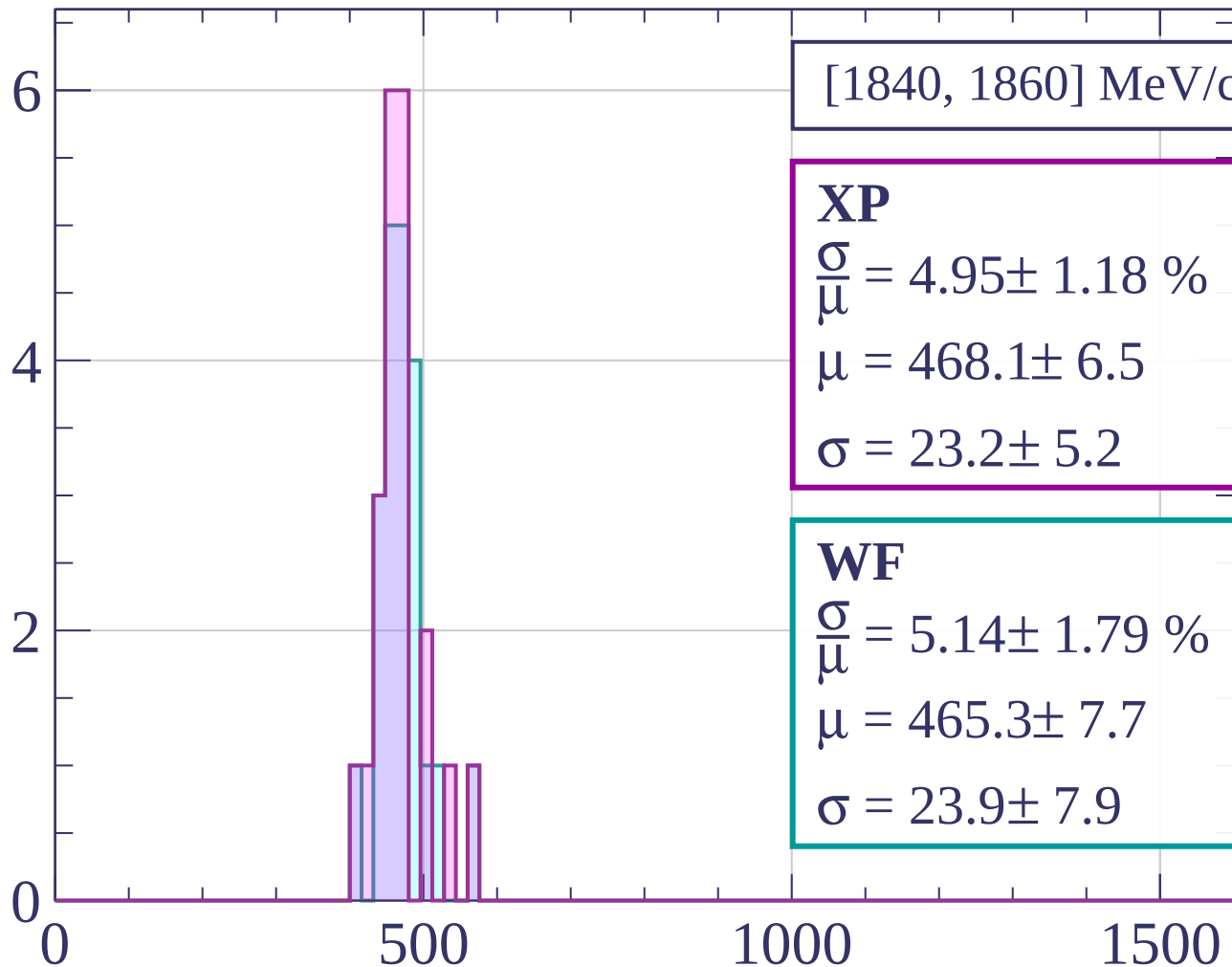
$$\frac{\sigma}{\mu} = 14.14 \pm 11.53 \%$$

$$\mu = 1014.5 \pm 351.9$$

$$\sigma = 143.5 \pm 67.2$$

dE/dx [ADC counts/cm]

Count



[1840, 1860] MeV/c

XP

$$\frac{\sigma}{\mu} = 4.95 \pm 1.18 \%$$

$$\mu = 468.1 \pm 6.5$$

$$\sigma = 23.2 \pm 5.2$$

WF

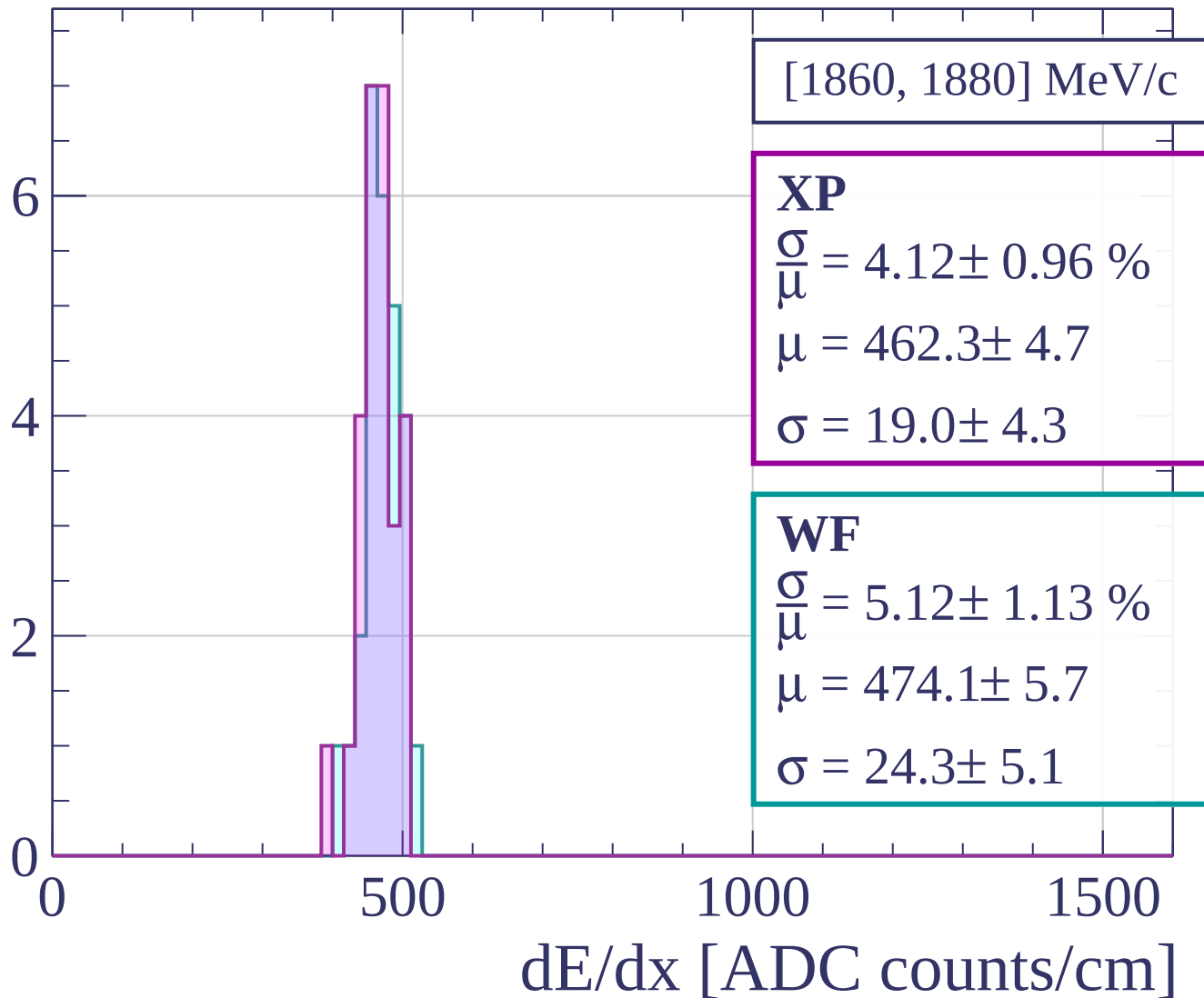
$$\frac{\sigma}{\mu} = 5.14 \pm 1.79 \%$$

$$\mu = 465.3 \pm 7.7$$

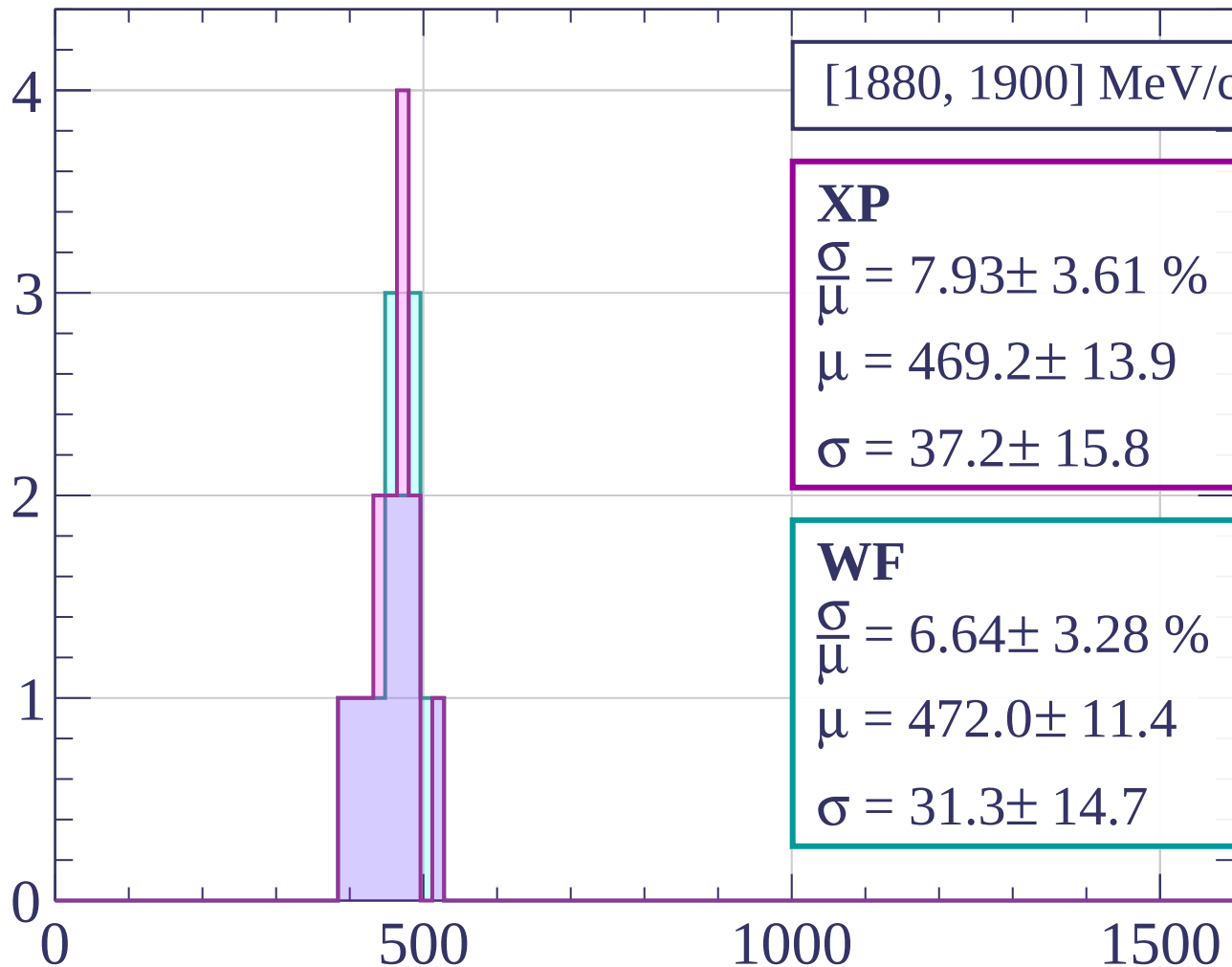
$$\sigma = 23.9 \pm 7.9$$

dE/dx [ADC counts/cm]

Count

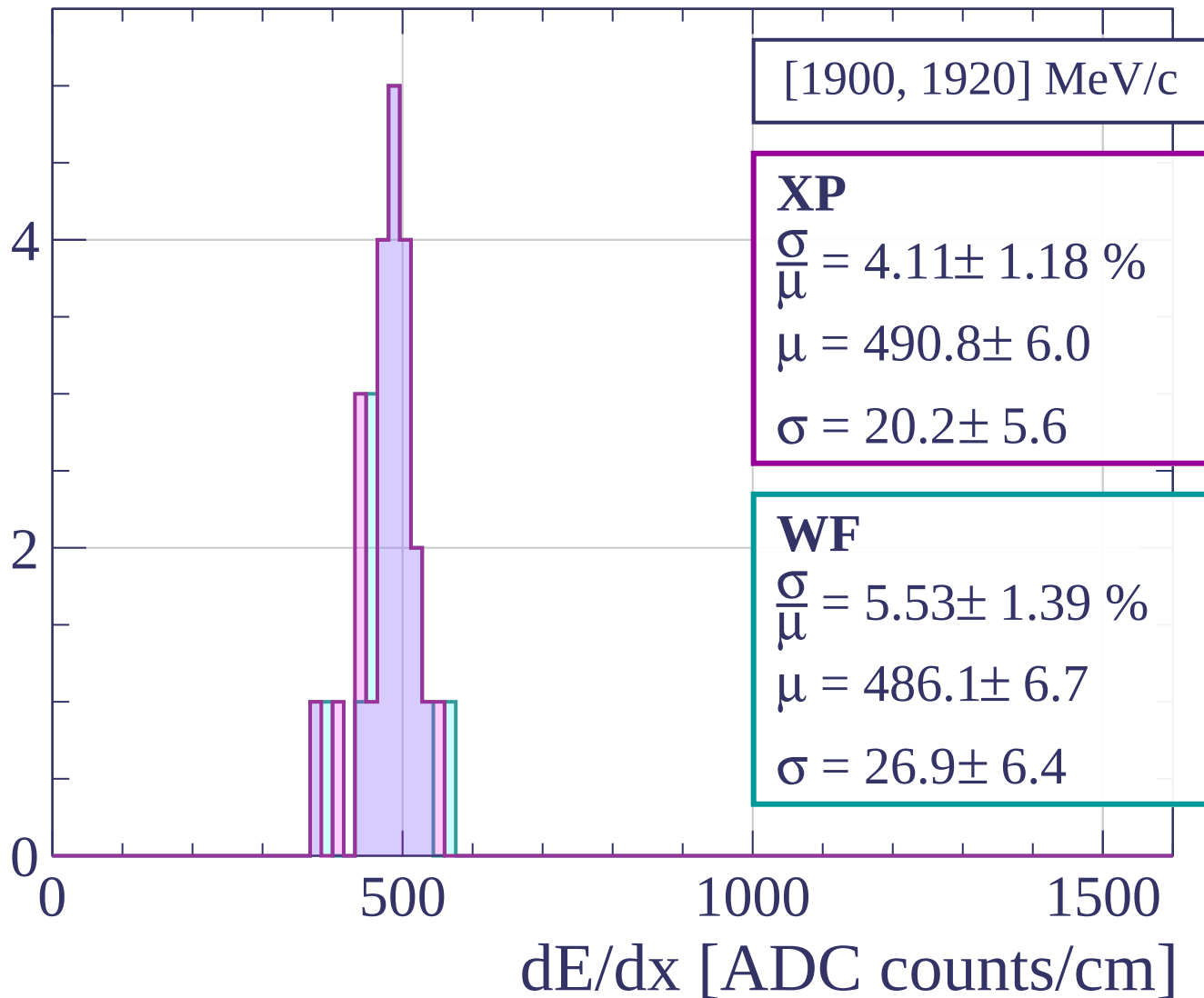


Count

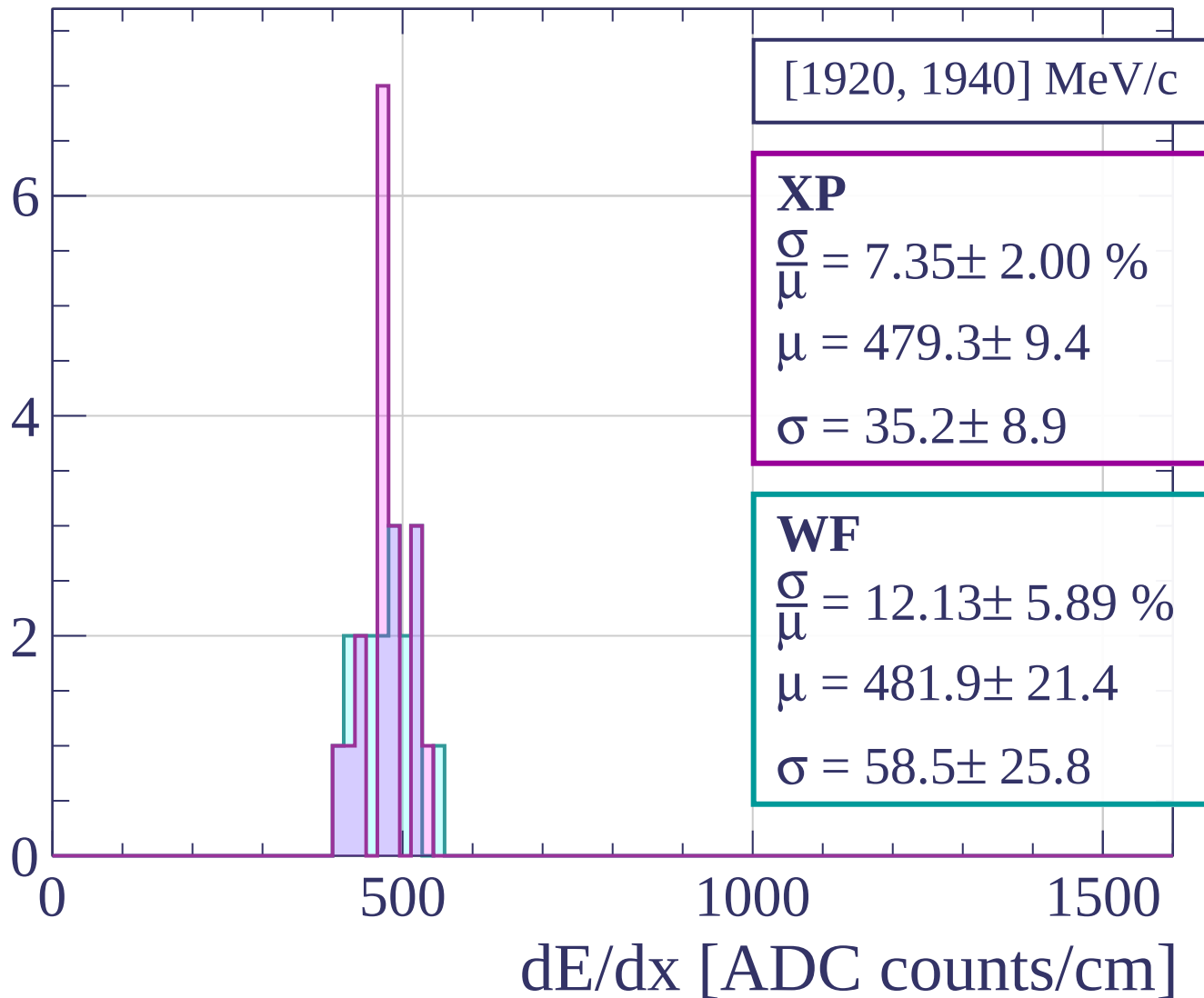


dE/dx [ADC counts/cm]

Count



Count



Count

[1940, 1960] MeV/c

XP

$$\frac{\sigma}{\mu} = 0.05 \pm 0.47 \%$$

$$\mu = 503.7 \pm 1.7$$

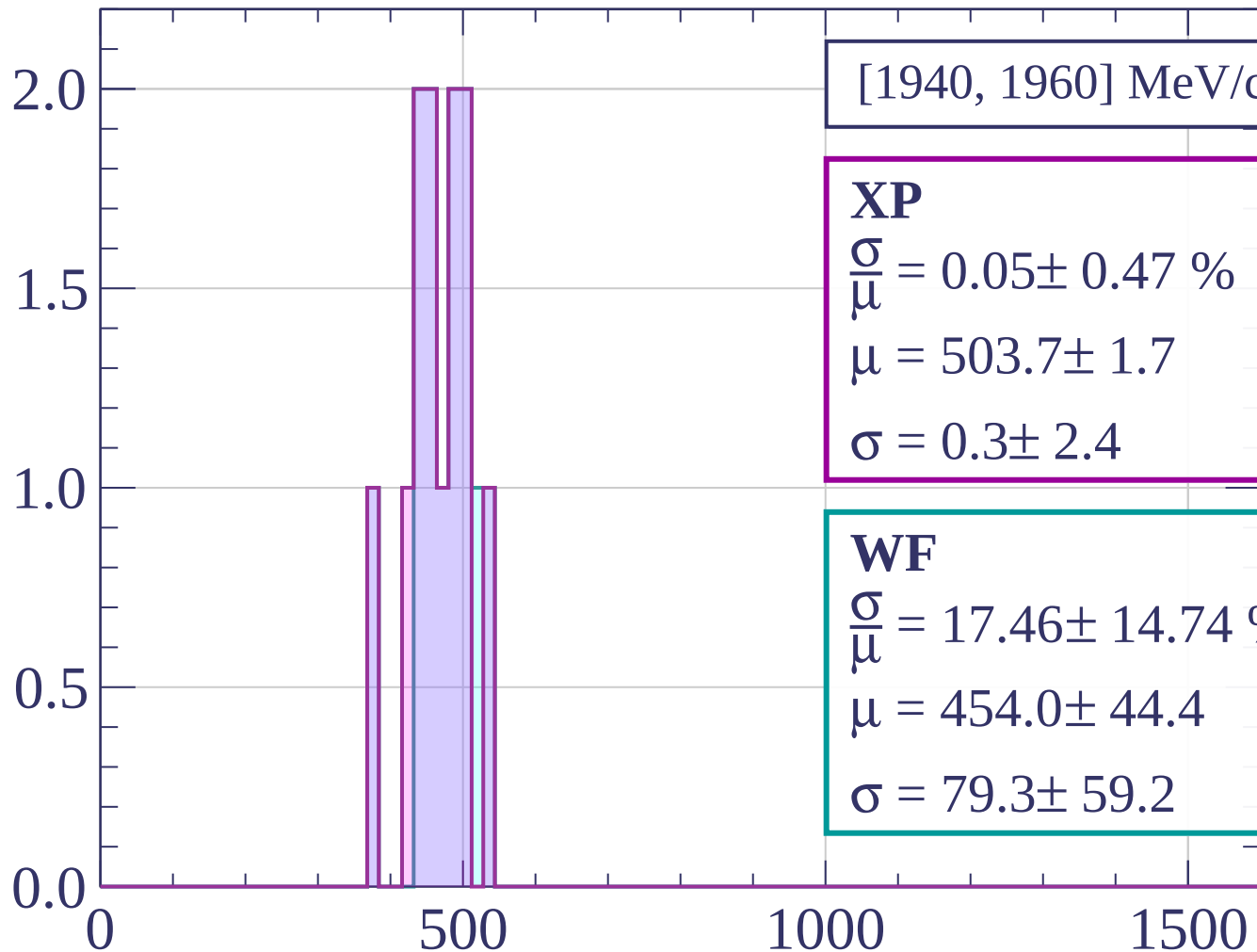
$$\sigma = 0.3 \pm 2.4$$

WF

$$\frac{\sigma}{\mu} = 17.46 \pm 14.74 \%$$

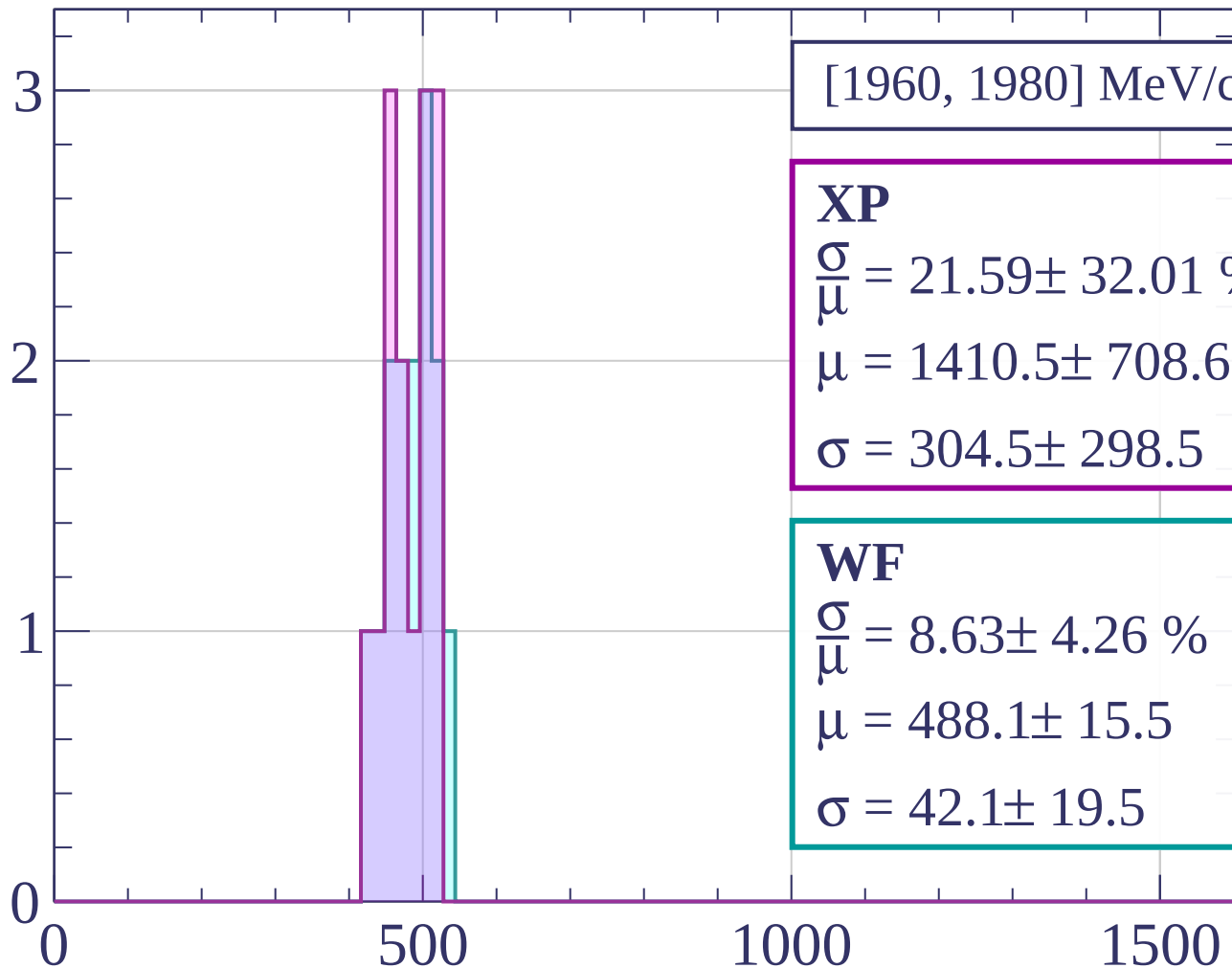
$$\mu = 454.0 \pm 44.4$$

$$\sigma = 79.3 \pm 59.2$$



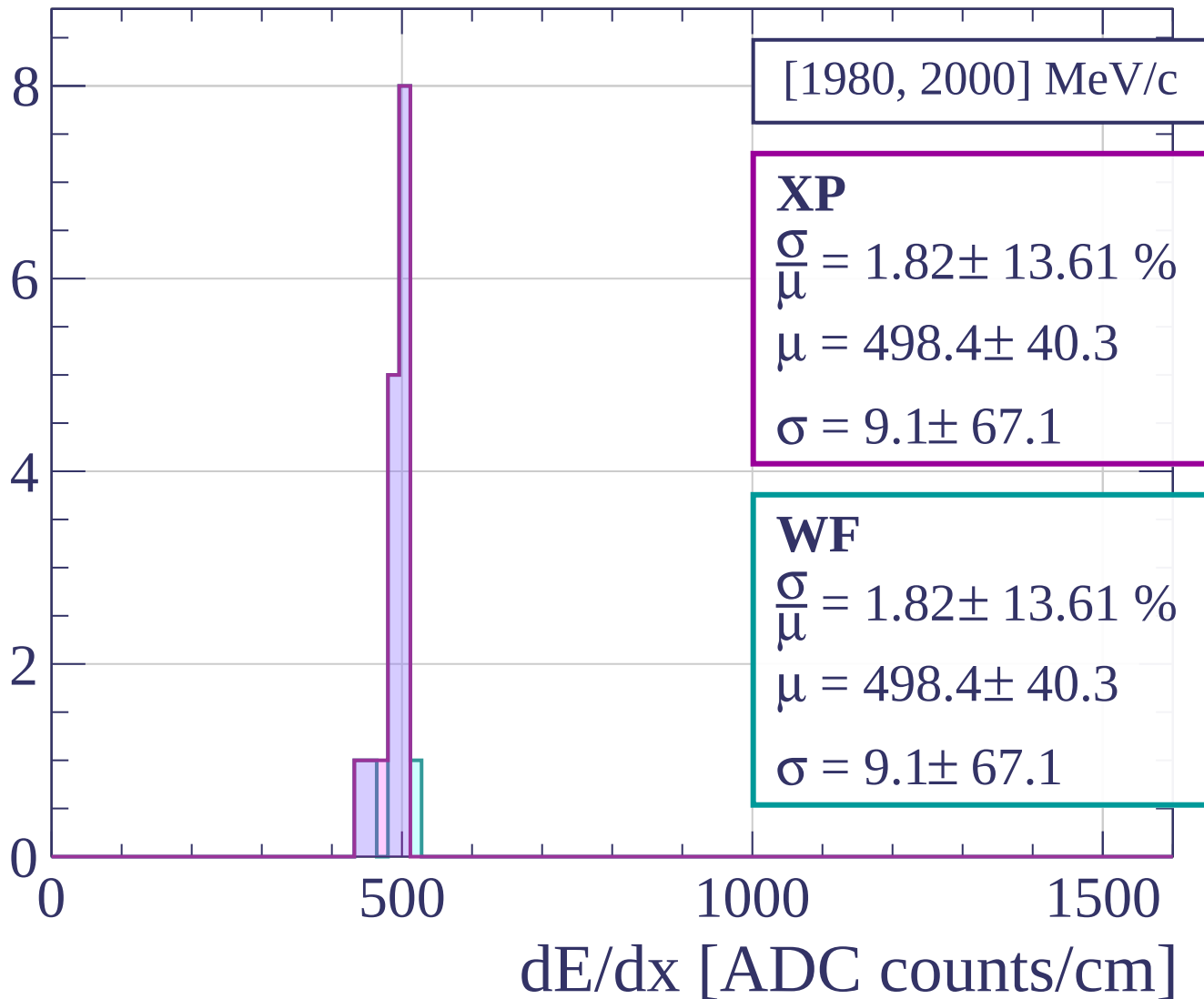
dE/dx [ADC counts/cm]

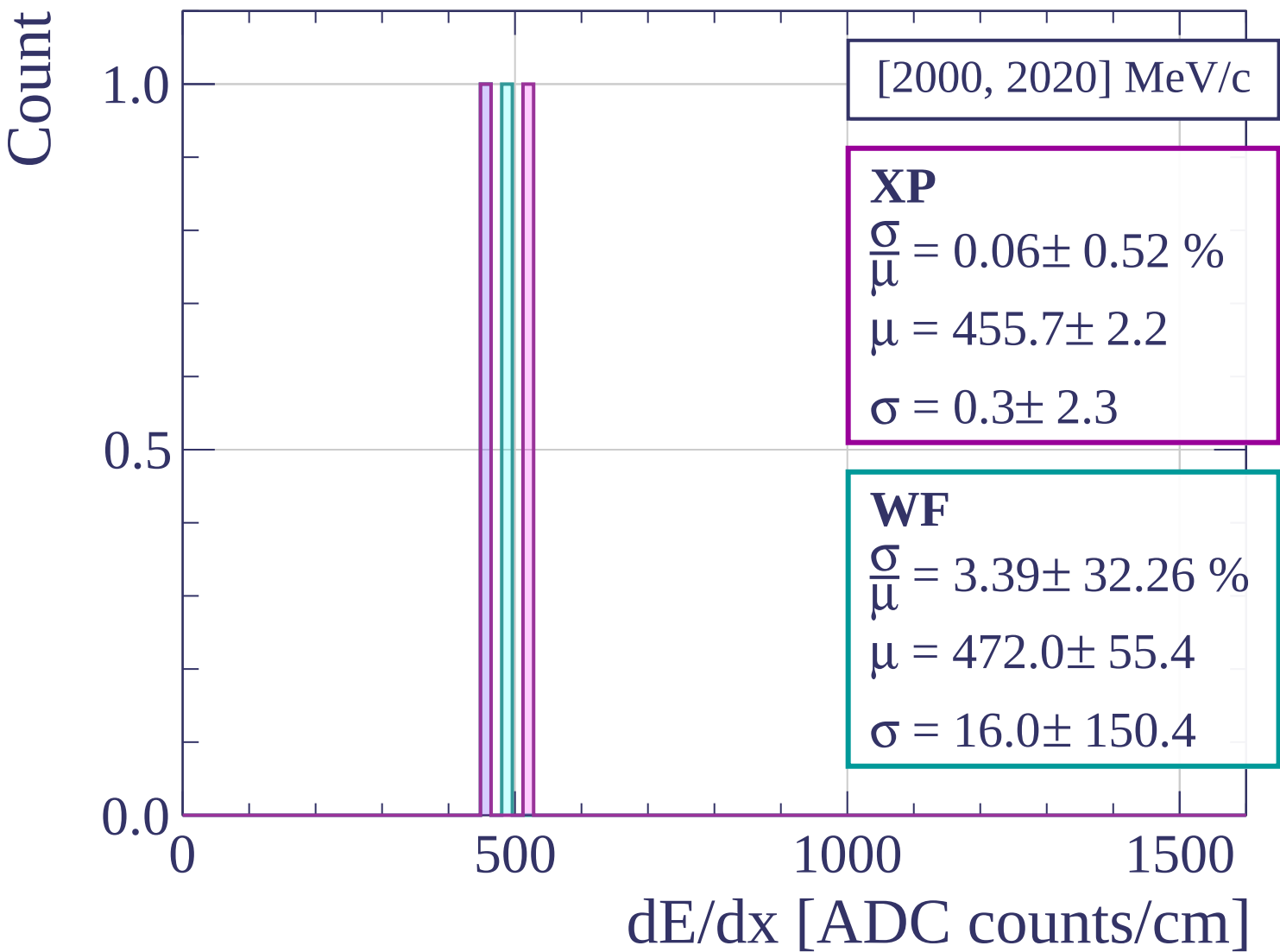
Count

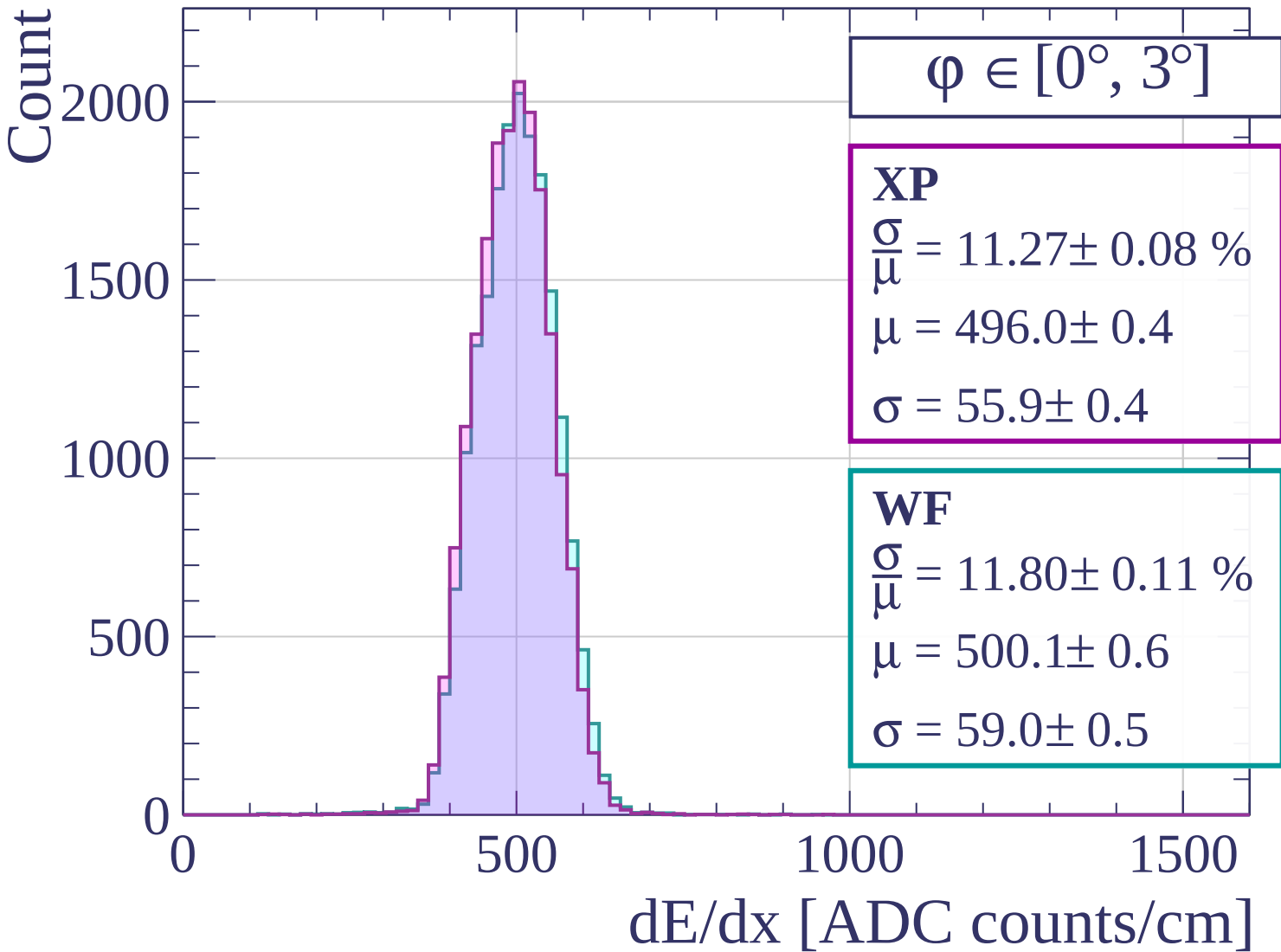


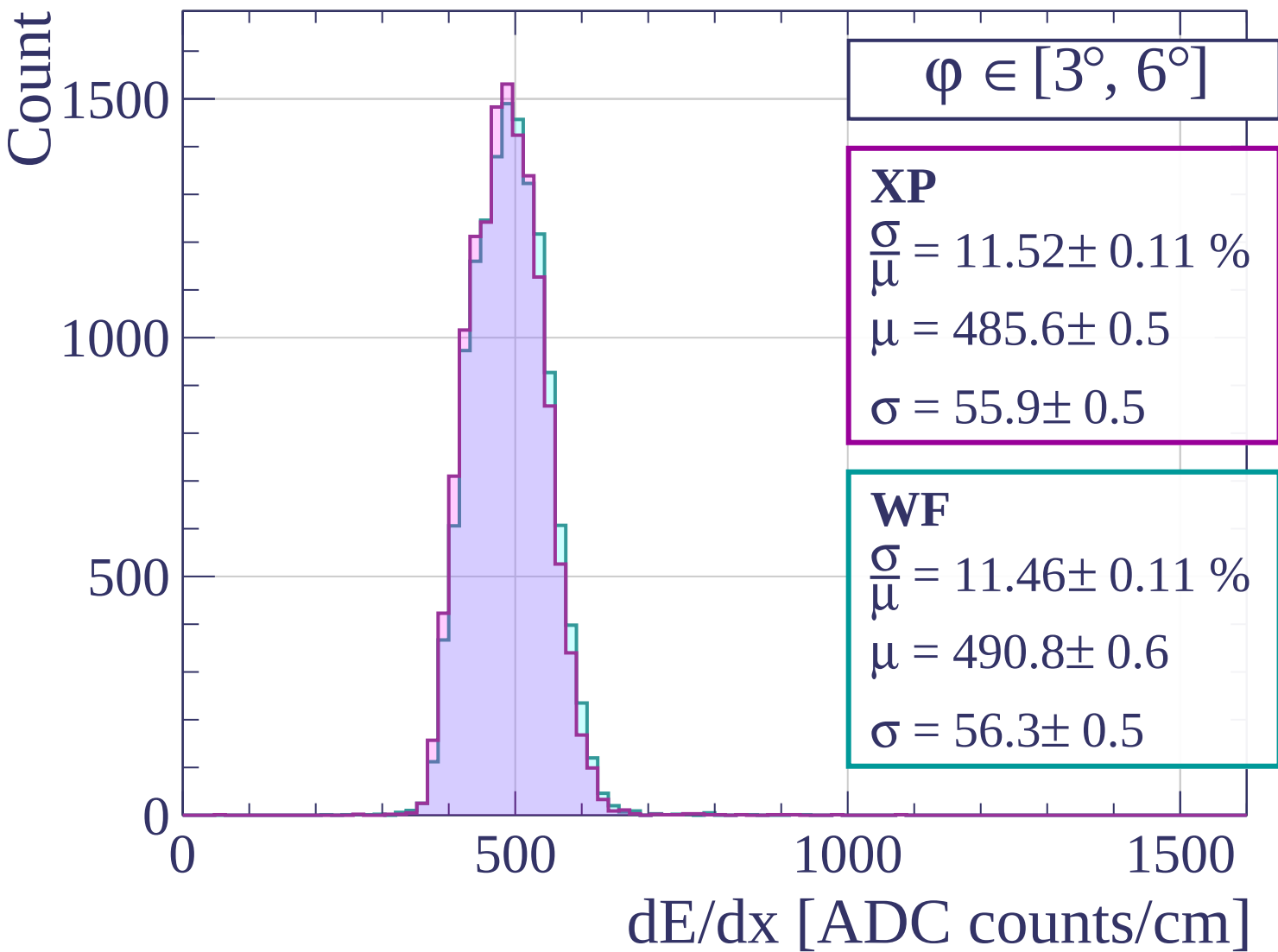
dE/dx [ADC counts/cm]

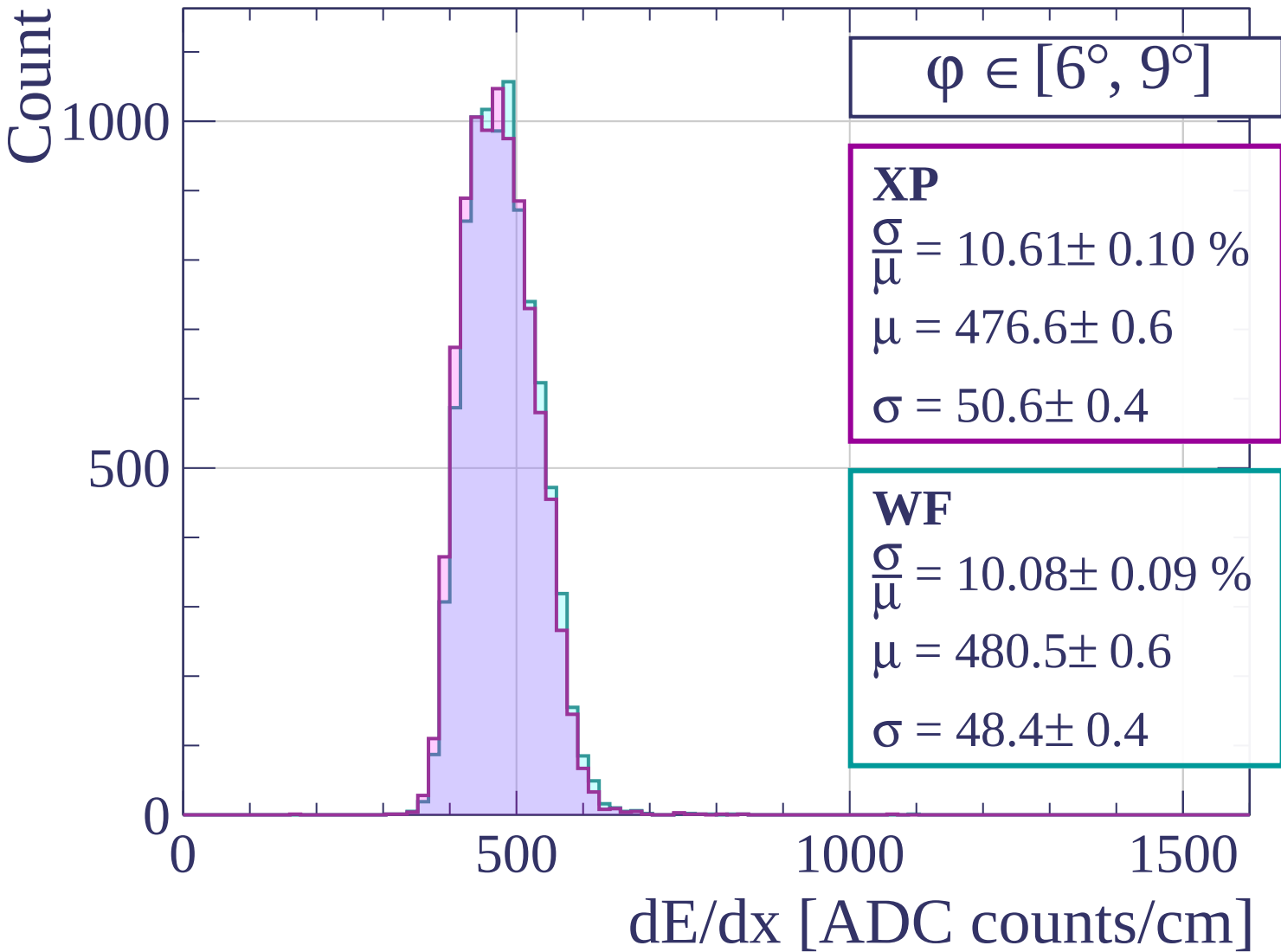
Count











Count

$$\varphi \in [9^\circ, 12^\circ]$$

XP

$$\frac{\sigma}{\mu} = 10.31 \pm 0.19 \%$$

$$\mu = 457.4 \pm 0.8$$

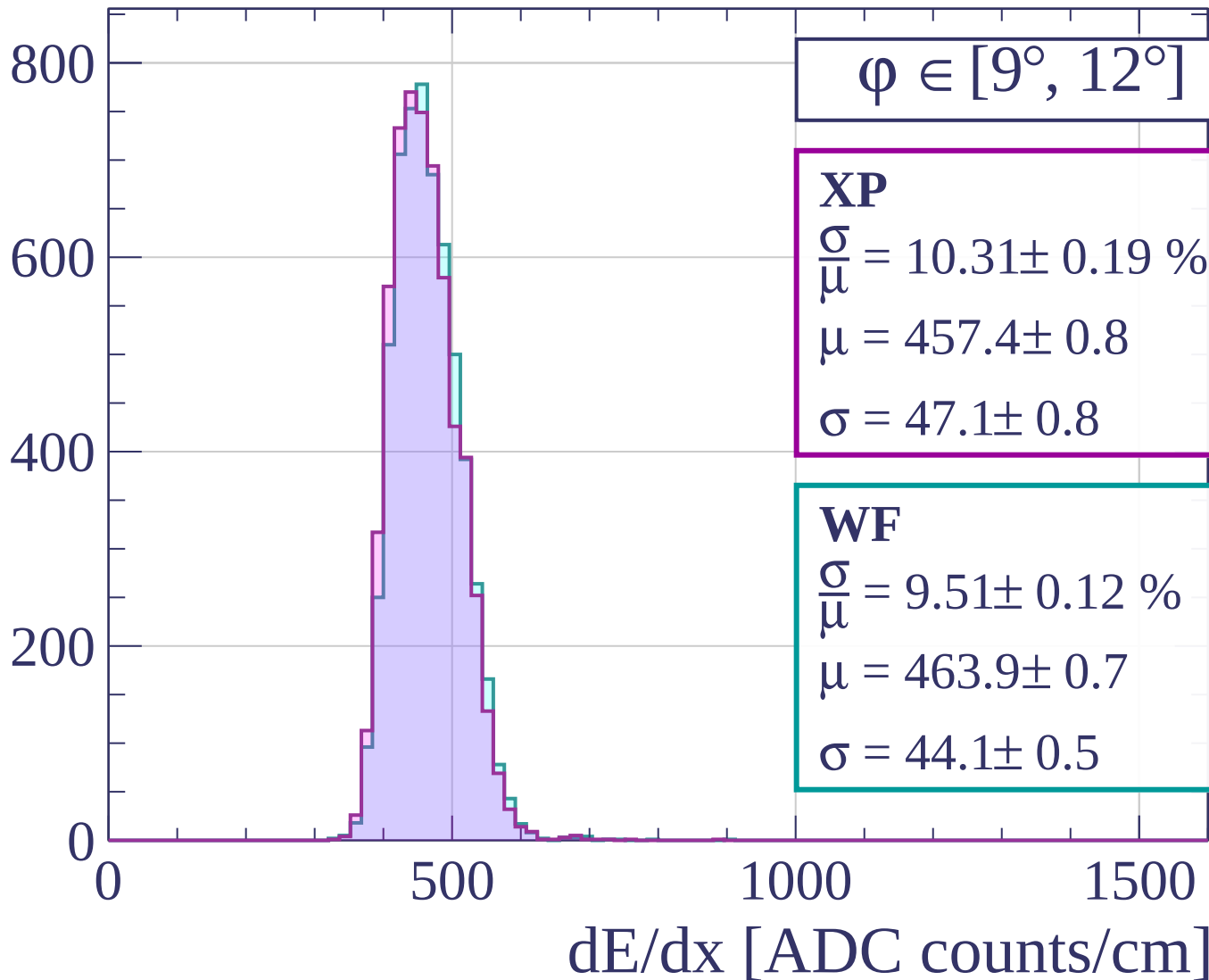
$$\sigma = 47.1 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.51 \pm 0.12 \%$$

$$\mu = 463.9 \pm 0.7$$

$$\sigma = 44.1 \pm 0.5$$



Count

$\phi \in [12^\circ, 15^\circ]$

XP

$$\frac{\sigma}{\mu} = 9.40 \pm 0.22 \%$$

$$\mu = 446.4 \pm 0.9$$

$$\sigma = 42.0 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.05 \pm 0.20 \%$$

$$\mu = 450.8 \pm 0.9$$

$$\sigma = 40.8 \pm 0.8$$

400

200

0

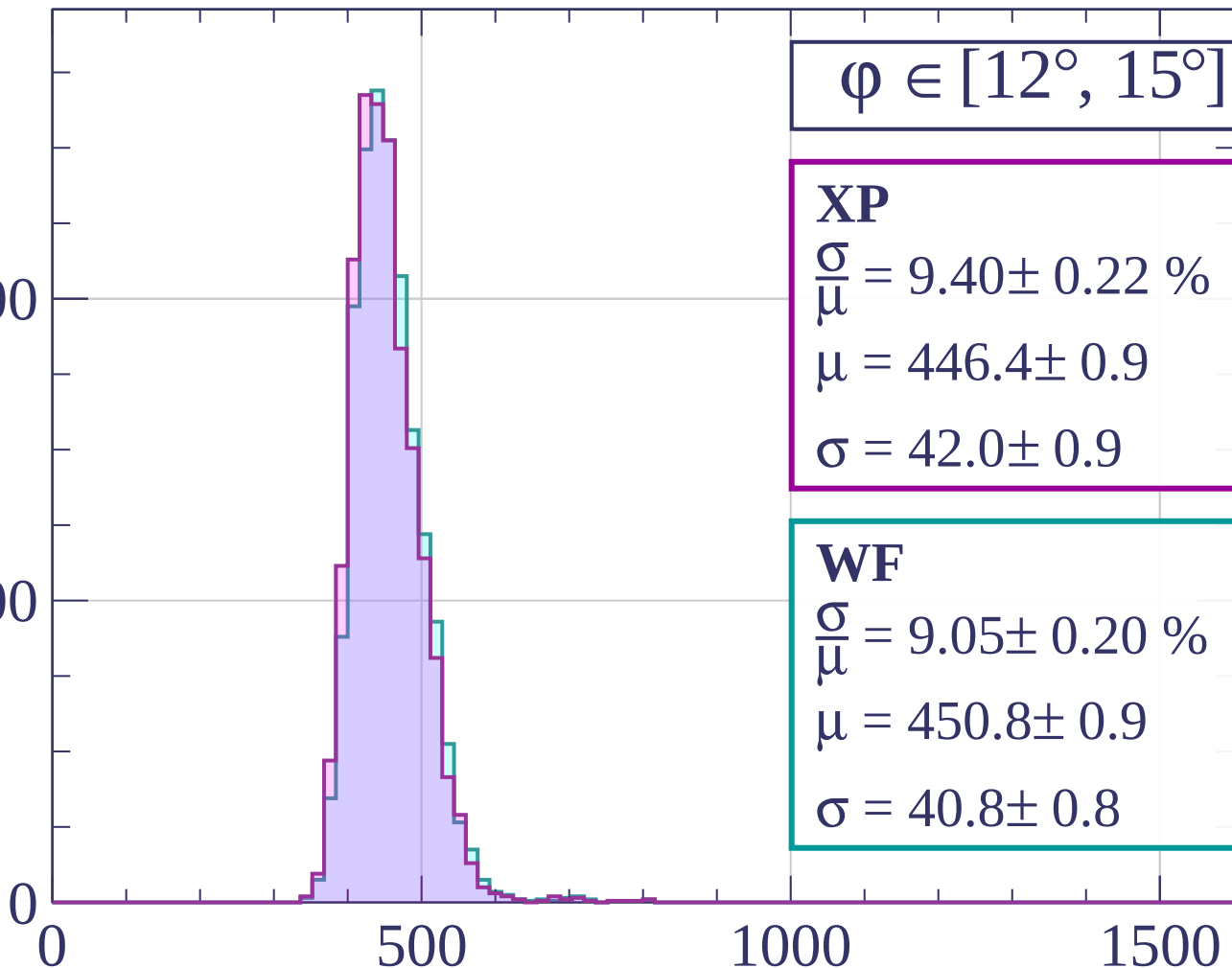
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\varphi \in [15^\circ, 18^\circ]$

XP

$$\frac{\sigma}{\mu} = 8.56 \pm 0.25 \%$$

$$\mu = 436.4 \pm 1.0$$

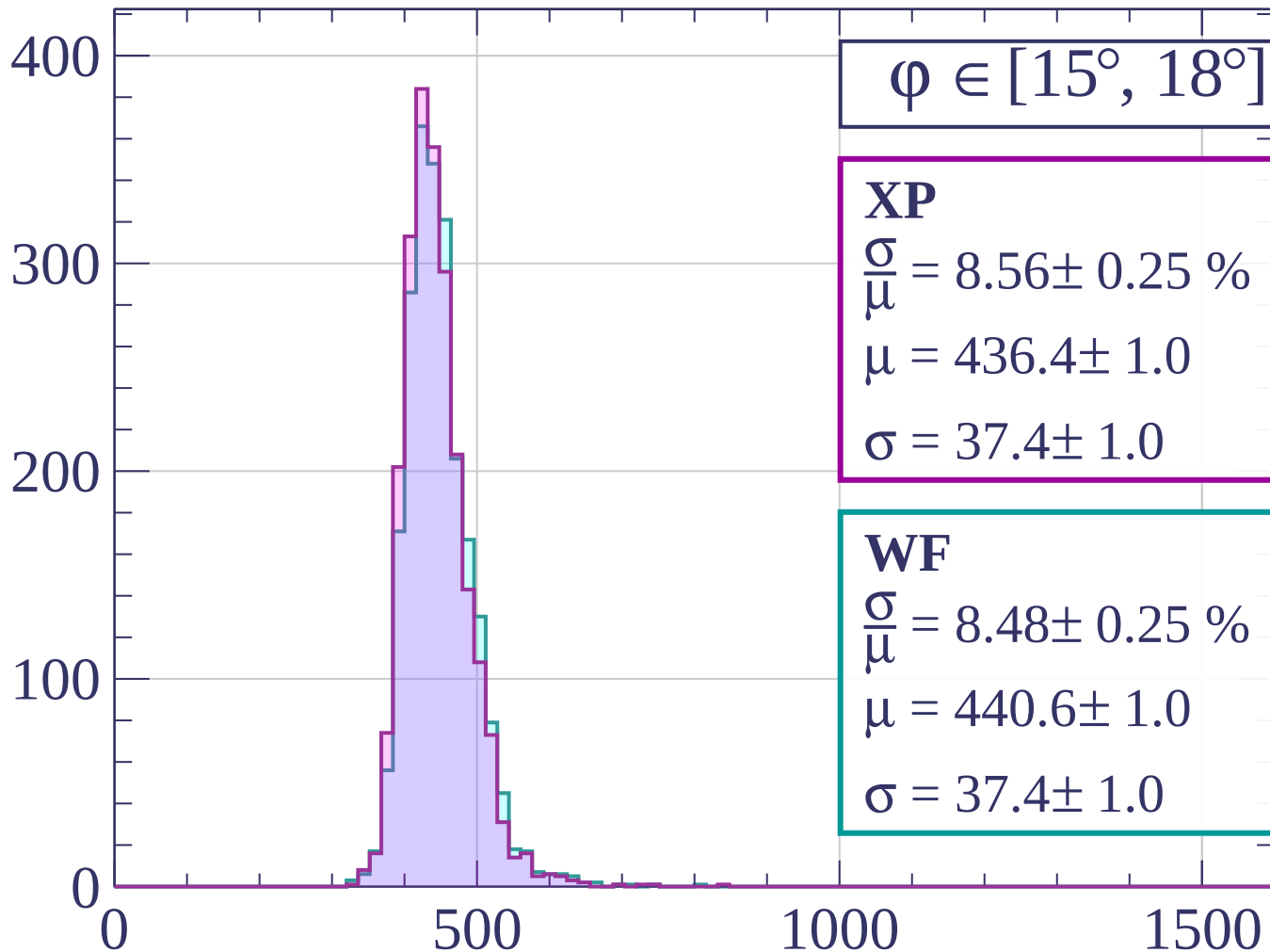
$$\sigma = 37.4 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 8.48 \pm 0.25 \%$$

$$\mu = 440.6 \pm 1.0$$

$$\sigma = 37.4 \pm 1.0$$



dE/dx [ADC counts/cm]

Count

$\varphi \in [18^\circ, 21^\circ]$

XP

$$\frac{\sigma}{\mu} = 7.78 \pm 0.29 \%$$

$$\mu = 429.7 \pm 1.1$$

$$\sigma = 33.4 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 7.74 \pm 0.29 \%$$

$$\mu = 433.2 \pm 1.2$$

$$\sigma = 33.5 \pm 1.1$$

200

100

0

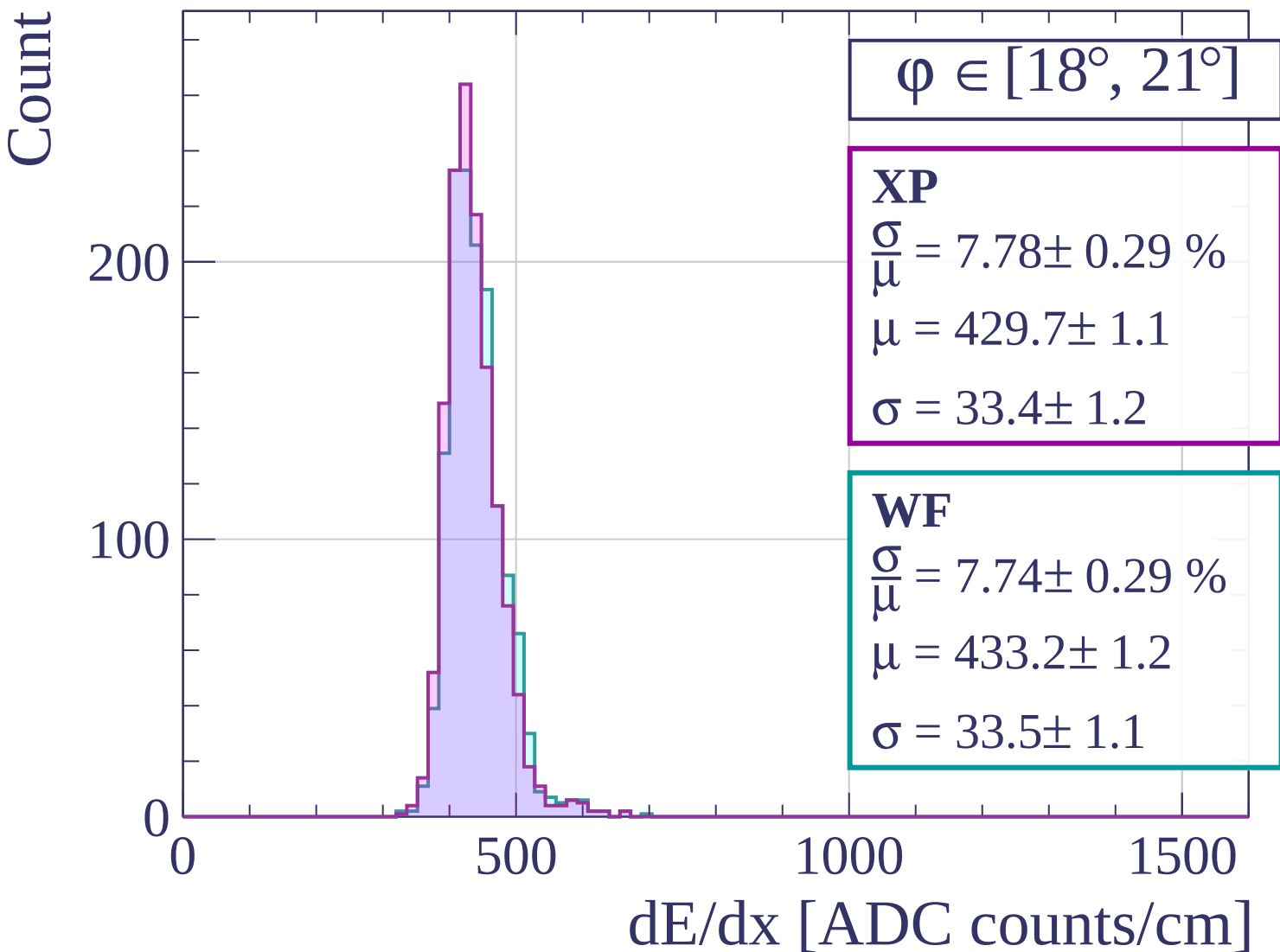
0

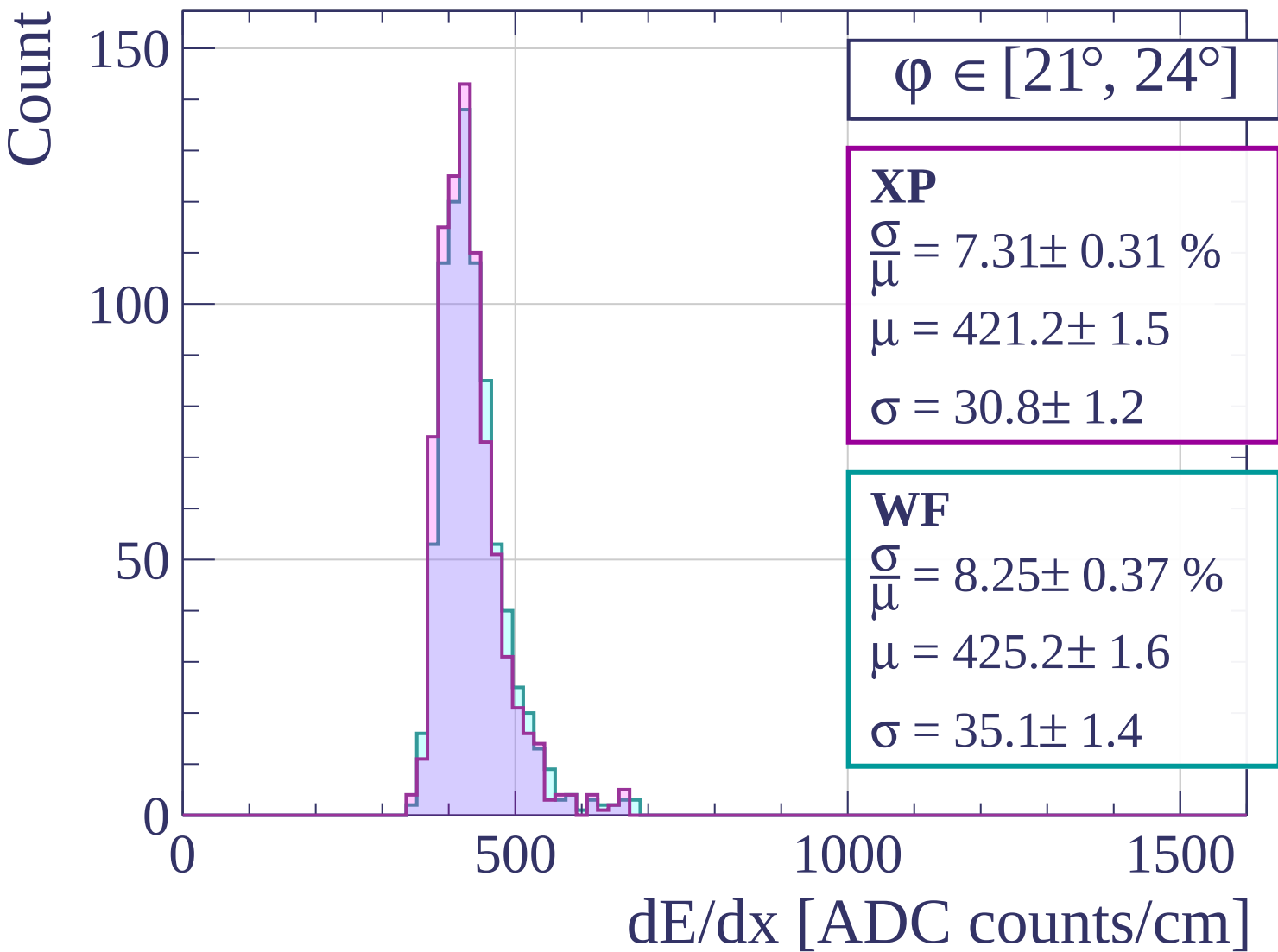
500

1000

1500

dE/dx [ADC counts/cm]





Count

$\phi \in [24^\circ, 27^\circ]$

XP

$$\frac{\sigma}{\mu} = 7.07 \pm 0.31 \%$$

$$\mu = 419.2 \pm 1.4$$

$$\sigma = 29.6 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 7.11 \pm 0.33 \%$$

$$\mu = 420.6 \pm 1.4$$

$$\sigma = 29.9 \pm 1.3$$

100

50

0

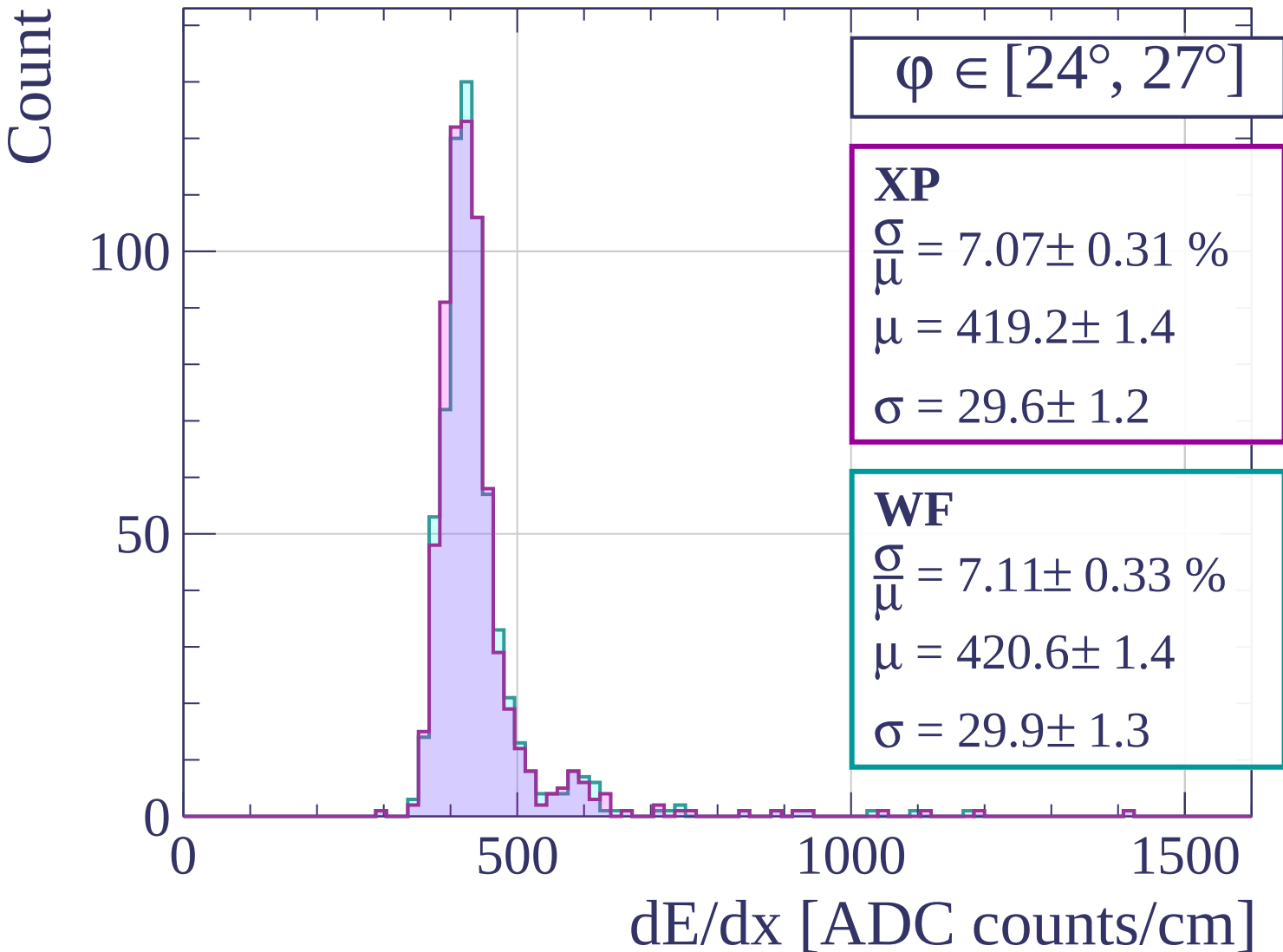
0

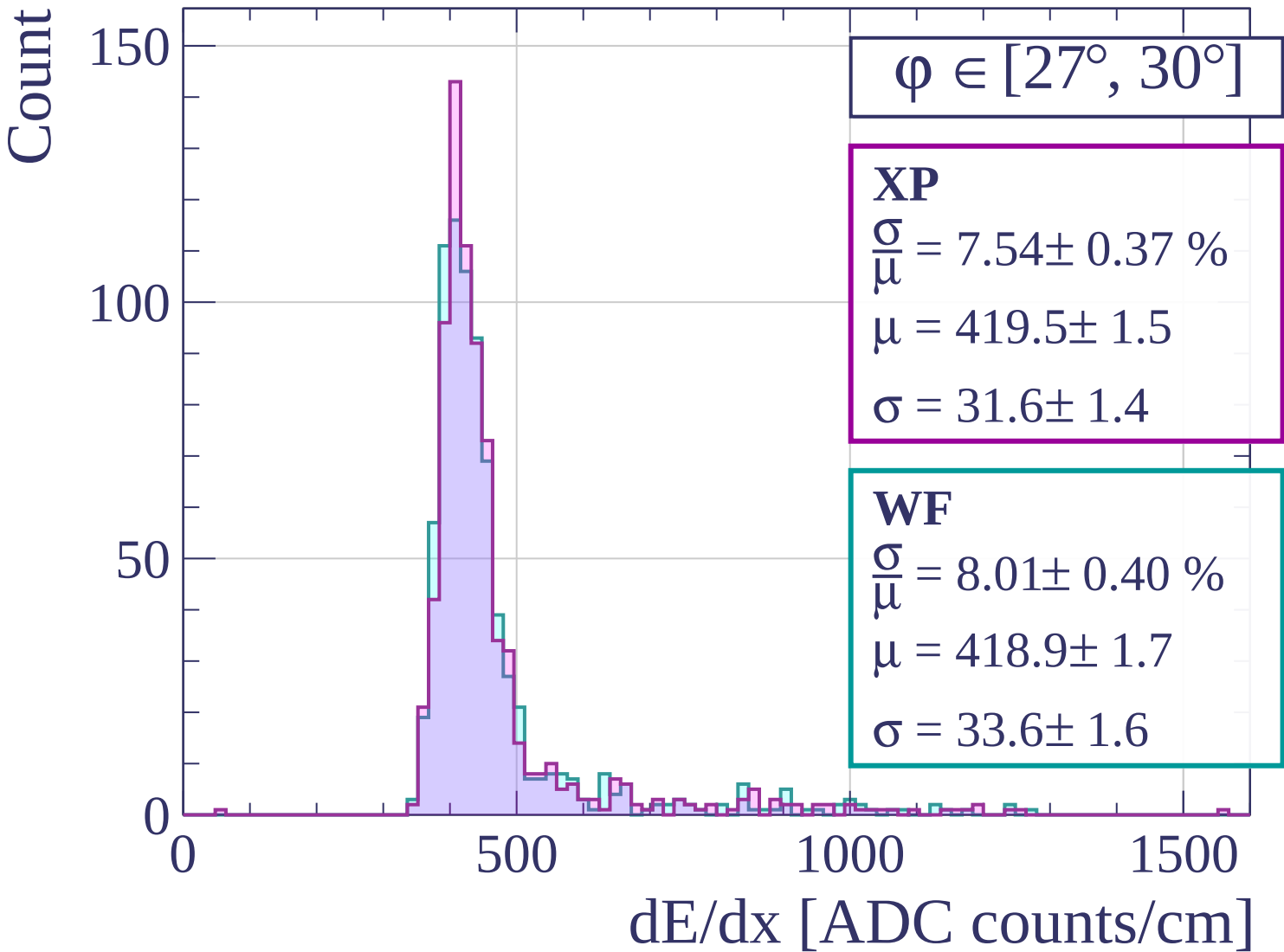
500

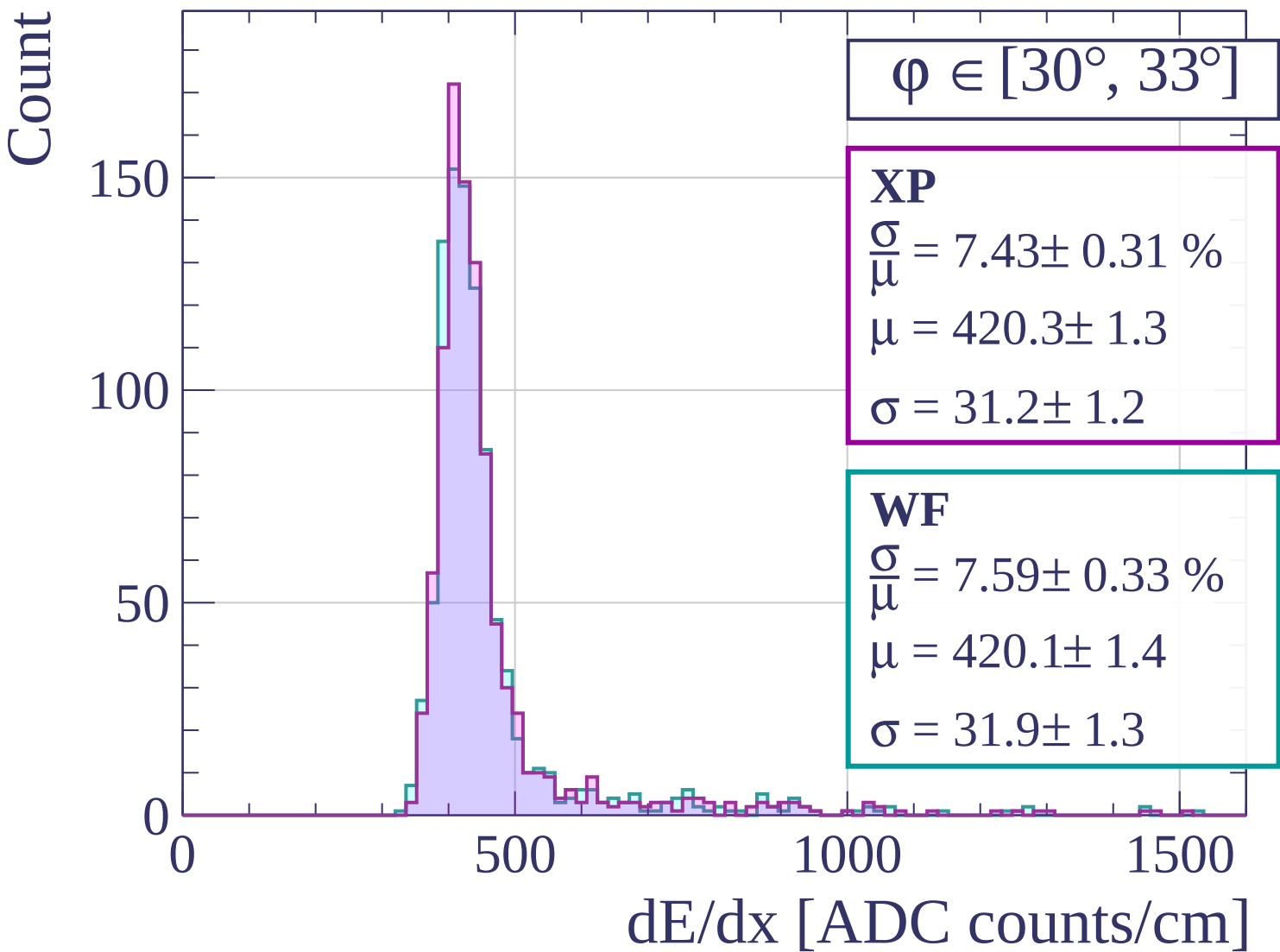
1000

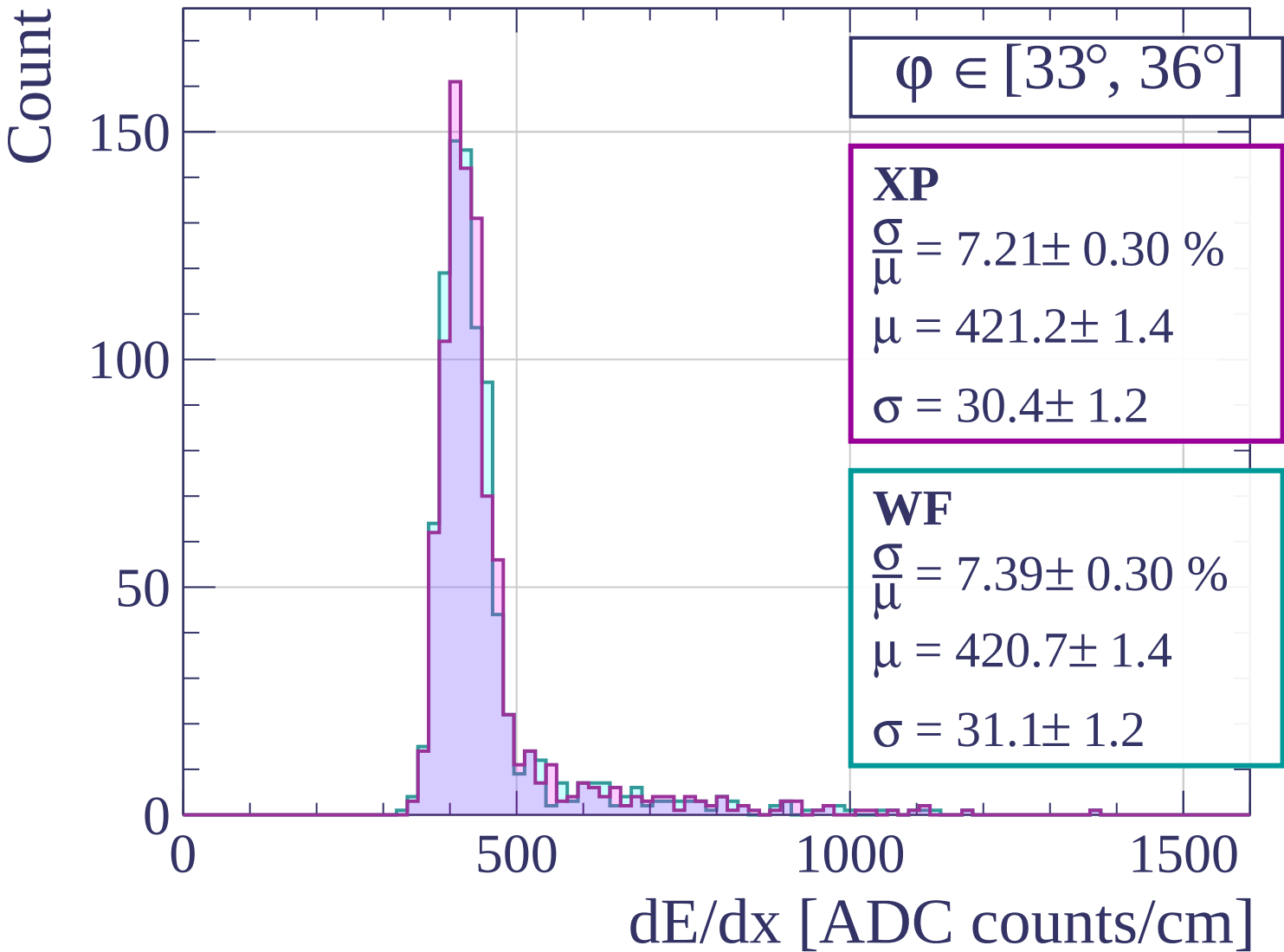
1500

dE/dx [ADC counts/cm]









Count

$\phi \in [36^\circ, 39^\circ]$

XP

$$\frac{\sigma}{\mu} = 8.05 \pm 0.36 \%$$

$$\mu = 421.4 \pm 1.5$$

$$\sigma = 33.9 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 8.12 \pm 0.36 \%$$

$$\mu = 421.7 \pm 1.5$$

$$\sigma = 34.3 \pm 1.4$$

100

50

0

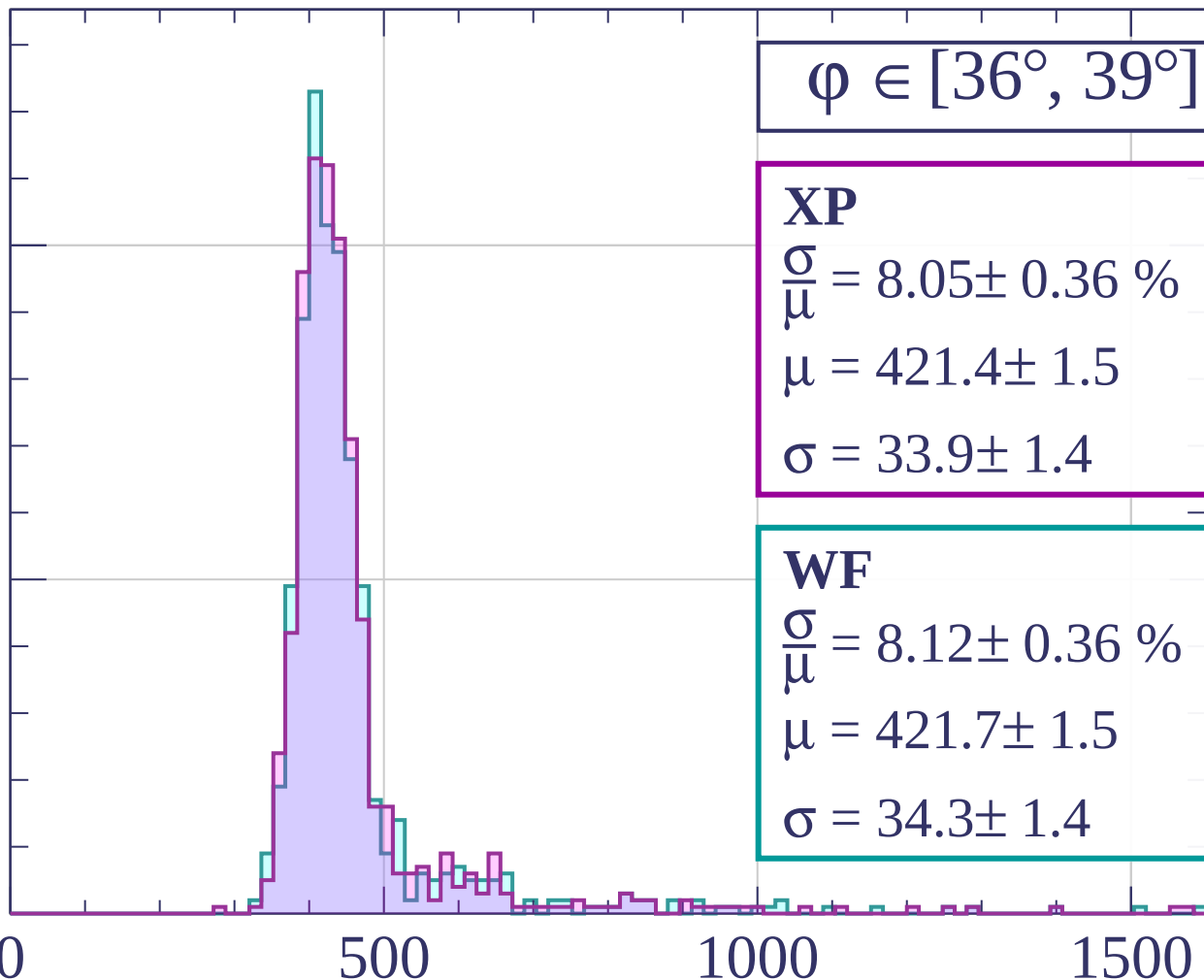
0

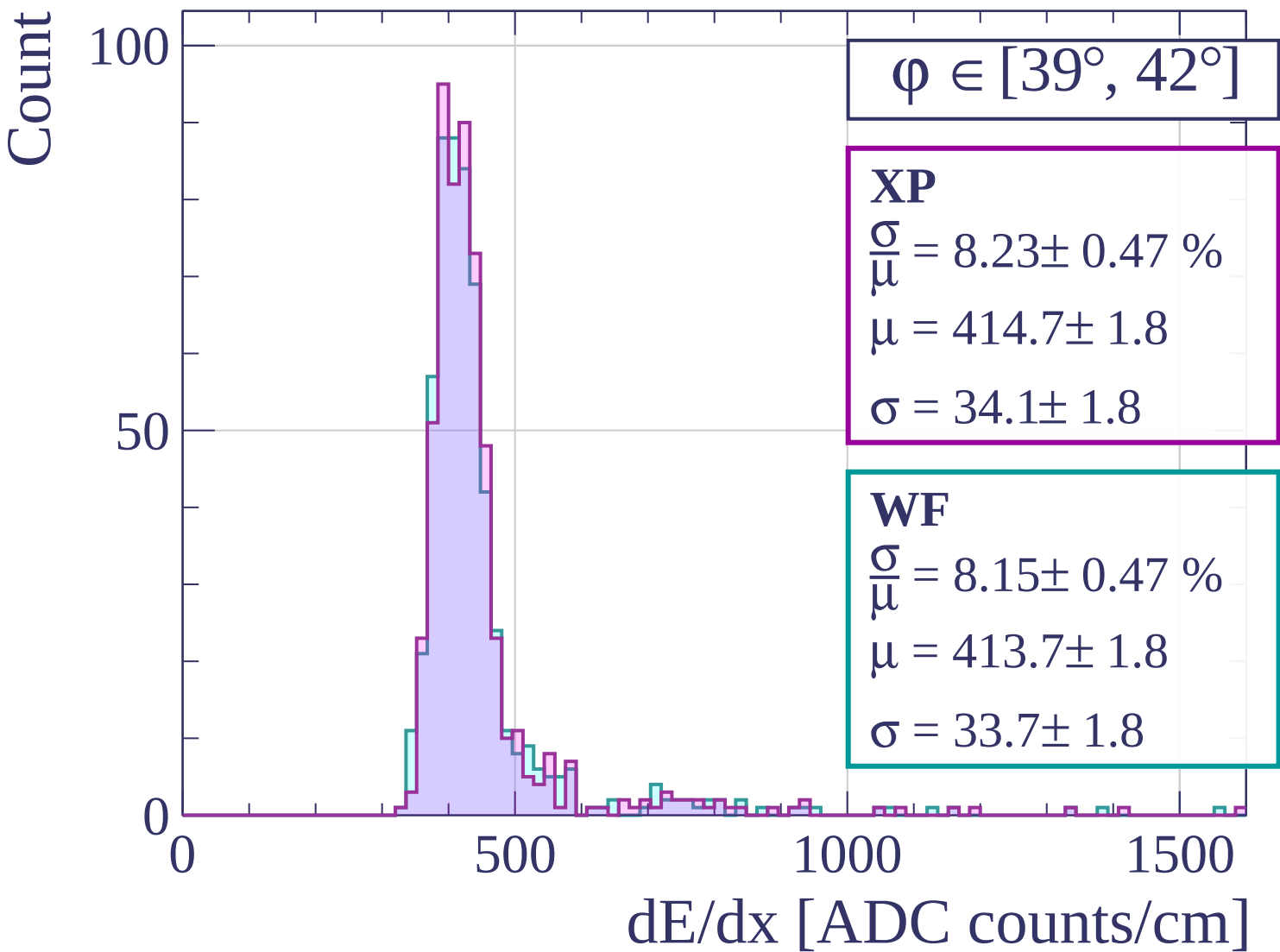
500

1000

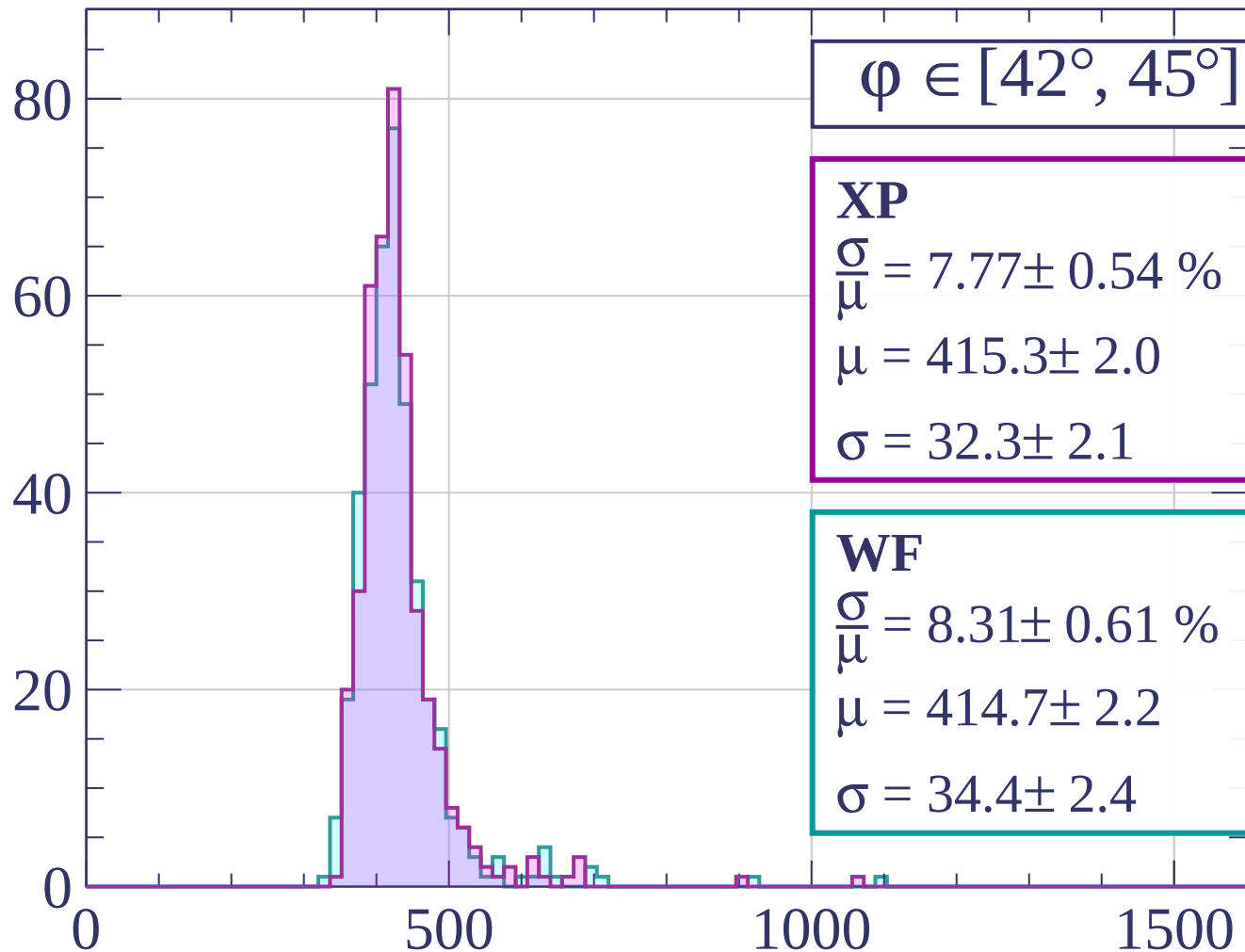
1500

dE/dx [ADC counts/cm]



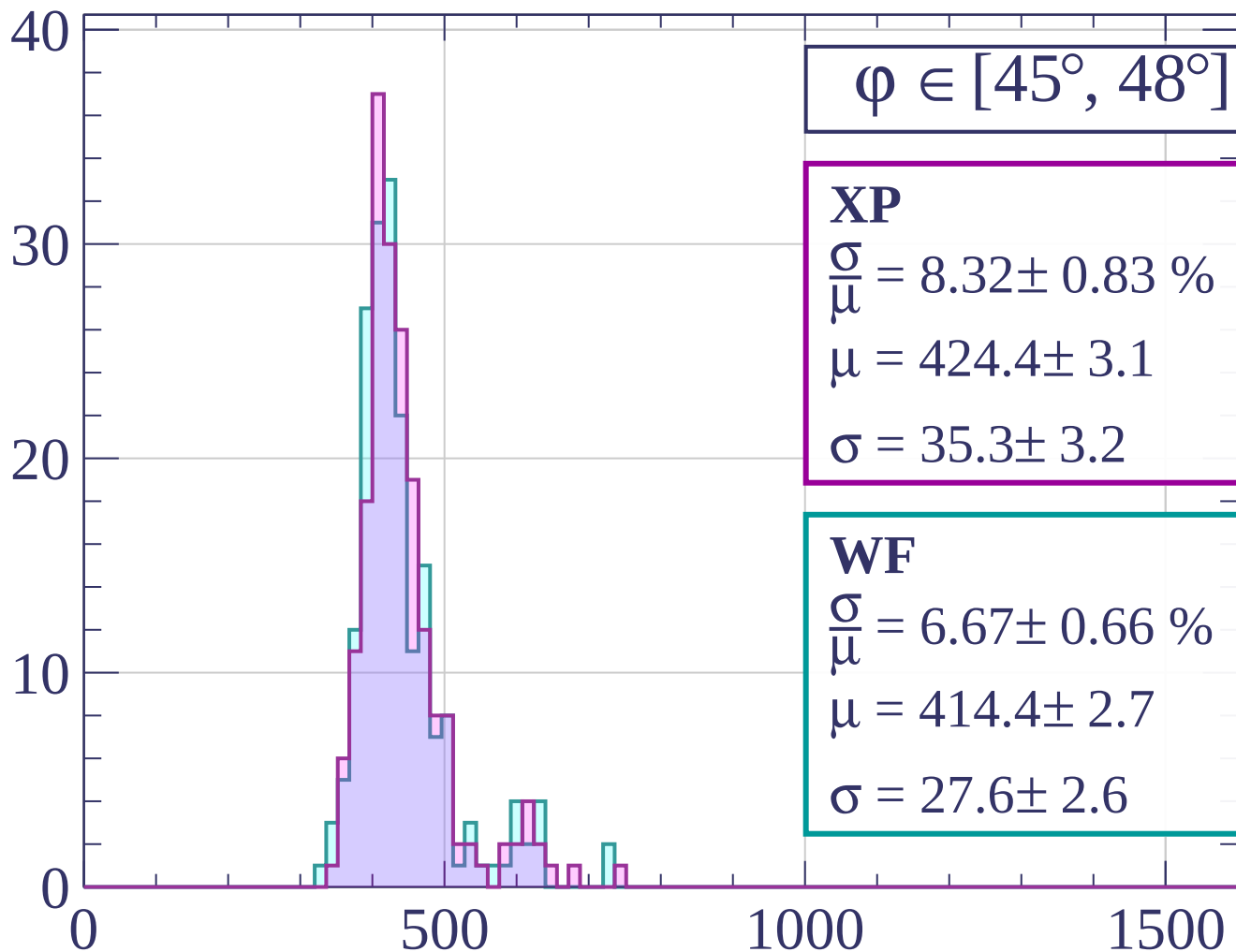


Count

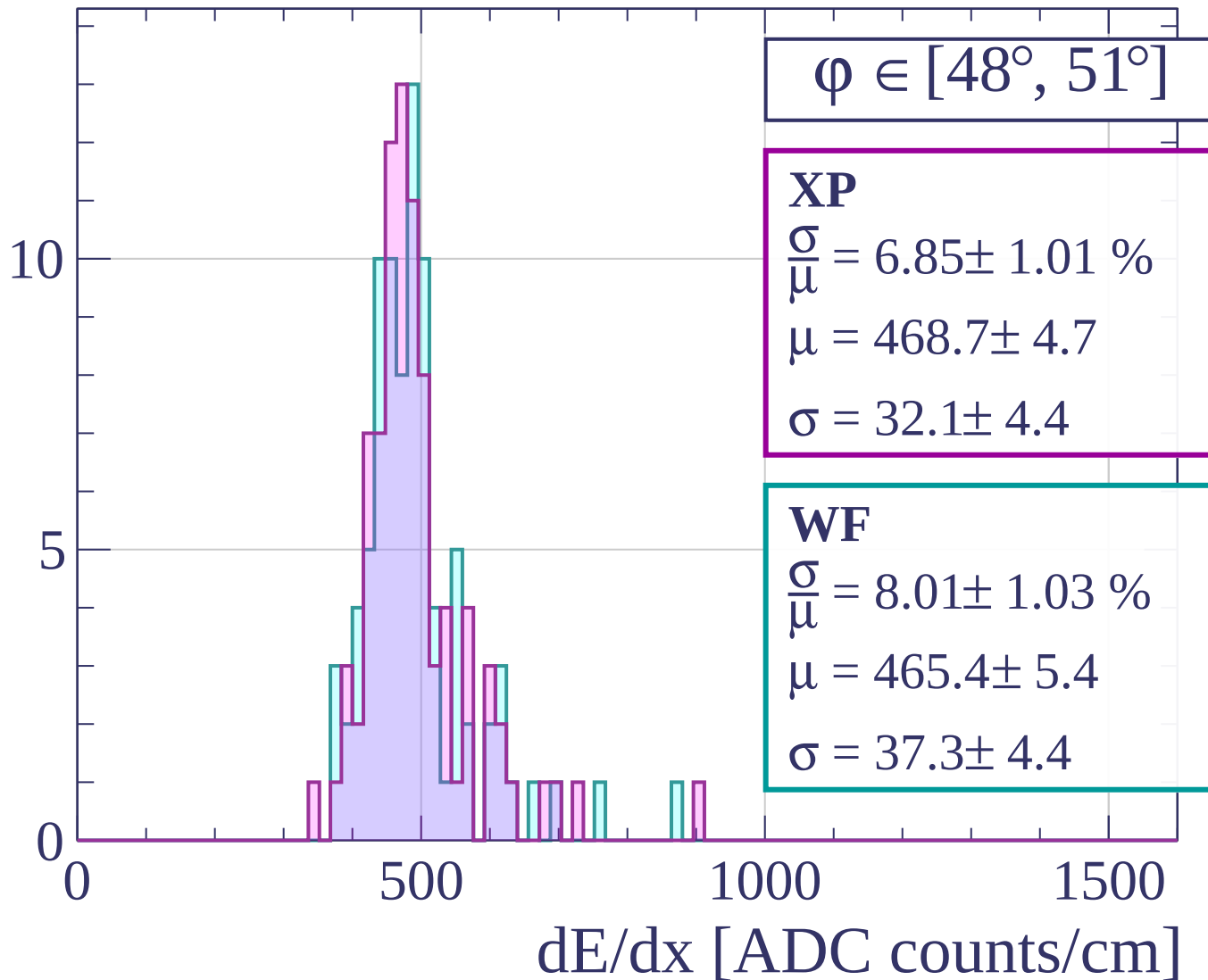


dE/dx [ADC counts/cm]

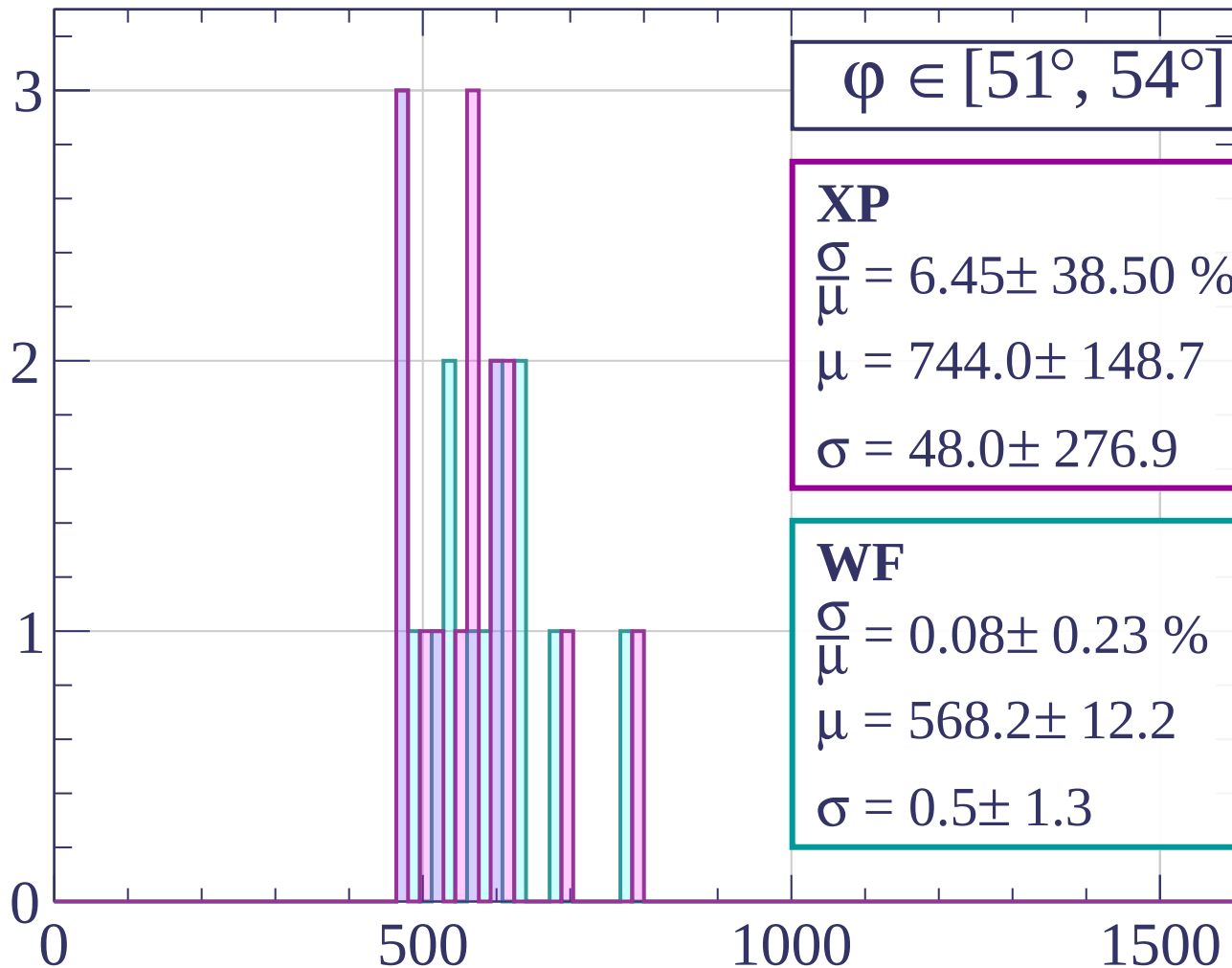
Count



Count



Count



$\varphi \in [51^\circ, 54^\circ]$

XP

$$\frac{\sigma}{\mu} = 6.45 \pm 38.50 \%$$

$$\mu = 744.0 \pm 148.7$$

$$\sigma = 48.0 \pm 276.9$$

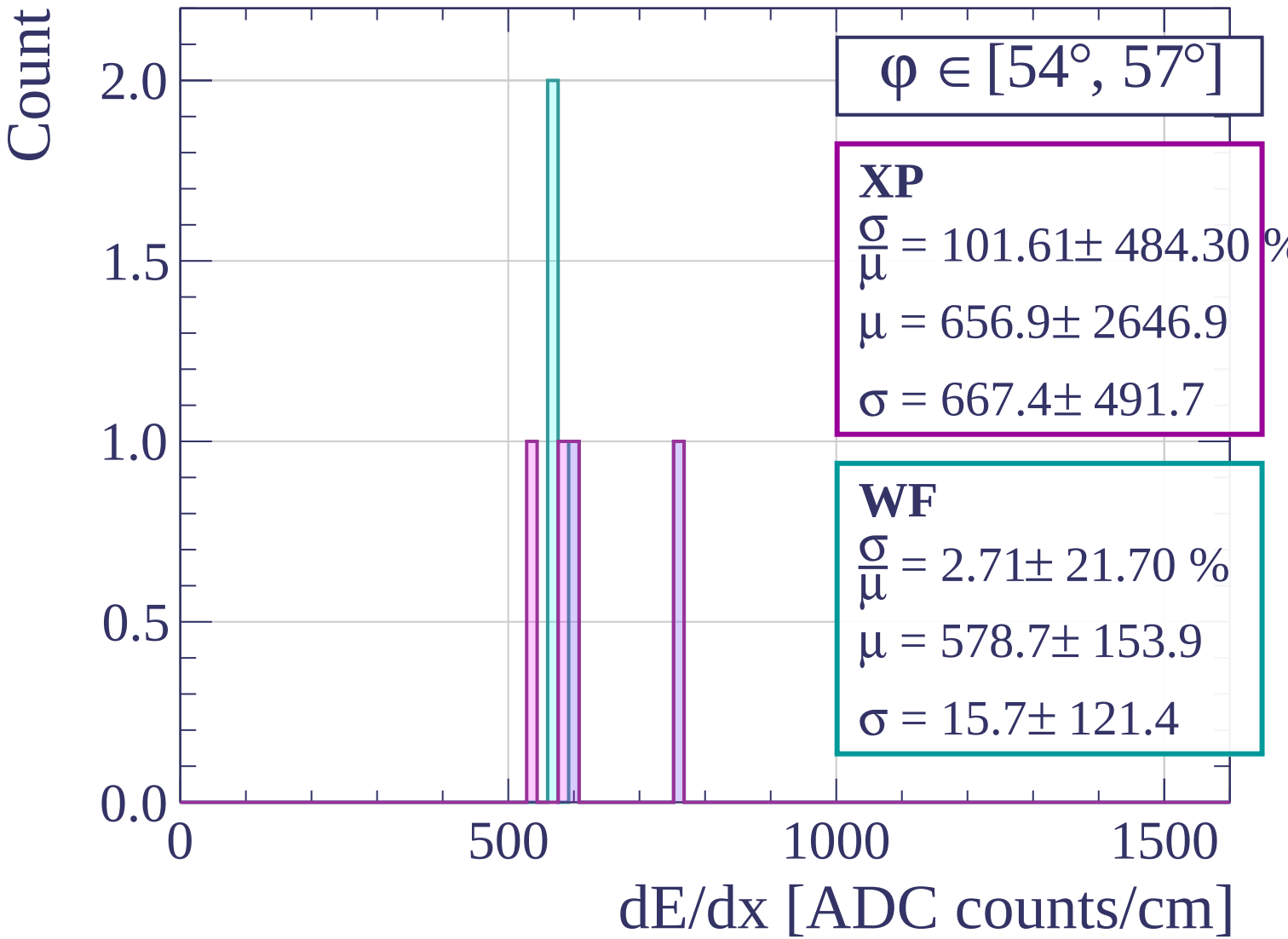
WF

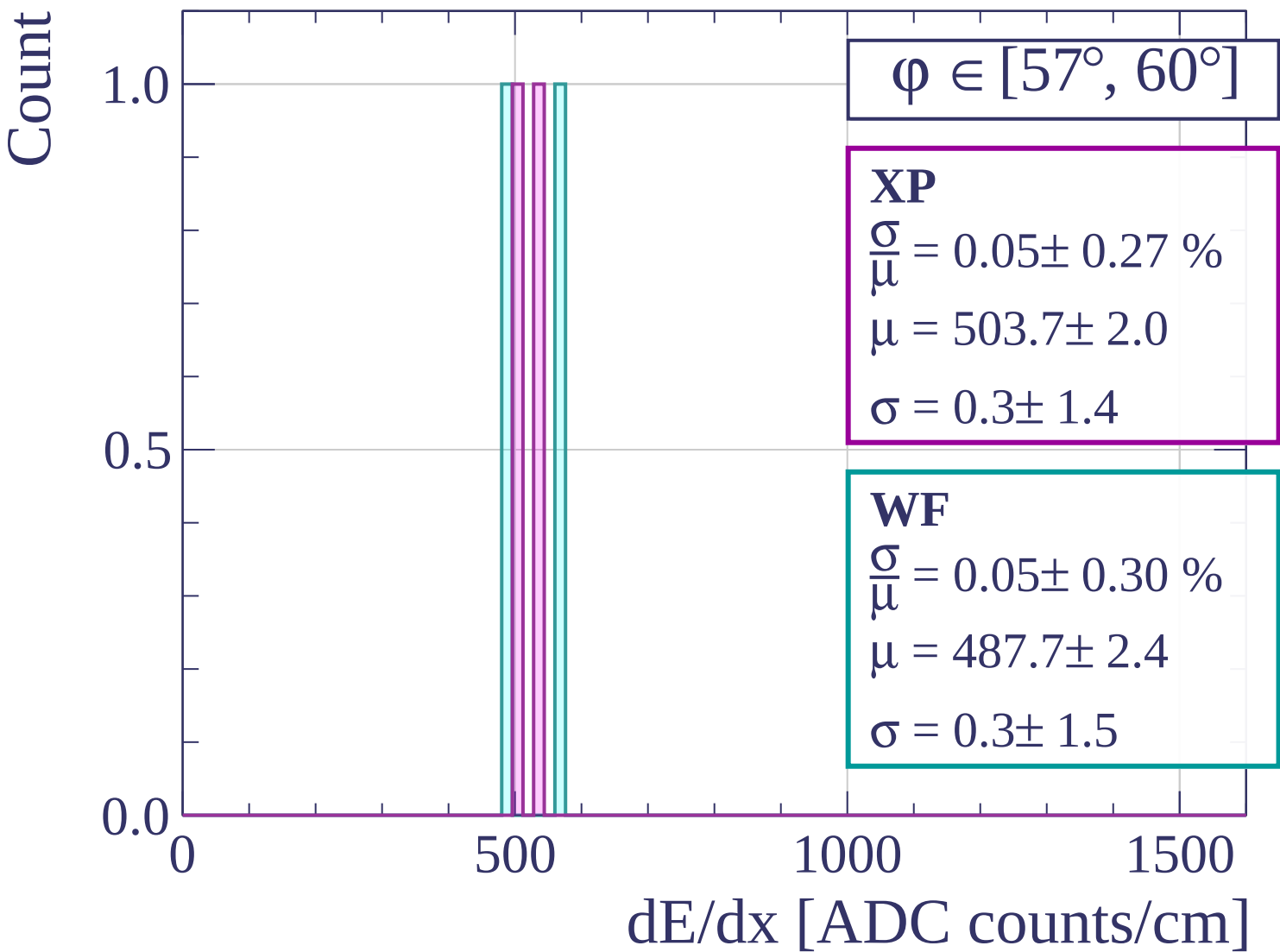
$$\frac{\sigma}{\mu} = 0.08 \pm 0.23 \%$$

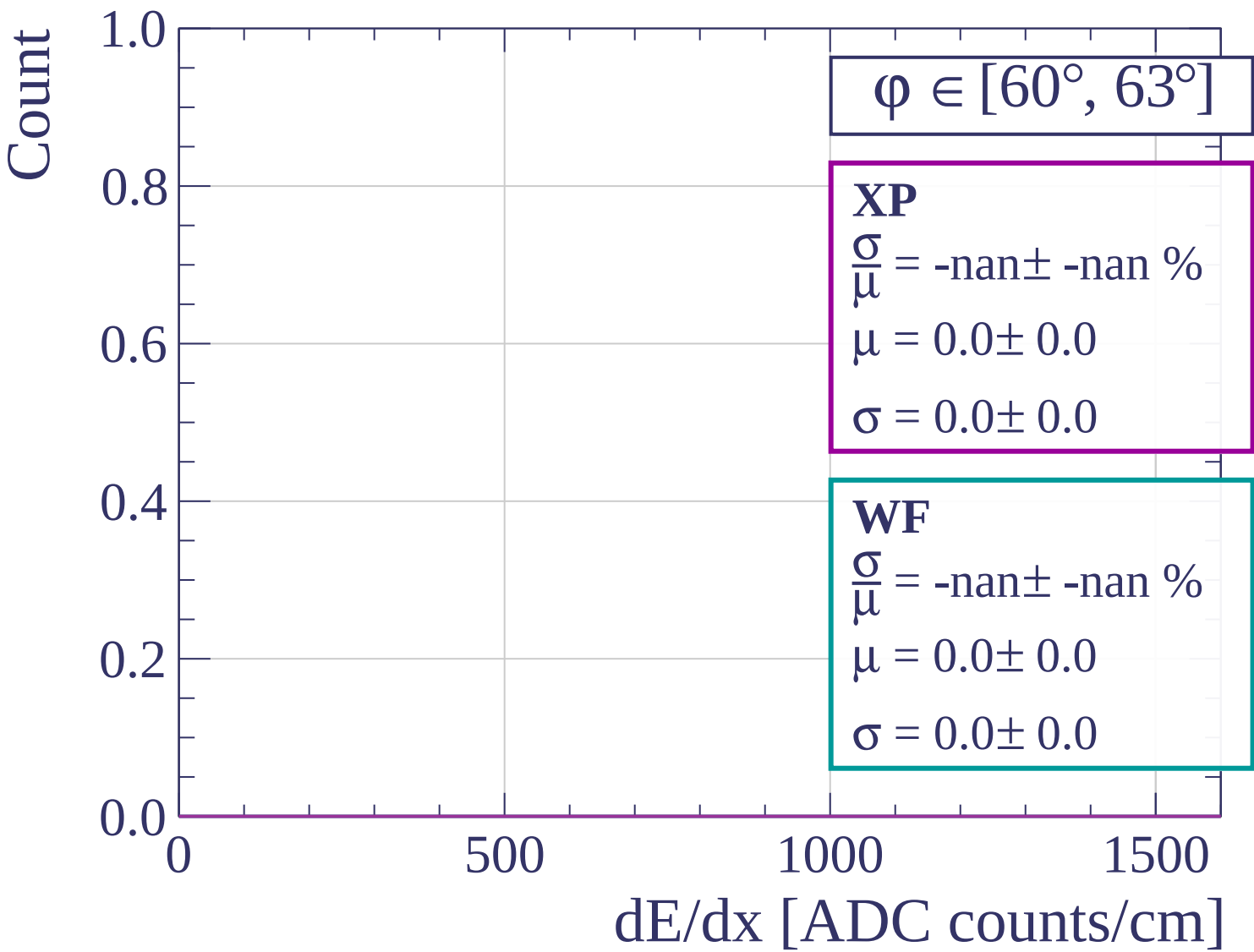
$$\mu = 568.2 \pm 12.2$$

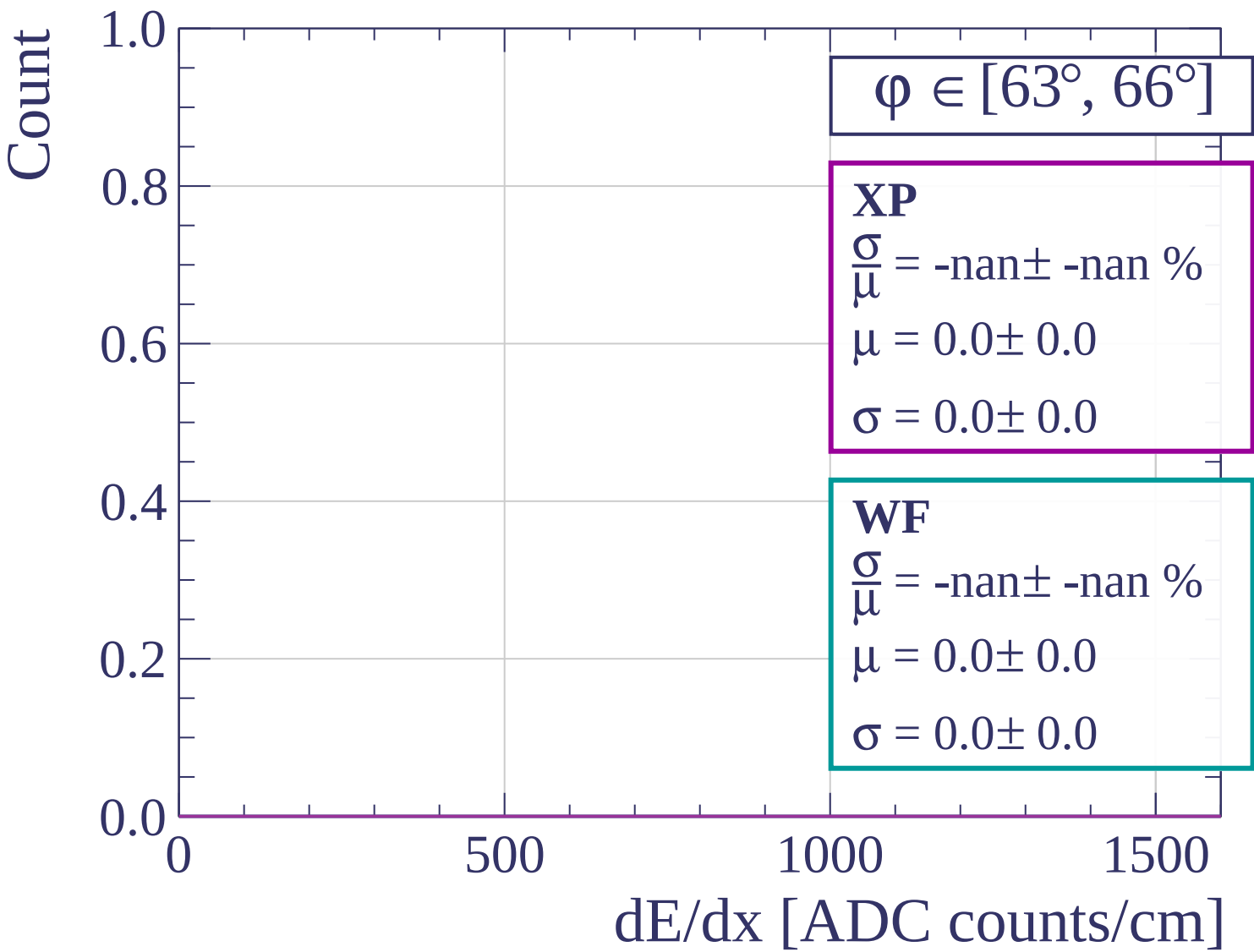
$$\sigma = 0.5 \pm 1.3$$

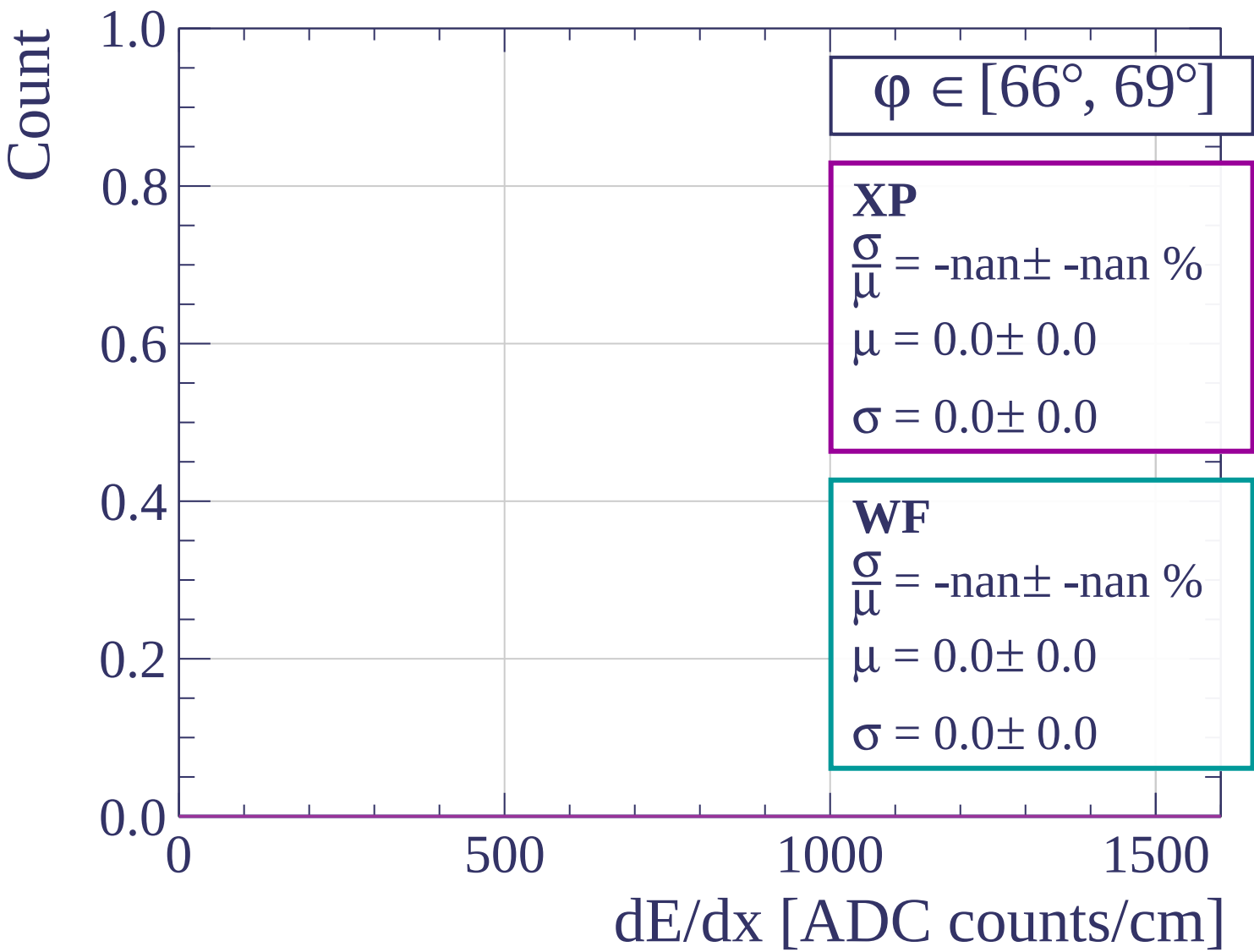
dE/dx [ADC counts/cm]

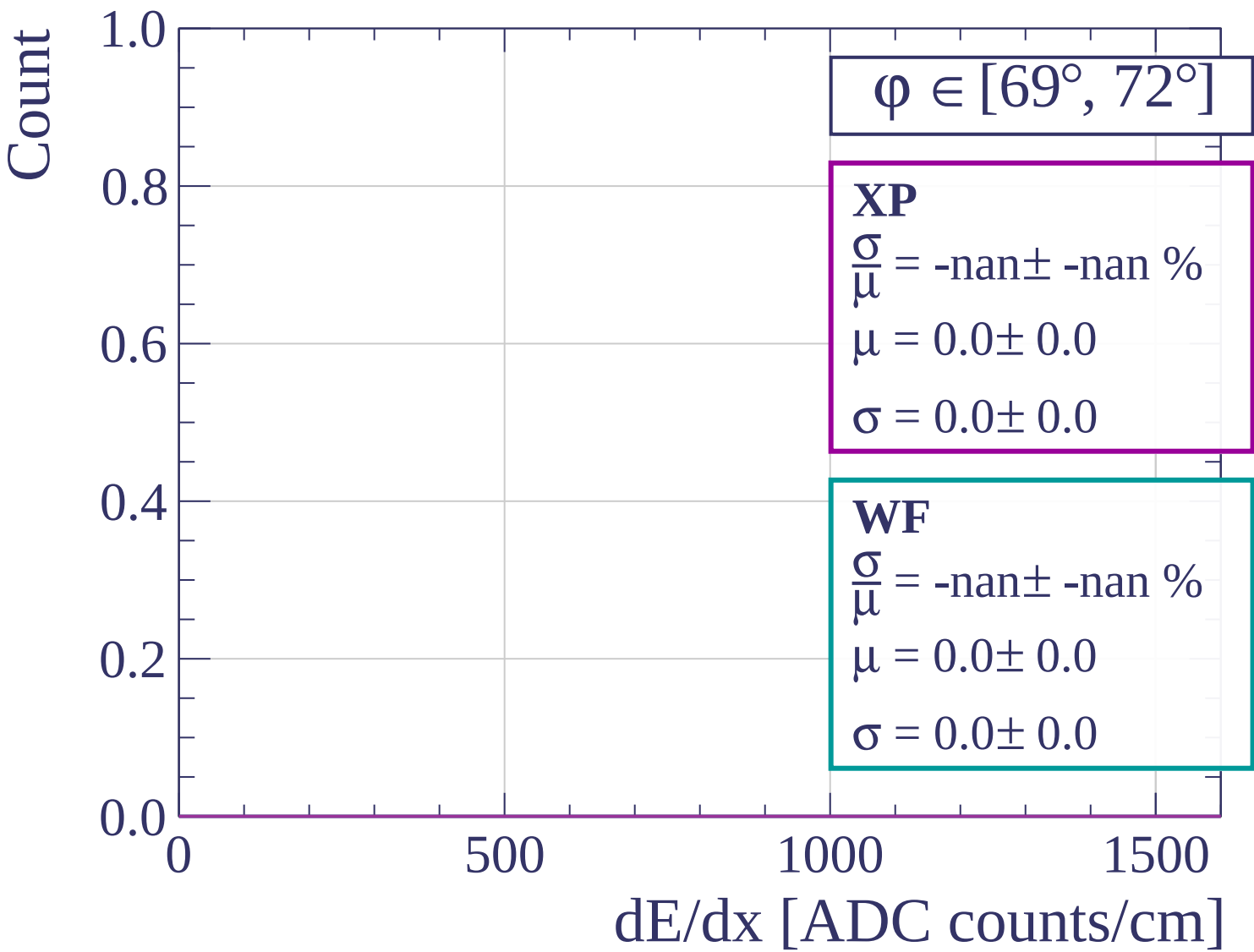


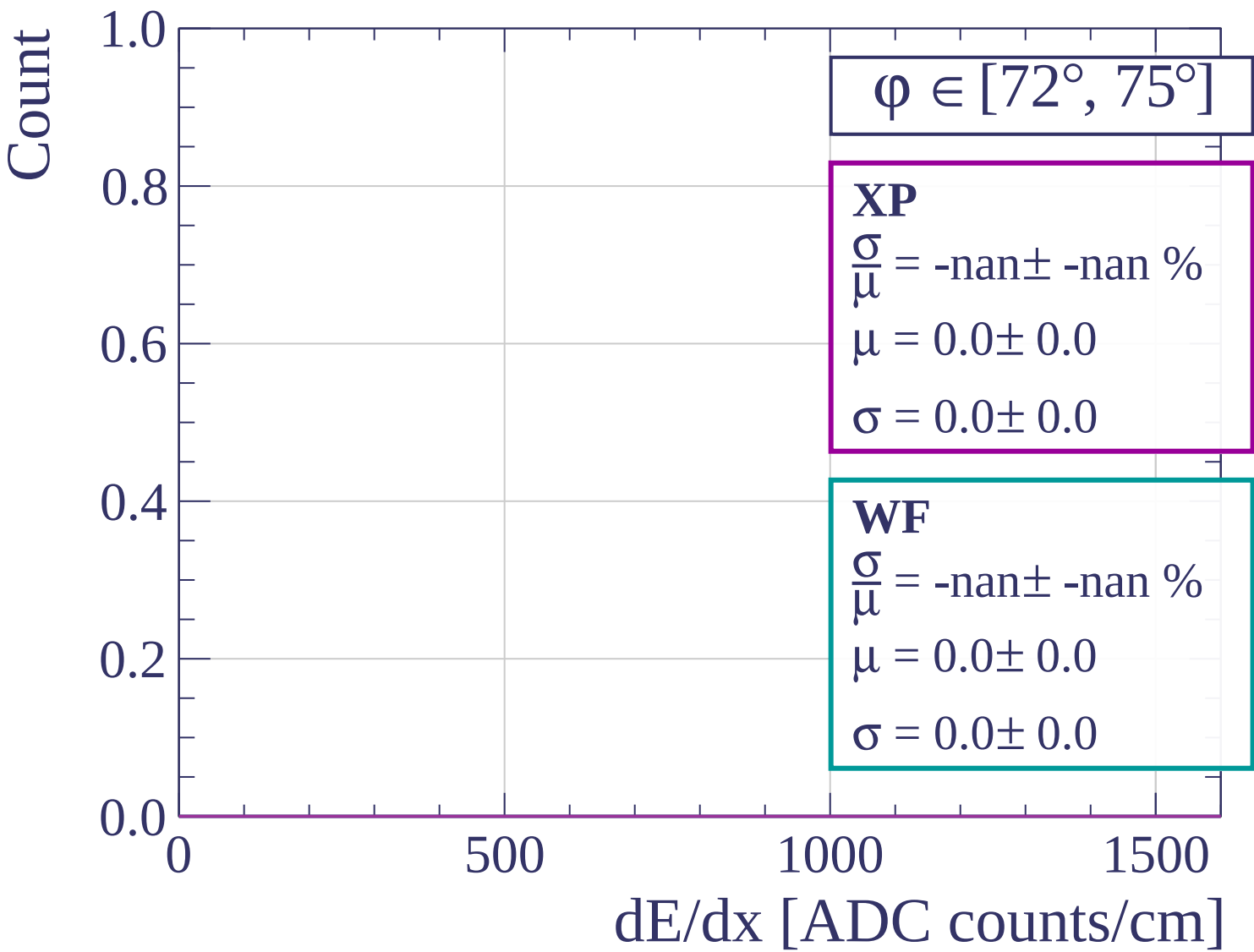


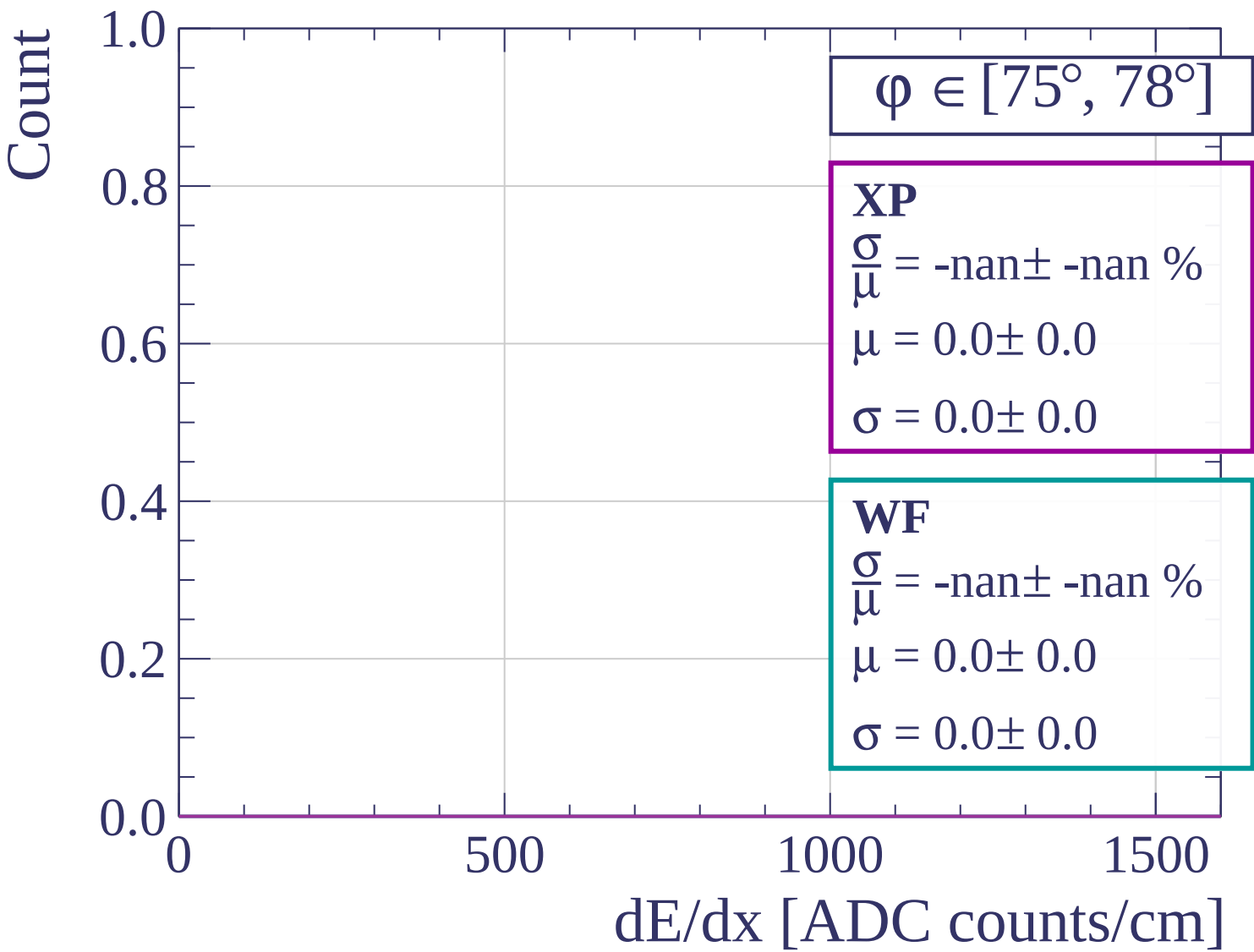




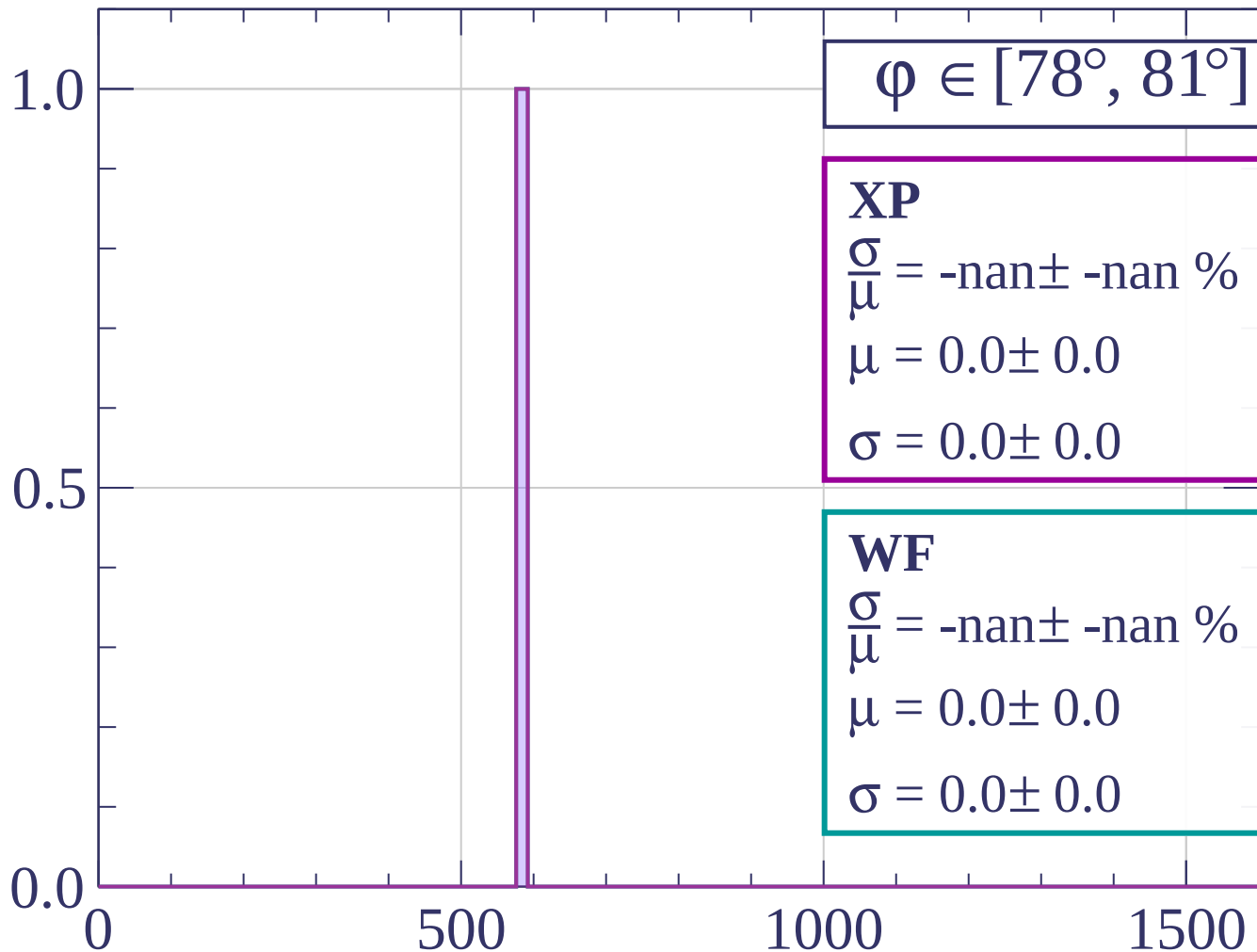




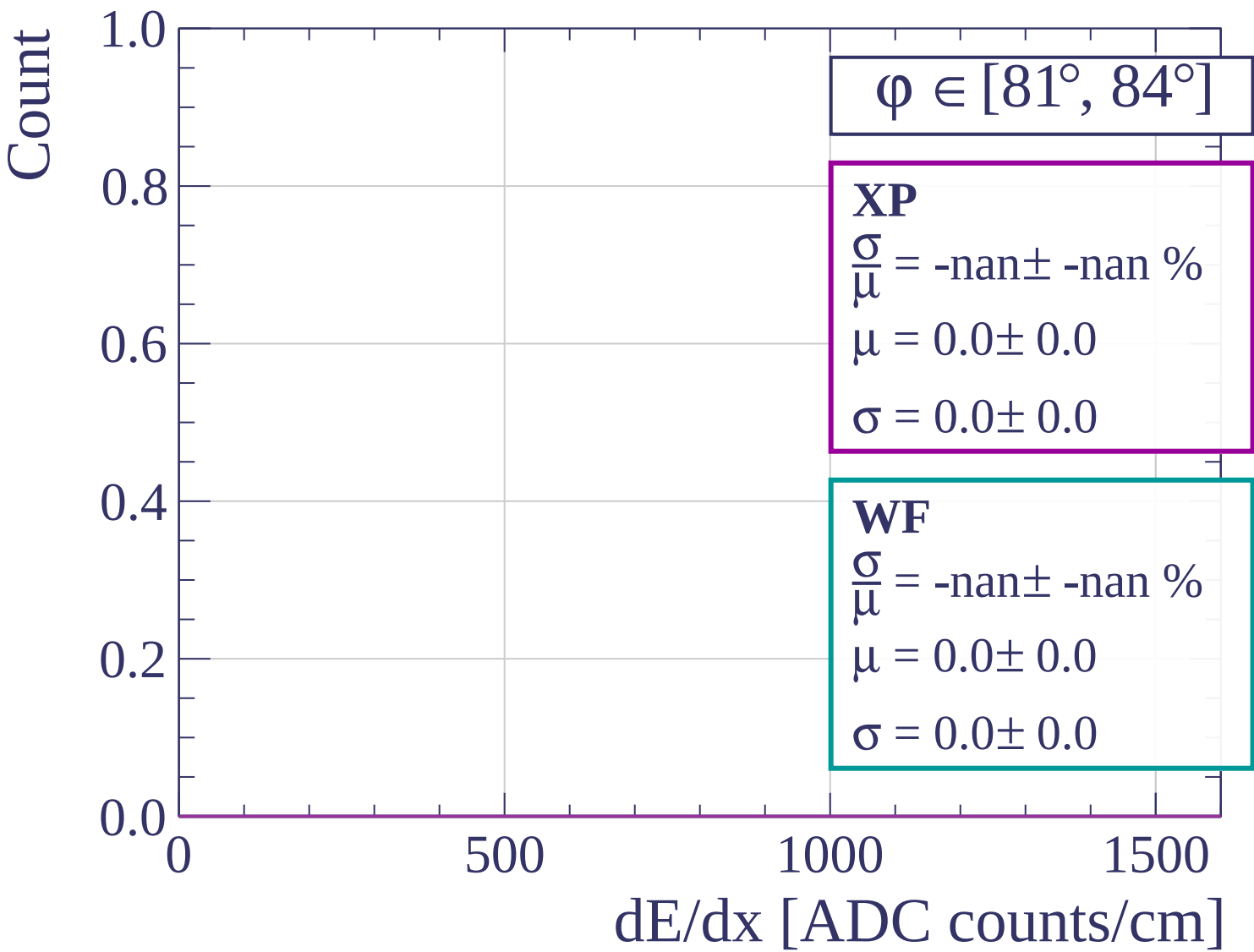


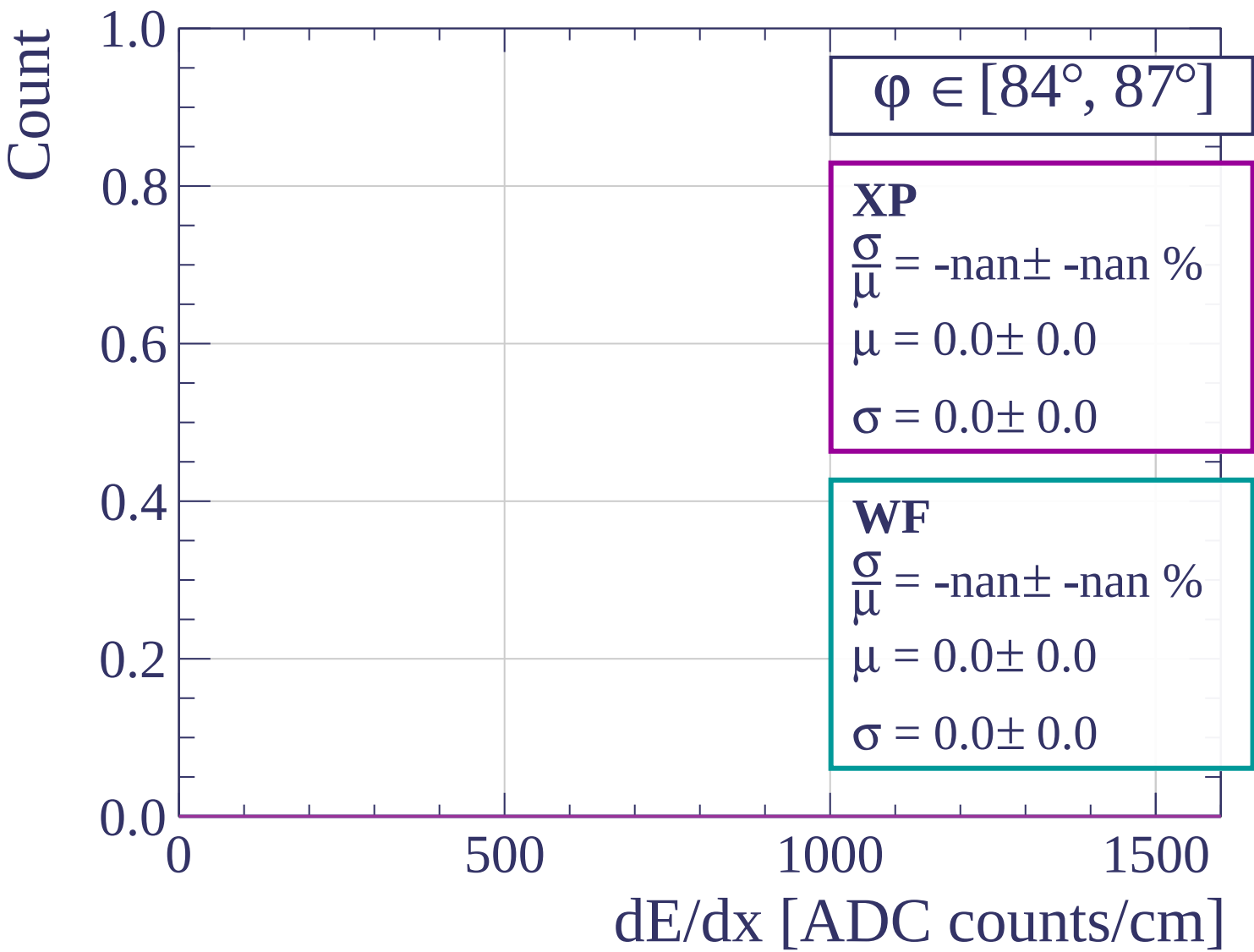


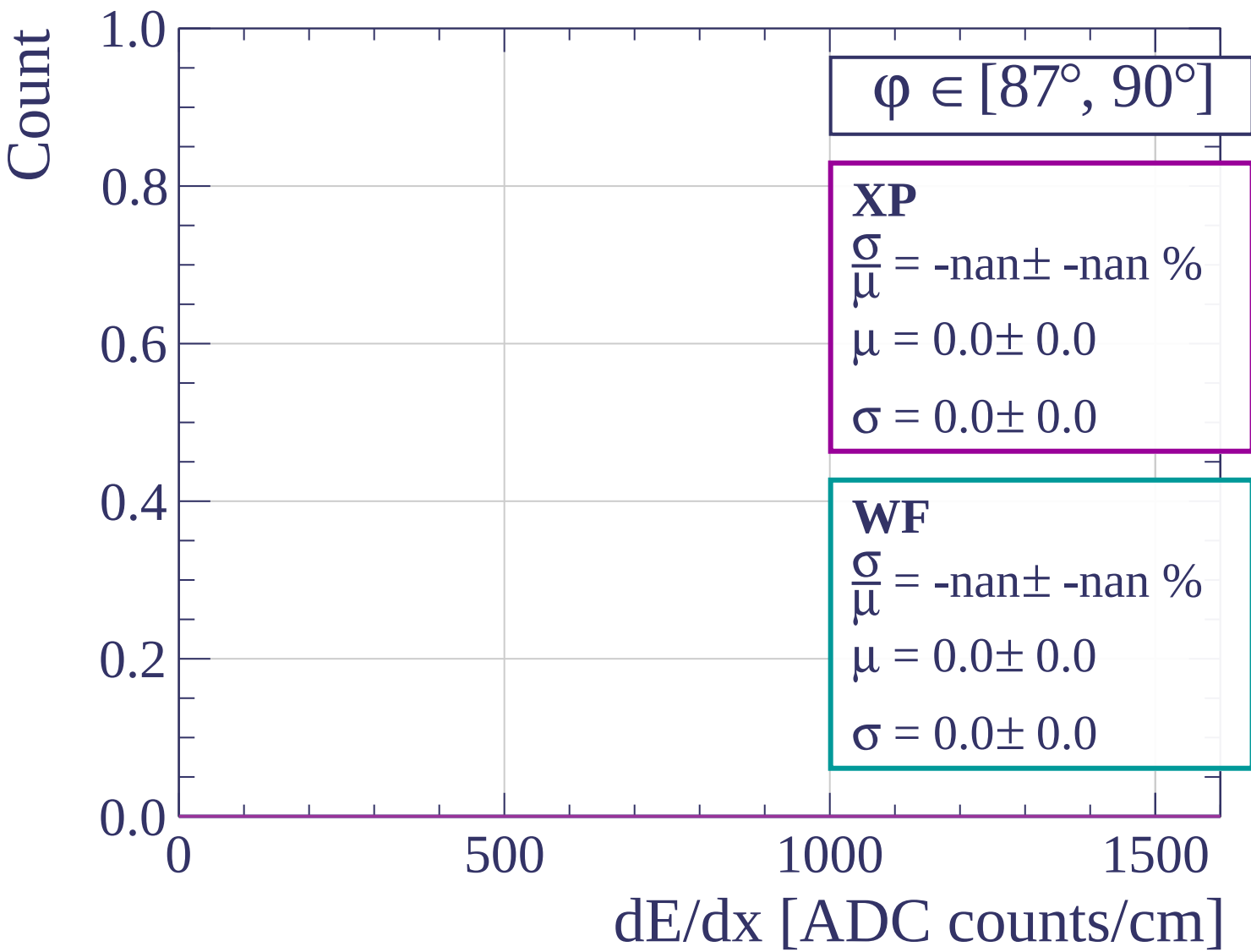
Count

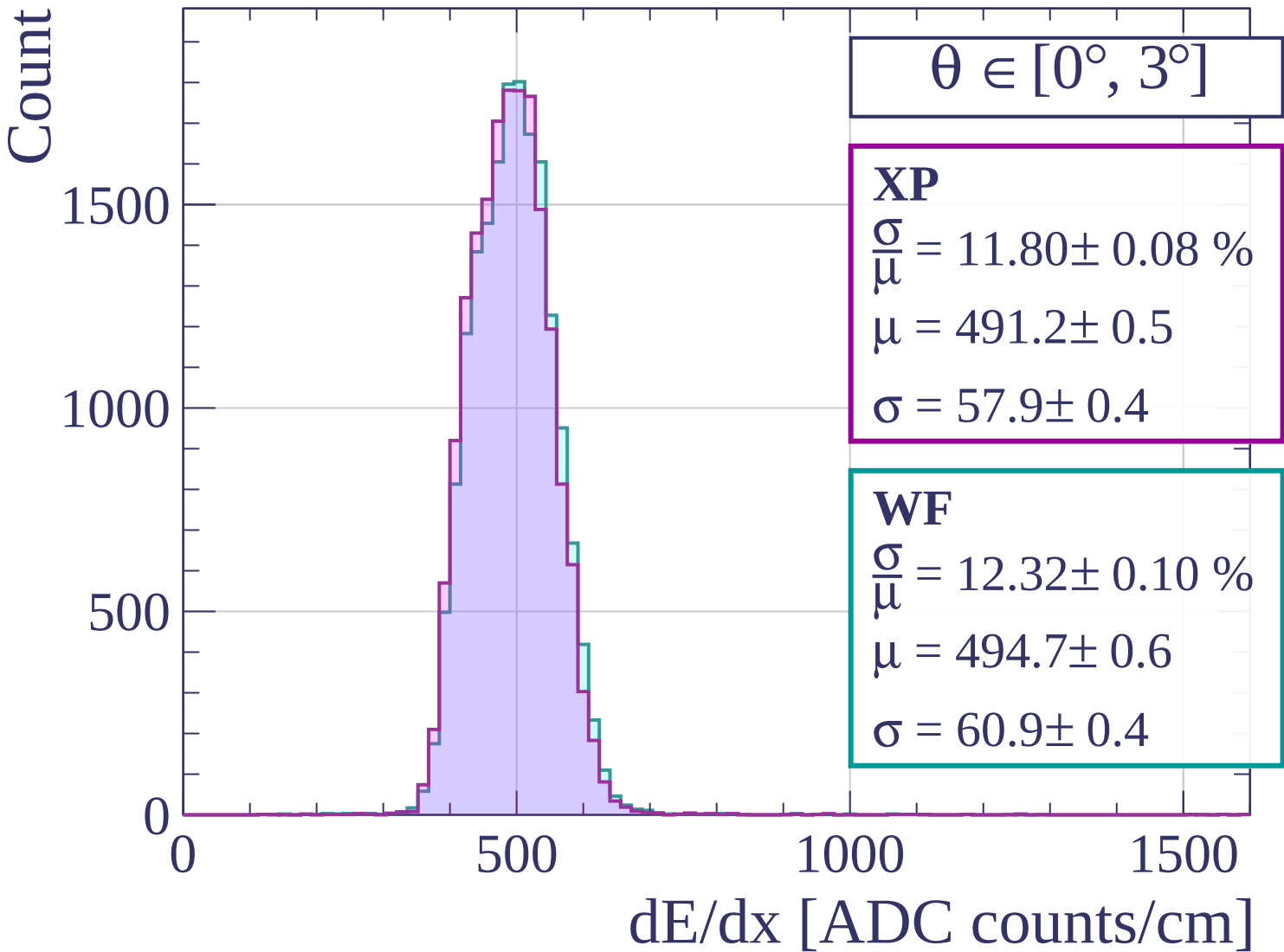


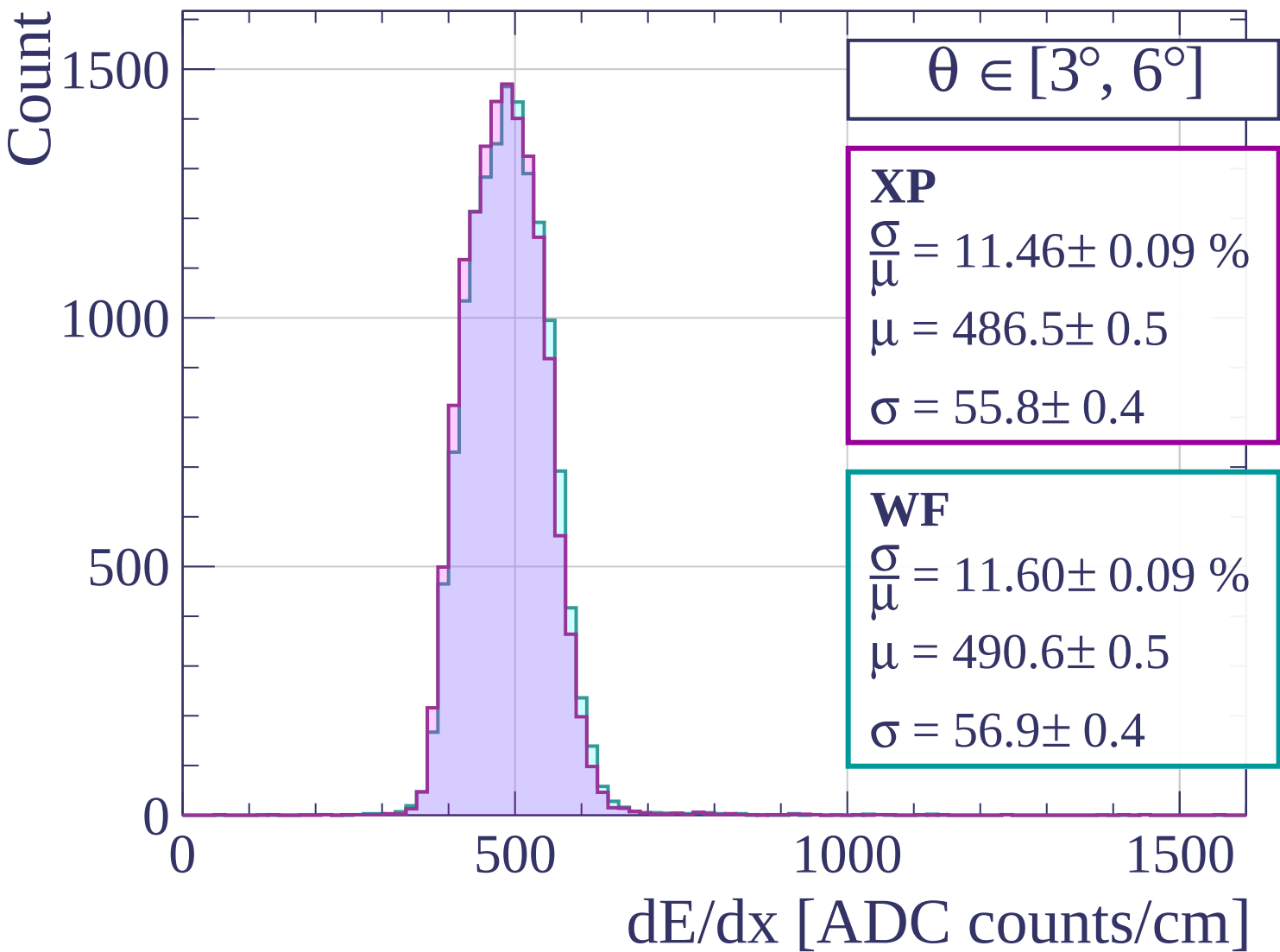
dE/dx [ADC counts/cm]

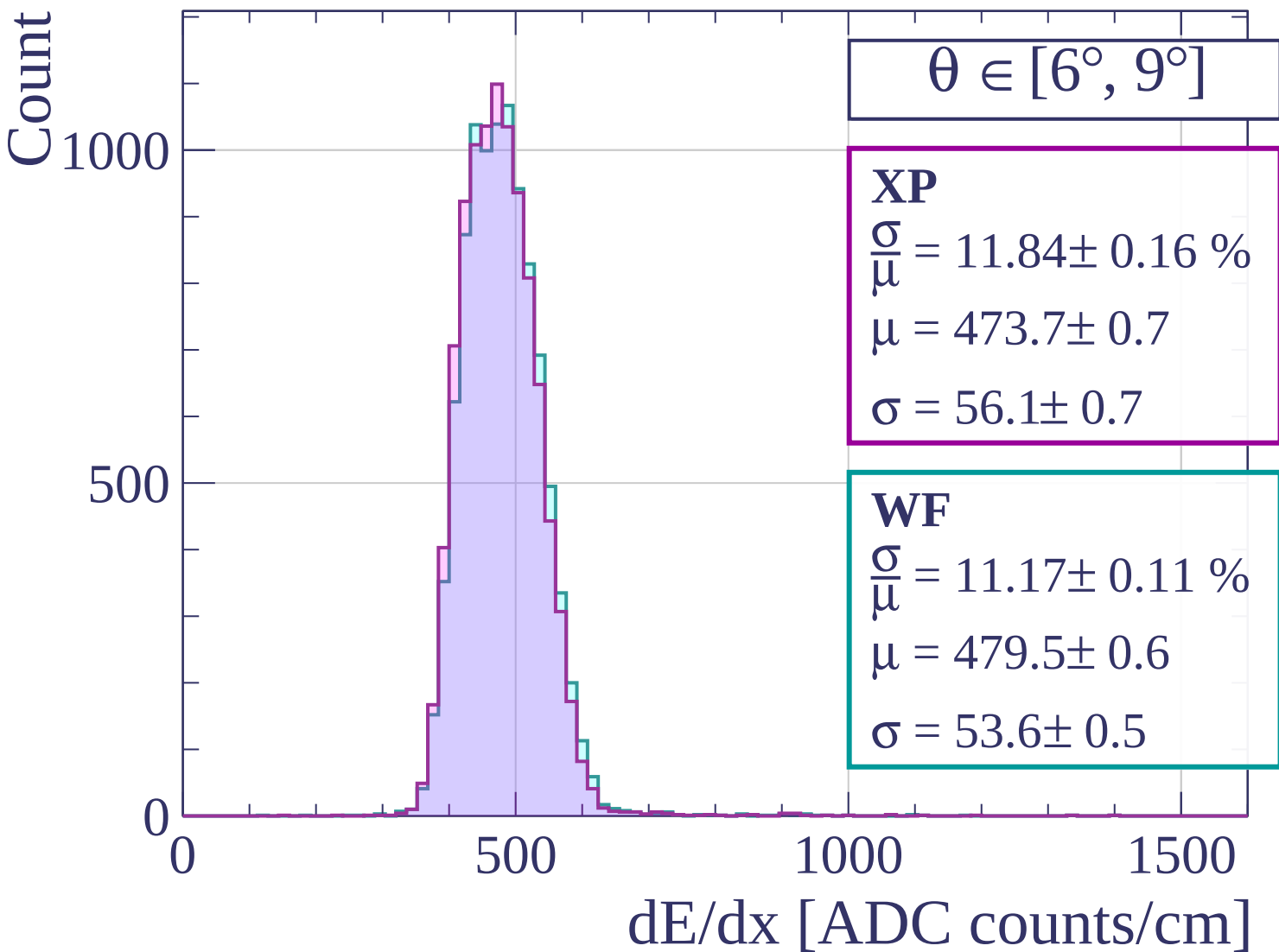












Count

$$\theta \in [9^\circ, 12^\circ]$$

XP

$$\frac{\sigma}{\mu} = 10.32 \pm 0.14 \%$$

$$\mu = 464.6 \pm 0.8$$

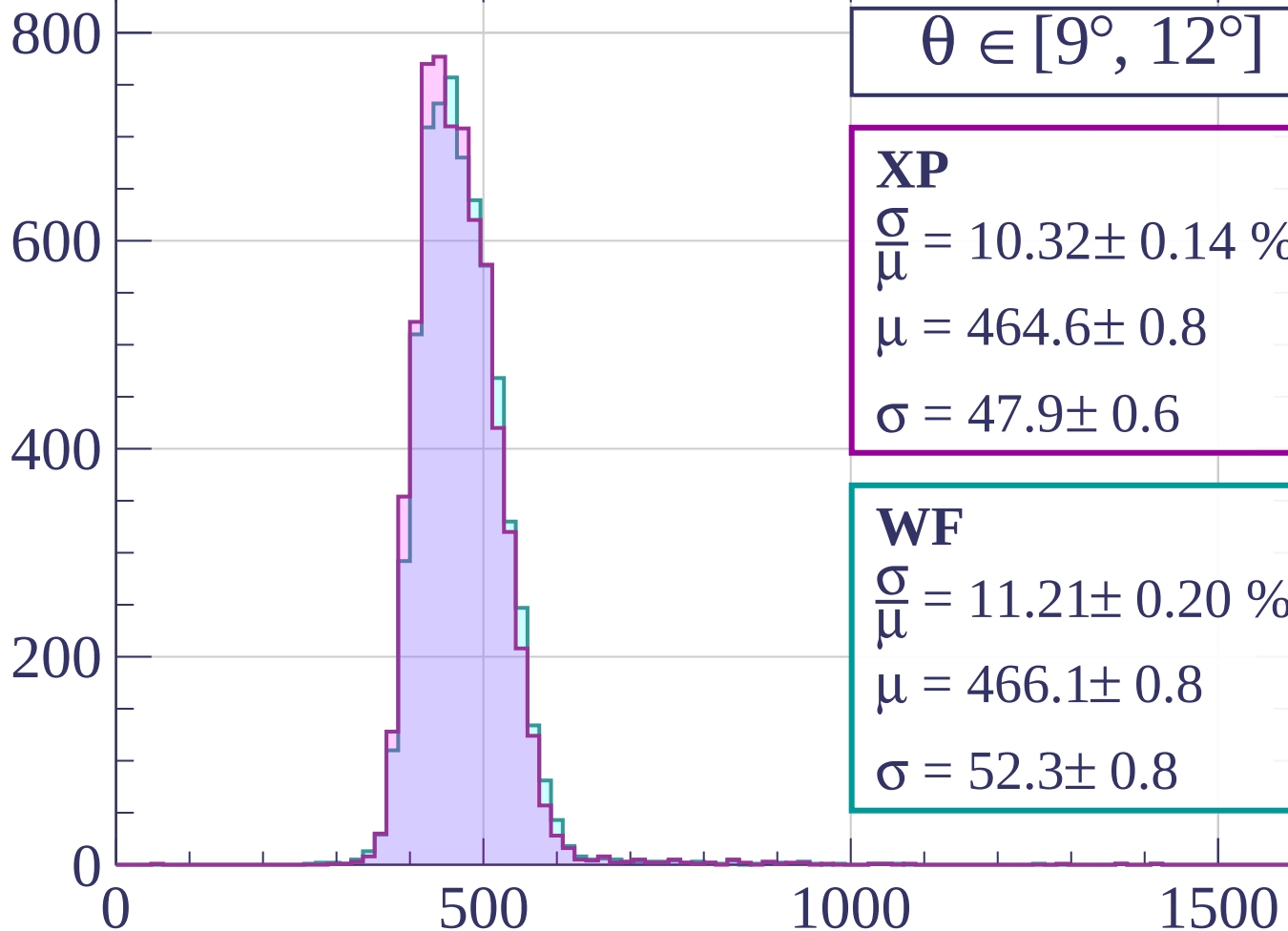
$$\sigma = 47.9 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 11.21 \pm 0.20 \%$$

$$\mu = 466.1 \pm 0.8$$

$$\sigma = 52.3 \pm 0.8$$



dE/dx [ADC counts/cm]

Count

$\theta \in [12^\circ, 15^\circ]$

XP

$$\frac{\sigma}{\mu} = 10.33 \pm 0.22 \%$$

$$\mu = 451.1 \pm 0.9$$

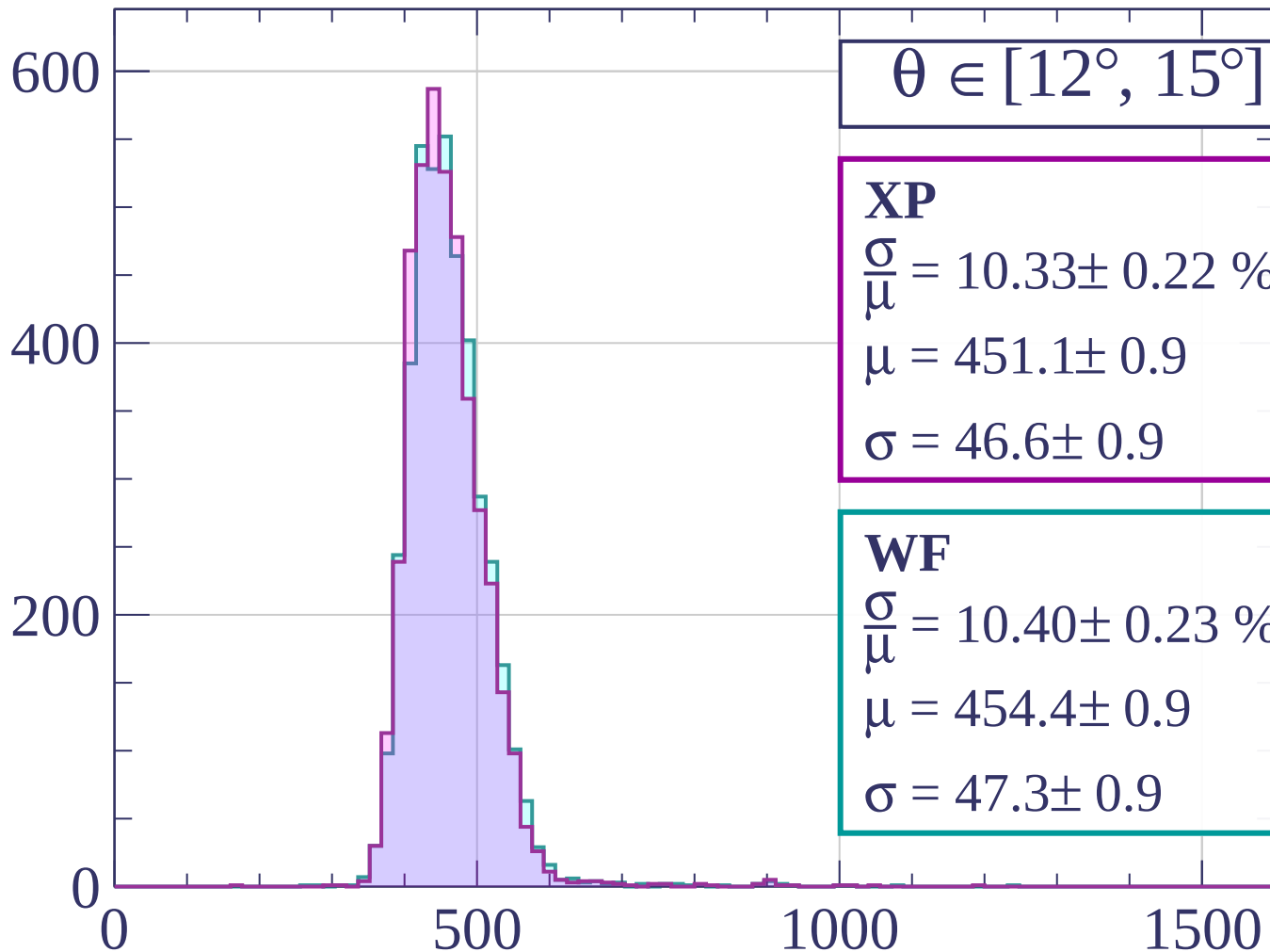
$$\sigma = 46.6 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 10.40 \pm 0.23 \%$$

$$\mu = 454.4 \pm 0.9$$

$$\sigma = 47.3 \pm 0.9$$



dE/dx [ADC counts/cm]

Count

$\theta \in [15^\circ, 18^\circ]$

XP

$$\frac{\sigma}{\mu} = 8.90 \pm 0.19 \%$$

$$\mu = 444.0 \pm 1.0$$

$$\sigma = 39.5 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.59 \pm 0.26 \%$$

$$\mu = 446.0 \pm 1.0$$

$$\sigma = 42.8 \pm 1.0$$

400

300

200

100

0

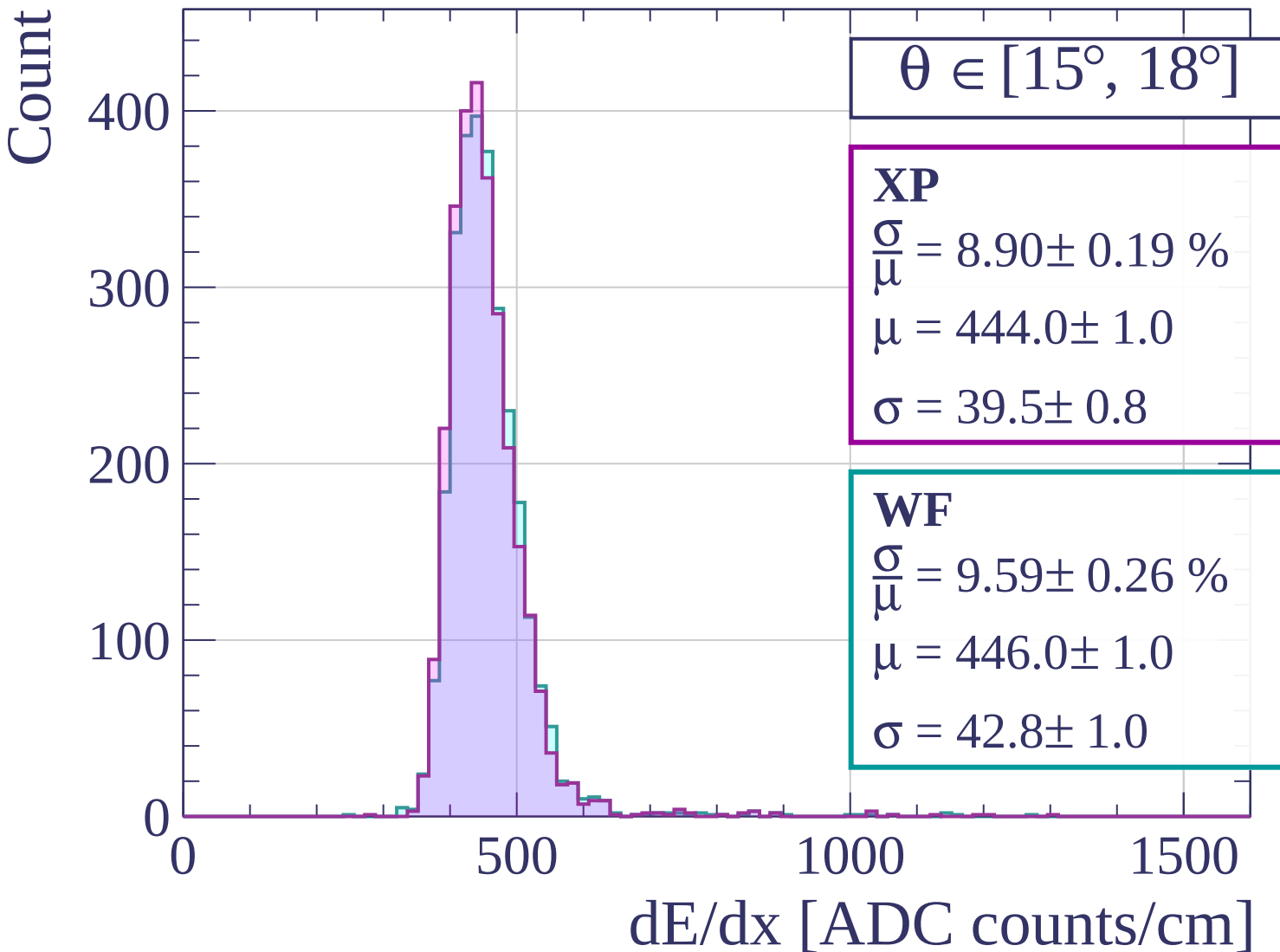
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\theta \in [18^\circ, 21^\circ]$

XP

$$\frac{\sigma}{\mu} = 7.96 \pm 0.27 \%$$

$$\mu = 434.4 \pm 1.1$$

$$\sigma = 34.6 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 8.61 \pm 0.28 \%$$

$$\mu = 438.6 \pm 1.1$$

$$\sigma = 37.8 \pm 1.1$$

300

200

100

0

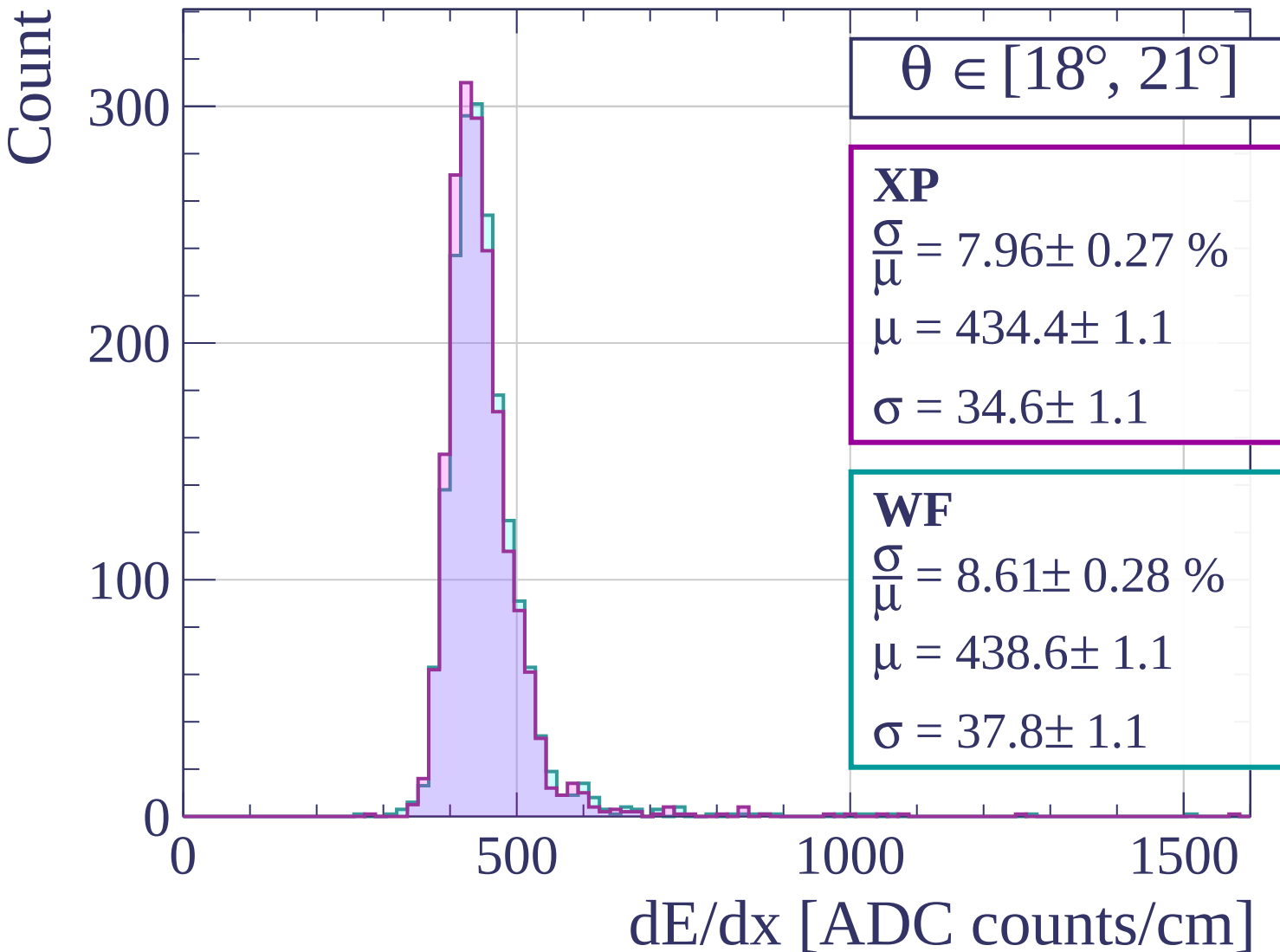
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\theta \in [21^\circ, 24^\circ]$

XP

$$\frac{\sigma}{\mu} = 8.14 \pm 0.33 \%$$

$$\mu = 428.4 \pm 1.3$$

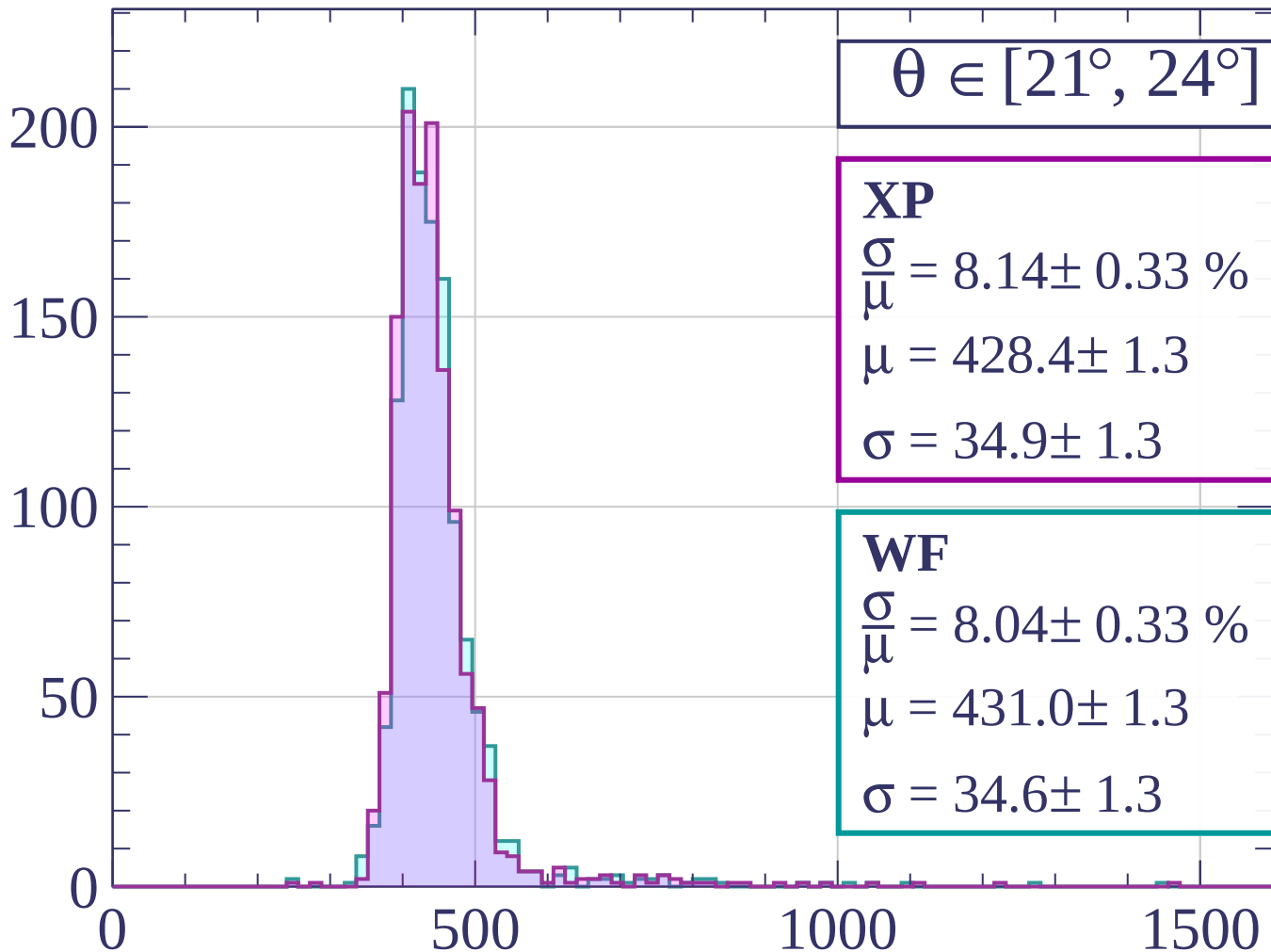
$$\sigma = 34.9 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 8.04 \pm 0.33 \%$$

$$\mu = 431.0 \pm 1.3$$

$$\sigma = 34.6 \pm 1.3$$



dE/dx [ADC counts/cm]

Count

$\theta \in [24^\circ, 27^\circ]$

XP

$$\frac{\sigma}{\mu} = 7.89 \pm 0.40 \%$$

$$\mu = 430.1 \pm 1.6$$

$$\sigma = 33.9 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 8.33 \pm 0.33 \%$$

$$\mu = 433.6 \pm 1.6$$

$$\sigma = 36.1 \pm 1.3$$

100

50

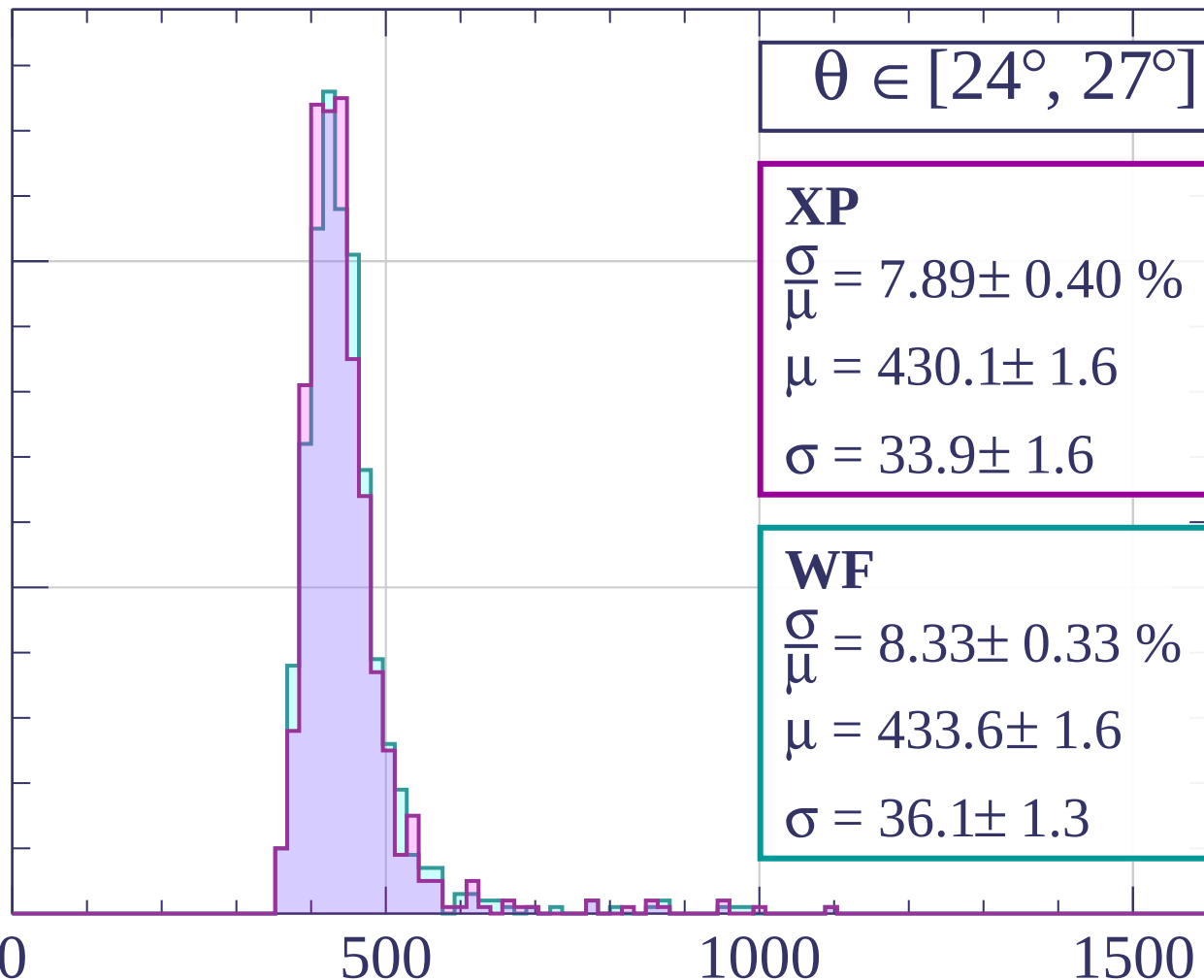
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\theta \in [27^\circ, 30^\circ]$

XP

$$\frac{\sigma}{\mu} = 6.87 \pm 0.35 \%$$

$$\mu = 423.1 \pm 1.6$$

$$\sigma = 29.0 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 7.21 \pm 0.47 \%$$

$$\mu = 424.7 \pm 1.8$$

$$\sigma = 30.6 \pm 1.9$$

100

50

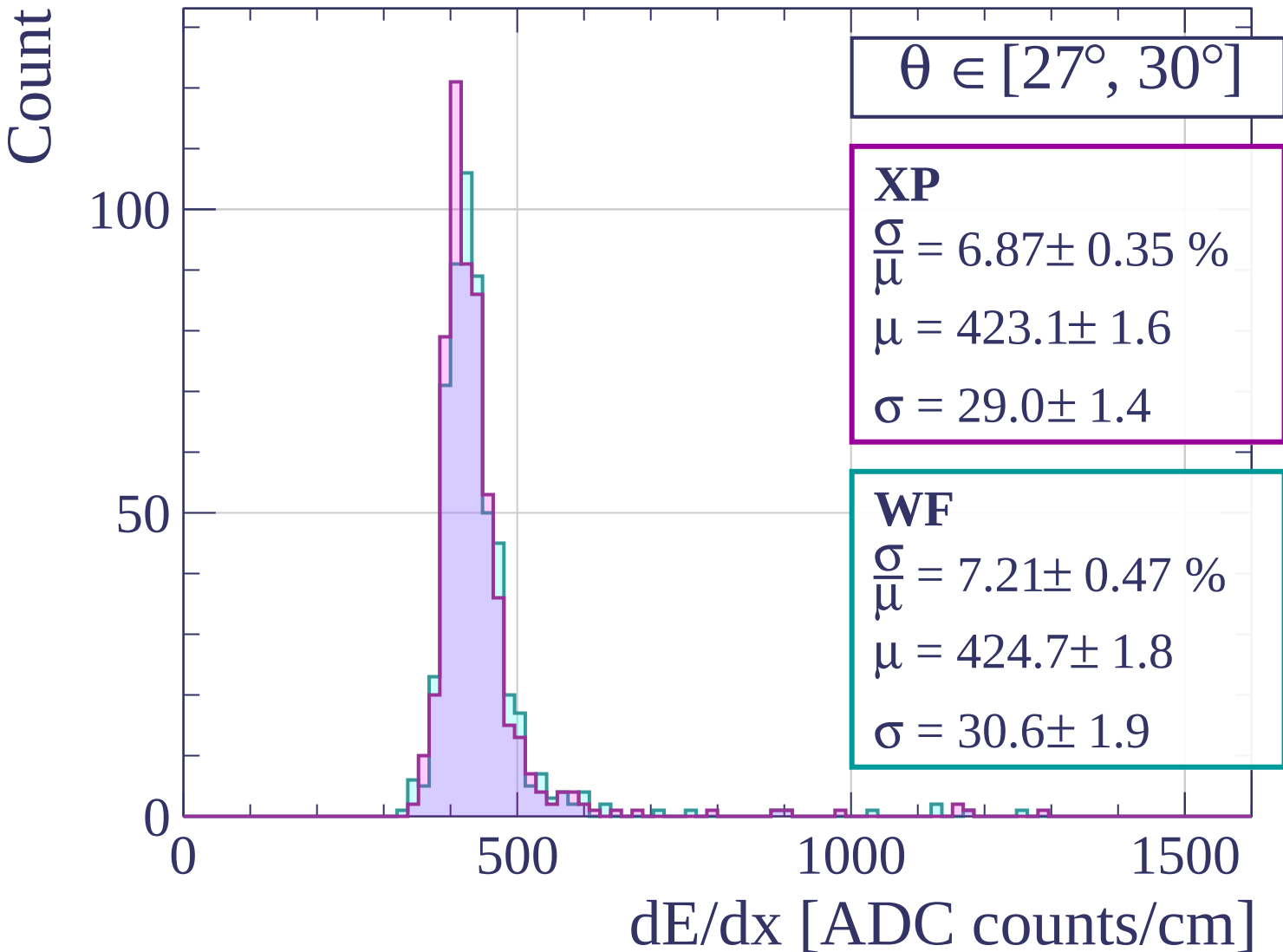
0

500

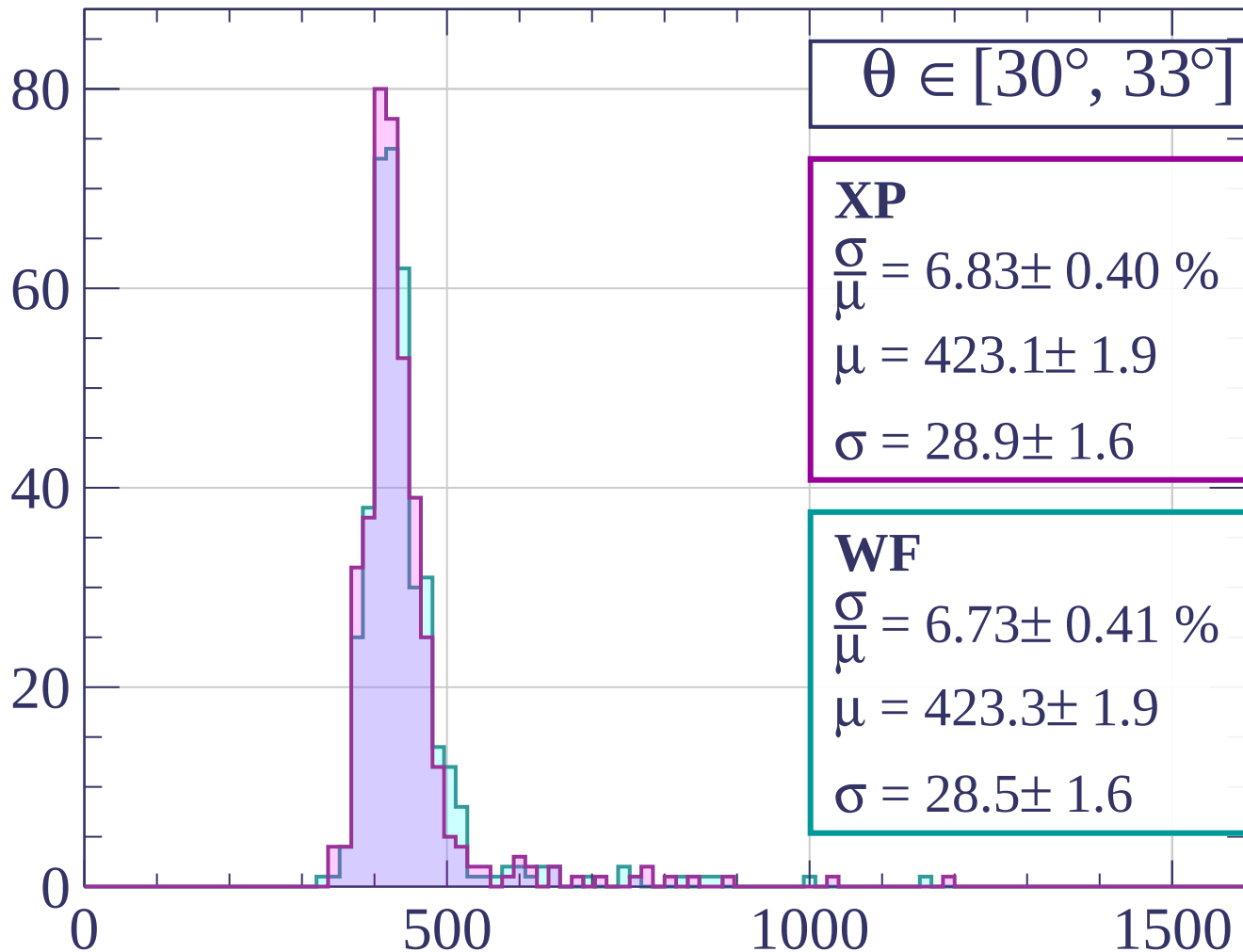
1000

1500

dE/dx [ADC counts/cm]



Count



$\theta \in [30^\circ, 33^\circ]$

XP

$$\frac{\sigma}{\mu} = 6.83 \pm 0.40 \%$$

$$\mu = 423.1 \pm 1.9$$

$$\sigma = 28.9 \pm 1.6$$

WF

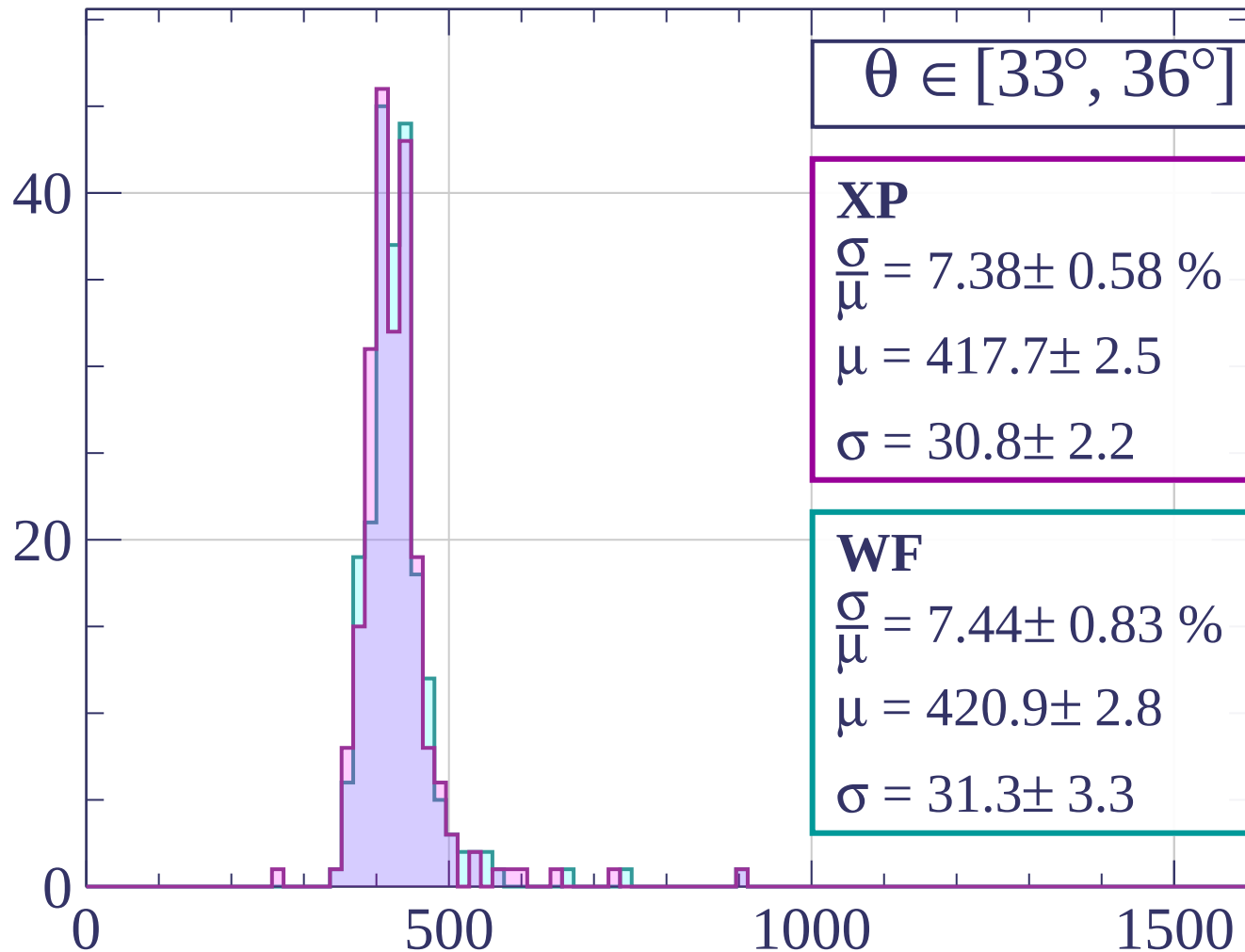
$$\frac{\sigma}{\mu} = 6.73 \pm 0.41 \%$$

$$\mu = 423.3 \pm 1.9$$

$$\sigma = 28.5 \pm 1.6$$

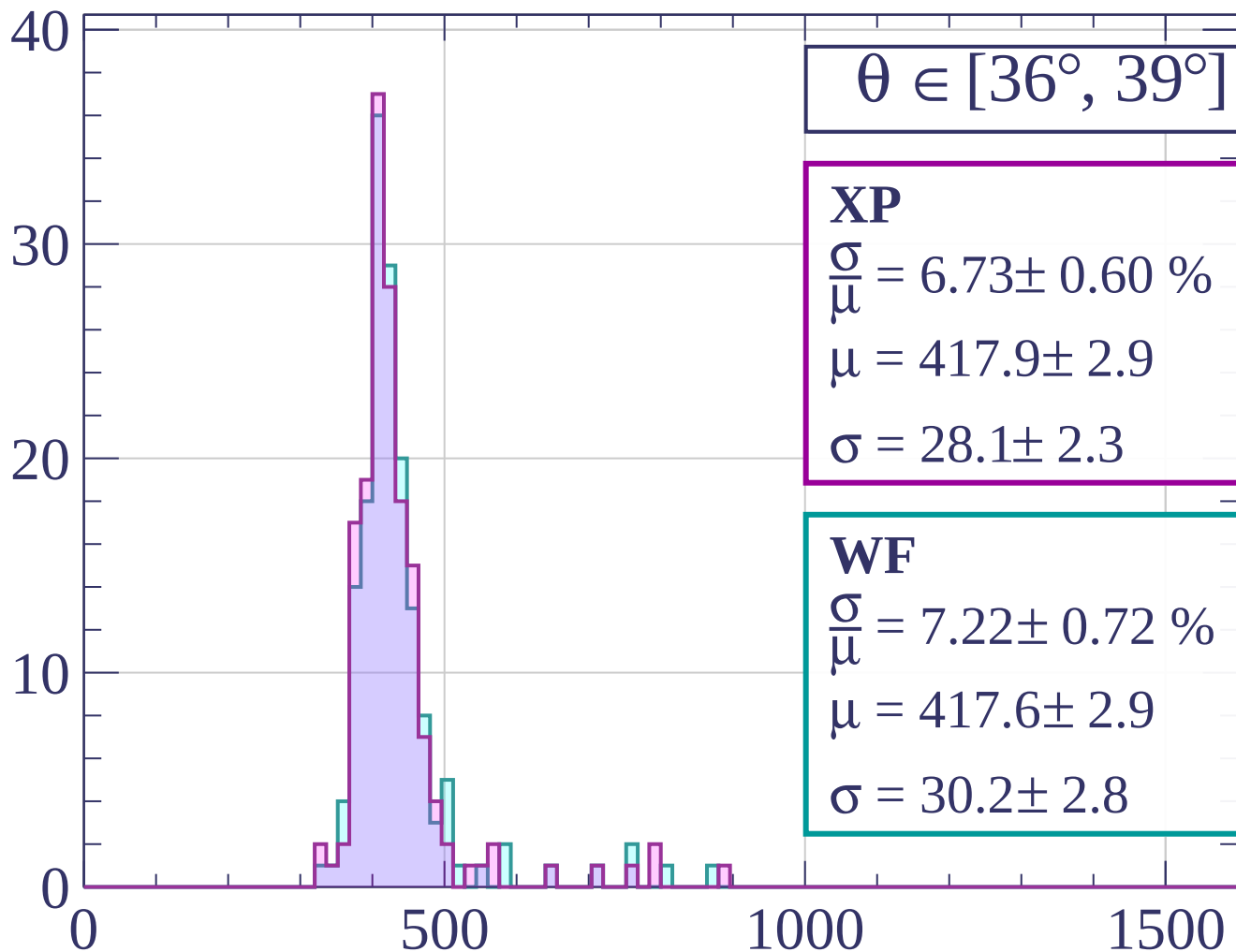
dE/dx [ADC counts/cm]

Count



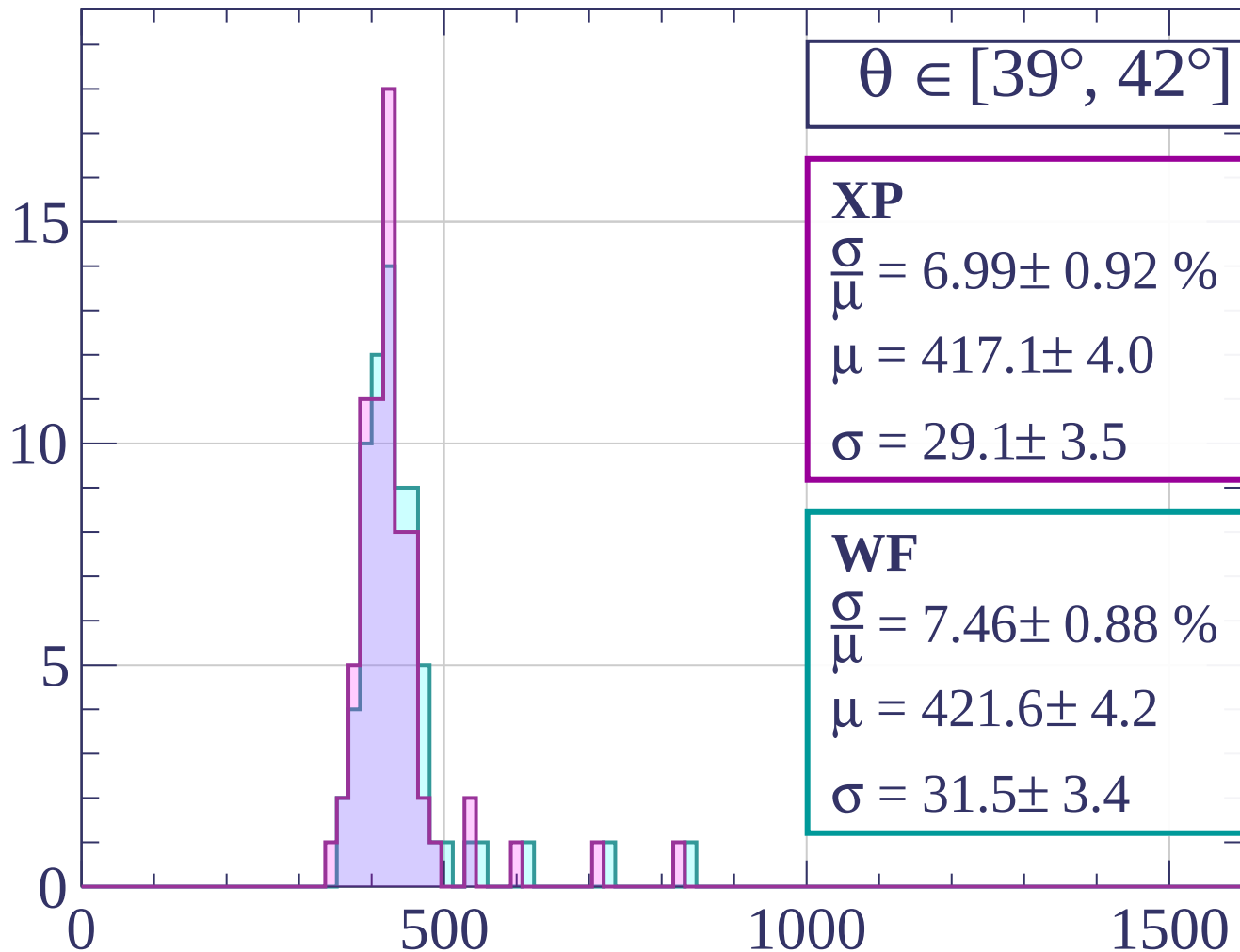
dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count



$\theta \in [39^\circ, 42^\circ]$

XP

$$\frac{\sigma}{\mu} = 6.99 \pm 0.92 \%$$

$$\mu = 417.1 \pm 4.0$$

$$\sigma = 29.1 \pm 3.5$$

WF

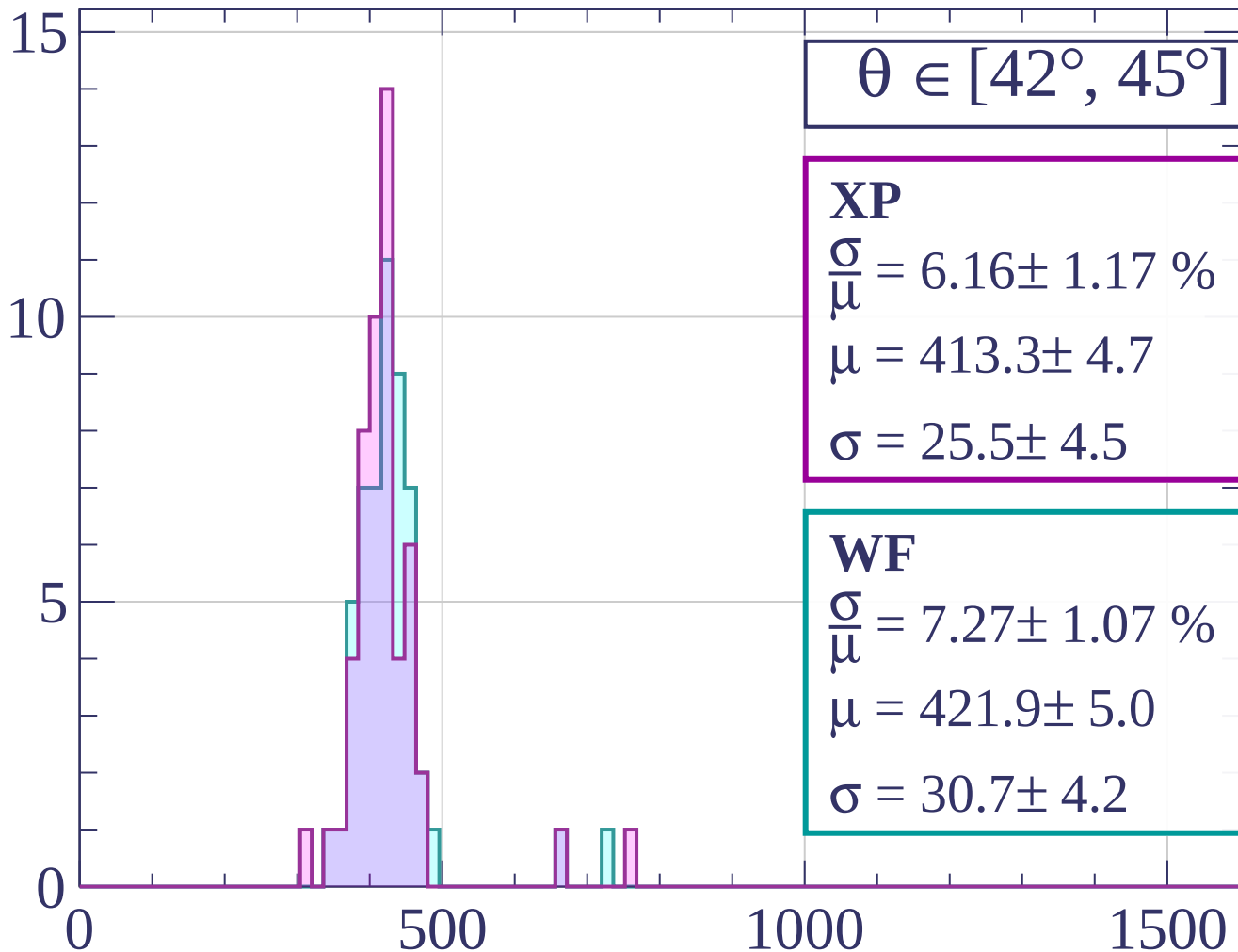
$$\frac{\sigma}{\mu} = 7.46 \pm 0.88 \%$$

$$\mu = 421.6 \pm 4.2$$

$$\sigma = 31.5 \pm 3.4$$

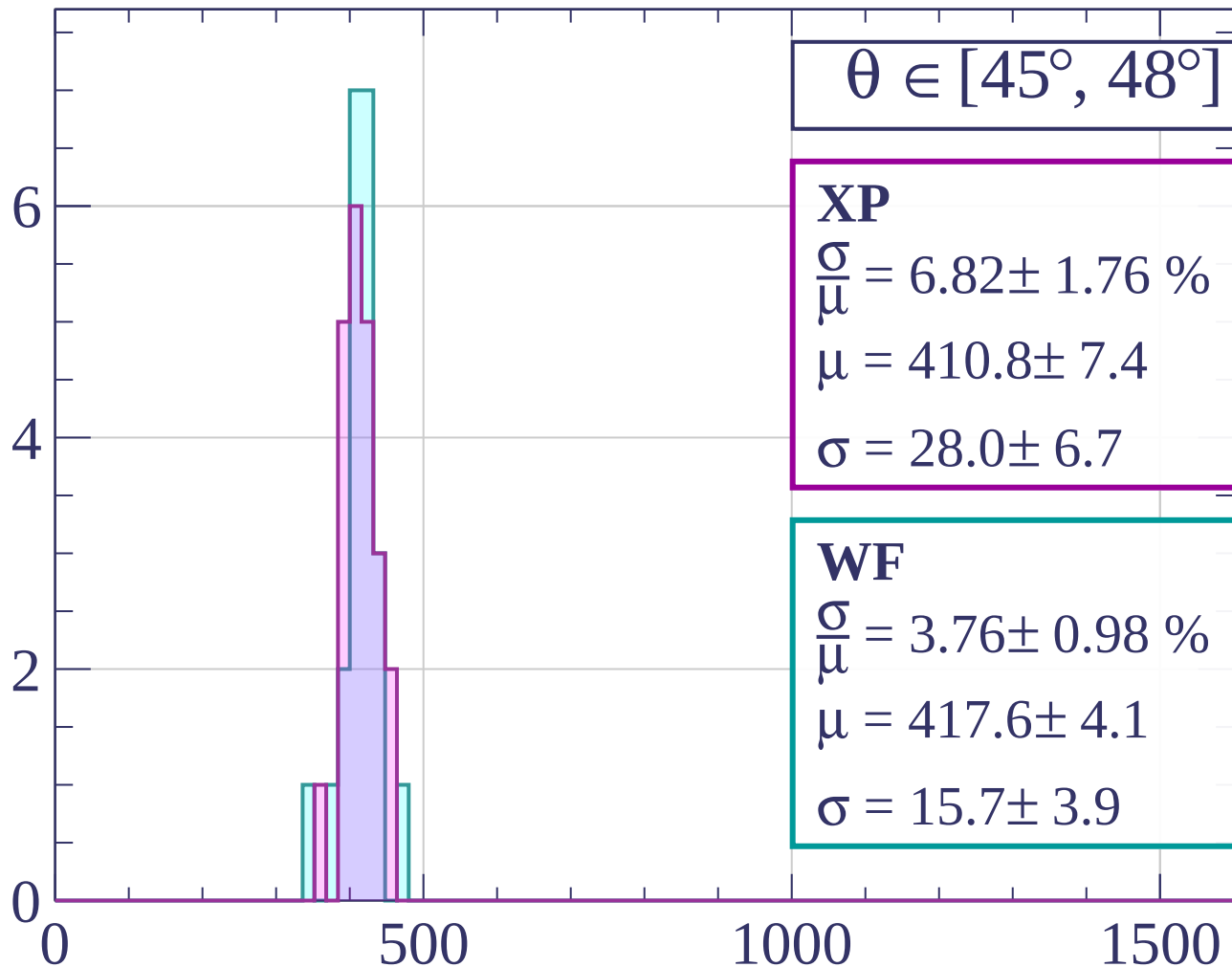
dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count

$\theta \in [48^\circ, 51^\circ]$

XP

$$\frac{\sigma}{\mu} = 14.83 \pm 10.38 \%$$

$$\mu = 365.9 \pm 23.6$$

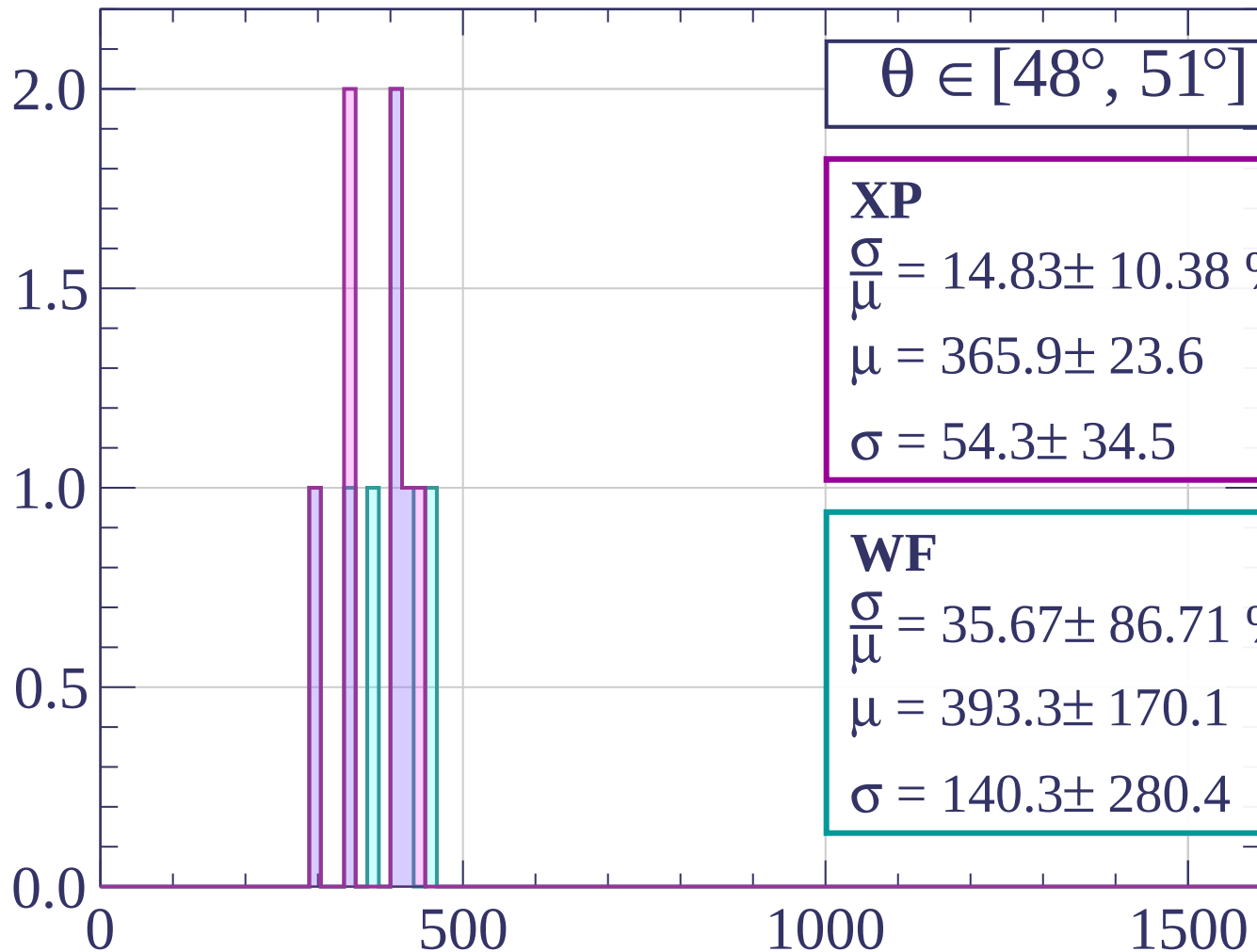
$$\sigma = 54.3 \pm 34.5$$

WF

$$\frac{\sigma}{\mu} = 35.67 \pm 86.71 \%$$

$$\mu = 393.3 \pm 170.1$$

$$\sigma = 140.3 \pm 280.4$$



dE/dx [ADC counts/cm]

Count

$\theta \in [51^\circ, 54^\circ]$

XP

$$\frac{\sigma}{\mu} = 19.29 \pm 40.88 \%$$

$$\mu = 886.3 \pm 975.4$$

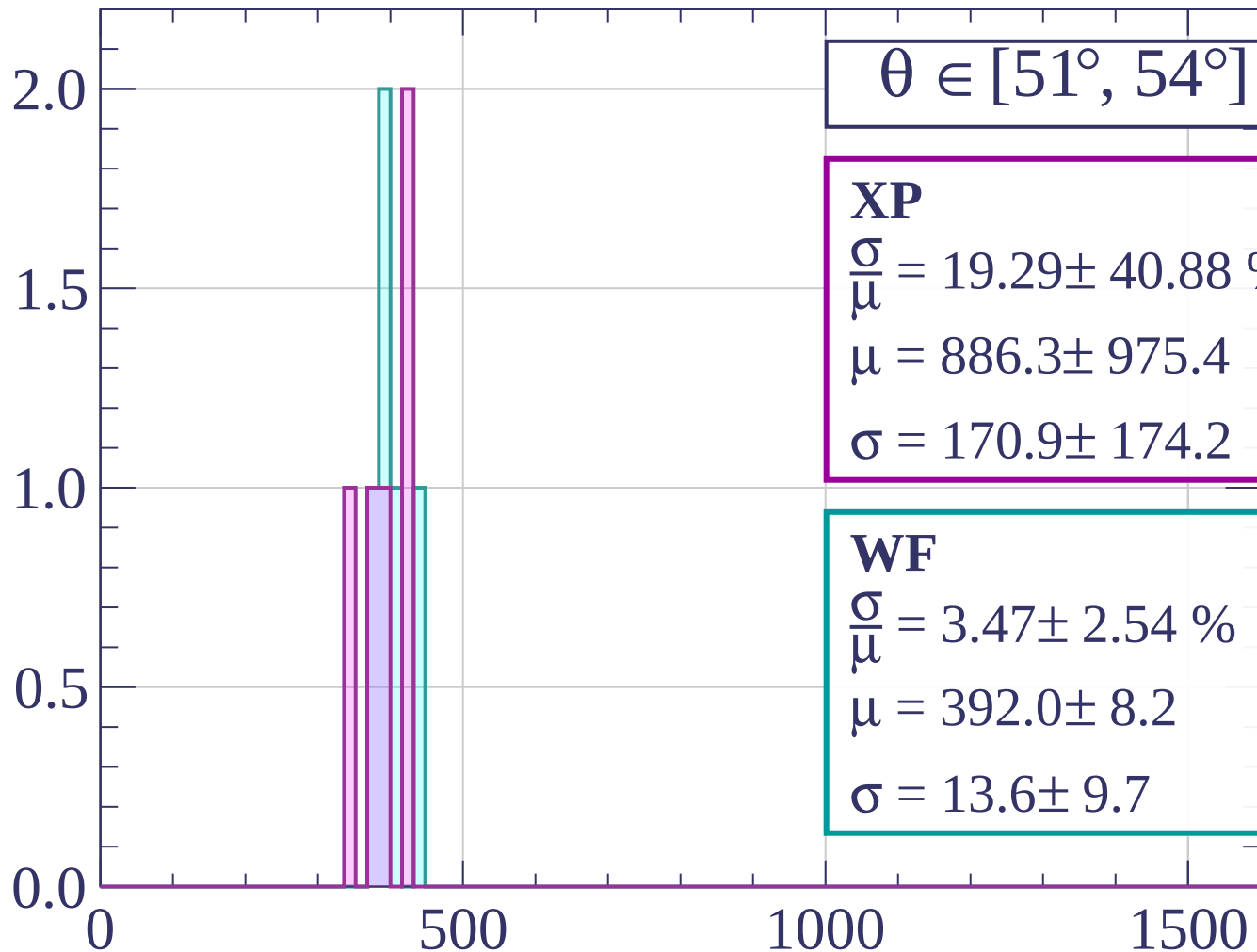
$$\sigma = 170.9 \pm 174.2$$

WF

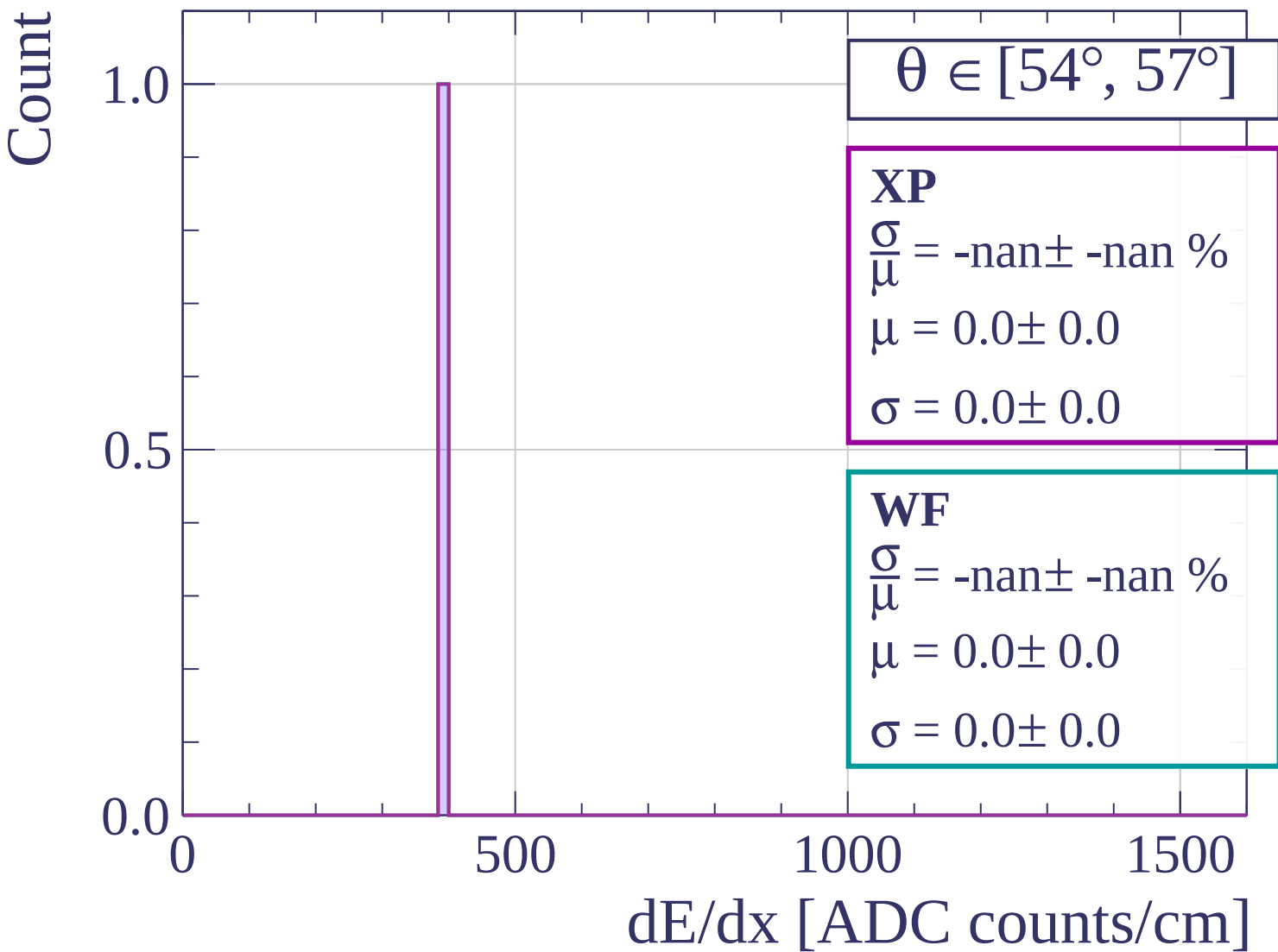
$$\frac{\sigma}{\mu} = 3.47 \pm 2.54 \%$$

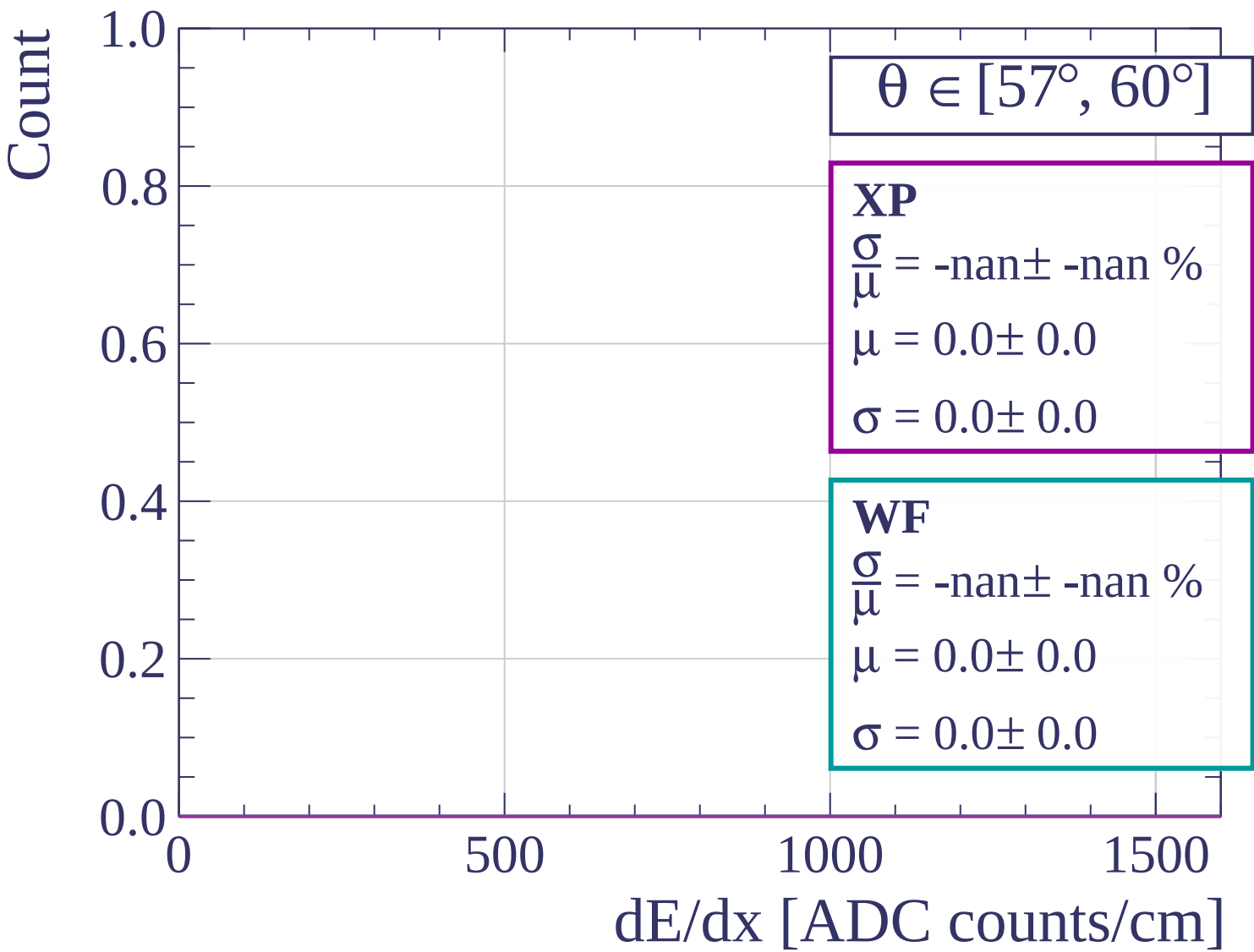
$$\mu = 392.0 \pm 8.2$$

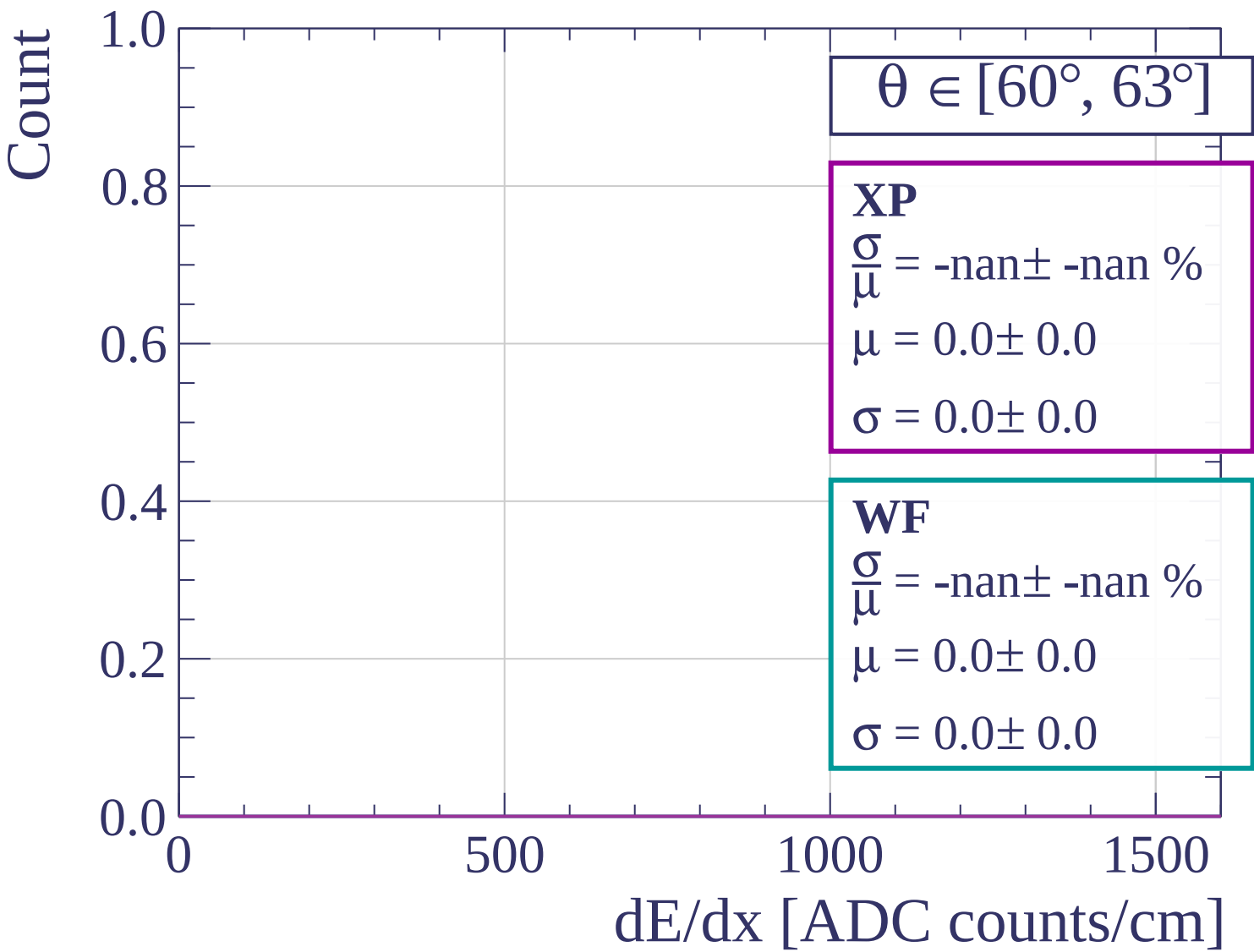
$$\sigma = 13.6 \pm 9.7$$

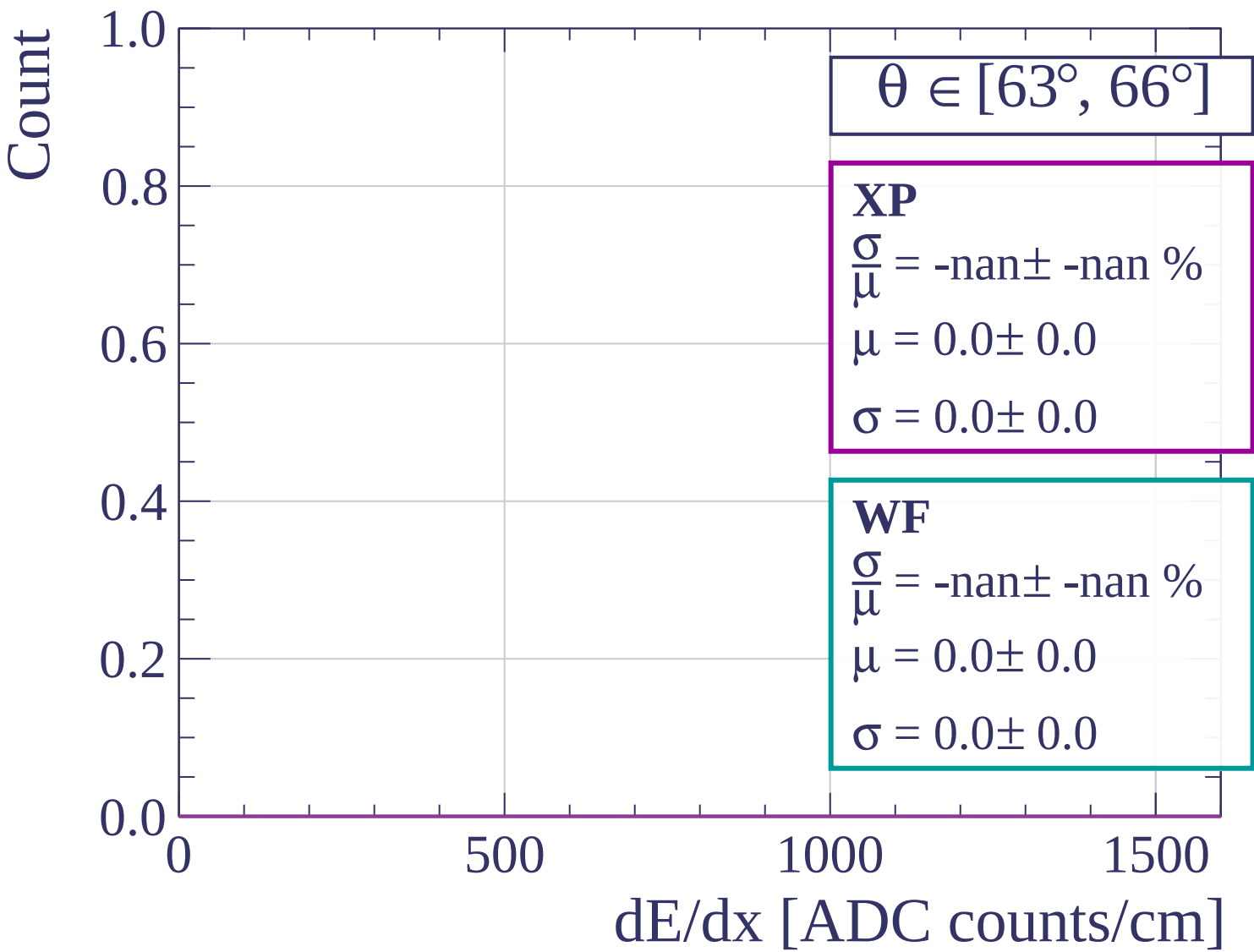


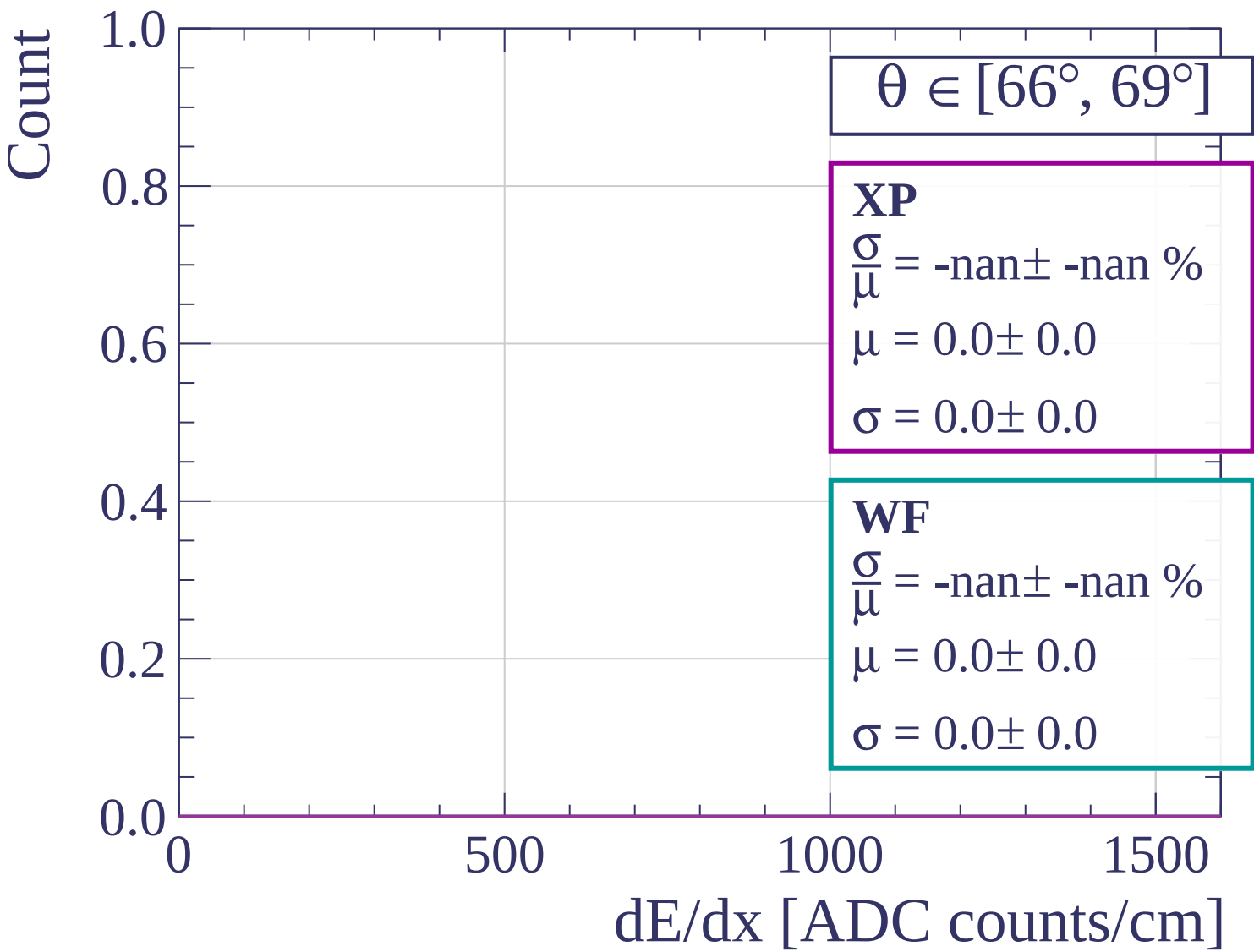
dE/dx [ADC counts/cm]

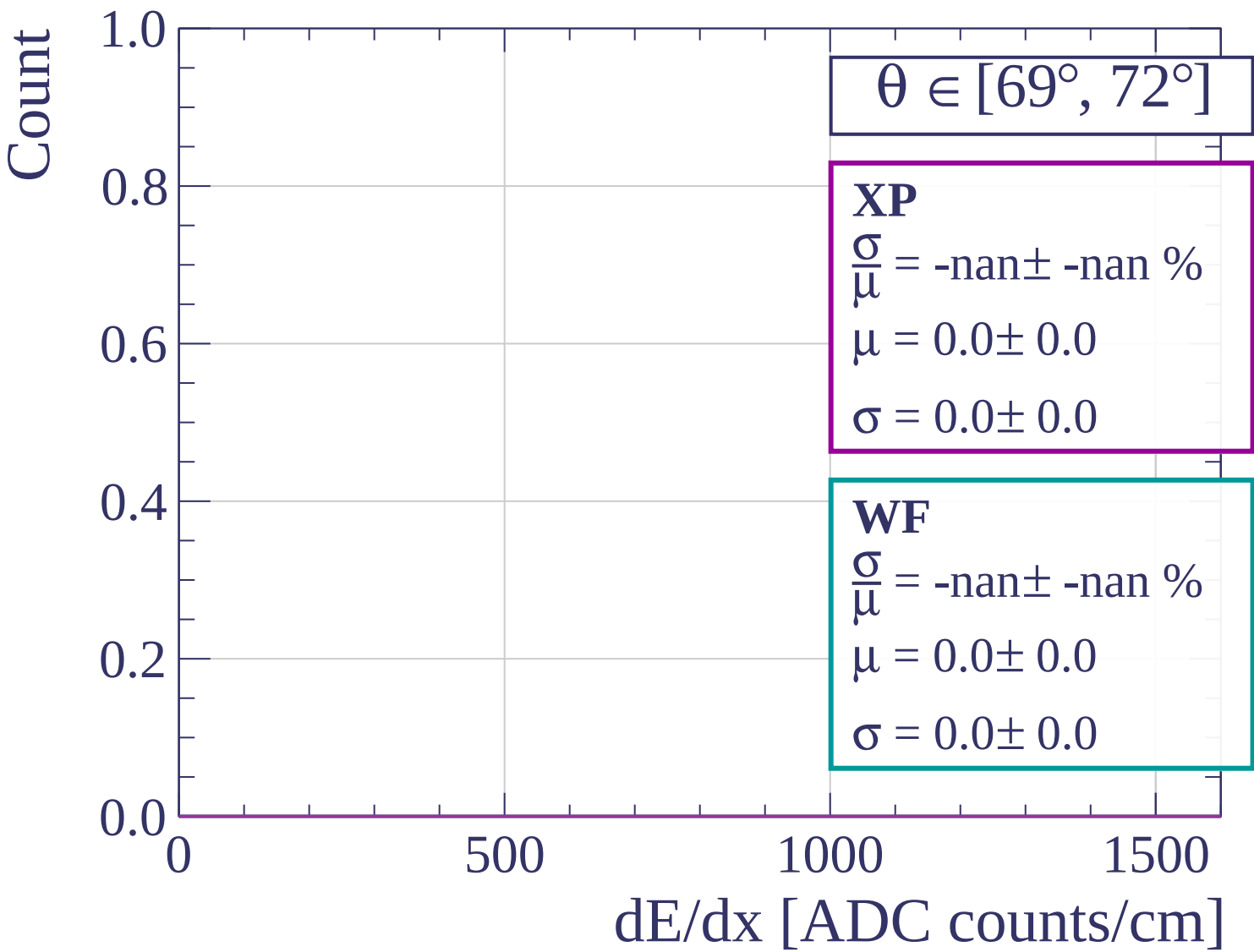


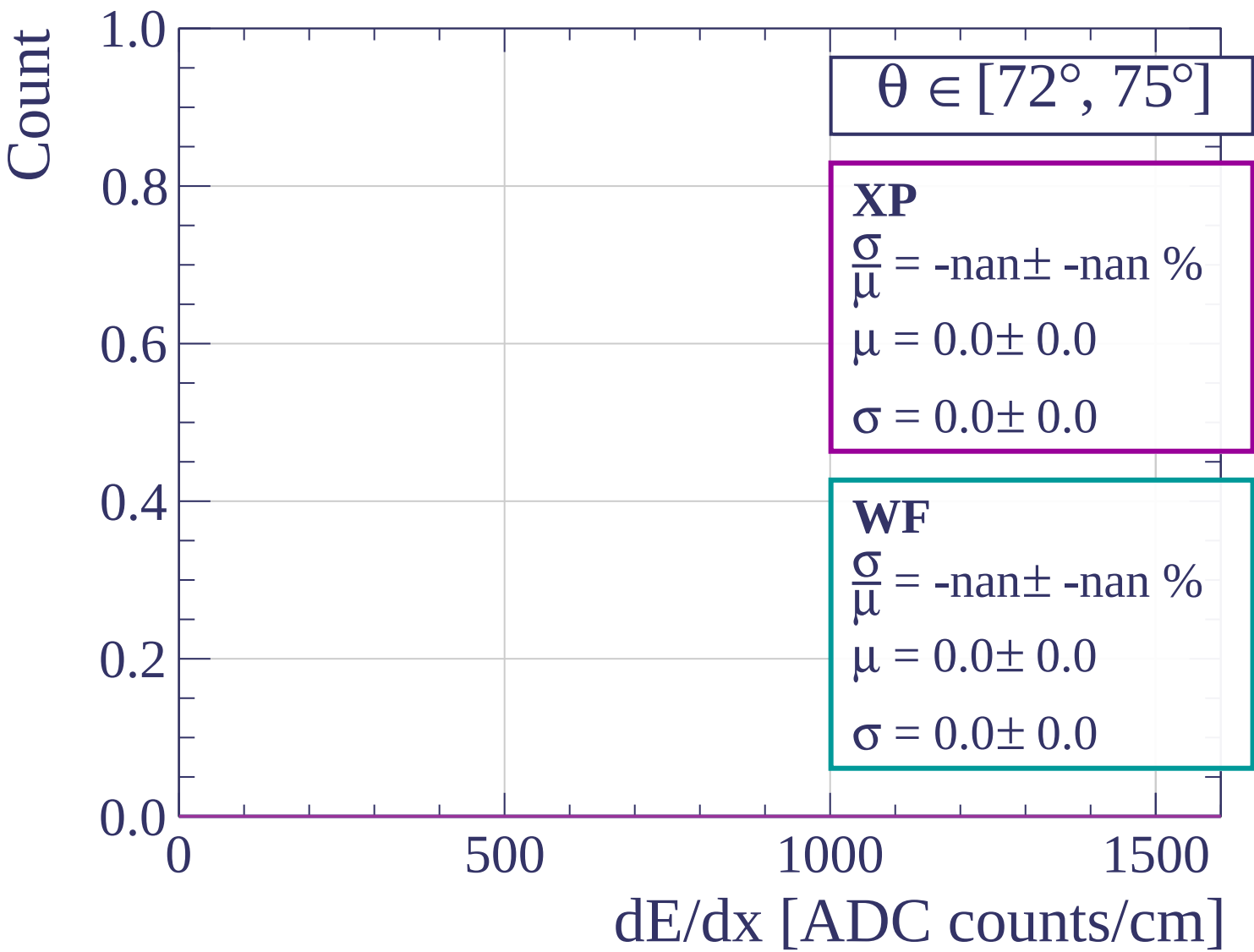


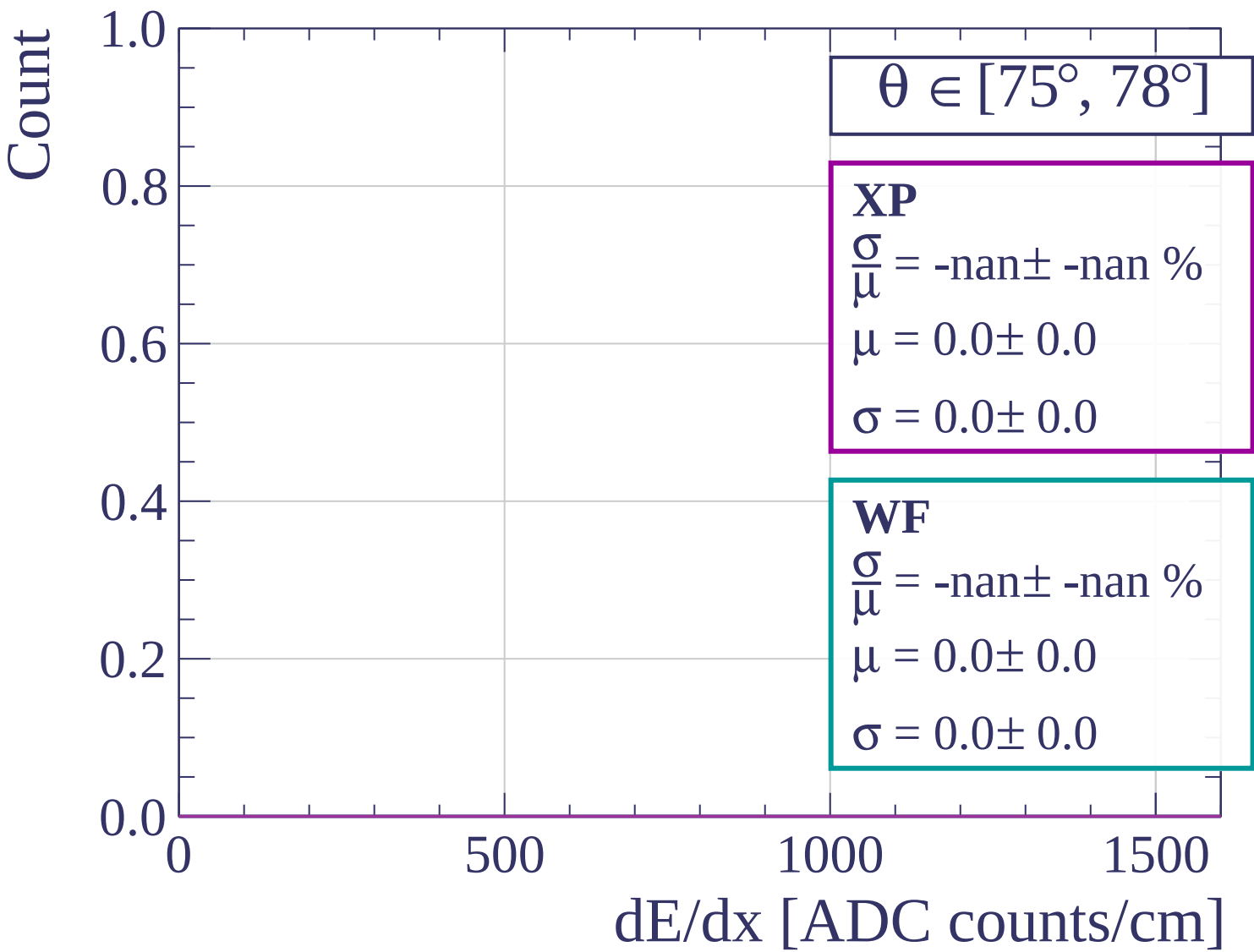


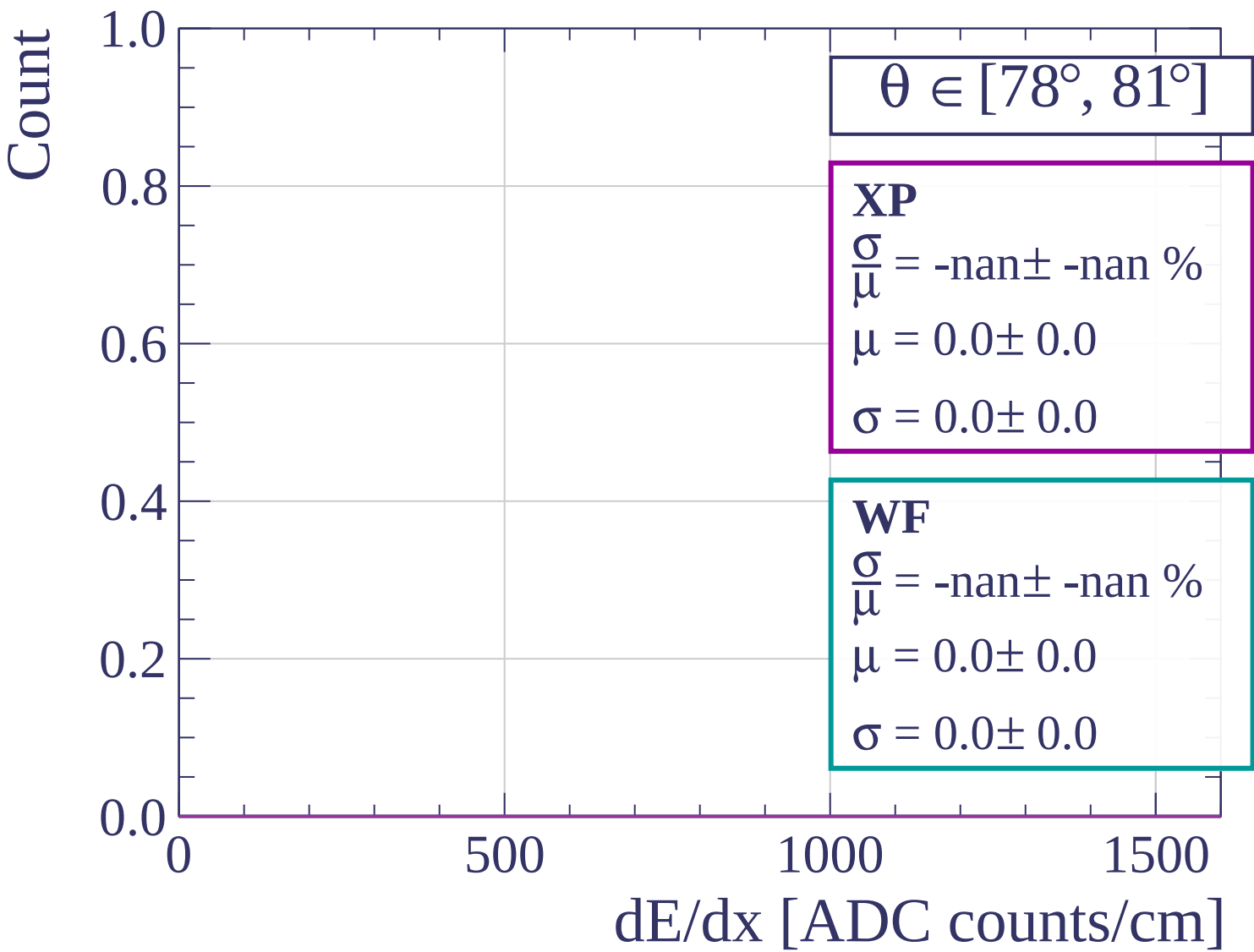


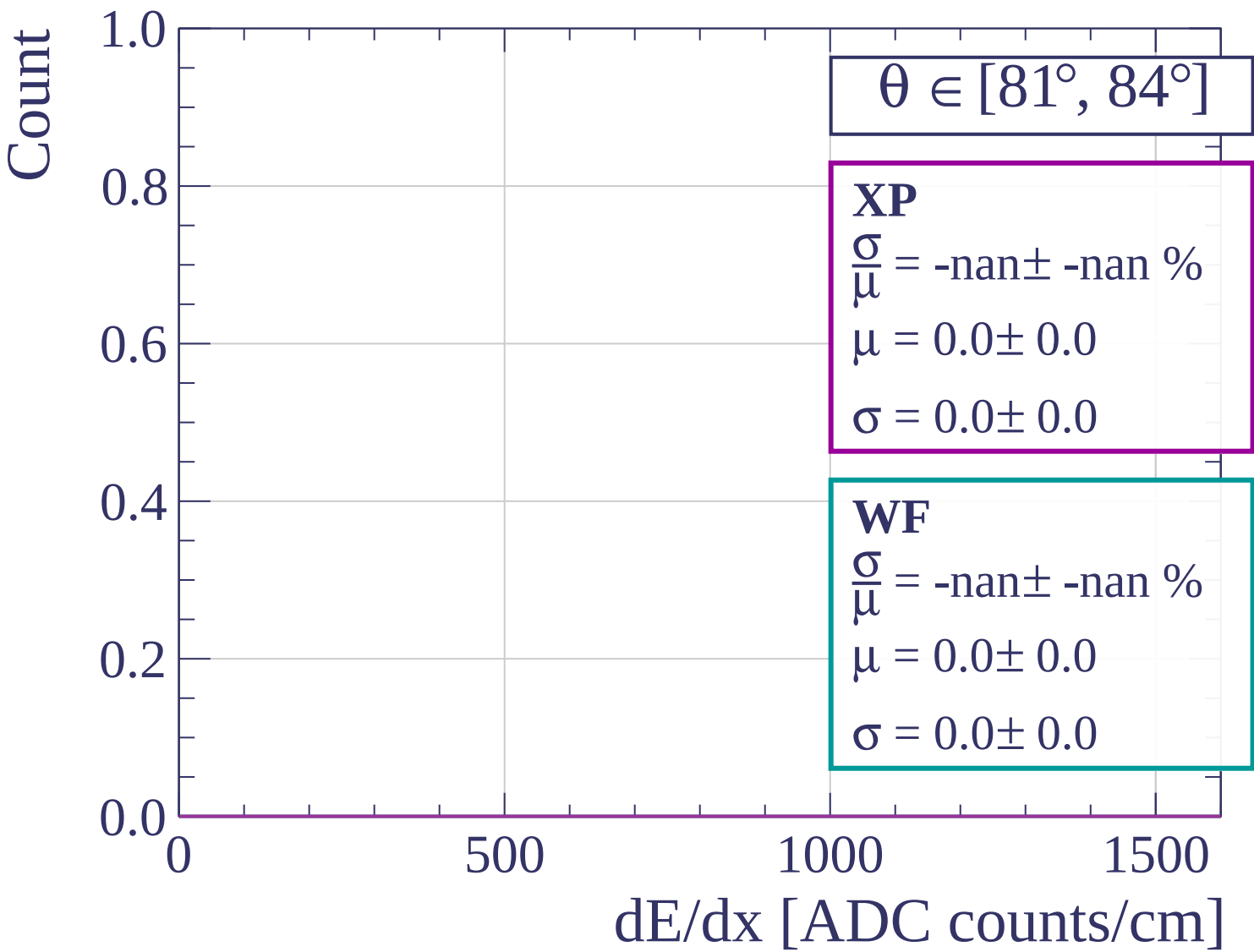


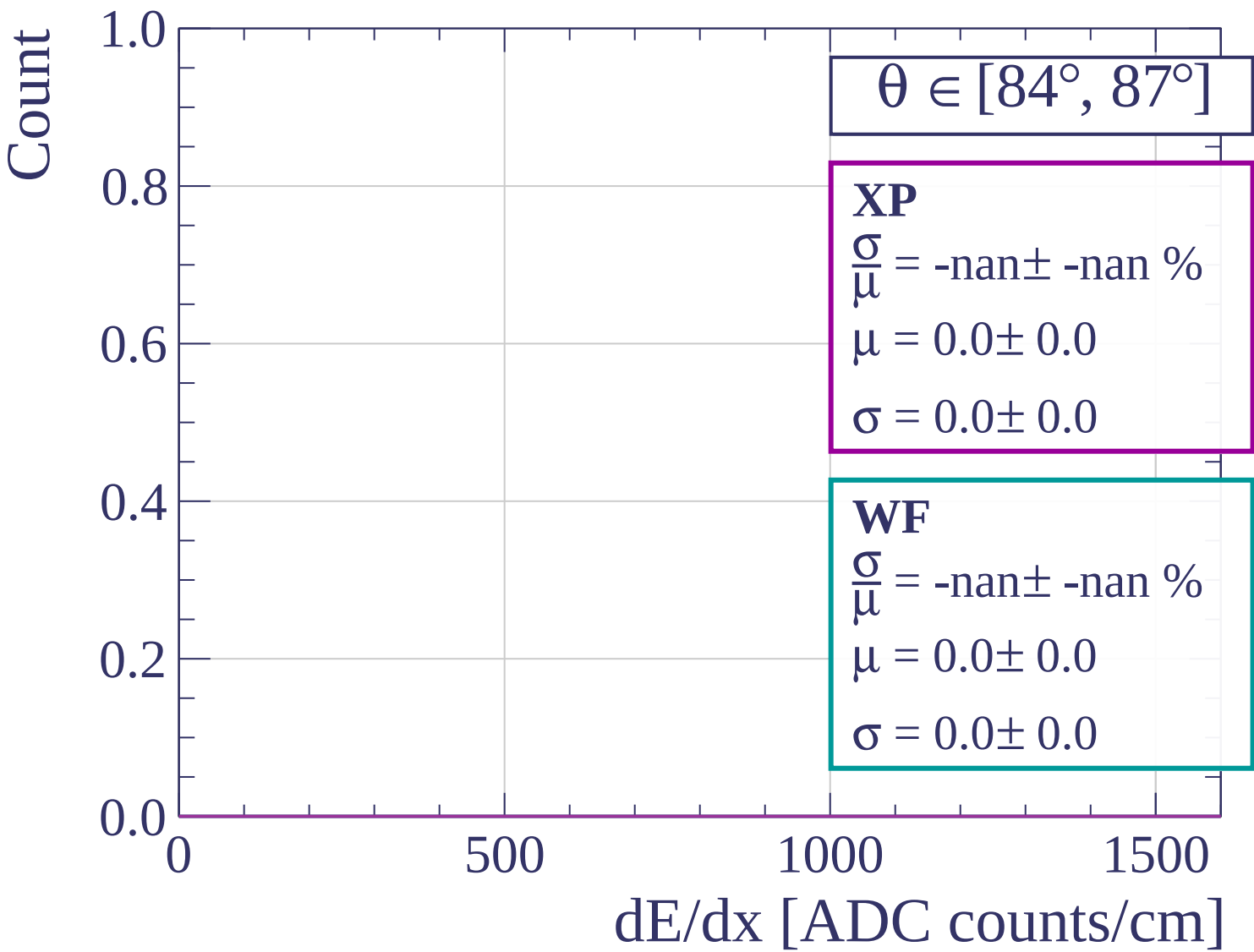


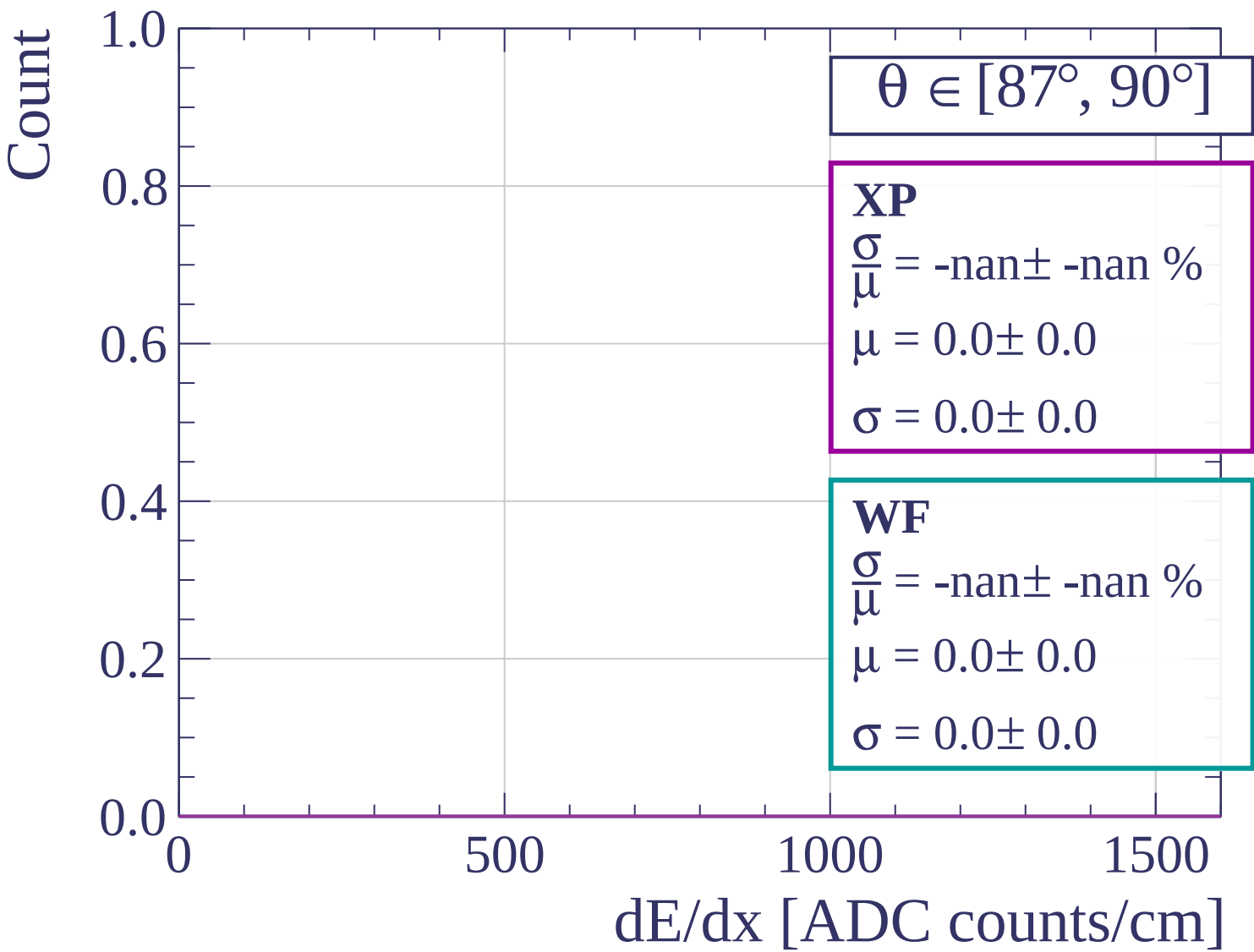




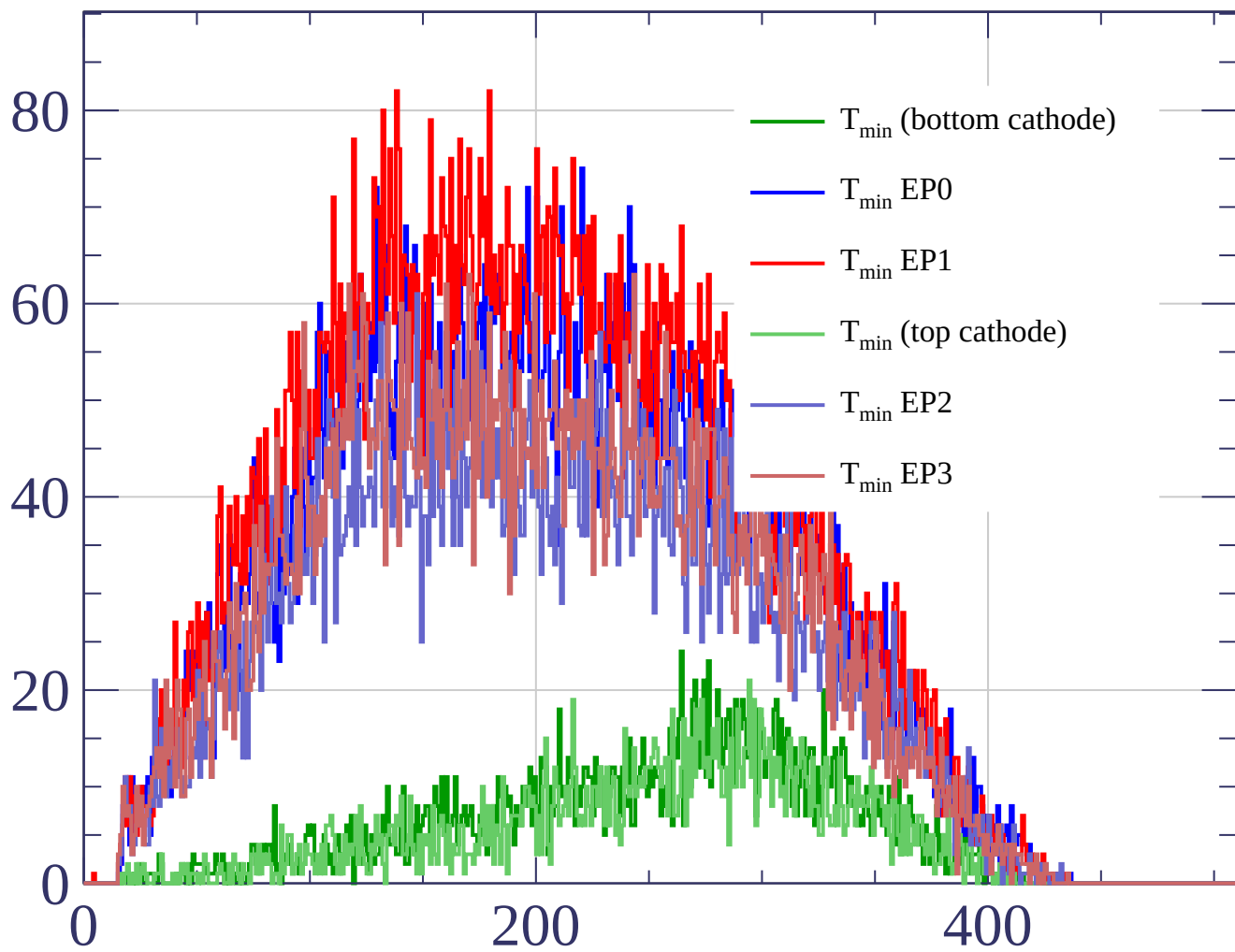






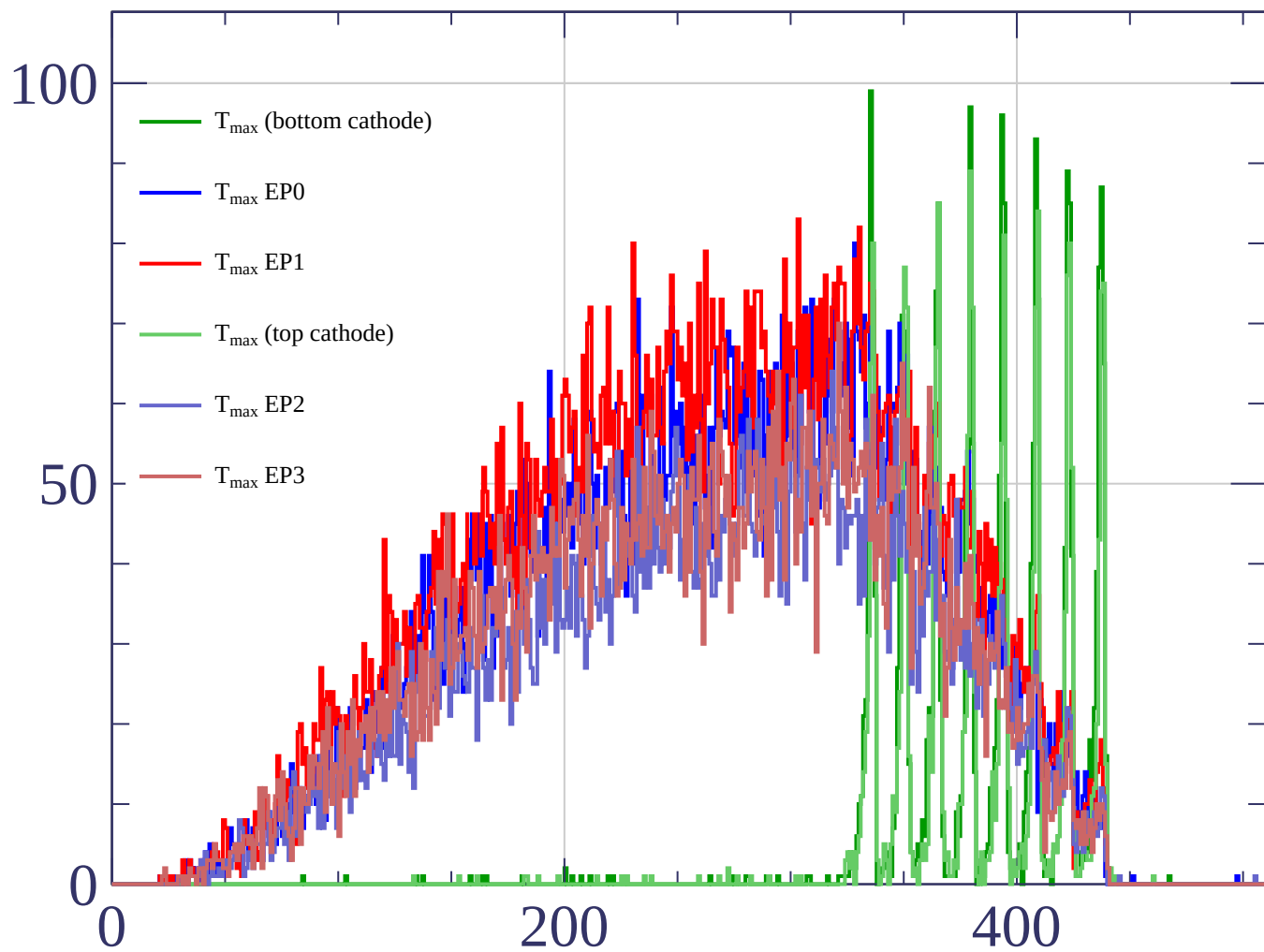


Count



time bin

Count



time bin